

Code Export

Source Code PDF Report

Generated 2026-05-05 10:18

Root directory C:\Users\LENOVO\Documents\Java_run\Smart_EmployeeManagementSystem\SmartEmployeeManagementSys

Files found 173

Total lines 21,035

Table of Contents

1. Parent: README.md (3 lines)
2. Current: .mvn\wrapper\maven-wrapper.properties (4 lines)
3. Current: pom.xml (138 lines)
4. Current: src\main\java\com\example\EmployeeManagementSystem\code\CodeToPdf.java (348 lines)
5. Current: src\main\java\com\example\EmployeeManagementSystem\config\BasicAuthConfig.java (52 lines)
6. Current: src\main\java\com\example\EmployeeManagementSystem\config\JWTauth\JWTAuthConfig.java (197 lines)
7. Current: src\main\java\com\example\EmployeeManagementSystem\config\LeaveTypeDataInitializer.java (44 lines)
8. Current: src\main\java\com\example\EmployeeManagementSystem\config\Seassion\SessionAuthConfig.java (32 lines)
9. Current: src\main\java\com\example\EmployeeManagementSystem\config\swaggerConfig.java (38 lines)
10. Current: src\main\java\com\example\EmployeeManagementSystem\Controller\AdminController.java (182 lines)
11. Current: src\main\java\com\example\EmployeeManagementSystem\Controller\ApiKeyController.java (37 lines)
12. Current: src\main\java\com\example\EmployeeManagementSystem\Controller\AuthController.java (184 lines)
13. Current: src\main\java\com\example\EmployeeManagementSystem\Controller\DeliveryController.java (33 lines)
14. Current: src\main\java\com\example\EmployeeManagementSystem\Controller\DeviceController.java (71 lines)
15. Current: src\main\java\com\example\EmployeeManagementSystem\Controller\EmployeeController.java (59 lines)
16. Current: src\main\java\com\example\EmployeeManagementSystem\Controller\GlobalExceptionHandler.java (132 lines)
17. Current: src\main\java\com\example\EmployeeManagementSystem\Controller\HolidayController.java (40 lines)
18. Current: src\main\java\com\example\EmployeeManagementSystem\Controller\LeaveDashboardController.java (119 lines)
19. Current: src\main\java\com\example\EmployeeManagementSystem\Controller\LeaveRequestController.java (72 lines)
20. Current: src\main\java\com\example\EmployeeManagementSystem\Controller>LoginController.java (17 lines)
21. Current: src\main\java\com\example\EmployeeManagementSystem\Controller\ManagerController.java (72 lines)
22. Current: src\main\java\com\example\EmployeeManagementSystem\Controller\NotificationController.java (60 lines)
23. Current: src\main\java\com\example\EmployeeManagementSystem\Controller\OauthController.java (52 lines)
24. Current: src\main\java\com\example\EmployeeManagementSystem\Controller\RestaurantController.java (124 lines)
25. Current: src\main\java\com\example\EmployeeManagementSystem\Controller\SchedulerController.java (103 lines)

26. Current: src\main\java\com\example\EmployeeManagementSystem\Controller\ServiceRequestController.java (74 lines)

27. Current: src\main\java\com\example\EmployeeManagementSystem\Controller\SessionAuthController.java (96 lines)

28. Current: src\main\java\com\example\EmployeeManagementSystem\Controller\SubscriptionController.java (73 lines)

29. Current: src\main\java\com\example\EmployeeManagementSystem\Controller\TotpController.java (134 lines)

30. Current: src\main\java\com\example\EmployeeManagementSystem\Controller\VendorController.java (59 lines)

31. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\AccessTokenRequest.java (18 lines)

32. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\ActionDTO.java (34 lines)

33. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\AdminActionDTO.java (34 lines)

34. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\AdminEmployeeDTO.java (61 lines)

35. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\AssignDeviceRequest.java (14 lines)

36. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\AuthRequest.java (28 lines)

37. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\AuthResponse.java (28 lines)

38. Current:
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39. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\DeviceDTO.java (45 lines)

40. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\DeviceResponseDTO.java (103 lines)

41. Current:
src\main\java\com\example\EmployeeManagementSystem\DTO\EmployeeAttendanceResponseDTO.java (43 lines)

42. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\EmployeeDetails.java (52 lines)

43. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\EmployeeDTO.java (61 lines)

44. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\EmployeeStatusDTO.java (71 lines)

45. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\HolidayDTO.java (50 lines)

46. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\LeaveDashboardResponse.java (122 lines)

47. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\LeaveRequestDTO.java (48 lines)

48. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\LeaveResponseDTO.java (82 lines)

49. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\ManagerDTO.java (43 lines)

50. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\ManagerEmployeeDetailsDTO.java (72 lines)

51. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\ManagerLeaveResponseDTO.java (73 lines)

52. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\MaternityLeaveRequestDTO.java (25 lines)

53. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\PromoteRequest.java (23 lines)

54. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\RepairDTO.java (34 lines)

55. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\RestaurantDTO.java (42 lines)

56. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\RestaurantRequest.java (46 lines)

57. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\ServiceRequestDTO.java (44 lines)

58. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\ServiceRequestResponseDTO.java (116 lines)

59. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\SubscriptionDTO.java (103 lines)

60. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\SubscriptionRequest.java (72 lines)

61. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\VendorDTO.java (26 lines)

62. Current: src\main\java\com\example\EmployeeManagementSystem\DTO\VendorRequest.java (47 lines)

63. Current:
src\main\java\com\example\EmployeeManagementSystem\EmployeeManagementSystemApplication.java (16 lines)

64. Current: src\main\java\com\example\EmployeeManagementSystem\Entity\ApiKey.java (79 lines)

65. Current: src\main\java\com\example\EmployeeManagementSystem\Entity\Delivery.java (93 lines)

66. Current: src\main\java\com\example\EmployeeManagementSystem\Entity\Device.java (107 lines)

67. Current: src\main\java\com\example\EmployeeManagementSystem\Entity\DeviceAssignment.java (68 lines)

68. Current: src\main\java\com\example\EmployeeManagementSystem\Entity\Employee.java (197 lines)

69. Current: src\main\java\com\example\EmployeeManagementSystem\Entity\FixedHoliday.java (61 lines)

70. Current: src\main\java\com\example\EmployeeManagementSystem\Entity\LeaveEntitlement.java (104 lines)

71. Current: src\main\java\com\example\EmployeeManagementSystem\Entity\LeaveRequest.java (105 lines)

72. Current: src\main\java\com\example\EmployeeManagementSystem\Entity\LeaveType.java (49 lines)

73. Current: src\main\java\com\example\EmployeeManagementSystem\Entity\LeaveWarning.java (58 lines)

74. Current: src\main\java\com\example\EmployeeManagementSystem\Entity\Notification.java (42 lines)

75. Current: src\main\java\com\example\EmployeeManagementSystem\Entity\RefreshToken.java (62 lines)

76. Current: src\main\java\com\example\EmployeeManagementSystem\Entity\Restaurant.java (58 lines)

77. Current: src\main\java\com\example\EmployeeManagementSystem\Entity\ServiceRequest.java (141 lines)

78. Current: src\main\java\com\example\EmployeeManagementSystem\Entity\Subscription.java (105 lines)

79. Current: src\main\java\com\example\EmployeeManagementSystem\Entity\Vendor.java (124 lines)

80. Current: src\main\java\com\example\EmployeeManagementSystem\Entity\YearEndCarryForwardLog.java (62 lines)

81. Current: src\main\java\com\example\EmployeeManagementSystem\Enum\AssignmentStatus.java (8 lines)

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83. Current: src\main\java\com\example\EmployeeManagementSystem\Enum\DeviceStatus.java (10 lines)

84. Current: src\main\java\com\example\EmployeeManagementSystem\Enum\DeviceType.java (15 lines)

85. Current: src\main\java\com\example\EmployeeManagementSystem\Enum\Gender.java (5 lines)

86. Current: src\main\java\com\example\EmployeeManagementSystem\Enum\LeaveStatus.java (9 lines)

87. Current: src\main\java\com\example\EmployeeManagementSystem\Enum\LeaveType.java (9 lines)

88. Current: src\main\java\com\example\EmployeeManagementSystem\Enum\MealSlot.java (8 lines)

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94. Current: src\main\java\com\example\EmployeeManagementSystem\Enum\Status.java (8 lines)

95. Current: src\main\java\com\example\EmployeeManagementSystem\Enum\SubscriptionStatus.java (8 lines)

96. Current: src\main\java\com\example\EmployeeManagementSystem\Exception\DuplicateRequestException.java (8 lines)

97. Current: src\main\java\com\example\EmployeeManagementSystem\Exception\EmployeeNotFound.java (10 lines)

98. Current: src\main\java\com\example\EmployeeManagementSystem\Exception\InvalidEndDateException.java (8 lines)

99. Current: src\main\java\com\example\EmployeeManagementSystem\Exception\InvalidManagerException.java (8 lines)

100. Current: src\main\java\com\example\EmployeeManagementSystem\Exception\InvalidStartDateException.java (8 lines)

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src\main\java\com\example\EmployeeManagementSystem\Exception\LeaveRequestNotFoundException.java (8 lines)

102. Current:
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103. Current:
src\main\java\com\example\EmployeeManagementSystem\Exception\RestaurantNotFoundException.java (8 lines)

104. Current: src\main\java\com\example\EmployeeManagementSystem\Exception\SubscriptionAlreadyExists.java (8 lines)

105. Current:
src\main\java\com\example\EmployeeManagementSystem\Exception\UnauthorizedAccessException.java (7 lines)

106. Current: src\main\java\com\example\EmployeeManagementSystem\Exception\VendorNotFoundException.java (7 lines)

107. Current: src\main\java\com\example\EmployeeManagementSystem\Filter\ApiKeyFilter.java (70 lines)

108. Current: src\main\java\com\example\EmployeeManagementSystem\Filter\JwtAuthFilter.java (83 lines)

109. Current: src\main\java\com\example\EmployeeManagementSystem\Repository\ApiKeyRepository.java (34 lines)

110. Current: src\main\java\com\example\EmployeeManagementSystem\Repository\DeliveryRepository.java (23 lines)

111. Current: src\main\java\com\example\EmployeeManagementSystem\Repository\DeviceAssignmentRepo.java (21 lines)

112. Current: src\main\java\com\example\EmployeeManagementSystem\Repository\DeviceRepository.java (13 lines)

113. Current: src\main\java\com\example\EmployeeManagementSystem\Repository\EmployeeRepo.java (26 lines)

114. Current: src\main\java\com\example\EmployeeManagementSystem\Repository\FixedHolidayRepository.java (13 lines)

115. Current:
src\main\java\com\example\EmployeeManagementSystem\Repository\LeaveEntitlementRepository.java (31 lines)

116. Current: src\main\java\com\example\EmployeeManagementSystem\Repository\LeaveRequestRepo.java (134 lines)

117. Current: src\main\java\com\example\EmployeeManagementSystem\Repository\LeaveTypeRepository.java (13 lines)

118. Current: src\main\java\com\example\EmployeeManagementSystem\Repository\LeaveWarningRepository.java (21 lines)

119. Current: src\main\java\com\example\EmployeeManagementSystem\Repository\NotificationRepository.java (12 lines)

120. Current: src\main\java\com\example\EmployeeManagementSystem\Repository\RefreshTokenRepository.java (19 lines)

121. Current: src\main\java\com\example\EmployeeManagementSystem\Repository\RestaurantRepository.java (31 lines)

122. Current:
src\main\java\com\example\EmployeeManagementSystem\Repository\ServiceRequestRepository.java (23 lines)

123. Current: src\main\java\com\example\EmployeeManagementSystem\Repository\SubscriptionRepository.java (32 lines)

124. Current: src\main\java\com\example\EmployeeManagementSystem\Repository\VendorRepo.java (13 lines)

125. Current:
src\main\java\com\example\EmployeeManagementSystem\Repository\YearEndCarryForwardLogRepository.java (18 lines)

126. Current: src\main\java\com\example\EmployeeManagementSystem\Scheduler\DeliveryScheduler.java (30 lines)

127. Current: src\main\java\com\example\EmployeeManagementSystem\Scheduler\LeaveAccrualScheduler.java (60 lines)

128. Current: src\main\java\com\example\EmployeeManagementSystem\Scheduler\LeaveStatusScheduler.java (107 lines)

129. Current:
src\main\java\com\example\EmployeeManagementSystem\security\ApiKeyAuthenticationProvider.java (85 lines)

130. Current: src\main\java\com\example\EmployeeManagementSystem\security\ApiKeyAuthenticationToken.java (84 lines)

131. Current: src\main\java\com\example\EmployeeManagementSystem\Service\ApiKeyService.java (52 lines)

132. Current:
src\main\java\com\example\EmployeeManagementSystem\Service\CarryForwardWarningService.java (126 lines)

133. Current: src\main\java\com\example\EmployeeManagementSystem\Service\CombinedUserDetailService.java (41 lines)

134. Current: src\main\java\com\example\EmployeeManagementSystem\Service\CustomOAuth2UserService.java (87 lines)

135. Current: src\main\java\com\example\EmployeeManagementSystem\Service\DeliveryService.java (178 lines)

136. Current: src\main\java\com\example\EmployeeManagementSystem\Service\DeliveryTimeService.java (74 lines)

137. Current: src\main\java\com\example\EmployeeManagementSystem\Service\DeviceService.java (193 lines)

138. Current: src\main\java\com\example\EmployeeManagementSystem\Service\EmployeeService.java (214 lines)

139. Current: src\main\java\com\example\EmployeeManagementSystem\Service\HolidayService.java (39 lines)

140. Current: src\main\java\com\example\EmployeeManagementSystem\Service\LeaveAccrualService.java (128 lines)

141. Current: src\main\java\com\example\EmployeeManagementSystem\Service\LeaveDashboardService.java (215 lines)

142. Current: src\main\java\com\example\EmployeeManagementSystem\Service\LeaveRequestService.java (542 lines)

143. Current: src\main\java\com\example\EmployeeManagementSystem\Service\ManagerService.java (219 lines)

144. Current: src\main\java\com\example\EmployeeManagementSystem\Service\MaternityLeaveService.java (129 lines)

145. Current: src\main\java\com\example\EmployeeManagementSystem\Service\MealSlotService.java (22 lines)

146. Current: src\main\java\com\example\EmployeeManagementSystem\Service\NotificationService.java (57 lines)

147. Current: src\main\java\com\example\EmployeeManagementSystem\Service\RefreshTokenService.java (41 lines)

148. Current: src\main\java\com\example\EmployeeManagementSystem\Service\RestaurantService.java (162 lines)

149. Current: src\main\java\com\example\EmployeeManagementSystem\Service\ServiceRequestService.java (209 lines)

150. Current: src\main\java\com\example\EmployeeManagementSystem\Service\SickLeaveResetService.java (67 lines)

151. Current: src\main\java\com\example\EmployeeManagementSystem\Service\SubscriptionService.java (228 lines)

152. Current: src\main\java\com\example\EmployeeManagementSystem\Service\TotpService.java (45 lines)

153. Current:
src\main\java\com\example\EmployeeManagementSystem\Service\YearEndCarryForwardService.java (224 lines)

154. Current: src\main\java\com\example\EmployeeManagementSystem\Util\AuthUtil.java (21 lines)

155. Current: src\main\java\com\example\EmployeeManagementSystem\Util\JWTUtil.java (76 lines)

156. Current: src\main\java\com\example\EmployeeManagementSystem\Util\OAuth2SuccessHandler.java (85 lines)

157. Current: src\main\resources\application.yaml (69 lines)

158. Current: src\main\resources\static\dashboard.html (2,061 lines)

159. Current: src\main\resources\static\employee-dashboard.html (1,596 lines)

160. Current: src\main\resources\static\google.html (764 lines)

161. Current: src\main\resources\static\leave_management.html (770 lines)

162. Current: src\main\resources\static\login.html (10 lines)

163. Current: src\main\resources\static\manager-dashboard.html (2,162 lines)

164. Current: src\main\resources\static\pending.html (76 lines)

165. Current: src\main\resources\static\tech-vendor-dashboard.html (991 lines)

166. Current: src\main\resources\static\otp-setup.html (442 lines)

167. Current: src\main\resources\static\vendor-dashboard.html (529 lines)

168. Current: src\main\resources\static\vendor-login.html (396 lines)

169. Current:
src\test\java\com\example\EmployeeManagementSystem\EmployeeManagementSystemApplicationTests.java (14 lines)

170. Current: src\test\java\com\example\EmployeeManagementSystem\service\LeaveAccrualServiceTest.java (112 lines)

171. Current: src\test\java\com\example\EmployeeManagementSystem\service\LeaveRequestServiceTest.java (86 lines)

172. Current: src\test\java\com\example\EmployeeManagementSystem\service\SickLeaveResetServiceTest.java (135 lines)

173. Current:
src\test\java\com\example\EmployeeManagementSystem\service\YearEndCarryForwardServiceTest.java (137 lines)

1 Parent: README.md

3 lines

```
1 # SmartEmployeeManagementSystem
2 A secure and scalable Spring Boot backend for employee management, leave tracking, and meal subscription
services with JWT, OAuth2, API key authentication, and role-based access control.
3
```

2 Current: .mvn\wrapper\maven-wrapper.properties

4 lines

```
1 wrapperVersion=3.3.4
2 distributionType=only-script
3 distributionUrl=https://repo.maven.apache.org/maven2/org/apache/maven/apache-maven/3.9.14/apache-maven-
3.9.14-bin.zip
4
```

```

1  <?xml version="1.0" encoding="UTF-8"?>
2  <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3      xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">
4      <modelVersion>4.0.0</modelVersion>
5      <parent>
6          <groupId>org.springframework.boot</groupId>
7          <artifactId>spring-boot-starter-parent</artifactId>
8          <version>4.0.5</version>
9          <relativePath/> <!-- lookup parent from repository -->
10     </parent>
11     <groupId>com.example</groupId>
12     <artifactId>EmployeeLeaveManagementSystem</artifactId>
13     <version>0.0.1-SNAPSHOT</version>
14     <name/>
15     <description/>
16     <url/>
17     <licenses>
18         <license/>
19     </licenses>
20     <developers>
21         <developer/>
22     </developers>
23     <scm>
24         <connection/>
25         <developerConnection/>
26         <tag/>
27         <url/>
28     </scm>
29     <properties>
30         <java.version>21</java.version>
31     </properties>
32     <dependencies>
33         <dependency>
34             <groupId>org.springframework.boot</groupId>
35             <artifactId>spring-boot-starter-data-jpa</artifactId>
36         </dependency>
37         <dependency>
38             <groupId>org.springframework.boot</groupId>
39             <artifactId>spring-boot-starter-webmvc</artifactId>
40         </dependency>
41         <dependency>
42             <groupId>org.springdoc</groupId>
43             <artifactId>springdoc-openapi-starter-webmvc-ui</artifactId>
44             <version>3.0.2</version>
45         </dependency>
46         <dependency>
47             <groupId>dev.samstevens.totp</groupId>
48             <artifactId>totp-spring-boot-starter</artifactId>
49             <version>1.7.1</version>
50         </dependency>
51         <dependency>
52             <groupId>org.springframework.session</groupId>
53             <artifactId>spring-session-core</artifactId>
54         </dependency>
55         <!-- TOTP -->
56         <dependency>
57             <groupId>dev.samstevens.totp</groupId>
58             <artifactId>totp-spring-boot-starter</artifactId>
59             <version>1.7.1</version>
60         </dependency>
61         <!-- -->
62         <dependency>
63             <groupId>org.springframework.boot</groupId>
64             <artifactId>spring-boot-devtools</artifactId>
65             <scope>runtime</scope>
66             <optional>true</optional>
67         </dependency>
68         <dependency>
69             <groupId>com.mysql</groupId>
70             <artifactId>mysql-connector-j</artifactId>
71             <scope>runtime</scope>
72     </dependencies>

```

```

73     <dependency>
74         <groupId>org.springframework.boot</groupId>
75         <artifactId>spring-boot-starter-data-jpa-test</artifactId>
76         <scope>test</scope>
77     </dependency>
78     <dependency>
79         <groupId>com.itextpdf</groupId>
80         <artifactId>layout</artifactId>
81         <version>7.2.5</version>
82     </dependency>
83     <dependency>
84         <groupId>org.springframework.boot</groupId>
85         <artifactId>spring-boot-starter-security</artifactId>
86     </dependency>
87
88     <dependency>
89         <groupId>org.springframework.boot</groupId>
90         <artifactId>spring-boot-starter-security-test</artifactId>
91         <scope>test</scope>
92     </dependency>
93     <dependency>
94         <groupId>org.springframework.boot</groupId>
95         <artifactId>spring-boot-starter-webmvc-test</artifactId>
96         <scope>test</scope>
97     </dependency>
98     <dependency>
99         <groupId>io.jsonwebtoken</groupId>
100        <artifactId>jjwt-api</artifactId>
101        <version>0.13.0</version>
102        <scope>compile</scope>
103    </dependency>
104    <dependency>
105        <groupId>io.jsonwebtoken</groupId>
106        <artifactId>jjwt-impl</artifactId>
107        <version>0.13.0</version>
108        <scope>runtime</scope>
109    </dependency>
110    <dependency>
111        <groupId>io.jsonwebtoken</groupId>
112        <artifactId>jjwt-jackson</artifactId>
113        <version>0.13.0</version>
114        <scope>runtime</scope>
115    </dependency>
116    <dependency>
117        <groupId>org.springframework.boot</groupId>
118        <artifactId>spring-boot-starter-security-oauth2-client</artifactId>
119    </dependency>
120
121    <dependency>
122        <groupId>org.springframework.boot</groupId>
123        <artifactId>spring-boot-starter-security-oauth2-client-test</artifactId>
124        <scope>test</scope>
125    </dependency>
126 </dependencies>
127
128 <build>
129     <plugins>
130         <plugin>
131             <groupId>org.springframework.boot</groupId>
132             <artifactId>spring-boot-maven-plugin</artifactId>
133         </plugin>
134     </plugins>
135 </build>
136
137 </project>
138

```

```

1 package com.example.EmployeeManagementSystem.code;
2
3 import com.itextpdf.kernel.colors.ColorConstants;
4 import com.itextpdf.kernel.colors.DeviceRgb;
5 import com.itextpdf.kernel.font.PdfFont;
6 import com.itextpdf.kernel.font.PdfFontFactory;
7 import com.itextpdf.kernel.geom.PageSize;
8 import com.itextpdf.kernel.pdf.PdfDocument;
9 import com.itextpdf.kernel.pdf.PdfWriter;
10 import com.itextpdf.layout.Document;
11 import com.itextpdf.layout.element.AreaBreak;
12 import com.itextpdf.layout.element.Cell;
13 import com.itextpdf.layout.element.Paragraph;
14 import com.itextpdf.layout.element.Table;
15 import com.itextpdf.layout.element.Text;
16 import com.itextpdf.layout.properties.AreaBreakType;
17 import com.itextpdf.layout.properties.UnitValue;
18 import com.itextpdf.io.font.constants.StandardFonts;
19
20 import java.io.File;
21 import java.io.IOException;
22 import java.nio.file.*;
23 import java.nio.file.attribute.BasicFileAttributes;
24 import java.time.LocalDateTime;
25 import java.time.format.DateTimeFormatter;
26 import java.util.*;
27
28 /**
29  * CodeToPdf - Extracts all source code files from the current, parent,
30  * and immediate child directories into a formatted PDF document.
31  *
32  * Dependencies (add to your pom.xml or build.gradle):
33  *   com.itextpdf:itext7-core:7.2.5 (or itext7 bundle)
34  *
35  * Compile:
36  *   javac -cp itext7-core-7.2.5.jar:. CodeToPdf.java
37  *
38  * Run:
39  *   java -cp itext7-core-7.2.5.jar:. CodeToPdf
40  *
41  * Or with Maven:
42  *   mvn compile exec:java -Dexec.mainClass=CodeToPdf
43  */
44 public class CodeToPdf {
45
46     // Configuration
47
48     /** Output PDF filename */
49     private static final String OUTPUT_FILE = "code_export.pdf";
50
51     /** Maximum file size to include (skip very large generated files) */
52     private static final long MAX_FILE_SIZE_BYTES = 500_000; // 500 KB
53
54     /** Source code file extensions to include */
55     private static final Set<String> CODE_EXTENSIONS = new LinkedHashSet<>(Arrays.asList(
56         // JVM languages
57         ".java", ".kt", ".groovy", ".scala", ".clj",
58         // Web
59         ".js", ".ts", ".jsx", ".tsx", ".html", ".css", ".scss", ".less",
60         // Python / Ruby / PHP
61         ".py", ".rb", ".php",
62         // Systems / low-level
63         ".c", ".cpp", ".cc", ".h", ".hpp", ".rs", ".go", ".swift",
64         // Data / config
65         ".xml", ".json", ".yaml", ".yml", ".toml", ".properties", ".env",
66         // Shell / scripts
67         ".sh", ".bash", ".zsh", ".bat", ".ps1",
68         // Build
69         ".gradle", ".cmake", "Makefile", "Dockerfile",
70         // Other
71         ".sql", ".md", ".txt"

```

```

72     });
73
74     /** Directories to skip entirely */
75     private static final Set<String> SKIP_DIRS = new HashSet<>(Arrays.asList(
76         ".git", ".svn", ".hg", "node_modules", "target", "build",
77         "out", ".idea", ".vscode", "__pycache__", ".gradle", "dist", "vendor"
78     ));
79
80     // Colors
81     private static final DeviceRgb COLOR_HEADER_BG = new DeviceRgb(0x18, 0x5F, 0xA5); // blue
82     private static final DeviceRgb COLOR_HEADER_TEXT = new DeviceRgb(0xFF, 0xFF, 0xFF);
83     private static final DeviceRgb COLOR_FILE_TITLE = new DeviceRgb(0x18, 0x5F, 0xA5);
84     private static final DeviceRgb COLOR_CODE_BG = new DeviceRgb(0xF7, 0xF7, 0xF7);
85     private static final DeviceRgb COLOR_LINE_NUM = new DeviceRgb(0x99, 0x99, 0x99);
86     private static final DeviceRgb COLOR_TOC_HEADER = new DeviceRgb(0x18, 0x5F, 0xA5);
87
88     // Font sizes
89     private static final float FONT_TITLE = 20f;
90     private static final float FONT_SECTION = 13f;
91     private static final float FONT_FILE = 11f;
92     private static final float FONT_CODE = 7.5f;
93     private static final float FONT_TOC = 10f;
94
95     // State
96     private final List<FileEntry> fileEntries = new ArrayList<>();
97     private int totalLines = 0;
98
99     //
100
101     public static void main(String[] args) throws Exception {
102         new CodeToPdf().run();
103     }
104
105     private void run() throws Exception {
106         Path currentDir = Paths.get("").toAbsolutePath();
107         Path parentDir = currentDir.getParent();
108
109         System.out.println("=== Code to PDF Extractor ===");
110         System.out.println("Current directory : " + currentDir);
111         System.out.println("Parent directory : " + (parentDir != null ? parentDir : "(none)"));
112         System.out.println();
113
114         // 1. Collect files
115
116         // Parent directory (top-level files only, not recursive)
117         if (parentDir != null && Files.exists(parentDir)) {
118             collectFiles(parentDir, false, "Parent");
119         }
120
121         // Current directory (recursive – includes all child subdirectories)
122         collectFiles(currentDir, true, "Current");
123
124         System.out.printf("%nFound %d code file(s) with %,d total lines.%n",
125             fileEntries.size(), totalLines);
126
127         if (fileEntries.isEmpty()) {
128             System.out.println("No code files found. Exiting.");
129             return;
130         }
131
132         // 2. Generate PDF
133         generatePdf(currentDir);
134
135         System.out.println("PDF saved to: " + Paths.get(OUTPUT_FILE).toAbsolutePath());
136     }
137
138     // File Collection
139
140     private void collectFiles(Path root, boolean recursive, String scope) throws IOException {
141         if (!Files.exists(root)) return;
142
143         Files.walkFileTree(root, new SimpleFileVisitor<>() {
144             @Override
145             public FileVisitResult preVisitDirectory(Path dir, BasicFileAttributes attrs) {
146                 String name = dir.getFileName() == null ? "" : dir.getFileName().toString();
147                 // Never recurse into skip-dirs; always allow the root itself

```

```

148         if (!dir.equals(root) && SKIP_DIRS.contains(name)) {
149             return FileVisitResult.SKIP_SUBTREE;
150         }
151         // If not recursive, skip any subdirectory of root
152         if (!recursive && !dir.equals(root)) {
153             return FileVisitResult.SKIP_SUBTREE;
154         }
155         return FileVisitResult.CONTINUE;
156     }
157
158     @Override
159     public FileVisitResult visitFile(Path file, BasicFileAttributes attrs) throws IOException {
160         if (isCodeFile(file) && attrs.size() <= MAX_FILE_SIZE_BYTES) {
161             String content = Files.readString(file);
162             int lines = content.split("\n", -1).length;
163             String relativePath = scope + ": " + root.relativize(file);
164             fileEntries.add(new FileEntry(relativePath, file, content, lines));
165             totalLines += lines;
166             System.out.println("  " + relativePath + " (" + lines + " lines)");
167         }
168         return FileVisitResult.CONTINUE;
169     }
170 });
171 }
172
173 private boolean isCodeFile(Path file) {
174     String name = file.getFileName().toString();
175     // Check for exact filename match (e.g. "Makefile", "Dockerfile")
176     if (CODE_EXTENSIONS.contains(name)) return true;
177     // Check extension
178     int dot = name.lastIndexOf('.');
179     if (dot >= 0) {
180         return CODE_EXTENSIONS.contains(name.substring(dot));
181     }
182     return false;
183 }
184
185 // PDF Generation
186
187 private void generatePdf(Path currentDir) throws IOException {
188     PdfWriter writer = new PdfWriter(OUTPUT_FILE);
189     PdfDocument pdfDoc = new PdfDocument(writer);
190     Document doc = new Document(pdfDoc, PageSize.A4);
191     doc.setMargins(40, 40, 40, 40);
192
193     PdfFont sansFont = PdfFontFactory.createFont(StandardFonts.HELVETICA);
194     PdfFont boldFont = PdfFontFactory.createFont(StandardFonts.HELVETICA_BOLD);
195     PdfFont monoFont = PdfFontFactory.createFont(StandardFonts.COURIER);
196
197     // Cover page
198     addCoverPage(doc, sansFont, boldFont, currentDir);
199     doc.add(new AreaBreak(AreaBreakType.NEXT_PAGE));
200
201     // Table of contents
202     addTableOfContents(doc, sansFont, boldFont);
203     doc.add(new AreaBreak(AreaBreakType.NEXT_PAGE));
204
205     // Code sections
206     for (int i = 0; i < fileEntries.size(); i++) {
207         FileEntry entry = fileEntries.get(i);
208         addCodeSection(doc, entry, i + 1, sansFont, boldFont, monoFont);
209         if (i < fileEntries.size() - 1) {
210             doc.add(new AreaBreak(AreaBreakType.NEXT_PAGE));
211         }
212     }
213
214     doc.close();
215 }
216
217 private void addCoverPage(Document doc, PdfFont sansFont, PdfFont boldFont, Path dir)
218     throws IOException {
219
220     doc.add(new Paragraph("\n\n\n"));
221
222     // Big title
223     doc.add(new Paragraph("Code Export"))

```

```

224         .setFont(boldFont).setFontSize(FONT_TITLE + 10)
225         .setFillColor(COLOR_FILE_TITLE)
226         .setMarginBottom(6));
227
228     doc.add(new Paragraph("Source Code PDF Report")
229         .setFont(sansFont).setFontSize(FONT_SECTION)
230         .setFillColor(ColorConstants.DARK_GRAY)
231         .setMarginBottom(24));
232
233     // Stats table
234     Table stats = new Table(UnitValue.createPercentArray(new float[]{40, 60}))
235         .setWidth(UnitValue.createPercentValue(80));
236     addStatRow(stats, "Generated",      LocalDateTime.now()
237         .format(DateTimeFormatter.ofPattern("yyyy-MM-dd HH:mm")), sansFont, boldFont);
238     addStatRow(stats, "Root directory", dir.toString(), sansFont, boldFont);
239     addStatRow(stats, "Files found",   String.valueOf(fileEntries.size()), sansFont, boldFont);
240     addStatRow(stats, "Total lines",   String.format("%,d", totalLines), sansFont, boldFont);
241     doc.add(stats);
242 }
243
244 private void addStatRow(Table t, String label, String value,
245     PdfFont sans, PdfFont bold) {
246     t.addCell(new Cell().add(new Paragraph(label)
247         .setFont(bold).setFontSize(10)
248         .setFillColor(ColorConstants.DARK_GRAY))
249         .setBorder(null).setPaddingBottom(6));
250     t.addCell(new Cell().add(new Paragraph(value)
251         .setFont(sans).setFontSize(10))
252         .setBorder(null).setPaddingBottom(6));
253 }
254
255 private void addTableOfContents(Document doc, PdfFont sansFont, PdfFont boldFont)
256     throws IOException {
257     doc.add(new Paragraph("Table of Contents")
258         .setFont(boldFont).setFontSize(FONT_SECTION + 2)
259         .setFillColor(COLOR_TOC_HEADER)
260         .setMarginBottom(12));
261
262     for (int i = 0; i < fileEntries.size(); i++) {
263         FileEntry entry = fileEntries.get(i);
264         String label = String.format("%d.  %s  (%,d lines)",
265             i + 1, entry.relativePath, entry.lineCount);
266         doc.add(new Paragraph(label)
267             .setFont(sansFont).setFontSize(FONT_TOC)
268             .setMarginBottom(3));
269     }
270 }
271
272 private void addCodeSection(Document doc, FileEntry entry, int index,
273     PdfFont sansFont, PdfFont boldFont, PdfFont monoFont)
274     throws IOException {
275
276     // Section header bar
277     Table header = new Table(UnitValue.createPercentArray(new float[]{5, 75, 20}))
278         .setWidth(UnitValue.createPercentValue(100))
279         .setMarginBottom(8);
280
281     header.addCell(new Cell().add(new Paragraph(String.valueOf(index))
282         .setFont(boldFont).setFontSize(FONT_FILE)
283         .setFillColor(COLOR_HEADER_TEXT))
284         .setBackgroundColor(COLOR_HEADER_BG)
285         .setBorder(null).setPadding(6));
286
287     header.addCell(new Cell().add(new Paragraph(entry.relativePath)
288         .setFont(boldFont).setFontSize(FONT_FILE)
289         .setFillColor(COLOR_HEADER_TEXT))
290         .setBackgroundColor(COLOR_HEADER_BG)
291         .setBorder(null).setPadding(6));
292
293     header.addCell(new Cell().add(new Paragraph(entry.lineCount + " lines")
294         .setFont(sansFont).setFontSize(FONT_FILE - 1)
295         .setFillColor(COLOR_HEADER_TEXT))
296         .setBackgroundColor(COLOR_HEADER_BG)
297         .setBorder(null).setPadding(6));
298
299     doc.add(header);

```



```

300
301 // Code block with line numbers
302 String[] lines = entry.content.split("\n", -1);
303 int padWidth = String.valueOf(lines.length).length();
304
305 for (int ln = 0; ln < lines.length; ln++) {
306     String lineNum = String.format("%" + padWidth + "d", ln + 1);
307     String codeLine = lines[ln]
308         .replace("\t", "    ") // expand tabs
309         .replace("\r", "");    // strip CR
310
311     // Truncate very long lines
312     if (codeLine.length() > 200) {
313         codeLine = codeLine.substring(0, 197) + "...";
314     }
315
316     Text numText = new Text(lineNum + " ").setFont(monoFont)
317         .setFontSize(FONT_CODE).setFontColor(COLOR_LINE_NUM);
318     Text codeText = new Text(codeLine).setFont(monoFont)
319         .setFontSize(FONT_CODE).setFontColor(ColorConstants.BLACK);
320
321     Paragraph p = new Paragraph().add(numText).add(codeText)
322         .setBackgroundColor(ln % 2 == 0 ? COLOR_CODE_BG : ColorConstants.WHITE)
323         .setMargin(0).setPaddingLeft(4).setPaddingRight(4)
324         .setMultipliedLeading(1.3f)
325         .setFixedLeading(FONT_CODE + 2.5f);
326
327     doc.add(p);
328 }
329 }
330
331 // Data class
332
333 private static class FileEntry {
334     final String relativePath;
335     final Path absolutePath;
336     final String content;
337     final int lineCount;
338
339     FileEntry(String relativePath, Path absolutePath, String content, int lineCount) {
340         this.relativePath = relativePath;
341         this.absolutePath = absolutePath;
342         this.content = content;
343         this.lineCount = lineCount;
344     }
345 }
346 }
347
348

```

```

1 //package com.example.EmployeeManagementSystem.config;
2 //
3 //import org.springframework.beans.factory.annotation.Qualifier;
4 //import org.springframework.context.annotation.Bean;
5 //import org.springframework.context.annotation.Configuration;
6 //import org.springframework.core.annotation.Order;
7 //import org.springframework.http.HttpMethod;
8 //import org.springframework.security.authentication.AuthenticationManager;
9 //import org.springframework.security.authentication.ProviderManager;
10 //import org.springframework.security.authentication.dao.DaoAuthenticationProvider;
11 //import org.springframework.security.config.Customizer;
12 //import org.springframework.security.config.annotation.method.configuration.EnableMethodSecurity;
13 //import org.springframework.security.config.annotation.web.builders.HttpSecurity;
14 //import org.springframework.security.core.userdetails.UserDetailsService;
15 //import org.springframework.security.crypto.bcrypt.BCryptPasswordEncoder;
16 //import org.springframework.security.crypto.password.PasswordEncoder;
17 //import org.springframework.security.web.SecurityFilterChain;
18 //
19 //
20 // @Configuration
21 // @EnableMethodSecurity
22 // public class BasicAuthConfig {
23 //     @Bean
24 //     @Order(3)
25 //     public SecurityFilterChain basicAuth(HttpSecurity security){
26 //         security
27 //             .csrf(csrf-> csrf.disable())
28 //             .authorizeHttpRequests(
29 //                 auth->auth
30 //                 .requestMatchers("/employee/register","/employee/register/manager").permitAll()
31 //                 .requestMatchers(HttpMethod.POST,"/vendors").permitAll()
32 //                 .anyRequest().authenticated()
33 //             )
34 //             .httpBasic(Customizer.withDefaults());
35 //         return security.build();
36 //     }
37 //
38 //     @Bean
39 //     public PasswordEncoder passwordEncoder(){
40 //         return new BCryptPasswordEncoder();
41 //     }
42 //
43 //     @Bean
44 //     public AuthenticationManager authenticationManager(@Qualifier("combinedUserDetailService")
45 // UserDetailsService userDetailsService,
46 // PasswordEncoder passwordEncoder){
47 //         DaoAuthenticationProvider employeeAuthenticationProvider=new
48 // DaoAuthenticationProvider(userDetailsService);
49 //         employeeAuthenticationProvider.setPasswordEncoder(passwordEncoder);
50 //         return new ProviderManager(employeeAuthenticationProvider);
51 //     }
52 // }

```

```

1 // src/main/java/com/example/EmployeeManagementSystem/config/JWTAuth/JWTAuthConfig.java
2 package com.example.EmployeeManagementSystem.config.JWTAuth;
3
4 import com.example.EmployeeManagementSystem.Filter.ApiKeyFilter;
5 import com.example.EmployeeManagementSystem.Filter.JwtAuthFilter;
6 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
7 import com.example.EmployeeManagementSystem.Repository.RefreshTokenRepository;
8 import com.example.EmployeeManagementSystem.Repository.VendorRepo;
9 import com.example.EmployeeManagementSystem.Service.CustomOAuth2UserService;
10 import com.example.EmployeeManagementSystem.Service.RefreshTokenService;
11 import com.example.EmployeeManagementSystem.Util.JWTUtil;
12 import com.example.EmployeeManagementSystem.Util.OAuth2SuccessHandler;
13 import com.example.EmployeeManagementSystem.security.ApiKeyAuthenticationProvider;
14 import org.springframework.beans.factory.annotation.Qualifier;
15 import org.springframework.context.annotation.Bean;
16 import org.springframework.context.annotation.Configuration;
17 import org.springframework.context.annotation.Primary;
18 import org.springframework.http.HttpMethod;
19 import org.springframework.security.authentication.AuthenticationManager;
20 import org.springframework.security.authentication.ProviderManager;
21 import org.springframework.security.authentication.dao.DaoAuthenticationProvider;
22 import org.springframework.security.config.annotation.method.configuration.EnableMethodSecurity;
23 import org.springframework.security.config.annotation.web.builders.HttpSecurity;
24 import org.springframework.security.config.annotation.web.configuration.EnableWebSecurity;
25 import org.springframework.security.config.http.SessionCreationPolicy;
26 import org.springframework.security.core.userdetails.UserDetailsService;
27 import org.springframework.security.crypto.bcrypt.BCryptPasswordEncoder;
28 import org.springframework.security.crypto.password.PasswordEncoder;
29 import org.springframework.security.web.SecurityFilterChain;
30 import org.springframework.security.web.authentication.UsernamePasswordAuthenticationFilter;
31 import org.springframework.security.web.context.DelegatingSecurityContextRepository;
32 import org.springframework.security.web.context.HttpSessionSecurityContextRepository;
33 import org.springframework.security.web.context.RequestAttributeSecurityContextRepository;
34 import org.springframework.web.client.RestTemplate;
35 import org.springframework.web.cors.CorsConfiguration;
36 import org.springframework.web.cors.CorsConfigurationSource;
37 import org.springframework.web.cors.UrlBasedCorsConfigurationSource;
38
39 import java.util.List;
40
41 @Configuration
42 @EnableWebSecurity
43 @EnableMethodSecurity
44 public class JWTAuthConfig {
45
46     private final JwtAuthFilter jwtAuthFilter;
47     private final ApiKeyAuthenticationProvider apiKeyAuthProvider;
48     private final EmployeeRepo employeeRepo;
49     private final VendorRepo vendorRepo;
50     private final RefreshTokenService refreshTokenService;
51     private final RefreshTokenRepository refreshTokenRepository;
52     private final JWTUtil jwtUtil;
53
54     public JWTAuthConfig(JwtAuthFilter jwtAuthFilter,
55                         ApiKeyAuthenticationProvider apiKeyAuthProvider,
56                         EmployeeRepo employeeRepo,
57                         VendorRepo vendorRepo,
58                         RefreshTokenService refreshTokenService,
59                         RefreshTokenRepository refreshTokenRepository,
60                         JWTUtil jwtUtil) {
61         this.jwtAuthFilter = jwtAuthFilter;
62         this.apiKeyAuthProvider = apiKeyAuthProvider;
63         this.employeeRepo = employeeRepo;
64         this.vendorRepo = vendorRepo;
65         this.refreshTokenService = refreshTokenService;
66         this.refreshTokenRepository = refreshTokenRepository;
67         this.jwtUtil = jwtUtil;
68     }
69
70     @Bean
71     public RestTemplate getRestTemplate() {

```

```

72     return new RestTemplate();
73 }
74
75 @Bean
76 public ApiKeyFilter apiKeyFilter(
77     @Qualifier("apiKeyAuthenticationManager") AuthenticationManager apiKeyAuthManager) {
78     return new ApiKeyFilter(apiKeyAuthManager);
79 }
80
81 @Bean
82 public PasswordEncoder passwordEncoder() {
83     return new BCryptPasswordEncoder();
84 }
85
86 @Bean
87 public AuthenticationManager authenticationManager(
88     @Qualifier("combinedUserDetailsService") UserDetailsService userDetailsService,
89     PasswordEncoder passwordEncoder) {
90     DaoAuthenticationProvider provider = new DaoAuthenticationProvider(userDetailsService);
91     provider.setPasswordEncoder(passwordEncoder);
92     return new ProviderManager(provider);
93 }
94
95 @Bean
96 public CustomOAuth2UserService customOAuth2UserService() {
97     return new CustomOAuth2UserService(employeeRepo, vendorRepo, passwordEncoder());
98 }
99
100 @Bean
101 public OAuth2SuccessHandler oAuth2SuccessHandler() {
102     return new OAuth2SuccessHandler(jwtUtil, employeeRepo, vendorRepo,
103         refreshTokenService, refreshTokenRepository);
104 }
105
106 @Bean
107 @Qualifier("apiKeyAuthenticationManager")
108 public AuthenticationManager apiKeyAuthenticationManager() {
109     return new ProviderManager(apiKeyAuthProvider);
110 }
111
112 @Bean
113 public SecurityFilterChain securityFilterChain(HttpSecurity http, ApiKeyFilter apiKeyFilter) throws
Exception {
114     http
115         .csrf(csrf -> csrf.disable())
116         .cors(cors -> cors.configurationSource(corsConfigurationSource()))
117         .securityContext(ctx -> ctx
118             .securityContextRepository(new DelegatingSecurityContextRepository(
119                 new HttpSessionSecurityContextRepository(),
120                 new RequestAttributeSecurityContextRepository()
121             ))
122         )
123         .sessionManagement(session -> session
124             .sessionCreationPolicy(SessionCreationPolicy.IF_REQUIRED)
125             .maximumSessions(1)
126             .maxSessionsPreventsLogin(false)
127         )
128         .authorizeHttpRequests(auth -> auth
129             // Auth endpoints
130             .requestMatchers("/Authenticate",
131                 "/Authenticate/totp", "/Authenticate/refresh").permitAll()
132             .requestMatchers("/session/login", "/session/logout").permitAll()
133             // OAuth2 endpoints
134             .requestMatchers("/oauth2/**", "/login/oauth2/**").permitAll()
135             .requestMatchers("/auth/google/init").permitAll()
136             // API docs
137             .requestMatchers("/swagger-ui/**", "/v3/api-docs/**").permitAll()
138             // Shared login page
139             .requestMatchers("/login.html", "/google.html").permitAll()
140             .requestMatchers("/error").permitAll()
141             // Dashboards – served as static HTML, actual data is protected by JWT
142             .requestMatchers("/login.html",

```

```

146         "/google.html",
147         "/vendor-login.html",
148         "/vendor-dashboard.html",
149         "/tech-vendor-dashboard.html",
150         "/employee-dashboard.html",
151         "/manager-dashboard.html",
152         "/dashboard.html",
153         "/pending.html",
154         "/leave_management.html",
155         "/totp-setup.html").permitAll()
156
157         // Vendor pages
158         .requestMatchers("/vendor-login.html").permitAll()
159         .requestMatchers("/vendor-dashboard.html").permitAll()
160
161         // API key generation requires authentication
162         .requestMatchers("/api-keys/generate").authenticated()
163         .requestMatchers("/manager/createAdmin").permitAll()
164         // totp
165         .requestMatchers("/totp/**").authenticated()
166         // notification
167         .requestMatchers(HttpMethod.GET, "/notifications").authenticated()
168         .requestMatchers(HttpMethod.GET, "/notifications/unread-count").authenticated()
169         .requestMatchers(HttpMethod.POST, "/notifications/mark-read").authenticated()
170         // Everything else requires a valid token
171         .anyRequest().authenticated()
172     )
173     .oauth2Login(oauth -> oauth
174         .userInfoEndpoint(userInfo -> userInfo
175             .userService(customOAuth2UserService()))
176         )
177     .successHandler(oAuth2SuccessHandler())
178 )
179 .addFilterBefore(jwtAuthFilter, UsernamePasswordAuthenticationFilter.class)
180 .addFilterBefore(apiKeyFilter, UsernamePasswordAuthenticationFilter.class);
181
182     return http.build();
183 }
184 @Bean
185 public CorsConfigurationSource corsConfigurationSource() {
186     CorsConfiguration config = new CorsConfiguration();
187     config.setAllowedOrigins(List.of("http://192.168.1.3:8080"));
188     config.setAllowedMethods(List.of("GET", "POST", "PUT", "DELETE", "OPTIONS"));
189     config.setAllowedHeaders(List.of("*"));
190     config.setAllowCredentials(true);
191
192     UrlBasedCorsConfigurationSource source = new UrlBasedCorsConfigurationSource();
193     source.registerCorsConfiguration("/**", config);
194     return source;
195 }
196 }
197

```

```
1 package com.example.EmployeeManagementSystem.config;
2
3 import com.example.EmployeeManagementSystem.Entity.LeaveType;
4 import com.example.EmployeeManagementSystem.Enum.Gender;
5 import com.example.EmployeeManagementSystem.Repository.LeaveTypeRepository;
6 import org.springframework.boot.CommandLineRunner;
7 import org.springframework.stereotype.Component;
8
9 @Component
10 public class LeaveTypeDataInitializer implements CommandLineRunner {
11
12     private final LeaveTypeRepository leaveTypeRepository;
13
14     public LeaveTypeDataInitializer(LeaveTypeRepository leaveTypeRepository) {
15         this.leaveTypeRepository = leaveTypeRepository;
16     }
17
18     @Override
19     public void run(String... args) {
20         ensureLeaveType("SICK", false, true, 0, false, null);
21         ensureLeaveType("CASUAL", true, false, 30, false, null);
22         ensureLeaveType("MATERNITY", false, false, 0, true, Gender.F);
23         ensureLeaveType("UNPAID", false, false, 0, false, null);
24     }
25
26     private void ensureLeaveType(String name,
27                                boolean carriesForward,
28                                boolean monthlyReset,
29                                int maxCarryForward,
30                                boolean genderRestricted,
31                                Gender genderRestriction) {
32         leaveTypeRepository.findByName(name).orElseGet(() -> {
33             LeaveType leaveType = new LeaveType();
34             leaveType.setName(name);
35             leaveType.setCarriesForward(carriesForward);
36             leaveType.setMonthlyReset(monthlyReset);
37             leaveType.setMaxCarryForward(maxCarryForward);
38             leaveType.setGenderRestricted(genderRestricted);
39             leaveType.setGenderRestriction(genderRestriction);
40             return leaveTypeRepository.save(leaveType);
41         });
42     }
43 }
44
```

8 Current:

src\main\java\com\example\EmployeeManagementSystem\config\Session\SessionAuthConfig.java

```
1 //package com.example.EmployeeManagementSystem.config.Session;
2 //
3 //import org.springframework.context.annotation.Bean;
4 //import org.springframework.context.annotation.Configuration;
5 //import org.springframework.core.annotation.Order;
6 //import org.springframework.security.config.annotation.web.builders.HttpSecurity;
7 //import org.springframework.security.web.SecurityFilterChain;
8 //import org.springframework.security.web.context.HttpSessionSecurityContextRepository;
9 //
10 //@Configuration
11 //public class SessionAuthConfig {
12 //
13 //    @Bean
14 //    @Order(1)
15 //    public SecurityFilterChain sessionFilterChain(HttpSecurity http) throws Exception {
16 //        http
17 //            .securityMatcher("/session/**", "/web/**")
18 //            .csrf(csrf -> csrf.disable())
19 //            .authorizeHttpRequests(auth -> auth
20 //                .requestMatchers("/session/login", "/session/logout").permitAll()
21 //                .anyRequest().authenticated()
22 //            )
23 //            .sessionManagement(session -> session
24 //                .maximumSessions(1) // one session per user
25 //                .maxSessionsPreventsLogin(false) // new login kicks old session
26 //            )
27 //            .securityContext(ctx -> ctx
28 //                .securityContextRepository(new HttpSessionSecurityContextRepository())
29 //            );
30 //        return http.build();
31 //    }
32 //}
```

```
1 package com.example.EmployeeManagementSystem.config;
2
3 import io.swagger.v3.oas.models.Components;
4 import io.swagger.v3.oas.models.OpenAPI;
5 import io.swagger.v3.oas.models.security.SecurityRequirement;
6 import io.swagger.v3.oas.models.security.SecurityScheme;
7 import org.springframework.context.annotation.Bean;
8 import org.springframework.context.annotation.Configuration;
9
10 @Configuration
11 public class swaggerConfig {
12     @Bean
13     public OpenAPI openAPI() {
14         return new OpenAPI()
15             .components(new Components()
16                 .addSecuritySchemes("bearerAuth",
17                     new SecurityScheme()
18                         .type(SecurityScheme.Type.HTTP)
19                         .scheme("bearer")
20                         .bearerFormat("JWT")
21                 )
22             )
23             .addSecurityItem(new SecurityRequirement().addList("bearerAuth"));
24     }
25     // @Bean
26     // public OpenAPI openAPI() {
27     //     return new OpenAPI()
28     //         .components(new Components()
29     //             .addSecuritySchemes("basicAuth",
30     //                 new SecurityScheme()
31     //                     .type(SecurityScheme.Type.HTTP)
32     //                     .scheme("basic")
33     //             )
34     //         )
35     //         .addSecurityItem(new SecurityRequirement().addList("basicAuth"));
36     // }
37 }
38
```



```

1 package com.example.EmployeeManagementSystem.Controller;
2
3 import com.example.EmployeeManagementSystem.DTO.*;
4 import com.example.EmployeeManagementSystem.Entity.Employee;
5 import com.example.EmployeeManagementSystem.Service.*;
6 import org.springframework.format.annotation.DateTimeFormat;
7 import org.springframework.http.HttpStatus;
8 import org.springframework.http.ResponseEntity;
9 import org.springframework.security.access.prepost.PreAuthorize;
10 import org.springframework.web.bind.annotation.*;
11
12 import java.time.LocalDate;
13 import java.util.List;
14
15 @RestController
16 @RequestMapping("/admin")
17 @PreAuthorize("hasRole('ADMIN')")
18 public class AdminController {
19
20     private final EmployeeService employeeService;
21     private final DeliveryService.VendorService vendorService;
22     private final LeaveRequestService leaveRequestService;
23     private final RestaurantService restaurantService;
24     private final SubscriptionService subscriptionService;
25     private final LeaveAccrualService accrualService;
26     private final SickLeaveResetService resetService;
27
28     public AdminController(
29         EmployeeService employeeService,
30         DeliveryService.VendorService vendorService,
31         LeaveRequestService leaveRequestService,
32         RestaurantService restaurantService,
33         SubscriptionService subscriptionService,
34         LeaveAccrualService accrualService, SickLeaveResetService resetService) {
35         this.employeeService = employeeService;
36         this.vendorService = vendorService;
37         this.leaveRequestService = leaveRequestService;
38         this.restaurantService = restaurantService;
39         this.subscriptionService = subscriptionService;
40         this.accrualService = accrualService;
41         this.resetService = resetService;
42     }
43
44     //
45     // EMPLOYEE BRANCH
46     //
47
48     /** List every employee in the system. */
49     @GetMapping("/employees")
50     public ResponseEntity<?> getAllEmployees() {
51         return ResponseEntity.ok(employeeService.getAllEmployees());
52     }
53
54     /** Create a regular employee. */
55     @PostMapping("/employees")
56     public ResponseEntity<Employee> createEmployee(@RequestBody AdminEmployeeDTO dto) {
57         return ResponseEntity.status(HttpStatus.CREATED)
58             .body(employeeService.createEmployee(dto));
59     }
60
61     @PreAuthorize("hasRole('ADMIN')")
62     @GetMapping("/managers")
63     public ResponseEntity<List<ManagerDTO>> getAllManagers(){
64         return employeeService.getAllManagers();
65     }
66
67
68     /** Promote / create a manager. */
69     @PostMapping("/employees/manager")
70     public ResponseEntity<Employee> createManager(@RequestBody EmployeeDTO dto) {
71         return ResponseEntity.status(HttpStatus.CREATED)

```

```

72         .body(employeeService.createManager(dto));
73     }
74
75     /** Update any employee record. */
76     @PutMapping("/employees/{id}")
77     public ResponseEntity<Employee> updateEmployee(
78         @PathVariable long id,
79         @RequestBody EmployeeDTO dto) {
80         return ResponseEntity.ok(employeeService.updateEmployee(id, dto));
81     }
82
83     /** Hard-delete an employee. */
84     @DeleteMapping("/employees/{id}")
85     public ResponseEntity<String> deleteEmployee(@PathVariable long id) {
86         employeeService.deleteEmployee(id);
87         return ResponseEntity.ok("Employee with id: " + id + " has been deleted.");
88     }
89
90     /** Soft-deactivate an employee (sets status to INACTIVE). */
91     @PutMapping("/employees/{id}/inactive")
92     public ResponseEntity<String> deactivateEmployee(@PathVariable long id) {
93         return employeeService.inactivateUser(id);
94     }
95
96     //
97     // VENDOR BRANCH (only admin can register vendors)
98     //
99
100    /** List every vendor. */
101    @GetMapping("/vendors")
102    public ResponseEntity<List<VendorDTO>> getAllVendors() {
103        return ResponseEntity.ok(vendorService.getAllVendors());
104    }
105
106    @PostMapping("/vendors")
107    public ResponseEntity<VendorDTO> registerVendor(@RequestBody VendorRequest request) {
108        return ResponseEntity.status(HttpStatus.CREATED)
109            .body(vendorService.registerVendor(request));
110    }
111
112    /** Remove a vendor by ID. */
113    @DeleteMapping("/vendors/{vendorId}")
114    public ResponseEntity<String> deleteVendor(@PathVariable Long vendorId) {
115        vendorService.deleteVendorById(vendorId);
116        return ResponseEntity.ok("Vendor with id: " + vendorId + " has been removed.");
117    }
118
119    //
120    // LEAVE MANAGEMENT BRANCH
121    //
122
123    /** Fetch all pending leave requests across the organisation. */
124    @GetMapping("/leaves/pending")
125    public ResponseEntity<?> getPendingLeaves() {
126        return ResponseEntity.ok(leaveRequestService.getPendingLeaves());
127    }
128
129    /** Approve a leave request. */
130    @PutMapping("/leaves/{id}/approve")
131    public ResponseEntity<?> approveLeave(@PathVariable Long id) {
132        return ResponseEntity.ok(leaveRequestService.approveLeave(id));
133    }
134
135    /** Reject a leave request. */
136    @PutMapping("/leaves/{id}/reject")
137    public ResponseEntity<?> rejectLeave(@PathVariable Long id) {
138        return ResponseEntity.ok(leaveRequestService.rejectLeave(id));
139    }
140
141    //
142    // RESTAURANT MANAGEMENT BRANCH
143    //
144
145    /** View all restaurants. */
146    @GetMapping("/restaurants")
147    public ResponseEntity<?> getAllRestaurants() {

```

```

148         return ResponseEntity.ok(restaurantService.getAllActiveRestaurants());
149     }
150
151     //
152     // SUBSCRIPTION MANAGEMENT BRANCH
153     //
154
155     /** View all active subscriptions. */
156     @GetMapping("/subscriptions")
157     public ResponseEntity<?> getAllSubscriptions() {
158         return ResponseEntity.ok(subscriptionService.getAllSubscriptions());
159     }
160
161     @PostMapping("/leave/accrue")
162     @PreAuthorize("hasRole('ADMIN')")
163     public ResponseEntity<String> triggerAccrual(
164         @RequestParam(required = false)
165         @DateTimeFormat(iso = DateTimeFormat.ISO.DATE) LocalDate date
166     ) {
167         LocalDate accrualDate = (date != null) ? date : LocalDate.now();
168         accrualService.runMonthlyAccrual(accrualDate);
169         return ResponseEntity.ok("Accrual complete for " + accrualDate);
170     }
171
172     @PostMapping("/reset-sick")
173     @PreAuthorize("hasRole('ADMIN')")
174     public ResponseEntity<String> triggerSickReset(
175         @RequestParam(required = false)
176         @DateTimeFormat(iso = DateTimeFormat.ISO.DATE) LocalDate date
177     ) {
178         LocalDate resetDate = (date != null) ? date : LocalDate.now();
179         resetService.runMonthlyReset(resetDate);
180         return ResponseEntity.ok("Sick leave reset complete for " + resetDate);
181     }
182 }

```

```
1 package com.example.EmployeeManagementSystem.Controller;
2
3 import com.example.EmployeeManagementSystem.Entity.ApiKey;
4 import com.example.EmployeeManagementSystem.Service.ApiKeyService;
5 import org.springframework.beans.factory.annotation.Autowired;
6 import org.springframework.http.ResponseEntity;
7 import org.springframework.security.access.prepost.PreAuthorize;
8 import org.springframework.web.bind.annotation.*;
9
10 @RestController
11 @RequestMapping("/api-keys")
12 public class ApiKeyController {
13
14     @Autowired
15     private ApiKeyService apiKeyService;
16
17     // Admin generates a key for a vendor
18     @PostMapping("/vendor/{vendorId}/generate")
19     @PreAuthorize("hasRole('ADMIN')")
20     public ResponseEntity<ApiKey> generateVendorKey(
21         @PathVariable Long vendorId,
22         @RequestParam String name,
23         @RequestParam(defaultValue = "VENDOR_WRITE") String permissions,
24         @RequestParam(defaultValue = "365") int validDays) {
25         return ResponseEntity.ok(
26             apiKeyService.generateForVendor(vendorId, name, permissions, validDays));
27     }
28
29     // Revoke
30     @DeleteMapping("/{keyId}")
31     @PreAuthorize("hasRole('ADMIN')")
32     public ResponseEntity<String> revokeKey(@PathVariable Long keyId) {
33         apiKeyService.revokeKey(keyId);
34         return ResponseEntity.ok("Key revoked");
35     }
36 }
37
```

```
1 package com.example.EmployeeManagementSystem.Controller;
2
3 import com.example.EmployeeManagementSystem.DTO.AccessTokenRequest;
4 import com.example.EmployeeManagementSystem.DTO.AuthRequest;
5 import com.example.EmployeeManagementSystem.DTO.AuthResponse;
6 import com.example.EmployeeManagementSystem.Entity.Employee;
7 import com.example.EmployeeManagementSystem.Entity.RefreshToken;
8 import com.example.EmployeeManagementSystem.Entity.Vendor;
9 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
10 import com.example.EmployeeManagementSystem.Repository.RefreshTokenRepository;
11 import com.example.EmployeeManagementSystem.Service.RefreshTokenService;
12 import com.example.EmployeeManagementSystem.Service.TotpService;
13 import com.example.EmployeeManagementSystem.Util.JWTUtil;
14 import org.springframework.http.ResponseEntity;
15 import org.springframework.security.authentication.AuthenticationManager;
16 import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
17 import org.springframework.security.core.Authentication;
18 import org.springframework.security.core.userdetails.UserDetails;
19 import org.springframework.web.bind.annotation.*;
20
21 import java.time.Instant;
22 import java.util.Map;
23 import java.util.UUID;
24 import java.util.concurrent.ConcurrentHashMap;
25
26 @RestController
27 @RequestMapping("/Authenticate")
28 public class AuthController {
29
30     private final AuthenticationManager authenticationManager;
31     private final JWTUtil jwtUtil;
32     private final RefreshTokenService refreshTokenService;
33     private final RefreshTokenRepository refreshTokenRepository;
34     private final TotpService totpService;
35     private final EmployeeRepo employeeRepo;
36
37     // Temporary store for pre-auth tokens (username -> pre-auth token)
38     // In production, use Redis or DB-backed store with TTL
39     private static final ConcurrentHashMap<String, PreAuthEntry> PRE_AUTH_STORE = new ConcurrentHashMap<>();
40
41     private record PreAuthEntry(String username, String role, long expiresAt) {}
42
43     public AuthController(AuthenticationManager authenticationManager,
44                           JWTUtil jwtUtil,
45                           RefreshTokenService refreshTokenService,
46                           RefreshTokenRepository refreshTokenRepository,
47                           TotpService totpService,
48                           EmployeeRepo employeeRepo) {
49         this.authenticationManager = authenticationManager;
50         this.jwtUtil = jwtUtil;
51         this.refreshTokenService = refreshTokenService;
52         this.refreshTokenRepository = refreshTokenRepository;
53         this.totpService = totpService;
54         this.employeeRepo = employeeRepo;
55     }
56
57     /**
58      * Step 1 of login. If TOTP is enabled, returns a pre-auth token instead of a JWT.
59      * The client must then call /Authenticate/totp with that pre-auth token + 6-digit code.
60      */
61     @PostMapping
62     public ResponseEntity<?> authenticateToken(@RequestBody AuthRequest authRequest) {
63         try {
64             Authentication auth = authenticationManager.authenticate(
65                 new UsernamePasswordAuthenticationToken(
66                     authRequest.getUsername(),
67                     authRequest.getPassword()
68                 )
69             );
70
71             UserDetails user = (UserDetails) auth.getPrincipal();
```

```

72
73     String role = user.getAuthorities().stream()
74         .map(a -> a.getAuthority())
75         .filter(a -> a.startsWith("ROLE_"))
76         .findFirst()
77         .map(a -> a.substring("ROLE_".length()))
78         .orElseThrow(() -> new RuntimeException("Role not found"));
79
80     // Handle Vendor login – no TOTP, issue tokens directly
81     if (user instanceof Vendor vendor) {
82         return ResponseEntity.ok(issueTokens(vendor, role));
83     }
84
85     // Handle Employee login
86     Employee employee = employeeRepo.findByEmail(authRequest.getUsername())
87         .orElseThrow(() -> new RuntimeException("Employee not found"));
88
89     if (employee.isTotpEnabled()) {
90         String preAuthToken = UUID.randomUUID().toString();
91         PRE_AUTH_STORE.put(preAuthToken, new PreAuthEntry(
92             employee.getEmail(),
93             role,
94             System.currentTimeMillis() + 5 * 60 * 1000L
95         ));
96         return ResponseEntity.ok(Map.of(
97             "requiresTotp", true,
98             "preAuthToken", preAuthToken
99         ));
100     }
101
102     return ResponseEntity.ok(issueTokens(user, role));
103
104 } catch (Exception e) {
105     throw new RuntimeException(e);
106 }
107 }
108
109 /**
110  * Step 2 of login when TOTP is enabled.
111  * Submit the pre-auth token + 6-digit TOTP code to get the real JWT.
112  */
113 @PostMapping("/totp")
114 public ResponseEntity<?> completeTotpLogin(@RequestBody Map<String, String> body) {
115     String preAuthToken = body.get("preAuthToken");
116     String code = body.get("code");
117
118     if (preAuthToken == null || code == null) {
119         return ResponseEntity.badRequest().body(Map.of("error", "preAuthToken and code are required"));
120     }
121
122     PreAuthEntry entry = PRE_AUTH_STORE.get(preAuthToken);
123     if (entry == null || entry.expiresAt() < System.currentTimeMillis()) {
124         PRE_AUTH_STORE.remove(preAuthToken);
125         return ResponseEntity.status(401).body(Map.of("error", "Pre-auth token expired or invalid.
Please login again."));
126     }
127
128     Employee employee = employeeRepo.findByEmail(entry.username())
129         .orElseThrow(() -> new RuntimeException("Employee not found"));
130
131     if (!totpService.verifyCode(employee.getTotpSecret(), code)) {
132         return ResponseEntity.status(401).body(Map.of("error", "Invalid TOTP code"));
133     }
134
135     // TOTP verified – consume the pre-auth token and issue JWT
136     PRE_AUTH_STORE.remove(preAuthToken);
137     return ResponseEntity.ok(issueTokens(employee, entry.role()));
138 }
139
140 /**
141  * Refresh token endpoint – unchanged from original.
142  */
143 @PostMapping("/refresh")
144 public AuthResponse refresh(@RequestBody AccessTokenRequest request) {
145     RefreshToken storedToken = refreshTokenRepository.findByToken(request.getRefreshToken())
146         .orElseThrow(() -> new RuntimeException("Invalid refresh token"));

```

```

147
148     if (storedToken.getExpiryTime().isBefore(Instant.now())) {
149         refreshTokenRepository.delete(storedToken);
150         throw new RuntimeException("Refresh token expired");
151     }
152
153     String username;
154     String role;
155     if (storedToken.getEmployee() != null) {
156         username = storedToken.getEmployee().getUsername();
157         role = storedToken.getEmployee().getRole().name();
158     } else if (storedToken.getVendor() != null) {
159         username = storedToken.getVendor().getUsername();
160         role = storedToken.getVendor().getRole().name();
161     } else {
162         throw new RuntimeException("Refresh token has no associated user");
163     }
164
165     String newAccessToken = jwtUtil.generateToken(username, role);
166     return new AuthResponse(newAccessToken, storedToken.getToken());
167 }
168
169 // ---- helper ----
170
171 private AuthResponse issueTokens(UserDetails user, String role) {
172     String accessToken = jwtUtil.generateToken(user.getUsername(), role);
173     RefreshToken refreshToken;
174     if (user instanceof Employee employee) {
175         refreshToken = refreshTokenService.createForEmployee(employee);
176     } else if (user instanceof Vendor vendor) {
177         refreshToken = refreshTokenService.createForVendor(vendor);
178     } else {
179         throw new RuntimeException("Unknown user type");
180     }
181     refreshTokenRepository.save(refreshToken);
182     return new AuthResponse(accessToken, refreshToken.getToken());
183 }
184 }

```

```
1 package com.example.EmployeeManagementSystem.Controller;
2
3 import com.example.EmployeeManagementSystem.Entity.Delivery;
4 import com.example.EmployeeManagementSystem.Enum.DeliveryStatus;
5 import com.example.EmployeeManagementSystem.Service.DeliveryService;
6 import org.springframework.http.ResponseEntity;
7 import org.springframework.web.bind.annotation.*;
8
9 import java.util.List;
10
11 @RestController
12 @RequestMapping("/delivery")
13 public class DeliveryController {
14     private final DeliveryService deliveryService;
15
16     public DeliveryController(DeliveryService deliveryService) {
17         this.deliveryService = deliveryService;
18     }
19
20     @PutMapping("/{id}/status")
21     public ResponseEntity<Delivery> updateDeliveryStatus(
22         @PathVariable Long id,
23         @RequestParam DeliveryStatus status) {
24         return ResponseEntity.ok(deliveryService.updateDeliveryStatus(id, status));
25     }
26
27     @GetMapping("/subscription/{subscriptionId}")
28     public ResponseEntity<List<Delivery>> getDeliveriesBySubscription(
29         @PathVariable Long subscriptionId) {
30         return ResponseEntity.ok(deliveryService.getDeliveriesBySubscription(subscriptionId));
31     }
32 }
33
```



```

1 package com.example.EmployeeManagementSystem.Controller;
2
3 import com.example.EmployeeManagementSystem.DTO.AssignDeviceRequest;
4 import com.example.EmployeeManagementSystem.DTO.DeviceAssignmentResponseDTO;
5 import com.example.EmployeeManagementSystem.DTO.DeviceDTO;
6 import com.example.EmployeeManagementSystem.DTO.DeviceResponseDTO;
7 import com.example.EmployeeManagementSystem.Entity.Device;
8 import com.example.EmployeeManagementSystem.Entity.DeviceAssignment;
9 import com.example.EmployeeManagementSystem.Service.DeviceService;
10 import org.springframework.http.HttpStatus;
11 import org.springframework.http.ResponseEntity;
12 import org.springframework.security.access.prepost.PreAuthorize;
13 import org.springframework.security.core.Authentication;
14 import org.springframework.web.bind.annotation.*;
15
16 import java.util.List;
17
18 @RestController
19 @RequestMapping("/devices")
20 public class DeviceController {
21     private final DeviceService deviceService;
22
23     public DeviceController(DeviceService deviceService) {
24         this.deviceService = deviceService;
25     }
26
27     @PostMapping
28     @PreAuthorize("hasRole('TECH_VENDOR')")
29     public ResponseEntity<Device> addDevice(@RequestBody DeviceDTO deviceDTO,
30                                             Authentication authentication) {
31         return ResponseEntity.status(HttpStatus.CREATED)
32             .body(deviceService.addDevice(deviceDTO, authentication));
33     }
34
35     @GetMapping("/vendor/me")
36     @PreAuthorize("hasRole('TECH_VENDOR')")
37     public ResponseEntity<List<DeviceResponseDTO>> getMyVendorDevices(Authentication authentication) {
38         return ResponseEntity.ok(deviceService.getDevicesForLoggedInVendor(authentication));
39     }
40
41     @GetMapping("/my")
42     @PreAuthorize("hasAnyRole('EMPLOYEE', 'MANAGER', 'ADMIN')")
43     public ResponseEntity<List<DeviceResponseDTO>> getMyAssignedDevices(Authentication authentication) {
44         return ResponseEntity.ok(deviceService.getDevicesForLoggedInEmployee(authentication));
45     }
46
47     @GetMapping("/unassigned")
48     @PreAuthorize("hasRole('ADMIN')")
49     public ResponseEntity<List<DeviceResponseDTO>> getUnassignedDevices() {
50         return ResponseEntity.ok(deviceService.getUnassignedDevices());
51     }
52
53     @PostMapping("/{deviceId}/assign")
54     @PreAuthorize("hasRole('ADMIN')")
55     public ResponseEntity<String> assignDevice(@PathVariable Long deviceId,
56                                              @RequestBody AssignDeviceRequest request) {
57         DeviceAssignment assignment = deviceService.assignDevice(deviceId, request);
58         return ResponseEntity.status(HttpStatus.CREATED)
59             .body("Device " + assignment.getDevice().getId()
60                 + " assigned to employee " + assignment.getAssignedTo().getEmployeeId());
61     }
62
63     @GetMapping("/assignments")
64     @PreAuthorize("hasRole('TECH_VENDOR')")
65     public List<DeviceAssignmentResponseDTO> getDeviceAssignmentOfVendor(Authentication authentication){
66         return deviceService.getDeviceAssignmentOfVendor(authentication);
67     }
68 }
69
70 }
71

```

```

1 package com.example.EmployeeManagementSystem.Controller;
2
3 import com.example.EmployeeManagementSystem.DTO.EmployeeAttendanceResponseDTO;
4 import com.example.EmployeeManagementSystem.DTO.EmployeeDTO;
5 import com.example.EmployeeManagementSystem.Entity.Employee;
6 import com.example.EmployeeManagementSystem.Service.EmployeeService;
7 import org.springframework.http.HttpStatus;
8 import org.springframework.http.ResponseEntity;
9 import org.springframework.security.access.prepost.PreAuthorize;
10 import org.springframework.security.core.Authentication;
11 import org.springframework.web.bind.annotation.*;
12 @RestController
13 @RequestMapping("/employee")
14 public class EmployeeController {
15     private final EmployeeService employeeService;
16
17     public EmployeeController(EmployeeService employeeService) {
18         this.employeeService = employeeService;
19     }
20
21     @GetMapping
22     @PreAuthorize("hasAnyRole('MANAGER', 'ADMIN', 'EMPLOYEE', 'VENDOR')")
23     public ResponseEntity<?> getAllEmployees(){
24         return ResponseEntity.ok(employeeService.getAllEmployees());
25     }
26
27     @GetMapping("/me")
28     @PreAuthorize("hasAnyRole('ADMIN', 'MANAGER', 'EMPLOYEE')")
29     public ResponseEntity<Employee> myAccount(Authentication authentication){
30         return ResponseEntity.ok(employeeService.getAccount(authentication));
31     }
32
33     @GetMapping("/attendance")
34     @PreAuthorize("hasAnyRole('ADMIN', 'MANAGER', 'EMPLOYEE', 'VENDOR')")
35     public ResponseEntity<EmployeeAttendanceResponseDTO> getAttendanceOverview() {
36         return ResponseEntity.ok(employeeService.getEmployeeAttendanceOverview());
37     }
38
39     @PutMapping("/{id}")
40     @PreAuthorize("hasRole('ADMIN')")
41     public ResponseEntity<Employee> updateEmployee(@PathVariable long id, @RequestBody EmployeeDTO
employeeDTO){
42         return ResponseEntity.ok(employeeService.updateEmployee(id, employeeDTO));
43     }
44
45     @DeleteMapping("/{id}")
46     @PreAuthorize("hasAnyRole('ADMIN', 'MANAGER')")
47     public ResponseEntity<?> deleteEmployee(@PathVariable long id){
48         employeeService.deleteEmployee(id);
49         return ResponseEntity.ok("Employee with id: "+id+" is deleted");
50     }
51
52     //should be done by manager
53     @PutMapping("/inactive")
54     @PreAuthorize("hasAnyRole('MANAGER', 'ADMIN')")
55     public ResponseEntity<String> inactivateEmployee(@RequestParam long employeeId) {
56         return employeeService.inactivateUser(employeeId);
57     }
58 }
59

```

16 Current:

src\main\java\com\example\EmployeeManagementSystem\Controller\GlobalExceptionHandler.java

```
1 package com.example.EmployeeManagementSystem.Controller;
2
3 import com.example.EmployeeManagementSystem.Exception.*;
4 import org.springframework.dao.DataIntegrityViolationException;
5 import org.springframework.http.HttpStatus;
6 import org.springframework.http.ResponseEntity;
7 import org.springframework.web.bind.annotation.ExceptionHandler;
8 import org.springframework.web.bind.annotation.RestControllerAdvice;
9 import org.springframework.web.context.request.WebRequest;
10 import org.springframework.security.access.AccessDeniedException;
11
12 import java.time.Instant;
13 import java.time.LocalDateTime;
14 import java.util.LinkedHashMap;
15 import java.util.Map;
16
17 @RestControllerAdvice
18 public class GlobalExceptionHandler {
19
20     private Map<String, Object> buildError(HttpStatus status, String message, WebRequest request) {
21         Map<String, Object> body = new LinkedHashMap<>();
22         body.put("timestamp", Instant.now());
23         body.put("status", status.value());
24         body.put("error", status.getReasonPhrase());
25         body.put("message", message);
26         body.put("path", request.getDescription(false).replace("uri=", ""));
27         return body;
28     }
29
30     @ExceptionHandler(EmployeeNotFound.class)
31     public ResponseEntity<Map<String, Object>> handleEmployeeNotFound(
32         EmployeeNotFound ex, WebRequest request) {
33         return new ResponseEntity<>(buildError(HttpStatus.NOT_FOUND, ex.getMessage(), request),
34             HttpStatus.NOT_FOUND);
35     }
36
37     @ExceptionHandler(InvalidStartDateException.class)
38     public ResponseEntity<Map<String, Object>> handleInvalidStartDate(
39         InvalidStartDateException ex, WebRequest request) {
40         return new ResponseEntity<>(buildError(HttpStatus.BAD_REQUEST, ex.getMessage(), request),
41             HttpStatus.BAD_REQUEST);
42     }
43
44     @ExceptionHandler(InvalidEndDateException.class)
45     public ResponseEntity<Map<String, Object>> handleInvalidEndDate(
46         InvalidEndDateException ex, WebRequest request) {
47         return new ResponseEntity<>(buildError(HttpStatus.BAD_REQUEST, ex.getMessage(), request),
48             HttpStatus.BAD_REQUEST);
49     }
50
51     @ExceptionHandler(OverlappingLeaveException.class)
52     public ResponseEntity<Map<String, Object>> handleOverlappingLeave(
53         OverlappingLeaveException ex, WebRequest request) {
54         return new ResponseEntity<>(buildError(HttpStatus.BAD_REQUEST, ex.getMessage(), request),
55             HttpStatus.BAD_REQUEST);
56     }
57
58     @ExceptionHandler(DuplicateRequestException.class)
59     public ResponseEntity<Map<String, Object>> handleDuplicateRequest(
60         DuplicateRequestException ex, WebRequest request) {
61         return new ResponseEntity<>(buildError(HttpStatus.BAD_REQUEST, ex.getMessage(), request),
62             HttpStatus.BAD_REQUEST);
63     }
64
65     @ExceptionHandler(LeaveRequestNotFoundException.class)
66     public ResponseEntity<Map<String, Object>> handleLeaveNotFound(
67         LeaveRequestNotFoundException ex, WebRequest request) {
68         return new ResponseEntity<>(buildError(HttpStatus.NOT_FOUND, ex.getMessage(), request),
69             HttpStatus.NOT_FOUND);
70     }
71 }
```

```

72     @ExceptionHandler(InvalidManagerException.class)
73     public ResponseEntity<Map<String, Object>> handleInvalidManager(
74         InvalidManagerException ex, WebRequest request) {
75         return new ResponseEntity<>(buildError(HttpStatus.FORBIDDEN, ex.getMessage(), request),
76             HttpStatus.FORBIDDEN);
77     }
78
79     @ExceptionHandler(SubscriptionAlreadyExists.class)
80     public ResponseEntity<Map<String, Object>> handleSubscriptionExists(
81         SubscriptionAlreadyExists ex, WebRequest request) {
82         return new ResponseEntity<>(buildError(HttpStatus.CONFLICT, ex.getMessage(), request),
83             HttpStatus.CONFLICT);
84     }
85
86
87     @ExceptionHandler(VendorNotFoundException.class)
88     public ResponseEntity<Map<String, Object>> handleVendorNotFound(
89         VendorNotFoundException ex, WebRequest request) {
90         return new ResponseEntity<>(buildError(HttpStatus.NOT_FOUND, ex.getMessage(), request),
91             HttpStatus.NOT_FOUND);
92     }
93
94     @ExceptionHandler(RestaurantNotFoundException.class)
95     public ResponseEntity<Map<String, Object>> handleRestaurantNotFound(
96         RestaurantNotFoundException ex, WebRequest request) {
97         return new ResponseEntity<>(buildError(HttpStatus.NOT_FOUND, ex.getMessage(), request),
98             HttpStatus.NOT_FOUND);
99     }
100
101     @ExceptionHandler(UnauthorizedAccessException.class)
102     public ResponseEntity<Map<String, Object>> handleUnauthorized(
103         UnauthorizedAccessException ex, WebRequest request) {
104         return new ResponseEntity<>(buildError(HttpStatus.FORBIDDEN, ex.getMessage(), request),
105             HttpStatus.FORBIDDEN);
106     }
107
108     @ExceptionHandler(IllegalArgumentException.class)
109     public ResponseEntity<Map<String, Object>> handleIllegalArgument(
110         IllegalArgumentException ex, WebRequest request) {
111         return new ResponseEntity<>(buildError(HttpStatus.BAD_REQUEST, ex.getMessage(), request),
112             HttpStatus.BAD_REQUEST);
113     }
114
115     @ExceptionHandler(DataIntegrityViolationException.class)
116     public ResponseEntity<Map<String, Object>> handleDataIntegrity(
117         DataIntegrityViolationException ex, WebRequest request) {
118         return new ResponseEntity<>(
119             buildError(HttpStatus.CONFLICT, "Cannot delete: related records exist", request),
120             HttpStatus.CONFLICT);
121     }
122
123     @ExceptionHandler(AccessDeniedException.class)
124     public ResponseEntity<Map<String, Object>> handleAccessDenied(AccessDeniedException ex) {
125         return ResponseEntity.status(HttpStatus.FORBIDDEN).body(Map.of(
126             "error", "Access denied",
127             "message", ex.getMessage(),
128             "timestamp", LocalDateTime.now().toString()
129         ));
130     }
131
132 }

```

```
1 package com.example.EmployeeManagementSystem.Controller;
2
3 import com.example.EmployeeManagementSystem.DTO.HolidayDTO;
4 import com.example.EmployeeManagementSystem.Service.HolidayService;
5 import org.springframework.http.ResponseEntity;
6 import org.springframework.security.access.prepost.PreAuthorize;
7 import org.springframework.web.bind.annotation.*;
8
9 import java.time.LocalDate;
10
11 @RestController
12 @RequestMapping("/holidays")
13 public class HolidayController {
14     private final HolidayService holidayService;
15
16     public HolidayController(HolidayService holidayService) {
17         this.holidayService = holidayService;
18     }
19
20     @PostMapping
21     @PreAuthorize("hasRole('ADMIN')")
22     public ResponseEntity<?> addHoliday(@RequestBody HolidayDTO dto) {
23         return ResponseEntity.ok(holidayService.addHoliday(dto));
24     }
25
26     @GetMapping
27     public ResponseEntity<?> getHolidays(
28         @RequestParam(defaultValue = "0") int year
29     ) {
30         int targetYear = (year == 0) ? LocalDate.now().getYear() : year;
31         return ResponseEntity.ok(holidayService.getHolidaysByYear(targetYear));
32     }
33
34     @DeleteMapping("/{id}")
35     @PreAuthorize("hasRole('ADMIN')")
36     public ResponseEntity<?> deleteHoliday(@PathVariable Long id) {
37         holidayService.deleteHoliday(id);
38         return ResponseEntity.ok("Holiday deleted");
39     }
40 }
```

18 Current:

src\main\java\com\example\EmployeeManagementSystem\Controller\LeaveDashboardController.java

```
1 package com.example.EmployeeManagementSystem.Controller;
2
3 import com.example.EmployeeManagementSystem.DTO.LeaveDashboardResponse;
4 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
5 import com.example.EmployeeManagementSystem.Service.LeaveDashboardService;
6 import org.springframework.http.ResponseEntity;
7 import org.springframework.security.core.Authentication;
8 import org.springframework.security.core.userdetails.UserDetails;
9 import org.springframework.security.oauth2.core.user.OAuth2User;
10 import org.springframework.web.server.ResponseStatusException;
11 import org.springframework.web.bind.annotation.*;
12
13 import java.util.List;
14
15 import static org.springframework.http.HttpStatus.UNAUTHORIZED;
16
17 @RestController
18 @RequestMapping("/api")
19 public class LeaveDashboardController {
20
21     private final LeaveDashboardService dashboardService;
22     private final EmployeeRepo employeeRepository;
23
24     public LeaveDashboardController(LeaveDashboardService dashboardService,
25                                     EmployeeRepo employeeRepository) {
26         this.dashboardService = dashboardService;
27         this.employeeRepository = employeeRepository;
28     }
29
30     // GET own dashboard (any authenticated employee)
31     // GET /api/employees/me/leave-dashboard
32     @GetMapping("/employees/me/leave-dashboard")
33     public ResponseEntity<LeaveDashboardResponse> getMyDashboard(
34         Authentication authentication) {
35
36         Long callerId = resolveEmployeeId(authentication);
37         return ResponseEntity.ok(
38             dashboardService.getDashboard(callerId, callerId, resolveRole(authentication)));
39     }
40
41     // GET any employee's dashboard (manager/admin only)
42     // GET /api/employees/5/leave-dashboard
43     @GetMapping("/employees/{id}/leave-dashboard")
44     public ResponseEntity<LeaveDashboardResponse> getEmployeeDashboard(
45         @PathVariable Long id,
46         Authentication authentication) {
47
48         Long callerId = resolveEmployeeId(authentication);
49         String callerRole = resolveRole(authentication);
50         return ResponseEntity.ok(
51             dashboardService.getDashboard(id, callerId, callerRole));
52     }
53
54     // GET team dashboard (manager/admin only)
55     // GET /api/employees/me/team-dashboard
56     @GetMapping("/employees/me/team-dashboard")
57     public ResponseEntity<List<LeaveDashboardResponse>> getTeamDashboard(
58         Authentication authentication) {
59
60         Long callerId = resolveEmployeeId(authentication);
61         String callerRole = resolveRole(authentication);
62         return ResponseEntity.ok(
63             dashboardService.getTeamDashboard(callerId, callerRole));
64     }
65
66     // PATCH mark warning read
67     // PATCH /api/warnings/42/read
68     @PatchMapping("/warnings/{warningId}/read")
69     public ResponseEntity<Void> markWarningRead(
70         @PathVariable Long warningId,
71         Authentication authentication) {
```

```

72
73     Long callerId = resolveEmployeeId(authentication);
74     String callerRole = resolveRole(authentication);
75     dashboardService.markWarningRead(warningId, callerId, callerRole);
76     return ResponseEntity.noContent().build();
77 }
78
79 // Supports both JWT/session UserDetails and Google OAuth2 principals.
80 private Long resolveEmployeeId(Authentication authentication) {
81     String email = resolveEmail(authentication);
82     return employeeRepository
83         .findByEmail(email)
84         .orElseThrow(() -> new IllegalStateException("Authenticated user not found"))
85         .getEmployeeId();
86 }
87
88 private String resolveRole(Authentication authentication) {
89     if (authentication == null || !authentication.isAuthenticated()) {
90         throw new ResponseStatusException(UNAUTHORIZED, "Authentication required");
91     }
92     return authentication.getAuthorities().stream()
93         .map(a -> a.getAuthority())
94         .filter(authority -> authority.startsWith("ROLE_"))
95         .findFirst()
96         .map(authority -> authority.replace("ROLE_", ""))
97         .orElse("EMPLOYEE");
98 }
99
100 private String resolveEmail(Authentication authentication) {
101     if (authentication == null || !authentication.isAuthenticated()) {
102         throw new ResponseStatusException(UNAUTHORIZED, "Authentication required");
103     }
104
105     Object principal = authentication.getPrincipal();
106     if (principal instanceof UserDetails userDetails) {
107         return userDetails.getUsername();
108     }
109     if (principal instanceof OAuth2User oAuth2User) {
110         String email = oAuth2User.getAttribute("email");
111         if (email != null && !email.isBlank()) {
112             return email;
113         }
114     }
115
116     throw new ResponseStatusException(UNAUTHORIZED, "Authenticated email not found");
117 }
118 }
119

```

19 Current:

src\main\java\com\example\EmployeeManagementSystem\Controller\LeaveRequestController.java

```
1 package com.example.EmployeeManagementSystem.Controller;
2
3 import com.example.EmployeeManagementSystem.DTO.ActionDTO;
4 import com.example.EmployeeManagementSystem.DTO.LeaveRequestDTO;
5 import com.example.EmployeeManagementSystem.DTO.LeaveResponseDTO;
6 import com.example.EmployeeManagementSystem.DTO.MaternityLeaveRequestDTO;
7 import com.example.EmployeeManagementSystem.Service.LeaveRequestService;
8 import com.example.EmployeeManagementSystem.Service.MaternityLeaveService;
9 import org.springframework.http.ResponseEntity;
10 import org.springframework.security.access.prepost.PreAuthorize;
11 import org.springframework.security.core.Authentication;
12 import org.springframework.web.bind.annotation.*;
13
14 import java.util.List;
15
16 @RestController
17 @RequestMapping("leave_requests")
18 public class LeaveRequestController {
19     private final LeaveRequestService leaveRequestService;
20     private final MaternityLeaveService maternityLeaveService;
21
22     public LeaveRequestController(LeaveRequestService leaveRequestService, MaternityLeaveService
23 maternityLeaveService) {
24         this.leaveRequestService = leaveRequestService;
25         this.maternityLeaveService = maternityLeaveService;
26     }
27
28     @GetMapping
29     @PreAuthorize("hasAnyRole('ADMIN','MANAGER')")
30     public ResponseEntity<List<LeaveResponseDTO>> getAllLeaveRequest() {
31         return ResponseEntity.ok(leaveRequestService.getAllTheLeaveRequest());
32     }
33
34     @PostMapping
35     @PreAuthorize("hasAnyRole('ADMIN','MANAGER','EMPLOYEE')")
36     public ResponseEntity<?> addLeaveRequest(Authentication authentication,@RequestBody LeaveRequestDTO
37 dto){
38         return leaveRequestService.createRequest(authentication,dto);
39     }
40
41     @GetMapping("/pending")
42     @PreAuthorize("hasAnyRole('ADMIN','MANAGER')")
43     public ResponseEntity<List<LeaveResponseDTO>> getAllTheLeaveRequests() {
44         return ResponseEntity.ok(leaveRequestService.getAllThePendingLeaveRequests());
45     }
46
47     @PutMapping("/approval")
48     @PreAuthorize("hasAnyRole('ADMIN','MANAGER')")
49     public ResponseEntity<?> updateLeaveRequestStatus(@RequestBody ActionDTO actionDTO,Authentication
50 authentication){
51         return leaveRequestService.updateLeaveRequestStatus(actionDTO,authentication);
52     }
53
54     @PutMapping("/cancel/{leaveId}")
55     @PreAuthorize("hasAuthority('LEAVE_CANCEL')")
56     public ResponseEntity<?> cancelLeaveRequest(Authentication authentication,@PathVariable long leaveId){
57         return leaveRequestService.cancelLeaveRequest(authentication, leaveId);
58     }
59
60     @GetMapping("/employee")
61     @PreAuthorize("hasAnyRole('ADMIN','MANAGER','EMPLOYEE')")
62     public ResponseEntity<?> getEmployeeLeaves(Authentication authentication) {
63         return leaveRequestService.getLeaveRequestsByEmployee(authentication);
64     }
65
66     @PostMapping("/maternity/apply")
67     @PreAuthorize("hasAnyRole('ADMIN','MANAGER','EMPLOYEE')")
68     public ResponseEntity<?> applyMaternityLeave(
69         Authentication authentication,
70         @RequestBody MaternityLeaveRequestDTO dto) {
71         return maternityLeaveService.applyMaternityLeave(authentication, dto);
72     }
73 }
```



```
69 }
```

```
70
```

```
71 }
```

```
72
```

```
1 package com.example.EmployeeManagementSystem.Controller;
2
3 import org.springframework.http.ResponseEntity;
4 import org.springframework.web.bind.annotation.GetMapping;
5 import org.springframework.web.bind.annotation.RestController;
6
7 import java.util.Map;
8
9 @RestController
10 public class LoginController {
11
12     @GetMapping("/login")
13     public ResponseEntity<?> login() {
14         return ResponseEntity.ok().body(Map.of("message", "Please use POST to /api/auth/login with
credentials"));
15     }
16 }
17
```

```

1  package com.example.EmployeeManagementSystem.Controller;
2
3  import com.example.EmployeeManagementSystem.DTO.*;
4  import com.example.EmployeeManagementSystem.Entity.Employee;
5  import com.example.EmployeeManagementSystem.Service.ManagerService;
6  import org.springframework.http.ResponseEntity;
7  import org.springframework.security.access.prepost.PreAuthorize;
8  import org.springframework.security.core.Authentication;
9  import org.springframework.web.bind.annotation.*;
10
11 import java.util.List;
12
13 @RestController
14 @RequestMapping("/manager")
15 public class ManagerController {
16     private final ManagerService manangerService;
17
18     public ManagerController(ManagerService manangerService) {
19         this.manangerService = manangerService;
20     }
21
22     @PostMapping("/createAdmin")
23     public Employee createAdmin(@RequestBody EmployeeDetails employeeDetails){
24         return manangerService.createAdmin(employeeDetails);
25     }
26
27     @PostMapping("/employees")
28     @PreAuthorize("hasAnyRole('MANAGER')")
29     public Employee createEmployee(@RequestBody EmployeeDetails employeeDetails, Authentication
authentication){
30         return manangerService.createEmp(employeeDetails,authentication);
31     }
32
33     @GetMapping("/employees")
34     @PreAuthorize("hasRole('MANAGER')")
35     public List<Employee> getAllEmployeeByManager(Authentication authentication){
36         return manangerService.getAllEmployeeByManager(authentication);
37     }
38
39     @GetMapping("/requests")
40     @PreAuthorize("hasRole('MANAGER')")
41     public List<ManagerLeaveResponseDTO> getAllLeaveRequestByManager(Authentication authentication){
42         return manangerService.getAllLeaveRequestByManager(authentication);
43     }
44
45     @GetMapping("/Users")
46     @PreAuthorize("hasAnyRole('MANAGER','ADMIN')")
47     public List<EmployeeDTO> getUsers(){
48         return manangerService.getUsers();
49     }
50
51     @PutMapping("/promote")
52     @PreAuthorize("hasRole('MANAGER')")
53     public ResponseEntity<?> promoteUser(Authentication authentication, @RequestBody PromoteRequest
promoteRequest){
54         return manangerService.promoteUser(authentication,promoteRequest);
55     }
56
57     @GetMapping("/pending")
58     @PreAuthorize("hasAnyRole('MANAGER')")
59     public ResponseEntity<List<LeaveResponseDTO>> getAllTheLeaveRequests(Authentication authentication) {
60         return ResponseEntity.ok(manangerService.getAllThePendingLeaveRequests(authentication));
61     }
62
63     @GetMapping("/employees/search")
64     @PreAuthorize("hasAnyRole('MANAGER')")
65     public ResponseEntity<ManagerEmployeeDetailsDTO> getEmployeeDetails( @RequestParam String name){
66         return ResponseEntity.ok(manangerService.getEmployeeDetailsByName(name));
67     }
68
69

```

70

71 }

72

```

1 package com.example.EmployeeManagementSystem.Controller;
2
3 import com.example.EmployeeManagementSystem.Entity.Employee;
4 import com.example.EmployeeManagementSystem.Entity.Notification;
5 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
6 import com.example.EmployeeManagementSystem.Service.NotificationService;
7 import org.springframework.http.ResponseEntity;
8 import org.springframework.security.core.Authentication;
9 import org.springframework.security.oauth2.core.user.OAuth2User;
10 import org.springframework.security.oauth2.client.authentication.OAuth2AuthenticationToken;
11 import org.springframework.web.bind.annotation.*;
12 import java.util.List;
13 import java.util.Map;
14
15 @RestController
16 @RequestMapping("/notifications")
17 public class NotificationController {
18
19     private final NotificationService notificationService;
20     private final EmployeeRepo employeeRepo;
21
22     public NotificationController(NotificationService notificationService,
23                                 EmployeeRepo employeeRepo) {
24         this.notificationService = notificationService;
25         this.employeeRepo = employeeRepo;
26     }
27
28     /** GET /notifications – all notifications for logged-in user */
29     @GetMapping
30     public ResponseEntity<List<Notification>> getAll(Authentication authentication) {
31         Employee employee = getEmployee(authentication);
32         return ResponseEntity.ok(notificationService.getForEmployee(employee));
33     }
34
35     /** GET /notifications/unread-count – for the badge number */
36     @GetMapping("/unread-count")
37     public ResponseEntity<?> unreadCount(Authentication authentication) {
38         Employee employee = getEmployee(authentication);
39         return ResponseEntity.ok(Map.of("count",
40                                         notificationService.getUnreadCount(employee)));
41     }
42
43     /** POST /notifications/mark-read – mark all as read when bell is opened */
44     @PostMapping("/mark-read")
45     public ResponseEntity<?> markRead(Authentication authentication) {
46         Employee employee = getEmployee(authentication);
47         notificationService.markAllRead(employee);
48         return ResponseEntity.ok(Map.of("message", "All notifications marked as read"));
49     }
50
51     private Employee getEmployee(Authentication authentication) {
52         final String email = (authentication instanceof OAuth2AuthenticationToken oauth)
53             ? oauth.getPrincipal().<String>getAttribute("email")
54             : authentication.getName(); // JWT path – getName() returns email
55
56         return employeeRepo.findByEmail(email)
57             .orElseThrow(() -> new RuntimeException("Employee not found: " + email));
58     }
59 }
60

```

```
1 package com.example.EmployeeManagementSystem.Controller;
2
3 import com.example.EmployeeManagementSystem.DTO.AuthResponse;
4 import com.example.EmployeeManagementSystem.Entity.Employee;
5 import com.example.EmployeeManagementSystem.Entity.RefreshToken;
6 import com.example.EmployeeManagementSystem.Entity.Vendor;
7 import com.example.EmployeeManagementSystem.Enum.Role;
8 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
9 import com.example.EmployeeManagementSystem.Repository.RefreshTokenRepository;
10 import com.example.EmployeeManagementSystem.Repository.VendorRepo;
11 import com.example.EmployeeManagementSystem.Service.CombinedUserDetailsService;
12 import com.example.EmployeeManagementSystem.Service.RefreshTokenService;
13 import com.example.EmployeeManagementSystem.Util.JWTUtil;
14 import jakarta.servlet.http.HttpServletRequest;
15 import jakarta.servlet.http.HttpServletResponse;
16 import jakarta.servlet.http.HttpSession;
17 import org.springframework.beans.factory.annotation.Autowired;
18 import org.springframework.beans.factory.annotation.Value;
19 import org.springframework.http.*;
20 import org.springframework.security.core.userdetails.UserDetails;
21 import org.springframework.security.core.userdetails.UsernameNotFoundException;
22 import org.springframework.security.crypto.password.PasswordEncoder;
23 import org.springframework.util.LinkedMultiValueMap;
24 import org.springframework.util.MultiValueMap;
25 import org.springframework.web.bind.annotation.GetMapping;
26 import org.springframework.web.bind.annotation.RequestMapping;
27 import org.springframework.web.bind.annotation.RequestParam;
28 import org.springframework.web.bind.annotation.RestController;
29 import org.springframework.web.client.RestTemplate;
30
31 import java.io.IOException;
32 import java.util.Collections;
33 import java.util.Map;
34 import java.util.Optional;
35 import java.util.UUID;
36
37 @RestController
38 @RequestMapping("/auth/google")
39 public class OAuthController {
40
41     @GetMapping("/init")
42     public void initOAuth(@RequestParam(defaultValue = "EMPLOYEE") String role,
43                          @RequestParam(defaultValue = "Asia/Kolkata") String timezone,
44                          HttpServletRequest request,
45                          HttpServletResponse response) throws IOException {
46
47         // Now hand off to Spring's built-in OAuth2 flow
48         System.out.println("Now hand off to Spring's built-in OAuth2 flow");
49         response.sendRedirect("/oauth2/authorization/google");
50     }
51 }
52
```

```

1 package com.example.EmployeeManagementSystem.Controller;
2
3 import com.example.EmployeeManagementSystem.DTO.RestaurantDTO;
4 import com.example.EmployeeManagementSystem.DTO.RestaurantRequest;
5 import com.example.EmployeeManagementSystem.Enum.MealSlot;
6 import com.example.EmployeeManagementSystem.Service.RestaurantService;
7 import org.springframework.http.HttpStatus;
8 import org.springframework.http.ResponseEntity;
9 import org.springframework.security.access.prepost.PreAuthorize;
10 import org.springframework.security.core.Authentication;
11 import org.springframework.web.bind.annotation.*;
12
13 import java.util.List;
14 import java.util.stream.Collectors;
15
16 /**
17  * Restaurant management endpoints.
18  *
19  * VENDOR-ONLY (write operations)
20  * POST /restaurants - Create a restaurant (vendor supplies vendorId in body)
21  * PUT /restaurants/{id} - Update restaurant details (vendor ownership verified)
22  * PUT /restaurants/{id}/deactivate - Soft-delete a restaurant (vendor ownership verified)
23  * GET /restaurants/vendor/{vendorId} - List MY restaurants (vendor views their own)
24  *
25  * EMPLOYEE / MANAGER (read-only browse)
26  * GET /restaurants - Browse all active restaurants
27  * GET /restaurants/{id} - Get a specific restaurant
28  * GET /restaurants/by-slot?slot=LUNCH - Find restaurants that serve a given meal slot
29  *
30  * Employees use the browse endpoints to pick a restaurant when creating a subscription.
31  * Employees and managers cannot create, update, or deactivate restaurants.
32  */
33 @RestController
34 @RequestMapping("/restaurants")
35 public class RestaurantController {
36
37     private final RestaurantService restaurantService;
38
39     public RestaurantController(RestaurantService restaurantService) {
40         this.restaurantService = restaurantService;
41     }
42
43     // VENDOR-ONLY write operations
44
45     /**
46      * Create a new restaurant.
47      * Request body must include vendorId – the service validates the vendor exists.
48      */
49     @PostMapping
50     @PreAuthorize("hasRole('FOOD_VENDOR')")
51     public ResponseEntity<RestaurantDTO> createRestaurant(@RequestBody RestaurantRequest
52 request, Authentication authentication) {
53         return ResponseEntity.status(HttpStatus.CREATED)
54             .body(restaurantService.createRestaurant(request, authentication));
55     }
56
57     /**
58      * Update a restaurant's name, address, description, or supported meal slots.
59      * vendorId in the request body is used to verify ownership before any change is made.
60      */
61     @PutMapping("/{id}")
62     @PreAuthorize("hasRole('FOOD_VENDOR')")
63     public ResponseEntity<RestaurantDTO> updateRestaurant(@PathVariable Long id,
64 @RequestBody RestaurantRequest request,
65 Authentication authentication) {
66         return ResponseEntity.ok(restaurantService.updateRestaurant(id, request, authentication));
67     }
68
69     /**
70      * Soft-deactivate a restaurant. Active subscriptions remain in the DB but
71      * new deliveries will stop since the restaurant is inactive.

```

```

71     * Pass vendorId as a query param so ownership can be verified.
72     */
73     @PutMapping("/{id}/deactivate")
74     @PreAuthorize("hasRole('FOOD_VENDOR')")
75     public ResponseEntity<String> deactivateRestaurant(@PathVariable Long id,
76                                                         Authentication authentication) {
77         restaurantService.deactivateRestaurant(id, authentication);
78         return ResponseEntity.ok("Restaurant " + id + " has been deactivated.");
79     }
80
81     /**
82     * List all active restaurants owned by a specific vendor.
83     * Only the vendor themselves should call this to see their portfolio.
84     */
85     @GetMapping("/vendor")
86     @PreAuthorize("hasRole('FOOD_VENDOR')")
87     public ResponseEntity<List<RestaurantDTO>> getRestaurantsByVendor(Authentication authentication) {
88         return ResponseEntity.ok(restaurantService.getRestaurantsByVendor(authentication));
89     }
90
91     // EMPLOYEE / MANAGER read-only browse
92
93     /**
94     * Browse all active restaurants.
95     * Employees use this to discover available restaurants before subscribing.
96     */
97     @GetMapping
98     @PreAuthorize("hasAnyRole('ADMIN') or hasAuthority('RESTAURANTS_READ')")
99     public ResponseEntity<List<RestaurantDTO>> getAllActiveRestaurants() {
100         return ResponseEntity.ok(restaurantService.getAllActiveRestaurants());
101     }
102
103     /**
104     * Get a specific restaurant by id.
105     */
106     @GetMapping("/{id}")
107     @PreAuthorize("hasAnyRole('ADMIN') or hasAuthority('RESTAURANTS_READ')")
108     public ResponseEntity<RestaurantDTO> getRestaurant(@PathVariable Long id) {
109         return ResponseEntity.ok(restaurantService.getRestaurant(id));
110     }
111
112     /**
113     * Find all active restaurants that serve a specific meal slot.
114     * Example: GET /restaurants/by-slot?slot=LUNCH
115     * Employees call this when building their subscription to see valid options for each slot.
116     */
117     @GetMapping("/by-slot")
118     @PreAuthorize("hasAnyRole('ADMIN', 'VENDOR', 'EMPLOYEE', 'MANAGER')")
119     public ResponseEntity<List<RestaurantDTO>> getRestaurantsByMealSlot(@RequestParam MealSlot slot) {
120         return ResponseEntity.ok(restaurantService.getRestaurantsByMealSlot(slot));
121     }
122
123 }
124

```



```

1 package com.example.EmployeeManagementSystem.Controller;
2
3 import com.example.EmployeeManagementSystem.Scheduler.LeaveStatusScheduler;
4 import com.example.EmployeeManagementSystem.Service.LeaveAccrualService;
5 import com.example.EmployeeManagementSystem.Service.SickLeaveResetService;
6 import org.slf4j.Logger;
7 import org.slf4j.LoggerFactory;
8 import org.springframework.http.ResponseEntity;
9 import org.springframework.security.access.prepost.PreAuthorize;
10 import org.springframework.web.bind.annotation.*;
11
12 import java.time.LocalDate;
13 import java.time.YearMonth;
14 import java.util.Map;
15
16 /**
17  * Controller for Scheduler Operations
18  */
19 @RestController
20 @RequestMapping("/scheduler")
21 public class SchedulerController {
22
23     private static final Logger log = LoggerFactory.getLogger(SchedulerController.class);
24
25     private final LeaveStatusScheduler leaveStatusScheduler;
26     private final SickLeaveResetService sickLeaveResetService; // FIX: was missing
27     private final LeaveAccrualService leaveAccrualService; // FIX: was missing
28
29     public SchedulerController(LeaveStatusScheduler leaveStatusScheduler,
30                               SickLeaveResetService sickLeaveResetService,
31                               LeaveAccrualService leaveAccrualService) {
32         this.leaveStatusScheduler = leaveStatusScheduler;
33         this.sickLeaveResetService = sickLeaveResetService;
34         this.leaveAccrualService = leaveAccrualService;
35     }
36
37     /**
38      * Manual trigger for leave status scheduler
39      */
40     @PostMapping("/run")
41     @PreAuthorize("hasRole('ADMIN') or hasAuthority('SCHEDULER_TRIGGER')")
42     public ResponseEntity<?> runScheduler() {
43         try {
44             leaveStatusScheduler.runManually();
45             return ResponseEntity.ok(Map.of(
46                 "status", "success",
47                 "message", "Leave status scheduler executed successfully"
48             ));
49         } catch (Exception e) {
50             log.error("Leave status scheduler failed", e);
51             return ResponseEntity.internalServerError().body(Map.of(
52                 "status", "error",
53                 "message", e.getMessage()
54             ));
55         }
56     }
57
58     /**
59      * FIX: Manual trigger for sick leave reset (missing from original).
60      * Useful for recovery and testing without waiting for the end-of-month cron.
61      */
62     @PostMapping("/sick-leave-reset")
63     @PreAuthorize("hasRole('ADMIN')")
64     public ResponseEntity<?> runSickLeaveReset() {
65         try {
66             LocalDate lastDayOfMonth = YearMonth.now().atEndOfMonth();
67             sickLeaveResetService.runMonthlyReset(lastDayOfMonth);
68             return ResponseEntity.ok(Map.of(
69                 "status", "success",
70                 "message", "Sick leave reset executed for " + lastDayOfMonth
71             ));

```

```

72         } catch (Exception e) {
73             log.error("Sick leave reset failed", e);
74             return ResponseEntity.internalServerError().body(Map.of(
75                 "status", "error",
76                 "message", e.getMessage()
77             ));
78         }
79     }
80
81     /**
82      * FIX: Manual trigger for monthly leave accrual (missing from original).
83      * Useful for onboarding new employees or recovering from a missed cron run.
84      */
85     @PostMapping("/accrue")
86     @PreAuthorize("hasRole('ADMIN')")
87     public ResponseEntity<?> runMonthlyAccrual() {
88         try {
89             LocalDate today = LocalDate.now();
90             leaveAccrualService.runMonthlyAccrual(today);
91             return ResponseEntity.ok(Map.of(
92                 "status", "success",
93                 "message", "Monthly leave accrual executed for " + today
94             ));
95         } catch (Exception e) {
96             log.error("Monthly accrual failed", e);
97             return ResponseEntity.internalServerError().body(Map.of(
98                 "status", "error",
99                 "message", e.getMessage()
100             ));
101         }
102     }
103 }

```

26 Current:

src\main\java\com\example\EmployeeManagementSystem\Controller\ServiceRequestController.java

```
1 package com.example.EmployeeManagementSystem.Controller;
2
3 import com.example.EmployeeManagementSystem.DTO.AdminActionDTO;
4 import com.example.EmployeeManagementSystem.DTO.RepairDTO;
5 import com.example.EmployeeManagementSystem.DTO.ServiceRequestDTO;
6 import com.example.EmployeeManagementSystem.DTO.ServiceRequestResponseDTO;
7 import com.example.EmployeeManagementSystem.Entity.ServiceRequest;
8 import com.example.EmployeeManagementSystem.Service.ServiceRequestService;
9 import org.springframework.http.HttpStatus;
10 import org.springframework.http.ResponseEntity;
11 import org.springframework.security.access.prepost.PreAuthorize;
12 import org.springframework.security.core.Authentication;
13 import org.springframework.web.bind.annotation.GetMapping;
14 import org.springframework.web.bind.annotation.PostMapping;
15 import org.springframework.web.bind.annotation.PutMapping;
16 import org.springframework.web.bind.annotation.RequestBody;
17 import org.springframework.web.bind.annotation.RequestMapping;
18 import org.springframework.web.bind.annotation.RestController;
19
20 import java.util.List;
21
22 @RestController
23 @RequestMapping("/service-requests")
24 public class ServiceRequestController {
25     private final ServiceRequestService serviceRequestService;
26
27     public ServiceRequestController(ServiceRequestService serviceRequestService) {
28         this.serviceRequestService = serviceRequestService;
29     }
30
31     @PostMapping
32     @PreAuthorize("hasAnyRole('EMPLOYEE', 'MANAGER', 'ADMIN')")
33     public ResponseEntity<ServiceRequest> createServiceRequest(
34         @RequestBody ServiceRequestDTO serviceRequestDTO,
35         Authentication authentication) {
36         return ResponseEntity.status(HttpStatus.CREATED)
37             .body(serviceRequestService.createServiceRequest(serviceRequestDTO, authentication));
38     }
39
40     @GetMapping("/my")
41     @PreAuthorize("hasAnyRole('EMPLOYEE', 'MANAGER', 'ADMIN')")
42     public ResponseEntity<List<ServiceRequestResponseDTO>> getMyServiceRequests(Authentication
43 authentication) {
44         return ResponseEntity.ok(serviceRequestService.getMyServiceRequests(authentication));
45     }
46
47     @GetMapping("/open")
48     @PreAuthorize("hasRole('ADMIN')")
49     public ResponseEntity<List<ServiceRequestResponseDTO>> getAllOpenServiceRequests() {
50         return ResponseEntity.ok(serviceRequestService.getAllOpenServiceRequests());
51     }
52
53     @PutMapping("/admin")
54     @PreAuthorize("hasRole('ADMIN')")
55     public ResponseEntity<ServiceRequestResponseDTO> updateServiceRequestByAdmin(
56         Authentication authentication,
57         @RequestBody AdminActionDTO actionDTO) {
58         return ResponseEntity.ok(serviceRequestService.updateServiceRequestByAdmin(authentication,
59 actionDTO));
60     }
61
62     @GetMapping("/vendor/me")
63     @PreAuthorize("hasRole('TECH_VENDOR')")
64     public ResponseEntity<List<ServiceRequestResponseDTO>> getAllServiceRequestByVendor(Authentication
65 authentication) {
66         return ResponseEntity.ok(serviceRequestService.getAllServiceRequestByVendor(authentication));
67     }
68
69     @PutMapping("/vendor")
70     @PreAuthorize("hasRole('TECH_VENDOR')")
71     public ResponseEntity<ServiceRequestResponseDTO> updateServiceRequestByVendor(
```

```
69         Authentication authentication,
70         @RequestBody RepairDTO repairDTO) {
71         return ResponseEntity.ok(serviceRequestService.updateServiceRequestByVendor(authentication,
72         repairDTO));
73     }
74 }
```

[illegible]

```
70         // Invalidate Spring Security context
71         SecurityContextHolder.clearContext();
72
73         // Invalidate HTTP session (kills OAuth2 session too)
74         session.invalidate();
75
76         // Clear the session cookie
77         Cookie cookie = new Cookie("JSESSIONID", null);
78         cookie.setMaxAge(0);
79         cookie.setPath("/");
80         response.addCookie(cookie);
81
82         return ResponseEntity.ok("Logged out");
83     }
84
85     @GetMapping("/me")
86     public ResponseEntity<?> me(Authentication authentication) {
87         if (authentication == null || !authentication.isAuthenticated()) {
88             return ResponseEntity.status(401).body(Map.of("error", "Not logged in"));
89         }
90         return ResponseEntity.ok(Map.of(
91             "username", authentication.getName(),
92             "roles", authentication.getAuthorities().toString(),
93             "authenticated", true
94         ));
95     }
96 }
```

```

1 package com.example.EmployeeManagementSystem.Controller;
2
3 import com.example.EmployeeManagementSystem.DTO.SubscriptionDTO;
4 import com.example.EmployeeManagementSystem.DTO.SubscriptionRequest;
5 import com.example.EmployeeManagementSystem.Entity.Subscription;
6 import com.example.EmployeeManagementSystem.Service.SubscriptionService;
7 import org.springframework.http.HttpStatus;
8 import org.springframework.http.ResponseEntity;
9 import org.springframework.security.access.prepost.PreAuthorize;
10 import org.springframework.security.core.Authentication;
11 import org.springframework.web.bind.annotation.*;
12
13 import java.util.List;
14
15 @RestController
16 @RequestMapping("/subscription")
17 public class SubscriptionController {
18     private final SubscriptionService subscriptionService;
19
20     public SubscriptionController(SubscriptionService subscriptionService) {
21         this.subscriptionService = subscriptionService;
22     }
23
24     //Authentication makes sure that whoever is logged can add subscription and cannot add other
25     //subscription
26     @PostMapping
27     @PreAuthorize("hasAnyRole('EMPLOYEE','MANAGER','ADMIN','VENDOR')")
28     public ResponseEntity<?> addSubscription(@RequestBody SubscriptionRequest request, Authentication
29     authentication) {
30         subscriptionService.addSubscription(request, authentication);
31         return ResponseEntity.status(HttpStatus.CREATED)
32             .body("Added new subscription for user: " + authentication.getName());
33     }
34
35     @GetMapping
36     @PreAuthorize("hasAuthority('SUBSCRIPTION_READ')")
37     public List<SubscriptionDTO> getAllSubscriptions() {
38         return subscriptionService.getAllSubscriptions();
39     }
40
41     @PutMapping("/{id}")
42     @PreAuthorize("hasAnyRole('EMPLOYEE','MANAGER')")
43     public ResponseEntity<Subscription> updateSubscription(
44         @PathVariable Long id,
45         @RequestBody SubscriptionRequest request) {
46         return ResponseEntity.ok(subscriptionService.updateSubscription(id, request));
47     }
48
49     @PutMapping("/{id}/pause")
50     @PreAuthorize("hasAnyRole('EMPLOYEE','MANAGER')")
51     public ResponseEntity<Subscription> pauseSubscription(@PathVariable Long id) {
52         return ResponseEntity.ok(subscriptionService.pauseSubscription(id));
53     }
54
55     @PutMapping("/{id}/resume")
56     @PreAuthorize("hasAnyRole('EMPLOYEE','MANAGER')")
57     public ResponseEntity<Subscription> resumeSubscription(@PathVariable Long id) {
58         return ResponseEntity.ok(subscriptionService.resumeSubscription(id));
59     }
60
61     @PutMapping("/{id}/expire")
62     @PreAuthorize("hasAnyRole('EMPLOYEE','MANAGER')")
63     public ResponseEntity<Subscription> expireSubscription(@PathVariable Long id) {
64         return ResponseEntity.ok(subscriptionService.expireSubscription(id));
65     }
66
67     @GetMapping("/user")
68     @PreAuthorize("hasAuthority('SUBSCRIPTION_READ')")
69     public ResponseEntity<List<SubscriptionDTO>> getSubscriptionOfUser(Authentication authentication){
70         return ResponseEntity.ok(subscriptionService.getSubscriptionOfUser(authentication));
71     }

```

```
70     }
```

```
71
```

```
72 }
```

```
73
```



```

1 package com.example.EmployeeManagementSystem.Controller;
2
3 import com.example.EmployeeManagementSystem.Entity.Employee;
4 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
5 import com.example.EmployeeManagementSystem.Service.TotpService;
6 import org.springframework.http.ResponseEntity;
7 import org.springframework.security.core.Authentication;
8 import org.springframework.web.bind.annotation.*;
9
10 import java.util.Map;
11
12 @RestController
13 @RequestMapping("/totp")
14 public class TotpController {
15
16     private final TotpService totpService;
17     private final EmployeeRepo employeeRepo;
18
19     public TotpController(TotpService totpService, EmployeeRepo employeeRepo) {
20         this.totpService = totpService;
21         this.employeeRepo = employeeRepo;
22     }
23
24     /**
25      * GET /totp/status
26      * Returns whether TOTP is currently enabled for the logged-in employee.
27      * Used by the frontend to show the correct UI state on load.
28      */
29     @GetMapping("/status")
30     public ResponseEntity<?> status(Authentication authentication) {
31         Employee employee = getEmployee(authentication);
32         return ResponseEntity.ok(Map.of(
33             "totpEnabled", employee.isTotpEnabled()
34         ));
35     }
36
37     /**
38      * POST /totp/setup
39      * Step 1 – Generate a new TOTP secret and return the QR code URI.
40      * Employee scans the QR with Google Authenticator / Authy.
41      * TOTP is NOT enabled yet until /totp/verify-setup succeeds.
42      */
43     @PostMapping("/setup")
44     public ResponseEntity<?> setup(Authentication authentication) {
45         Employee employee = getEmployee(authentication);
46
47         String secret = totpService.generateSecret();
48         employee.setTotpSecret(secret);
49         employee.setTotpEnabled(false); // not active until confirmed
50         employeeRepo.save(employee);
51
52         String qrUri = totpService.getQrCodeUri(secret, employee.getName());
53
54         return ResponseEntity.ok(Map.of(
55             "secret", secret, // backup code for manual entry
56             "qrCodeUri", qrUri // embed in QR image on frontend
57         ));
58     }
59
60     /**
61      * POST /totp/verify-setup
62      * Step 2 – Employee scans QR and types their first code to confirm setup.
63      * Only after this succeeds is totpEnabled set to true.
64      * Body: { "code": "123456" }
65      */
66     @PostMapping("/verify-setup")
67     public ResponseEntity<?> verifySetup(@RequestBody Map<String, String> body,
68                                         Authentication authentication) {
69         Employee employee = getEmployee(authentication);
70
71         if (employee.getTotpSecret() == null) {

```

```

72         return ResponseEntity.badRequest().body(Map.of(
73             "error", "No TOTP secret found. Call /totp/setup first.");
74     }
75
76     String code = body.get("code");
77     if (code == null || code.isBlank()) {
78         return ResponseEntity.badRequest().body(Map.of("error", "code is required"));
79     }
80
81     if (!totpService.verifyCode(employee.getTotpSecret(), code)) {
82         return ResponseEntity.badRequest().body(Map.of("error", "Invalid code. Try again.));
83     }
84
85     employee.setTotpEnabled(true);
86     employeeRepo.save(employee);
87
88     return ResponseEntity.ok(Map.of("message", "TOTP 2FA enabled successfully"));
89 }
90
91 /**
92  * POST /totp/validate
93  * Step 3 – On every login when totpEnabled=true, call this after password auth.
94  * In the new flow this is handled by /Authenticate/totp using a pre-auth token.
95  * This endpoint is kept for compatibility / manual use.
96  * Body: { "code": "123456" }
97  */
98 @PostMapping("/validate")
99 public ResponseEntity<?> validate(@RequestBody Map<String, String> body,
100     Authentication authentication) {
101     Employee employee = getEmployee(authentication);
102
103     if (!employee.isTotpEnabled()) {
104         return ResponseEntity.ok(Map.of("message", "TOTP not enabled for this user"));
105     }
106
107     String code = body.get("code");
108     if (!totpService.verifyCode(employee.getTotpSecret(), code)) {
109         return ResponseEntity.status(401).body(Map.of("error", "Invalid TOTP code"));
110     }
111
112     return ResponseEntity.ok(Map.of("message", "TOTP verified", "authenticated", true));
113 }
114
115 /**
116  * DELETE /totp/disable
117  * Disable TOTP for the logged-in employee.
118  */
119 @DeleteMapping("/disable")
120 public ResponseEntity<?> disable(Authentication authentication) {
121     Employee employee = getEmployee(authentication);
122     employee.setTotpSecret(null);
123     employee.setTotpEnabled(false);
124     employeeRepo.save(employee);
125     return ResponseEntity.ok(Map.of("message", "TOTP disabled"));
126 }
127
128 // ---- helper ----
129
130 private Employee getEmployee(Authentication authentication) {
131     return employeeRepo.findByEmail(authentication.getName())
132         .orElseThrow(() -> new RuntimeException("Employee not found"));
133 }
134 }

```

```
1 package com.example.EmployeeManagementSystem.Controller;
2
3 import com.example.EmployeeManagementSystem.DTO.VendorDTO;
4 import com.example.EmployeeManagementSystem.DTO.VendorRequest;
5 import com.example.EmployeeManagementSystem.Service.DeliveryService;
6 import org.springframework.http.HttpStatus;
7 import org.springframework.http.ResponseEntity;
8 import org.springframework.security.access.prepost.PreAuthorize;
9 import org.springframework.security.core.Authentication;
10 import org.springframework.web.bind.annotation.*;
11
12 import java.util.List;
13
14 @RestController
15 @RequestMapping("/vendors")
16 public class VendorController {
17
18     private final DeliveryService.VendorService vendorService;
19
20     public VendorController(DeliveryService.VendorService vendorService) {
21         this.vendorService = vendorService;
22     }
23
24     @PostMapping("/register")
25     @PreAuthorize("hasRole('ADMIN')")
26     public ResponseEntity<VendorDTO> registerVendor(@RequestBody VendorRequest request) {
27         return ResponseEntity.status(HttpStatus.CREATED)
28             .body(vendorService.registerVendor(request));
29     }
30
31     @GetMapping
32     @PreAuthorize("hasAnyRole('FOOD_VENDOR', 'ADMIN', 'TECH_VENDOR')")
33     public ResponseEntity<List<VendorDTO>> getAllVendors() {
34         return ResponseEntity.ok(vendorService.getAllVendors());
35     }
36
37     @GetMapping("/me")
38     @PreAuthorize("hasAuthority('VENDOR_WRITE')")
39     public ResponseEntity<VendorDTO> getVendor(Authentication authentication) {
40         return ResponseEntity.ok(vendorService.getVendor(authentication));
41     }
42
43
44     @PutMapping("/update")
45     @PreAuthorize("hasAuthority('VENDOR_WRITE')")
46     public ResponseEntity<VendorDTO> updateVendor(
47         @RequestBody VendorRequest request,
48         Authentication authentication) {
49         return ResponseEntity.ok(vendorService.updateVendor(request, authentication));
50     }
51
52     @DeleteMapping
53     @PreAuthorize("hasRole('ADMIN')")
54     public ResponseEntity<String> deleteVendor(Authentication authentication) {
55         vendorService.deleteVendor(authentication);
56         return ResponseEntity.ok(
57             "Vendor account for " + authentication.getName() + " has been closed.");
58     }
59 }
```

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 public class AccessTokenRequest {
4     private String refreshToken;
5
6     public AccessTokenRequest(String refreshToken) {
7         this.refreshToken = refreshToken;
8     }
9
10    public void setRefreshToken(String refreshToken) {
11        this.refreshToken = refreshToken;
12    }
13
14    public String getRefreshToken() {
15        return refreshToken;
16    }
17 }
18
```

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3
4 public class ActionDTO {
5     private long leaveRequestId;
6     private String action;
7     private String remarks;
8
9     public long getLeaveRequestId() {
10         return leaveRequestId;
11     }
12
13     public void setLeaveRequestId(long leaveRequestId) {
14         this.leaveRequestId = leaveRequestId;
15     }
16
17
18     public String getAction() {
19         return action;
20     }
21
22     public void setAction(String action) {
23         this.action = action;
24     }
25
26     public String getRemarks() {
27         return remarks;
28     }
29
30     public void setRemarks(String remarks) {
31         this.remarks = remarks;
32     }
33 }
34
```

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import com.example.EmployeeManagementSystem.Enum.ServiceRequestStatus;
4
5 public class AdminActionDTO {
6     private Long serviceRequestId;
7     private ServiceRequestStatus status;
8     private String adminRemarks;
9
10    public Long getServiceRequestId() {
11        return serviceRequestId;
12    }
13
14    public void setServiceRequestId(Long serviceRequestId) {
15        this.serviceRequestId = serviceRequestId;
16    }
17
18    public String getAdminRemarks() {
19        return adminRemarks;
20    }
21
22    public void setAdminRemarks(String adminRemarks) {
23        this.adminRemarks = adminRemarks;
24    }
25
26    public ServiceRequestStatus getStatus() {
27        return status;
28    }
29
30    public void setStatus(ServiceRequestStatus status) {
31        this.status = status;
32    }
33 }
34
```

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import com.example.EmployeeManagementSystem.Enum.Gender;
4
5 public class AdminEmployeeDTO {
6     private String name;
7     private String email;
8     private String password;
9     private Gender gender;
10    private String timezone;
11    private long managerId;
12
13    public String getName() {
14        return name;
15    }
16
17    public String getPassword() {
18        return password;
19    }
20
21    public String getEmail() {
22        return email;
23    }
24
25    public String getTimezone() {
26        return timezone;
27    }
28
29    public Gender getGender() {
30        return gender;
31    }
32
33    public long getManagerId() {
34        return managerId;
35    }
36
37    public void setEmail(String email) {
38        this.email = email;
39    }
40
41    public void setPassword(String password) {
42        this.password = password;
43    }
44
45    public void setManagerId(long managerId) {
46        this.managerId = managerId;
47    }
48
49    public void setTimezone(String timezone) {
50        this.timezone = timezone;
51    }
52
53    public void setGender(Gender gender) {
54        this.gender = gender;
55    }
56
57    public void setName(String name) {
58        this.name = name;
59    }
60 }
61
```

```
1 package com.example.EmployeeManagementSystem.DTO;  
2  
3 public class AssignDeviceRequest {  
4     private Long employeeId;  
5  
6     public Long getEmployeeId() {  
7         return employeeId;  
8     }  
9  
10    public void setEmployeeId(Long employeeId) {  
11        this.employeeId = employeeId;  
12    }  
13 }  
14
```



```
1 package com.example.EmployeeManagementSystem.DTO;
2
3
4 import com.fasterxml.jackson.annotation.JsonProperty;
5
6 public class AuthRequest {
7     @JsonProperty("username")
8     private String Username;
9     @JsonProperty("password")
10    private String Password;
11
12    public String getPassword() {
13        return Password;
14    }
15
16    public void setPassword(String password) {
17        Password = password;
18    }
19
20    public String getUsername() {
21        return Username;
22    }
23
24    public void setUsername(String username) {
25        Username = username;
26    }
27 }
28
```

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 public class AuthResponse {
4     private String accessToken;
5     private String refreshToken;
6
7     public AuthResponse(String accessToken, String token) {
8         this.accessToken=accessToken;
9         this.refreshToken=token;
10    }
11
12    public String getAccessToken() {
13        return accessToken;
14    }
15
16    public void setAccessToken(String accessToken) {
17        this.accessToken = accessToken;
18    }
19
20    public String getRefreshToken() {
21        return refreshToken;
22    }
23
24    public void setRefreshToken(String refreshToken) {
25        this.refreshToken = refreshToken;
26    }
27 }
28
```

38 Current:

src\main\java\com\example\EmployeeManagementSystem\DTO\DeviceAssignmentResponseDTO.java

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import com.example.EmployeeManagementSystem.Enum.AssignmentStatus;
4 import com.example.EmployeeManagementSystem.Enum.DeviceType;
5
6 import java.time.LocalDate;
7
8 public class DeviceAssignmentResponseDTO {
9     private Long id;
10    private Long deviceId;
11    private String deviceName;
12    private String deviceBrand;
13    private DeviceType deviceType;
14    private Long employeeId;
15    private String employeeName;
16    private AssignmentStatus status;
17    private LocalDate assignedDate;
18
19    public Long getId() {
20        return id;
21    }
22
23    public void setId(Long id) {
24        this.id = id;
25    }
26
27    public String getDeviceName() {
28        return deviceName;
29    }
30
31    public void setDeviceName(String deviceName) {
32        this.deviceName = deviceName;
33    }
34
35    public Long getDeviceId() {
36        return deviceId;
37    }
38
39    public void setDeviceId(Long deviceId) {
40        this.deviceId = deviceId;
41    }
42
43    public DeviceType getDeviceType() {
44        return deviceType;
45    }
46
47    public void setDeviceType(DeviceType deviceType) {
48        this.deviceType = deviceType;
49    }
50
51    public String getDeviceBrand() {
52        return deviceBrand;
53    }
54
55    public void setDeviceBrand(String deviceBrand) {
56        this.deviceBrand = deviceBrand;
57    }
58
59    public String getEmployeeName() {
60        return employeeName;
61    }
62
63    public void setEmployeeName(String employeeName) {
64        this.employeeName = employeeName;
65    }
66
67    public Long getEmployeeId() {
68        return employeeId;
69    }
70
71    public void setEmployeeId(Long employeeId) {
```

```
72         this.employeeId = employeeId;
73     }
74
75     public AssignmentStatus getStatus() {
76         return status;
77     }
78
79     public void setStatus(AssignmentStatus status) {
80         this.status = status;
81     }
82
83     public LocalDate getAssignedDate() {
84         return assignedDate;
85     }
86
87     public void setAssignedDate(LocalDate assignedDate) {
88         this.assignedDate = assignedDate;
89     }
90 }
91
```

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import com.example.EmployeeManagementSystem.Enum.DeviceType;
4
5 import java.time.LocalDate;
6
7 public class DeviceDTO {
8     private String deviceName;        // e.g. "Dell XPS 15"
9     private String brand;
10    private DeviceType deviceType;
11    private LocalDate warrantyExpiryDate;
12
13    public String getDeviceName() {
14        return deviceName;
15    }
16
17    public void setDeviceName(String deviceName) {
18        this.deviceName = deviceName;
19    }
20
21    public String getBrand() {
22        return brand;
23    }
24
25    public void setBrand(String brand) {
26        this.brand = brand;
27    }
28
29    public DeviceType getDeviceType() {
30        return deviceType;
31    }
32
33    public void setDeviceType(DeviceType deviceType) {
34        this.deviceType = deviceType;
35    }
36
37    public LocalDate getWarrantyExpiryDate() {
38        return warrantyExpiryDate;
39    }
40
41    public void setWarrantyExpiryDate(LocalDate warrantyExpiryDate) {
42        this.warrantyExpiryDate = warrantyExpiryDate;
43    }
44 }
45
```

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import com.example.EmployeeManagementSystem.Entity.Employee;
4 import com.example.EmployeeManagementSystem.Entity.Vendor;
5 import com.example.EmployeeManagementSystem.Enum.DeviceStatus;
6 import com.example.EmployeeManagementSystem.Enum.DeviceType;
7
8 import java.time.LocalDate;
9
10 public class DeviceResponseDTO {
11     private Long id;
12     private String deviceName;
13     private String brand;
14     private DeviceType deviceType;
15     private DeviceStatus deviceStatus;
16     private String vendorName;
17     private long assignedEmployeeId;
18     private String assignedEmployeeName;
19     private LocalDate assignedDate;
20
21     private LocalDate warrantyExpiryDate;
22
23     public Long getId() {
24         return id;
25     }
26
27     public void setId(Long id) {
28         this.id = id;
29     }
30
31     public String getDeviceName() {
32         return deviceName;
33     }
34
35     public void setDeviceName(String deviceName) {
36         this.deviceName = deviceName;
37     }
38
39     public String getBrand() {
40         return brand;
41     }
42
43     public void setBrand(String brand) {
44         this.brand = brand;
45     }
46
47     public DeviceType getDeviceType() {
48         return deviceType;
49     }
50
51     public void setDeviceType(DeviceType deviceType) {
52         this.deviceType = deviceType;
53     }
54
55     public DeviceStatus getDeviceStatus() {
56         return deviceStatus;
57     }
58
59     public void setDeviceStatus(DeviceStatus deviceStatus) {
60         this.deviceStatus = deviceStatus;
61     }
62
63     public String getVendorName() {
64         return vendorName;
65     }
66
67     public void setVendorName(String vendorName) {
68         this.vendorName = vendorName;
69     }
70
71     public long getAssignedEmployeeId() {
```

```
72         return assignedEmployeeId;
73     }
74
75     public void setAssignedEmployeeId(long assignedEmployeeId) {
76         this.assignedEmployeeId = assignedEmployeeId;
77     }
78
79     public String getAssignedEmployeeName() {
80         return assignedEmployeeName;
81     }
82
83     public void setAssignedEmployeeName(String assignedEmployeeName) {
84         this.assignedEmployeeName = assignedEmployeeName;
85     }
86
87     public LocalDate getAssignedDate() {
88         return assignedDate;
89     }
90
91     public void setAssignedDate(LocalDate assignedDate) {
92         this.assignedDate = assignedDate;
93     }
94
95     public LocalDate getWarrantyExpiryDate() {
96         return warrantyExpiryDate;
97     }
98
99     public void setWarrantyExpiryDate(LocalDate warrantyExpiryDate) {
100         this.warrantyExpiryDate = warrantyExpiryDate;
101     }
102 }
103
```

41 Current:

src\main\java\com\example\EmployeeManagementSystem\DTO\EmployeeAttendanceResponseDTO.

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import java.util.List;
4
5 public class EmployeeAttendanceResponseDTO {
6     private long workingCount;
7     private long onLeaveCount;
8     private List<EmployeeStatusDTO> workingEmployees;
9     private List<EmployeeStatusDTO> onLeaveEmployees;
10
11     public long getWorkingCount() {
12         return workingCount;
13     }
14
15     public void setWorkingCount(long workingCount) {
16         this.workingCount = workingCount;
17     }
18
19     public long getOnLeaveCount() {
20         return onLeaveCount;
21     }
22
23     public void setOnLeaveCount(long onLeaveCount) {
24         this.onLeaveCount = onLeaveCount;
25     }
26
27     public List<EmployeeStatusDTO> getWorkingEmployees() {
28         return workingEmployees;
29     }
30
31     public void setWorkingEmployees(List<EmployeeStatusDTO> workingEmployees) {
32         this.workingEmployees = workingEmployees;
33     }
34
35     public List<EmployeeStatusDTO> getOnLeaveEmployees() {
36         return onLeaveEmployees;
37     }
38
39     public void setOnLeaveEmployees(List<EmployeeStatusDTO> onLeaveEmployees) {
40         this.onLeaveEmployees = onLeaveEmployees;
41     }
42 }
43
```



```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import com.example.EmployeeManagementSystem.Enum.Gender;
4
5 public class EmployeeDetails {
6     private String name;
7     private String email;
8     private String password;
9     private String timezone;
10    private Gender gender;
11
12    public String getName() {
13        return name;
14    }
15
16    public void setName(String name) {
17        this.name = name;
18    }
19
20    public String getPassword() {
21        return password;
22    }
23
24    public void setPassword(String password) {
25        this.password = password;
26    }
27
28    public String getEmail() {
29        return email;
30    }
31
32    public void setEmail(String email) {
33        this.email = email;
34    }
35
36    public String getTimezone() {
37        return timezone;
38    }
39
40    public void setTimezone(String timezone) {
41        this.timezone = timezone;
42    }
43
44    public Gender getGender() {
45        return gender;
46    }
47
48    public void setGender(Gender gender) {
49        this.gender = gender;
50    }
51 }
52
```

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3
4 import com.example.EmployeeManagementSystem.Enum.Gender;
5
6 public class EmployeeDTO {
7     private String name;
8     private String email;
9     private String dept;
10    private String password;
11    private Gender gender;
12    private String timezone;
13
14    public String getTimezone() {
15        return timezone;
16    }
17
18    public void setTimezone(String timezone) {
19        this.timezone = timezone;
20    }
21
22    public String getName() {
23        return name;
24    }
25
26    public void setName(String name) {
27        this.name = name;
28    }
29
30    public String getEmail() {
31        return email;
32    }
33
34    public void setEmail(String email) {
35        this.email = email;
36    }
37
38    public String getDept() {
39        return dept;
40    }
41
42    public void setDept(String dept) {
43        this.dept = dept;
44    }
45
46    public String getPassword() {
47        return password;
48    }
49
50    public void setPassword(String password) {
51        this.password = password;
52    }
53
54    public Gender getGender() {
55        return gender;
56    }
57
58    public void setGender(Gender gender) {
59        this.gender = gender;
60    }
61 }
```

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import com.example.EmployeeManagementSystem.Enum.Role;
4 import com.example.EmployeeManagementSystem.Enum.Status;
5
6 public class EmployeeStatusDTO {
7     private long employeeId;
8     private String name;
9     private String email;
10    private String dept;
11    private Role role;
12    private Status status;
13    private String timezone;
14
15    public long getEmployeeId() {
16        return employeeId;
17    }
18
19    public void setEmployeeId(long employeeId) {
20        this.employeeId = employeeId;
21    }
22
23    public String getName() {
24        return name;
25    }
26
27    public void setName(String name) {
28        this.name = name;
29    }
30
31    public String getEmail() {
32        return email;
33    }
34
35    public void setEmail(String email) {
36        this.email = email;
37    }
38
39    public String getDept() {
40        return dept;
41    }
42
43    public void setDept(String dept) {
44        this.dept = dept;
45    }
46
47    public Role getRole() {
48        return role;
49    }
50
51    public void setRole(Role role) {
52        this.role = role;
53    }
54
55    public Status getStatus() {
56        return status;
57    }
58
59    public void setStatus(Status status) {
60        this.status = status;
61    }
62
63    public String getTimezone() {
64        return timezone;
65    }
66
67    public void setTimezone(String timezone) {
68        this.timezone = timezone;
69    }
70 }
71
```

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 public class HolidayDTO {
4     private Long id;
5     private String name;
6     private String date;
7     private String description;
8     private int year;
9
10    public Long getId() {
11        return id;
12    }
13
14    public void setId(Long id) {
15        this.id = id;
16    }
17
18    public String getName() {
19        return name;
20    }
21
22    public void setName(String name) {
23        this.name = name;
24    }
25
26    public String getDate() {
27        return date;
28    }
29
30    public void setDate(String date) {
31        this.date = date;
32    }
33
34    public String getDescription() {
35        return description;
36    }
37
38    public void setDescription(String description) {
39        this.description = description;
40    }
41
42    public int getYear() {
43        return year;
44    }
45
46    public void setYear(int year) {
47        this.year = year;
48    }
49 }
50
```

```

1 package com.example.EmployeeManagementSystem.DTO;
2
3 import java.math.BigDecimal;
4 import java.util.List;
5 import java.util.Map;
6
7 public class LeaveDashboardResponse {
8
9     private Long    employeeId;
10    private String  employeeName;
11    private int     year;
12
13    // Individual leave balances
14    private LeaveBalanceDto sick;
15    private LeaveBalanceDto casual;
16    private LeaveBalanceDto maternity; // null for non-female employees
17
18    // Carry-forward meter (casual + maternity combined)
19    private CarryForwardMeterDto carryForwardMeter;
20
21    // Generic balances keyed by leave type name, e.g. SICK, CASUAL, MATERNITY.
22    private Map<String, LeaveBalanceDto> leaveBalances;
23
24    // Active warnings (unread)
25    private List<WarningDto> warnings;
26
27    // nested DTOs
28
29    public static class LeaveBalanceDto {
30        private String    leaveType;
31        private BigDecimal opening;
32        private BigDecimal accrued;
33        private BigDecimal used;
34        private BigDecimal available;
35        private boolean    carriesForward;
36
37        public LeaveBalanceDto(String leaveType, BigDecimal opening,
38                                BigDecimal accrued, BigDecimal used,
39                                BigDecimal available, boolean carriesForward) {
40            this.leaveType    = leaveType;
41            this.opening      = opening;
42            this.accrued      = accrued;
43            this.used         = used;
44            this.available    = available;
45            this.carriesForward = carriesForward;
46        }
47
48        public String    getLeaveType()    { return leaveType; }
49        public BigDecimal getOpening()    { return opening; }
50        public BigDecimal getAccrued()    { return accrued; }
51        public BigDecimal getUsed()       { return used; }
52        public BigDecimal getAvailable()  { return available; }
53        public boolean    isCarriesForward() { return carriesForward; }
54    }
55
56    public static class CarryForwardMeterDto {
57        private BigDecimal current; // e.g. 28.00
58        private int        cap;    // always 30
59        private String     label;  // "28/30"
60        private String     status; // SAFE | WARNING | CRITICAL | CAPPED
61
62        public CarryForwardMeterDto(BigDecimal current, int cap) {
63            this.current = current;
64            this.cap     = cap;
65            this.label   = current.toPlainString() + "/" + cap;
66            this.status  = resolveStatus(current, cap);
67        }
68
69        private String resolveStatus(BigDecimal current, int cap) {
70            int cmp = current.compareTo(BigDecimal.valueOf(cap));
71            if (cmp >= 0) return "CAPPED";

```

```

72         if (current.compareTo(BigDecimal.valueOf(cap - 1)) >= 0) return "CRITICAL";
73         if (current.compareTo(BigDecimal.valueOf(cap - 2)) >= 0) return "WARNING";
74         return "SAFE";
75     }
76
77     public BigDecimal getCurrent() { return current; }
78     public int         getCap()      { return cap; }
79     public String       getLabel()    { return label; }
80     public String       getStatus()   { return status; }
81 }
82
83 public static class WarningDto {
84     private Long id;
85     private String type;
86     private String message;
87     private String date;
88
89     public WarningDto(Long id, String type, String message, String date) {
90         this.id = id; this.type = type;
91         this.message = message; this.date = date;
92     }
93
94     public Long getId()      { return id; }
95     public String getType()  { return type; }
96     public String getMessage() { return message; }
97     public String getDate()  { return date; }
98 }
99
100 // root getters & setters
101
102 public Long         getEmployeeId()      { return employeeId; }
103 public String       getEmployeeName()    { return employeeName; }
104 public int          getYear()             { return year; }
105 public LeaveBalanceDto getSick()          { return sick; }
106 public LeaveBalanceDto getCasual()        { return casual; }
107 public LeaveBalanceDto getMaternity()     { return maternity; }
108 public CarryForwardMeterDto getCarryForwardMeter(){ return carryForwardMeter; }
109 public Map<String, LeaveBalanceDto> getLeaveBalances() { return leaveBalances; }
110 public List<WarningDto> getWarnings()      { return warnings; }
111
112 public void setEmployeeId(Long id)        { this.employeeId = id; }
113 public void setEmployeeName(String n)     { this.employeeName = n; }
114 public void setYear(int y)                { this.year = y; }
115 public void setSick(LeaveBalanceDto b)     { this.sick = b; }
116 public void setCasual(LeaveBalanceDto b)   { this.casual = b; }
117 public void setMaternity(LeaveBalanceDto b) { this.maternity = b; }
118 public void setCarryForwardMeter(CarryForwardMeterDto m) { this.carryForwardMeter = m; }
119 public void setLeaveBalances(Map<String, LeaveBalanceDto> b) { this.leaveBalances = b; }
120 public void setWarnings(List<WarningDto> w) { this.warnings = w; }
121 }
122

```

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import com.example.EmployeeManagementSystem.Enum.LeaveType;
4
5 import java.time.LocalDate;
6
7 // DTO used when employee applies for leave.
8 // Contains only input fields required from employee.
9 public class LeaveRequestDTO {
10     private LeaveType leaveType;
11     private LocalDate startDate;
12     private LocalDate endDate;
13     private String reason;
14
15
16     public LeaveType getLeaveType() {
17         return leaveType;
18     }
19
20     public void setLeaveType(LeaveType leaveType) {
21         this.leaveType = leaveType;
22     }
23
24     public LocalDate getStartDate() {
25         return startDate;
26     }
27
28     public void setStartDate(LocalDate startDate) {
29         this.startDate = startDate;
30     }
31
32     public LocalDate getEndDate() {
33         return endDate;
34     }
35
36     public void setEndDate(LocalDate endDate) {
37         this.endDate = endDate;
38     }
39
40     public String getReason() {
41         return reason;
42     }
43
44     public void setReason(String reason) {
45         this.reason = reason;
46     }
47 }
48
```

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import com.example.EmployeeManagementSystem.Enum.LeaveStatus;
4 import com.example.EmployeeManagementSystem.Enum.LeaveType;
5
6 import java.time.LocalDate;
7
8 public class LeaveResponseDTO {
9     private long leaveRequestId;
10    private String employeeName;
11    private LeaveType leaveType;
12    private LocalDate startDate;
13    private LocalDate endDate;
14    private long numberOfDays;
15    private String reason;
16    private LeaveStatus status;
17
18    public long getLeaveRequestId() {
19        return leaveRequestId;
20    }
21
22    public void setLeaveRequestId(long leaveRequestId) {
23        this.leaveRequestId = leaveRequestId;
24    }
25
26    public String getEmployeeName() {
27        return employeeName;
28    }
29
30    public void setEmployeeName(String employeeName) {
31        this.employeeName = employeeName;
32    }
33
34    public LocalDate getStartDate() {
35        return startDate;
36    }
37
38    public void setStartDate(LocalDate startDate) {
39        this.startDate = startDate;
40    }
41
42    public LeaveType getLeaveType() {
43        return leaveType;
44    }
45
46    public void setLeaveType(LeaveType leaveType) {
47        this.leaveType = leaveType;
48    }
49
50    public LocalDate getEndDate() {
51        return endDate;
52    }
53
54    public void setEndDate(LocalDate endDate) {
55        this.endDate = endDate;
56    }
57
58    public long getNumberOfDays() {
59        return numberOfDays;
60    }
61
62    public void setNumberOfDays(long numberOfDays) {
63        this.numberOfDays = numberOfDays;
64    }
65
66    public String getReason() {
67        return reason;
68    }
69
70    public void setReason(String reason) {
71        this.reason = reason;
```



```
72     }
73
74     public LeaveStatus getStatus() {
75         return status;
76     }
77
78     public void setStatus(L leaveStatus status) {
79         this.status = status;
80     }
81 }
82
```

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import org.bouncycastle.crypto.signers.ISOTrailers;
4
5 public class ManagerDTO {
6     private long managerId;
7     private String name;
8     private String email;
9     private String dept;
10
11     public long getManagerId() {
12         return managerId;
13     }
14
15     public void setManagerId(long managerId) {
16         this.managerId = managerId;
17     }
18
19     public String getName() {
20         return name;
21     }
22
23     public void setName(String name) {
24         this.name = name;
25     }
26
27     public String getEmail() {
28         return email;
29     }
30
31     public void setEmail(String email) {
32         this.email = email;
33     }
34
35     public String getDept() {
36         return dept;
37     }
38
39     public void setDept(String dept) {
40         this.dept = dept;
41     }
42 }
43
```

50 Current:

src\main\java\com\example\EmployeeManagementSystem\DTO\ManagerEmployeeDetailsDTO.java

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import com.example.EmployeeManagementSystem.Enum.Status;
4
5 import java.util.List;
6
7 public class ManagerEmployeeDetailsDTO {
8     private Long employeeId;
9     private String name;
10    private String email;
11    private String dept;
12    private Status status;
13    private String attendance;
14    private List<DeviceResponseDTO> devices;
15
16    public Long getEmployeeId() {
17        return employeeId;
18    }
19
20    public void setEmployeeId(Long employeeId) {
21        this.employeeId = employeeId;
22    }
23
24    public String getName() {
25        return name;
26    }
27
28    public void setName(String name) {
29        this.name = name;
30    }
31
32    public String getEmail() {
33        return email;
34    }
35
36    public void setEmail(String email) {
37        this.email = email;
38    }
39
40    public String getDept() {
41        return dept;
42    }
43
44    public void setDept(String dept) {
45        this.dept = dept;
46    }
47
48    public Status getStatus() {
49        return status;
50    }
51
52    public void setStatus(Status status) {
53        this.status = status;
54    }
55
56    public String getAttendance() {
57        return attendance;
58    }
59
60    public void setAttendance(String attendance) {
61        this.attendance = attendance;
62    }
63
64    public List<DeviceResponseDTO> getDevices() {
65        return devices;
66    }
67
68    public void setDevices(List<DeviceResponseDTO> devices) {
69        this.devices = devices;
70    }
71 }
```


51 Current:

src\main\java\com\example\EmployeeManagementSystem\DTO\ManagerLeaveResponseDTO.java

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import com.example.EmployeeManagementSystem.Enum.LeaveStatus;
4 import com.example.EmployeeManagementSystem.Enum.LeaveType;
5
6 import java.time.LocalDate;
7
8 public class ManagerLeaveResponseDTO {
9     private String employeeName;
10    private LeaveType leaveType;
11    private LocalDate startDate;
12    private LocalDate endDate;
13    private long numberOfDays;
14    private String reason;
15    private LeaveStatus status;
16
17    public String getEmployeeName() {
18        return employeeName;
19    }
20
21    public void setEmployeeName(String employeeName) {
22        this.employeeName = employeeName;
23    }
24
25    public LeaveType getLeaveType() {
26        return leaveType;
27    }
28
29    public void setLeaveType(LeaveType leaveType) {
30        this.leaveType = leaveType;
31    }
32
33    public LocalDate getStartDate() {
34        return startDate;
35    }
36
37    public void setStartDate(LocalDate startDate) {
38        this.startDate = startDate;
39    }
40
41    public LocalDate getEndDate() {
42        return endDate;
43    }
44
45    public void setEndDate(LocalDate endDate) {
46        this.endDate = endDate;
47    }
48
49    public long getNumberOfDays() {
50        return numberOfDays;
51    }
52
53    public void setNumberOfDays(long numberOfDays) {
54        this.numberOfDays = numberOfDays;
55    }
56
57    public String getReason() {
58        return reason;
59    }
60
61    public void setReason(String reason) {
62        this.reason = reason;
63    }
64
65    public LeaveStatus getStatus() {
66        return status;
67    }
68
69    public void setStatus(LeaveStatus status) {
70        this.status = status;
71    }
```



```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import java.time.LocalDate;
4
5 public class MaternityLeaveRequestDTO {
6     private LocalDate startDate; // only start date – end is always fixed
7     private String reason;
8
9     public LocalDate getStartDate() {
10         return startDate;
11     }
12
13     public void setStartDate(LocalDate startDate) {
14         this.startDate = startDate;
15     }
16
17     public String getReason() {
18         return reason;
19     }
20
21     public void setReason(String reason) {
22         this.reason = reason;
23     }
24 }
25
```

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 public class PromoteRequest {
4     private String email;
5     private String password;
6
7     public String getEmail() {
8         return email;
9     }
10
11     public void setEmail(String email) {
12         this.email = email;
13     }
14
15     public String getPassword() {
16         return password;
17     }
18
19     public void setPassword(String password) {
20         this.password = password;
21     }
22 }
23
```



```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import com.example.EmployeeManagementSystem.Enum.ServiceRequestStatus;
4
5 public class RepairDTO {
6     private Long requestId;
7     private String resolution;
8     private ServiceRequestStatus status;
9
10    public Long getRequestId() {
11        return requestId;
12    }
13
14    public void setRequestId(Long requestId) {
15        this.requestId = requestId;
16    }
17
18    public String getResolution() {
19        return resolution;
20    }
21
22    public void setResolution(String resolution) {
23        this.resolution = resolution;
24    }
25
26    public ServiceRequestStatus getStatus() {
27        return status;
28    }
29
30    public void setStatus(ServiceRequestStatus status) {
31        this.status = status;
32    }
33 }
34
```

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import com.example.EmployeeManagementSystem.Enum.MealSlot;
4
5 import java.util.Set;
6
7 public class RestaurantDTO {
8     private Long id;
9     private String name;
10    private String address;
11    private String description;
12    private Set<MealSlot> supportedMealSlots;
13    private boolean active;
14    private Long vendorId;
15    private String vendorName;
16
17    public Long getId() { return id; }
18    public void setId(Long id) { this.id = id; }
19
20    public String getName() { return name; }
21    public void setName(String name) { this.name = name; }
22
23    public String getAddress() { return address; }
24    public void setAddress(String address) { this.address = address; }
25
26    public String getDescription() { return description; }
27    public void setDescription(String description) { this.description = description; }
28
29    public Set<MealSlot> getSupportedMealSlots() { return supportedMealSlots; }
30    public void setSupportedMealSlots(Set<MealSlot> supportedMealSlots) {
31        this.supportedMealSlots = supportedMealSlots;
32    }
33
34    public boolean isActive() { return active; }
35    public void setActive(boolean active) { this.active = active; }
36
37    public Long getVendorId() { return vendorId; }
38    public void setVendorId(Long vendorId) { this.vendorId = vendorId; }
39
40    public String getVendorName() { return vendorName; }
41    public void setVendorName(String vendorName) { this.vendorName = vendorName; }
42 }
```

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import com.example.EmployeeManagementSystem.Enum.MealSlot;
4 import jakarta.validation.constraints.NotBlank;
5 import jakarta.validation.constraints.NotEmpty;
6 import jakarta.validation.constraints.NotNull;
7
8 import java.util.Set;
9
10 public class RestaurantRequest {
11
12     @NotBlank
13     private String name;
14
15     @NotBlank
16     private String address;
17
18     private String description;
19
20     /**
21      * Which meal slots this restaurant serves.
22      * Must be at least one of: BREAKFAST, LUNCH, DINNER.
23      */
24     @NotNull
25     @NotEmpty
26     private Set<MealSlot> supportedMealSlots;
27
28     /**
29      * The vendor ID of the owner making this request.
30      * Used to verify ownership on updates/deletes.
31      */
32
33     public String getName() { return name; }
34     public void setName(String name) { this.name = name; }
35
36     public String getAddress() { return address; }
37     public void setAddress(String address) { this.address = address; }
38
39     public String getDescription() { return description; }
40     public void setDescription(String description) { this.description = description; }
41
42     public Set<MealSlot> getSupportedMealSlots() { return supportedMealSlots; }
43     public void setSupportedMealSlots(Set<MealSlot> supportedMealSlots) {
44         this.supportedMealSlots = supportedMealSlots;
45     }
46 }
```

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import com.example.EmployeeManagementSystem.Enum.ServiceRequestType;
4
5 public class ServiceRequestDTO {
6
7     private Long deviceId;
8     private ServiceRequestType requestType;
9     private String issueDescription;
10    private boolean urgent;
11
12    public Long getDeviceId() {
13        return deviceId;
14    }
15
16    public void setDeviceId(Long deviceId) {
17        this.deviceId = deviceId;
18    }
19
20    public ServiceRequestType getRequestType() {
21        return requestType;
22    }
23
24    public void setRequestType(ServiceRequestType requestType) {
25        this.requestType = requestType;
26    }
27
28    public String getIssueDescription() {
29        return issueDescription;
30    }
31
32    public void setIssueDescription(String issueDescription) {
33        this.issueDescription = issueDescription;
34    }
35
36    public boolean isUrgent() {
37        return urgent;
38    }
39
40    public void setUrgent(boolean urgent) {
41        this.urgent = urgent;
42    }
43 }
44
```

58 Current:**src\main\java\com\example\EmployeeManagementSystem\DTO\ServiceRequestResponseDTO.java**

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import java.time.LocalDateTime;
4
5 public class ServiceRequestResponseDTO
6 {
7     private Long id;
8     private String requestType;
9     private String status;
10    private String issueDescription;
11    private String adminRemarks;
12    private String resolution;
13    private boolean urgent;
14    private LocalDateTime raisedAt;
15    private LocalDateTime resolvedAt;
16    private String deviceName;
17    private String raisedByName;
18    private String reviewedByName;
19
20    public String getDeviceName() {
21        return deviceName;
22    }
23
24    public void setDeviceName(String deviceName) {
25        this.deviceName = deviceName;
26    }
27
28    public String getReviewedByName() {
29        return reviewedByName;
30    }
31
32    public void setReviewedByName(String reviewedByName) {
33        this.reviewedByName = reviewedByName;
34    }
35
36    public LocalDateTime getResolvedAt() {
37        return resolvedAt;
38    }
39
40    public void setResolvedAt(LocalDateTime resolvedAt) {
41        this.resolvedAt = resolvedAt;
42    }
43
44    public LocalDateTime getRaisedAt() {
45        return raisedAt;
46    }
47
48    public void setRaisedAt(LocalDateTime raisedAt) {
49        this.raisedAt = raisedAt;
50    }
51
52    public boolean isUrgent() {
53        return urgent;
54    }
55
56    public void setUrgent(boolean urgent) {
57        this.urgent = urgent;
58    }
59
60    public String getResolution() {
61        return resolution;
62    }
63
64    public void setResolution(String resolution) {
65        this.resolution = resolution;
66    }
67
68    public Long getId() {
69        return id;
70    }
71}
```

```
72     public void setId(Long id) {
73         this.id = id;
74     }
75
76     public String getRequestType() {
77         return requestType;
78     }
79
80     public void setRequestType(String requestType) {
81         this.requestType = requestType;
82     }
83
84     public String getStatus() {
85         return status;
86     }
87
88     public void setStatus(String status) {
89         this.status = status;
90     }
91
92     public String getAdminRemarks() {
93         return adminRemarks;
94     }
95
96     public void setAdminRemarks(String adminRemarks) {
97         this.adminRemarks = adminRemarks;
98     }
99
100    public String getIssueDescription() {
101        return issueDescription;
102    }
103
104    public void setIssueDescription(String issueDescription) {
105        this.issueDescription = issueDescription;
106    }
107
108    public String getRaisedByName() {
109        return raisedByName;
110    }
111
112    public void setRaisedByName(String raisedByName) {
113        this.raisedByName = raisedByName;
114    }
115 }
116
```

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import com.example.EmployeeManagementSystem.Enum.MealSlot;
4 import com.example.EmployeeManagementSystem.Enum.ScheduleType;
5 import com.example.EmployeeManagementSystem.Enum.SubscriptionStatus;
6
7 import java.time.DayOfWeek;
8 import java.time.ZonedDateTime;
9
10 public class SubscriptionDTO {
11     private long subscriptionId;
12     private long userId;
13     private MealSlot mealSlot;
14     private ScheduleType scheduleType;
15     private DayOfWeek dayOfWeek;
16     private SubscriptionStatus status;
17     private ZonedDateTime nextDeliveryTime;
18
19     private Long restaurantId;
20     private String restaurantName;
21     private String restaurantAddress;
22
23     public void setRestaurantId(Long restaurantId) {
24         this.restaurantId = restaurantId;
25     }
26
27     public void setRestaurantName(String restaurantName) {
28         this.restaurantName = restaurantName;
29     }
30
31     public void setRestaurantAddress(String restaurantAddress) {
32         this.restaurantAddress = restaurantAddress;
33     }
34
35     public Long getRestaurantId() {
36         return restaurantId;
37     }
38
39     public String getRestaurantName() {
40         return restaurantName;
41     }
42
43     public String getRestaurantAddress() {
44         return restaurantAddress;
45     }
46
47     public long getSubscriptionId() {
48         return subscriptionId;
49     }
50
51     public void setSubscriptionId(long subscriptionId) {
52         this.subscriptionId = subscriptionId;
53     }
54
55     public long getUserId() {
56         return userId;
57     }
58
59     public void setUserId(long userId) {
60         this.userId = userId;
61     }
62
63     public MealSlot getMealSlot() {
64         return mealSlot;
65     }
66
67     public void setMealSlot(MealSlot mealSlot) {
68         this.mealSlot = mealSlot;
69     }
70
71     public ScheduleType getScheduleType() {
```

```
72         return scheduleType;
73     }
74
75     public void setScheduleType(ScheduleType scheduleType) {
76         this.scheduleType = scheduleType;
77     }
78
79     public DayOfWeek getDayOfWeek() {
80         return dayOfWeek;
81     }
82
83     public void setDayOfWeek(DayOfWeek dayOfWeek) {
84         this.dayOfWeek = dayOfWeek;
85     }
86
87     public SubscriptionStatus getStatus() {
88         return status;
89     }
90
91     public void setStatus(SubscriptionStatus status) {
92         this.status = status;
93     }
94
95     public ZonedDateTime getNextDeliveryTime() {
96         return nextDeliveryTime;
97     }
98
99     public void setNextDeliveryTime(ZonedDateTime nextDeliveryTime) {
100         this.nextDeliveryTime = nextDeliveryTime;
101     }
102 }
103
```



```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import com.example.EmployeeManagementSystem.Enum.MealSlot;
4 import com.example.EmployeeManagementSystem.Enum.ScheduleType;
5 import jakarta.validation.constraints.NotEmpty;
6 import jakarta.validation.constraints.NotNull;
7
8 import java.time.DayOfWeek;
9 import java.util.List;
10
11 public class SubscriptionRequest {
12     private ScheduleType scheduleType;
13     private DayOfWeek dayOfWeek;
14     @NotNull
15     @NotEmpty
16     private List<MealSlot> mealSlots;
17     @NotNull
18     private List<SlotOrder> slotOrders;
19     public static class SlotOrder {
20         @NotNull
21         private MealSlot mealSlot;
22
23         @NotNull
24         private Long restaurantId;
25
26         public MealSlot getMealSlot() {
27             return mealSlot;
28         }
29
30         public void setMealSlot(MealSlot mealSlot) {
31             this.mealSlot = mealSlot;
32         }
33
34         public Long getRestaurantId() {
35             return restaurantId;
36         }
37
38         public void setRestaurantId(Long restaurantId) {
39             this.restaurantId = restaurantId;
40         }
41     }
42
43     public ScheduleType getScheduleType() {
44         return scheduleType;
45     }
46
47     public void setScheduleType(ScheduleType scheduleType) {
48         this.scheduleType = scheduleType;
49     }
50
51     public List<MealSlot> getMealSlots() {
52         return mealSlots;
53     }
54
55     public void setMealSlots(List<MealSlot> mealSlots) {
56         this.mealSlots = mealSlots;
57     }
58
59     public DayOfWeek getDayOfWeek() {
60         return dayOfWeek;
61     }
62
63     public void setDayOfWeek(DayOfWeek dayOfWeek) {
64         this.dayOfWeek = dayOfWeek;
65     }
66
67
68
69     public List<SlotOrder> getSlotOrders() { return slotOrders; }
70     public void setSlotOrders(List<SlotOrder> slotOrders) { this.slotOrders = slotOrders; }
71 }
```



```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import java.time.LocalDate;
4
5 public class VendorDTO {
6     private Long id;
7     private String name;
8     private String email;
9     private String phone;
10    private LocalDate registeredAt;
11
12    public Long getId() { return id; }
13    public void setId(Long id) { this.id = id; }
14
15    public String getName() { return name; }
16    public void setName(String name) { this.name = name; }
17
18    public String getEmail() { return email; }
19    public void setEmail(String email) { this.email = email; }
20
21    public String getPhone() { return phone; }
22    public void setPhone(String phone) { this.phone = phone; }
23
24    public LocalDate getRegisteredAt() { return registeredAt; }
25    public void setRegisteredAt(LocalDate registeredAt) { this.registeredAt = registeredAt; }
26 }
```

```
1 package com.example.EmployeeManagementSystem.DTO;
2
3 import com.example.EmployeeManagementSystem.Enum.Role;
4 import jakarta.validation.constraints.Email;
5 import jakarta.validation.constraints.NotBlank;
6
7 public class VendorRequest {
8
9     @NotBlank
10     private String name;
11
12     @Email
13     @NotBlank
14     private String email;
15
16     @NotBlank
17     private String phone;
18
19     private String password;
20
21     private Role role;
22
23     public String getName() { return name; }
24     public void setName(String name) { this.name = name; }
25
26     public String getEmail() { return email; }
27     public void setEmail(String email) { this.email = email; }
28
29     public String getPhone() { return phone; }
30     public void setPhone(String phone) { this.phone = phone; }
31
32     public String getPassword() {
33         return password;
34     }
35
36     public void setPassword(String password) {
37         this.password = password;
38     }
39
40     public Role getRole() {
41         return role;
42     }
43
44     public void setRole(Role role) {
45         this.role = role;
46     }
47 }
```

63 Current:

src\main\java\com\example\EmployeeManagementSystem\EmployeeManagementSystemApplication

```
1 package com.example.EmployeeManagementSystem;
2
3 import org.springframework.boot.SpringApplication;
4 import org.springframework.boot.autoconfigure.SpringBootApplication;
5 import org.springframework.scheduling.annotation.EnableScheduling;
6
7 @SpringBootApplication
8 @EnableScheduling
9 public class EmployeeManagementSystemApplication {
10
11     public static void main(String[] args) {
12         SpringApplication.run(EmployeeManagementSystemApplication.class, args);
13     }
14
15 }
16
```

```

1 // src/main/java/com/example/EmployeeManagementSystem/Entity/ApiKey.java
2 package com.example.EmployeeManagementSystem.Entity;
3
4 import jakarta.persistence.*;
5 import java.time.Instant;
6 import java.util.UUID;
7
8 @Entity
9 @Table(name = "api_keys")
10 public class ApiKey {
11
12     @Id
13     @GeneratedValue(strategy = GenerationType.IDENTITY)
14     private Long id;
15
16     @Column(nullable = false, unique = true)
17     private String apiKey;
18
19     @Column(nullable = false)
20     private String keyName;
21
22     @ManyToOne
23     @JoinColumn(name = "employee_id")
24     private Employee employee;
25
26     @ManyToOne
27     @JoinColumn(name = "vendor_id")
28     private Vendor vendor;
29
30     @Column(nullable = false)
31     private String permissions; // Comma-separated permissions
32
33     @Column(nullable = false)
34     private Instant createdAt;
35
36     private Instant expiresAt;
37
38     private Instant lastUsedAt;
39
40     @Column(nullable = false)
41     private boolean active = true;
42
43     @PrePersist
44     public void prePersist() {
45         this.apiKey = "ems_" + UUID.randomUUID().toString().replace("-", "");
46         this.createdAt = Instant.now();
47     }
48
49     // Getters and setters
50     public Long getId() { return id; }
51     public void setId(Long id) { this.id = id; }
52
53     public String getApiKey() { return apiKey; }
54     public void setApiKey(String apiKey) { this.apiKey = apiKey; }
55
56     public String getKeyName() { return keyName; }
57     public void setKeyName(String keyName) { this.keyName = keyName; }
58
59     public Employee getEmployee() { return employee; }
60     public void setEmployee(Employee employee) { this.employee = employee; }
61
62     public Vendor getVendor() { return vendor; }
63     public void setVendor(Vendor vendor) { this.vendor = vendor; }
64
65     public String getPermissions() { return permissions; }
66     public void setPermissions(String permissions) { this.permissions = permissions; }
67
68     public Instant getCreatedAt() { return createdAt; }
69     public void setCreatedAt(Instant createdAt) { this.createdAt = createdAt; }
70
71     public Instant getExpiresAt() { return expiresAt; }

```

```
72     public void setExpiresAt(Instant expiresAt) { this.expiresAt = expiresAt; }
73
74     public Instant getLastUsedAt() { return lastUsedAt; }
75     public void setLastUsedAt(Instant lastUsedAt) { this.lastUsedAt = lastUsedAt; }
76
77     public boolean isActive() { return active; }
78     public void setActive(boolean active) { this.active = active; }
79 }
```

```
1 package com.example.EmployeeManagementSystem.Entity;
2
3 import com.example.EmployeeManagementSystem.Enum.DeliveryStatus;
4 import com.example.EmployeeManagementSystem.Enum.MealSlot;
5 import jakarta.persistence.*;
6
7 import java.time.Instant;
8
9 @Entity
10 @Table(name = "delivery", uniqueConstraints = { @UniqueConstraint(name = "uq_subscription_scheduled_time",
11     columnNames = {"subscription_id", "scheduledDeliveryTime"})})
12 public class Delivery {
13     @Id
14     @GeneratedValue(strategy = GenerationType.IDENTITY)
15     private Long id;
16
17     @ManyToOne
18     @JoinColumn(name = "subscription_id")
19     private Subscription subscription;
20     private MealSlot mealSlot;
21     private Instant scheduledDeliveryTime;
22     private Instant actualDeliveryTime;
23
24     @Enumerated(EnumType.STRING)
25     private DeliveryStatus status;
26
27     private Instant createdAt;
28
29     @PrePersist
30     public void prePersist() {
31         createdAt = Instant.now();
32         if (status == null) {
33             status = DeliveryStatus.SCHEDULED;
34         }
35     }
36
37     public MealSlot getMealSlot() {
38         return mealSlot;
39     }
40
41     public void setMealSlot(MealSlot mealSlot) {
42         this.mealSlot = mealSlot;
43     }
44
45     public Subscription getSubscription() {
46         return subscription;
47     }
48
49     public void setSubscription(Subscription subscription) {
50         this.subscription = subscription;
51     }
52
53     public Long getId() {
54         return id;
55     }
56
57     public void setId(Long id) {
58         this.id = id;
59     }
60
61     public Instant getScheduledDeliveryTime() {
62         return scheduledDeliveryTime;
63     }
64
65     public void setScheduledDeliveryTime(Instant scheduledDeliveryTime) {
66         this.scheduledDeliveryTime = scheduledDeliveryTime;
67     }
68
69     public Instant getActualDeliveryTime() {
70         return actualDeliveryTime;
71     }
```



```
72
73     public void setActualDeliveryTime(Instant actualDeliveryTime) {
74         this.actualDeliveryTime = actualDeliveryTime;
75     }
76
77     public DeliveryStatus getStatus() {
78         return status;
79     }
80
81     public void setStatus(DeliveryStatus status) {
82         this.status = status;
83     }
84
85     public Instant getCreatedAt() {
86         return createdAt;
87     }
88
89     public void setCreatedAt(Instant createdAt) {
90         this.createdAt = createdAt;
91     }
92 }
93
```

```
1 package com.example.EmployeeManagementSystem.Entity;
2
3
4 import com.example.EmployeeManagementSystem.Enum.DeviceStatus;
5 import com.example.EmployeeManagementSystem.Enum.DeviceType;
6 import jakarta.persistence.*;
7
8 import java.time.LocalDate;
9 import java.time.LocalDateTime;
10
11 @Entity
12 public class Device {
13     @Id
14     @GeneratedValue(strategy = GenerationType.AUTO)
15     private Long id;
16
17     private String deviceName;        // e.g. "Dell XPS 15"
18     private String brand;
19
20     @ManyToOne
21     @JoinColumn(name = "vendor_id")
22     private Vendor techVendor;
23
24     @OneToOne(mappedBy = "device")
25     private DeviceAssignment currentAssignment;
26
27     private DeviceStatus deviceStatus;
28
29     private DeviceType deviceType;
30
31     private LocalDateTime createdAt;
32
33     private LocalDate warrantyExpiryDate;
34
35     public Long getId() {
36         return id;
37     }
38
39     public void setId(Long id) {
40         this.id = id;
41     }
42
43     public String getDeviceName() {
44         return deviceName;
45     }
46
47     public void setDeviceName(String deviceName) {
48         this.deviceName = deviceName;
49     }
50
51     public String getBrand() {
52         return brand;
53     }
54
55     public void setBrand(String brand) {
56         this.brand = brand;
57     }
58
59     public Vendor getTechVendor() {
60         return techVendor;
61     }
62
63     public void setTechVendor(Vendor techVendor) {
64         this.techVendor = techVendor;
65     }
66
67     public DeviceType getDeviceType() {
68         return deviceType;
69     }
70
71     public void setDeviceType(DeviceType deviceType) {
```

```
72         this.deviceType = deviceType;
73     }
74
75     public LocalDateTime getCreatedAt() {
76         return createdAt;
77     }
78
79     public void setCreatedAt(LocalDateTime createdAt) {
80         this.createdAt = createdAt;
81     }
82
83     public DeviceStatus getDeviceStatus() {
84         return deviceStatus;
85     }
86
87     public void setDeviceStatus(DeviceStatus deviceStatus) {
88         this.deviceStatus = deviceStatus;
89     }
90
91     public DeviceAssignment getCurrentAssignment() {
92         return currentAssignment;
93     }
94
95     public void setCurrentAssignment(DeviceAssignment currentAssignment) {
96         this.currentAssignment = currentAssignment;
97     }
98
99     public LocalDate getWarrantyExpiryDate() {
100         return warrantyExpiryDate;
101     }
102
103     public void setWarrantyExpiryDate(LocalDate warrantyExpiryDate) {
104         this.warrantyExpiryDate = warrantyExpiryDate;
105     }
106 }
107
```

```
1 package com.example.EmployeeManagementSystem.Entity;
2
3 import com.example.EmployeeManagementSystem.Enum.AssignmentStatus;
4 import jakarta.persistence.*;
5
6 import java.time.LocalDate;
7
8 @Entity
9 public class DeviceAssignment {
10
11     @Id
12     @GeneratedValue(strategy = GenerationType.IDENTITY)
13     private Long id;
14
15     @OneToOne
16     @JoinColumn(name = "device_id")
17     private Device device;
18
19     @ManyToOne
20     @JoinColumn(name = "employee_id")
21     private Employee assignedTo;
22
23     @Enumerated(EnumType.STRING)
24     private AssignmentStatus status;
25
26     private LocalDate assignedDate;
27
28     public Long getId() {
29         return id;
30     }
31
32     public void setId(Long id) {
33         this.id = id;
34     }
35
36     public Device getDevice() {
37         return device;
38     }
39
40     public void setDevice(Device device) {
41         this.device = device;
42     }
43
44     public Employee getAssignedTo() {
45         return assignedTo;
46     }
47
48     public void setAssignedTo(Employee assignedTo) {
49         this.assignedTo = assignedTo;
50     }
51
52     public LocalDate getAssignedDate() {
53         return assignedDate;
54     }
55
56     public void setAssignedDate(LocalDate assignedDate) {
57         this.assignedDate = assignedDate;
58     }
59
60     public AssignmentStatus getStatus() {
61         return status;
62     }
63
64     public void setStatus(AssignmentStatus status) {
65         this.status = status;
66     }
67 }
68
```

```
1 package com.example.EmployeeManagementSystem.Entity;
2
3 import com.example.EmployeeManagementSystem.Enum.Gender;
4 import com.example.EmployeeManagementSystem.Enum.Role;
5 import com.example.EmployeeManagementSystem.Enum.Status;
6 import com.fasterxml.jackson.annotation.JsonIgnore;
7 import jakarta.persistence.*;
8 import org.springframework.security.core.GrantedAuthority;
9 import org.springframework.security.core.authority.SimpleGrantedAuthority;
10 import org.springframework.security.core.userdetails.UserDetails;
11
12 import java.time.LocalDate;
13 import java.util.Collection;
14 import java.util.HashSet;
15 import java.util.List;
16 import java.util.Set;
17 import java.util.stream.Collectors;
18
19 @Entity
20 public class Employee implements UserDetails {
21     @Id
22     @GeneratedValue(strategy = GenerationType.IDENTITY)
23     private long employeeId;
24     @Column(nullable = true, unique = false)
25     private String name;
26     @Column(unique = true, nullable = false)
27     private String email;
28     @Column(nullable = true)
29     private String dept;
30     @Enumerated(EnumType.STRING)
31     private Status status;
32     private String password;
33
34     // Self-referential – replaces `String manager`
35     @ManyToOne(fetch = FetchType.LAZY)
36     @JoinColumn(name = "manager_id", nullable = true)
37     @JsonIgnore // prevent infinite recursion in serialization
38     private Employee manager;
39
40
41     private LocalDate joined_at;
42     @Enumerated(EnumType.STRING)
43     private Role role;
44     @Column(nullable = true)
45     private String totpSecret;
46     private boolean totpEnabled = false;
47     /** IANA timezone id, e.g. "Asia/Kolkata", "America/New_York". Defaults to UTC. */
48     @Column(nullable = false)
49     private String timezone = "UTC";
50     @Enumerated(EnumType.STRING)
51     @Column(nullable = true)
52     private Gender gender;
53
54     public String getTimezone() {
55         return timezone;
56     }
57
58     public void setTimezone(String timezone) {
59         this.timezone = timezone;
60     }
61
62     public Role getRole() {
63         return role;
64     }
65
66     public void setRole(Role role) {
67         this.role = role;
68     }
69
70     public long getEmployeeId() {
71         return employeeId;
```

```

72     }
73
74     public void setEmployeeId(long employeeId) {
75         this.employeeId = employeeId;
76     }
77
78     public String getName() {
79         return name;
80     }
81
82     public void setName(String name) {
83         this.name = name;
84     }
85
86     public String getEmail() {
87         return email;
88     }
89
90     public void setEmail(String email) {
91         this.email = email;
92     }
93
94     public String getDept() {
95         return dept;
96     }
97
98     public void setDept(String dept) {
99         this.dept = dept;
100    }
101
102    public Status getStatus() {
103        return status;
104    }
105
106    public void setStatus(Status status) {
107        this.status = status;
108    }
109
110    public LocalDate getJoined_at() {
111        return joined_at;
112    }
113
114    public void setJoined_at(LocalDate joined_at) {
115        this.joined_at = joined_at;
116    }
117
118    public Gender getGender() { return gender; }
119    public void setGender(Gender gender) { this.gender = gender; }
120
121    @PrePersist
122    public void initialSetup() {
123        this.status = Status.ACTIVE;
124        this.joined_at = LocalDate.now();
125        if (this.timezone == null || this.timezone.isBlank()) {
126            this.timezone = "UTC";
127        }
128    }
129
130    @Override
131    public Collection<? extends GrantedAuthority> getAuthorities() {
132        Set<SimpleGrantedAuthority> authorities=new HashSet<>();
133        authorities.add(new SimpleGrantedAuthority("ROLE_"+role.name()));
134        authorities.addAll(role.getPermissions().stream()
135            .map(permissions -> new SimpleGrantedAuthority(permissions.name()))
136            .collect(Collectors.toSet()));
137        return authorities;
138    }
139
140    public String getPassword() {
141        return password;
142    }
143
144    @Override
145    public String getUsername() {
146        return email;
147    }

```

```
148
149     @Override
150     public boolean isAccountNonExpired() {
151         return UserDetails.super.isAccountNonExpired();
152     }
153
154     @Override
155     public boolean isAccountNonLocked() {
156         return UserDetails.super.isAccountNonLocked();
157     }
158
159     @Override
160     public boolean isCredentialsNonExpired() {
161         return UserDetails.super.isCredentialsNonExpired();
162     }
163
164     @Override
165     public boolean isEnabled() {
166         return UserDetails.super.isEnabled();
167     }
168
169     public void setPassword(String password) {
170         this.password = password;
171     }
172
173     public String getTotpSecret() {
174         return totpSecret;
175     }
176
177     public boolean isTotpEnabled() {
178         return totpEnabled;
179     }
180
181     public void setTotpSecret(String totpSecret) {
182         this.totpSecret = totpSecret;
183     }
184
185     public void setTotpEnabled(boolean totpEnabled) {
186         this.totpEnabled = totpEnabled;
187     }
188
189     public Employee getManager() {
190         return manager;
191     }
192
193     public void setManager(Employee manager) {
194         this.manager = manager;
195     }
196
197 }
```

```
1 package com.example.EmployeeManagementSystem.Entity;
2
3 import jakarta.persistence.*;
4
5 import java.time.LocalDate;
6
7 @Entity
8 public class FixedHoliday {
9
10
11     @Id
12     @GeneratedValue(strategy = GenerationType.IDENTITY)
13     private Long id;
14
15     private String name;           // "Diwali", "Christmas"
16     private LocalDate date;       // 2026-10-20
17     private String description;   // optional note
18     private int year;
19
20
21     public Long getId() {
22         return id;
23     }
24
25     public void setId(Long id) {
26         this.id = id;
27     }
28
29     public String getName() {
30         return name;
31     }
32
33     public void setName(String name) {
34         this.name = name;
35     }
36
37     public LocalDate getDate() {
38         return date;
39     }
40
41     public void setDate(LocalDate date) {
42         this.date = date;
43     }
44
45     public int getYear() {
46         return year;
47     }
48
49     public void setYear(int year) {
50         this.year = year;
51     }
52
53     public String getDescription() {
54         return description;
55     }
56
57     public void setDescription(String description) {
58         this.description = description;
59     }
60 }
61
```



```
1 package com.example.EmployeeManagementSystem.Entity;
2
3 import jakarta.persistence.*;
4
5 import java.math.BigDecimal;
6
7 @Entity
8 @Table(name = "leave_entitlement", uniqueConstraints = @UniqueConstraint(columnNames = {"employee_id",
9 "leave_type_id", "year"}))
10 public class LeaveEntitlement {
11     @Id
12     @GeneratedValue(strategy = GenerationType.IDENTITY)
13     private Long id;
14
15     @ManyToOne(fetch = FetchType.LAZY, optional = false)
16     @JoinColumn(name = "employee_id", nullable = false)
17     private Employee employee;
18
19     @ManyToOne(fetch = FetchType.LAZY, optional = false)
20     @JoinColumn(name = "leave_type_id", nullable = false)
21     private LeaveType leaveType;
22
23     @Column(nullable = false)
24     private Integer year;
25
26     @Column(nullable = false, precision = 5, scale = 2)
27     private BigDecimal openingBalance = BigDecimal.ZERO;
28
29     @Column(nullable = false, precision = 5, scale = 2)
30     private BigDecimal accruedThisYear = BigDecimal.ZERO;
31
32     @Column(nullable = false, precision = 5, scale = 2)
33     private BigDecimal usedThisYear = BigDecimal.ZERO;
34
35     @Column(nullable = false, precision = 5, scale = 2)
36     private BigDecimal closingBalance = BigDecimal.ZERO;
37
38     // Convenience: current available = opening + accrued - used
39     @Transient
40     public BigDecimal getAvailableBalance() {
41         return openingBalance.add(accruedThisYear).subtract(usedThisYear);
42     }
43
44     // --- getters & setters ---
45     public Long getId() {
46         return id;
47     }
48
49     public Employee getEmployee() {
50         return employee;
51     }
52
53     public LeaveType getLeaveType() {
54         return leaveType;
55     }
56
57     public Integer getYear() {
58         return year;
59     }
60
61     public BigDecimal getOpeningBalance() {
62         return openingBalance;
63     }
64
65     public BigDecimal getAccruedThisYear() {
66         return accruedThisYear;
67     }
68
69     public BigDecimal getUsedThisYear() {
70         return usedThisYear;
```

```
71     }
72
73     public BigDecimal getClosingBalance() {
74         return closingBalance;
75     }
76
77     public void setEmployee(Employee e) {
78         this.employee = e;
79     }
80
81     public void setLeaveType(LeaveType lt) {
82         this.leaveType = lt;
83     }
84
85     public void setYear(Integer y) {
86         this.year = y;
87     }
88
89     public void setOpeningBalance(BigDecimal b) {
90         this.openingBalance = b;
91     }
92
93     public void setAccruedThisYear(BigDecimal a) {
94         this.accruedThisYear = a;
95     }
96
97     public void setUsedThisYear(BigDecimal u) {
98         this.usedThisYear = u;
99     }
100
101     public void setClosingBalance(BigDecimal c) {
102         this.closingBalance = c;
103     }
104 }
```

```
1 package com.example.EmployeeManagementSystem.Entity;
2
3 import com.example.EmployeeManagementSystem.Enum.LeaveStatus;
4 import com.example.EmployeeManagementSystem.Enum.LeaveType;
5 import jakarta.persistence.*;
6
7 import java.time.LocalDate;
8
9 @Entity
10 public class LeaveRequest {
11     @Id
12     @GeneratedValue(strategy = GenerationType.IDENTITY)
13     private long id;
14     @Enumerated(EnumType.STRING)
15     private LeaveType leaveType;
16     private LocalDate startDate;
17     private LocalDate endDate;
18     private String reason;
19     @Enumerated(EnumType.STRING)
20     private LeaveStatus status;
21     private String Manager;
22     private String remarks;
23     @ManyToOne
24     @JoinColumn(name = "employee_id")
25     private Employee employee;
26
27
28     public long getId() {
29         return id;
30     }
31
32     public void setId(long id) {
33         this.id = id;
34     }
35
36     public LeaveType getLeaveType() {
37         return leaveType;
38     }
39
40     public void setLeaveType(LeaveType leaveType) {
41         this.leaveType = leaveType;
42     }
43
44     public LocalDate getStartDate() {
45         return startDate;
46     }
47
48     public void setStartDate(LocalDate startDate) {
49         this.startDate = startDate;
50     }
51
52     public LocalDate getEndDate() {
53         return endDate;
54     }
55
56     public void setEndDate(LocalDate endDate) {
57         this.endDate = endDate;
58     }
59
60     public LeaveStatus getStatus() {
61         return status;
62     }
63
64     public void setStatus(LeaveStatus status) {
65         this.status = status;
66     }
67
68     public String getManager() {
69         return Manager;
70     }
71 }
```

```
72     public void setManager(String manager) {
73         Manager = manager;
74     }
75
76     public String getRemarks() {
77         return remarks;
78     }
79
80     public void setRemarks(String remarks) {
81         this.remarks = remarks;
82     }
83
84     public Employee getEmployee() {
85         return employee;
86     }
87
88     public void setEmployee(Employee employee) {
89         this.employee = employee;
90     }
91
92     public String getReason() {
93         return reason;
94     }
95
96     public void setReason(String reason) {
97         this.reason = reason;
98     }
99
100     @PrePersist
101     public void init(){
102         this.status=LeaveStatus.PENDING;
103     }
104 }
105
```

```
1 package com.example.EmployeeManagementSystem.Entity;
2
3 import com.example.EmployeeManagementSystem.Enum.Gender;
4 import jakarta.persistence.*;
5
6 @Entity
7 @Table(name = "leave_type")
8 public class LeaveType {
9
10     @Id
11     @GeneratedValue(strategy = GenerationType.IDENTITY)
12     private Integer id;
13
14     @Column(nullable = false, unique = true, length = 50)
15     private String name;           // SICK, CASUAL, MATERNITY
16
17     @Column(nullable = false)
18     private boolean carriesForward;
19
20     @Column(nullable = false)
21     private boolean monthlyReset;   // true = sick leave wipes at month end
22
23     @Column(nullable = false)
24     private int maxCarryForward;    // 30 for carry-forwardable types
25
26     @Column(nullable = false)
27     private boolean genderRestricted;
28
29     @Enumerated(EnumType.STRING)
30     @Column(nullable = true)
31     private Gender genderRestriction; // null = open to all
32
33     // --- getters & setters ---
34     public Integer getId() { return id; }
35     public String getName() { return name; }
36     public boolean isCarriesForward() { return carriesForward; }
37     public boolean isMonthlyReset() { return monthlyReset; }
38     public int getMaxCarryForward() { return maxCarryForward; }
39     public boolean isGenderRestricted() { return genderRestricted; }
40     public Gender getGenderRestriction() { return genderRestriction; }
41
42     public void setId(Integer id) { this.id = id; }
43     public void setName(String name) { this.name = name; }
44     public void setCarriesForward(boolean cf) { this.carriesForward = cf; }
45     public void setMonthlyReset(boolean mr) { this.monthlyReset = mr; }
46     public void setMaxCarryForward(int max) { this.maxCarryForward = max; }
47     public void setGenderRestricted(boolean gr) { this.genderRestricted = gr; }
48     public void setGenderRestriction(Gender g) { this.genderRestriction = g; }
49 }
```

```
1 package com.example.EmployeeManagementSystem.Entity;
2
3
4 import jakarta.persistence.*;
5 import java.math.BigDecimal;
6 import java.time.LocalDate;
7
8 @Entity
9 @Table(name = "leave_warning")
10 public class LeaveWarning {
11
12     public enum WarningType {
13         APPROACHING_28, // 2 months before cap
14         APPROACHING_29, // 1 month before cap
15         CAP_REACHED,    // exactly at 30
16         DAY_WASTED      // already at 30, day lost
17     }
18
19     @Id
20     @GeneratedValue(strategy = GenerationType.IDENTITY)
21     private Long id;
22
23     @ManyToOne(fetch = FetchType.LAZY, optional = false)
24     @JoinColumn(name = "employee_id", nullable = false)
25     private Employee employee;
26
27     @Enumerated(EnumType.STRING)
28     @Column(nullable = false, length = 30)
29     private WarningType warningType;
30
31     @Column(nullable = false, columnDefinition = "TEXT")
32     private String message;
33
34     @Column(nullable = false, precision = 5, scale = 2)
35     private BigDecimal carryForwardBalance;
36
37     @Column(nullable = false)
38     private LocalDate warningDate;
39
40     @Column(nullable = false)
41     private boolean isRead = false;
42
43     // --- getters & setters ---
44     public Long getId() { return id; }
45     public Employee getEmployee() { return employee; }
46     public WarningType getWarningType() { return warningType; }
47     public String getMessage() { return message; }
48     public BigDecimal getCarryForwardBalance() { return carryForwardBalance; }
49     public LocalDate getWarningDate() { return warningDate; }
50     public boolean isRead() { return isRead; }
51
52     public void setEmployee(Employee e) { this.employee = e; }
53     public void setWarningType(WarningType wt) { this.warningType = wt; }
54     public void setMessage(String m) { this.message = m; }
55     public void setCarryForwardBalance(BigDecimal b) { this.carryForwardBalance = b; }
56     public void setWarningDate(LocalDate d) { this.warningDate = d; }
57     public void setRead(boolean r) { this.isRead = r; }
58 }
```

```
1 package com.example.EmployeeManagementSystem.Entity;
2
3 import jakarta.persistence.*;
4 import java.time.LocalDateTime;
5
6 @Entity
7 @Table(name = "notification")
8 public class Notification {
9
10     @Id
11     @GeneratedValue(strategy = GenerationType.IDENTITY)
12     private Long id;
13
14     @ManyToOne(fetch = FetchType.LAZY)
15     @JoinColumn(name = "employee_id", nullable = false)
16     private Employee recipient;
17
18     @Column(nullable = false)
19     private String message;
20
21     @Column(nullable = false)
22     private String type; // LEAVE_APPROVED, LEAVE_REJECTED, LEAVE_REQUEST, HOLIDAY_ADDED
23
24     @Column(nullable = false)
25     private boolean isRead = false;
26
27     @Column(nullable = false)
28     private LocalDateTime createdAt = LocalDateTime.now();
29
30     // getters and setters
31     public Long getId() { return id; }
32     public Employee getRecipient() { return recipient; }
33     public void setRecipient(Employee recipient) { this.recipient = recipient; }
34     public String getMessage() { return message; }
35     public void setMessage(String message) { this.message = message; }
36     public String getType() { return type; }
37     public void setType(String type) { this.type = type; }
38     public boolean isRead() { return isRead; }
39     public void setRead(boolean read) { isRead = read; }
40     public LocalDateTime getCreatedAt() { return createdAt; }
41 }
42
```

```
1 package com.example.EmployeeManagementSystem.Entity;
2
3 import jakarta.persistence.*;
4
5 import java.time.Instant;
6 @Entity
7 public class RefreshToken {
8     @Id
9     @GeneratedValue(strategy = GenerationType.IDENTITY)
10    private Long id;
11
12    @Column(nullable = false, unique = true)
13    private String token;
14
15    @Column(nullable = false)
16    private Instant expiryTime;
17
18    @ManyToOne
19    @JoinColumn(name = "employee_id")
20    private Employee employee;
21
22    @ManyToOne
23    @JoinColumn(name = "vendor_id")
24    private Vendor vendor;
25
26    public Long getId() {
27        return id;
28    }
29
30    public String getToken() {
31        return token;
32    }
33
34    public void setToken(String token) {
35        this.token = token;
36    }
37
38    public Instant getExpiryTime() {
39        return expiryTime;
40    }
41
42    public void setExpiryTime(Instant expiryTime) {
43        this.expiryTime = expiryTime;
44    }
45
46    public Employee getEmployee() {
47        return employee;
48    }
49
50    public void setEmployee(Employee employee) {
51        this.employee = employee;
52    }
53
54    public Vendor getVendor() {
55        return vendor;
56    }
57
58    public void setVendor(Vendor vendor) {
59        this.vendor = vendor;
60    }
61 }
62
```



```
1 package com.example.EmployeeManagementSystem.Entity;
2
3 import com.example.EmployeeManagementSystem.Enum.MealSlot;
4 import jakarta.persistence.*;
5 import java.util.Set;
6
7
8 @Entity
9 public class Restaurant {
10
11     @Id
12     @GeneratedValue(strategy = GenerationType.IDENTITY)
13     private Long id;
14
15     @Column(nullable = false)
16     private String name;
17
18     @Column(nullable = false)
19     private String address;
20
21     private String description;
22
23     @ElementCollection(targetClass = MealSlot.class, fetch = FetchType.EAGER)
24     @CollectionTable(name = "restaurant_meal_slots",
25         joinColumns = @JoinColumn(name = "restaurant_id"))
26     @Enumerated(EnumType.STRING)
27     @Column(name = "meal_slot")
28     private Set<MealSlot> supportedMealSlots;
29
30     private boolean active = true;
31
32     @ManyToOne(fetch = FetchType.LAZY)
33     @JoinColumn(name = "vendor_id", nullable = false)
34     private Vendor vendor;
35
36     public Long getId() { return id; }
37     public void setId(Long id) { this.id = id; }
38
39     public String getName() { return name; }
40     public void setName(String name) { this.name = name; }
41
42     public String getAddress() { return address; }
43     public void setAddress(String address) { this.address = address; }
44
45     public String getDescription() { return description; }
46     public void setDescription(String description) { this.description = description; }
47
48     public Set<MealSlot> getSupportedMealSlots() { return supportedMealSlots; }
49     public void setSupportedMealSlots(Set<MealSlot> supportedMealSlots) {
50         this.supportedMealSlots = supportedMealSlots;
51     }
52
53     public boolean isActive() { return active; }
54     public void setActive(boolean active) { this.active = active; }
55
56     public Vendor getVendor() { return vendor; }
57     public void setVendor(Vendor vendor) { this.vendor = vendor; }
58 }
```

```
1 package com.example.EmployeeManagementSystem.Entity;
2
3 import com.example.EmployeeManagementSystem.Enum.ServiceRequestStatus;
4 import com.example.EmployeeManagementSystem.Enum.ServiceRequestType;
5 import jakarta.persistence.*;
6
7 import java.time.LocalDateTime;
8
9 @Entity
10 public class ServiceRequest {
11     @Id
12     @GeneratedValue(strategy = GenerationType.IDENTITY)
13     private Long id;
14
15     @ManyToOne
16     @JoinColumn(name = "device_id")
17     private Device device;
18
19     @ManyToOne
20     @JoinColumn(name = "raised_by")
21     private Employee raisedBy; // employee reporting issue
22
23     @ManyToOne
24     @JoinColumn(name = "reviewed_by")
25     private Employee reviewedBy; // admin who reviewed
26
27     @Enumerated(EnumType.STRING)
28     private ServiceRequestType requestType;
29
30     @Enumerated(EnumType.STRING)
31     private ServiceRequestStatus status;
32
33     private String issueDescription; // "screen cracked", "won't boot"
34     private String adminRemarks; // admin notes on decision
35     private String resolution; // what was done finally
36
37     private boolean urgent; // employee can flag as urgent
38
39     private LocalDateTime raisedAt;
40     private LocalDateTime resolvedAt;
41
42     @PrePersist
43     void onCreate() { this.raisedAt = LocalDateTime.now(); }
44
45     public Employee getRaisedBy() {
46         return raisedBy;
47     }
48
49     public void setRaisedBy(Employee raisedBy) {
50         this.raisedBy = raisedBy;
51     }
52
53     public Device getDevice() {
54         return device;
55     }
56
57     public void setDevice(Device device) {
58         this.device = device;
59     }
60
61     public Long getId() {
62         return id;
63     }
64
65     public void setId(Long id) {
66         this.id = id;
67     }
68
69     public Employee getReviewedBy() {
70         return reviewedBy;
71     }
```

```
72
73     public void setReviewedBy(Employee reviewedBy) {
74         this.reviewedBy = reviewedBy;
75     }
76
77     public ServiceRequestStatus getStatus() {
78         return status;
79     }
80
81     public void setStatus(ServiceRequestStatus status) {
82         this.status = status;
83     }
84
85     public ServiceRequestType getRequestType() {
86         return requestType;
87     }
88
89     public void setRequestType(ServiceRequestType requestType) {
90         this.requestType = requestType;
91     }
92
93     public String getIssueDescription() {
94         return issueDescription;
95     }
96
97     public void setIssueDescription(String issueDescription) {
98         this.issueDescription = issueDescription;
99     }
100
101     public String getAdminRemarks() {
102         return adminRemarks;
103     }
104
105     public void setAdminRemarks(String adminRemarks) {
106         this.adminRemarks = adminRemarks;
107     }
108
109     public String getResolution() {
110         return resolution;
111     }
112
113     public void setResolution(String resolution) {
114         this.resolution = resolution;
115     }
116
117     public boolean isUrgent() {
118         return urgent;
119     }
120
121     public void setUrgent(boolean urgent) {
122         this.urgent = urgent;
123     }
124
125     public LocalDateTime getRaisedAt() {
126         return raisedAt;
127     }
128
129     public void setRaisedAt(LocalDateTime raisedAt) {
130         this.raisedAt = raisedAt;
131     }
132
133     public LocalDateTime getResolvedAt() {
134         return resolvedAt;
135     }
136
137     public void setResolvedAt(LocalDateTime resolvedAt) {
138         this.resolvedAt = resolvedAt;
139     }
140 }
141
```

```
1 package com.example.EmployeeManagementSystem.Entity;
2
3 import com.example.EmployeeManagementSystem.Enum.MealSlot;
4 import com.example.EmployeeManagementSystem.Enum.ScheduleType;
5 import com.example.EmployeeManagementSystem.Enum.SubscriptionStatus;
6 import jakarta.persistence.*;
7
8 import java.time.DayOfWeek;
9 import java.time.Instant;
10 import java.time.LocalDateTime;
11 import java.time.LocalTime;
12 @Entity
13 public class Subscription {
14     @Id
15     @GeneratedValue(strategy = GenerationType.IDENTITY)
16     private long id;
17     @Enumerated(EnumType.STRING)
18     private MealSlot slot;
19     private ScheduleType scheduleType;
20     private DayOfWeek dayOfWeek;
21     @Enumerated(EnumType.STRING)
22     private SubscriptionStatus status;
23     private Instant nextDeliveryTime;
24     private LocalTime created_at;
25
26     @ManyToOne
27     @JoinColumn(name="employee_id")
28     private Employee employee;
29
30     @ManyToOne(fetch = FetchType.EAGER)
31     @JoinColumn(name = "restaurant_id", nullable = false)
32     private Restaurant restaurant;
33
34     @PrePersist
35     public void init(){
36         this.created_at=LocalTime.now();
37     }
38
39     public long getId() {
40         return id;
41     }
42
43     public void setId(long id) {
44         this.id = id;
45     }
46
47     public MealSlot getSlot() {
48         return slot;
49     }
50
51     public void setSlot(MealSlot slot) {
52         this.slot = slot;
53     }
54
55     public ScheduleType getScheduleType() {
56         return scheduleType;
57     }
58
59     public void setScheduleType(ScheduleType scheduleType) {
60         this.scheduleType = scheduleType;
61     }
62
63     public DayOfWeek getDayOfWeek() {
64         return dayOfWeek;
65     }
66
67     public void setDayOfWeek(DayOfWeek dayOfWeek) {
68         this.dayOfWeek = dayOfWeek;
69     }
70
71     public SubscriptionStatus getStatus() {
```

```
72         return status;
73     }
74
75     public void setStatus(SubscriptionStatus status) {
76         this.status = status;
77     }
78
79     public LocalTime getCreated_at() {
80         return created_at;
81     }
82
83     public void setCreated_at(LocalTime created_at) {
84         this.created_at = created_at;
85     }
86
87     public Instant getNextDeliveryTime() {
88         return nextDeliveryTime;
89     }
90
91     public void setNextDeliveryTime(Instant nextDeliveryTime) {
92         this.nextDeliveryTime = nextDeliveryTime;
93     }
94
95     public Employee getEmployee() {
96         return employee;
97     }
98
99     public void setEmployee(Employee employee) {
100         this.employee = employee;
101     }
102     public Restaurant getRestaurant() { return restaurant; }
103     public void setRestaurant(Restaurant restaurant) { this.restaurant = restaurant; }
104 }
105
```

```
1 package com.example.EmployeeManagementSystem.Entity;
2
3 import com.example.EmployeeManagementSystem.Enum.Role;
4 import jakarta.persistence.*;
5 import jakarta.xml.bind.annotation.XmlAttribute;
6 import org.jspecify.annotations.Nullable;
7 import org.springframework.security.core.GrantedAuthority;
8 import org.springframework.security.core.authority.SimpleGrantedAuthority;
9 import org.springframework.security.core.userdetails.UserDetails;
10
11 import java.time.LocalDate;
12 import java.util.Collection;
13 import java.util.HashSet;
14 import java.util.List;
15 import java.util.Set;
16 import java.util.stream.Collectors;
17
18 /**
19  * Represents a restaurant vendor/owner.
20  * Vendors can register and manage their restaurants.
21  * They cannot access employee, leave, or subscription APIs.
22  */
23 @Entity
24 @Table(name = "vendor")
25 public class Vendor implements UserDetails {
26
27     @Id
28     @GeneratedValue(strategy = GenerationType.IDENTITY)
29     private Long id;
30
31     @Column(nullable = false)
32     private String name;
33
34     @Column(unique = true, nullable = false)
35     private String email;
36
37     @Column(nullable = false)
38     private String phone;
39
40     @Column(nullable = false)
41     private String password;
42
43     @Enumerated(EnumType.STRING)
44     private Role role;
45
46     private LocalDate registeredAt;
47
48     @PrePersist
49     public void init() {
50         if (this.registeredAt == null) {
51             this.registeredAt = LocalDate.now();
52         }
53     }
54
55     public Long getId() { return id; }
56     public void setId(Long id) { this.id = id; }
57
58     public String getName() { return name; }
59     public void setName(String name) { this.name = name; }
60
61     public String getEmail() { return email; }
62     public void setEmail(String email) { this.email = email; }
63
64     public String getPhone() { return phone; }
65     public void setPhone(String phone) { this.phone = phone; }
66
67     public LocalDate getRegisteredAt() { return registeredAt; }
68     public void setRegisteredAt(LocalDate registeredAt) { this.registeredAt = registeredAt; }
69
70
71     @Override
```

```

72     public Collection<? extends GrantedAuthority> getAuthorities() {
73         Role effectiveRole = role;
74         Set<SimpleGrantedAuthority> authorities=new HashSet<>();
75         authorities.add(new SimpleGrantedAuthority("ROLE_"+effectiveRole.name()));
76         authorities.addAll(effectiveRole.getPermissions().stream()
77             .map(permissions -> new SimpleGrantedAuthority(permissions.name()))
78             .collect(Collectors.toSet()));
79         return authorities;
80     }
81
82     @Override
83     public @Nullable String getPassword() {
84         return password;
85     }
86
87     @Override
88     public String getUsername() {
89         return email;
90     }
91
92     @Override
93     public boolean isAccountNonExpired() {
94         return UserDetails.super.isAccountNonExpired();
95     }
96
97     @Override
98     public boolean isAccountNonLocked() {
99         return UserDetails.super.isAccountNonLocked();
100    }
101
102    @Override
103    public boolean isCredentialsNonExpired() {
104        return UserDetails.super.isCredentialsNonExpired();
105    }
106
107    @Override
108    public boolean isEnabled() {
109        return UserDetails.super.isEnabled();
110    }
111
112    public void setPassword(String password) {
113        this.password = password;
114    }
115
116    public Role getRole() {
117        return role;
118    }
119
120    public void setRole(Role role) {
121        this.role = role;
122    }
123 }
124

```

```

1 package com.example.EmployeeManagementSystem.Entity;
2
3
4 import jakarta.persistence.*;
5 import java.math.BigDecimal;
6 import java.time.LocalDateTime;
7
8 @Entity
9 @Table(name = "year_end_carry_forward_log")
10 public class YearEndCarryForwardLog {
11
12     @Id
13     @GeneratedValue(strategy = GenerationType.IDENTITY)
14     private Long id;
15
16     @ManyToOne(fetch = FetchType.LAZY, optional = false)
17     @JoinColumn(name = "employee_id", nullable = false)
18     private Employee employee;
19
20     @Column(nullable = false)
21     private Integer processedYear;
22
23     @ManyToOne(fetch = FetchType.LAZY, optional = false)
24     @JoinColumn(name = "leave_type_id", nullable = false)
25     private LeaveType leaveType;
26
27     @Column(nullable = false, precision = 5, scale = 2)
28     private BigDecimal closingBalance; // raw closing before cap
29
30     @Column(nullable = false, precision = 5, scale = 2)
31     private BigDecimal cappedBalance; // after global 30-day cap applied
32
33     @Column(nullable = false, precision = 5, scale = 2)
34     private BigDecimal daysWasted = BigDecimal.ZERO;
35
36     @Column(nullable = false, precision = 5, scale = 2)
37     private BigDecimal openingNextYear; // written into next year's entitlement
38
39     @Column(nullable = false)
40     private LocalDateTime processedAt = LocalDateTime.now();
41
42     // --- getters & setters ---
43     public Long getId() { return id; }
44     public Employee getEmployee() { return employee; }
45     public Integer getProcessedYear() { return processedYear; }
46     public LeaveType getLeaveType() { return leaveType; }
47     public BigDecimal getClosingBalance() { return closingBalance; }
48     public BigDecimal getCappedBalance() { return cappedBalance; }
49     public BigDecimal getDaysWasted() { return daysWasted; }
50     public BigDecimal getOpeningNextYear() { return openingNextYear; }
51     public LocalDateTime getProcessedAt() { return processedAt; }
52
53     public void setEmployee(Employee e) { this.employee = e; }
54     public void setProcessedYear(Integer y) { this.processedYear = y; }
55     public void setLeaveType(LeaveType lt) { this.leaveType = lt; }
56     public void setClosingBalance(BigDecimal b) { this.closingBalance = b; }
57     public void setCappedBalance(BigDecimal b) { this.cappedBalance = b; }
58     public void setDaysWasted(BigDecimal d) { this.daysWasted = d; }
59     public void setOpeningNextYear(BigDecimal b) { this.openingNextYear = b; }
60     public void setProcessedAt(LocalDateTime dt) { this.processedAt = dt; }
61 }
62

```


81 Current:

src\main\java\com\example\EmployeeManagementSystem\Enum\AssignmentStatus.java

8
lines

```
1 package com.example.EmployeeManagementSystem.Enum;
```

```
2
```

```
3 public enum AssignmentStatus {
```

```
4     ACTIVE,
```

```
5     RETURNED,
```

```
6     REVOKED
```

```
7 }
```

```
8
```

82

Current:

9

src\main\java\com\example\EmployeeManagementSystem\Enum\DeliveryStatus.java

lines

```
1 package com.example.EmployeeManagementSystem.Enum;  
2  
3 public enum DeliveryStatus {  
4     SCHEDULED,  
5     IN_PROGRESS,  
6     DELIVERED,  
7     MISSED  
8 }  
9
```

83

Current:

10

src\main\java\com\example\EmployeeManagementSystem\Enum\DeviceStatus.java

lines

```
1 package com.example.EmployeeManagementSystem.Enum;  
2  
3 public enum DeviceStatus {  
4     AVAILABLE,  
5     ASSIGNED,  
6     UNDER_REPAIR,  
7     CONDEMNED,  
8     RETURNED_TO_VENDOR  
9 }  
10
```

```
1 package com.example.EmployeeManagementSystem.Enum;  
2  
3 public enum DeviceType {  
4     LAPTOP,  
5     DESKTOP,  
6     PHONE,  
7     TABLET,  
8     MONITOR,  
9     KEYBOARD,  
10    MOUSE,  
11    HEADSET,  
12    OTHER  
13  
14 }  
15
```

85

Current:

src\main\java\com\example\EmployeeManagementSystem\Enum\Gender.java

5

lines

```
1 package com.example.EmployeeManagementSystem.Enum;  
2  
3 public enum Gender {  
4     M, F, O  
5 }
```

86

Current:

9

src\main\java\com\example\EmployeeManagementSystem\Enum\LeaveStatus.java

lines

```
1 package com.example.EmployeeManagementSystem.Enum;  
2  
3 public enum LeaveStatus {  
4     PENDING,  
5     APPROVED,  
6     REJECTED,  
7     CANCELLED  
8 }  
9
```

87

Current:

9

src\main\java\com\example\EmployeeManagementSystem\Enum\LeaveType.java

lines

```
1 package com.example.EmployeeManagementSystem.Enum;  
2  
3 public enum LeaveType {  
4     SICK,  
5     CASUAL,  
6     MATERNITY,  
7     UNPAID  
8 }  
9
```

88

Current:

8

src\main\java\com\example\EmployeeManagementSystem\Enum\MealSlot.java

lines

```
1 package com.example.EmployeeManagementSystem.Enum;
```

```
2
```

```
3 public enum MealSlot {
```

```
4     BREAKFAST,
```

```
5     LUNCH,
```

```
6     DINNER
```

```
7 }
```

```
8
```



```
1 package com.example.EmployeeManagementSystem.Enum;  
2  
3 import java.util.Set;  
4  
5 public enum Permission {  
6     LEAVE_READ,  
7     LEAVE_WRITE,  
8     LEAVE_UPDATE,  
9     LEAVE_CANCEL,  
10    SUBSCRIPTION_WRITE,  
11    SUBSCRIPTION_READ,  
12    SUBSCRIPTION_UPDATE,  
13    VENDOR_UPDATE,  
14    VENDOR_DELETE,  
15    RESTAURANTS_ADD,  
16    RESTAURANTS_READ,  
17    RESTAURANTS_DELETE,  
18    PROFILE_READ,  
19    VENDOR_WRITE;  
20 }  
21
```

```
1 package com.example.EmployeeManagementSystem.Enum;
2
3 import java.util.Collection;
4 import java.util.Set;
5
6 import static com.example.EmployeeManagementSystem.Enum.Permission.*;
7
8
9 public enum Role {
10     EMPLOYEE(Set.of(LEAVE_WRITE,
11         LEAVE_CANCEL,
12         SUBSCRIPTION_WRITE,
13         SUBSCRIPTION_UPDATE,
14         RESTAURANTS_READ,
15         SUBSCRIPTION_READ,
16         PROFILE_READ)),
17     MANAGER(Set.of(LEAVE_UPDATE,
18         LEAVE_WRITE,
19         SUBSCRIPTION_WRITE,
20         RESTAURANTS_READ,
21         SUBSCRIPTION_UPDATE,
22         LEAVE_READ,
23         SUBSCRIPTION_READ,
24         PROFILE_READ)),
25     FOOD_VENDOR(Set.of(
26         VENDOR_WRITE,
27         VENDOR_DELETE,
28         RESTAURANTS_ADD,
29         RESTAURANTS_DELETE,
30         RESTAURANTS_READ,
31         PROFILE_READ
32     )),
33     ADMIN(Set.of(
34         LEAVE_READ,
35         LEAVE_WRITE,
36         LEAVE_UPDATE,
37         LEAVE_CANCEL,
38         RESTAURANTS_READ,
39         SUBSCRIPTION_READ,
40         SUBSCRIPTION_WRITE,
41         SUBSCRIPTION_UPDATE,
42         VENDOR_WRITE,
43         VENDOR_UPDATE,
44         VENDOR_DELETE,
45         PROFILE_READ
46     )),
47     TECH_VENDOR(Set.of(VENDOR_WRITE)),
48     USER(Set.of(PROFILE_READ));
49     private final Set<Permission> permissions;
50
51     Role(Set<Permission> permissions) {
52         this.permissions = permissions;
53     }
54
55     public Set<Permission> getPermissions() {
56         return permissions;
57     }
58 }
59
60
```

91

Current:

7

src\main\java\com\example\EmployeeManagementSystem\Enum\ScheduleType.java

lines

```
1 package com.example.EmployeeManagementSystem.Enum;  
2  
3 public enum ScheduleType {  
4     DAILY,  
5     WEEKLY  
6 }  
7
```

```
1 package com.example.EmployeeManagementSystem.Enum;  
2  
3 public enum ServiceRequestStatus {  
4     OPEN,  
5     SENT_FOR_REPAIR,  
6     REPAIR_DONE,  
7     REJECTED  
8 }  
9
```

93 Current:

src\main\java\com\example\EmployeeManagementSystem\Enum\ServiceRequestType.java

10
lines

```
1 package com.example.EmployeeManagementSystem.Enum;  
2  
3 public enum ServiceRequestType {  
4     REPAIR,  
5     REPLACEMENT,  
6     UPGRADE,  
7     MAINTENANCE,  
8     RETURN  
9 }  
10
```

```
1 package com.example.EmployeeManagementSystem.Enum;  
2  
3 public enum Status {  
4     ACTIVE,  
5     INACTIVE,  
6     ON_LEAVE  
7 }  
8
```

95 Current:

src\main\java\com\example\EmployeeManagementSystem\Enum\SubscriptionStatus.java

8

lines

```
1 package com.example.EmployeeManagementSystem.Enum;
```

```
2
```

```
3 public enum SubscriptionStatus {
```

```
4     ACTIVE,
```

```
5     PAUSED,
```

```
6     EXPIRED
```

```
7 }
```

```
8
```

96 Current:

src\main\java\com\example\EmployeeManagementSystem\Exception\DuplicateRequestException.java

```
1 package com.example.EmployeeManagementSystem.Exception;
2
3 public class DuplicateRequestException extends RuntimeException {
4     public DuplicateRequestException(String message) {
5         super(message);
6     }
7 }
8
```



```
1 package com.example.EmployeeManagementSystem.Exception;
2
3 import org.springframework.security.core.userdetails.UsernameNotFoundException;
4
5 public class EmployeeNotFound extends UsernameNotFoundException {
6     public EmployeeNotFound(String message) {
7         super(message);
8     }
9 }
10
```

98 Current:**src\main\java\com\example\EmployeeManagementSystem\Exception\InvalidEndDateException.java**

```
1 package com.example.EmployeeManagementSystem.Exception;
2
3 public class InvalidEndDateException extends RuntimeException{
4     public InvalidEndDateException(String message) {
5         super(message);
6     }
7 }
8
```

99 Current:

src\main\java\com\example\EmployeeManagementSystem\Exception\InvalidManagerException.java

```
1 package com.example.EmployeeManagementSystem.Exception;
2
3 public class InvalidManagerException extends RuntimeException{
4     public InvalidManagerException(String message) {
5         super(message);
6     }
7 }
8
```

100 Current:

src\main\java\com\example\EmployeeManagementSystem\Exception\InvalidStartDateException.java

```
1 package com.example.EmployeeManagementSystem.Exception;
2
3 public class InvalidStartDateException extends RuntimeException{
4     public InvalidStartDateException(String message) {
5         super(message);
6     }
7 }
8
```

101 Current:

src\main\java\com\example\EmployeeManagementSystem\Exception\LeaveRequestNotFoundExcep

```
1 package com.example.EmployeeManagementSystem.Exception;
2
3 public class LeaveRequestNotFoundException extends RuntimeException{
4     public LeaveRequestNotFoundException(String message) {
5         super(message);
6     }
7 }
8
```

102 Current:

src\main\java\com\example\EmployeeManagementSystem\Exception\OverlappingLeaveException.j

```
1 package com.example.EmployeeManagementSystem.Exception;
2
3 public class OverlappingLeaveException extends RuntimeException{
4     public OverlappingLeaveException(String message) {
5         super(message);
6     }
7 }
8
```

103 Current:

src\main\java\com\example\EmployeeManagementSystem\Exception\RestaurantNotFoundException

```
1 package com.example.EmployeeManagementSystem.Exception;
2
3 public class RestaurantNotFoundException extends RuntimeException {
4     public RestaurantNotFoundException(String message) {
5         super(message);
6     }
7 }
8
```

104 Current:

src\main\java\com\example\EmployeeManagementSystem\Exception\SubscriptionAlreadyExists.java

```
1 package com.example.EmployeeManagementSystem.Exception;
2
3 public class SubscriptionAlreadyExists extends RuntimeException{
4     public SubscriptionAlreadyExists(String message) {
5         super(message);
6     }
7 }
8
```


105 Current:

src\main\java\com\example\EmployeeManagementSystem\Exception\UnauthorizedAccessException

```
1 package com.example.EmployeeManagementSystem.Exception;
2
3 public class UnauthorizedAccessException extends RuntimeException {
4     public UnauthorizedAccessException(String message) {
5         super(message);
6     }
7 }
```

106 Current:

src\main\java\com\example\EmployeeManagementSystem\Exception\VendorNotFoundException.java

```
1 package com.example.EmployeeManagementSystem.Exception;
2
3 public class VendorNotFoundException extends RuntimeException {
4     public VendorNotFoundException(String message) {
5         super(message);
6     }
7 }
```

```

1 // src/main/java/com/example/EmployeeManagementSystem/Filter/ApiKeyFilter.java
2 package com.example.EmployeeManagementSystem.Filter;
3
4 import com.example.EmployeeManagementSystem.security.ApiKeyAuthenticationToken;
5 import jakarta.servlet.FilterChain;
6 import jakarta.servlet.ServletException;
7 import jakarta.servlet.http.HttpServletRequest;
8 import jakarta.servlet.http.HttpServletResponse;
9 import org.springframework.beans.factory.annotation.Autowired;
10 import org.springframework.beans.factory.annotation.Qualifier;
11 import org.springframework.security.authentication.AuthenticationManager;
12 import org.springframework.security.core.Authentication;
13 import org.springframework.security.core.context.SecurityContextHolder;
14 import org.springframework.stereotype.Component;
15 import org.springframework.web.filter.OncePerRequestFilter;
16
17 import java.io.IOException;
18 public class ApiKeyFilter extends OncePerRequestFilter {
19
20     private static final String API_KEY_HEADER = "X-API-Key";
21     private final AuthenticationManager authenticationManager;
22
23     public ApiKeyFilter(AuthenticationManager authenticationManager) {
24         this.authenticationManager = authenticationManager;
25     }
26     // @Autowired
27     // @Qualifier("apiKeyAuthenticationManager") // or just use ProviderManager directly
28     // private AuthenticationManager authenticationManager;
29
30     @Override
31     protected void doFilterInternal(HttpServletRequest request,
32                                     HttpServletResponse response,
33                                     FilterChain filterChain) throws ServletException, IOException {
34
35         String apiKey = request.getHeader(API_KEY_HEADER);
36
37         if (apiKey != null && !apiKey.isEmpty()) {
38             try {
39                 // Create the unauthenticated token
40                 ApiKeyAuthenticationToken authRequest = new ApiKeyAuthenticationToken(apiKey);
41                 // Run it through the provider – this hits the DB, checks expiry, loads roles
42                 Authentication authenticated = authenticationManager.authenticate(authRequest);
43                 // Only set it if validation passed
44                 SecurityContextHolder.getContext().setAuthentication(authenticated);
45             } catch (RuntimeException e) {
46                 // Invalid/expired key – clear context and return 401
47                 SecurityContextHolder.clearContext();
48                 response.setStatus(HttpServletResponse.SC_UNAUTHORIZED);
49                 response.setContentType("application/json");
50                 response.getWriter().write("{\"error\": \"" + e.getMessage() + "\"}");
51                 return;
52             }
53         }
54
55         filterChain.doFilter(request, response);
56     }
57
58     @Override
59     protected boolean shouldNotFilter(HttpServletRequest request) {
60         String path = request.getRequestURI();
61         return path.startsWith("/Authenticate") ||
62             path.startsWith("/session/login") ||
63             path.startsWith("/session/logout") ||
64             path.startsWith("/oauth2") ||
65             path.startsWith("/login/oauth2") ||
66             path.startsWith("/swagger-ui") ||
67             path.startsWith("/v3/api-docs") ||
68             path.startsWith("/api-keys/generate"); // Allow API key generation without API key
69     }
70 }

```

```
1 package com.example.EmployeeManagementSystem.Filter;
2
3 import com.example.EmployeeManagementSystem.Util.JWTUtil;
4 import io.jsonwebtoken.ExpiredJwtException;
5 import io.jsonwebtoken.MalformedJwtException;
6 import io.jsonwebtoken.UnsupportedJwtException;
7 import jakarta.servlet.FilterChain;
8 import jakarta.servlet.ServletException;
9 import jakarta.servlet.http.HttpServletRequest;
10 import jakarta.servlet.http.HttpServletResponse;
11 import org.springframework.beans.factory.annotation.Autowired;
12 import org.springframework.beans.factory.annotation.Qualifier;
13 import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
14 import org.springframework.security.core.context.SecurityContextHolder;
15 import org.springframework.security.core.userdetails.UserDetails;
16 import org.springframework.security.core.userdetails.UserDetailsService;
17 import org.springframework.stereotype.Component;
18 import org.springframework.web.filter.OncePerRequestFilter;
19
20 import java.io.IOException;
21
22 @Component
23 public class JwtAuthFilter extends OncePerRequestFilter {
24
25     @Autowired
26     @Qualifier("combinedUserDetailsService")
27     private UserDetailsService customUserDetailsService;
28
29     @Autowired
30     private JWTUtil jwtUtil;
31
32     @Override
33     protected void doFilterInternal(HttpServletRequest request,
34                                     HttpServletResponse response,
35                                     FilterChain filterChain) throws ServletException, IOException {
36         String token = null;
37         String username = null;
38
39         String authHeader = request.getHeader("Authorization");
40
41         if (authHeader != null && authHeader.startsWith("Bearer ")) {
42             token = authHeader.substring(7).trim();
43
44             // Reject obviously invalid values before hitting JWT
45             if (token.isEmpty() || token.equals("null") || token.equals("undefined")) {
46                 filterChain.doFilter(request, response);
47                 return;
48             }
49
50             try {
51                 username = jwtUtil.extractUsernameFromToken(token);
52             } catch (ExpiredJwtException e) {
53                 response.setStatus(HttpServletResponse.SC_UNAUTHORIZED);
54                 response.setContentType("application/json");
55                 response.getWriter().write("{\""
56                     {
57                         "error": "TOKEN_EXPIRED",
58                         "message": "JWT expired. Please login again to get a new token."
59                     }
60                     "\"");
61                 return;
62             } catch (MalformedJwtException | UnsupportedJwtException | IllegalArgumentException e) {
63                 // Token is garbage (e.g. "null" string, garbled value) –
64                 // don't crash. OAuth2 session auth will still work fine.
65                 logger.debug("JwtAuthFilter: skipping malformed token – " + e.getMessage());
66                 filterChain.doFilter(request, response);
67                 return;
68             }
69         }
70
71         if (username != null && SecurityContextHolder.getContext().getAuthentication() == null) {
```

```
72         UserDetails userDetails = customUserDetailsService.loadUserByUsername(username);
73         if (userDetails != null && jwtUtil.validateToken(username, userDetails, token)) {
74             UsernamePasswordAuthenticationToken authenticationToken =
75                 new UsernamePasswordAuthenticationToken(
76                     userDetails, null, userDetails.getAuthorities());
77             SecurityContextHolder.getContext().setAuthentication(authenticationToken);
78         }
79     }
80
81     filterChain.doFilter(request, response);
82 }
83 }
```

```
1 // src/main/java/com/example/EmployeeManagementSystem/Repository/ApiKeyRepository.java
2 package com.example.EmployeeManagementSystem.Repository;
3
4 import com.example.EmployeeManagementSystem.Entity.ApiKey;
5 import org.springframework.data.jpa.repository.JpaRepository;
6 import org.springframework.data.jpa.repository.Modifying;
7 import org.springframework.data.jpa.repository.Query;
8 import org.springframework.data.repository.query.Param;
9 import org.springframework.transaction.annotation.Transactional;
10
11 import java.time.Instant;
12 import java.util.List;
13 import java.util.Optional;
14
15 public interface ApiKeyRepository extends JpaRepository<ApiKey, Long> {
16
17     Optional<ApiKey> findByApiKeyAndActiveTrue(String apiKey);
18
19
20     List<ApiKey> findByEmployee_EmployeeIdAndActiveTrue(Long employeeId);
21
22
23     List<ApiKey> findByVendor_IdAndActiveTrue(Long vendorId);
24
25     @Modifying
26     @Transactional
27     @Query("UPDATE ApiKey a SET a.lastUsedAt = :now WHERE a.apiKey = :apiKey")
28     void updateLastUsed(@Param("apiKey") String apiKey, @Param("now") Instant now);
29
30     @Modifying
31     @Transactional
32     @Query("UPDATE ApiKey a SET a.active = false WHERE a.expiresAt < :now AND a.active = true")
33     int expireOldKeys(@Param("now") Instant now);
34 }
```

```
1 package com.example.EmployeeManagementSystem.Repository;
2
3 import com.example.EmployeeManagementSystem.Entity.Delivery;
4 import com.example.EmployeeManagementSystem.Entity.Subscription;
5 import com.example.EmployeeManagementSystem.Enum.DeliveryStatus;
6 import org.springframework.data.jpa.repository.JpaRepository;
7 import org.springframework.data.jpa.repository.Query;
8 import org.springframework.data.repository.query.Param;
9
10 import java.time.Instant;
11 import java.util.List;
12
13 public interface DeliveryRepository extends JpaRepository<Delivery, Long> {
14     List<Delivery> findByStatusAndScheduledDeliveryTimeLessThanEqual(
15         DeliveryStatus status, Instant time);
16
17     @Query("SELECT d FROM Delivery d WHERE d.subscription = :subscription " +
18         "AND d.status != :status ORDER BY d.scheduledDeliveryTime DESC")
19     List<Delivery> findRecentDeliveriesBySubscription(
20         @Param("subscription") Subscription subscription,
21         @Param("status") DeliveryStatus status);
22 }
23
```

111 Current:

src\main\java\com\example\EmployeeManagementSystem\Repository\DeviceAssignmentRepo.java

```
1 package com.example.EmployeeManagementSystem.Repository;
2
3 import com.example.EmployeeManagementSystem.Entity.DeviceAssignment;
4 import com.example.EmployeeManagementSystem.Enum.AssignmentStatus;
5 import org.springframework.data.jpa.repository.JpaRepository;
6 import org.springframework.data.jpa.repository.Query;
7 import org.springframework.data.repository.query.Param;
8
9 import java.util.List;
10 import java.util.Optional;
11
12 public interface DeviceAssignmentRepo extends JpaRepository<DeviceAssignment, Long> {
13     Optional<DeviceAssignment> findByDeviceIdAndStatus(Long deviceId, AssignmentStatus status);
14     List<DeviceAssignment> findByAssignedToEmployeeIdAndStatus(Long employeeId, AssignmentStatus status);
15
16     @Query(value = "SELECT da.* FROM device_assignment da JOIN device d ON da.device_id = d.id WHERE d.vendor_id = :vendorId", nativeQuery = true)
17     List<DeviceAssignment> getAllAssignmentsByVendorId(@Param("vendorId") long vendorId);
18
19
20 }
21
```



```
1 package com.example.EmployeeManagementSystem.Repository;
2
3 import com.example.EmployeeManagementSystem.Entity.Device;
4 import com.example.EmployeeManagementSystem.Enum.DeviceStatus;
5 import org.springframework.data.jpa.repository.JpaRepository;
6
7 import java.util.List;
8
9 public interface DeviceRepository extends JpaRepository<Device, Long> {
10     List<Device> findByTechVendorId(Long vendorId);
11     List<Device> findByDeviceStatus(DeviceStatus deviceStatus);
12 }
13
```

```
1 package com.example.EmployeeManagementSystem.Repository;
2
3 import com.example.EmployeeManagementSystem.Entity.Employee;
4 import com.example.EmployeeManagementSystem.Enum.Role;
5 import com.example.EmployeeManagementSystem.Enum.Status;
6 import org.springframework.data.jpa.repository.JpaRepository;
7 import org.springframework.data.jpa.repository.Query;
8 import org.springframework.data.repository.query.Param;
9 import org.springframework.stereotype.Repository;
10
11 import java.util.List;
12 import java.util.Optional;
13
14 @Repository
15 public interface EmployeeRepo extends JpaRepository<Employee, Long> {
16     Optional<Employee> findByEmail(String email);
17     Optional<Employee> findByName(String name);
18     long countByRole(Role role);
19
20     List<Employee> findByManager(Employee manager);
21     List<Employee> findByStatus(Status status);
22     List<Employee> findByRole(Role role);
23     @Query("SELECT e FROM Employee e WHERE e.manager.employeeId = :managerId")
24     List<Employee> findByManagerEmployeeId(@Param("managerId") Long managerId);
25 }
26
```

114 Current:

src\main\java\com\example\EmployeeManagementSystem\Repository\FixedHolidayRepository.java

```
1 package com.example.EmployeeManagementSystem.Repository;
2
3 import com.example.EmployeeManagementSystem.Entity.FixedHoliday;
4 import org.springframework.data.jpa.repository.JpaRepository;
5
6 import java.time.LocalDate;
7 import java.util.List;
8
9 public interface FixedHolidayRepository extends JpaRepository<FixedHoliday, Long> {
10     List<FixedHoliday> findByYear(int year);
11     List<FixedHoliday> findByDateBetween(LocalDate start, LocalDate end);
12 }
13
```

115 Current:

src\main\java\com\example\EmployeeManagementSystem\Repository\LeaveEntitlementRepository.

```
1 package com.example.EmployeeManagementSystem.Repository;
2
3 import com.example.EmployeeManagementSystem.Entity.LeaveEntitlement;
4 import org.springframework.data.jpa.repository.JpaRepository;
5 import org.springframework.data.jpa.repository.Query;
6 import org.springframework.data.repository.query.Param;
7
8 import java.math.BigDecimal;
9 import java.util.List;
10 import java.util.Optional;
11
12 public interface LeaveEntitlementRepository extends JpaRepository<LeaveEntitlement, Long> {
13
14     Optional<LeaveEntitlement> findByEmployeeEmployeeIdAndLeaveTypeIdAndYear(
15         Long employeeId, Integer leaveTypeId, Integer year);
16
17     List<LeaveEntitlement> findByEmployeeEmployeeIdAndYear(Long employeeId, Integer year);
18
19     @Query("""
20         SELECT COALESCE(SUM(e.closingBalance), 0)
21         FROM LeaveEntitlement e
22         WHERE e.employee.employeeId = :empId
23             AND e.leaveType.carriesForward = true
24             AND e.year IN :years
25         """)
26     BigDecimal sumCarryForwardBalances(
27         @Param("empId") Long employeeId,
28         @Param("years") List<Integer> years);
29
30     List<LeaveEntitlement> findByLeaveTypeIdAndYear(Integer leaveTypeId, Integer year);
31 }
```

```

1 package com.example.EmployeeManagementSystem.Repository;
2
3 import com.example.EmployeeManagementSystem.Entity.Employee;
4 import com.example.EmployeeManagementSystem.Entity.LeaveRequest;
5 import com.example.EmployeeManagementSystem.Enum.LeaveStatus;
6 import com.example.EmployeeManagementSystem.Enum.LeaveType;
7 import org.springframework.data.jpa.repository.JpaRepository;
8 import org.springframework.data.jpa.repository.Query;
9 import org.springframework.data.repository.query.Param;
10
11 import java.time.LocalDate;
12 import java.util.List;
13
14 public interface LeaveRequestRepo extends JpaRepository<LeaveRequest, Long> {
15
16     List<LeaveRequest> findByStatus(LeaveStatus status);
17
18     /**
19      * Find leave requests by start date and status
20      * Used by scheduler to find leaves starting today
21      */
22     List<LeaveRequest> findByStartDateAndStatus(LocalDate startDate, LeaveStatus status);
23
24     // Fix the checkDuplicate query (fix parameter names)
25     @Query(value = "SELECT COUNT(*) " +
26         "FROM leave_request l " +
27         "WHERE l.employee_id = :employeeId " +
28         "AND l.start_date = :startDate " +
29         "AND l.end_date = :endDate " +
30         "AND l.status=:status",nativeQuery = true)
31     long checkDuplicate(@Param("employeeId") long employeeId,
32         @Param("startDate") LocalDate startDate,
33         @Param("endDate") LocalDate endDate,
34         @Param("status") String status);
35
36     // Fix overlapping leave query (was commented in your code)
37     @Query(value = "select COUNT(*) " +
38         "from leave_request l " +
39         "where l.employee_id=:employeeId " +
40         "AND l.status IN ('APPROVED','PENDING') " +
41         "AND l.start_date <= :endDate " +
42         "AND l.end_date >= :startDate",nativeQuery = true)
43     long countOverlappingLeave(
44         @Param("employeeId") long employeeId,
45         @Param("startDate") LocalDate startDate,
46         @Param("endDate") LocalDate endDate
47     );
48
49     // Add missing methods
50     List<LeaveRequest> findByEmployeeOrderByStartDateDesc(Employee employee);
51
52
53     List<LeaveRequest> findByEndDateAndStatus(LocalDate endDate, LeaveStatus status);
54
55     // Add for leave duration validation
56     @Query("SELECT COALESCE(SUM(DATEDIFF(l.endDate, l.startDate) + 1), 0) " +
57         "FROM LeaveRequest l WHERE l.employee = :employee " +
58         "AND l.leaveType = :leaveType " +
59         "AND YEAR(l.startDate) = :year " +
60         "AND l.status IN ('APPROVED', 'PENDING'))")
61     long countDaysByEmployeeAndLeaveTypeAndYear(@Param("employee") Employee employee,
62         @Param("leaveType") LeaveType leaveType,
63         @Param("year") int year);
64
65     @Query("SELECT lr FROM LeaveRequest lr JOIN FETCH lr.employee WHERE lr.status = :status")
66     List<LeaveRequest> findApprovedLeavesWithEmployee(LeaveStatus status);
67
68
69     List<LeaveRequest> findByEmployee_Manager(Employee manager);
70
71     List<LeaveRequest> findByStatusAndEmployee_Manager(LeaveStatus status, Employee manager);

```

```

72
73     @Query("""
74     SELECT COALESCE(SUM(DATEDIFF(lr.endDate, lr.startDate) + 1), 0)
75     FROM LeaveRequest lr
76     WHERE lr.employee = :employee
77           AND lr.leaveType = com.example.EmployeeManagementSystem.Enum.LeaveType.SICK
78           AND lr.status = com.example.EmployeeManagementSystem.Enum.LeaveStatus.APPROVED
79           AND YEAR(lr.startDate) = :year
80           AND MONTH(lr.startDate) = :month
81     """)
82     long countApprovedSickDaysInMonth(
83         @Param("employee") Employee employee,
84         @Param("year") int year,
85         @Param("month") int month);
86
87     @Query("""
88     SELECT COALESCE(SUM(DATEDIFF(lr.endDate, lr.startDate) + 1), 0)
89     FROM LeaveRequest lr
90     WHERE lr.employee = :employee
91           AND lr.leaveType = com.example.EmployeeManagementSystem.Enum.LeaveType.CASUAL
92           AND lr.status IN (
93               com.example.EmployeeManagementSystem.Enum.LeaveStatus.APPROVED,
94               com.example.EmployeeManagementSystem.Enum.LeaveStatus.PENDING
95           )
96           AND YEAR(lr.startDate) = :year
97           AND MONTH(lr.startDate) = :month
98     """)
99     long countCasualDaysInMonth(
100         @Param("employee") Employee employee,
101         @Param("year") int year,
102         @Param("month") int month
103     );
104
105     @Query("SELECT COUNT(l) > 0 FROM LeaveRequest l " +
106         "WHERE l.employee = :employee " +
107         "AND l.leaveType = :leaveType " +
108         "AND YEAR(l.startDate) = :year " +
109         "AND l.status NOT IN (:statuses)")
110     boolean existsMaternityLeaveForYear(
111         @Param("employee") Employee employee,
112         @Param("leaveType") LeaveType leaveType,
113         @Param("year") int year,
114         @Param("statuses") List<LeaveStatus> statuses
115     );
116     @Query("""
117     SELECT COALESCE(SUM(DATEDIFF(lr.endDate, lr.startDate) + 1), 0)
118     FROM LeaveRequest lr
119     WHERE lr.employee = :employee
120           AND lr.leaveType = com.example.EmployeeManagementSystem.Enum.LeaveType.UNPAID
121           AND lr.status IN (
122               com.example.EmployeeManagementSystem.Enum.LeaveStatus.APPROVED,
123               com.example.EmployeeManagementSystem.Enum.LeaveStatus.PENDING
124           )
125           AND YEAR(lr.startDate) = :year
126           AND MONTH(lr.startDate) = :month
127     """)
128     long countUnpaidDaysInMonth(
129         @Param("employee") Employee employee,
130         @Param("year") int year,
131         @Param("month") int month
132     );
133 }
134

```

117 Current:

src\main\java\com\example\EmployeeManagementSystem\Repository\LeaveTypeRepository.java

```
1 package com.example.EmployeeManagementSystem.Repository;
2
3 import com.example.EmployeeManagementSystem.Entity.LeaveEntitlement;
4 import com.example.EmployeeManagementSystem.Entity.LeaveType;
5 import org.springframework.data.jpa.repository.JpaRepository;
6
7 import java.util.List;
8 import java.util.Optional;
9
10 public interface LeaveTypeRepository extends JpaRepository<LeaveType, Integer> {
11     Optional<LeaveType> findByName(String name);
12     List<LeaveType> findByMonthlyResetTrue();
13 }
```

118 Current:

src\main\java\com\example\EmployeeManagementSystem\Repository\LeaveWarningRepository.java

```
1 package com.example.EmployeeManagementSystem.Repository;
2
3 import com.example.EmployeeManagementSystem.Entity.LeaveWarning;
4 import org.springframework.data.jpa.repository.JpaRepository;
5
6 import java.time.LocalDate;
7 import java.util.List;
8
9 public interface LeaveWarningRepository extends JpaRepository<LeaveWarning, Long> {
10
11     // All unread warnings for an employee – shown on dashboard
12     List<LeaveWarning> findByEmployeeEmployeeIdAndIsReadFalseOrderByWarningDateDesc(Long employeeId);
13
14     // All warnings for an employee (read + unread) – for history view
15     List<LeaveWarning> findByEmployeeEmployeeIdOrderByWarningDateDesc(Long employeeId);
16
17     // Prevent duplicate warnings for same type on same date
18     boolean existsByEmployeeEmployeeIdAndWarningTypeAndWarningDate(
19         Long employeeId, LeaveWarning.WarningType warningType, LocalDate date
20     );
21 }
```


119 Current:

src\main\java\com\example\EmployeeManagementSystem\Repository\NotificationRepository.java

```
1 package com.example.EmployeeManagementSystem.Repository;
2
3 import com.example.EmployeeManagementSystem.Entity.Notification;
4 import com.example.EmployeeManagementSystem.Entity.Employee;
5 import org.springframework.data.jpa.repository.JpaRepository;
6 import java.util.List;
7
8 public interface NotificationRepository extends JpaRepository<Notification, Long> {
9     List<Notification> findByRecipientOrderByCreatedAtDesc(Employee recipient);
10    long countByRecipientAndIsReadFalse(Employee recipient);
11 }
12
```

120 Current:

src\main\java\com\example\EmployeeManagementSystem\Repository\RefreshTokenRepository.java

```
1 package com.example.EmployeeManagementSystem.Repository;
2
3 import com.example.EmployeeManagementSystem.Entity.Employee;
4 import com.example.EmployeeManagementSystem.Entity.RefreshToken;
5 import com.example.EmployeeManagementSystem.Entity.Vendor;
6 import org.springframework.data.jpa.repository.JpaRepository;
7
8 import java.util.Optional;
9
10 public interface RefreshTokenRepository extends JpaRepository<RefreshToken, Long> {
11
12     Optional<RefreshToken> findByToken(String token);
13
14     // Needed so EmployeeService can clean up tokens before deleting an employee
15     void deleteByEmployee(Employee employee);
16
17     // Needed so VendorService can clean up tokens before deleting a vendor
18     void deleteByVendor(Vendor vendor);
19 }
```

121 Current:

src\main\java\com\example\EmployeeManagementSystem\Repository\RestaurantRepository.java

```
1 package com.example.EmployeeManagementSystem.Repository;
2
3 import com.example.EmployeeManagementSystem.Entity.Restaurant;
4 import com.example.EmployeeManagementSystem.Enum.MealSlot;
5 import org.springframework.data.jpa.repository.JpaRepository;
6 import org.springframework.data.jpa.repository.Query;
7 import org.springframework.data.repository.query.Param;
8 import org.springframework.stereotype.Repository;
9
10 import java.util.List;
11
12 @Repository
13 public interface RestaurantRepository extends JpaRepository<Restaurant, Long> {
14
15     /** All active restaurants owned by a vendor. */
16     List<Restaurant> findByVendor_EmailAndActiveTrue(String email);
17
18     /** All active restaurants (for employees browsing available options). */
19     List<Restaurant> findByActiveTrue();
20
21     /**
22      * Active restaurants that support a given meal slot.
23      * Employees use this to discover which restaurants they can subscribe to for a given slot.
24      */
25     @Query("SELECT r FROM Restaurant r JOIN r.supportedMealSlots s " +
26           "WHERE r.active = true AND s = :mealSlot")
27     List<Restaurant> findActiveByMealSlot(@Param("mealSlot") MealSlot mealSlot);
28
29     /** Check if a restaurant belongs to a vendor (ownership guard). */
30     boolean existsByIdAndVendor_Email(Long restaurantId, String email);
31 }
```

122 Current:

src\main\java\com\example\EmployeeManagementSystem\Repository\ServiceRequestRepository.java

```
1 package com.example.EmployeeManagementSystem.Repository;
2
3
4 import com.example.EmployeeManagementSystem.Entity.ServiceRequest;
5 import com.example.EmployeeManagementSystem.Entity.Employee;
6 import com.example.EmployeeManagementSystem.Entity.Vendor;
7 import com.example.EmployeeManagementSystem.Enum.ServiceRequestStatus;
8 import jdk.dynalink.linker.LinkerServices;
9 import org.springframework.data.jpa.repository.JpaRepository;
10
11 import java.util.List;
12
13 public interface ServiceRequestRepository extends JpaRepository<ServiceRequest, Long> {
14
15     List<ServiceRequest> findByStatus(ServiceRequestStatus status);
16     List<ServiceRequest> findByRaisedByOrderByRaisedAtDesc(Employee employee);
17     List<ServiceRequest> findByDeviceTechVendorAndStatus(
18         Vendor vendor,
19         ServiceRequestStatus status
20     );
21
22 }
23
```

123 Current:

src\main\java\com\example\EmployeeManagementSystem\Repository\SubscriptionRepository.java

```
1 package com.example.EmployeeManagementSystem.Repository;
2
3 import com.example.EmployeeManagementSystem.Entity.Subscription;
4 import org.springframework.data.jpa.repository.JpaRepository;
5 import org.springframework.data.jpa.repository.Query;
6 import org.springframework.data.repository.query.Param;
7 import org.springframework.stereotype.Repository;
8
9 import java.time.Instant;
10 import java.util.List;
11
12 @Repository
13 public interface SubscriptionRepository extends JpaRepository<Subscription, Long> {
14     @Query(value = "select count(*) from subscription s where s.employee_id=:employeeId and s.slot=:slot and s.status=:status", nativeQuery = true)
15     int checkSubscriptionExists(
16         @Param("employeeId") long employeeId,
17         @Param("slot") String slot,
18         @Param("status") String status
19     );
20
21     List<Subscription> findByEmployee_employeeId(long id);
22
23     @Query(value = "SELECT * FROM subscription s WHERE s.status = :status AND s.next_delivery_time IS NOT NULL AND s.next_delivery_time <= :currentTime", nativeQuery = true)
24     List<Subscription> findDueSubscriptions(
25         @Param("status") String status,
26         @Param("currentTime") Instant currentTime // was LocalDateTime
27     );
28
29     List<Subscription> findByEmployee_name(String name);
30     List<Subscription> findByEmployee_Email(String email);
31 }
32
```

```
1 package com.example.EmployeeManagementSystem.Repository;
2
3 import com.example.EmployeeManagementSystem.Entity.Vendor;
4 import org.springframework.data.jpa.repository.JpaRepository;
5 import org.springframework.stereotype.Repository;
6
7 import java.util.Optional;
8
9 @Repository
10 public interface VendorRepo extends JpaRepository<Vendor, Long> {
11     Optional<Vendor> findByEmail(String email);
12     boolean existsByEmail(String email);
13 }
```

125 Current:

src\main\java\com\example\EmployeeManagementSystem\Repository\YearEndCarryForwardLogRe

```
1 package com.example.EmployeeManagementSystem.Repository;
2
3
4 import com.example.EmployeeManagementSystem.Entity.YearEndCarryForwardLog;
5 import org.springframework.data.jpa.repository.JpaRepository;
6
7 import java.util.List;
8
9 public interface YearEndCarryForwardLogRepository
10     extends JpaRepository<YearEndCarryForwardLog, Long> {
11
12     List<YearEndCarryForwardLog> findByEmployeeEmployeeIdAndProcessedYear(
13         Long employeeId, Integer year
14     );
15
16     // Check if year-end has already been run for this year (idempotency guard)
17     boolean existsByProcessedYear(Integer year);
18 }
```

```
1 package com.example.EmployeeManagementSystem.Scheduler;
2
3 import com.example.EmployeeManagementSystem.Service.DeliveryService;
4 import org.slf4j.Logger;
5 import org.slf4j.LoggerFactory;
6 import org.springframework.scheduling.annotation.Scheduled;
7 import org.springframework.stereotype.Component;
8
9 @Component
10 public class DeliveryScheduler {
11     private static final Logger logger = LoggerFactory.getLogger(DeliveryScheduler.class);
12     private final DeliveryService deliveryService;
13
14     public DeliveryScheduler(DeliveryService deliveryService) {
15         this.deliveryService = deliveryService;
16     }
17
18     // Runs every hour
19     @Scheduled(cron = "0 0 * * * *")
20     public void processDeliveries() {
21         logger.info("Starting delivery processing scheduler...");
22         try {
23             deliveryService.processDueDeliveries();
24             logger.info("Delivery processing completed successfully");
25         } catch (Exception e) {
26             logger.error("Error processing deliveries: ", e);
27         }
28     }
29 }
30
```



```

1  package com.example.EmployeeManagementSystem.Scheduler;
2
3  import com.example.EmployeeManagementSystem.Service.CarryForwardWarningService;
4  import com.example.EmployeeManagementSystem.Service.LeaveAccrualService;
5  import com.example.EmployeeManagementSystem.Service.SickLeaveResetService;
6  import com.example.EmployeeManagementSystem.Service.YearEndCarryForwardService;
7  import org.slf4j.Logger;
8  import org.slf4j.LoggerFactory;
9  import org.springframework.scheduling.annotation.Scheduled;
10 import org.springframework.stereotype.Component;
11
12 import java.time.LocalDate;
13 import java.time.YearMonth;
14
15 @Component
16 public class LeaveAccrualScheduler {
17
18     private static final Logger log = LoggerFactory.getLogger(LeaveAccrualScheduler.class);
19
20     private final LeaveAccrualService accrualService;
21     private final SickLeaveResetService resetService;
22     private final CarryForwardWarningService warningService;
23     private final YearEndCarryForwardService yearEndService;
24
25     public LeaveAccrualScheduler(LeaveAccrualService accrualService,
26                                 SickLeaveResetService resetService,
27                                 CarryForwardWarningService warningService,
28                                 YearEndCarryForwardService yearEndService) {
29         this.accrualService = accrualService;
30         this.resetService = resetService;
31         this.warningService = warningService;
32         this.yearEndService = yearEndService;
33     }
34
35     // 1st of month at 00:05 – accrue 1 day for all eligible employees, then warn
36     @Scheduled(cron = "0 5 0 1 * ?")
37     public void runMonthlyAccrual() {
38         LocalDate today = LocalDate.now();
39         accrualService.runMonthlyAccrual(today);
40         warningService.runWarningCheck(today);
41     }
42
43     // Last day of month at 23:55 – reset unused sick leave
44     // FIX: use YearMonth to compute the actual last day of the current month,
45     //      so if the scheduler fires even a minute late (past midnight) on the
46     //      1st, we still reset the correct month and log the correct date.
47     @Scheduled(cron = "0 55 23 L * ?")
48     public void runSickLeaveReset() {
49         LocalDate lastDayOfMonth = YearMonth.now().atEndOfMonth();
50         log.info("Running sick leave reset for end-of-month: {}", lastDayOfMonth);
51         resetService.runMonthlyReset(lastDayOfMonth);
52     }
53
54     // Dec 31 at 22:00 – year-end carry-forward cap
55     @Scheduled(cron = "0 0 22 31 12 ?")
56     public void runYearEnd() {
57         int closingYear = LocalDate.now().getYear();
58         yearEndService.runYearEnd(closingYear);
59     }
60 }

```

```

1 package com.example.EmployeeManagementSystem.Scheduler;
2
3 import com.example.EmployeeManagementSystem.Entity.Employee;
4 import com.example.EmployeeManagementSystem.Entity.LeaveRequest;
5 import com.example.EmployeeManagementSystem.Enum.LeaveStatus;
6 import com.example.EmployeeManagementSystem.Enum.Status;
7 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
8 import com.example.EmployeeManagementSystem.Repository.LeaveRequestRepo;
9 import org.slf4j.Logger;
10 import org.slf4j.LoggerFactory;
11 import org.springframework.scheduling.annotation.Scheduled;
12 import org.springframework.stereotype.Component;
13 import org.springframework.transaction.annotation.Transactional;
14
15 import java.time.LocalDate;
16 import java.time.ZoneId;
17 import java.util.List;
18 import java.util.Map;
19 import java.util.stream.Collectors;
20
21 @Component
22 public class LeaveStatusScheduler {
23
24     private static final Logger log = LoggerFactory.getLogger(LeaveStatusScheduler.class);
25
26     private final LeaveRequestRepo leaveRequestRepo;
27     private final EmployeeRepo employeeRepo;
28
29     public LeaveStatusScheduler(LeaveRequestRepo leaveRequestRepo,
30                               EmployeeRepo employeeRepo) {
31         this.leaveRequestRepo = leaveRequestRepo;
32         this.employeeRepo = employeeRepo;
33     }
34
35     /**
36      * Runs every day at midnight UTC
37      */
38     @Scheduled(cron = "0 0 0 * * *", zone = "UTC")
39     @Transactional
40     public void updateEmployeeLeaveStatus() {
41
42         log.info("=== Starting Timezone-Aware Leave Scheduler ===");
43
44         List<LeaveRequest> leaves =
45             leaveRequestRepo.findApprovedLeavesWithEmployee(LeaveStatus.APPROVED);
46
47         Map<Long, List<LeaveRequest>> employeeLeavesMap = leaves.stream()
48             .collect(Collectors.groupingBy(lr -> lr.getEmployee().getEmployeeId()));
49
50         int onLeaveCount = 0;
51         int activeCount = 0;
52
53         for (Map.Entry<Long, List<LeaveRequest>> entry : employeeLeavesMap.entrySet()) {
54
55             Employee employee = entry.getValue().get(0).getEmployee();
56
57             ZoneId zone = ZoneId.of(
58                 employee.getTimezone() != null ? employee.getTimezone() : "UTC"
59             );
60
61             LocalDate today = LocalDate.now(zone);
62
63             boolean isOnLeaveToday = entry.getValue().stream().anyMatch(leave ->
64                 !today.isBefore(leave.getStartDate()) &&
65                 !today.isAfter(leave.getEndDate())
66             );
67
68             // CASE 1: Should be ON_LEAVE
69             if (isOnLeaveToday && employee.getStatus() != Status.ON_LEAVE) {
70
71                 employee.setStatus(Status.ON_LEAVE);

```

```

72         employeeRepo.save(employee);
73         onLeaveCount++;
74
75         log.info("Employee {} ({} marked ON_LEAVE | TZ: {}",
76                 employee.getEmployeeId(),
77                 employee.getName(),
78                 zone);
79     }
80
81     // CASE 2: Should be ACTIVE
82     if (!isOnLeaveToday && employee.getStatus() == Status.ON_LEAVE) {
83
84         employee.setStatus(Status.ACTIVE);
85         employeeRepo.save(employee);
86         activeCount++;
87
88         log.info("Employee {} ({} marked ACTIVE | TZ: {}",
89                 employee.getEmployeeId(),
90                 employee.getName(),
91                 zone);
92     }
93 }
94
95     log.info("=== Scheduler Completed ===");
96     log.info("ON_LEAVE updated: {}", onLeaveCount);
97     log.info("ACTIVE updated: {}", activeCount);
98 }
99
100 /**
101  * Manual trigger (for testing)
102  */
103 public void runManually() {
104     log.info("Manual scheduler trigger started...");
105     updateEmployeeLeaveStatus();
106 }
107 }

```

```

1 // src/main/java/com/example/EmployeeManagementSystem/security/ApiKeyAuthenticationProvider.java
2 package com.example.EmployeeManagementSystem.security;
3
4 import com.example.EmployeeManagementSystem.Entity.ApiKey;
5 import com.example.EmployeeManagementSystem.Entity.Employee;
6 import com.example.EmployeeManagementSystem.Entity.Vendor;
7 import com.example.EmployeeManagementSystem.Repository.ApiKeyRepository;
8 import org.springframework.security.authentication.AuthenticationProvider;
9 import org.springframework.security.core.Authentication;
10 import org.springframework.security.core.AuthenticationException;
11 import org.springframework.security.core.GrantedAuthority;
12 import org.springframework.security.core.authority.SimpleGrantedAuthority;
13 import org.springframework.stereotype.Component;
14 import org.springframework.transaction.annotation.Transactional;
15
16 import java.time.Instant;
17 import java.util.ArrayList;
18 import java.util.Arrays;
19 import java.util.Collection;
20 import java.util.List;
21 import java.util.stream.Collectors;
22
23 @Component
24 public class ApiKeyAuthenticationProvider implements AuthenticationProvider {
25
26     private final ApiKeyRepository apiKeyRepository;
27
28     public ApiKeyAuthenticationProvider(ApiKeyRepository apiKeyRepository) {
29         this.apiKeyRepository = apiKeyRepository;
30     }
31
32     @Override
33     @Transactional
34     public Authentication authenticate(Authentication authentication) throws AuthenticationException {
35         String apiKey = null;
36
37         // Extract API key from the authentication object
38         if (authentication instanceof ApiKeyAuthenticationToken) {
39             apiKey = ((ApiKeyAuthenticationToken) authentication).getApiKey();
40         }
41
42         if (apiKey == null || apiKey.isEmpty()) {
43             throw new RuntimeException("Invalid API key");
44         }
45
46         ApiKey key = apiKeyRepository.findByApiKeyAndActiveTrue(apiKey)
47             .orElseThrow(() -> new RuntimeException("Invalid or inactive API key"));
48
49         // Check expiration
50         if (key.getExpiresAt() != null && key.getExpiresAt().isBefore(Instant.now())) {
51             throw new RuntimeException("API key has expired");
52         }
53
54         // Update last used timestamp
55         apiKeyRepository.updateLastUsed(apiKey, Instant.now());
56
57         // Build authorities from permissions
58         List<GrantedAuthority> authorities = new ArrayList<>();
59
60         if (key.getPermissions() != null && !key.getPermissions().isEmpty()) {
61             authorities.addAll(Arrays.stream(key.getPermissions().split(","))
62                 .map(String::trim)
63                 .map(SimpleGrantedAuthority::new)
64                 .collect(Collectors.toList()));
65         }
66
67         Object principal;
68         if (key.getEmployee() != null) {
69             principal = key.getEmployee();
70             authorities.add(new SimpleGrantedAuthority("ROLE_" + key.getEmployee().getRole().name()));
71         } else if (key.getVendor() != null) {

```

```
72         principal = key.getVendor();
73         authorities.add(new SimpleGrantedAuthority("ROLE_VENDOR"));
74     } else {
75         throw new RuntimeException("No user associated with API key");
76     }
77
78     return new ApiKeyAuthenticationToken(principal, apiKey, authorities);
79 }
80
81 @Override
82 public boolean supports(Class<?> authentication) {
83     return ApiKeyAuthenticationToken.class.isAssignableFrom(authentication);
84 }
85 }
```

```
1 // src/main/java/com/example/EmployeeManagementSystem/security/ApiKeyAuthenticationToken.java
2 package com.example.EmployeeManagementSystem.security;
3
4 import org.springframework.security.core.Authentication;
5 import org.springframework.security.core.GrantedAuthority;
6 import java.util.Collection;
7
8 public class ApiKeyAuthenticationToken implements Authentication {
9
10     private static final long serialVersionUID = 1L;
11
12     private final Object principal;
13     private final Object credentials;
14     private final String apiKey;
15     private final Collection<? extends GrantedAuthority> authorities;
16     private boolean authenticated;
17
18     // Constructor for unauthenticated API key (before validation)
19     public ApiKeyAuthenticationToken(String apiKey) {
20         this.apiKey = apiKey;
21         this.principal = null;
22         this.credentials = null;
23         this.authorities = null;
24         this.authenticated = false;
25     }
26
27     // Constructor for authenticated API key (after validation)
28     public ApiKeyAuthenticationToken(Object principal, Object credentials,
29                                     Collection<? extends GrantedAuthority> authorities) {
30         this.apiKey = null;
31         this.principal = principal;
32         this.credentials = credentials;
33         this.authorities = authorities;
34         this.authenticated = true;
35     }
36
37     @Override
38     public Collection<? extends GrantedAuthority> getAuthorities() {
39         return authorities;
40     }
41
42     @Override
43     public Object getCredentials() {
44         return credentials;
45     }
46
47     @Override
48     public Object getDetails() {
49         return null;
50     }
51
52     @Override
53     public Object getPrincipal() {
54         return principal;
55     }
56
57     @Override
58     public boolean isAuthenticated() {
59         return authenticated;
60     }
61
62     @Override
63     public void setAuthenticated(boolean isAuthenticated) throws IllegalArgumentException {
64         this.authenticated = isAuthenticated;
65     }
66
67     @Override
68     public String getName() {
69         if (principal instanceof org.springframework.security.core.userdetails.UserDetails) {
70             return ((org.springframework.security.core.userdetails.UserDetails) principal).getUsername();
71         }
72     }
```

```
72         if (principal instanceof com.example.EmployeeManagementSystem.Entity.Employee) {
73             return ((com.example.EmployeeManagementSystem.Entity.Employee) principal).getUsername();
74         }
75         if (principal instanceof com.example.EmployeeManagementSystem.Entity.Vendor) {
76             return ((com.example.EmployeeManagementSystem.Entity.Vendor) principal).getUsername();
77         }
78         return principal != null ? principal.toString() : apiKey;
79     }
80
81     public String getApiKey() {
82         return apiKey;
83     }
84 }
```

```
1 package com.example.EmployeeManagementSystem.Service;
2
3 import com.example.EmployeeManagementSystem.Entity.ApiKey;
4 import com.example.EmployeeManagementSystem.Entity.Vendor;
5 import com.example.EmployeeManagementSystem.Repository.ApiKeyRepository;
6 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
7 import com.example.EmployeeManagementSystem.Repository.VendorRepo;
8 import org.springframework.beans.factory.annotation.Autowired;
9 import org.springframework.stereotype.Service;
10 import org.springframework.transaction.annotation.Transactional;
11
12 import java.time.Instant;
13 import java.time.temporal.ChronoUnit;
14 import java.util.List;
15
16 @Service
17 public class ApiKeyService {
18
19     @Autowired
20     private ApiKeyRepository apiKeyRepository;
21     @Autowired private EmployeeRepo employeeRepo;
22     @Autowired private VendorRepo vendorRepo;
23
24     @Transactional
25     public ApiKey generateForVendor(Long vendorId, String name,
26                                     String permissions, int validDays) {
27         Vendor vendor = vendorRepo.findById(vendorId)
28             .orElseThrow(() -> new RuntimeException("Vendor not found"));
29
30         ApiKey key = new ApiKey(); // constructor auto-generates "ems_<uuid>"
31         key.setVendor(vendor);
32         key.setKeyName(name);
33         key.setPermissions(permissions); // e.g. "VENDOR_WRITE" or "TECH_VENDOR"
34         key.setExpiresAt(Instant.now().plus(validDays, ChronoUnit.DAYS));
35         key.setActive(true);
36         return apiKeyRepository.save(key);
37     }
38
39     // Same for employee-issued keys if needed
40     public List<ApiKey> getActiveKeysForVendor(Long vendorId) {
41         return apiKeyRepository.findByVendor_IdAndActiveTrue(vendorId);
42     }
43
44     @Transactional
45     public void revokeKey(Long keyId) {
46         ApiKey key = apiKeyRepository.findById(keyId)
47             .orElseThrow(() -> new RuntimeException("Key not found"));
48         key.setActive(false);
49         apiKeyRepository.save(key);
50     }
51 }
52
```



```

1 package com.example.EmployeeManagementSystem.Service;
2
3 import com.example.EmployeeManagementSystem.Entity.Employee;
4 import com.example.EmployeeManagementSystem.Entity.LeaveEntitlement;
5 import com.example.EmployeeManagementSystem.Entity.LeaveWarning;
6 import com.example.EmployeeManagementSystem.Enum.Status;
7 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
8 import com.example.EmployeeManagementSystem.Repository.LeaveEntitlementRepository;
9 import com.example.EmployeeManagementSystem.Repository.LeaveWarningRepository;
10 import org.slf4j.Logger;
11 import org.slf4j.LoggerFactory;
12 import org.springframework.stereotype.Service;
13 import org.springframework.transaction.annotation.Transactional;
14
15 import java.math.BigDecimal;
16 import java.time.LocalDate;
17 import java.util.List;
18
19 @Service
20 public class CarryForwardWarningService {
21
22     private static final Logger log = LoggerFactory.getLogger(CarryForwardWarningService.class);
23
24     private static final BigDecimal CAP = BigDecimal.valueOf(30);
25     private static final BigDecimal WARN_AT_29 = BigDecimal.valueOf(29);
26     private static final BigDecimal WARN_AT_28 = BigDecimal.valueOf(28);
27
28     private final EmployeeRepo employeeRepository;
29     private final LeaveEntitlementRepository entitlementRepository;
30     private final LeaveWarningRepository warningRepository;
31
32     public CarryForwardWarningService(EmployeeRepo employeeRepository,
33                                     LeaveEntitlementRepository entitlementRepository,
34                                     LeaveWarningRepository warningRepository) {
35         this.employeeRepository = employeeRepository;
36         this.entitlementRepository = entitlementRepository;
37         this.warningRepository = warningRepository;
38     }
39
40     @Transactional
41     public void runWarningCheck(LocalDate checkDate) {
42         List<Employee> activeEmployees = employeeRepository.findByStatus(Status.ACTIVE);
43
44         for (Employee employee : activeEmployees) {
45             BigDecimal carryForwardTotal = calculateCarryForwardBalance(employee, checkDate);
46             evaluateAndWarn(employee, carryForwardTotal, checkDate);
47         }
48
49         log.info("Warning check complete for {} – {} employees evaluated",
50               checkDate, activeEmployees.size());
51     }
52
53     // -----
54     // Sum available balance across all carry-forwardable types
55     // for the current year (this is what will carry forward if unused)
56     // -----
57     private BigDecimal calculateCarryForwardBalance(Employee employee, LocalDate date) {
58         int year = date.getYear();
59
60         List<LeaveEntitlement> entitlements =
61             entitlementRepository.findByEmployeeEmployeeIdAndYear(employee.getEmployeeId(), year);
62
63         return entitlements.stream()
64             .filter(e -> e.getLeaveType().isCarriesForward())
65             .map(LeaveEntitlement::getAvailableBalance)
66             .reduce(BigDecimal.ZERO, BigDecimal::add);
67     }
68
69     // -----
70     // Decide which warning to issue based on carry-forward total
71     // -----

```

```

72     private void evaluateAndWarn(Employee employee, BigDecimal balance, LocalDate date) {
73         int cmp30 = balance.compareTo(CAP);
74         int cmp29 = balance.compareTo(WARN_AT_29);
75         int cmp28 = balance.compareTo(WARN_AT_28);
76
77         if (cmp30 > 0) {
78             // Balance EXCEEDS cap – a day was wasted (cap enforcement happens in Phase 5,
79             // but we flag it here so the warning appears immediately after accrual)
80             issueWarning(employee, LeaveWarning.WarningType.DAY_WASTED, balance, date,
81                 "A carry-forward day was wasted this month because your balance exceeded the 30-day
82 cap.");
83         } else if (cmp30 == 0) {
84             issueWarning(employee, LeaveWarning.WarningType.CAP_REACHED, balance, date,
85                 "Carry-forward cap reached (30 days). Next month's unused leave will be wasted.");
86         } else if (cmp29 >= 0) {
87             issueWarning(employee, LeaveWarning.WarningType.APPROACHING_29, balance, date,
88                 "You have " + balance.toPlainString() + " carry-forward days. "
89                 + "1 month until cap. Use leave before it stops accumulating.");
90         } else if (cmp28 >= 0) {
91             issueWarning(employee, LeaveWarning.WarningType.APPROACHING_28, balance, date,
92                 "You have " + balance.toPlainString() + " carry-forward days. "
93                 + "2 months until cap (30). Plan to use leave soon.");
94         }
95         // Below 28 – no warning needed
96     }
97 }
98
99 // -----
100 // Persist warning – skip if same type already issued today
101 // -----
102 private void issueWarning(Employee employee, LeaveWarning.WarningType type,
103     BigDecimal balance, LocalDate date, String message) {
104     boolean alreadyIssued = warningRepository
105         .existsByEmployeeEmployeeIdAndWarningTypeAndWarningDate(employee.getEmployeeId(), type,
106 date);
107     if (alreadyIssued) {
108         log.debug("Warning {} already issued for employee {} on {}", type, employee.getEmployeeId(),
109 date);
110         return;
111     }
112
113     LeaveWarning warning = new LeaveWarning();
114     warning.setEmployee(employee);
115     warning.setWarningType(type);
116     warning.setMessage(message);
117     warning.setCarryForwardBalance(balance);
118     warning.setWarningDate(date);
119
120     warningRepository.save(warning);
121
122     log.info("Warning issued – employee={} type={} balance={} date={}",
123         employee.getEmployeeId(), type, balance, date);
124 }
125 }
126 }

```

133 Current:

src\main\java\com\example\EmployeeManagementSystem\Service\CombinedUserDetailService.java

```
1 package com.example.EmployeeManagementSystem.Service;
2
3 import com.example.EmployeeManagementSystem.Entity.Employee;
4 import com.example.EmployeeManagementSystem.Entity.Vendor;
5 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
6 import com.example.EmployeeManagementSystem.Repository.VendorRepo;
7 import org.springframework.security.core.userdetails.UserDetails;
8 import org.springframework.security.core.userdetails.UserDetailsService;
9 import org.springframework.security.core.userdetails.UsernameNotFoundException;
10 import org.springframework.stereotype.Service;
11
12 import java.util.Optional;
13
14 @Service("combinedUserDetailsService")
15 public class CombinedUserDetailsService implements UserDetailsService {
16
17     private final VendorRepo vendorRepository;
18     private final EmployeeRepo employeeRepository;
19
20     public CombinedUserDetailsService(VendorRepo vendorRepository, EmployeeRepo employeeRepository) {
21         this.vendorRepository = vendorRepository;
22         this.employeeRepository = employeeRepository;
23     }
24
25     @Override
26     public UserDetails loadUserByUsername(String username) throws UsernameNotFoundException {
27         Optional<Employee> employee = employeeRepository.findByEmail(username);
28         if (employee.isPresent()) {
29             return employee.get(); // must implement UserDetails
30         }
31
32         // Then try Vendor
33         Optional<Vendor> vendor = vendorRepository.findByEmail(username);
34         if (vendor.isPresent()) {
35             return vendor.get();
36         }
37
38         throw new UsernameNotFoundException("User not found: " + username);
39     }
40 }
41
```

```

1 package com.example.EmployeeManagementSystem.Service;
2
3 import com.example.EmployeeManagementSystem.Entity.Employee;
4 import com.example.EmployeeManagementSystem.Enum.Role;
5 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
6 import com.example.EmployeeManagementSystem.Repository VendorRepo;
7 import org.springframework.security.core.GrantedAuthority;
8 import org.springframework.security.core.authority.SimpleGrantedAuthority;
9 import org.springframework.security.crypto.password.PasswordEncoder;
10 import org.springframework.security.oauth2.client.userinfo.DefaultOAuth2UserService;
11 import org.springframework.security.oauth2.client.userinfo.OAuth2UserRequest;
12 import org.springframework.security.oauth2.core.user.DefaultOAuth2User;
13 import org.springframework.security.oauth2.core.user.OAuth2User;
14 import org.springframework.stereotype.Service;
15
16 import java.util.ArrayList;
17 import java.util.List;
18 import java.util.UUID;
19
20 @Service
21 public class CustomOAuth2UserService extends DefaultOAuth2UserService {
22
23     private final EmployeeRepo employeeRepo;
24     private final VendorRepo vendorRepo;
25     private final PasswordEncoder passwordEncoder;
26
27     public CustomOAuth2UserService(EmployeeRepo employeeRepo,
28                                     VendorRepo vendorRepo,
29                                     PasswordEncoder passwordEncoder) {
30         this.employeeRepo = employeeRepo;
31         this.vendorRepo = vendorRepo;
32         this.passwordEncoder = passwordEncoder;
33     }
34
35     @Override
36     public OAuth2User loadUser(OAuth2UserRequest userRequest) {
37         System.out.println("=== CustomOAuth2UserService.loadUser() called ===");
38         OAuth2User oAuth2User = super.loadUser(userRequest);
39
40         String email = oAuth2User.getAttribute("email");
41         String name = oAuth2User.getAttribute("name");
42         System.out.println("OAuth email: " + email);
43
44         boolean employeeExists = employeeRepo.findByEmail(email).isPresent();
45         boolean vendorExists = vendorRepo.findByEmail(email).isPresent();
46         System.out.println("employeeExists=" + employeeExists + ", vendorExists=" + vendorExists);
47
48         List<GrantedAuthority> authorities = new ArrayList<>(oAuth2User.getAuthorities());
49
50         if (employeeExists) {
51             Employee employee = employeeRepo.findByEmail(email).get();
52             authorities.add(new SimpleGrantedAuthority("ROLE_" + employee.getRole().name()));
53
54             // ADD THIS – loads LEAVE_CANCEL, SUBSCRIPTION_READ, etc. into the session
55             employee.getRole().getPermissions().forEach(permission ->
56                 authorities.add(new SimpleGrantedAuthority(permission.name()))
57             );
58
59             System.out.println("Returning existing employee with role: " + employee.getRole());
60             return new DefaultOAuth2User(authorities, oAuth2User.getAttributes(), "sub");
61         }
62
63         if (vendorExists) {
64             // add vendor role if needed
65             authorities.add(new SimpleGrantedAuthority("ROLE_VENDOR"));
66             System.out.println("Returning existing vendor");
67             return new DefaultOAuth2User(authorities, oAuth2User.getAttributes(), "sub");
68         }
69
70         // New user
71         registerAsUser(email, name);

```

```
72     authorities.add(new SimpleGrantedAuthority("ROLE_USER"));
73     return new DefaultOAuth2User(authorities, oAuth2User.getAttributes(), "sub");
74 }
75
76 private void registerAsUser(String email, String name) {
77     Employee e = new Employee();
78     e.setEmail(email);
79     e.setName(name);
80     e.setPassword(passwordEncoder.encode(UUID.randomUUID().toString()));
81     e.setRole(Role.USER);
82     e.setTimezone("Asia/Kolkata");
83     e.setDept("UNASSIGNED");
84     employeeRepo.saveAndFlush(e); // flush immediately so OAuth2SuccessHandler sees the record
85     System.out.println("New USER registered and flushed for: " + email);
86 }
87 }
```

```

1 package com.example.EmployeeManagementSystem.Service;
2
3 import com.example.EmployeeManagementSystem.DTO.VendorDTO;
4 import com.example.EmployeeManagementSystem.DTO.VendorRequest;
5 import com.example.EmployeeManagementSystem.Entity.Delivery;
6 import com.example.EmployeeManagementSystem.Entity.Subscription;
7 import com.example.EmployeeManagementSystem.Entity.Vendor;
8 import com.example.EmployeeManagementSystem.Enum.DeliveryStatus;
9 import com.example.EmployeeManagementSystem.Enum.Role;
10 import com.example.EmployeeManagementSystem.Enum.SubscriptionStatus;
11 import com.example.EmployeeManagementSystem.Exception.VendorNotFoundException;
12 import com.example.EmployeeManagementSystem.Repository.DeliveryRepository;
13 import com.example.EmployeeManagementSystem.Repository.SubscriptionRepository;
14 import com.example.EmployeeManagementSystem.Repository.VendorRepo;
15 import com.example.EmployeeManagementSystem.Util.AuthUtil;
16 import org.slf4j.Logger;
17 import org.slf4j.LoggerFactory;
18 import org.springframework.dao.DataIntegrityViolationException;
19 import org.springframework.security.core.Authentication;
20 import org.springframework.security.crypto.password.PasswordEncoder;
21 import org.springframework.stereotype.Service;
22 import org.springframework.transaction.annotation.Transactional;
23
24 import java.time.Instant;
25 import java.util.List;
26 import java.util.stream.Collectors;
27
28 @Service
29 public class DeliveryService {
30     private final SubscriptionRepository subscriptionRepository;
31     private final DeliveryRepository deliveryRepository;
32     private final DeliveryTimeService deliveryTimeService;
33     private static final Logger logger = LoggerFactory.getLogger(DeliveryService.class);
34
35     public DeliveryService(SubscriptionRepository subscriptionRepository,
36                           DeliveryRepository deliveryRepository,
37                           DeliveryTimeService deliveryTimeService) {
38         this.subscriptionRepository = subscriptionRepository;
39         this.deliveryRepository = deliveryRepository;
40         this.deliveryTimeService = deliveryTimeService;
41     }
42
43     // AFTER
44     @Transactional
45     public void processDueDeliveries() {
46         Instant now = Instant.now();
47         List<Subscription> dueSubscriptions = subscriptionRepository
48             .findDueSubscriptions(SubscriptionStatus.ACTIVE.name(), now);
49
50         for (Subscription subscription : dueSubscriptions) {
51             try {
52                 Delivery delivery = new Delivery();
53                 delivery.setSubscription(subscription);
54                 delivery.setMealSlot(subscription.getSlot());
55                 delivery.setScheduledDeliveryTime(subscription.getNextDeliveryTime());
56                 delivery.setStatus(DeliveryStatus.IN_PROGRESS);
57                 deliveryRepository.save(delivery);
58
59                 Instant nextDelivery = deliveryTimeService.getNextDeliveryTime(subscription);
60                 subscription.setNextDeliveryTime(nextDelivery);
61                 subscriptionRepository.save(subscription);
62
63             } catch (DataIntegrityViolationException e) {
64                 // Duplicate delivery already exists – just advance the next delivery time
65                 logger.warn("Duplicate delivery skipped for subscription ID: {}", subscription.getId());
66                 Instant nextDelivery = deliveryTimeService.getNextDeliveryTime(subscription);
67                 subscription.setNextDeliveryTime(nextDelivery);
68                 subscriptionRepository.save(subscription);
69             }
70         }
71     }

```

```

72
73     @Transactional
74     public Delivery updateDeliveryStatus(Long deliveryId, DeliveryStatus status) {
75         Delivery delivery = deliveryRepository.findById(deliveryId)
76             .orElseThrow(() -> new RuntimeException("Delivery not found"));
77
78         delivery.setStatus(status);
79         if (status == DeliveryStatus.DELIVERED) {
80             delivery.setActualDeliveryTime(Instant.now());
81         }
82
83         return deliveryRepository.save(delivery);
84     }
85
86     public List<Delivery> getDeliveriesBySubscription(Long subscriptionId) {
87         Subscription subscription = subscriptionRepository.findById(subscriptionId)
88             .orElseThrow(() -> new RuntimeException("Subscription not found"));
89         return deliveryRepository.findRecentDeliveriesBySubscription(subscription,
DeliveryStatus.SCHEDULED);
90     }
91
92     @Service
93     public static class VendorService {
94
95         private static final Logger log = LoggerFactory.getLogger(VendorService.class);
96         private final VendorRepo vendorRepo;
97         private final PasswordEncoder passwordEncoder;
98
99         public VendorService(VendorRepo vendorRepo, PasswordEncoder passwordEncoder) {
100             this.vendorRepo = vendorRepo;
101             this.passwordEncoder = passwordEncoder;
102         }
103
104         public VendorDTO registerVendor(VendorRequest request) {
105             if (vendorRepo.existsByEmail(request.getEmail())) {
106                 throw new IllegalArgumentException(
107                     "A vendor with email " + request.getEmail() + " already exists");
108             }
109             Vendor vendor = new Vendor();
110             vendor.setName(request.getName());
111             vendor.setEmail(request.getEmail());
112             vendor.setPhone(request.getPhone());
113             vendor.setRole(request.getRole());
114             vendor.setPassword(passwordEncoder.encode(request.getPassword()));
115             Vendor saved = vendorRepo.save(vendor);
116             log.info("Vendor registered: {} (id={})", saved.getName(), saved.getId());
117             return toDTO(saved);
118         }
119
120         public VendorDTO getVendor(Authentication authentication) {
121             String email= AuthUtil.extractEmail(authentication);
122             return toDTO(vendorRepo.findById(email).orElseThrow(
123                 ()->new VendorNotFoundException("Cannot get others account")
124             ));
125         }
126
127         public List<VendorDTO> getAllVendors() {
128             return vendorRepo.findAll().stream()
129                 .map(this::toDTO)
130                 .collect(Collectors.toList());
131         }
132
133         public VendorDTO updateVendor(VendorRequest request, Authentication authentication) {
134             String email= AuthUtil.extractEmail(authentication);
135             Vendor vendor = vendorRepo.findById(email).orElseThrow(
136                 ()->new VendorNotFoundException("Cannot update others account")
137             );
138             if (request.getName() != null) vendor.setName(request.getName());
139             if (request.getPhone() != null) vendor.setPhone(request.getPhone());
140             if (request.getPassword() != null)
vendor.setPassword(passwordEncoder.encode(request.getPassword()));
141             // Email update not allowed to avoid breaking restaurant ownership references
142             return toDTO(vendorRepo.save(vendor));
143         }
144
145         public void deleteVendor(Authentication authentication) {

```

```
146         String email= AuthUtil.extractEmail(authentication);
147         Vendor vendor = vendorRepo.findByEmail(email).orElseThrow(
148             ()->new VendorNotFoundException("Cannot delete others account")
149         );
150         vendorRepo.delete(vendor);
151         log.info("Vendor deleted: email={}", email);
152     }
153
154     public void deleteVendorById(Long vendorId) {
155         Vendor vendor = findOrThrow(vendorId);
156         vendorRepo.delete(vendor);
157         log.info("Vendor deleted by admin: id={}, email={}", vendor.getId(), vendor.getEmail());
158     }
159
160     // helpers
161
162     private Vendor findOrThrow(Long id) {
163         return vendorRepo.findById(id)
164             .orElseThrow(() -> new VendorNotFoundException("Vendor with id " + id + " not found"));
165     }
166
167     private VendorDTO toDTO(Vendor vendor) {
168         VendorDTO dto = new VendorDTO();
169         dto.setId(vendor.getId());
170         dto.setName(vendor.getName());
171         dto.setEmail(vendor.getEmail());
172         dto.setPhone(vendor.getPhone());
173         dto.setRegisteredAt(vendor.getRegisteredAt());
174         return dto;
175     }
176 }
177 }
178
```



```

1 package com.example.EmployeeManagementSystem.Service;
2
3 import com.example.EmployeeManagementSystem.DTO.SubscriptionRequest;
4 import com.example.EmployeeManagementSystem.Entity.Employee;
5 import com.example.EmployeeManagementSystem.Entity.Subscription;
6 import com.example.EmployeeManagementSystem.Enum.MealSlot;
7 import com.example.EmployeeManagementSystem.Enum.ScheduleType;
8 import org.springframework.stereotype.Service;
9
10 import java.time.*;
11
12 @Service
13 public class DeliveryTimeService {
14
15     private final MealSlotService mealSlotService;
16
17     public DeliveryTimeService(MealSlotService mealSlotService) {
18         this.mealSlotService = mealSlotService;
19     }
20     public Instant calculateNextDelivery(SubscriptionsRequest request, MealSlot slot, Employee employee) {
21         ZoneId userZone=ZoneId.of(employee.getTimezone());
22         if (request.getScheduleType() == ScheduleType.DAILY) {
23             return calculateDailyNextDelivery(slot,userZone);
24         } else {
25             return calculateWeeklyNextDelivery(slot, request.getDayOfWeek(),userZone);
26         }
27     }
28
29     private Instant calculateDailyNextDelivery(MealSlot slot, ZoneId userZone) {
30         LocalDateTime slotTime = mealSlotService.getTime(slot);
31         ZonedDateTime now = ZonedDateTime.now(userZone);
32         ZonedDateTime deliveryTime;
33
34         if (now.toLocalTime().isBefore(slotTime)) {
35             deliveryTime=ZonedDateTime.of(now.toLocalDate(),slotTime,userZone);
36         } else {
37             deliveryTime=ZonedDateTime.of(now.toLocalDate().plusDays(1), slotTime,userZone);
38         }
39         return deliveryTime.toInstant();
40     }
41
42     private Instant calculateWeeklyNextDelivery(MealSlot slot, DayOfWeek dayOfWeek,ZoneId userZone) {
43         LocalDateTime slotTime = mealSlotService.getTime(slot);
44         ZonedDateTime now=ZonedDateTime.now(userZone);
45         DayOfWeek today = now.getDayOfWeek();
46
47         int daysToAdd = dayOfWeek.getValue() - today.getValue();
48
49         if (daysToAdd == 0) {
50             if (now.toLocalTime().isBefore(slotTime)) {
51                 return ZonedDateTime.of(now.toLocalDate(), slotTime,userZone).toInstant();
52             } else {
53                 return ZonedDateTime.of(now.toLocalDate().plusDays(7), slotTime,userZone).toInstant();
54             }
55         }
56         if (daysToAdd < 0) {
57             daysToAdd += 7;
58         }
59
60         ZonedDateTime deliveryTime = ZonedDateTime.of(
61             now.toLocalDate().plusDays(daysToAdd),slotTime,userZone);
62         return deliveryTime.toInstant();
63     }
64
65     public Instant getNextDeliveryTime(Subscriptions subscription) {
66         ZoneId userZone=ZoneId.of(subscription.getEmployee().getTimezone());
67         if (subscription.getScheduleType() == ScheduleType.DAILY) {
68             return calculateDailyNextDelivery(subscription.getSlot(),userZone);
69         } else {
70             return calculateWeeklyNextDelivery(subscription.getSlot(),
71 subscription.getDayOfWeek(),userZone);

```

```
71     }
```

```
72 }
```

```
73 }
```

```
74
```

```

1  package com.example.EmployeeManagementSystem.Service;
2
3  import com.example.EmployeeManagementSystem.DTO.AssignDeviceRequest;
4  import com.example.EmployeeManagementSystem.DTO.DeviceAssignmentResponseDTO;
5  import com.example.EmployeeManagementSystem.DTO.DeviceDTO;
6  import com.example.EmployeeManagementSystem.DTO.DeviceResponseDTO;
7  import com.example.EmployeeManagementSystem.Entity.Device;
8  import com.example.EmployeeManagementSystem.Entity.DeviceAssignment;
9  import com.example.EmployeeManagementSystem.Entity.Employee;
10 import com.example.EmployeeManagementSystem.Entity.Vendor;
11 import com.example.EmployeeManagementSystem.Enum.AssignmentStatus;
12 import com.example.EmployeeManagementSystem.Enum.DeviceStatus;
13 import com.example.EmployeeManagementSystem.Enum.Role;
14 import com.example.EmployeeManagementSystem.Enum.Status;
15 import com.example.EmployeeManagementSystem.Exception.VendorNotFoundException;
16 import com.example.EmployeeManagementSystem.Repository.DeviceAssignmentRepo;
17 import com.example.EmployeeManagementSystem.Repository.DeviceRepository;
18 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
19 import com.example.EmployeeManagementSystem.Repository.VendorRepo;
20 import com.example.EmployeeManagementSystem.Util.AuthUtil;
21 import org.springframework.security.core.Authentication;
22 import org.springframework.stereotype.Service;
23 import org.springframework.transaction.annotation.Transactional;
24
25 import java.time.LocalDate;
26 import java.time.LocalDateTime;
27 import java.util.List;
28 import java.util.Set;
29
30 @Service
31 public class DeviceService {
32     private final DeviceRepository deviceRepository;
33     private final VendorRepo vendorRepo;
34     private final DeviceAssignmentRepo deviceAssignmentRepo;
35     private final EmployeeRepo employeeRepo;
36
37     public DeviceService(DeviceRepository deviceRepository,
38                         VendorRepo vendorRepo,
39                         DeviceAssignmentRepo deviceAssignmentRepo,
40                         EmployeeRepo employeeRepo) {
41         this.deviceRepository = deviceRepository;
42         this.vendorRepo = vendorRepo;
43         this.deviceAssignmentRepo = deviceAssignmentRepo;
44         this.employeeRepo = employeeRepo;
45     }
46
47     public Device addDevice(DeviceDTO deviceDTO, Authentication authentication) {
48         String email = AuthUtil.extractEmail(authentication);
49         Vendor vendor = vendorRepo.findByEmail(email)
50             .orElseThrow(() -> new RuntimeException("Vendor not found"));
51
52         Device device = new Device();
53         device.setDeviceName(deviceDTO.getDeviceName());
54         device.setWarrantyExpiryDate(deviceDTO.getWarrantyExpiryDate());
55         device.setBrand(deviceDTO.getBrand());
56         device.setDeviceStatus(DeviceStatus.AVAILABLE);
57         device.setDeviceType(deviceDTO.getDeviceType());
58         device.setTechVendor(vendor);
59         device.setCreatedAt(LocalDateTime.now());
60         return deviceRepository.save(device);
61     }
62
63     public List<DeviceResponseDTO> getDevicesForLoggedInVendor(Authentication authentication) {
64         String email = AuthUtil.extractEmail(authentication);
65         Vendor vendor = vendorRepo.findByEmail(email)
66             .orElseThrow(() -> new RuntimeException("Vendor not found"));
67
68         return deviceRepository.findByTechVendorId(vendor.getId()).stream()
69             .map(this::toDeviceResponse)
70             .toList();
71     }

```

```

72
73     public List<DeviceResponseDTO> getDevicesForLoggedInEmployee(Authentication authentication) {
74         String email = AuthUtil.extractEmail(authentication);
75         Employee employee = employeeRepo.findByEmail(email)
76             .orElseThrow(() -> new RuntimeException("Employee not found"));
77
78         return deviceAssignmentRepo.findByAssignedToEmployeeIdAndStatus(employee.getEmployeeId(),
79 AssignmentStatus.ACTIVE)
80             .stream()
81             .map(DeviceAssignment::getDevice)
82             .map(this::toDeviceResponse)
83             .toList();
84     }
85
86     public List<DeviceResponseDTO> getUnassignedDevices() {
87         return deviceRepository.findByDeviceStatus(DeviceStatus.AVAILABLE).stream()
88             .map(this::toDeviceResponse)
89             .toList();
90     }
91
92     @Transactional
93     public DeviceAssignment assignDevice(Long deviceId, AssignDeviceRequest request) {
94         if (request.getEmployeeId() == null) {
95             throw new IllegalArgumentException("employeeId is required");
96         }
97
98         Device device = deviceRepository.findById(deviceId)
99             .orElseThrow(() -> new RuntimeException("Device not found"));
100
101         Employee employee = employeeRepo.findById(request.getEmployeeId())
102             .orElseThrow(() -> new RuntimeException("Employee not found"));
103
104         if (employee.getStatus() != Status.ACTIVE) {
105             throw new IllegalArgumentException("Device can only be assigned to an active user");
106         }
107
108         if (!Set.of(Role.EMPLOYEE, Role.MANAGER, Role.ADMIN).contains(employee.getRole())) {
109             throw new IllegalArgumentException("Devices can only be assigned to employees, managers, or
110 admins");
111         }
112
113         if (device.getDeviceStatus() != DeviceStatus.AVAILABLE) {
114             throw new IllegalArgumentException("Only available devices can be assigned");
115         }
116
117         deviceAssignmentRepo.findByDeviceIdAndStatus(deviceId, AssignmentStatus.ACTIVE)
118             .ifPresent(existingAssignment -> {
119                 throw new IllegalArgumentException("Device is already assigned");
120             });
121
122         DeviceAssignment assignment = new DeviceAssignment();
123         assignment.setDevice(device);
124         assignment.setAssignedTo(employee);
125         assignment.setStatus(AssignmentStatus.ACTIVE);
126         assignment.setAssignedDate(LocalDate.now());
127
128         DeviceAssignment savedAssignment = deviceAssignmentRepo.save(assignment);
129         device.setCurrentAssignment(savedAssignment);
130         device.setDeviceStatus(DeviceStatus.ASSIGNED);
131         deviceRepository.save(device);
132
133         return savedAssignment;
134     }
135
136     private DeviceResponseDTO toDeviceResponse(Device device) {
137         DeviceResponseDTO response = new DeviceResponseDTO();
138         response.setId(device.getId());
139         response.setDeviceName(device.getDeviceName());
140         response.setBrand(device.getBrand());
141         response.setWarrantyExpiryDate(device.getWarrantyExpiryDate());
142         response.setDeviceType(device.getDeviceType());
143         response.setDeviceStatus(device.getDeviceStatus());
144
145         if (device.getTechVendor() != null) {

```

```

146         response.setVendorName(device.getTechVendor().getName());
147     }
148
149     DeviceAssignment currentAssignment = device.getCurrentAssignment();
150     if (currentAssignment != null) {
151         response.setAssignedDate(currentAssignment.getAssignedDate());
152
153         Employee assignedEmployee = currentAssignment.getAssignedTo();
154         if (assignedEmployee != null) {
155             response.setAssignedEmployeeId(assignedEmployee.getEmployeeId());
156             response.setAssignedEmployeeName(assignedEmployee.getName());
157         }
158     }
159
160     return response;
161 }
162
163 public List<DeviceAssignmentResponseDTO> getDeviceAssignmentOfVendor(Authentication authentication) {
164     String email=AuthUtil.extractEmail(authentication);
165     Vendor vendor=vendorRepo.findByEmail(email).orElseThrow(
166         ()->new VendorNotFoundException("Vendor:"+email+" not found")
167     );
168     List<DeviceAssignment>
deviceAssignments=deviceAssignmentRepo.getAllAssignmentsByVendorId(vendor.getId());
169     return deviceAssignments.stream().map(deviceAssignment -> toAssignmentDTO(deviceAssignment)).toList();
170 }
171
172 private DeviceAssignmentResponseDTO toAssignmentDTO(DeviceAssignment a) {
173     DeviceAssignmentResponseDTO dto = new DeviceAssignmentResponseDTO();
174     dto.setId(a.getId());
175     dto.setStatus(a.getStatus());
176     dto.setAssignedDate(a.getAssignedDate());
177
178     if (a.getDevice() != null) {
179         dto.setDeviceId(a.getDevice().getId());
180         dto.setDeviceName(a.getDevice().getDeviceName());
181         dto.setDeviceBrand(a.getDevice().getBrand());
182         dto.setDeviceType(a.getDevice().getDeviceType());
183     }
184
185     if (a.getAssignedTo() != null) {
186         dto.setEmployeeId(a.getAssignedTo().getEmployeeId());
187         dto.setEmployeeName(a.getAssignedTo().getName()); // adjust to your Employee field
188     }
189
190     return dto;
191 }
192 }
193

```

```

1 package com.example.EmployeeManagementSystem.Service;
2
3 import com.example.EmployeeManagementSystem.DTO.*;
4 import com.example.EmployeeManagementSystem.Entity.Employee;
5 import com.example.EmployeeManagementSystem.Entity.LeaveRequest;
6 import com.example.EmployeeManagementSystem.Enum.Role;
7 import com.example.EmployeeManagementSystem.Enum.Status;
8 import com.example.EmployeeManagementSystem.Exception.EmployeeNotFound;
9 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
10 import com.example.EmployeeManagementSystem.Repository.LeaveRequestRepo;
11 import com.example.EmployeeManagementSystem.Repository.RefreshTokenRepository;
12 import com.example.EmployeeManagementSystem.Util.AuthUtil;
13 import org.slf4j.Logger;
14 import org.slf4j.LoggerFactory;
15 import org.springframework.http.ResponseEntity;
16 import org.springframework.security.core.Authentication;
17 import org.springframework.security.crypto.password.PasswordEncoder;
18 import org.springframework.stereotype.Service;
19 import org.springframework.transaction.annotation.Transactional;
20
21 import java.time.ZoneId;
22 import java.util.List;
23 import java.util.stream.Collectors;
24
25 @Service
26 public class EmployeeService {
27
28     private static final Logger log = LoggerFactory.getLogger(EmployeeService.class);
29
30     private final EmployeeRepo employeeRepo;
31     private final LeaveRequestRepo leaveRequestRepo;
32     private final PasswordEncoder passwordEncoder;
33     private final RefreshTokenRepository refreshTokenRepository; // FIX: added
34
35     public EmployeeService(EmployeeRepo employeeRepo,
36                             LeaveRequestRepo leaveRequestRepo,
37                             PasswordEncoder passwordEncoder,
38                             RefreshTokenRepository refreshTokenRepository) {
39         this.employeeRepo = employeeRepo;
40         this.leaveRequestRepo = leaveRequestRepo;
41         this.passwordEncoder = passwordEncoder;
42         this.refreshTokenRepository = refreshTokenRepository; // FIX: injected
43     }
44
45     public List<Employee> getAllEmployees() {
46         return employeeRepo.findAll();
47     }
48
49     public Employee createEmployee(AdminEmployeeDTO employeeDTO) {
50         log.info("Creating new employee with email: {}", employeeDTO.getEmail());
51         var employee = new Employee();
52         if (employeeDTO.getName() != null) employee.setName(employeeDTO.getName());
53         if (employeeDTO.getEmail() != null) employee.setEmail(employeeDTO.getEmail());
54         if (employeeDTO.getTimezone() != null && isValidTimezone(employeeDTO.getTimezone())) {
55             employee.setTimezone(employeeDTO.getTimezone());
56         } else {
57             employee.setTimezone("UTC");
58         }
59         employee.setRole(Role.EMPLOYEE);
60         if (employeeDTO.getPassword() == null) {
61             throw new RuntimeException("Password is required");
62         }
63         employee.setPassword(passwordEncoder.encode(employeeDTO.getPassword()));
64         employee.setGender(employeeDTO.getGender());
65         Employee manager=employeeRepo.findById(employeeDTO.getManagerId()).orElseThrow(
66             ()->new EmployeeNotFound("Employee with not found"+ employeeDTO.getManagerId())
67         );
68         employee.setDept(manager.getDept());
69         employee.setManager(manager);
70         return employeeRepo.save(employee);
71     }

```

```

72
73     public Employee createManager(EmployeeDTO employeeDTO) {
74         log.info("Creating new manager with email: {}", employeeDTO.getEmail());
75         var employee = new Employee();
76         if (employeeDTO.getName() != null) employee.setName(employeeDTO.getName());
77         if (employeeDTO.getEmail() != null) employee.setEmail(employeeDTO.getEmail());
78         if (employeeDTO.getDept() != null) employee.setDept(employeeDTO.getDept());
79         if (employeeDTO.getTimezone() != null && isValidTimezone(employeeDTO.getTimezone())) {
80             employee.setTimezone(employeeDTO.getTimezone());
81         } else {
82             employee.setTimezone("UTC");
83         }
84         employee.setRole(Role.MANAGER);
85         if (employeeDTO.getPassword() == null) {
86             throw new RuntimeException("Password is required");
87         }
88         employee.setPassword(passwordEncoder.encode(employeeDTO.getPassword()));
89         employee.setGender(employeeDTO.getGender());
90         return employeeRepo.save(employee);
91     }
92
93     public Employee updateEmployee(long id, EmployeeDTO employee) {
94         var updateEmployee = employeeRepo.findById(id).orElseThrow(
95             () -> new EmployeeNotFound("Employee with id:" + id + " not found")
96         );
97         if (employee.getName() != null)
98             updateEmployee.setName(employee.getName());
99         if (employee.getEmail() != null)
100             updateEmployee.setEmail(employee.getEmail());
101         if (employee.getDept() != null)
102             updateEmployee.setDept(employee.getDept());
103         if (employee.getTimezone() != null && isValidTimezone(employee.getTimezone())) {
104             updateEmployee.setTimezone(employee.getTimezone());
105         }
106         return employeeRepo.save(updateEmployee);
107     }
108
109     /**
110      * FIX: Delete order must be:
111      * 1. refresh_token (FK employee)
112      * 2. leave_request (FK employee)
113      * 3. employee
114      *
115      * Previously only leave_request was deleted first, causing a
116      * FK constraint failure on refresh_token when deleting the employee.
117      */
118     @Transactional
119     public void deleteEmployee(long id) {
120         var employee = employeeRepo.findById(id).orElseThrow(
121             () -> new EmployeeNotFound("Employee with id:" + id + " not found")
122         );
123
124         // Step 1: delete refresh tokens for this employee
125         refreshTokenRepository.deleteByEmployee(employee);
126
127         // Step 2: delete leave requests for this employee
128         List<LeaveRequest> leaveRequests =
129             leaveRequestRepo.findByEmployeeOrderByStartDateDesc(employee);
130         leaveRequestRepo.deleteAll(leaveRequests);
131
132         // Step 3: delete the employee
133         employeeRepo.delete(employee);
134     }
135
136     public ResponseEntity<String> inactivateUser(long employeeId) {
137         var employee = employeeRepo.findById(employeeId).orElseThrow(
138             () -> new EmployeeNotFound("Employee with id:" + employeeId + " not found")
139         );
140         if (employee.getStatus() == Status.INACTIVE) {
141             return ResponseEntity.badRequest()
142                 .body("Status of the employee with id: " + employeeId + " is already set to inactive");
143         }
144         employee.setStatus(Status.INACTIVE);
145         employeeRepo.save(employee);
146         return ResponseEntity.ok("Status of the employee with id: " + employeeId + " is set to inactive");
147     }

```

```

148
149  /** Validate that the given string is a recognized IANA timezone id. */
150  private boolean isValidTimezone(String tz) {
151      try {
152          ZoneId.of(tz);
153          return true;
154      } catch (Exception e) {
155          return false;
156      }
157  }
158
159  public Employee getAccount(Authentication authentication) {
160      String email = AuthUtil.extractEmail(authentication);
161      Employee employee = employeeRepo.findByEmail(email).orElseThrow(
162          () -> new EmployeeNotFound("Employee :" + email + " NOT FOUND")
163      );
164      return employee;
165  }
166
167  public EmployeeAttendanceResponseDTO getEmployeeAttendanceOverview() {
168      List<Employee> employees = employeeRepo.findAll();
169
170      List<EmployeeStatusDTO> workingEmployees = employees.stream()
171          .filter(e -> e.getStatus() == Status.ACTIVE)
172          .map(this::toStatusDTO)
173          .collect(Collectors.toList());
174
175      List<EmployeeStatusDTO> onLeaveEmployees = employees.stream()
176          .filter(e -> e.getStatus() == Status.ON_LEAVE)
177          .map(this::toStatusDTO)
178          .collect(Collectors.toList());
179
180      EmployeeAttendanceResponseDTO response = new EmployeeAttendanceResponseDTO();
181      response.setWorkingCount(workingEmployees.size());
182      response.setOnLeaveCount(onLeaveEmployees.size());
183      response.setWorkingEmployees(workingEmployees);
184      response.setOnLeaveEmployees(onLeaveEmployees);
185      return response;
186  }
187
188  private EmployeeStatusDTO toStatusDTO(Employee employee) {
189      EmployeeStatusDTO dto = new EmployeeStatusDTO();
190      dto.setEmployeeId(employee.getEmployeeId());
191      dto.setName(employee.getName());
192      dto.setEmail(employee.getEmail());
193      dto.setDept(employee.getDept());
194      dto.setRole(employee.getRole());
195      return dto;
196  }
197
198  public ResponseEntity<List<ManagerDTO>> getAllManagers() {
199      List<ManagerDTO> managers = employeeRepo.findByRole(Role.MANAGER).stream()
200          .map(this::toManagerDTO)
201          .collect(Collectors.toList());
202      return ResponseEntity.ok(managers);
203  }
204
205  private ManagerDTO toManagerDTO(Employee employee) {
206      ManagerDTO dto = new ManagerDTO();
207      dto.setManagerId(employee.getEmployeeId());
208      dto.setName(employee.getName());
209      dto.setEmail(employee.getEmail());
210      dto.setDept(employee.getDept());
211      return dto;
212  }
213 }
214

```



```
1 package com.example.EmployeeManagementSystem.Service;
2
3 import com.example.EmployeeManagementSystem.DTO.HolidayDTO;
4 import com.example.EmployeeManagementSystem.Entity.FixedHoliday;
5 import com.example.EmployeeManagementSystem.Repository.FixedHolidayRepository;
6 import org.springframework.stereotype.Service;
7
8 import java.time.LocalDate;
9 import java.util.List;
10 @Service
11 public class HolidayService {
12     private final FixedHolidayRepository holidayRepository;
13
14     public HolidayService(FixedHolidayRepository holidayRepository) {
15         this.holidayRepository = holidayRepository;
16     }
17
18     // Admin adds a holiday
19     public FixedHoliday addHoliday(HolidayDTO dto) {
20         FixedHoliday h = new FixedHoliday();
21         h.setName(dto.getName());
22         h.setDate(LocalDate.parse(dto.getDate()));
23         h.setDescription(dto.getDescription());
24         h.setYear(LocalDate.parse(dto.getDate()).getYear());
25         return holidayRepository.save(h);
26     }
27
28     // Everyone can view holidays
29     public List<FixedHoliday> getHolidaysByYear(int year) {
30         return holidayRepository.findByYear(year);
31     }
32
33     // Admin deletes a holiday
34     public void deleteHoliday(Long id) {
35         holidayRepository.deleteById(id);
36     }
37 }
38
39
```

```

1 package com.example.EmployeeManagementSystem.Service;
2
3 import com.example.EmployeeManagementSystem.Entity.Employee;
4 import com.example.EmployeeManagementSystem.Entity.LeaveEntitlement;
5 import com.example.EmployeeManagementSystem.Entity.LeaveType;
6 import com.example.EmployeeManagementSystem.Enum.Gender;
7 import com.example.EmployeeManagementSystem.Enum.Status;
8 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
9 import com.example.EmployeeManagementSystem.Repository.LeaveEntitlementRepository;
10 import com.example.EmployeeManagementSystem.Repository.LeaveTypeRepository;
11 import org.slf4j.Logger;
12 import org.slf4j.LoggerFactory;
13 import org.springframework.stereotype.Service;
14 import org.springframework.transaction.annotation.Transactional;
15
16 import java.math.BigDecimal;
17 import java.time.LocalDate;
18 import java.util.List;
19 import java.util.Set;
20
21 @Service
22 public class LeaveAccrualService {
23
24     private static final Logger log = LoggerFactory.getLogger(LeaveAccrualService.class);
25     private static final BigDecimal ONE_DAY = BigDecimal.ONE;
26     private static final Set<String> SUPPORTED_LEAVE_TYPES = Set.of("SICK", "CASUAL");
27
28     private final EmployeeRepo employeeRepository;
29     private final LeaveTypeRepository leaveTypeRepository;
30     private final LeaveEntitlementRepository entitlementRepository;
31
32     public LeaveAccrualService(EmployeeRepo employeeRepository,
33                               LeaveTypeRepository leaveTypeRepository,
34                               LeaveEntitlementRepository entitlementRepository) {
35         this.employeeRepository = employeeRepository;
36         this.leaveTypeRepository = leaveTypeRepository;
37         this.entitlementRepository = entitlementRepository;
38     }
39
40     @Transactional
41     public void runMonthlyAccrual(LocalDate accrualDate) {
42         int year = accrualDate.getYear();
43
44         List<Employee> activeEmployees = employeeRepository.findByStatus(Status.ACTIVE);
45         List<LeaveType> allLeaveTypes = leaveTypeRepository.findAll();
46
47         log.info("Starting monthly accrual for {} - {} active employees",
48                 accrualDate, activeEmployees.size());
49
50         for (Employee employee : activeEmployees) {
51             for (LeaveType leaveType : allLeaveTypes) {
52
53                 if (!SUPPORTED_LEAVE_TYPES.contains(leaveType.getName())) {
54                     continue;
55                 }
56                 if (!isEligible(employee, leaveType, accrualDate)) {
57                     continue;
58                 }
59
60                 // Find or create entitlement row for this employee/type/year
61                 LeaveEntitlement entitlement = entitlementRepository
62                     .findByEmployeeIdAndLeaveTypeIdAndYear(
63                         employee.getEmployeeId(), leaveType.getId(), year)
64                     .orElseGet(() -> createNewEntitlement(employee, leaveType, year));
65
66                 // Add 1 day accrual
67                 entitlement.setAccruedThisYear(
68                     entitlement.getAccruedThisYear().add(ONE_DAY)
69                 );
70
71                 entitlementRepository.save(entitlement);

```

```

72         log.debug("Accrued 1 {} day for employee {} (year {})",
73             leaveType.getName(), employee.getEmployeeId(), year);
74     }
75 }
76
77     log.info("Monthly accrual complete for {}", accrualDate);
78 }
79
80 // -----
81 // Eligibility check – single place for all rules
82 // -----
83 private boolean isEligible(Employee employee, LeaveType leaveType, LocalDate accrualDate) {
84
85     // 1. Gender restriction (Maternity = female only)
86     if (leaveType.isGenderRestricted()) {
87         Gender restriction = leaveType.getGenderRestriction();
88         if (restriction != null && restriction != employee.getGender()) {
89             return false;
90         }
91     }
92
93     // 2. Employee must have joined ON OR BEFORE the accrual date
94     // (no accrual for employees joining after the 1st of this month)
95     if (employee.getJoined_at() != null && employee.getJoined_at().isAfter(accrualDate)) {
96         return false;
97     }
98
99     return true;
100 }
101
102 // -----
103 // Creates a fresh entitlement row, carrying forward previous year's
104 // closing balance if this is the first row of the year
105 // -----
106 private LeaveEntitlement createNewEntitlement(Employee employee,
107     LeaveType leaveType,
108     int year) {
109     LeaveEntitlement entitlement = new LeaveEntitlement();
110     entitlement.setEmployee(employee);
111     entitlement.setLeaveType(leaveType);
112     entitlement.setYear(year);
113
114     // Carry forward last year's closing balance (if the type supports it)
115     if (leaveType.isCarriesForward()) {
116         BigDecimal lastYearClosing = entitlementRepository
117             .findByEmployeeEmployeeIdAndLeaveTypeIdAndYear(
118                 employee.getEmployeeId(), leaveType.getId(), year - 1)
119             .map(LeaveEntitlement::getClosingBalance)
120             .orElse(BigDecimal.ZERO);
121
122         entitlement.setOpeningBalance(lastYearClosing);
123     }
124
125     return entitlementRepository.save(entitlement);
126 }
127 }
128

```

```

1 package com.example.EmployeeManagementSystem.Service;
2
3 import com.example.EmployeeManagementSystem.DTO.LeaveDashboardResponse;
4 import com.example.EmployeeManagementSystem.Entity.Employee;
5 import com.example.EmployeeManagementSystem.Entity.LeaveEntitlement;
6 import com.example.EmployeeManagementSystem.Entity.LeaveType;
7 import com.example.EmployeeManagementSystem.Entity.LeaveWarning;
8 import com.example.EmployeeManagementSystem.Enum.Gender;
9 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
10 import com.example.EmployeeManagementSystem.Repository.LeaveEntitlementRepository;
11 import com.example.EmployeeManagementSystem.Repository.LeaveTypeRepository;
12 import com.example.EmployeeManagementSystem.Repository.LeaveWarningRepository;
13 import org.springframework.security.access.AccessDeniedException;
14 import org.springframework.stereotype.Service;
15 import org.springframework.transaction.annotation.Transactional;
16
17 import java.math.BigDecimal;
18 import java.time.LocalDate;
19 import java.util.Set;
20 import java.util.LinkedHashMap;
21 import java.util.List;
22 import java.util.Map;
23
24 @Service
25 public class LeaveDashboardService {
26
27     private static final Set<String> SUPPORTED_LEAVE_TYPES = Set.of("SICK", "CASUAL", "MATERNITY");
28
29     private final EmployeeRepo employeeRepository;
30     private final LeaveEntitlementRepository leaveEntitlementRepository;
31     private final LeaveTypeRepository leaveTypeRepository;
32     private final LeaveWarningRepository warningRepository;
33
34     public LeaveDashboardService(EmployeeRepo employeeRepository,
35                                 LeaveEntitlementRepository leaveEntitlementRepository,
36                                 LeaveTypeRepository leaveTypeRepository,
37                                 LeaveWarningRepository warningRepository) {
38         this.employeeRepository = employeeRepository;
39         this.leaveEntitlementRepository = leaveEntitlementRepository;
40         this.leaveTypeRepository = leaveTypeRepository;
41         this.warningRepository = warningRepository;
42     }
43
44     @Transactional
45     public LeaveDashboardResponse getDashboard(Long requestedEmployeeId,
46                                              Long callerEmployeeId,
47                                              String callerRole) {
48         assertAccess(requestedEmployeeId, callerEmployeeId, callerRole);
49
50         Employee employee = employeeRepository.findById(requestedEmployeeId)
51             .orElseThrow(() -> new IllegalArgumentException(
52                 "Employee not found: " + requestedEmployeeId));
53
54         int year = LocalDate.now().getYear();
55
56         ensureCurrentYearEntitlements(employee, year);
57         List<LeaveEntitlement> entitlements =
58             leaveEntitlementRepository.findByEmployeeEmployeeIdAndYear(requestedEmployeeId, year);
59
60         List<LeaveWarning> warnings =
61             warningRepository.findByEmployeeEmployeeIdAndIsReadFalseOrderByWarningDateDesc(
62                 requestedEmployeeId);
63
64         LeaveDashboardResponse.LeaveBalanceDto sick = buildBalanceDto(entitlements, "SICK");
65         LeaveDashboardResponse.LeaveBalanceDto casual = buildBalanceDto(entitlements, "CASUAL");
66         LeaveDashboardResponse.LeaveBalanceDto maternity = buildBalanceDto(entitlements, "MATERNITY");
67
68         Map<String, LeaveDashboardResponse.LeaveBalanceDto> leaveBalances = buildBalanceMap(entitlements);
69         BigDecimal carryTotal = entitlements.stream()
70             .filter(e -> isSupportedLeaveType(e.getLeaveType()))
71             .filter(e -> e.getLeaveType().isCarriesForward())

```

```

72         .map(LeaveEntitlement::getAvailableBalance)
73         .reduce(BigDecimal.ZERO, BigDecimal::add);
74
75     LeaveDashboardResponse response = new LeaveDashboardResponse();
76     response.setEmployeeId(employee.getEmployeeId());
77     response.setEmployeeName(employee.getName());
78     response.setYear(year);
79     response.setSick(sick);
80     response.setCasual(casual);
81     response.setMaternity(maternity);
82     response.setLeaveBalances(leaveBalances);
83     response.setCarryForwardMeter(new LeaveDashboardResponse.CarryForwardMeterDto(carryTotal, 30));
84     response.setWarnings(warnings.stream()
85         .map(w -> new LeaveDashboardResponse.WarningDto(
86             w.getId(),
87             w.getWarningType().name(),
88             w.getMessage(),
89             w.getWarningDate().toString()))
90         .toList());
91
92     return response;
93 }
94
95 // Mark warning read
96
97 @Transactional
98 public void markWarningRead(Long warningId, Long callerEmployeeId, String callerRole) {
99     LeaveWarning warning = warningRepository.findById(warningId)
100         .orElseThrow(() -> new IllegalArgumentException("Warning not found: " + warningId));
101
102     assertAccess(warning.getEmployee().getEmployeeId(), callerEmployeeId, callerRole);
103
104     warning.setRead(true);
105     warningRepository.save(warning);
106 }
107
108 // Team dashboard (manager/admin)
109
110 @Transactional
111 public List<LeaveDashboardResponse> getTeamDashboard(Long managerId, String callerRole) {
112     if (!"MANAGER".equals(callerRole) && !"ADMIN".equals(callerRole)) {
113         throw new AccessDeniedException("Only managers and admins can view team dashboards");
114     }
115
116     List<Employee> team = employeeRepository.findByManagerEmployeeId(managerId);
117     return team.stream()
118         .map(emp -> getDashboard(emp.getEmployeeId(), managerId, callerRole))
119         .toList();
120 }
121
122 // Helpers
123
124 private void ensureCurrentYearEntitlements(Employee employee, int year) {
125     List<LeaveType> leaveTypes = leaveTypeRepository.findAll();
126
127     for (LeaveType leaveType : leaveTypes) {
128         if (!isSupportedLeaveType(leaveType)) {
129             continue;
130         }
131         if (!isEligible(employee, leaveType)) {
132             continue;
133         }
134
135         leaveEntitlementRepository
136             .findByEmployeeEmployeeIdAndLeaveTypeIdAndYear(
137                 employee.getEmployeeId(), leaveType.getId(), year)
138             .orElseGet(() -> createEntitlement(employee, leaveType, year));
139     }
140 }
141
142 private boolean isEligible(Employee employee, LeaveType leaveType) {
143     if (!leaveType.isGenderRestricted()) {
144         return true;
145     }
146
147     Gender restriction = leaveType.getGenderRestriction();

```

```

148         return restriction == null || restriction == employee.getGender();
149     }
150
151     private LeaveEntitlement createEntitlement(Employee employee, LeaveType leaveType, int year) {
152         LeaveEntitlement entitlement = new LeaveEntitlement();
153         entitlement.setEmployee(employee);
154         entitlement.setLeaveType(leaveType);
155         entitlement.setYear(year);
156
157         if (leaveType.isCarriesForward()) {
158             BigDecimal lastYearClosing = leaveEntitlementRepository
159                 .findByEmployeeIdAndLeaveTypeIdAndYear(
160                     employee.getId(), leaveType.getId(), year - 1)
161                 .map(LeaveEntitlement::getClosingBalance)
162                 .orElse(BigDecimal.ZERO);
163             entitlement.setOpeningBalance(lastYearClosing);
164         }
165
166         return leaveEntitlementRepository.save(entitlement);
167     }
168
169     private Map<String, LeaveDashboardResponse.LeaveBalanceDto> buildBalanceMap(
170         List<LeaveEntitlement> entitlements) {
171         Map<String, LeaveDashboardResponse.LeaveBalanceDto> balances = new LinkedHashMap<>();
172         for (LeaveEntitlement entitlement : entitlements) {
173             String typeName = entitlement.getLeaveType().getName();
174             if (!SUPPORTED_LEAVE_TYPES.contains(typeName)) {
175                 continue;
176             }
177             balances.put(typeName, toBalanceDto(entitlement, typeName));
178         }
179         return balances;
180     }
181
182     private LeaveDashboardResponse.LeaveBalanceDto buildBalanceDto(List<LeaveEntitlement> entitlements,
183                                                                     String typeName) {
184         return entitlements.stream()
185             .filter(e -> typeName.equalsIgnoreCase(e.getLeaveType().getName()))
186             .findFirst()
187             .map(e -> toBalanceDto(e, typeName))
188             .orElse(null); // null = leave type not applicable for this employee
189     }
190
191     private LeaveDashboardResponse.LeaveBalanceDto toBalanceDto(LeaveEntitlement entitlement,
192                                                                 String typeName) {
193         return new LeaveDashboardResponse.LeaveBalanceDto(
194             typeName,
195             entitlement.getOpeningBalance(),
196             entitlement.getAccruedThisYear(),
197             entitlement.getUsedThisYear(),
198             entitlement.getAvailableBalance(),
199             entitlement.getLeaveType().isCarriesForward());
200     }
201
202     private boolean isSupportedLeaveType(LeaveType leaveType) {
203         return leaveType != null && SUPPORTED_LEAVE_TYPES.contains(leaveType.getName());
204     }
205
206     private void assertAccess(Long targetId, Long callerId, String role) {
207         boolean isSelf = targetId.equals(callerId);
208         boolean isManager = "MANAGER".equals(role) || "ADMIN".equals(role);
209         if (!isSelf && !isManager) {
210             throw new AccessDeniedException(
211                 "Employee " + callerId + " cannot view dashboard for " + targetId);
212         }
213     }
214 }
215

```

```

1 package com.example.EmployeeManagementSystem.Service;
2
3 import com.example.EmployeeManagementSystem.DTO.*;
4 import com.example.EmployeeManagementSystem.Entity.Employee;
5 import com.example.EmployeeManagementSystem.Entity.LeaveRequest;
6 import com.example.EmployeeManagementSystem.Enum.*;
7 import com.example.EmployeeManagementSystem.Exception.*;
8 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
9 import com.example.EmployeeManagementSystem.Repository.LeaveEntitlementRepository;
10 import com.example.EmployeeManagementSystem.Repository.LeaveRequestRepo;
11 import com.example.EmployeeManagementSystem.Repository.LeaveTypeRepository;
12 import com.example.EmployeeManagementSystem.Util.AuthUtil;
13 import org.slf4j.Logger;
14 import com.example.EmployeeManagementSystem.Enum.LeaveType;
15 import org.slf4j.LoggerFactory;
16 import org.springframework.http.HttpStatus;
17 import org.springframework.http.ResponseEntity;
18 import org.springframework.security.core.Authentication;
19 import org.springframework.stereotype.Service;
20 import org.springframework.transaction.annotation.Transactional;
21
22 import java.math.BigDecimal;
23 import java.time.LocalDate;
24 import java.time.ZoneId;
25 import java.time.temporal.ChronoUnit;
26 import java.util.ArrayList;
27 import java.util.List;
28 import java.util.Map;
29 import java.util.stream.Collectors;
30
31 @Service
32 public class LeaveRequestService {
33     private final LeaveRequestRepo leaveRequestRepo;
34     private final EmployeeRepo employeeRepo;
35     private final LeaveTypeRepository leaveTypeRepository;
36     private final LeaveEntitlementRepository leaveEntitlementRepository;
37     private final NotificationService notificationService;
38     private static final Logger log = LoggerFactory.getLogger(LeaveRequestService.class);
39
40     public LeaveRequestService(LeaveRequestRepo leaveRequestRepo,
41                               EmployeeRepo employeeRepo,
42                               LeaveTypeRepository leaveTypeRepository,
43                               LeaveEntitlementRepository leaveEntitlementRepository,
44                               NotificationService notificationService) {
45         this.leaveRequestRepo = leaveRequestRepo;
46         this.employeeRepo = employeeRepo;
47         this.leaveTypeRepository = leaveTypeRepository;
48         this.leaveEntitlementRepository = leaveEntitlementRepository;
49         this.notificationService=notificationService;
50     }
51
52     public List<LeaveResponseDTO> getAllTheLeaveRequest(){
53         List<LeaveRequest> requestList=leaveRequestRepo.findAll();
54         List<LeaveResponseDTO> dtos=new ArrayList<>();
55         for(LeaveRequest request:requestList){
56             dtos.add(convertToDTO(request));
57         }
58         return dtos;
59     }
60
61
62     public ResponseEntity<?> createRequest(Authentication authentication, LeaveRequestDTO requestDTO){
63         var leaveRequest=new LeaveRequest();
64         String email= AuthUtil.extractEmail(authentication);
65         //Validate employee exists and is active
66         var employee=employeeRepo.findByEmail(email).orElseThrow(()->
67             new EmployeeNotFound("Employee with name: "+email+" is not found")
68         );
69         if (employee.getStatus() != Status.ACTIVE) {
70             return ResponseEntity.badRequest()
71                 .body(Map.of("error", "Only active employees can apply for leave"));

```

```

72     }
73
74     if (requestDTO.getLeaveType() == LeaveType.MATERNITY) {
75         if (employee.getGender() == Gender.M) {
76             return ResponseEntity.status(HttpStatus.BAD_REQUEST)
77                 .body("Maternity leave is not applicable");
78         }
79         return ResponseEntity.status(HttpStatus.BAD_REQUEST)
80             .body("Please use the maternity leave application process.");
81     }
82
83     leaveRequest.setEmployee(employee);
84     //Checking if the start date is before today
85     String timezone = employee.getTimezone();
86     ZoneId zoneId = ZoneId.of(timezone != null && !timezone.isBlank()
87         ? timezone
88         : "UTC");
89     LocalDate today = LocalDate.now(zoneId);
90     if (requestDTO.getStartDate().isBefore(today)) {
91         throw new InvalidStartDateException("Start date cannot be before current date");
92     }
93     //Checking if the end date is before start date
94     if (requestDTO.getEndDate().isBefore(requestDTO.getStartDate())) {
95         throw new InvalidEndDateException("End date must be equal to or greater than start date");
96     }
97     //optional
98     long daysRequested = ChronoUnit.DAYS.between(requestDTO.getStartDate(), requestDTO.getEndDate()) +
99     1;
100    if (daysRequested > 30) {
101        return ResponseEntity.badRequest()
102            .body(Map.of("error", "Leave cannot exceed 30 consecutive days"));
103    }
104    // ----- SICK -----
105    if (requestDTO.getLeaveType() == LeaveType.SICK) {
106        if (daysRequested > 1) {
107            return ResponseEntity.badRequest()
108                .body(Map.of("error", "Sick leave is limited to 1 day per month."));
109        }
110
111        long sickDaysUsedThisMonth = leaveRequestRepo.countApprovedSickDaysInMonth(
112            employee, today.getYear(), today.getMonthValue());
113
114        var sickLeaveType = leaveTypeRepository.findByName("SICK")
115            .orElseThrow(() -> new RuntimeException("SICK leave type not configured"));
116
117        var entitlement = leaveEntitlementRepository
118            .findByEmployeeEmployeeIdAndLeaveTypeIdAndYear(
119                employee.getEmployeeId(),
120                sickLeaveType.getId(),
121                today.getYear())
122            .orElse(null);
123
124        boolean noBalance = (entitlement == null) ||
125            entitlement.getClosingBalance().compareTo(BigDecimal.ZERO) <= 0;
126
127        if (sickDaysUsedThisMonth >= 1 || noBalance) {
128            requestDTO.setLeaveType(LeaveType.UNPAID);
129        }
130    }
131
132    // ----- CASUAL -----
133    if (requestDTO.getLeaveType() == LeaveType.CASUAL) {
134        if (daysRequested > 1) {
135            return ResponseEntity.badRequest()
136                .body(Map.of("error", "Casual leave is limited to 1 day per month."));
137        }
138
139        long casualDaysUsedThisMonth = leaveRequestRepo.countCasualDaysInMonth(
140            employee, today.getYear(), today.getMonthValue());
141
142        var casualLeaveType = leaveTypeRepository.findByName("CASUAL")
143            .orElseThrow(() -> new RuntimeException("CASUAL leave type not configured"));
144    }
145
146

```



```

147         var entitlement = leaveEntitlementRepository
148             .findByEmployeeEmployeeIdAndLeaveTypeIdAndYear(
149                 employee.getEmployeeId(),
150                 casualLeaveType.getId(),
151                 today.getYear())
152             .orElse(null);
153
154         boolean noBalance = (entitlement == null) ||
155             entitlement.getClosingBalance().compareTo(BigDecimal.ZERO) <= 0;
156
157         if (casualDaysUsedThisMonth >= 1 || noBalance) {
158             requestDTO.setLeaveType(LeaveType.UNPAID);
159         }
160     }
161
162     // ----- UNPAID cap -----
163     // An employee may take at most 1 unpaid day per month – the same quota
164     // as the leave type it replaces. This applies whether the employee
165     // submitted UNPAID directly or was auto-downgraded from SICK/CASUAL.
166     if (requestDTO.getLeaveType() == LeaveType.UNPAID) {
167         long unpaidDaysThisMonth = leaveRequestRepo.countUnpaidDaysInMonth(
168             employee, today.getYear(), today.getMonthValue());
169         if (unpaidDaysThisMonth >= 1) {
170             return ResponseEntity.badRequest()
171                 .body(Map.of("error",
172                     "Unpaid leave limit (1 day/month) already reached. "
173                     + "No further leave can be taken this month."));
174         }
175     }
176
177     long duplicateCount = leaveRequestRepo.checkDuplicate(
178         employee.getEmployeeId(),
179         requestDTO.getStartDate(),
180         requestDTO.getEndDate(),
181         LeaveStatus.PENDING.name()
182     );
183     if (duplicateCount > 0) {
184         throw new DuplicateRequestException("Duplicate leave request");
185     }
186
187     //checking if there is any overlapping leave request of that employee
188     long count = leaveRequestRepo.countOverlappingLeave(
189         employee.getEmployeeId(),
190         requestDTO.getStartDate(),
191         requestDTO.getEndDate()
192     );
193     if (count > 0) {
194         throw new OverlappingLeaveException("Overlapping leave exists");
195     }
196     leaveRequest.setStartDate(requestDTO.getStartDate());
197     leaveRequest.setEndDate(requestDTO.getEndDate());
198
199     if (requestDTO.getLeaveType() == null) {
200         return ResponseEntity.status(HttpStatus.BAD_REQUEST)
201             .body("Leave type is required..");
202     }
203     LeaveType originalLeaveType = requestDTO.getLeaveType();
204     leaveRequest.setLeaveType(requestDTO.getLeaveType());
205     if (requestDTO.getReason() != null) {
206         leaveRequest.setReason(requestDTO.getReason());
207     }
208     LeaveRequest savedRequest = leaveRequestRepo.save(leaveRequest);
209
210     Employee manager = employee.getManager(); // assuming getManager() exists
211     if (manager != null) {
212         notificationService.notify(
213             manager,
214             employee.getName() + " submitted a " + savedRequest.getLeaveType() ,
215             "LEAVE_REQUEST"
216         );
217     }
218     // After the existing manager notification block
219     employeeRepo.findByRole(Role.ADMIN).forEach(admin ->
220         notificationService.notify(
221             admin,
222             employee.getName() + " submitted a " + requestDTO.getLeaveType() + " leave request

```

```

(" +
223         requestDTO.getStartDate() + " " + requestDTO.getEndDate() + ")",
224         "LEAVE_REQUEST"
225     )
226 );
227 LeaveResponseDTO responseDTO = convertToDTO(savedRequest);
228 if (savedRequest.getLeaveType() == LeaveType.UNPAID && savedRequest.getLeaveType() !=
originalLeaveType) {
229     Map<String, Object> body = new java.util.LinkedHashMap<>();
230     body.put("leave", responseDTO);
231     body.put("warning", "Your " + originalLeaveType.name().toLowerCase() +
232         " leave quota for this month is exhausted. Request recorded as Unpaid leave.");
233     return ResponseEntity.status(HttpStatus.CREATED).body(body);
234 }
235 return ResponseEntity.status(HttpStatus.CREATED).body(responseDTO);
236 }
237
238 //Only for manager
239 public List<LeaveResponseDTO> getAllThePendingLeaveRequests(){
240     List<LeaveRequest> requestList=leaveRequestRepo.findByStatus(LeaveStatus.PENDING);
241     List<LeaveResponseDTO> responseDTOS=new ArrayList<>();
242     for(LeaveRequest request:requestList){
243         responseDTOS.add(convertToDTO(request));
244     }
245     return responseDTOS;
246 }
247
248 @Transactional
249 public ResponseEntity<?> updateLeaveRequestStatus(ActionDTO actionDTO,Authentication authentication){
250     String email= AuthUtil.extractEmail(authentication);
251     Employee manager=employeeRepo.findByEmail(email).orElseThrow(
252         ()->new EmployeeNotFound("Manager with name: "+email+" Not Found")
253     );
254     if (actionDTO.getAction() == null) {
255         return ResponseEntity.badRequest().body("Action is required");
256     }
257     log.info("Updating leave request status for id: {}", actionDTO.getLeaveRequestId());
258
259     var leaveRequest=leaveRequestRepo.findById(actionDTO.getLeaveRequestId()).orElseThrow(
260         ()-> new LeaveRequestNotFoundException("LeaveRequest with
id:"+actionDTO.getLeaveRequestId()+" not found")
261     );
262
263     if (leaveRequest.getStatus() != LeaveStatus.PENDING) {
264         return ResponseEntity.badRequest()
265             .body("Only PENDING requests can be approved or rejected. Current status: "
266                 + leaveRequest.getStatus());
267     }
268
269     if(actionDTO.getAction().equalsIgnoreCase("approved")){
270         leaveRequest.setStatus(LeaveStatus.APPROVED);
271         applyApprovedLeaveToEntitlement(leaveRequest);
272         notificationService.notify(
273             leaveRequest.getEmployee(),
274             "Your leave request from " + leaveRequest.getStartDate() + " to " +
leaveRequest.getEndDate() + " has been approved.",
275             "LEAVE_APPROVED"
276         );
277         employeeRepo.findByRole(Role.ADMIN).forEach(admin ->
278             notificationService.notify(
279                 admin,
280                 leaveRequest.getEmployee().getName() + "'s leave request has been approved by
" + manager.getName(),
281                 "LEAVE_APPROVED"
282             )
283         );
284     }
285     else if(actionDTO.getAction().equalsIgnoreCase("rejected")){
286         leaveRequest.setStatus(LeaveStatus.REJECTED);
287         notificationService.notify(
288             leaveRequest.getEmployee(),
289             "Your leave request from " + leaveRequest.getStartDate() + " to " +
leaveRequest.getEndDate() + " was rejected." +
290             (actionDTO.getRemarks() != null ? " Reason: " + actionDTO.getRemarks() : ""),
291             "LEAVE_REJECTED"
292         );

```

```

293         employeeRepo.findByRole(Role.ADMIN).forEach(admin ->
294             notificationService.notify(
295                 admin,
296                 leaveRequest.getEmployee().getName() + "'s leave request has been rejected by
" + manager.getName(),
297                 "LEAVE_REJECTED"
298             )
299         );
300     }
301     else{
302         return ResponseEntity.badRequest().body("Invalid action,Use APPROVED or REJECTED");
303     }
304     leaveRequest.setManager(manager.getEmail());
305     if(actionDTO.getRemarks()!=null){
306         leaveRequest.setRemarks(actionDTO.getRemarks());
307     }
308     return ResponseEntity.ok(leaveRequestRepo.save(leaveRequest));
309 }
310
311 @Transactional
312 public ResponseEntity<?> cancelLeaveRequest(Authentication authentication,long leaveId){
313     String email= AuthUtil.extractEmail(authentication);
314     var employee=employeeRepo.findByEmail(email).orElseThrow(
315         ()->new EmployeeNotFound("Employee with email:"+email+" not found")
316     );
317     var leaveRequest=leaveRequestRepo.findById(leaveId).orElseThrow(
318         ()->new LeaveRequestNotFoundException("Leave request with id:"+leaveId+" is not found")
319     );
320     if(employee.getEmployeeId()!=leaveRequest.getEmployee().getEmployeeId()){
321         return ResponseEntity.badRequest().body("You cannot cancel others Leave request..");
322     }
323     String timezone= employee.getTimezone();
324     ZoneId zoneId=ZoneId.of(timezone != null && !timezone.isBlank()
325         ? timezone
326         : "UTC");
327     LocalDate today = LocalDate.now(zoneId);
328     if(leaveRequest.getStartDate().isEqual(today)||leaveRequest.getStartDate().isBefore(today)){
329         return ResponseEntity.badRequest().body("You cannot cancel leave request after the leave has
started");
330     }
331     LeaveStatus previousStatus = leaveRequest.getStatus();
332     leaveRequest.setStatus(LeaveStatus.CANCELLED);
333     if (previousStatus == LeaveStatus.APPROVED) {
334         reverseApprovedLeaveFromEntitlement(leaveRequest);
335     }
336     leaveRequestRepo.save(leaveRequest);
337     return ResponseEntity.ok("Leave Request with id:"+ leaveId+" Successfully cancelled");
338 }
339
340 private LeaveResponseDTO convertToDTO(LeaveRequest request){
341     var responseDTO=new LeaveResponseDTO();
342     responseDTO.setLeaveRequestId(request.getId());
343     responseDTO.setEmployeeName(request.getEmployee().getName());
344     responseDTO.setLeaveType(request.getLeaveType());
345     responseDTO.setStartDate(request.getStartDate());
346     responseDTO.setEndDate(request.getEndDate());
347     responseDTO.setNumberOfDays(requestedDays(request).longValue());
348     responseDTO.setReason(request.getReason());
349     responseDTO.setStatus(request.getStatus());
350     return responseDTO;
351 }
352
353 public ResponseEntity<?> getLeaveRequestsByEmployee(Authentication authentication) {
354     String email= AuthUtil.extractEmail(authentication);
355     log.info("Fetching leave requests for employee: {}", email);
356
357     Employee employee = employeeRepo.findByEmail(email)
358         .orElseThrow(() -> new EmployeeNotFound("Employee not found: " +email ));
359
360     List<LeaveRequest> requests = leaveRequestRepo.findByEmployeeOrderByStartDateDesc(employee);
361     List<LeaveResponseDTO> response = requests.stream()
362         .map(this::convertToDTO)
363         .collect(Collectors.toList());
364
365     return ResponseEntity.ok(Map.of(
366         "employeeName", email,

```

```

367         "totalRequests", response.size(),
368         "requests", response
369     ));
370 }
371 public List<LeaveResponseDTO> getPendingLeaves() {
372     List<LeaveRequest> pendingRequests = leaveRequestRepo.findByStatus(LeaveStatus.PENDING);
373     return pendingRequests.stream()
374         .map(this::convertToDTO)
375         .collect(Collectors.toList());
376 }
377
378 /**
379  * Approve a leave request by ID (simplified version for AdminController).
380  * Returns LeaveResponseDTO directly.
381  */
382 @Transactional
383 public LeaveResponseDTO approveLeave(Long id) {
384     LeaveRequest leaveRequest = leaveRequestRepo.findById(id)
385         .orElseThrow(() -> new LeaveRequestNotFoundException("Leave request with id: " + id + " not
found"));
386
387     if (leaveRequest.getStatus() != LeaveStatus.PENDING) {
388         throw new IllegalStateException("Only PENDING requests can be approved. Current status: " +
leaveRequest.getStatus());
389     }
390
391     leaveRequest.setStatus(LeaveStatus.APPROVED);
392     leaveRequest.setManager("ADMIN"); // Or get from security context if needed
393     applyApprovedLeaveToEntitlement(leaveRequest);
394     notificationService.notify(
395         leaveRequest.getEmployee(),
396         "Your leave request from " + leaveRequest.getStartDate() + " to " +
leaveRequest.getEndDate() + " has been approved by admin.",
397         "LEAVE_APPROVED"
398     );
399
400     LeaveRequest saved = leaveRequestRepo.save(leaveRequest);
401     return convertToDTO(saved);
402 }
403
404 /**
405  * Reject a leave request by ID (simplified version for AdminController).
406  * Returns LeaveResponseDTO directly.
407  */
408 @Transactional
409 public LeaveResponseDTO rejectLeave(Long id) {
410     LeaveRequest leaveRequest = leaveRequestRepo.findById(id)
411         .orElseThrow(() -> new LeaveRequestNotFoundException("Leave request with id: " + id + " not
found"));
412
413     if (leaveRequest.getStatus() != LeaveStatus.PENDING) {
414         throw new IllegalStateException("Only PENDING requests can be rejected. Current status: " +
leaveRequest.getStatus());
415     }
416
417     leaveRequest.setStatus(LeaveStatus.REJECTED);
418     leaveRequest.setManager("ADMIN");
419     notificationService.notify(
420         leaveRequest.getEmployee(),
421         "Your leave request from " + leaveRequest.getStartDate() + " to " +
leaveRequest.getEndDate() + " was rejected by admin.",
422         "LEAVE_REJECTED"
423     );
424
425     LeaveRequest saved = leaveRequestRepo.save(leaveRequest);
426     return convertToDTO(saved);
427 }
428
429 private void applyApprovedLeaveToEntitlement(LeaveRequest leaveRequest) {
430     if (leaveRequest.getLeaveType() == LeaveType.MATERNITY) {
431         grantAndDeductMaternity(leaveRequest); // special maternity flow
432     } else {
433         updateUsedLeave(leaveRequest, requestedDays(leaveRequest)); // existing flow
434     }
435 }
436

```

```

437     private void grantAndDeductMaternity(LeaveRequest leaveRequest) {
438         int year = leaveRequest.getStartDate().getYear();
439         Employee employee = leaveRequest.getEmployee();
440
441         com.example.EmployeeManagementSystem.Entity.LeaveType maternityType =
442             leaveTypeRepository.findByName("MATERNITY")
443                 .orElseThrow(() -> new IllegalStateException(
444                     "Maternity leave type not configured"));
445
446         var entitlement = leaveEntitlementRepository
447             .findByEmployeeEmployeeIdAndLeaveTypeIdAndYear(
448                 employee.getEmployeeId(), maternityType.getId(), year)
449             .orElseGet(() -> createEntitlement(employee, maternityType, year));
450
451         // Grant 180 days one time on approval
452         entitlement.setAccruedThisYear(BigDecimal.valueOf(180));
453
454         // Deduct the days she is taking (always 180)
455         entitlement.setUsedThisYear(requestedDays(leaveRequest));
456
457         leaveEntitlementRepository.save(entitlement);
458     }
459
460     private void reverseApprovedLeaveFromEntitlement(LeaveRequest leaveRequest) {
461         if (leaveRequest.getLeaveType() == LeaveType.MATERNITY) {
462             reverseMaternityEntitlement(leaveRequest); // wipe the grant entirely
463         } else {
464             updateUsedLeave(leaveRequest, requestedDays(leaveRequest).negate());
465         }
466     }
467
468     private void reverseMaternityEntitlement(LeaveRequest leaveRequest) {
469         int year = leaveRequest.getStartDate().getYear();
470         Employee employee = leaveRequest.getEmployee();
471
472         com.example.EmployeeManagementSystem.Entity.LeaveType maternityType =
473             leaveTypeRepository.findByName("MATERNITY")
474                 .orElseThrow(() -> new IllegalStateException(
475                     "Maternity leave type not configured"));
476
477         leaveEntitlementRepository
478             .findByEmployeeEmployeeIdAndLeaveTypeIdAndYear(
479                 employee.getEmployeeId(), maternityType.getId(), year)
480             .ifPresent(entitlement -> {
481                 // Wipe both the grant and the usage
482                 entitlement.setAccruedThisYear(BigDecimal.ZERO);
483                 entitlement.setUsedThisYear(BigDecimal.ZERO);
484                 leaveEntitlementRepository.save(entitlement);
485             });
486     }
487
488     private void updateUsedLeave(LeaveRequest leaveRequest, BigDecimal delta) {
489         int year = leaveRequest.getStartDate().getYear();
490         String leaveTypeName = leaveRequest.getLeaveType().name();
491         var leaveType = leaveTypeRepository
492             .findByName(leaveTypeName)
493             .orElseThrow(() -> new IllegalStateException(
494                 "Leave type not configured: " + leaveTypeName));
495
496         var entitlement = leaveEntitlementRepository
497             .findByEmployeeEmployeeIdAndLeaveTypeIdAndYear(
498                 leaveRequest.getEmployee().getEmployeeId(), leaveType.getId(), year)
499             .orElseGet(() -> createEntitlement(leaveRequest.getEmployee(), leaveType, year));
500
501         BigDecimal newUsed = entitlement.getUsedThisYear().add(delta);
502
503         // Prevent negative
504         newUsed = newUsed.max(BigDecimal.ZERO);
505
506         // Prevent exceeding available balance
507         BigDecimal closing = entitlement.getClosingBalance();
508         if (newUsed.compareTo(closing) > 0) {
509             newUsed = closing;
510         }
511
512         entitlement.setUsedThisYear(newUsed);

```

```
513         leaveEntitlementRepository.save(entitlement);
514     }
515
516     private com.example.EmployeeManagementSystem.Entity.LeaveEntitlement createEntitlement(
517         Employee employee,
518         com.example.EmployeeManagementSystem.Entity.LeaveType leaveType,
519         int year) {
520         var entitlement = new com.example.EmployeeManagementSystem.Entity.LeaveEntitlement();
521         entitlement.setEmployee(employee);
522         entitlement.setLeaveType(leaveType);
523         entitlement.setYear(year);
524
525         if (leaveType.isCarriesForward()) {
526             BigDecimal lastYearClosing = leaveEntitlementRepository
527                 .findByEmployeeEmployeeIdAndLeaveTypeIdAndYear(
528                     employee.getEmployeeId(), leaveType.getId(), year - 1)
529                 .map(com.example.EmployeeManagementSystem.Entity.LeaveEntitlement::getClosingBalance)
530                 .orElse(BigDecimal.ZERO);
531             entitlement.setOpeningBalance(lastYearClosing);
532         }
533
534         return leaveEntitlementRepository.save(entitlement);
535     }
536
537     private BigDecimal requestedDays(LeaveRequest leaveRequest) {
538         long days = ChronoUnit.DAYS.between(leaveRequest.getStartDate(), leaveRequest.getEndDate()) + 1;
539         return BigDecimal.valueOf(days);
540     }
541 }
542
```

```

1 package com.example.EmployeeManagementSystem.Service;
2
3 import com.example.EmployeeManagementSystem.DTO.*;
4 import com.example.EmployeeManagementSystem.Entity.Device;
5 import com.example.EmployeeManagementSystem.Entity.DeviceAssignment;
6 import com.example.EmployeeManagementSystem.Entity.Employee;
7 import com.example.EmployeeManagementSystem.Entity.LeaveRequest;
8 import com.example.EmployeeManagementSystem.Enum.AssignmentStatus;
9 import com.example.EmployeeManagementSystem.Enum.LeaveStatus;
10 import com.example.EmployeeManagementSystem.Enum.Role;
11 import com.example.EmployeeManagementSystem.Enum.Status;
12 import com.example.EmployeeManagementSystem.Exception.EmployeeNotFound;
13 import com.example.EmployeeManagementSystem.Repository.DeviceAssignmentRepo;
14 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
15 import com.example.EmployeeManagementSystem.Repository.LeaveRequestRepo;
16 import com.example.EmployeeManagementSystem.Util.AuthUtil;
17 import org.springframework.http.ResponseEntity;
18 import org.springframework.security.core.Authentication;
19 import org.springframework.security.crypto.password.PasswordEncoder;
20 import org.springframework.stereotype.Service;
21
22 import java.time.temporal.ChronoUnit;
23 import java.util.List;
24 import java.util.stream.Collectors;
25
26 @Service
27 public class ManagerService {
28     private final EmployeeRepo employeeRepo;
29     private final PasswordEncoder passwordEncoder;
30     private final LeaveRequestRepo leaveRequestRepo;
31     private final DeviceAssignmentRepo deviceAssignmentRepo;
32
33     public ManagerService(EmployeeRepo employeeRepo,
34                           PasswordEncoder passwordEncoder,
35                           LeaveRequestRepo leaveRequestRepo,
36                           DeviceAssignmentRepo deviceAssignmentRepo) {
37         this.employeeRepo = employeeRepo;
38         this.passwordEncoder = passwordEncoder;
39         this.leaveRequestRepo = leaveRequestRepo;
40         this.deviceAssignmentRepo = deviceAssignmentRepo;
41     }
42
43     public Employee createEmp(EmployeeDetails employeeDetails, Authentication authentication) {
44         String email= AuthUtil.extractEmail(authentication);
45         Employee manager= employeeRepo.findByEmail(email).orElseThrow(
46             ()->new EmployeeNotFound("Manager :"+email+"Not found")
47         );
48         Employee employee=new Employee();
49         employee.setName(employeeDetails.getName());
50         employee.setEmail(employeeDetails.getEmail());
51         employee.setPassword(passwordEncoder.encode(employeeDetails.getPassword()));
52         employee.setGender(employeeDetails.getGender());
53         employee.setDept(manager.getDept());
54         employee.setRole(Role.EMPLOYEE);
55         employee.setManager(manager);
56         employee.setTimezone(employeeDetails.getTimezone());
57         return employeeRepo.save(employee);
58     }
59
60
61     public List<Employee> getAllEmployeeByManager(Authentication authentication) {
62         String email=AuthUtil.extractEmail(authentication);
63         Employee manager=employeeRepo.findByEmail(email).orElseThrow(
64             ()->new EmployeeNotFound("Manger:"+email+" not found")
65         );
66         return employeeRepo.findByManager(manager);
67     }
68
69     public List<ManagerLeaveResponseDTO> getAllLeaveRequestByManager(Authentication authentication) {
70         String email=AuthUtil.extractEmail(authentication);
71         Employee manager=employeeRepo.findByEmail(email).orElseThrow(

```

```

72         ()->new EmployeeNotFound("Manager with "+email+" not found")
73     );
74     List<LeaveRequest> requests=leaveRequestRepo.findByEmployee_Manager(manager);
75     return requests.stream().map(request->toManagerLeaveResponseDTO(request)).toList();
76 }
77
78 public List<EmployeeDTO> getUsers(){
79     List<Employee> employeeList=employeeRepo.findByRole(Role.USER);
80     return employeeList.stream().map(this::toDTO).toList();
81 }
82
83 public ResponseEntity<?> promoteUser(Authentication authentication, PromoteRequest promoteRequest){
84     Employee user=employeeRepo.findByEmail(promoteRequest.getEmail()).orElseThrow(
85         ()->new EmployeeNotFound("Employee:"+promoteRequest.getEmail()+" not found")
86     );
87     if(user.getRole()!=Role.USER){
88         return ResponseEntity.ok("Employee already promoted");
89     }
90     String managerEmail=AuthUtil.extractEmail(authentication);
91     Employee manager=employeeRepo.findByEmail(managerEmail).orElseThrow(
92         ()->new EmployeeNotFound("Manager:"+managerEmail+" not found")
93     );
94
95     user.setRole(Role.EMPLOYEE);
96     user.setManager(manager);
97     user.setDept(manager.getDept());
98     user.setPassword(passwordEncoder.encode(promoteRequest.getPassword()));
99     Employee promotedEmployee=employeeRepo.save(user);
100    return ResponseEntity.ok(toDTO(promotedEmployee));
101 }
102
103 public EmployeeDTO toDTO(Employee employee){
104     EmployeeDTO employeeDTO=new EmployeeDTO();
105     employeeDTO.setName(employee.getName());
106     employeeDTO.setEmail(employee.getEmail());
107     employeeDTO.setDept(employee.getDept());
108     employeeDTO.setPassword(employee.getPassword());
109     employeeDTO.setTimezone(employee.getTimezone());
110     return employeeDTO;
111 }
112
113 public Employee createAdmin(EmployeeDetails employeeDetails) {
114     Employee employee=new Employee();
115     employee.setName(employeeDetails.getName());
116     employee.setEmail(employeeDetails.getEmail());
117     employee.setPassword(passwordEncoder.encode(employeeDetails.getPassword()));
118     employee.setDept("Main");
119     employee.setRole(Role.ADMIN);
120     employee.setTimezone(employeeDetails.getTimezone());
121     return employeeRepo.save(employee);
122 }
123
124 public List<LeaveResponseDTO> getAllThePendingLeaveRequests(Authentication authentication) {
125     String email=AuthUtil.extractEmail(authentication);
126     Employee manager=employeeRepo.findByEmail(email).orElseThrow(
127         ()->new EmployeeNotFound("Manager with "+email+" not found")
128     );
129     List<LeaveRequest>
leaveRequests=leaveRequestRepo.findByStatusAndEmployee_Manager(LeaveStatus.PENDING,manager);
130     return leaveRequests.stream()
131         .map(this::convertToLeaveResponseDTO)
132         .collect(Collectors.toList());
133 }
134
135 public ManagerEmployeeDetailsDTO getEmployeeDetailsByName( String name) {
136
137     Employee employee = employeeRepo.findByName(name).orElseThrow(
138         () -> new EmployeeNotFound("Employee:" + name + " not found")
139     );
140
141
142
143     ManagerEmployeeDetailsDTO dto = new ManagerEmployeeDetailsDTO();
144     dto.setEmployeeId(employee.getEmployeeId());
145     dto.setName(employee.getName());
146     dto.setEmail(employee.getEmail());

```



```

147         dto.setDept(employee.getDept());
148         dto.setStatus(employee.getStatus());
149         dto.setAttendance(toAttendance(employee.getStatus()));
150         dto.setDevices(deviceAssignmentRepo
151             .findByAssignedToEmployeeIdAndStatus(employee.getEmployeeId(), AssignmentStatus.ACTIVE)
152             .stream()
153             .map(DeviceAssignment::getDevice)
154             .map(this::toDeviceResponseDTO)
155             .toList());
156         return dto;
157     }
158
159     private LeaveResponseDTO convertToLeaveResponseDTO(LeaveRequest leaveRequest) {
160         LeaveResponseDTO dto = new LeaveResponseDTO();
161         dto.setLeaveRequestId(leaveRequest.getId());
162         dto.setEmployeeName(leaveRequest.getEmployee().getName());
163         dto.setLeaveType(leaveRequest.getLeaveType());
164         dto.setStartDate(leaveRequest.getStartDate());
165         dto.setEndDate(leaveRequest.getEndDate());
166         dto.setReason(leaveRequest.getReason());
167         dto.setStatus(leaveRequest.getStatus());
168         return dto;
169     }
170
171     private ManagerLeaveResponseDTO toManagerLeaveResponseDTO(LeaveRequest leaveRequest){
172         ManagerLeaveResponseDTO managerLeaveResponseDTO=new ManagerLeaveResponseDTO();
173         managerLeaveResponseDTO.setEmployeeName(leaveRequest.getEmployee().getName());
174         managerLeaveResponseDTO.setLeaveType(leaveRequest.getLeaveType());
175         managerLeaveResponseDTO.setStartDate(leaveRequest.getStartDate());
176         managerLeaveResponseDTO.setEndDate(leaveRequest.getEndDate());
177         managerLeaveResponseDTO.setNumberOfDays(
178             ChronoUnit.DAYS.between(leaveRequest.getStartDate(), leaveRequest.getEndDate()) + 1);
179         managerLeaveResponseDTO.setStatus(leaveRequest.getStatus());
180         managerLeaveResponseDTO.setReason(leaveRequest.getReason());
181         return managerLeaveResponseDTO;
182     }
183
184     private String toAttendance(Status status) {
185         if (status == Status.ON_LEAVE) {
186             return "ON_LEAVE";
187         }
188         if (status == Status.ACTIVE) {
189             return "PRESENT";
190         }
191         return status.name();
192     }
193
194     private DeviceResponseDTO toDeviceResponseDTO(Device device) {
195         DeviceResponseDTO dto = new DeviceResponseDTO();
196         dto.setId(device.getId());
197         dto.setDeviceName(device.getDeviceName());
198         dto.setBrand(device.getBrand());
199         dto.setDeviceType(device.getDeviceType());
200         dto.setDeviceStatus(device.getDeviceStatus());
201         dto.setWarrantyExpiryDate(device.getWarrantyExpiryDate());
202
203         if (device.getTechVendor() != null) {
204             dto.setVendorName(device.getTechVendor().getName());
205         }
206
207         DeviceAssignment assignment = device.getCurrentAssignment();
208         if (assignment != null && assignment.getStatus() == AssignmentStatus.ACTIVE) {
209             dto.setAssignedDate(assignment.getAssignedDate());
210             if (assignment.getAssignedTo() != null) {
211                 dto.setAssignedEmployeeId(assignment.getAssignedTo().getEmployeeId());
212                 dto.setAssignedEmployeeName(assignment.getAssignedTo().getName());
213             }
214         }
215
216         return dto;
217     }
218 }
219

```

```

1 package com.example.EmployeeManagementSystem.Service;
2
3 import com.example.EmployeeManagementSystem.DTO.LeaveResponseDTO;
4 import com.example.EmployeeManagementSystem.DTO.MaternityLeaveRequestDTO;
5 import com.example.EmployeeManagementSystem.Entity.Employee;
6 import com.example.EmployeeManagementSystem.Entity.LeaveRequest;
7 import com.example.EmployeeManagementSystem.Enum.*;
8 import com.example.EmployeeManagementSystem.Exception.EmployeeNotFound;
9 import com.example.EmployeeManagementSystem.Exception.InvalidStartDateException;
10 import com.example.EmployeeManagementSystem.Exception.OverlappingLeaveException;
11 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
12 import com.example.EmployeeManagementSystem.Repository.LeaveRequestRepo;
13 import com.example.EmployeeManagementSystem.Util.AuthUtil;
14 import org.springframework.http.HttpStatus;
15 import org.springframework.http.ResponseEntity;
16 import org.springframework.security.core.Authentication;
17 import org.springframework.stereotype.Service;
18
19 import java.math.BigDecimal;
20 import java.time.LocalDate;
21 import java.time.temporal.ChronoUnit;
22 import java.util.List;
23 import java.util.Map;
24
25 @Service
26 public class MaternityLeaveService {
27
28     private final EmployeeRepo employeeRepo;
29     private final LeaveRequestRepo leaveRequestRepo;
30     private final NotificationService notificationService;
31
32     public MaternityLeaveService(EmployeeRepo employeeRepo, LeaveRequestRepo leaveRequestRepo,
33 NotificationService notificationService) {
34         this.employeeRepo = employeeRepo;
35         this.leaveRequestRepo = leaveRequestRepo;
36         this.notificationService = notificationService;
37     }
38
39     public ResponseEntity<?> applyMaternityLeave(Authentication authentication,
40 MaternityLeaveRequestDTO maternityLeaveRequestDTO){
41         String email = AuthUtil.extractEmail(authentication);
42         Employee employee = employeeRepo.findByEmail(email)
43             .orElseThrow(() -> new EmployeeNotFound("Employee not found: " + email));
44
45         if (employee.getStatus() != Status.ACTIVE) {
46             return ResponseEntity.badRequest()
47                 .body(Map.of("error", "Only active employees can apply"));
48         }
49
50         if (employee.getGender() != Gender.F) {
51             return ResponseEntity.badRequest()
52                 .body(Map.of("error", "Only female employees can apply for maternity leave"));
53         }
54
55         boolean alreadyApplied = leaveRequestRepo
56             .existsMaternityLeaveForYear(
57                 employee,
58                 LeaveType.MATERNITY,
59                 maternityLeaveRequestDTO.getStartDate().getYear(),
60                 List.of(LeaveStatus.CANCELLED, LeaveStatus.REJECTED));
61         if (alreadyApplied) {
62             return ResponseEntity.badRequest()
63                 .body(Map.of("error", "Maternity leave already applied for this year"));
64         }
65
66         LocalDate today = LocalDate.now();
67         if (maternityLeaveRequestDTO.getStartDate().isBefore(today)) {
68             throw new InvalidStartDateException("Start date cannot be before today");
69         }
70     }

```

```

71
72     // 5. Overlapping check
73     LocalDate endDate = maternityLeaveRequestDTO.getStartDate().plusDays(179); // always 180 days
74     long overlapping = leaveRequestRepo.countOverlappingLeave(
75         employee.getId(),
76         maternityLeaveRequestDTO.getStartDate(),
77         endDate);
78     if (overlapping > 0) {
79         throw new OverlappingLeaveException("Overlapping leave exists");
80     }
81
82     LeaveRequest request = new LeaveRequest();
83     request.setEmployee(employee);
84     request.setLeaveType(LeaveType.MATERNITY);
85     request.setStartDate(maternityLeaveRequestDTO.getStartDate());
86     request.setEndDate(endDate); // forced 180 days
87     request.setStatus(LeaveStatus.PENDING);
88     request.setReason(maternityLeaveRequestDTO.getReason());
89
90     LeaveRequest saved = leaveRequestRepo.save(request);
91     Employee manager = employee.getManager();
92     if (manager != null) {
93         notificationService.notify(
94             manager,
95             employee.getName() + " has applied for maternity leave.",
96             "MATERNITY_LEAVE_REQUEST"
97         );
98     }
99     employeeRepo.findByRole(Role.ADMIN).forEach(admin ->
100         notificationService.notify(
101             admin,
102             employee.getName() + " applied for maternity leave (" +
103                 maternityLeaveRequestDTO.getStartDate() + " " + endDate + ")",
104             "MATERNITY_LEAVE_REQUEST"
105         )
106     );
107
108     return ResponseEntity.status(HttpStatus.CREATED).body(convertToDTO(saved));
109 }
110
111 private LeaveResponseDTO convertToDTO(LeaveRequest request){
112     var responseDTO=new LeaveResponseDTO();
113     responseDTO.setLeaveRequestId(request.getId());
114     responseDTO.setEmployeeName(request.getEmployee().getName());
115     responseDTO.setLeaveType(request.getLeaveType());
116     responseDTO.setStartDate(request.getStartDate());
117     responseDTO.setEndDate(request.getEndDate());
118     responseDTO.setNumberOfDays(requestedDays(request).longValue());
119     responseDTO.setReason(request.getReason());
120     responseDTO.setStatus(request.getStatus());
121     return responseDTO;
122 }
123
124 private BigDecimal requestedDays(LeaveRequest leaveRequest) {
125     long days = ChronoUnit.DAYS.between(leaveRequest.getStartDate(), leaveRequest.getEndDate()) + 1;
126     return BigDecimal.valueOf(days);
127 }
128 }
129

```

```
1 package com.example.EmployeeManagementSystem.Service;
2
3 import com.example.EmployeeManagementSystem.Enum.MealSlot;
4 import org.springframework.stereotype.Service;
5
6 import java.time.LocalTime;
7 import java.util.Map;
8
9 @Service
10 public class MealSlotService {
11
12     private final Map<MealSlot, LocalTime> slotLocalTimeMap = Map.of(
13         MealSlot.BREAKFAST, LocalTime.of(8, 0),
14         MealSlot.LUNCH, LocalTime.of(13, 0),
15         MealSlot.DINNER, LocalTime.of(21, 0)
16     );
17
18     public LocalTime getTime(MealSlot slot) {
19         return slotLocalTimeMap.get(slot);
20     }
21 }
22
```

```
1 package com.example.EmployeeManagementSystem.Service;
2
3 import com.example.EmployeeManagementSystem.Entity.Employee;
4 import com.example.EmployeeManagementSystem.Entity.Notification;
5 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
6 import com.example.EmployeeManagementSystem.Repository.NotificationRepository;
7 import org.springframework.stereotype.Service;
8 import java.util.List;
9
10 @Service
11 public class NotificationService {
12
13     private final NotificationRepository notificationRepo;
14     private final EmployeeRepo employeeRepo;
15
16     public NotificationService(NotificationRepository notificationRepo,
17                               EmployeeRepo employeeRepo) {
18         this.notificationRepo = notificationRepo;
19         this.employeeRepo = employeeRepo;
20     }
21
22     /** Create a notification for a specific employee */
23     public void notify(Employee recipient, String message, String type) {
24         Notification n = new Notification();
25         n.setRecipient(recipient);
26         n.setMessage(message);
27         n.setType(type);
28         notificationRepo.save(n);
29     }
30
31     /** Notify every employee in the system (e.g. new public holiday) */
32     public void notifyAll(String message, String type) {
33         List<Employee> all = employeeRepo.findAll();
34         for (Employee e : all) {
35             notify(e, message, type);
36         }
37     }
38
39     public List<Notification> getForEmployee(Employee employee) {
40         return notificationRepo.findByRecipientOrderByCreatedAtDesc(employee);
41     }
42
43     public long getUnreadCount(Employee employee) {
44         return notificationRepo.countByRecipientAndIsReadFalse(employee);
45     }
46
47     public void markAllRead(Employee employee) {
48         List<Notification> unread = notificationRepo
49             .findByRecipientOrderByCreatedAtDesc(employee)
50             .stream()
51             .filter(n -> !n.isRead())
52             .toList();
53         unread.forEach(n -> n.setRead(true));
54         notificationRepo.saveAll(unread);
55     }
56 }
57
```

```
1 package com.example.EmployeeManagementSystem.Service;
2
3 import com.example.EmployeeManagementSystem.Entity.Employee;
4 import com.example.EmployeeManagementSystem.Entity.RefreshToken;
5 import com.example.EmployeeManagementSystem.Entity.Vendor;
6 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
7 import com.example.EmployeeManagementSystem.Repository.RefreshTokenRepository;
8 import com.example.EmployeeManagementSystem.Repository.VendorRepo;
9 import org.springframework.stereotype.Service;
10
11 import java.time.Instant;
12 import java.time.temporal.ChronoUnit;
13 import java.util.UUID;
14
15 @Service
16 public class RefreshTokenService {
17     private final EmployeeRepo employeeRepo;
18     private final VendorRepo vendorRepo;
19
20     public RefreshTokenService(RefreshTokenRepository refreshTokenRepository, EmployeeRepo employeeRepo,
21     VendorRepo vendorRepo) {
22         this.employeeRepo = employeeRepo;
23         this.vendorRepo = vendorRepo;
24     }
25
26     public RefreshToken createForEmployee(Employee employee) {
27         RefreshToken refreshToken = new RefreshToken();
28         refreshToken.setToken(UUID.randomUUID().toString());
29         refreshToken.setExpiryTime(Instant.now().plus(7, ChronoUnit.DAYS));
30         refreshToken.setEmployee(employee);
31         return refreshToken;
32     }
33
34     public RefreshToken createForVendor(Vendor vendor) {
35         RefreshToken refreshToken = new RefreshToken();
36         refreshToken.setToken(UUID.randomUUID().toString());
37         refreshToken.setExpiryTime(Instant.now().plus(7, ChronoUnit.DAYS));
38         refreshToken.setVendor(vendor);
39         return refreshToken;
40     }
41 }
```

```

1 package com.example.EmployeeManagementSystem.Service;
2
3 import com.example.EmployeeManagementSystem.DTO.RestaurantDTO;
4 import com.example.EmployeeManagementSystem.DTO.RestaurantRequest;
5 import com.example.EmployeeManagementSystem.Entity.Restaurant;
6 import com.example.EmployeeManagementSystem.Entity.Vendor;
7 import com.example.EmployeeManagementSystem.Enum.MealsSlot;
8 import com.example.EmployeeManagementSystem.Exception.RestaurantNotFoundException;
9 import com.example.EmployeeManagementSystem.Exception.UnauthorizedAccessException;
10 import com.example.EmployeeManagementSystem.Exception.VendorNotFoundException;
11 import com.example.EmployeeManagementSystem.Repository.RestaurantRepository;
12 import com.example.EmployeeManagementSystem.Repository.VendorRepo;
13 import com.example.EmployeeManagementSystem.Util.AuthUtil;
14 import org.slf4j.Logger;
15 import org.slf4j.LoggerFactory;
16 import org.springframework.security.core.Authentication;
17 import org.springframework.stereotype.Service;
18
19 import java.util.List;
20 import java.util.stream.Collectors;
21
22 @Service
23 public class RestaurantService {
24
25     private static final Logger log = LoggerFactory.getLogger(RestaurantService.class);
26
27     private final RestaurantRepository restaurantRepository;
28     private final VendorRepo vendorRepo;
29
30     public RestaurantService(RestaurantRepository restaurantRepository, VendorRepo vendorRepo) {
31         this.restaurantRepository = restaurantRepository;
32         this.vendorRepo = vendorRepo;
33     }
34
35     /**
36      * Create a new restaurant.
37      * Only the owning vendor (matched by vendorId in the request) may do this.
38      */
39     public RestaurantDTO createRestaurant(RestaurantRequest request, Authentication authentication) {
40         String email = AuthUtil.extractEmail(authentication);
41         Vendor vendor = vendorRepo.findByEmail(email)
42             .orElseThrow(() -> new VendorNotFoundException(
43                 "Vendor with email " + email + " not found"));
44
45         Restaurant restaurant = new Restaurant();
46         restaurant.setName(request.getName());
47         restaurant.setAddress(request.getAddress());
48         restaurant.setDescription(request.getDescription());
49         restaurant.setSupportedMealsSlots(request.getSupportedMealsSlots());
50         restaurant.setVendor(vendor);
51         restaurant.setActive(true);
52
53         Restaurant saved = restaurantRepository.save(restaurant);
54         log.info("Restaurant created: {} by vendor {}", saved.getName(), vendor.getId());
55         return toDTO(saved);
56     }
57
58     /**
59      * Update restaurant details.
60      * Vendor must own this restaurant – otherwise 403.
61      */
62     public RestaurantDTO updateRestaurant(Long restaurantId, RestaurantRequest request, Authentication authentication) {
63         Restaurant restaurant = findOrThrow(restaurantId);
64         String email = AuthUtil.extractEmail(authentication);
65         Vendor vendor = vendorRepo.findByEmail(email).orElseThrow(
66             () -> new VendorNotFoundException("Vendor Not found")
67         );
68         assertOwnership(restaurantId, vendor.getEmail());
69
70         if (request.getName() != null) restaurant.setName(request.getName());

```

```

71         if (request.getAddress() != null) restaurant.setAddress(request.getAddress());
72         if (request.getDescription() != null) restaurant.setDescription(request.getDescription());
73         if (request.getSupportedMealSlots() != null && !request.getSupportedMealSlots().isEmpty()) {
74             restaurant.setSupportedMealSlots(request.getSupportedMealSlots());
75         }
76         return toDTO(restaurantRepository.save(restaurant));
77     }
78
79     /**
80      * Deactivate a restaurant (soft delete).
81      * Only the owning vendor may do this.
82      * Active subscriptions to this restaurant will no longer receive new deliveries
83      * – the Subscription status should be handled separately.
84      */
85     public void deactivateRestaurant(Long restaurantId, Authentication authentication) {
86         Restaurant restaurant = findOrThrow(restaurantId);
87         String email= AuthUtil.extractEmail(authentication);
88         Vendor vendor=vendorRepo.findByEmail(email).orElseThrow(
89             ()->new VendorNotFoundException("Vendor Not found")
90         );
91         assertOwnership(restaurantId, vendor.getEmail());
92         restaurant.setActive(false);
93         restaurantRepository.save(restaurant);
94         log.info("Restaurant deactivated: id={}", restaurantId);
95     }
96
97     /**
98      * Get all restaurants owned by a vendor.
99      * VENDOR-only endpoint.
100     */
101     public List<RestaurantDTO> getRestaurantsByVendor(Authentication authentication) {
102         String email= AuthUtil.extractEmail(authentication);
103         vendorRepo.findByEmail(email)
104             .orElseThrow(() -> new VendorNotFoundException("Vendor " + email + " not found"));
105         return restaurantRepository.findByVendor_EmailAndActiveTrue(email)
106             .stream().map(this::toDTO).collect(Collectors.toList());
107     }
108
109     /**
110      * Get all active restaurants.
111      * Available to EMPLOYEE so they can browse and choose.
112      */
113     public List<RestaurantDTO> getAllActiveRestaurants() {
114         return restaurantRepository.findByActiveTrue()
115             .stream().map(this::toDTO).collect(Collectors.toList());
116     }
117
118     /**
119      * Get active restaurants that serve a specific meal slot.
120      * Employees call this when subscribing, to see valid restaurant options for a slot.
121      */
122     public List<RestaurantDTO> getRestaurantsByMealSlot(MealSlot mealSlot) {
123         return restaurantRepository.findActiveByMealSlot(mealSlot)
124             .stream().map(this::toDTO).collect(Collectors.toList());
125     }
126
127     public RestaurantDTO getRestaurant(Long id) {
128         return toDTO(findOrThrow(id));
129     }
130
131     // helpers
132
133     private Restaurant findOrThrow(Long id) {
134         return restaurantRepository.findById(id)
135             .orElseThrow(() -> new RestaurantNotFoundException(
136                 "Restaurant with id " + id + " not found"));
137     }
138
139     /**
140      * Throws UnauthorizedAccessException if the restaurant does not belong to the vendor.
141      * This is the core access-control guard for all mutating vendor operations.
142      */
143     private void assertOwnership(Long restaurantId, String email) {
144         if (!restaurantRepository.existsByIdAndVendor_Email(restaurantId, email)) {
145             throw new UnauthorizedAccessException(
146                 "Vendor " + email + " does not own restaurant " + restaurantId);

```



```
147     }
148 }
149
150 public RestaurantDTO toDTO(Restaurant r) {
151     RestaurantDTO dto = new RestaurantDTO();
152     dto.setId(r.getId());
153     dto.setName(r.getName());
154     dto.setAddress(r.getAddress());
155     dto.setDescription(r.getDescription());
156     dto.setSupportedMealSlots(r.getSupportedMealSlots());
157     dto.setActive(r.isActive());
158     dto.setVendorId(r.getVendor().getId());
159     dto.setVendorName(r.getVendor().getName());
160     return dto;
161 }
162 }
```

```

1 package com.example.EmployeeManagementSystem.Service;
2
3 import com.example.EmployeeManagementSystem.DTO.AdminActionDTO;
4 import com.example.EmployeeManagementSystem.DTO.RepairDTO;
5 import com.example.EmployeeManagementSystem.DTO.ServiceRequestDTO;
6 import com.example.EmployeeManagementSystem.DTO.ServiceRequestResponseDTO;
7 import com.example.EmployeeManagementSystem.Entity.Device;
8 import com.example.EmployeeManagementSystem.Entity.Employee;
9 import com.example.EmployeeManagementSystem.Entity.ServiceRequest;
10 import com.example.EmployeeManagementSystem.Entity.Vendor;
11 import com.example.EmployeeManagementSystem.Enum.DeviceStatus;
12 import com.example.EmployeeManagementSystem.Enum.ServiceRequestStatus;
13 import com.example.EmployeeManagementSystem.Exception.EmployeeNotFound;
14 import com.example.EmployeeManagementSystem.Exception.VendorNotFoundException;
15 import com.example.EmployeeManagementSystem.Repository.DeviceRepository;
16 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
17 import com.example.EmployeeManagementSystem.Repository.ServiceRequestRepository;
18 import com.example.EmployeeManagementSystem.Repository.VendorRepo;
19 import com.example.EmployeeManagementSystem.Util.AuthUtil;
20 import org.springframework.security.core.Authentication;
21 import org.springframework.stereotype.Service;
22 import org.springframework.transaction.annotation.Transactional;
23 import org.w3c.dom.stylesheets.LinkStyle;
24
25 import java.time.LocalDateTime;
26 import java.util.List;
27 import java.util.Vector;
28
29 @Service
30 public class ServiceRequestService {
31     private final ServiceRequestRepository serviceRequestRepository;
32     private final EmployeeRepo employeeRepo;
33     private final DeviceRepository deviceRepository;
34     private final VendorRepo vendorRepo;
35
36     public ServiceRequestService(ServiceRequestRepository serviceRequestRepository, EmployeeRepo employeeRepo, DeviceRepository deviceRepository, VendorRepo vendorRepo) {
37         this.serviceRequestRepository = serviceRequestRepository;
38         this.employeeRepo = employeeRepo;
39         this.deviceRepository = deviceRepository;
40         this.vendorRepo = vendorRepo;
41     }
42
43     public ServiceRequest createServiceRequest(ServiceRequestDTO dto, Authentication authentication) {
44         String email= AuthUtil.extractEmail(authentication);
45         Employee employee = employeeRepo.findByEmail(email)
46             .orElseThrow(() -> new RuntimeException("Employee not found"));
47
48         // 2. Fetch device
49         Device device = deviceRepository.findById(dto.getDeviceId())
50             .orElseThrow(() -> new RuntimeException("Device not found"));
51
52         // 3. Validate device is assigned to employee
53         if (device.getCurrentAssignment() == null ||
54             !device.getCurrentAssignment().getAssignedTo().equals(employee)) {
55             throw new RuntimeException("Device not assigned to this employee");
56         }
57
58         // 4. Validate device status
59         if (device.getDeviceStatus() != DeviceStatus.ASSIGNED) {
60             throw new RuntimeException("Device is not in usable state");
61         }
62
63         // 5. Create Service Request
64         ServiceRequest request = new ServiceRequest();
65         request.setDevice(device);
66         request.setRaisedBy(employee);
67         request.setRequestType(dto.getRequestType());
68         request.setIssueDescription(dto.getIssueDescription());
69         request.setUrgent(dto.isUrgent());
70         request.setStatus(ServiceRequestStatus.OPEN);

```

```

71         return serviceRequestRepository.save(request);
72     }
73
74
75     public List<ServiceRequestResponseDTO> getAllOpenServiceRequests(){
76         List<ServiceRequest>
serviceRequestList=serviceRequestRepository.findByStatus(ServiceRequestStatus.OPEN);
77         return serviceRequestList.stream().map(serviceRequest ->
toServiceRequestResponseDTO(serviceRequest)).toList();
78
79     }
80
81     public List<ServiceRequestResponseDTO> getMyServiceRequests(Authentication authentication) {
82         String email = AuthUtil.extractEmail(authentication);
83         Employee employee = employeeRepo.findByEmail(email)
84             .orElseThrow(() -> new EmployeeNotFound("Employee not found: " + email));
85
86         List<ServiceRequest> serviceRequestList =
serviceRequestRepository.findByRaisedByOrderByRaisedAtDesc(employee);
87         return serviceRequestList.stream()
88             .map(this::toServiceRequestResponseDTO)
89             .toList();
90     }
91
92     private ServiceRequestDTO toServiceRequestDTO(ServiceRequest serviceRequest){
93         ServiceRequestDTO requestDTO=new ServiceRequestDTO();
94         requestDTO.setDeviceId(serviceRequest.getDevice().getId());
95         requestDTO.setRequestType(serviceRequest.getRequestType());
96         requestDTO.setIssueDescription(serviceRequest.getIssueDescription());
97         requestDTO.setUrgent(serviceRequest.isUrgent());
98         return requestDTO;
99     }
100
101     private ServiceRequestResponseDTO toServiceRequestResponseDTO(ServiceRequest serviceRequest) {
102         ServiceRequestResponseDTO responseDTO = new ServiceRequestResponseDTO();
103         responseDTO.setId(serviceRequest.getId());
104         responseDTO.setRequestType(serviceRequest.getRequestType() != null
105             ? serviceRequest.getRequestType().name()
106             : null);
107         responseDTO.setStatus(serviceRequest.getStatus() != null
108             ? serviceRequest.getStatus().name()
109             : null);
110         responseDTO.setIssueDescription(serviceRequest.getIssueDescription());
111         responseDTO.setAdminRemarks(serviceRequest.getAdminRemarks());
112         responseDTO.setResolution(serviceRequest.getResolution());
113         responseDTO.setUrgent(serviceRequest.isUrgent());
114         responseDTO.setRaisedAt(serviceRequest.getRaisedAt());
115         responseDTO.setResolvedAt(serviceRequest.getResolvedAt());
116         responseDTO.setDeviceName(serviceRequest.getDevice() != null
117             ? serviceRequest.getDevice().getDeviceName()
118             : null);
119         responseDTO.setReviewedByName(serviceRequest.getReviewedBy() != null
120             ? serviceRequest.getReviewedBy().getName()
121             : null);
122         responseDTO.setRaisedByName(serviceRequest.getRaisedBy().getName());
123         return responseDTO;
124     }
125
126
127     @Transactional
128     public ServiceRequestResponseDTO updateServiceRequestByAdmin(Authentication authentication,
AdminActionDTO actionDTO){
129         ServiceRequest
serviceRequest=serviceRequestRepository.findById(actionDTO.getServiceRequestId()).orElseThrow(
130             ()->new RuntimeException("Service Request Not Found")
131         );
132         String email=AuthUtil.extractEmail(authentication);
133         Employee admin=employeeRepo.findByEmail(email).orElseThrow(
134             ()->new EmployeeNotFound("Admin not found:"+email)
135         );
136         Device device=serviceRequest.getDevice();
137         if (actionDTO.getStatus() == ServiceRequestStatus.SENT_FOR_REPAIR) {
138             device.setDeviceStatus(DeviceStatus.UNDER_REPAIR);
139             deviceRepository.save(device);
140         } else if (actionDTO.getStatus() == ServiceRequestStatus.REJECTED) {
141             device.setDeviceStatus(DeviceStatus.ASSIGNED);

```

```

142     }
143     serviceRequest.setReviewedBy(admin);
144     serviceRequest.setAdminRemarks(actionDTO.getAdminRemarks());
145     serviceRequest.setStatus(actionDTO.getStatus());
146     ServiceRequest saved=serviceRequestRepository.save(serviceRequest);
147     return toServiceRequestResponseDTO(saved);
148 }
149
150 public List<ServiceRequestResponseDTO> getAllServiceRequestByVendor(Authentication authentication){
151     String email=AuthUtil.extractEmail(authentication);
152     Vendor vendor=vendorRepo.findByEmail(email).orElseThrow(
153         ()->new VendorNotFoundException("Vendor not found: "+email)
154     );
155
156     List<ServiceRequest>
serviceRequestList=serviceRequestRepository.findByDeviceTechVendorAndStatus(vendor, ServiceRequestStatus.SENT_FOR
_REPAIR);
157     return serviceRequestList.stream()
158         .map(serviceRequest -> toServiceRequestResponseDTO(serviceRequest))
159         .toList();
160 }
161
162 public ServiceRequestResponseDTO updateServiceRequestByVendor(
163     Authentication authentication, RepairDTO repairDTO
164 ) {
165
166     // 1. Get vendor
167     String email = AuthUtil.extractEmail(authentication);
168     Vendor vendor = vendorRepo.findByEmail(email)
169         .orElseThrow(() -> new RuntimeException("Vendor not found"));
170
171     // 2. Get request
172     ServiceRequest request = serviceRequestRepository.findById(repairDTO.getRequestId())
173         .orElseThrow(() -> new RuntimeException("Request not found"));
174
175     // 3. Validate request belongs to vendor
176     if (!request.getDevice().getTechVendor().getId().equals(vendor.getId())) {
177         throw new RuntimeException("Unauthorized access");
178     }
179
180     // 4. Validate status
181     if (request.getStatus() != ServiceRequestStatus.SENT_FOR_REPAIR) {
182         throw new RuntimeException("Request is not in repair stage");
183     }
184
185     Device device = request.getDevice();
186
187     // 5. Update based on repair result
188     if (repairDTO.getStatus()==ServiceRequestStatus.REPAIR_DONE) {
189         request.setResolution(repairDTO.getResolution());
190         request.setStatus(repairDTO.getStatus());
191
192         device.setDeviceStatus(DeviceStatus.ASSIGNED); // back to employee
193     } else {
194         request.setResolution(repairDTO.getResolution());
195         request.setStatus(ServiceRequestStatus.REPAIR_DONE);
196
197         device.setDeviceStatus(DeviceStatus.CONDEMNED); // not usable
198     }
199
200     request.setResolvedAt(LocalDateTime.now());
201
202     deviceRepository.save(device);
203     ServiceRequest serviceRequest=serviceRequestRepository.save(request);
204     return toServiceRequestResponseDTO(serviceRequest);
205 }
206 }
207
208
209

```

```

1 package com.example.EmployeeManagementSystem.Service;
2
3 import com.example.EmployeeManagementSystem.Entity.LeaveEntitlement;
4 import com.example.EmployeeManagementSystem.Entity.LeaveType;
5 import com.example.EmployeeManagementSystem.Repository.LeaveEntitlementRepository;
6 import com.example.EmployeeManagementSystem.Repository.LeaveTypeRepository;
7 import org.slf4j.Logger;
8 import org.slf4j.LoggerFactory;
9 import org.springframework.stereotype.Service;
10 import org.springframework.transaction.annotation.Transactional;
11
12 import java.math.BigDecimal;
13 import java.time.LocalDate;
14 import java.util.List;
15
16 @Service
17 public class SickLeaveResetService {
18
19     private static final Logger log = LoggerFactory.getLogger(SickLeaveResetService.class);
20
21     private final LeaveTypeRepository leaveTypeRepository;
22     private final LeaveEntitlementRepository entitlementRepository;
23
24     public SickLeaveResetService(LeaveTypeRepository leaveTypeRepository,
25                               LeaveEntitlementRepository entitlementRepository) {
26         this.leaveTypeRepository = leaveTypeRepository;
27         this.entitlementRepository = entitlementRepository;
28     }
29
30     @Transactional
31     public void runMonthlyReset(LocalDate resetDate) {
32         // resetDate = last day of the month being reset (e.g. 2025-06-30)
33         int year = resetDate.getYear();
34         int month = resetDate.getMonthValue(); // FIX: scope reset to this month only
35
36         // Find all leave types that reset monthly (only SICK in your design)
37         List<LeaveType> resetTypes = leaveTypeRepository.findByMonthlyResetTrue();
38
39         for (LeaveType leaveType : resetTypes) {
40             // FIX: was findByLeaveTypeIdAndYear – fetched ALL entitlements for the year
41             // and re-reset them every month. Now scoped to current month's accruals.
42             List<LeaveEntitlement> entitlements =
43                 entitlementRepository.findByLeaveTypeIdAndYear(leaveType.getId(), year);
44
45             for (LeaveEntitlement entitlement : entitlements) {
46
47                 BigDecimal available = entitlement.getAvailableBalance();
48
49                 if (available.compareTo(BigDecimal.ZERO) > 0) {
50                     // Wipe unused days – set accrued = used so available = 0
51                     // Preserves the audit trail of what was used
52                     entitlement.setAccruedThisYear(entitlement.getUsedThisYear());
53
54                     log.info("Sick leave reset for employee {} – {} unused day(s) expired at end of {}-{}",
55                             entitlement.getEmployee().getEmployeeId(), available, year, month);
56                 } else {
57                     log.debug("Sick leave reset for employee {} – nothing to expire (month {}-{})",
58                             entitlement.getEmployee().getEmployeeId(), year, month);
59                 }
60
61                 entitlementRepository.save(entitlement);
62             }
63         }
64
65         log.info("Sick leave monthly reset complete for {}", resetDate);
66     }
67 }

```

```
1 package com.example.EmployeeManagementSystem.Service;
2
3 import com.example.EmployeeManagementSystem.DTO.SubscriptionDTO;
4 import com.example.EmployeeManagementSystem.DTO.SubscriptionRequest;
5 import com.example.EmployeeManagementSystem.Entity.Employee;
6 import com.example.EmployeeManagementSystem.Entity.Restaurant;
7 import com.example.EmployeeManagementSystem.Entity.Subscription;
8 import com.example.EmployeeManagementSystem.Enum.MealSlot;
9 import com.example.EmployeeManagementSystem.Enum.ScheduleType;
10 import com.example.EmployeeManagementSystem.Enum.SubscriptionStatus;
11 import com.example.EmployeeManagementSystem.Exception.EmployeeNotFound;
12 import com.example.EmployeeManagementSystem.Exception.RestaurantNotFoundException;
13 import com.example.EmployeeManagementSystem.Exception.SubscriptionAlreadyExists;
14 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
15 import com.example.EmployeeManagementSystem.Repository.RestaurantRepository;
16 import com.example.EmployeeManagementSystem.Repository.SubscriptionRepository;
17 import com.example.EmployeeManagementSystem.Util.AuthUtil;
18 import org.springframework.security.core.Authentication;
19 import org.springframework.stereotype.Service;
20
21 import java.time.ZoneId;
22 import java.time.ZonedDateTime;
23 import java.util.ArrayList;
24 import java.util.List;
25
26 @Service
27 public class SubscriptionService {
28
29     private final SubscriptionRepository subscriptionRepository;
30     private final EmployeeRepo employeeRepo;
31     private final DeliveryTimeService deliveryTimeService;
32     private final RestaurantRepository restaurantRepository;
33
34     public SubscriptionService(SubscriptionRepository subscriptionRepository,
35                               EmployeeRepo employeeRepo,
36                               DeliveryTimeService deliveryTimeService,
37                               RestaurantRepository restaurantRepository) {
38         this.subscriptionRepository = subscriptionRepository;
39         this.employeeRepo = employeeRepo;
40         this.deliveryTimeService = deliveryTimeService;
41         this.restaurantRepository = restaurantRepository;
42     }
43
44     /**
45      * Add subscriptions for an employee.
46      *
47      * Each SlotOrder specifies a meal slot AND the restaurant for that slot,
48      * so an employee can have breakfast from Restaurant A and lunch from Restaurant B.
49      *
50      * Validations:
51      * - Employee must exist and be ACTIVE
52      * - Restaurant must exist and be active
53      * - Restaurant must support the requested meal slot
54      * - No duplicate active subscription for the same employee + slot combination
55      */
56     public void addSubscription(SubscriptionsRequest request, Authentication authentication) {
57         String email = AuthUtil.extractEmail(authentication);
58         Employee employee = employeeRepo.findByEmail(email)
59             .orElseThrow(() -> new EmployeeNotFound(
60                 "Employee with id " + email + " not found"));
61
62         if (request.getSlotOrders() == null || request.getSlotOrders().isEmpty()) {
63             throw new IllegalArgumentException("slotOrders cannot be null or empty");
64         }
65
66         if (request.getScheduleType() == ScheduleType.DAILY) {
67             request.setDayOfWeek(null);
68         } else if (request.getScheduleType() == ScheduleType.WEEKLY) {
69             if (request.getDayOfWeek() == null) {
70                 throw new IllegalArgumentException("dayOfWeek is required for WEEKLY schedule");
71             }
72         }
73     }
74 }
```

```

72     }
73
74     for (SubscriptionRequest.SlotOrder slotOrder : request.getSlotOrders()) {
75         MealSlot slot = slotOrder.getMealSlot();
76         Long restaurantId = slotOrder.getRestaurantId();
77
78         // 1. Validate restaurant exists and is active
79         Restaurant restaurant = restaurantRepository.findById(restaurantId)
80             .orElseThrow(() -> new RestaurantNotFoundException(
81                 "Restaurant with id " + restaurantId + " not found"));
82         if (!restaurant.isActive()) {
83             throw new IllegalArgumentException(
84                 "Restaurant '" + restaurant.getName() + "' is not currently active");
85         }
86
87         // 2. Validate the restaurant actually serves this meal slot
88         if (!restaurant.getSupportedMealSlots().contains(slot)) {
89             throw new IllegalArgumentException(
90                 "Restaurant '" + restaurant.getName() + "' does not serve " + slot +
91                 ". Supported slots: " + restaurant.getSupportedMealSlots());
92         }
93
94         // 3. Check for duplicate active subscription (same employee + slot)
95         if (subscriptionRepository.checkSubscriptionExists(
96             employee.getEmployeeId(), slot.name(), SubscriptionStatus.ACTIVE.name()) > 0) {
97             throw new SubscriptionAlreadyExists(
98                 "Employee " + employee.getEmployeeId() +
99                 " already has an active " + slot + " subscription. " +
100                 "Cancel the existing one before subscribing to a new restaurant.");
101         }
102
103         // 4. Create subscription
104         Subscription subscription = new Subscription();
105         subscription.setEmployee(employee);
106         subscription.setSlot(slot);
107         subscription.setRestaurant(restaurant);
108         subscription.setScheduleType(request.getScheduleType());
109         subscription.setDayOfWeek(request.getDayOfWeek());
110         subscription.setStatus(SubscriptionStatus.ACTIVE);
111         subscription.setNextDeliveryTime(
112             deliveryTimeService.calculateNextDelivery(request, slot, employee));
113
114         subscriptionRepository.save(subscription);
115     }
116 }
117
118 public List<SubscriptionDTO> getAllSubscriptions() {
119     List<Subscription> subscriptions = subscriptionRepository.findAll();
120     List<SubscriptionDTO> dtoList = new ArrayList<>();
121     for (Subscription s : subscriptions) {
122         dtoList.add(convertToDTO(s));
123     }
124     return dtoList;
125 }
126
127 public Subscription updateSubscription(Long id, SubscriptionRequest request) {
128     Subscription subscription = subscriptionRepository.findById(id)
129         .orElseThrow(() -> new RuntimeException("Subscription not found"));
130     Employee employee = subscription.getEmployee();
131
132     if (request.getScheduleType() == ScheduleType.DAILY) {
133         subscription.setDayOfWeek(null);
134     } else if (request.getScheduleType() == ScheduleType.WEEKLY) {
135         if (request.getDayOfWeek() == null) {
136             throw new IllegalArgumentException("dayOfWeek is required for WEEKLY schedule");
137         }
138         subscription.setDayOfWeek(request.getDayOfWeek());
139     }
140     subscription.setScheduleType(request.getScheduleType());
141
142     // Update slot & restaurant if a single slotOrder is provided
143     if (request.getSlotOrders() != null && !request.getSlotOrders().isEmpty()) {
144         if (request.getSlotOrders().size() > 1) {
145             throw new IllegalArgumentException(
146                 "Update supports one slot at a time. Use separate subscriptions for multiple
slots.");

```

```

147     }
148     SubscriptionRequest.SlotOrder slotOrder = request.getSlotOrders().get(0);
149     MealSlot newSlot = slotOrder.getMealSlot();
150     Restaurant restaurant = restaurantRepository.findById(slotOrder.getRestaurantId())
151         .orElseThrow(() -> new RestaurantNotFoundException(
152             "Restaurant " + slotOrder.getRestaurantId() + " not found"));
153     if (!restaurant.isActive()) {
154         throw new IllegalArgumentException("Restaurant '" + restaurant.getName() + "' is not
active");
155     }
156     if (!restaurant.getSupportedMealSlots().contains(newSlot)) {
157         throw new IllegalArgumentException(
158             "Restaurant '" + restaurant.getName() + "' does not serve " + newSlot);
159     }
160     subscription.setSlot(newSlot);
161     subscription.setRestaurant(restaurant);
162 }
163
164     subscription.setNextDeliveryTime(
165         deliveryTimeService.calculateNextDelivery(request, subscription.getSlot(), employee));
166
167     return subscriptionRepository.save(subscription);
168 }
169
170 public Subscription pauseSubscription(Long id) {
171     Subscription subscription = subscriptionRepository.findById(id)
172         .orElseThrow(() -> new RuntimeException("Subscription not found"));
173     subscription.setStatus(SubscriptionStatus.PAUSED);
174     subscription.setNextDeliveryTime(null);
175     return subscriptionRepository.save(subscription);
176 }
177
178 public Subscription resumeSubscription(Long id) {
179     Subscription subscription = subscriptionRepository.findById(id)
180         .orElseThrow(() -> new RuntimeException("Subscription not found"));
181     subscription.setStatus(SubscriptionStatus.ACTIVE);
182     subscription.setNextDeliveryTime(deliveryTimeService.getNextDeliveryTime(subscription));
183     return subscriptionRepository.save(subscription);
184 }
185
186 public Subscription expireSubscription(Long id) {
187     Subscription subscription = subscriptionRepository.findById(id)
188         .orElseThrow(() -> new RuntimeException("Subscription not found"));
189     subscription.setStatus(SubscriptionStatus.EXPIRED);
190     subscription.setNextDeliveryTime(null);
191     return subscriptionRepository.save(subscription);
192 }
193
194 public List<SubscriptionDTO> getSubscriptionOfUser(Authentication authentication) {
195     String email= AuthUtil.extractEmail(authentication);
196     List<Subscription> subscriptions = subscriptionRepository.findByEmployee_Email(email);
197     List<SubscriptionDTO> dtoList = new ArrayList<>();
198     for (Subscription s : subscriptions) {
199         dtoList.add(convertToDTO(s));
200     }
201     return dtoList;
202 }
203
204 // helpers
205
206 private SubscriptionDTO convertToDTO(Subscription s) {
207     SubscriptionDTO dto = new SubscriptionDTO();
208     dto.setSubscriptionId(s.getId());
209     dto.setUserId(s.getEmployee().getEmployeeId());
210     dto.setMealSlot(s.getSlot());
211     dto.setScheduleType(s.getScheduleType());
212     if (s.getScheduleType() == ScheduleType.WEEKLY) {
213         dto.setDayOfWeek(s.getDayOfWeek());
214     }
215     dto.setStatus(s.getStatus());
216     if (s.getNextDeliveryTime() != null) {
217         ZoneId userZone = ZoneId.of(s.getEmployee().getTimezone());
218         dto.setNextDeliveryTime(s.getNextDeliveryTime().atZone(userZone));
219     }
220     // Restaurant info
221     if (s.getRestaurant() != null) {

```



```
222         dto.setRestaurantId(s.getRestaurant().getId());
223         dto.setRestaurantName(s.getRestaurant().getName());
224         dto.setRestaurantAddress(s.getRestaurant().getAddress());
225     }
226     return dto;
227 }
228 }
```

```
1 package com.example.EmployeeManagementSystem.Service;
2
3 import dev.samstevens.totp.code.*;
4 import dev.samstevens.totp.secret.DefaultSecretGenerator;
5 import dev.samstevens.totp.secret.SecretGenerator;
6 import dev.samstevens.totp.time.SystemTimeProvider;
7 import dev.samstevens.totp.time.TimeProvider;
8 import org.springframework.stereotype.Service;
9
10 @Service
11 public class TotpService {
12
13     private final SecretGenerator secretGenerator = new DefaultSecretGenerator();
14     private final TimeProvider timeProvider = new SystemTimeProvider();
15     private final CodeGenerator codeGenerator = new DefaultCodeGenerator();
16     private final CodeVerifier codeVerifier = new DefaultCodeVerifier(codeGenerator, timeProvider);
17
18     /** Generate a new Base32 secret for a user */
19     public String generateSecret() {
20         return secretGenerator.generate();
21     }
22
23     /**
24      * Generate a QR code URI – user scans this with Google Authenticator / Authy.
25      * Format: otpauth://totp/{issuer}:{username}?secret={secret}&issuer={issuer}
26      */
27     public String getQrCodeUri(String secret, String username) {
28         return String.format(
29             "otpauth://totp/EmployeeMS:%s?secret=%s&issuer=EmployeeManagementSystem",
30             username, secret
31         );
32     }
33
34     /**
35      * Verify the 6-digit code the user typed.
36      * DefaultCodeVerifier already allows ±1 time step (±30 seconds) for clock skew.
37      */
38     public boolean verifyCode(String secret, String code) {
39         try {
40             return codeVerifier.isValidCode(secret, code);
41         } catch (Exception e) {
42             return false;
43         }
44     }
45 }
```

```

1 package com.example.EmployeeManagementSystem.Service;
2
3 import com.example.EmployeeManagementSystem.Entity.*;
4 import com.example.EmployeeManagementSystem.Enum.Status;
5 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
6 import com.example.EmployeeManagementSystem.Repository.LeaveEntitlementRepository;
7 import com.example.EmployeeManagementSystem.Repository.LeaveWarningRepository;
8 import com.example.EmployeeManagementSystem.Repository.YearEndCarryForwardLogRepository;
9 import org.slf4j.Logger;
10 import org.slf4j.LoggerFactory;
11 import org.springframework.stereotype.Service;
12 import org.springframework.transaction.annotation.Transactional;
13
14 import java.math.BigDecimal;
15 import java.math.RoundingMode;
16 import java.time.LocalDate;
17 import java.util.List;
18 import java.util.Map;
19 import java.util.Set;
20 import java.util.stream.Collectors;
21
22 @Service
23 public class YearEndCarryForwardService {
24
25     private static final Logger log = LoggerFactory.getLogger(YearEndCarryForwardService.class);
26     private static final BigDecimal CAP = BigDecimal.valueOf(30);
27     private static final Set<String> SUPPORTED_LEAVE_TYPES = Set.of("SICK", "CASUAL", "MATERNITY");
28
29     private final EmployeeRepo employeeRepository;
30     private final LeaveEntitlementRepository entitlementRepository;
31     private final YearEndCarryForwardLogRepository logRepository;
32     private final LeaveWarningRepository warningRepository;
33
34     public YearEndCarryForwardService(
35         EmployeeRepo employeeRepository,
36         LeaveEntitlementRepository entitlementRepository,
37         YearEndCarryForwardLogRepository logRepository,
38         LeaveWarningRepository warningRepository) {
39         this.employeeRepository = employeeRepository;
40         this.entitlementRepository = entitlementRepository;
41         this.logRepository = logRepository;
42         this.warningRepository = warningRepository;
43     }
44
45     // -----
46     // Main entry point – called by scheduler on Dec 31 at 22:00
47     // -----
48     @Transactional
49     public void runYearEnd(int closingYear) {
50
51         // Idempotency guard – never run twice for same year
52         if (logRepository.existsByProcessedYear(closingYear)) {
53             log.warn("Year-end carry-forward already processed for {}. Skipping.", closingYear);
54             return;
55         }
56
57         List<Employee> activeEmployees = employeeRepository.findByStatus(Status.ACTIVE);
58         log.info("Starting year-end carry-forward for {} – {} employees", closingYear,
59             activeEmployees.size());
60         for (Employee employee : activeEmployees) {
61             processEmployeeYearEnd(employee, closingYear);
62         }
63
64         log.info("Year-end carry-forward complete for {}", closingYear);
65     }
66
67     // -----
68     // Per-employee processing
69     // -----
70     private void processEmployeeYearEnd(Employee employee, int closingYear) {

```

```

71     int nextYear = closingYear + 1;
72
73     // Fetch all entitlements for the closing year
74     List<LeaveEntitlement> entitlements =
75         entitlementRepository.findByEmployeeEmployeeIdAndYear(employee.getEmployeeId(),
closingYear);
76
77     // Separate carry-forwardable from non-carry-forwardable
78     List<LeaveEntitlement> carryForwardable = entitlements.stream()
79         .filter(e -> SUPPORTED_LEAVE_TYPES.contains(e.getLeaveType().getName()))
80         .filter(e -> e.getLeaveType().isCarriesForward())
81         .collect(Collectors.toList());
82
83     // Step 1 – compute raw closing balance for each type
84     //     closing = opening + accrued - used
85     Map<LeaveEntitlement, BigDecimal> rawClosing = carryForwardable.stream()
86         .collect(Collectors.toMap(
87             e -> e,
88             e -> e.getOpeningBalance()
89                 .add(e.getAccruedThisYear())
90                 .subtract(e.getUsedThisYear())
91                 .max(BigDecimal.ZERO) // never negative
92         ));
93
94     // Step 2 – sum all carry-forwardable closing balances
95     BigDecimal totalRaw = rawClosing.values().stream()
96         .reduce(BigDecimal.ZERO, BigDecimal::add);
97
98     // Step 3 – apply cap proportionally if total exceeds 30
99     Map<LeaveEntitlement, BigDecimal> cappedClosing = applyCap(rawClosing, totalRaw);
100
101     // Step 4 – persist: update closing balance on current year row,
102     //     create/update opening balance on next year row, write log
103     for (LeaveEntitlement entitlement : carryForwardable) {
104         BigDecimal raw = rawClosing.get(entitlement);
105         BigDecimal capped = cappedClosing.get(entitlement);
106         BigDecimal wasted = raw.subtract(capped);
107
108         // Update closing balance on the current year's row
109         entitlement.setClosingBalance(capped);
110         entitlementRepository.save(entitlement);
111
112         // Write or update next year's opening balance
113         LeaveEntitlement nextYearRow = entitlementRepository
114             .findByEmployeeEmployeeIdAndLeaveTypeIdAndYear(
115                 employee.getEmployeeId(), entitlement.getLeaveType().getId(), nextYear)
116             .orElseGet(() -> buildNextYearRow(employee, entitlement.getLeaveType(), nextYear));
117
118         nextYearRow.setOpeningBalance(capped);
119         entitlementRepository.save(nextYearRow);
120
121         // Write audit log
122         writeLog(employee, closingYear, entitlement.getLeaveType(), raw, capped, wasted);
123
124         // Issue DAY_WASTED warning if any days were lost
125         if (wasted.compareTo(BigDecimal.ZERO) > 0) {
126             issueWastedWarning(employee, wasted, capped);
127             log.info("Employee {} lost {} day(s) of {} to cap at year end",
128                 employee.getEmployeeId(), wasted, entitlement.getLeaveType().getName());
129         }
130     }
131
132     // Step 5 – zero out sick leave closing balance (it never carries)
133     entitlements.stream()
134         .filter(e -> SUPPORTED_LEAVE_TYPES.contains(e.getLeaveType().getName()))
135         .filter(e -> !e.getLeaveType().isCarriesForward())
136         .forEach(e -> {
137             e.setClosingBalance(BigDecimal.ZERO);
138             entitlementRepository.save(e);
139         });
140
141     log.info("Year-end processed for employee {} – total raw={}, capped={},",
142         employee.getEmployeeId(), totalRaw, CAP.min(totalRaw));
143 }
144
145 // -----

```

```

146 // Cap logic – distribute cap proportionally across leave types
147 //
148 // Example: casual=20, maternity=16 total=36, over cap by 6
149 // casual gets: 20/36 * 30 = 16.67
150 // maternity gets: 16/36 * 30 = 13.33
151 // total = 30 exactly
152 // -----
153 private Map<LeaveEntitlement, BigDecimal> applyCap(
154     Map<LeaveEntitlement, BigDecimal> rawClosing,
155     BigDecimal totalRaw) {
156
157     if (totalRaw.compareTo(CAP) <= 0) {
158         // Under cap – no adjustment needed
159         return rawClosing;
160     }
161
162     return rawClosing.entrySet().stream()
163         .collect(Collectors.toMap(
164             Map.Entry::getKey,
165             entry -> {
166                 BigDecimal raw = entry.getValue();
167                 // proportion = raw / totalRaw * CAP
168                 return raw
169                     .divide(totalRaw, 10, RoundingMode.HALF_UP)
170                     .multiply(CAP)
171                     .setScale(2, RoundingMode.HALF_UP);
172             }
173         ));
174 }
175
176 // -----
177 // Build a blank next-year entitlement row
178 // -----
179 private LeaveEntitlement buildNextYearRow(Employee employee, LeaveType leaveType, int year) {
180     LeaveEntitlement row = new LeaveEntitlement();
181     row.setEmployee(employee);
182     row.setLeaveType(leaveType);
183     row.setYear(year);
184     row.setOpeningBalance(BigDecimal.ZERO);
185     row.setAccruedThisYear(BigDecimal.ZERO);
186     row.setUsedThisYear(BigDecimal.ZERO);
187     row.setClosingBalance(BigDecimal.ZERO);
188     return entitlementRepository.save(row);
189 }
190
191 // -----
192 // Write audit log entry
193 // -----
194 private void writeLog(Employee employee, int year, LeaveType leaveType,
195     BigDecimal raw, BigDecimal capped, BigDecimal wasted) {
196     YearEndCarryForwardLog entry = new YearEndCarryForwardLog();
197     entry.setEmployee(employee);
198     entry.setProcessedYear(year);
199     entry.setLeaveType(leaveType);
200     entry.setClosingBalance(raw);
201     entry.setCappedBalance(capped);
202     entry.setDaysWasted(wasted);
203     entry.setOpeningNextYear(capped);
204     logRepository.save(entry);
205 }
206
207 // -----
208 // Issue a DAY_WASTED warning on the dashboard
209 // -----
210 private void issueWastedWarning(Employee employee, BigDecimal wasted, BigDecimal newBalance) {
211     LeaveWarning warning = new LeaveWarning();
212     warning.setEmployee(employee);
213     warning.setWarningType(LeaveWarning.WarningType.DAY_WASTED);
214     warning.setCarryForwardBalance(newBalance);
215     warning.setWarningDate(LocalDate.now());
216     warning.setMessage(
217         wasted.toPlainString() + " carry-forward day(s) were lost at year-end " +
218         "because your balance exceeded the 30-day cap. " +
219         "Your opening balance for next year is " + newBalance.toPlainString() + " days."
220     );
221     warningRepository.save(warning);

```

```
222     }  
223 }  
224
```

```
1 package com.example.EmployeeManagementSystem.Util;
2
3 import org.springframework.security.core.Authentication;
4 import org.springframework.security.oauth2.core.user.OAuth2User;
5 import org.springframework.stereotype.Component;
6
7 @Component
8 public class AuthUtil {
9     public static String extractEmail(Authentication authentication) {
10         if (authentication == null) return null;
11
12         // OAuth2 login (session-based) – principal is OAuth2User
13         if (authentication.getPrincipal() instanceof OAuth2User oAuth2User) {
14             return oAuth2User.getAttribute("email"); // always the actual email
15         }
16
17         // JWT or session username/password login – getName() is already the email
18         return authentication.getName();
19     }
20 }
21
```

```
1 package com.example.EmployeeManagementSystem.Util;
2
3 import com.example.EmployeeManagementSystem.Entity.Employee;
4 import com.example.EmployeeManagementSystem.Entity.Vendor;
5 import com.example.EmployeeManagementSystem.Enum.Role;
6 import com.example.EmployeeManagementSystem.Exception.EmployeeNotFound;
7 import com.example.EmployeeManagementSystem.Exception.VendorNotFoundException;
8 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
9 import com.example.EmployeeManagementSystem.Repository.VendorRepo;
10 import io.jsonwebtoken.Claims;
11 import io.jsonwebtoken.Jwts;
12 import io.jsonwebtoken.security.Keys;
13 import jakarta.annotation.PostConstruct;
14 import org.springframework.beans.factory.annotation.Value;
15 import org.springframework.security.core.userdetails.UserDetails;
16 import org.springframework.stereotype.Component;
17
18 import javax.crypto.SecretKey;
19 import java.util.Date;
20 import java.util.Optional;
21
22 @Component
23 public class JWTUtil {
24
25     private final EmployeeRepo employeeRepo;
26     private final VendorRepo vendorRepo;
27
28     // FIX 1: Was 1000*60*60 (1 hour) – caused token expiry mid-session.
29     // Extended to 24 hours to match the session timeout in application.yaml.
30     private final long Expiration_time = 1000L * 60 * 60 * 24;
31
32     @Value("${jwt.secret}")
33     private String secret;
34     private SecretKey key;
35
36     public JWTUtil(EmployeeRepo employeeRepo, VendorRepo vendorRepo) {
37         this.employeeRepo = employeeRepo;
38         this.vendorRepo = vendorRepo;
39     }
40
41     @PostConstruct
42     public void init() {
43         key = Keys.hmacShaKeyFor(secret.getBytes());
44     }
45
46     public String generateToken(String username, String role) {
47         return Jwts.builder()
48             .subject(username)
49             .claim("role", role)
50             .issuedAt(new Date())
51             .expiration(new Date(System.currentTimeMillis() + Expiration_time))
52             .signWith(key)
53             .compact();
54     }
55
56     public String extractUsernameFromToken(String token) {
57         Claims body = extractClaims(token);
58         return body.getSubject();
59     }
60
61     private Claims extractClaims(String token) {
62         return Jwts.parser()
63             .verifyWith(key)
64             .build()
65             .parseSignedClaims(token)
66             .getPayload();
67     }
68
69     public boolean validateToken(String username, UserDetails userDetails, String token) {
70         return username.equals(userDetails.getUsername()) && !isTokenExpired(token);
71     }
```



```
72
73     private boolean isTokenExpired(String token) {
74         return extractClaims(token).getExpiration().before(new Date());
75     }
76 }
```

```

1 package com.example.EmployeeManagementSystem.Util;
2
3 import com.example.EmployeeManagementSystem.Entity.RefreshToken;
4 import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
5 import com.example.EmployeeManagementSystem.Repository.RefreshTokenRepository;
6 import com.example.EmployeeManagementSystem.Repository.VendorRepo;
7 import com.example.EmployeeManagementSystem.Service.RefreshTokenService;
8 import jakarta.servlet.http.HttpServletRequest;
9 import jakarta.servlet.http.HttpServletResponse;
10 import org.springframework.security.core.Authentication;
11 import org.springframework.security.oauth2.core.user.OAuth2User;
12 import org.springframework.security.web.authentication.SimpleUrlAuthenticationSuccessHandler;
13 import org.springframework.stereotype.Service;
14
15 import java.io.IOException;
16 import java.net.URLEncoder;
17 import java.nio.charset.StandardCharsets;
18
19 @Service
20 public class OAuth2SuccessHandler extends SimpleUrlAuthenticationSuccessHandler{
21     private final JWTUtil jwtUtil;
22     private final EmployeeRepo employeeRepo;
23     private final VendorRepo vendorRepo;
24     private final RefreshTokenService refreshTokenService;
25     private final RefreshTokenRepository refreshTokenRepository;
26
27     public OAuth2SuccessHandler(JWTUtil jwtUtil,
28                               EmployeeRepo employeeRepo,
29                               VendorRepo vendorRepo,
30                               RefreshTokenService refreshTokenService,
31                               RefreshTokenRepository refreshTokenRepository) {
32         this.jwtUtil = jwtUtil;
33         this.employeeRepo = employeeRepo;
34         this.vendorRepo = vendorRepo;
35         this.refreshTokenService = refreshTokenService;
36         this.refreshTokenRepository = refreshTokenRepository;
37     }
38
39     @Override
40     public void onAuthenticationSuccess(HttpServletRequest request,
41                                       HttpServletResponse response,
42                                       Authentication authentication) throws IOException {
43
44         try{
45             OAuth2User oAuth2User = (OAuth2User) authentication.getPrincipal();
46             String email = oAuth2User.getAttribute("email");
47
48             // Generate refresh token – check which entity type owns this email
49             RefreshToken refreshToken;
50             String role;
51             var employeeOpt = employeeRepo.findByEmail(email);
52             var vendorOpt = vendorRepo.findByEmail(email);
53
54             if (employeeOpt.isPresent()) {
55                 role = employeeOpt.get().getRole().name();
56                 refreshToken = refreshTokenService.createForEmployee(employeeOpt.get());
57             } else if (vendorOpt.isPresent()) {
58                 role = vendorOpt.get().getRole().name();
59                 refreshToken = refreshTokenService.createForVendor(vendorOpt.get());
60             } else {
61                 response.sendError(HttpServletResponse.SC_INTERNAL_SERVER_ERROR, "User not found post-
62 OAuth");
63                 return;
64             }
65             String accessToken = jwtUtil.generateToken(email, role);
66             refreshTokenRepository.save(refreshToken);
67
68             // Redirect frontend to a callback page with tokens as query params
69             // (or use a fragment # so tokens don't hit server logs)
70             String redirectUrl = "http://localhost:8080/google.html" // your frontend page

```

```
71         + "?accessToken=" + URLEncoder.encode(accessToken, StandardCharsets.UTF_8)
72         + "&refreshToken=" + URLEncoder.encode(refreshToken.getToken(),
StandardCharsets.UTF_8);
73
74         getRedirectStrategy().sendRedirect(request, response, redirectUrl);
75     }
76     catch (Exception e) {
77         // This will now print the REAL error in your console
78         System.err.println("=== OAuth2SuccessHandler FAILED ===");
79         e.printStackTrace();
80         response.sendError(HttpServletResponse.SC_INTERNAL_SERVER_ERROR,
81             "OAuth2 login failed: " + e.getMessage());
82     }
83 }
84 }
85
```

```
1  spring:
2    application:
3      name: EmployeeManagementSystem
4
5    datasource :
6      url : jdbc:mysql://localhost:3306/employee_management_db
7      username : ${USER_NAME}
8      password : ${PASSWORD}
9
10   jpa:
11     hibernate :
12       ddl-auto: update
13     show-sql: true
14
15   springdoc:
16     api-docs:
17       path: /v3/api-docs
18       enabled: true
19     swagger-ui:
20       path: /swagger-ui.html
21       enabled: true
22       operations-sorter: method
23       tags-sorter: alpha
24       try-it-out-enabled: true
25       filter: true
26     show-actuator: true
27   mvc:
28     static-path-pattern: /**
29   session:
30     tracking-modes: COOKIE
31   web:
32     resources:
33       static-locations: classpath:/static/
34       add-mappings: true
35   session:
36     timeout: 86400s # 24 hours in seconds (was 1440m which is incorrect format)
37     store-type: none # Use default cookie-based session
38
39   servlet:
40     session:
41       cookie:
42         http-only: true
43         same-site: strict
44         secure: false # set true in production with HTTPS
45
46   security:
47     debug: true
48     oauth2:
49       client:
50         registration:
51           google:
52             client-id: ${client_id}
53             client-secret: ${client_secret}
54             scope:
55               - email
56               - profile
57
58   logging:
59     level:
60       org.springframework.security: DEBUG
61       com.example.EmployeeManagementSystem: DEBUG
62   jwt:
63     secret: ${JWT_SECRET}
64     # Add this to your application.yaml
65   api:
66     key:
67       enabled: true
68       header-name: X-API-Key
69
```

```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>EMS - Admin Console</title>
7      <link
href="https://fonts.googleapis.com/css2?family=DM+Sans:wght@400;500;600;700&family=DM+Mono:wght@400;500&display=
swap" rel="stylesheet">
8      <link href="https://cdn.jsdelivr.net/npm/fullcalendar@6.1.8/index.global.min.css" rel="stylesheet">
9      <script src="https://cdn.jsdelivr.net/npm/fullcalendar@6.1.8/index.global.min.js"></script>
10     <style>
11         * { margin: 0; padding: 0; box-sizing: border-box; }
12         :root {
13             --bg: #0d0f14; --bg2: #131720; --bg3: #1a2030;
14             --panel: #1e2436; --panel2: #252c40;
15             --border: rgba(255,255,255,0.07); --border2: rgba(255,255,255,0.12);
16             --text: #e0e0e0; --text2: #8b92a8; --text3: #5557fa;
17             --accent: #4f8cff; --accent-dim: rgba(79,140,255,0.12);
18             --green: #36d399; --green-dim: rgba(54,211,153,0.12);
19             --amber: #f9c74f; --amber-dim: rgba(249,199,79,0.12);
20             --red: #f87171; --red-dim: rgba(248,113,113,0.12);
21             --violet: #a78bfa; --violet-dim: rgba(167,139,250,0.12);
22             --font: 'DM Sans', sans-serif; --mono: 'DM Mono', monospace;
23         }
24         body { font-family: var(--font); background: var(--bg); color: var(--text); font-size: 14px; line-
height: 1.5; overflow: hidden; }
25         .app { display: flex; height: 100vh; }
26         .sidebar { width: 260px; background: var(--bg2); border-right: 1px solid var(--border); display:
flex; flex-direction: column; overflow-y: auto; }
27         .logo { padding: 20px; border-bottom: 1px solid var(--border); }
28         .logo h2 { font-size: 20px; color: var(--text); }
29         .logo span { color: var(--accent); }
30         .logo p { font-size: 11px; color: var(--text3); margin-top: 4px; }
31         .nav-item { padding: 12px 20px; margin: 4px 12px; border-radius: 8px; cursor: pointer; transition:
all 0.2s; color: var(--text2); display: flex; align-items: center; gap: 10px; }
32         .nav-item:hover, .nav-item.active { background: var(--accent-dim); color: var(--accent); }
33         .main-content { flex: 1; display: flex; flex-direction: column; overflow: hidden; }
34         .topbar { background: var(--bg2); padding: 16px 24px; display: flex; justify-content: space-between;
align-items: center; border-bottom: 1px solid var(--border); }
35         .page-title { font-size: 20px; font-weight: 600; }
36         .user-info { display: flex; align-items: center; gap: 16px; }
37         .logout-btn { padding: 8px 16px; background: var(--red-dim); border: 1px solid
rgba(248,113,113,0.3); border-radius: 6px; color: var(--red); cursor: pointer; }
38         .content { flex: 1; overflow-y: auto; padding: 24px; }
39         .view { display: none; }
40         .view.active { display: block; }
41         .stats-grid { display: grid; grid-template-columns: repeat(4, 1fr); gap: 16px; margin-bottom: 24px;
}
42         .stat-card { background: var(--panel); padding: 20px; border-radius: 12px; border: 1px solid var(--
border); }
43         .stat-label { font-size: 11px; color: var(--text3); text-transform: uppercase; margin-bottom: 8px; }
44         .stat-value { font-size: 28px; font-weight: 600; }
45         .form-card { background: var(--panel); border-radius: 12px; border: 1px solid var(--border);
padding: 24px; max-width: 600px; }
46         .form-grid { display: grid; grid-template-columns: 1fr 1fr; gap: 16px; }
47         .form-group { display: flex; flex-direction: column; gap: 6px; }
48         .form-group.full { grid-column: 1 / -1; }
49         label { font-size: 11px; font-weight: 600; color: var(--text3); text-transform: uppercase; }
50         input, select, textarea { padding: 10px 12px; background: var(--bg3); border: 1px solid var(--
border2); border-radius: 8px; color: var(--text); font-family: var(--font); font-size: 13px; }
51         input:focus, select:focus, textarea:focus { outline: none; border-color: var(--accent); }
52         .table-wrap { background: var(--panel); border-radius: 12px; border: 1px solid var(--border);
overflow-x: auto; }
53         table { width: 100%; border-collapse: collapse; }
54         th { text-align: left; padding: 12px 16px; background: var(--bg3); color: var(--text3); font-size:
11px; font-weight: 600; text-transform: uppercase; }
55         td { padding: 12px 16px; border-top: 1px solid var(--border); }
56         .chip { padding: 4px 10px; border-radius: 20px; font-size: 11px; font-weight: 600; display: inline-b
lock; }
57         .chip.active, .chip.approved, .chip.delivered, .chip.repair_done { background: var(--green-dim);
color: var(--green); }

```

```

58     .chip.pending, .chip.scheduled, .chip.sent_for_repair { background: var(--amber-dim); color: var(--
amber); }
59     .chip.open { background: var(--accent-dim); color: var(--accent); }
60     .chip.inactive, .chip.rejected, .chip.cancelled, .chip.expired { background: var(--red-dim); color:
var(--red); }
61     .chip.on_leave { background: var(--violet-dim); color: var(--violet); }
62     .chip.employee { background: var(--accent-dim); color: var(--accent); }
63     .chip.manager { background: var(--violet-dim); color: var(--violet); }
64     .chip.vendor { background: var(--red-dim); color: var(--red); }
65     .chip.under_repair { background: var(--amber-dim); color: var(--amber); }
66     .chip.unpaid, .chip.UNPAID { background: var(--violet-dim); color: var(--violet); }
67     .chip.assigned { background: var(--green-dim); color: var(--green); }
68     .chip.condemned { background: var(--red-dim); color: var(--red); }
69     .chip.admin { background: var(--amber-dim); color: var(--amber); }
70     .btn { padding: 8px 16px; border-radius: 6px; border: none; font-family: var(--font); font-size:
13px; font-weight: 500; cursor: pointer; transition: all 0.2s; }
71     .btn-primary { background: var(--accent); color: white; }
72     .btn-primary:hover { background: #6ba0ff; }
73     .btn-secondary { background: var(--bg3); color: var(--text2); border: 1px solid var(--border2); }
74     .btn-danger { background: var(--red-dim); color: var(--red); border: 1px solid
rgba(248,113,113,0.3); }
75     .btn-sm { padding: 4px 10px; font-size: 11px; margin: 2px; }
76     .section-head { display: flex; justify-content: space-between; align-items: center; margin-bottom:
16px; }
77     .section-title { font-size: 16px; font-weight: 600; }
78     .empty-state { text-align: center; padding: 40px; color: var(--text3); }
79     .modal-overlay { display: none; position: fixed; top: 0; left: 0; right: 0; bottom: 0; background:
rgba(0,0,0,0.7); z-index: 1000; align-items: center; justify-content: center; }
80     .modal-overlay.open { display: flex; }
81     .modal { background: var(--panel); border-radius: 12px; padding: 24px; width: 450px; max-width: 90%;
}
82     .modal-title { font-size: 18px; font-weight: 600; margin-bottom: 16px; }
83     .modal-actions { display: flex; gap: 12px; margin-top: 20px; justify-content: flex-end; }
84     .alert { position: fixed; top: 20px; right: 20px; padding: 12px 20px; border-radius: 8px; z-index:
1100; animation: slideIn 0.3s ease; }
85     .alert-success { background: var(--green-dim); color: var(--green); border: 1px solid var(--green);
}
86     .alert-error { background: var(--red-dim); color: var(--red); border: 1px solid var(--red); }
87     @keyframes slideIn { from { transform: translateX(100%); opacity: 0; } to { transform:
translateX(0); opacity: 1; } }
88     .action-row { display: flex; gap: 6px; flex-wrap: wrap; }
89     .notif-wrapper { position: relative; cursor: pointer; }
90
91     .notif-badge {
92         position: absolute; top: -5px; right: -6px;
93         background: #e53e3e; color: white;
94         font-size: 10px; font-weight: 600;
95         min-width: 16px; height: 16px;
96         border-radius: 50%; display: flex;
97         align-items: center; justify-content: center;
98         padding: 0 3px;
99     }
100     .notif-panel {
101         position: absolute; top: 48px; right: 0;
102         width: 320px; max-height: 400px; overflow-y: auto;
103         background: var(--panel);
104         border: 1px solid var(--border2);
105         border-radius: 10px;
106         box-shadow: 0 4px 24px rgba(0,0,0,0.4);
107         z-index: 999;
108     }
109     .notif-header {
110         padding: 12px 16px; font-weight: 600;
111         color: var(--text);
112         border-bottom: 1px solid var(--border);
113     }
114     .notif-item {
115         padding: 12px 16px; font-size: 13px;
116         color: var(--text2);
117         border-bottom: 1px solid var(--border);
118     }
119     .notif-item.unread { background: var(--accent-dim); color: var(--text); }
120     .notif-time { font-size: 11px; color: var(--text3); margin-top: 4px; }
121
122     /* ===== CALENDAR STYLES ===== */
123     .leave-balance-grid {

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124     display: grid;
125     grid-template-columns: repeat(auto-fit, minmax(160px, 1fr));
126     gap: 14px; padding: 20px;
127     border-bottom: 1px solid var(--border);
128 }
129 .leave-bal-card {
130     background: var(--bg3); border: 1px solid var(--border);
131     border-radius: 10px; padding: 16px 18px;
132     display: flex; flex-direction: column; gap: 6px;
133     position: relative; overflow: hidden; transition: border-color 0.2s;
134 }
135 .leave-bal-card:hover { border-color: var(--border2); }
136 .leave-bal-card::before {
137     content: ''; position: absolute;
138     left: 0; top: 0; bottom: 0; width: 3px;
139     border-radius: 3px 0 0 3px;
140 }
141 .leave-bal-card.sick::before { background: var(--red); }
142 .leave-bal-card.casual::before { background: var(--amber); }
143 .leave-bal-card.mat::before { background: var(--accent); }
144 .leave-bal-card.total::before { background: var(--green); }
145 .lbc-icon { font-size: 18px; }
146 .lbc-label { font-size: 10px; color: var(--text3); text-transform: uppercase; letter-spacing: 1px; font-weight: 600; }
147 .lbc-count { font-size: 24px; font-weight: 700; color: var(--text); font-family: var(--mono); line-height: 1; }
148 .lbc-sub { font-size: 11px; color: var(--text3); }
149 .cal-legend {
150     display: flex; align-items: center; gap: 20px;
151     padding: 12px 20px; border-bottom: 1px solid var(--border); flex-wrap: wrap;
152 }
153 .legend-label { font-size: 11px; color: var(--text3); font-weight: 600; text-transform: uppercase; letter-spacing: 0.8px; margin-right: 4px; }
154 .legend-item { display: flex; align-items: center; gap: 6px; font-size: 12px; color: var(--text2); }
155 .legend-dot { width: 10px; height: 10px; border-radius: 50%; flex-shrink: 0; }
156 .cal-section { background: var(--panel); border: 1px solid var(--border); border-radius: 12px; margin-bottom: 20px; overflow: hidden; }
157 .cal-section-head { padding: 16px 20px; border-bottom: 1px solid var(--border); display: flex; justify-content: space-between; align-items: center; }
158 .cal-section-title { font-size: 16px; font-weight: 600; color: var(--text); }
159 .cal-section-sub { font-size: 12px; color: var(--text2); margin-top: 2px; }
160 #employee-calendar { padding: 16px 20px 20px; }
161 #employee-calendar .fc .fc-toolbar-title { font-family: var(--font); font-size: 15px; font-weight: 600; color: var(--text); }
162 #employee-calendar .fc .fc-button {
163     background: var(--panel2); border: 1px solid var(--border2);
164     color: var(--text2); font-family: var(--font); font-size: 12px;
165     font-weight: 600; border-radius: 6px; padding: 5px 12px;
166     transition: all 0.2s; box-shadow: none; text-transform: capitalize;
167 }
168 #employee-calendar .fc .fc-button:hover,
169 #employee-calendar .fc .fc-button-active,
170 #employee-calendar .fc .fc-button:not(:disabled):active {
171     background: var(--accent-dim); border-color: var(--accent); color: var(--accent); box-shadow: none;
172 }
173 #employee-calendar .fc .fc-button:focus { box-shadow: none; outline: none; }
174 #employee-calendar .fc-theme-standard td,
175 #employee-calendar .fc-theme-standard th,
176 #employee-calendar .fc-theme-standard .fc-scrollgrid { border-color: var(--border); }
177 #employee-calendar .fc .fc-col-header-cell-cushion {
178     font-family: var(--font); font-size: 11px; font-weight: 600;
179     color: var(--text3); text-transform: uppercase; letter-spacing: 0.8px; padding: 8px 4px;
180 }
181 #employee-calendar .fc .fc-daygrid-day-number { font-family: var(--mono); font-size: 12px; color: var(--text3); padding: 6px 8px; }
182 #employee-calendar .fc .fc-daygrid-day.fc-day-today { background: var(--accent-dim) !important; }
183 #employee-calendar .fc .fc-daygrid-day.fc-day-today .fc-daygrid-day-number { color: var(--accent); font-weight: 700; }
184 #employee-calendar .fc-daygrid-day-frame { background: var(--bg3); }
185 #employee-calendar .fc .fc-day-other .fc-daygrid-day-frame { background: var(--bg2); }
186 #employee-calendar .fc-event {
187     border: none !important; border-radius: 4px;
188     font-family: var(--font); font-size: 11px; font-weight: 600; padding: 2px 6px; cursor: pointer;
189 }
190 .cal-tooltip {
191     position: fixed; background: var(--panel2); border: 1px solid var(--border2);

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192     border-radius: 8px; padding: 10px 14px; font-size: 12px; color: var(--text);
193     z-index: 9999; pointer-events: none; max-width: 200px;
194     box-shadow: 0 8px 24px rgba(0,0,0,0.4);
195 }
196 .cal-tooltip .tt-title { font-weight: 600; margin-bottom: 4px; font-size: 13px; }
197 .cal-tooltip .tt-row { color: var(--text2); margin-top: 2px; }
198     </style>
199 </head>
200 <body>
201 <div class="app">
202     <!-- Sidebar -->
203     <div class="sidebar">
204         <div class="logo">
205             <h2>E<span>M</span>S</h2>
206             <p>Admin Control Center</p>
207         </div>
208         <div style="flex: 1;">
209             <div class="nav-item active" onclick="showView('dashboard', event)">&#9632; Dashboard</div>
210             <div class="nav-item" onclick="showView('employees', event)">&#9632; Employees</div>
211             <div class="nav-item" onclick="showView('add-employee', event)">&#9632; Add Employee</div>
212             <div class="nav-item" onclick="showView('add-manager', event)">&#9632; Add Manager</div>
213             <div class="nav-item" onclick="showView('add-vendor', event)">&#9632; Add Vendor</div>
214             <div class="nav-item" onclick="window.location.href='/leave_management.html'">&#9632; Leave
Management</div>
215             <div class="nav-item" onclick="showView('holidays', event)">&#9632; Holidays</div>
216             <div class="nav-item" onclick="showView('restaurants', event)">&#9632; Restaurants</div>
217             <div class="nav-item" onclick="showView('subscriptions', event)">&#9632; Subscriptions</div>
218             <div class="nav-item" onclick="showView('add-subscription', event)">&#9632; Add
Subscription</div>
219             <div class="nav-item" onclick="showView('vendors', event)">&#9632; Vendors</div>
220             <div class="nav-item" onclick="showView('devices', event)">&#9632; Devices</div>
221             <div class="nav-item" onclick="showView('my-devices', event)">&#9632; My Devices</div>
222             <div class="nav-item" onclick="showView('service-requests', event)">&#9632; Service
Requests</div>
223             <div class="nav-item" onclick="showView('admin-service-requests', event)">&#9632; All Service
Requests</div>
224             <div class="nav-item" onclick="showView('my-calendar', event)">&#9632; My Calendar</div>
225             <div class="nav-item" onclick="showView('my-profile', event)">&#9632; My Profile</div>
226
227         </div>
228     </div>
229
230     <!-- Main Content -->
231     <div class="main-content">
232         <div class="topbar">
233             <div class="page-title" id="pageTitle">Dashboard</div>
234             <!-- Bell icon with badge -->
235             <div class="notif-wrapper" onclick="toggleNotifPanel()">
236                 <svg width="22" height="22" viewBox="0 0 24 24" fill="none"
237                     stroke="currentColor" stroke-width="2" stroke-linecap="round">
238                     <path d="M18 8A6 6 0 0 0 6 8c0 7-3 9-3 9h18s-3-2-3-9"/>
239                     <path d="M13.73 21a2 2 0 0 1-3.46 0"/>
240                 </svg>
241                 <span id="notif-badge" class="notif-badge" style="display:none"></span>
242             </div>
243
244             <!-- Dropdown panel -->
245             <div id="notif-panel" class="notif-panel" style="display:none">
246                 <div class="notif-header">Notifications</div>
247                 <div id="notif-list"></div>
248             </div>
249             <div class="user-info">
250                 <span id="username-display"></span>
251                 <button class="logout-btn" onclick="logout()">Logout</button>
252             </div>
253         </div>
254         <div class="content" id="content">
255
256             <!-- DASHBOARD VIEW -->
257             <div id="dashboard-view" class="view active">
258                 <div class="stats-grid">
259                     <div class="stat-card"><div class="stat-label">Total Employees</div><div class="stat-
value" id="total-employees">--</div></div>
260                     <div class="stat-card"><div class="stat-label">Active</div><div class="stat-value"
id="active-employees">--</div></div>
261                     <div class="stat-card"><div class="stat-label">On Leave</div><div class="stat-value"

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id="onleave-employees">--</div></div>
262         <div class="stat-card"><div class="stat-label">Pending Leaves</div><div class="stat-
value" id="pending-leaves">--</div></div>
263         </div>
264         <div class="section-head"><div class="section-title">Recent Leave Requests</div><button
class="btn btn-secondary btn-sm" onclick="loadDashboard()">Refresh</button></div>
265         <div class="table-
wrap"><table><thead><tr><th>Employee</th><th>Type</th><th>Start</th><th>End</th><th>Status</th></tr></thead><tbo
dy id="recent-leaves"></tbody></table></div>
266         </div>
267
268         <!-- EMPLOYEES VIEW -->
269         <div id="employees-view" class="view">
270             <div class="section-head"><div class="section-title">All Employees</div><button class="btn
btn-primary btn-sm" onclick="loadEmployees()">Refresh</button></div>
271             <div class="table-
wrap"><table><thead><tr><th>ID</th><th>Name</th><th>Email</th><th>Department</th><th>Role</th><th>Status</th><th>
>Timezone</th><th>Actions</th></tr></thead><tbody i...
272             </div>
273
274             <div id="add-employee-view" class="view">
275                 <div class="section-title" style="margin-bottom: 20px;">Add Employee</div>
276                 <div class="form-card">
277                     <div class="form-grid">
278                         <div class="form-group"><label>Name</label><input type="text" id="employee-name"
placeholder="Employee name"></div>
279                         <div class="form-group"><label>Email</label><input type="email" id="employee-email"
placeholder="employee@company.com"></div>
280                         <div class="form-group"><label>Password</label><input type="password" id="employee-
password" placeholder="Temporary password"></div>
281                         <div class="form-group">
282                             <label>Gender</label>
283                             <select id="employee-gender">
284                                 <option value="M">Male</option>
285                                 <option value="F">Female</option>
286                                 <option value="O">Other</option>
287                             </select>
288                         </div>
289                         <div class="form-group full">
290                             <label>Manager</label>
291                             <select id="employee-manager-select">
292                                 <option value="">-- Select Manager --</option>
293                             </select>
294                         </div>
295                         <div class="form-group full"><label>Timezone</label>
296                             <select id="employee-timezone">
297                                 <option value="Asia/Kolkata">India – IST UTC+5:30</option>
298                                 <option value="UTC">UTC</option>
299                                 <option value="America/New_York">New York – EST UTC-5</option>
300                                 <option value="America/Los_Angeles">Los Angeles – PST UTC-8</option>
301                                 <option value="Europe/London">London – GMT UTC+0</option>
302                                 <option value="Europe/Paris">Paris – CET UTC+1</option>
303                                 <option value="Asia/Tokyo">Tokyo – JST UTC+9</option>
304                                 <option value="Asia/Dubai">Dubai – GST UTC+4</option>
305                                 <option value="Australia/Sydney">Sydney – AEDT UTC+11</option>
306                                 <option value="Asia/Singapore">Singapore – SGT UTC+8</option>
307                             </select>
308                         </div>
309                     </div>
310                     <div class="modal-actions" style="margin-top: 20px;">
311                         <button class="btn btn-primary" onclick="createEmployee()">Create Employee</button>
312                         <button class="btn btn-secondary"
onclick="clearEmployeeForm('employee')">Clear</button>
313                     </div>
314                 </div>
315             </div>
316
317             <div id="add-manager-view" class="view">
318                 <div class="section-title" style="margin-bottom: 20px;">Add Manager</div>
319                 <div class="form-card">
320                     <div class="form-grid">
321                         <div class="form-group"><label>Name</label><input type="text" id="manager-name-
create" placeholder="Manager name"></div>
322                         <div class="form-group"><label>Email</label><input type="email" id="manager-email-
create" placeholder="manager@company.com"></div>
323                         <div class="form-group"><label>Password</label><input type="password" id="manager-

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password-create" placeholder="Temporary password"></div>
324         <div class="form-group"><label>Department</label><input type="text" id="manager-
dept-create" placeholder="Department"></div>
325         <div class="form-group full"><label>Timezone</label>
326             <select id="manager-timezone-create">
327                 <option value="Asia/Kolkata">India – IST UTC+5:30</option>
328                 <option value="UTC">UTC</option>
329                 <option value="America/New_York">New York – EST UTC-5</option>
330                 <option value="America/Los_Angeles">Los Angeles – PST UTC-8</option>
331                 <option value="Europe/London">London – GMT UTC+0</option>
332                 <option value="Europe/Paris">Paris – CET UTC+1</option>
333                 <option value="Asia/Tokyo">Tokyo – JST UTC+9</option>
334                 <option value="Asia/Dubai">Dubai – GST UTC+4</option>
335                 <option value="Australia/Sydney">Sydney – AEDT UTC+11</option>
336                 <option value="Asia/Singapore">Singapore – SGT UTC+8</option>
337             </select>
338         </div>
339         <div class="form-group"><label>Gender</label>
340             <select id="manager-gender">
341                 <option value="M">Male</option>
342                 <option value="F">Female</option>
343                 <option value="O">Other</option>
344             </select>
345         </div>
346     </div>
347     <div class="modal-actions" style="margin-top: 20px;"><button class="btn btn-primary"
onclick="createManager()">Create Manager</button><button class="btn btn-secondary" onclick="...
348     </div>
349 </div>
350
351     <div id="add-vendor-view" class="view">
352         <div class="section-title" style="margin-bottom: 20px;">Add Vendor</div>
353         <div class="form-card">
354             <div class="form-grid">
355                 <div class="form-group"><label>Name</label><input type="text" id="vendor-name-
create" placeholder="Vendor name"></div>
356                 <div class="form-group"><label>Email</label><input type="email" id="vendor-email-
create" placeholder="vendor@company.com"></div>
357                 <div class="form-group"><label>Password</label><input type="password" id="vendor-
password-create" placeholder="Temporary password"></div>
358                 <div class="form-group"><label>Phone</label><input type="text" id="vendor-phone-
create" placeholder="+91 90000 00000"></div>
359                 <div class="form-group">
360                     <label>Role</label>
361                     <select id="vendor-role">
362                         <option value="TECH_VENDOR">tech vendor</option>
363                         <option value="FOOD_VENDOR">Food Vendor</option>
364                     </select>
365                 </div>
366             </div>
367             <div class="modal-actions" style="margin-top: 20px;"><button class="btn btn-primary"
onclick="createVendor()">Create Vendor</button><button class="btn btn-secondary" onclick="cl...
368             </div>
369         </div>
370
371     <!-- APPLY LEAVE VIEW -->
372     <div id="apply-leave-view" class="view">
373         <div class="section-title" style="margin-bottom: 20px;">Apply for Leave</div>
374         <div class="form-card">
375             <div class="form-grid">
376                 <div class="form-group full"><label>Logged In User</label><input type="email"
id="leave-email" placeholder="your.email@company.com" readonly></div>
377                 <div class="form-group"><label>Leave Type</label><select id="leave-type"><option
value="SICK">Sick Leave</option><option value="CASUAL">Casual Leave</option><option value="M...
378                     Sick leave is limited to <strong>1 day per month</strong>. If already used
this month, please select Unpaid Leave.
379                 </div>
380                 <div id="casual-leave-hint" style="display:none; margin-top:6px; font-
size:11px; color:var(--amber); background:var(--amber-dim); border:1px solid rgba(251,191,36,0.2); ...
381                     Casual leave is limited to <strong>1 day per month</strong>. If already
used this month, it will be automatically recorded as <strong>Unpaid Leave</strong>.
382                 </div>
383             </div>
384             <div class="form-group"><label>Start Date</label><input type="date" id="leave-
start"></div>
385             <div class="form-group"><label>End Date</label><input type="date" id="leave-

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end"></div>
386         <div class="form-group full"><label>Reason</label><textarea id="leave-reason"
rows="3" placeholder="Please provide reason for leave..."></textarea></div>
387     </div>
388     <div class="modal-actions" style="margin-top: 20px;"><button class="btn btn-primary"
onclick="applyLeave()">Submit Leave Request</button><button class="btn btn-secondary" onclie...
389     </div>
390 </div>
391
392 <!-- HOLIDAYS VIEW -->
393 <div id="holidays-view" class="view">
394     <div class="section-head">
395         <div class="section-title">Fixed Holidays</div>
396         <button class="btn btn-primary btn-sm" onclick="loadHolidays()">Refresh</button>
397     </div>
398
399 <!-- Add Holiday Form -->
400 <div class="form-card" style="margin-bottom: 24px;">
401     <div class="form-grid">
402         <div class="form-group">
403             <label>Holiday Name</label>
404             <input type="text" id="h-name" placeholder="e.g. Diwali, Christmas">
405         </div>
406         <div class="form-group">
407             <label>Date</label>
408             <input type="date" id="h-date">
409         </div>
410         <div class="form-group full">
411             <label>Description (optional)</label>
412             <input type="text" id="h-desc" placeholder="Brief description">
413         </div>
414     </div>
415     <div class="modal-actions" style="margin-top: 20px;">
416         <button class="btn btn-primary" onclick="addHoliday()">Add Holiday</button>
417         <button class="btn btn-secondary" onclick="clearHolidayForm()">Clear</button>
418     </div>
419 </div>
420
421 <!-- Holidays Table -->
422 <div class="table-wrap">
423     <table>
424         <thead>
425             <tr>
426                 <th>ID</th>
427                 <th>Name</th>
428                 <th>Date</th>
429                 <th>Description</th>
430                 <th>Action</th>
431             </tr>
432         </thead>
433         <tbody id="holidays-list">
434             <tr><td colspan="5" class="empty-state">Loading...</td></tr>
435         </tbody>
436     </table>
437 </div>
438 </div>
439
440 <!-- MY LEAVES VIEW -->
441 <div id="my-leaves-view" class="view">
442     <div class="section-head"><div class="section-title">My Leave Requests</div><button
class="btn btn-primary btn-sm" onclick="loadMyLeaves()">Refresh</button></div>
443     <div class="table-wrap"><table><thead><tr><th>ID</th><th>Type</th><th>Start
Date</th><th>End Date</th><th>Status</th><th>Reason</th><th>Action</th></tr></thead><tbody id="my-leaves-...
444     </div>
445
446 <!-- PENDING LEAVES VIEW -->
447 <div id="pending-leaves-view" class="view">
448     <div class="section-head"><div class="section-title">Pending Leave Approvals</div><button
class="btn btn-primary btn-sm" onclick="loadPendingLeaves()">Refresh</button></div>
449     <div class="table-
wrap"><table><thead><tr><th>ID</th><th>Employee</th><th>Type</th><th>Start</th><th>End</th><th>Reason</th><th>Ac
tions</th></tr></thead><tbody id="pending-leaves-li...
450     </div>
451
452 <!-- RESTAURANTS VIEW -->
453 <div id="restaurants-view" class="view">

```

```

454         <div class="section-head"><div class="section-title">Restaurants</div><div><select
id="slot-filter" onchange="loadRestaurants()"><option value="">All Slots</option><option value="BR...
455         <div class="table-wrap"><table><thead><tr><th>Name</th><th>Address</th><th>Meal
Slots</th><th>Status</th><th>Vendor</th></tr></thead><tbody id="restaurants-list"></tbody></table></div>
456     </div>
457
458     <!-- MY SUBSCRIPTIONS VIEW -->
459     <div id="subscriptions-view" class="view">
460         <div class="section-head"><div class="section-title">My Meal Subscriptions</div><button
class="btn btn-primary btn-sm" onclick="loadSubscriptions()">Refresh</button></div>
461         <div class="table-wrap"><table><thead><tr><th>ID</th><th>Restaurant</th><th>Meal
Slot</th><th>Schedule</th><th>Status</th><th>Next Delivery</th><th>Actions</th></tr></thead><tbody i...
462     </div>
463
464     <!-- ADD SUBSCRIPTION VIEW -->
465     <div id="add-subscription-view" class="view">
466         <div class="section-title" style="margin-bottom: 20px;">Create Meal Subscription</div>
467         <div class="form-card">
468             <div class="form-grid">
469                 <div class="form-group full">
470                     <label>Employee Email</label>
471                     <input type="email" id="sub-email" placeholder="employee@company.com">
472                 </div>
473                 <div class="form-group full">
474                     <label>Restaurant</label>
475                     <select id="sub-restaurant-select">
476                         <option value="">-- Select Restaurant --</option>
477                     </select>
478                 </div>
479                 <div class="form-group full">
480                     <label>Meal Slots</label>
481                     <div id="meal-slot-checkboxes" style="display:flex; gap:16px; flex-wrap:wrap;
margin-top:6px;">
482                         <label style="display:flex;align-items:center;gap:6px;text-
transform:none;font-size:13px;font-weight:400;color:var(--text);">
483                             <input type="checkbox" value="BREAKFAST"
style="width:15px;height:15px;padding:0;background:var(--bg3);"> Breakfast
484                         </label>
485                         <label style="display:flex;align-items:center;gap:6px;text-
transform:none;font-size:13px;font-weight:400;color:var(--text);">
486                             <input type="checkbox" value="LUNCH"
style="width:15px;height:15px;padding:0;background:var(--bg3);"> Lunch
487                         </label>
488                         <label style="display:flex;align-items:center;gap:6px;text-
transform:none;font-size:13px;font-weight:400;color:var(--text);">
489                             <input type="checkbox" value="DINNER"
style="width:15px;height:15px;padding:0;background:var(--bg3);"> Dinner
490                         </label>
491                     </div>
492                 </div>
493                 <div class="form-group">
494                     <label>Schedule Type</label>
495                     <select id="sub-schedule">
496                         <option value="DAILY">Daily</option>
497                         <option value="WEEKLY">Weekly</option>
498                     </select>
499                 </div>
500                 <div class="form-group" id="day-of-week-group" style="display:none;">
501                     <label>Day of Week</label>
502                     <select id="sub-day">
503                         <option value="MONDAY">Monday</option>
504                         <option value="TUESDAY">Tuesday</option>
505                         <option value="WEDNESDAY">Wednesday</option>
506                         <option value="THURSDAY">Thursday</option>
507                         <option value="FRIDAY">Friday</option>
508                         <option value="SATURDAY">Saturday</option>
509                         <option value="SUNDAY">Sunday</option>
510                     </select>
511                 </div>
512             </div>
513             <div class="modal-actions" style="margin-top: 20px;">
514                 <button class="btn btn-primary" onclick="addSubscription()">Create
Subscription</button>
515                 <button class="btn btn-secondary" onclick="clearSubscriptionForm()">Clear</button>
516             </div>
517         </div>

```

```

518         </div>
519
520         <!-- DELIVERIES VIEW -->
521         <div id="deliveries-view" class="view">
522             <div class="section-head"><div class="section-title">My Deliveries</div><button class="btn
523 btn-primary btn-sm" onclick="loadDeliveries()">Refresh</button></div>
524             <div class="table-wrap"><table><thead><tr><th>ID</th><th>Subscription ID</th><th>Meal
525 Slot</th><th>Scheduled Time</th><th>Status</th></tr></thead><tbody id="deliveries-list"></tbody>...
526         </div>
527
528         <!-- VENDORS VIEW -->
529         <div id="vendors-view" class="view">
530             <div class="section-head"><div class="section-title">Vendors</div><button class="btn btn-
531 primary btn-sm" onclick="loadVendors()">Refresh</button></div>
532             <div class="table-
533 wrap"><table><thead><tr><th>ID</th><th>Name</th><th>Email</th><th>Phone</th><th>Registered</th><th>Actions</th><
534 /tr></thead><tbody id="vendors-list"></tbody></table>...
535         </div>
536
537         <!-- DEVICES VIEW -->
538         <div id="devices-view" class="view">
539             <div class="section-head">
540                 <div class="section-title">Unassigned Devices</div>
541                 <button class="btn btn-primary btn-sm"
542 onclick="loadUnassignedDevices()">Refresh</button>
543             </div>
544             <div class="table-wrap">
545                 <table>
546                     <thead>
547                         <tr>
548                             <th>ID</th>
549                             <th>Device Name</th>
550                             <th>Brand</th>
551                             <th>Type</th>
552                             <th>Warranty</th>
553                             <th>Vendor</th>
554                             <th>Actions</th>
555                         </tr>
556                     </thead>
557                     <tbody id="devices-list"></tbody>
558                 </table>
559             </div>
560         </div>
561
562         <!-- ASSIGN DEVICE MODAL -->
563         <div id="assign-modal" class="modal-overlay">
564             <div class="modal">
565                 <div class="modal-title">Assign Device to Employee</div>
566                 <input type="hidden" id="assign-device-id">
567                 <div class="form-group" style="margin-bottom:12px">
568                     <label>Device</label>
569                     <input type="text" id="assign-device-name" readonly style="color:var(--text2)">
570                 </div>
571                 <div class="form-group" style="margin-bottom:12px">
572                     <label>Employee</label>
573                     <select id="assign-employee-select">
574                         <option value="">-- Select Employee --</option>
575                     </select>
576                 </div>
577                 <div class="modal-actions">
578                     <button class="btn btn-primary" onclick="confirmAssign()">Assign</button>
579                     <button class="btn btn-secondary" onclick="closeAssignModal()">Cancel</button>
580                 </div>
581             </div>
582         </div>
583
584         <!-- MY DEVICES VIEW -->
585         <div id="my-devices-view" class="view">
586             <div class="section-head">
587                 <div class="section-title">My Assigned Devices</div>
588                 <button class="btn btn-primary btn-sm"
589 onclick="loadMyAssignedDevices()">Refresh</button>
590             </div>
591             <div class="table-wrap">
592                 <table>
593                     <thead>

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```

587         <tr>
588             <th>ID</th>
589             <th>Device Name</th>
590             <th>Brand</th>
591             <th>Type</th>
592             <th>Warranty</th>
593             <th>Assigned On</th>
594             <th>Status</th>
595         </tr>
596     </thead>
597     <tbody id="my-devices-list"></tbody>
598 </table>
599 </div>
600 </div>
601
602 <!-- MY SERVICE REQUESTS VIEW -->
603 <div id="service-requests-view" class="view">
604     <div class="section-head">
605         <div class="section-title">My Service Requests</div>
606         <button class="btn btn-primary btn-sm" onclick="loadServiceRequests()">Refresh</button>
607     </div>
608     <div class="table-wrap" style="margin-bottom:24px">
609         <table>
610             <thead><tr>
611                 <th>ID</th><th>Device</th><th>Type</th>
612                 <th>Description</th><th>Urgent</th><th>Status</th><th>Raised At</th>
613             </tr></thead>
614             <tbody id="service-requests-list"></tbody>
615         </table>
616     </div>
617
618     <div class="section-title" style="margin-bottom:16px">Raise a New Request</div>
619     <div class="form-card">
620         <div class="form-grid">
621             <div class="form-group full">
622                 <label>Device</label>
623                 <select id="sr-device-select">
624                     <option value="">-- Select your device --</option>
625                 </select>
626             </div>
627             <div class="form-group">
628                 <label>Request Type</label>
629                 <select id="sr-type">
630                     <option value="REPAIR">Repair</option>
631                     <option value="REPLACEMENT">Replacement</option>
632                     <option value="UPGRADE">Upgrade</option>
633                     <option value="MAINTENANCE">Maintenance</option>
634                     <option value="RETURN">Return</option>
635                 </select>
636             </div>
637             <div class="form-group" style="display:flex;align-items:center;gap:10px;padding-top:20px">
638                 <input type="checkbox" id="sr-urgent" style="width:16px;height:16px;padding:0">
639                 <label style="text-transform:none;font-size:13px;font-weight:400;color:var(--text);cursor:pointer" for="sr-urgent">Mark as Urgent</label>
640             </div>
641             <div class="form-group full">
642                 <label>Issue Description</label>
643                 <textarea id="sr-description" rows="3" placeholder="Describe the issue in detail..."></textarea>
644             </div>
645         </div>
646         <div class="modal-actions" style="margin-top:20px">
647             <button class="btn btn-primary" onclick="submitServiceRequest()">Submit
Request</button>
648             <button class="btn btn-secondary"
onclick="clearServiceRequestForm()">Clear</button>
649         </div>
650     </div>
651 </div>
652
653 <!-- MY CALENDAR VIEW -->
654 <div id="my-calendar-view" class="view">
655     <div class="cal-section">
656         <div class="cal-section-head">
657             <div>

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658             <div class="cal-section-title">My Leave Calendar</div>
659             <div class="cal-section-sub">All approved leaves and public holidays</div>
660         </div>
661         <button class="btn btn-secondary btn-sm" onclick="refreshCalendar()">
Refresh</button>
662     </div>
663
664     <!-- Balance Cards -->
665     <div class="leave-balance-grid">
666         <div class="leave-bal-card sick">
667             <span class="lbc-icon"></span>
668             <span class="lbc-label">Sick Leave</span>
669             <span class="lbc-count" id="cal-sick-remaining">--</span>
670             <span class="lbc-sub" id="cal-sick-sub">loading...</span>
671         </div>
672         <div class="leave-bal-card casual">
673             <span class="lbc-icon"></span>
674             <span class="lbc-label">Casual Leave</span>
675             <span class="lbc-count" id="cal-casual-remaining">--</span>
676             <span class="lbc-sub" id="cal-casual-sub">loading...</span>
677         </div>
678         <div class="leave-bal-card mat">
679             <span class="lbc-icon"></span>
680             <span class="lbc-label">Maternity Leave</span>
681             <span class="lbc-count" id="cal-mat-remaining">--</span>
682             <span class="lbc-sub" id="cal-mat-sub">loading...</span>
683         </div>
684         <div class="leave-bal-card total">
685             <span class="lbc-icon"></span>
686             <span class="lbc-label">Total Used</span>
687             <span class="lbc-count" id="cal-total-used">--</span>
688             <span class="lbc-sub" id="cal-total-sub">loading...</span>
689         </div>
690     </div>
691
692     <!-- Legend -->
693     <div class="cal-legend">
694         <span class="legend-label">Legend:</span>
695         <div class="legend-item"><div class="legend-dot"
style="background:#36d399"></div>Approved</div>
696         <div class="legend-item"><div class="legend-dot"
style="background:#f9c74f"></div>Pending</div>
697         <div class="legend-item"><div class="legend-dot"
style="background:#f87171"></div>Rejected</div>
698         <div class="legend-item"><div class="legend-dot"
style="background:#a78bfa"></div>Holiday</div>
699     </div>
700
701     <!-- Calendar -->
702     <div id="employee-calendar"></div>
703 </div>
704 </div>
705
706 <!-- MY PROFILE VIEW -->
707 <div id="my-profile-view" class="view">
708     <div class="section-title" style="margin-bottom:20px">My Profile</div>
709
710     <!-- Profile Info -->
711     <div class="form-card" style="max-width:100%;margin-bottom:24px">
712         <div class="form-grid">
713             <div class="form-group">
714                 <label>Name</label>
715                 <input type="text" id="p-name" readonly style="color:var(--text2)">
716             </div>
717             <div class="form-group">
718                 <label>Email</label>
719                 <input type="text" id="p-email" readonly style="color:var(--text2)">
720             </div>
721             <div class="form-group">
722                 <label>Department</label>
723                 <input type="text" id="p-dept" readonly style="color:var(--text2)">
724             </div>
725             <div class="form-group">
726                 <label>Role</label>
727                 <input type="text" id="p-role" readonly style="color:var(--text2)">
728             </div>

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729         <div class="form-group">
730             <label>Status</label>
731             <input type="text" id="p-status" readonly style="color:var(--text2)">
732         </div>
733         <div class="form-group">
734             <label>Timezone</label>
735             <input type="text" id="p-tz" readonly style="color:var(--text2)">
736         </div>
737     </div>
738 </div>
739
740 <!-- Maternity Section – only visible for female admins -->
741 <div id="maternity-apply-btn" style="display:none">
742     <div class="form-card" style="max-width:100%">
743         <div class="section-title" style="margin-bottom:20px"> Maternity Leave</div>
744         <div class="form-grid">
745             <div class="form-group">
746                 <label>Start Date</label>
747                 <input type="date" id="mat-start-date">
748             </div>
749             <div class="form-group">
750                 <label>End Date (Auto)</label>
751                 <input type="text" id="mat-end-date"
752                     placeholder="Auto: start + 180 days"
753                     disabled style="color:var(--text2)">
754             </div>
755             <div class="form-group full">
756                 <label>Reason / Notes</label>
757                 <textarea id="mat-reason" rows="3"
758                     placeholder="Optional notes..."></textarea>
759             </div>
760             <div class="form-group full">
761                 <div style="background:var(--accent-dim);
762                     border:1px solid rgba(79,140,255,0.2);
763                     border-radius:8px;padding:12px 16px;
764                     font-size:12px;color:var(--text2)">
765                     Maternity leave is fixed at
766                     <strong style="color:var(--accent)">180 days</strong>.
767                     End date is auto-calculated. You can apply
768                     <strong style="color:var(--accent)">once per year</strong>.
769                 </div>
770             </div>
771         </div>
772         <div class="modal-actions" style="margin-top:20px">
773             <button class="btn btn-primary" onclick="applyMaternityLeave()">
774                 Submit Maternity Leave
775             </button>
776         </div>
777     </div>
778 </div>
779 </div>
780
781 <!-- ===== ADMIN SERVICE REQUESTS VIEW ===== -->
782 <div id="admin-service-requests-view" class="view">
783     <div class="section-head">
784         <div class="section-title">All Open Service Requests</div>
785         <button class="btn btn-primary btn-sm"
786             onclick="loadAdminServiceRequests()">Refresh</button>
787     </div>
788     <div class="table-wrap">
789         <table>
790             <thead>
791                 <tr>
792                     <th>ID</th>
793                     <th>Employee</th>
794                     <th>Device</th>
795                     <th>Type</th>
796                     <th>Description</th>
797                     <th>Urgent</th>
798                     <th>Status</th>
799                     <th>Raised At</th>
800                     <th>Actions</th>
801                 </tr>
802             </thead>
803             <tbody id="admin-service-requests-list"></tbody>
804         </table>

```



```

804         </div>
805     </div>
806
807 </div>
808 </div>
809 </div>
810
811 <!-- Approve/Reject Modal -->
812 <div id="action-modal" class="modal-overlay">
813     <div class="modal">
814         <div class="modal-title" id="modal-title">Approve Leave Request</div>
815         <input type="hidden" id="action-leave-id">
816         <input type="hidden" id="action-type">
817         <div class="form-group"><label>Reviewed By</label><input type="email" id="manager-email"
placeholder="admin@company.com" readonly></div>
818         <div class="form-group"><label>Remarks</label><textarea id="action-remarks" rows="3"
placeholder="Add remarks..."></textarea></div>
819         <div class="modal-actions"><button class="btn btn-primary" id="action-btn"
onclick="submitApproval()">Approve</button><button class="btn btn-secondary"
onclick="closeModal()">Cancel</button>...
820     </div>
821 </div>
822
823 <!-- Cancel Leave Modal -->
824 <div id="cancel-modal" class="modal-overlay">
825     <div class="modal">
826         <div class="modal-title">Cancel Leave Request</div>
827         <input type="hidden" id="cancel-leave-id">
828         <div class="form-group"><label>Cancelled By</label><input type="email" id="cancel-email"
placeholder="admin@company.com" readonly></div>
829         <div class="modal-actions"><button class="btn btn-danger" onclick="confirmCancelLeave()">Cancel
Leave</button><button class="btn btn-secondary" onclick="closeCancelModal()">Close</button></div>
830     </div>
831 </div>
832
833 <!-- Admin Service Request Action Modal -->
834 <div id="admin-sr-modal" class="modal-overlay">
835     <div class="modal">
836         <div class="modal-title">Update Service Request <span id="admin-sr-modal-id" style="color:var(--
accent)"></span></div>
837         <input type="hidden" id="admin-sr-id">
838
839         <div class="form-group" style="margin-bottom:14px">
840             <label>Device</label>
841             <input type="text" id="admin-sr-device-display" readonly style="color:var(--text2)">
842         </div>
843         <div class="form-group" style="margin-bottom:14px">
844             <label>Employee</label>
845             <input type="text" id="admin-sr-employee-display" readonly style="color:var(--text2)">
846         </div>
847         <div class="form-group" style="margin-bottom:14px">
848             <label>Update Status</label>
849             <select id="admin-sr-status">
850                 <option value="SENT_FOR_REPAIR">Sent for Repair – device will be marked Under
Repair</option>
851                 <option value="REPAIR_DONE">Repair Done</option>
852                 <option value="REJECTED">Rejected</option>
853             </select>
854         </div>
855         <div class="form-group" style="margin-bottom:4px">
856             <label>Admin Remarks</label>
857             <textarea id="admin-sr-remarks" rows="3" placeholder="Add remarks or instructions for the
vendor..."></textarea>
858         </div>
859         <div id="admin-sr-repair-hint" style="font-size:11px;color:var(--amber);margin-
top:6px;display:none;">
860             Selecting "Sent for Repair" will automatically mark the device as <strong>Under
Repair</strong>.
861         </div>
862
863         <div class="modal-actions">
864             <button class="btn btn-primary" onclick="submitAdminSRUpdate()">Update Request</button>
865             <button class="btn btn-secondary" onclick="closeAdminSRModal()">Cancel</button>
866         </div>
867     </div>
868 </div>

```

```

869
870 <div class="cal-tooltip" id="cal-tooltip" style="display:none"></div>
871
872 <script>
873     const API = 'http://localhost:8080';
874     let currentUser = null;
875     const ROLE_DASHBOARDS = {
876         EMPLOYEE: '/employee-dashboard.html',
877         MANAGER: '/manager-dashboard.html',
878         VENDOR: '/vendor-dashboard.html',
879         ADMIN: '/dashboard.html'
880     };
881
882     (async function initAuth() {
883         const token = localStorage.getItem('jwt_token');
884         const authType = localStorage.getItem('auth_type');
885         if (!token && authType !== 'session') {
886             window.location.href = '/login.html';
887             return;
888         }
889
890         try {
891             const response = await fetch(`${API}/employee/me`, {
892                 headers: getAuthHeader(),
893                 credentials: 'include'
894             });
895
896             if (response.status === 401) { window.location.href = '/login.html'; return; }
897             if (!response.ok) { window.location.href = '/login.html'; return; }
898
899             const employee = await response.json();
900             const role = employee.role || 'EMPLOYEE';
901             if (role !== 'ADMIN') {
902                 window.location.href = ROLE_DASHBOARDS[role] || '/employee-dashboard.html';
903                 return;
904             }
905
906             currentUser = employee;
907             localStorage.setItem('last_role', role);
908             document.getElementById('username-display').textContent = employee.email || employee.name ||
909 'admin';
910             document.getElementById('leave-email').value = employee.email || '';
911             document.getElementById('manager-email').value = employee.email || '';
912             document.getElementById('cancel-email').value = employee.email || '';
913             loadDashboard();
914         } catch (e) {
915             window.location.href = '/login.html';
916         }
917     })();
918
919     function getAuthHeader() {
920         const token = localStorage.getItem('jwt_token');
921         return token ? { 'Authorization': `Bearer ${token}` } : {};
922     }
923
924     async function apiFetch(path, opts = {}) {
925         try {
926             const response = await fetch(`${API}${path}`, {
927                 ...opts,
928                 headers: {
929                     'Content-Type': 'application/json',
930                     ...getAuthHeader(),
931                     ...opts.headers
932                 },
933                 credentials: 'include'
934             });
935
936             if (response.status === 401) {
937                 localStorage.removeItem('jwt_token');
938                 localStorage.removeItem('auth_type');
939                 showAlert('Session expired. Please login again.', 'error');
940                 setTimeout(() => window.location.href = '/login.html', 1500);
941                 throw new Error('Unauthorized');
942             }
943
944             if (!response.ok) {

```

```

944         const error = await response.text();
945         throw new Error(error || 'Request failed');
946     }
947
948     const text = await response.text();
949     if (!text) return null;
950     try { return JSON.parse(text); } catch { return text; }
951 } catch (e) {
952     if (!e.message.includes('Unauthorized')) showAlert(e.message, 'error');
953     throw e;
954 }
955 }
956
957 function showAlert(message, type) {
958     const alert = document.createElement('div');
959     alert.className = `alert alert-${type}`;
960     alert.textContent = message;
961     document.body.appendChild(alert);
962     setTimeout(() => alert.remove(), 3000);
963 }
964
965 function showView(viewName, event) {
966     document.querySelectorAll('.view').forEach(v => v.classList.remove('active'));
967     document.getElementById(`${viewName}-view`).classList.add('active');
968     document.querySelectorAll('.nav-item').forEach(n => n.classList.remove('active'));
969     if (event && event.target) event.target.classList.add('active');
970     document.getElementById('pageTitle').textContent = viewName.split('-').map(w =>
971         w.charAt(0).toUpperCase() + w.slice(1)).join(' ');
972     const loaders = {
973         'dashboard': loadDashboard,
974         'employees': loadEmployees,
975         'add-employee': loadManagersDropdown,
976         'holidays': loadHolidays,
977         'add-manager': () => {},
978         'add-vendor': () => {},
979         'my-leaves': loadMyLeaves,
980         'pending-leaves': loadPendingLeaves,
981         'restaurants': loadRestaurants,
982         'subscriptions': loadSubscriptions,
983         'add-subscription': loadRestaurantDropdown,
984         'deliveries': loadDeliveries,
985         'vendors': loadVendors,
986         'devices': loadUnassignedDevices,
987         'my-devices': loadMyAssignedDevices,
988         'service-requests': loadServiceRequests,
989         'admin-service-requests': loadAdminServiceRequests,
990         'my-profile': loadAdminProfile,
991         'my-calendar': loadMyCalendar// ADD THIS
992     };
993     if (loaders[viewName]) loaders[viewName]();
994 }
995
996 // ===== DASHBOARD =====
997 async function loadDashboard() {
998     try {
999         const [employees, leaves] = await Promise.all([
1000             apiFetch('/employee'),
1001             apiFetch('/leave_requests')
1002         ]);
1003         document.getElementById('total-employees').textContent = employees.length;
1004         document.getElementById('active-employees').textContent = employees.filter(e => e.status ===
1005 'ACTIVE').length;
1006         document.getElementById('onleave-employees').textContent = employees.filter(e => e.status ===
1007 'ON_LEAVE').length;
1008         document.getElementById('pending-leaves').textContent = leaves.filter(l => l.status ===
1009 'PENDING').length;
1010         const recent = leaves.slice(-5).reverse();
1011         const tbody = document.getElementById('recent-leaves');
1012         if (!recent.length) {
1013             tbody.innerHTML = '<tr><td colspan="5" class="empty-state">No leave requests</td></tr>';
1014         } else {
1015             tbody.innerHTML = recent.map(l => `<tr>
1016                 <td>${l.employeeId || 'N/A'}</td>
1017                 <td><span class="chip">${l.leaveType || 'N/A'}</span></td>
1018                 <td>${l.startDate || 'N/A'}</td>
1019                 <td>${l.endDate || 'N/A'}</td>

```

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1017         <td><span class="chip ${l.status?.toLowerCase()}">${l.status || 'N/A'}</span></td>
1018     </tr>`).join('');
1019     }
1020     } catch (e) { console.error(e); }
1021 }
1022
1023 // ===== EMPLOYEES =====
1024 async function loadEmployees() {
1025     try {
1026         const employees = await apiFetch('/employee');
1027         const tbody = document.getElementById('employees-list');
1028         if (!employees.length) {
1029             tbody.innerHTML = '<tr><td colspan="8" class="empty-state">No employees found</td></tr>';
1030         } else {
1031             tbody.innerHTML = employees.map(e => `<tr>
1032                 <td>${e.employeeId}</td>
1033                 <td>${e.name || 'N/A'}</td>
1034                 <td>${e.email || 'N/A'}</td>
1035                 <td>${e.dept || 'N/A'}</td>
1036                 <td><span class="chip ${e.role?.toLowerCase()}">${e.role || 'N/A'}</span></td>
1037                 <td><span class="chip ${e.status?.toLowerCase()}">${e.status || 'N/A'}</span></td>
1038                 <td>${e.timezone || 'N/A'}</td>
1039                 <td><button class="btn btn-danger btn-sm"
onclick="deleteEmployee(${e.employeeId})">Delete</button></td>
1040             </tr>`).join('');
1041         }
1042     } catch (e) { console.error(e); }
1043 }
1044
1045 async function loadManagersDropdown() {
1046     try {
1047         const managers = await apiFetch('/admin/managers');
1048         const select = document.getElementById('employee-manager-select');
1049         select.innerHTML = '<option value="">-- Select Manager --</option>';
1050         (managers || []).forEach(m => {
1051             const opt = document.createElement('option');
1052             opt.value = m.managerId; // matches ManagerDTO field
1053             opt.textContent = `${m.name} - ${m.dept || 'N/A'} (${m.email})`;
1054             select.appendChild(opt);
1055         });
1056     } catch (e) {
1057         showAlert('Failed to load managers', 'error');
1058     }
1059 }
1060
1061
1062
1063 async function createEmployee() {
1064     const managerId = document.getElementById('employee-manager-select').value;
1065     const body = {
1066         name: document.getElementById('employee-name').value.trim(),
1067         email: document.getElementById('employee-email').value.trim(),
1068         password: document.getElementById('employee-password').value,
1069         gender: document.getElementById('employee-gender').value,
1070         timezone: document.getElementById('employee-timezone').value || 'UTC',
1071         managerId: parseInt(managerId)
1072     };
1073     if (!body.name || !body.email || !body.password) {
1074         showAlert('Please fill all employee fields', 'error');
1075         return;
1076     }
1077     if (!managerId) {
1078         showAlert('Please select a manager', 'error');
1079         return;
1080     }
1081     try {
1082         await apiFetch('/admin/employees', { method: 'POST', body: JSON.stringify(body) });
1083         showAlert('Employee created successfully!', 'success');
1084         clearEmployeeForm('employee');
1085         loadEmployees();
1086     } catch (e) {
1087         showAlert('Failed to create employee: ' + e.message, 'error');
1088     }
1089 }
1090
1091 async function createManager() {

```

```

1092     const body = {
1093         name: document.getElementById('manager-name-create').value.trim(),
1094         email: document.getElementById('manager-email-create').value.trim(),
1095         password: document.getElementById('manager-password-create').value,
1096         dept: document.getElementById('manager-dept-create').value.trim(),
1097         gender: document.getElementById('manager-gender').value,
1098         timezone: document.getElementById('manager-timezone-create').value || 'UTC'
1099     };
1100     if (!body.name || !body.email || !body.password || !body.dept || !body.gender) {
1101         showAlert('Please fill all manager fields', 'error');
1102         return;
1103     }
1104     try {
1105         await apiFetch('/admin/employees/manager', { method: 'POST', body: JSON.stringify(body) });
1106         showAlert('Manager created successfully!', 'success');
1107         clearEmployeeForm('manager');
1108         loadEmployees();
1109     } catch (e) { showAlert('Failed to create manager: ' + e.message, 'error'); }
1110 }
1111
1112 function clearEmployeeForm(type) {
1113     const fields = type === 'manager'
1114         ? ['manager-name-create', 'manager-email-create', 'manager-password-create', 'manager-dept-
1115 create', 'manager-timezone-create']
1116         : ['employee-name', 'employee-email', 'employee-password', 'employee-timezone'];
1117     fields.forEach(id => document.getElementById(id).value = '');
1118     if (type === 'employee') {
1119         document.getElementById('employee-manager-select').value = '';
1120     }
1121 }
1122
1123 async function deleteEmployee(employeeId) {
1124     if (!confirm('Delete employee ${employeeId}?')) return;
1125     try {
1126         await apiFetch(`/admin/employee/${employeeId}`, { method: 'DELETE' });
1127         showAlert('Employee deleted successfully!', 'success');
1128         loadEmployees();
1129         loadDashboard();
1130     } catch (e) { showAlert('Failed to delete employee: ' + e.message, 'error'); }
1131 }
1132
1133 // ===== LEAVE REQUESTS =====
1134 async function applyLeave() {
1135     const leaveType = document.getElementById('leave-type').value;
1136     const startDate = document.getElementById('leave-start').value;
1137     const endDate = document.getElementById('leave-end').value;
1138     const reason = document.getElementById('leave-reason').value.trim();
1139     if (!startDate || !endDate) { showAlert('Please fill all required fields', 'error'); return; }
1140     try {
1141         const data = await apiFetch('/leave_requests', {
1142             method: 'POST',
1143             body: JSON.stringify({ leaveType, startDate, endDate, reason })
1144         });
1145         if (data && data.warning) {
1146             showAlert(data.warning, 'warning'); // amber toast
1147         } else {
1148             showAlert('Leave request submitted successfully!', 'success');
1149         }
1150         clearLeaveForm();
1151         loadMyLeaves();
1152     } catch (e) { showAlert('Failed to submit: ' + e.message, 'error'); }
1153 }
1154
1155 document.getElementById('leave-type').addEventListener('change', function() {
1156     document.getElementById('sick-leave-hint').style.display =
1157         this.value === 'SICK' ? 'block' : 'none';
1158     document.getElementById('casual-leave-hint').style.display =
1159         this.value === 'CASUAL' ? 'block' : 'none';
1160 });
1161
1162 async function loadMyLeaves() {
1163     try {
1164         const data = await apiFetch('/leave_requests/employee');
1165         const leaves = data.requests || [];
1166         const tbody = document.getElementById('my-leaves-list');
1167         if (!leaves.length) {
1168             tbody.innerHTML = '<tr><td colspan="7" class="empty-state">No leave requests

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found</td></tr>';
1167         } else {
1168             tbody.innerHTML = leaves.map(l => `<tr>
1169                 <td>${l.leaveRequestId}</td>
1170                 <td><span class="chip">${l.leaveType}</span></td>
1171                 <td>${l.startDate}</td>
1172                 <td>${l.endDate}</td>
1173                 <td><span class="chip ${l.status?.toLowerCase()}>${l.status}</span></td>
1174                 <td style="max-width:200px;overflow:hidden;text-overflow:ellipsis;">${l.reason || '-
1175                 <td>${l.status === 'PENDING' ? `<button class="btn btn-danger btn-sm"
onclick="openCancelModal(${l.leaveRequestId})">Cancel</button>` : '-'}</td>
1176                 </tr>`).join('');
1177         }
1178     } catch (e) { console.error(e); }
1179 }
1180
1181 async function loadPendingLeaves() {
1182     try {
1183         const leaves = await apiFetch('/leave_requests/pending');
1184         const tbody = document.getElementById('pending-leaves-list');
1185         if (!leaves.length) {
1186             tbody.innerHTML = `<tr><td colspan="7" class="empty-state">No pending leave
requests</td></tr>`;
1187         } else {
1188             tbody.innerHTML = leaves.map(l => `<tr>
1189                 <td>${l.leaveRequestId}</td>
1190                 <td>${l.employeeName}</td>
1191                 <td><span class="chip">${l.leaveType}</span></td>
1192                 <td>${l.startDate}</td>
1193                 <td>${l.endDate}</td>
1194                 <td style="max-width:150px;overflow:hidden;text-overflow:ellipsis;">${l.reason || '-
1195                 <td class="action-row">
1196                     <button class="btn btn-primary btn-sm"
onclick="openActionModal(${l.leaveRequestId}, 'APPROVED')">Approve</button>
1197                     <button class="btn btn-danger btn-sm"
onclick="openActionModal(${l.leaveRequestId}, 'REJECTED')">Reject</button>
1198                 </td>
1199                 </tr>`).join('');
1200         }
1201     } catch (e) { console.error(e); }
1202 }
1203
1204 function openActionModal(leaveId, action) {
1205     document.getElementById('action-leave-id').value = leaveId;
1206     document.getElementById('action-type').value = action;
1207     document.getElementById('modal-title').textContent = `${action} Leave Request #${leaveId}`;
1208     document.getElementById('action-btn').textContent = action;
1209     document.getElementById('action-btn').className = action === 'APPROVED' ? 'btn btn-primary' : 'btn
btn-danger';
1210     document.getElementById('action-modal').classList.add('open');
1211 }
1212
1213 async function submitApproval() {
1214     const leaveRequestId = parseInt(document.getElementById('action-leave-id').value);
1215     const action = document.getElementById('action-type').value;
1216     const remarks = document.getElementById('action-remarks').value;
1217     try {
1218         await apiFetch('/leave_requests/approval', {
1219             method: 'PUT',
1220             body: JSON.stringify({ leaveRequestId, action, remarks })
1221         });
1222         showAlert(`Leave request ${action.toLowerCase()} successfully!`, 'success');
1223         closeModal();
1224         loadPendingLeaves();
1225     } catch (e) { showAlert('Failed: ' + e.message, 'error'); }
1226 }
1227
1228 function openCancelModal(leaveId) {
1229     document.getElementById('cancel-leave-id').value = leaveId;
1230     document.getElementById('cancel-modal').classList.add('open');
1231 }
1232
1233 async function confirmCancelLeave() {
1234     const leaveId = document.getElementById('cancel-leave-id').value;

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1235     try {
1236         await apiFetch(`/leave_requests/cancel/${leaveId}`, { method: 'PUT' });
1237         showAlert('Leave cancelled successfully!', 'success');
1238         closeCancelModal();
1239         loadMyLeaves();
1240     } catch (e) { showAlert('Failed: ' + e.message, 'error'); }
1241 }
1242
1243 function clearLeaveForm() {
1244     document.getElementById('leave-start').value = '';
1245     document.getElementById('leave-end').value = '';
1246     document.getElementById('leave-reason').value = '';
1247 }
1248
1249 // ===== RESTAURANTS =====
1250 async function loadRestaurants() {
1251     try {
1252         const slot = document.getElementById('slot-filter').value;
1253         const path = slot ? `/restaurants/by-slot?slot=${slot}` : '/restaurants';
1254         const restaurants = await apiFetch(path);
1255         const tbody = document.getElementById('restaurants-list');
1256         if (!restaurants.length) {
1257             tbody.innerHTML = '<tr><td colspan="5" class="empty-state">No restaurants
found</td></tr>';
1258         } else {
1259             tbody.innerHTML = restaurants.map(r => `<tr>
1260                 <td><strong>${r.name}</strong></td>
1261                 <td>${r.address || 'N/A'}</td>
1262                 <td>${r.supportedMealSlots ? r.supportedMealSlots.map(s => `<span
class="chip">${s}</span>`) : 'N/A'}</td>
1263                 <td><span class="chip ${r.active ? 'active' : 'inactive'}">${r.active ? 'ACTIVE' :
'INACTIVE'}</span></td>
1264                 <td>${r.vendorName || r.vendorId || 'N/A'}</td>
1265             </tr>`).join('');
1266         }
1267     } catch (e) { console.error(e); }
1268 }
1269
1270 // ===== SUBSCRIPTIONS =====
1271 async function loadRestaurantDropdown() {
1272     try {
1273         const restaurants = await apiFetch('/restaurants');
1274         const arr = Array.isArray(restaurants) ? restaurants : [];
1275         const select = document.getElementById('sub-restaurant-select');
1276         select.innerHTML = '<option value="">-- Select Restaurant --</option>';
1277         arr.filter(r => r.active).forEach(r => {
1278             const opt = document.createElement('option');
1279             opt.value = r.id || r.restaurantId;
1280             opt.textContent = r.name;
1281             opt.dataset.slots = JSON.stringify(r.supportedMealSlots || []);
1282             select.appendChild(opt);
1283         });
1284     } catch (e) { console.error('Failed to load restaurants:', e); }
1285 }
1286
1287 document.getElementById('sub-restaurant-select')?.addEventListener('change', function () {
1288     const selected = this.options[this.selectedIndex];
1289     const supported = selected?.dataset?.slots ? JSON.parse(selected.dataset.slots) : [];
1290     const boxes = document.querySelectorAll('#meal-slot-checkboxes input[type=checkbox]');
1291     boxes.forEach(cb => {
1292         const wrapper = cb.parentElement;
1293         if (!supported.length || supported.includes(cb.value)) {
1294             wrapper.style.display = 'flex'; wrapper.style.opacity = '1'; cb.disabled = false;
1295         } else {
1296             cb.checked = false; cb.disabled = true; wrapper.style.opacity = '0.3';
1297         }
1298     });
1299 });
1300
1301 async function loadSubscriptions() {
1302     try {
1303         const subscriptions = await apiFetch('/subscription/user');
1304         const arr = Array.isArray(subscriptions) ? subscriptions : [];
1305         const tbody = document.getElementById('subscriptions-list');
1306         tbody.innerHTML = arr.length
1307             ? arr.map(s => `<tr>

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1308         <td>${s.subscriptionId}</td>
1309         <td>${s.restaurantName || 'N/A'}</td>
1310         <td><span class="chip">${s.mealSlot}</span></td>
1311         <td>${s.scheduleType}</td>
1312         <td><span class="chip ${s.status?.toLowerCase()}>${s.status}</span></td>
1313         <td>${s.nextDeliveryTime ? new Date(s.nextDeliveryTime).toLocaleDateString() : 'Not
scheduled'}</td>
1314         <td class="action-row">
1315             ${s.status === 'ACTIVE' ? `<button class="btn btn-secondary btn-sm"
onclick="pauseSubscription(${s.subscriptionId})">Pause</button>` : ''}
1316             ${s.status === 'PAUSED' ? `<button class="btn btn-primary btn-sm"
onclick="resumeSubscription(${s.subscriptionId})">Resume</button>` : ''}
1317             <button class="btn btn-danger btn-sm"
onclick="expireSubscription(${s.subscriptionId})">Expire</button>
1318         </td>
1319     </tr>`).join('')
1320     : `<tr><td colspan="7" class="empty-state">No subscriptions found</td></tr>`;
1321 } catch (e) { console.error(e); }
1322 }
1323
1324 async function addSubscription() {
1325     const email = document.getElementById('sub-email').value.trim();
1326     const restaurantId = parseInt(document.getElementById('sub-restaurant-select').value);
1327     const scheduleType = document.getElementById('sub-schedule').value;
1328     const dayOfWeek = document.getElementById('sub-day').value;
1329     const selectedSlots = [...document.querySelectorAll('#meal-slot-checkboxes
input[type=checkbox]:checked')].map(cb => cb.value);
1330     if (!email) { showAlert('Please enter employee email', 'error'); return; }
1331     if (!restaurantId) { showAlert('Please select a restaurant', 'error'); return; }
1332     if (!selectedSlots.length) { showAlert('Please select at least one meal slot', 'error'); return; }
1333     const body = {
1334         scheduleType,
1335         mealSlots: selectedSlots,
1336         slotOrders: selectedSlots.map(slot => ({ mealSlot: slot, restaurantId }))
1337     };
1338     if (scheduleType === 'WEEKLY') body.dayOfWeek = dayOfWeek;
1339     try {
1340         await apiFetch('/subscription', { method: 'POST', body: JSON.stringify(body) });
1341         showAlert('Subscription created successfully!', 'success');
1342         clearSubscriptionForm();
1343         loadSubscriptions();
1344     } catch (e) { showAlert('Failed: ' + e.message, 'error'); }
1345 }
1346
1347 async function pauseSubscription(id) {
1348     try { await apiFetch(`/subscription/${id}/pause`, { method: 'PUT' }); showAlert('Subscription
paused', 'success'); loadSubscriptions(); }
1349     catch (e) { showAlert('Failed: ' + e.message, 'error'); }
1350 }
1351
1352 async function resumeSubscription(id) {
1353     try { await apiFetch(`/subscription/${id}/resume`, { method: 'PUT' }); showAlert('Subscription
resumed', 'success'); loadSubscriptions(); }
1354     catch (e) { showAlert('Failed: ' + e.message, 'error'); }
1355 }
1356
1357 async function expireSubscription(id) {
1358     try { await apiFetch(`/subscription/${id}/expire`, { method: 'PUT' }); showAlert('Subscription
expired', 'success'); loadSubscriptions(); }
1359     catch (e) { showAlert('Failed: ' + e.message, 'error'); }
1360 }
1361
1362 function clearSubscriptionForm() {
1363     document.getElementById('sub-email').value = '';
1364     document.getElementById('sub-restaurant-select').value = '';
1365     document.getElementById('sub-schedule').value = 'DAILY';
1366     document.getElementById('day-of-week-group').style.display = 'none';
1367     document.querySelectorAll('#meal-slot-checkboxes input[type=checkbox]').forEach(cb => {
1368         cb.checked = false; cb.disabled = false;
1369         cb.parentElement.style.opacity = '1'; cb.parentElement.style.display = 'flex';
1370     });
1371 }
1372
1373 document.getElementById('sub-schedule')?.addEventListener('change', function () {
1374     document.getElementById('day-of-week-group').style.display = this.value === 'WEEKLY' ? 'block' :
'none';

```



```

1375     });
1376
1377     // ===== DELIVERIES =====
1378     async function loadDeliveries() {
1379         try {
1380             const subscriptions = await apiFetch('/subscription/user');
1381             let allDeliveries = [];
1382             for (const sub of subscriptions) {
1383                 try {
1384                     const deliveries = await apiFetch(`/delivery/subscription/${sub.subscriptionId}`);
1385                     allDeliveries.push(...deliveries);
1386                 } catch (e) {}
1387             }
1388             const tbody = document.getElementById('deliveries-list');
1389             if (!allDeliveries.length) {
1390                 tbody.innerHTML = '<tr><td colspan="5" class="empty-state">No deliveries found</td></tr>';
1391             } else {
1392                 tbody.innerHTML = allDeliveries.map(d => `<tr>
1393                     <td>${d.id}</td>
1394                     <td>${d.subscription?.id || 'N/A'}</td>
1395                     <td><span class="chip">${d.mealSlot || 'N/A'}</span></td>
1396                     <td>${d.scheduledDeliveryTime ? new Date(d.scheduledDeliveryTime).toLocaleString() :
1397 'N/A'}</td>
1398                     <td><span class="chip ${d.status?.toLowerCase()}>${d.status ||
1399 'SCHEDULED'}</span></td>
1400                 </tr>`).join('');
1401             } catch (e) { console.error(e); }
1402         }
1403
1404     // ===== VENDORS =====
1405     async function loadVendors() {
1406         try {
1407             const vendors = await apiFetch('/vendors');
1408             const tbody = document.getElementById('vendors-list');
1409             if (!vendors.length) {
1410                 tbody.innerHTML = '<tr><td colspan="6" class="empty-state">No vendors found</td></tr>';
1411             } else {
1412                 tbody.innerHTML = vendors.map(v => `<tr>
1413                     <td>${v.id}</td>
1414                     <td><strong>${v.name}</strong></td>
1415                     <td>${v.email}</td>
1416                     <td>${v.phone || 'N/A'}</td>
1417                     <td>${v.registeredAt || 'N/A'}</td>
1418                     <td><button class="btn btn-danger btn-sm"
1419 onclick="deleteVendor(${v.id})">Delete</button></td>
1420                 </tr>`).join('');
1421             } catch (e) { console.error(e); }
1422         }
1423
1424     async function createVendor() {
1425         const body = {
1426             name: document.getElementById('vendor-name-create').value.trim(),
1427             email: document.getElementById('vendor-email-create').value.trim(),
1428             password: document.getElementById('vendor-password-create').value,
1429             phone: document.getElementById('vendor-phone-create').value.trim(),
1430             role: document.getElementById('vendor-role').value
1431         };
1432         if (!body.name || !body.email || !body.password || !body.phone) {
1433             showAlert('Please fill all vendor fields', 'error'); return;
1434         }
1435         try {
1436             await apiFetch('/vendors/register', { method: 'POST', body: JSON.stringify(body) });
1437             showAlert('Vendor created successfully!', 'success');
1438             clearVendorForm();
1439             loadVendors();
1440         } catch (e) { showAlert('Failed to create vendor: ' + e.message, 'error'); }
1441     }
1442
1443     function clearVendorForm() {
1444         ['vendor-name-create', 'vendor-email-create', 'vendor-password-create', 'vendor-phone-create']
1445             .forEach(id => document.getElementById(id).value = '');
1446     }
1447
1448     async function deleteVendor(vendorId) {

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1448         if (!confirm(`Delete vendor #${vendorId}?`)) return;
1449         try {
1450             await apiFetch(`/vendors/${vendorId}`, { method: 'DELETE' });
1451             showAlert('Vendor deleted successfully!', 'success');
1452             loadVendors();
1453         } catch (e) { showAlert('Failed to delete vendor: ' + e.message, 'error'); }
1454     }
1455
1456     // ===== MODAL HELPERS =====
1457     function closeModal() {
1458         document.getElementById('action-modal').classList.remove('open');
1459         document.getElementById('action-remarks').value = '';
1460         document.getElementById('manager-email').value = currentUser?.email || '';
1461     }
1462
1463     function closeCancelModal() {
1464         document.getElementById('cancel-modal').classList.remove('open');
1465         document.getElementById('cancel-email').value = currentUser?.email || '';
1466     }
1467
1468     // ===== DEVICES =====
1469     async function loadUnassignedDevices() {
1470         try {
1471             const devices = await apiFetch('/devices/unassigned');
1472             const tbody = document.getElementById('devices-list');
1473             if (!devices || !devices.length) {
1474                 tbody.innerHTML = `<tr><td colspan="7" class="empty-state">No unassigned
devices</td></tr>`;
1475                 return;
1476             }
1477             const today = new Date();
1478             tbody.innerHTML = devices.map(d => {
1479                 const warrantyExpiry = d.warrantyExpiryDate ? new Date(d.warrantyExpiryDate) : null;
1480                 const warrantyDisplay = warrantyExpiry
1481                     ? (warrantyExpiry > today
1482                         ? `<span style="color:var(--green);font-size:11px">
${d.warrantyExpiryDate}</span>`
1483                         : `<span style="color:var(--red);font-size:11px"> ${d.warrantyExpiryDate}</span>`)
1484                     : `<span style="color:var(--text3);font-size:11px">--</span>`;
1485                 return `<tr>
<td>#${d.id}</td>
<td><strong>${d.deviceName || '-'}</strong></td>
<td>${d.brand || '-'}</td>
<td><span class="chip">${d.deviceType || '-'}</span></td>
<td>${warrantyDisplay}</td>
<td>${d.vendorName || '-'}</td>
<td><button class="btn btn-primary btn-sm" onclick="openAssignModal(${d.id},
'${d.deviceName}')">Assign</button></td>
</tr>`;
1486             }).join('');
1487         } catch (e) { console.error(e); }
1488     }
1489
1490     async function openAssignModal(deviceId, deviceName) {
1491         document.getElementById('assign-device-id').value = deviceId;
1492         document.getElementById('assign-device-name').value = deviceName;
1493         try {
1494             const employees = await apiFetch('/employee');
1495             const select = document.getElementById('assign-employee-select');
1496             select.innerHTML = `<option value="">-- Select Employee --</option>`;
1497             employees.filter(e => e.status === 'ACTIVE').forEach(e => {
1498                 const opt = document.createElement('option');
1499                 opt.value = e.employeeId;
1500                 opt.textContent = `${e.name} (${e.email}) - ${e.dept || 'N/A'}`;
1501                 select.appendChild(opt);
1502             });
1503         } catch (e) { showAlert('Failed to load employees', 'error'); return; }
1504         document.getElementById('assign-modal').classList.add('open');
1505     }
1506
1507     async function confirmAssign() {
1508         const deviceId = document.getElementById('assign-device-id').value;
1509         const employeeId = document.getElementById('assign-employee-select').value;
1510         if (!employeeId) { showAlert('Please select an employee', 'error'); return; }
1511         try {
1512             const message = await apiFetch(`/devices/${deviceId}/assign`, {

```

```

1521         method: 'POST',
1522         body: JSON.stringify({ employeeId: parseInt(employeeId) })
1523     });
1524     showAlert(message || 'Device assigned successfully!', 'success');
1525     closeAssignModal();
1526     loadUnassignedDevices();
1527 } catch (e) { showAlert('Failed: ' + e.message, 'error'); }
1528 }
1529
1530 function closeAssignModal() {
1531     document.getElementById('assign-modal').classList.remove('open');
1532     document.getElementById('assign-employee-select').value = '';
1533     document.getElementById('assign-device-id').value = '';
1534     document.getElementById('assign-device-name').value = '';
1535 }
1536
1537 // ===== MY DEVICES =====
1538 async function loadMyAssignedDevices() {
1539     try {
1540         const devices = await apiFetch('/devices/my');
1541         const tbody = document.getElementById('my-devices-list');
1542         if (!devices || !devices.length) {
1543             tbody.innerHTML = '<tr><td colspan="7" class="empty-state">No devices assigned to
you</td></tr>';
1544             return;
1545         }
1546         const today = new Date();
1547         tbody.innerHTML = devices.map(d => {
1548             const warrantyExpiry = d.warrantyExpiryDate ? new Date(d.warrantyExpiryDate) : null;
1549             const warrantyDisplay = warrantyExpiry
1550                 ? (warrantyExpiry > today
1551                     ? `<span style="color:var(--green);font-size:11px">
${d.warrantyExpiryDate}</span>`
1552                     : `<span style="color:var(--red);font-size:11px"> Expired
${d.warrantyExpiryDate}</span>`)
1553                 : `<span style="color:var(--text3);font-size:11px">--</span>`;
1554             return `<tr>
1555                 <td>#${d.id}</td>
1556                 <td><strong>${d.deviceName || '-'}</strong></td>
1557                 <td>${d.brand || '-'}</td>
1558                 <td><span class="chip">${(d.deviceType || '').toLowerCase()}</span></td>
1559                 <td>${warrantyDisplay}</td>
1560                 <td style="font-size:12px;color:var(--text2)">${d.assignedDate || '-'}</td>
1561                 <td><span class="chip ${((d.deviceStatus || '').toLowerCase())}>${d.deviceStatus || '-'}</span></td>
1562             </tr>`;
1563         }).join('');
1564     } catch (e) { console.error(e); }
1565 }
1566
1567 // ===== MY SERVICE REQUESTS =====
1568 async function loadServiceRequests() {
1569     await Promise.all([loadServiceRequestsList(), loadMyDevicesForSR()]);
1570 }
1571
1572 async function loadServiceRequestsList() {
1573     try {
1574         const requests = await apiFetch('/service-requests/my');
1575         const tbody = document.getElementById('service-requests-list');
1576         if (!requests || !requests.length) {
1577             tbody.innerHTML = '<tr><td colspan="7" class="empty-state">No service requests
yet</td></tr>';
1578             return;
1579         }
1580         tbody.innerHTML = requests.map(r => `<tr>
1581             <td>#${r.id}</td>
1582             <td>${r.deviceName || '-'}</td>
1583             <td><span class="chip">${r.requestType}</span></td>
1584             <td style="max-width:200px;overflow:hidden;text-overflow:ellipsis;white-
space:nowrap">${r.issueDescription || '-'}</td>
1585             <td>${r.urgent ? `<span class="chip pending">YES</span>` : `<span style="color:var(--
text2)">No</span>`}</td>
1586             <td><span class="chip ${r.status?.toLowerCase()}>${r.status}</span></td>
1587             <td style="font-size:12px;color:var(--text2)">${r.raisedAt ? new
Date(r.raisedAt).toLocaleDateString() : '-'}</td>
1588         </tr>`).join('');
1589     } catch (e) { console.error(e); }

```

```

1590     }
1591
1592     async function loadMyDevicesForSR() {
1593         try {
1594             const devices = await apiFetch('/devices/my');
1595             const select = document.getElementById('sr-device-select');
1596             select.innerHTML = '<option value="">-- Select your device --</option>';
1597             (devices || []).forEach(d => {
1598                 const opt = document.createElement('option');
1599                 opt.value = d.id;
1600                 opt.textContent = `${d.deviceName} (${d.brand || d.deviceType || 'Device'})`;
1601                 select.appendChild(opt);
1602             });
1603         } catch (e) { console.error('Failed to load devices for SR', e); }
1604     }
1605
1606     async function submitServiceRequest() {
1607         const deviceId = parseInt(document.getElementById('sr-device-select').value);
1608         const requestType = document.getElementById('sr-type').value;
1609         const issueDescription = document.getElementById('sr-description').value.trim();
1610         const urgent = document.getElementById('sr-urgent').checked;
1611         if (!deviceId) { showAlert('Please select a device', 'error'); return; }
1612         if (!issueDescription) { showAlert('Please describe the issue', 'error'); return; }
1613         try {
1614             await apiFetch('/service-requests', {
1615                 method: 'POST',
1616                 body: JSON.stringify({ deviceId, requestType, issueDescription, urgent })
1617             });
1618             showAlert('Service request submitted!', 'success');
1619             clearServiceRequestForm();
1620             loadServiceRequestsList();
1621         } catch (e) { showAlert('Failed: ' + e.message, 'error'); }
1622     }
1623
1624     function clearServiceRequestForm() {
1625         document.getElementById('sr-device-select').value = '';
1626         document.getElementById('sr-type').value = 'REPAIR';
1627         document.getElementById('sr-description').value = '';
1628         document.getElementById('sr-urgent').checked = false;
1629     }
1630
1631     // ===== ADMIN SERVICE REQUESTS =====
1632     async function loadAdminServiceRequests() {
1633         try {
1634             const requests = await apiFetch('/service-requests/open');
1635             const tbody = document.getElementById('admin-service-requests-list');
1636             if (!requests || !requests.length) {
1637                 tbody.innerHTML = '<tr><td colspan="9" class="empty-state">No open service
requests</td></tr>';
1638                 return;
1639             }
1640             tbody.innerHTML = requests.map(r => `<tr>
1641                 <td>#${r.id}</td>
1642                 <td style="font-size:12px;color:var(--text2)">${r.raisedByName || r.employeeId ||
'-'}</td>
1643                 <td><strong>${r.deviceName || '-'}</strong></td>
1644                 <td><span class="chip">${r.requestType}</span></td>
1645                 <td style="max-width:180px;overflow:hidden;text-overflow:ellipsis;white-space:nowrap;font-
size:12px;color:var(--text2)" title="${r.issueDescription || ''}">${r.issueDescription || '...
1646                 <td>${r.urgent ? '<span class="chip pending">YES</span>' : '<span style="color:var(--
text2);font-size:12px">No</span>'}</td>
1647                 <td><span class="chip ${r.status?.toLowerCase()}>${r.status}</span></td>
1648                 <td style="font-size:12px;color:var(--text2)">${r.raisedAt ? new
Date(r.raisedAt).toLocaleDateString() : '-'}</td>
1649                 <td>
1650                     <button class="btn btn-primary btn-sm" onclick="openAdminSRModal(${r.id},
'${(r.deviceName || '').replace(/'/g, '\\')}', '${(r.raisedByName || r.employeeId || '').replace(/'/g,...
1651                     Update
1652                     </button>
1653                 </td>
1654             </tr>`).join('');
1655         } catch (e) { console.error(e); }
1656     }
1657
1658     function openAdminSRModal(requestId, deviceName, employeeName) {
1659         document.getElementById('admin-sr-id').value = requestId;

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1660     document.getElementById('admin-sr-modal-id').textContent = `#${requestId}`;
1661     document.getElementById('admin-sr-device-display').value = deviceName;
1662     document.getElementById('admin-sr-employee-display').value = employeeName;
1663     document.getElementById('admin-sr-status').value = 'SENT_FOR_REPAIR';
1664     document.getElementById('admin-sr-remarks').value = '';
1665     // Show hint for default selection
1666     document.getElementById('admin-sr-repair-hint').style.display = 'block';
1667     document.getElementById('admin-sr-modal').classList.add('open');
1668 }
1669
1670 // Show/hide the repair hint based on status selection
1671 document.getElementById('admin-sr-status')?.addEventListener('change', function () {
1672     document.getElementById('admin-sr-repair-hint').style.display =
1673         this.value === 'SENT_FOR_REPAIR' ? 'block' : 'none';
1674 });
1675
1676 async function submitAdminSRUpdate() {
1677     const serviceRequestId = parseInt(document.getElementById('admin-sr-id').value);
1678     const status = document.getElementById('admin-sr-status').value;
1679     const adminRemarks = document.getElementById('admin-sr-remarks').value.trim();
1680
1681     if (!adminRemarks) {
1682         showAlert('Please add remarks before updating', 'error');
1683         return;
1684     }
1685
1686     try {
1687         await apiFetch('/service-requests/admin', {
1688             method: 'PUT',
1689             body: JSON.stringify({ serviceRequestId, status, adminRemarks })
1690         });
1691         const statusLabel = status.replace(/_/g, ' ').toLowerCase();
1692         showAlert('Service request marked as "${statusLabel}" successfully!', 'success');
1693         closeAdminSRModal();
1694         loadAdminServiceRequests();
1695     } catch (e) { showAlert('Failed: ' + e.message, 'error'); }
1696 }
1697
1698 function closeAdminSRModal() {
1699     document.getElementById('admin-sr-modal').classList.remove('open');
1700     document.getElementById('admin-sr-id').value = '';
1701     document.getElementById('admin-sr-remarks').value = '';
1702     document.getElementById('admin-sr-repair-hint').style.display = 'none';
1703 }
1704
1705 // ===== LOGOUT =====
1706 function logout() {
1707     localStorage.removeItem('jwt_token');
1708     localStorage.removeItem('refresh_token');
1709     localStorage.removeItem('auth_type');
1710     localStorage.removeItem('last_role');
1711     fetch(`${API}/session/logout`, { method: 'POST', credentials: 'include' })
1712         .finally(() => {
1713             document.cookie = 'JSESSIONID=; expires=Thu, 01 Jan 1970 00:00:00 UTC; path=/;';
1714             window.location.href = '/google.html';
1715         });
1716 }
1717
1718 async function loadNotifCount() {
1719     try {
1720         const data = await apiFetch('/notifications/unread-count');
1721         const badge = document.getElementById('notif-badge');
1722
1723         // Handles: plain number, { count: N }, or { unreadCount: N }
1724         const count = typeof data === 'number'
1725             ? data
1726             : (data?.count ?? data?.unreadCount ?? 0);
1727
1728         if (count > 0) {
1729             badge.textContent = count;
1730             badge.style.display = 'flex';
1731         } else {
1732             badge.style.display = 'none';
1733         }
1734     } catch (e) { console.warn('Notif count error', e); }
1735 }

```

```

1736     async function toggleNotifPanel() {
1737         const panel = document.getElementById('notif-panel');
1738         if (panel.style.display === 'none') {
1739             panel.style.display = 'block';
1740             await loadNotifications();
1741             try { await apiFetch('/notifications/mark-read', { method: 'POST' }); } catch(e) {}
1742             document.getElementById('notif-badge').style.display = 'none';
1743         } else {
1744             panel.style.display = 'none';
1745         }
1746     }
1747
1748     async function loadNotifications() {
1749         try {
1750             const data = await apiFetch('/notifications');
1751             const list = document.getElementById('notif-list');
1752             list.innerHTML = (data && data.length)
1753                 ? data.map(n => `
1754                     <div class="notif-item ${n.read ? '' : 'unread'}">
1755                         <div>${n.message}</div>
1756                         <div class="notif-time">${new Date(n.createdAt).toLocaleString()}</div>
1757                     </div>`).join('')
1758                 : '<div class="notif-item">No notifications yet.</div>';
1759         } catch(e) { console.warn('Notif load error', e); }
1760     }
1761
1762     loadNotifCount();
1763     setInterval(loadNotifCount, 30000);
1764
1765     document.addEventListener('click', e => {
1766         const panel = document.getElementById('notif-panel');
1767         const wrapper = document.querySelector('.notif-wrapper');
1768         if (panel && wrapper && !panel.contains(e.target) && !wrapper.contains(e.target)) {
1769             panel.style.display = 'none';
1770         }
1771     });
1772
1773     // Set today's date as default for leave form
1774     document.getElementById('leave-start').valueAsDate = new Date();
1775     document.getElementById('leave-end').valueAsDate = new Date();
1776
1777     // ===== HOLIDAYS =====
1778     async function loadHolidays() {
1779         try {
1780             const year = new Date().getFullYear();
1781             const holidays = await apiFetch(`/holidays?year=${year}`);
1782             const tbody = document.getElementById('holidays-list');
1783             if (!holidays || !holidays.length) {
1784                 tbody.innerHTML = '<tr><td colspan="5" class="empty-state">No holidays added yet</td></tr>';
1785                 return;
1786             }
1787             tbody.innerHTML = holidays.map(h => `<tr>
1788                 <td>#${h.id}</td>
1789                 <td><strong>${h.name}</strong></td>
1790                 <td>${h.date}</td>
1791                 <td>${h.description || '-'}</td>
1792                 <td>
1793                     <button class="btn btn-danger btn-sm" onclick="deleteHoliday(${h.id})">Delete</button>
1794                 </td>
1795             </tr>`).join('');
1796         } catch(e) { console.error(e); }
1797     }
1798
1799     async function addHoliday() {
1800         const name = document.getElementById('h-name').value.trim();
1801         const date = document.getElementById('h-date').value;
1802         const description = document.getElementById('h-desc').value.trim();
1803         if (!name || !date) { showAlert('Please enter name and date', 'error'); return; }
1804         try {
1805             await apiFetch('/holidays', {
1806                 method: 'POST',
1807                 body: JSON.stringify({ name, date, description })
1808             });
1809             showAlert('Holiday added!', 'success');
1810             clearHolidayForm();
1811             loadHolidays();

```

```

1812     } catch(e) { showAlert('Failed: ' + e.message, 'error'); }
1813 }
1814
1815 async function deleteHoliday(id) {
1816     if (!confirm('Delete this holiday?')) return;
1817     try {
1818         await apiFetch(`/holidays/${id}`, { method: 'DELETE' });
1819         showAlert('Holiday deleted', 'success');
1820         loadHolidays();
1821     } catch(e) { showAlert('Failed: ' + e.message, 'error'); }
1822 }
1823
1824 function clearHolidayForm() {
1825     document.getElementById('h-name').value = '';
1826     document.getElementById('h-date').value = '';
1827     document.getElementById('h-desc').value = '';
1828 }
1829
1830 // ===== MY PROFILE =====
1831 async function loadAdminProfile() {
1832     try {
1833         const me = await apiFetch('/employee/me');
1834         if(me) {
1835             document.getElementById('p-name').value = me.name || '-';
1836             document.getElementById('p-email').value = me.email || '-';
1837             document.getElementById('p-dept').value = me.dept || '-';
1838             document.getElementById('p-role').value = me.role || 'ADMIN';
1839             document.getElementById('p-status').value = me.status || '-';
1840             document.getElementById('p-tz').value = me.timezone || '-';
1841
1842             // Show maternity section only for female admins
1843             document.getElementById('maternity-apply-btn').style.display =
1844                 me.gender === 'F' ? 'block' : 'none';
1845         }
1846     } catch(e) {
1847         showAlert('Could not load profile', 'error');
1848     }
1849 }
1850
1851 // Auto calculate maternity end date
1852 document.getElementById('mat-start-date').addEventListener('change', function(){
1853     if(this.value) {
1854         const end = new Date(this.value);
1855         end.setDate(end.getDate() + 179);
1856         document.getElementById('mat-end-date').value =
1857             end.toISOString().split('T')[0];
1858     } else {
1859         document.getElementById('mat-end-date').value = '';
1860     }
1861 });
1862
1863 // Submit maternity leave
1864 async function applyMaternityLeave() {
1865     const startDate = document.getElementById('mat-start-date').value;
1866     const reason = document.getElementById('mat-reason').value.trim();
1867
1868     if(!startDate) {
1869         showAlert('Please select a start date', 'error');
1870         return;
1871     }
1872
1873     try {
1874         await apiFetch('/leave_requests/maternity/apply', {
1875             method: 'POST',
1876             body: JSON.stringify({ startDate, reason })
1877         });
1878         showAlert('Maternity leave submitted successfully!', 'success');
1879         document.getElementById('mat-start-date').value = '';
1880         document.getElementById('mat-end-date').value = '';
1881         document.getElementById('mat-reason').value = '';
1882         loadMyLeaves();
1883     } catch(e) {
1884         showAlert('Failed: ' + e.message, 'error');
1885     }
1886 }
1887

```

```

1888 // ===== CALENDAR =====
1889 let employeeCalendar = null;
1890 let calendarInitialized = false;
1891
1892 const STATUS_COLORS = {
1893   APPROVED: { bg:'rgba(54,211,153,0.25)', border:'#36d399', text:'#36d399' },
1894   PENDING: { bg:'rgba(249,199,79,0.25)', border:'#f9c74f', text:'#f9c74f' },
1895   REJECTED: { bg:'rgba(248,113,113,0.25)', border:'#f87171', text:'#f87171' },
1896   HOLIDAY: { bg:'rgba(167,139,250,0.25)', border:'#a78bfa', text:'#a78bfa' },
1897 };
1898 const TYPE_LABEL = {
1899   SICK:' Sick', CASUAL:' Casual',
1900   MATERNITY:' Maternity', UNPAID:' Unpaid'
1901 };
1902
1903 async function loadMyCalendar() {
1904   await loadCalendarBalanceCards();
1905   if(!calendarInitialized) {
1906     initFullCalendar();
1907   } else {
1908     await refreshCalendarEvents();
1909   }
1910 }
1911
1912 async function loadCalendarBalanceCards() {
1913   ['cal-sick-remaining','cal-casual-remaining','cal-mat-remaining','cal-total-used']
1914     .forEach(id => document.getElementById(id).textContent = '--');
1915   try {
1916     const dashboard = await apiFetch('/api/employees/me/leave-dashboard');
1917     const balances = dashboard?.leaveBalances || {};
1918     let totalUsed = 0, totalAllowed = 0;
1919
1920     const sick = balances['SICK'] || {};
1921     const sickRem = Number(sick.available || 0);
1922     const sickUsed = Number(sick.total || 0) - sickRem;
1923     document.getElementById('cal-sick-remaining').textContent = sickRem;
1924     document.getElementById('cal-sick-sub').textContent = `${sickUsed} used of
1925     ${Number(sick.total||0)} days`;
1926
1927     const cas = balances['CASUAL'] || {};
1928     const casRem = Number(cas.available || 0);
1929     const casUsed = Number(cas.total || 0) - casRem;
1930     document.getElementById('cal-casual-remaining').textContent = casRem;
1931     document.getElementById('cal-casual-sub').textContent = `${casUsed} used of
1932     ${Number(cas.total||0)} days`;
1933
1934     const mat = balances['MATERNITY'] || {};
1935     const matRem = Number(mat.available || 0);
1936     const matUsed = Number(mat.total || 0) - matRem;
1937     document.getElementById('cal-mat-remaining').textContent = matRem;
1938     document.getElementById('cal-mat-sub').textContent = `${matUsed} used of ${Number(mat.total||0)}
1939     days`;
1940
1941     Object.values(balances).forEach(b => {
1942       totalUsed += (Number(b.total||0) - Number(b.available||0));
1943       totalAllowed += Number(b.total||0);
1944     });
1945     document.getElementById('cal-total-used').textContent = totalUsed;
1946     document.getElementById('cal-total-sub').textContent = `out of ${totalAllowed} days total`;
1947   } catch(e) {
1948     ['cal-sick-sub','cal-casual-sub','cal-mat-sub','cal-total-sub']
1949       .forEach(id => document.getElementById(id).textContent = 'unavailable');
1950   }
1951 }
1952
1953 async function buildCalendarEvents() {
1954   const events = [];
1955   try {
1956     const data = await apiFetch('/leave_requests/employee');
1957     const leaves = data?.requests || data || [];
1958     leaves.forEach(l => {
1959       const s = (l.status || 'PENDING').toUpperCase();
1960       const c = STATUS_COLORS[s] || STATUS_COLORS.PENDING;
1961       let endDate = l.endDate;
1962       if(endDate) {

```



```

1961         const d = new Date(endDate);
1962         d.setDate(d.getDate() + 1);
1963         endDate = d.toISOString().split('T')[0];
1964     }
1965     events.push({
1966         id: `leave-${l.leaveRequestId}`,
1967         title: `${TYPE_LABEL[l.leaveType] || l.leaveType}
1968         (${s.charAt(0)+s.slice(1).toLowerCase()})`,
1969         start: l.startDate, end: endDate, allDay: true,
1970         backgroundColor: c.bg, borderColor: c.border, textColor: c.text,
1971         extendedProps: { type:'leave', status:s, reason:l.reason||'-', leaveType:l.leaveType }
1972     });
1973 } catch(e) { console.warn('Could not load leaves for calendar'); }
1974
1975 try {
1976     const holidays = await apiFetch('/holidays');
1977     (holidays || []).forEach(h => {
1978         const c = STATUS_COLORS.HOLIDAY;
1979         events.push({
1980             id: `holiday-${h.id||Math.random()}`,
1981             title: ` ${h.name || 'Holiday'}`,
1982             start: h.date || h.holidayDate, allDay: true,
1983             backgroundColor: c.bg, borderColor: c.border, textColor: c.text,
1984             extendedProps: { type:'holiday', description: h.description||'' }
1985         });
1986     });
1987 } catch(e) { console.warn('Could not load holidays for calendar'); }
1988
1989 return events;
1990 }
1991
1992 function initFullCalendar() {
1993     employeeCalendar = new FullCalendar.Calendar(
1994         document.getElementById('employee-calendar'), {
1995             initialView: 'dayGridMonth',
1996             height: 'auto',
1997             headerToolbar: {
1998                 left:'prev,next today', center:'title', right:'dayGridMonth,dayGridWeek,listMonth'
1999             },
2000             buttonText: { today:'Today', month:'Month', week:'Week', listMonth:'List' },
2001             dayMaxEvents: 3,
2002             navLinks: true,
2003             events: [],
2004             eventMouseEnter: function(info) {
2005                 const ep = info.event.extendedProps;
2006                 const tt = document.getElementById('cal-tooltip');
2007                 let html = `<div class="tt-title">${info.event.title}</div>`;
2008                 if(ep.type === 'leave') {
2009                     html += `<div class="tt-row"> ${info.event.startStr}</div>`;
2010                     if(ep.reason && ep.reason !== '-')
2011                         html += `<div class="tt-row"> ${ep.reason.substring(0,60)}</div>`;
2012                     html += `<div class="tt-row">Status: <strong
2013 style="color:${STATUS_COLORS[ep.status]?.text||'#fff'}">${ep.status}</strong></div>`;
2014                 } else if(ep.description) {
2015                     html += `<div class="tt-row">${ep.description}</div>`;
2016                 }
2017                 tt.innerHTML = html;
2018                 tt.style.display = 'block';
2019             },
2020             eventMouseLeave: function() {
2021                 document.getElementById('cal-tooltip').style.display = 'none';
2022             },
2023             eventClick: function(info) {
2024                 if(info.event.extendedProps.type === 'leave') {
2025                     showView('my-leaves', null);
2026                 }
2027             }
2028         });
2029     employeeCalendar.render();
2030     calendarInitialized = true;
2031     refreshCalendarEvents();
2032 }
2033
2034 async function refreshCalendarEvents() {
2035     if(!employeeCalendar) return;

```

```
2035     const events = await buildCalendarEvents();
2036     employeeCalendar.removeAllEvents();
2037     events.forEach(e => employeeCalendar.addEvent(e));
2038 }
2039
2040 async function refreshCalendar() {
2041     await loadCalendarBalanceCards();
2042     await refreshCalendarEvents();
2043     showAlert('Calendar refreshed!', 'success');
2044 }
2045
2046 document.addEventListener('mousemove', function(e) {
2047     const tt = document.getElementById('cal-tooltip');
2048     if(tt && tt.style.display === 'block') {
2049         tt.style.left = (e.clientX + 14) + 'px';
2050         tt.style.top = (e.clientY + 14) + 'px';
2051         const rect = tt.getBoundingClientRect();
2052         if(rect.right > window.innerWidth)
2053             tt.style.left = (e.clientX - rect.width - 10) + 'px';
2054         if(rect.bottom > window.innerHeight)
2055             tt.style.top = (e.clientY - rect.height - 10) + 'px';
2056     }
2057 });
2058 </script>
2059 </body>
2060 </html>
2061
```

```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>EMS – Employee Portal</title>
7      <link
href="https://fonts.googleapis.com/css2?family=Sora:wght@300;400;500;600;700&family=JetBrains+Mono:wght@400;500&
display=swap" rel="stylesheet">
8      <!-- FullCalendar CDN -->
9      <link href="https://cdn.jsdelivr.net/npm/fullcalendar@6.1.8/index.global.min.css" rel="stylesheet">
10     <script src="https://cdn.jsdelivr.net/npm/fullcalendar@6.1.8/index.global.min.js"></script>
11     <style>
12         *{margin:0;padding:0;box-sizing:border-box}
13         :root{
14             --bg:#080c10;--bg2:#0d1117;--bg3:#111820;
15             --panel:#141c26;--panel2:#1a2332;
16             --border:rgba(99,179,237,0.08);--border2:rgba(99,179,237,0.15);
17             --text:#d4e6f1;--text2:#7a9bb5;--text3:#4a6d87;
18             --accent:#38bdf8;--accent-dim:rgba(56,189,248,0.1);
19             --green:#34d399;--green-dim:rgba(52,211,153,0.1);
20             --amber:#fbbf24;--amber-dim:rgba(251,191,36,0.1);
21             --red:#f87171;--red-dim:rgba(248,113,113,0.1);
22             --font:'Sora', sans-serif;--mono:'JetBrains Mono', monospace;
23             --sidebar:240px;
24         }
25         body{font-family:var(--font);background:var(--bg);color:var(--text);font-
size:14px;overflow:hidden;height:100vh}
26         .app{display:flex;height:100vh}
27
28         /* SIDEBAR */
29         .sidebar{width:var(--sidebar);background:var(--bg2);border-right:1px solid var(--
border);display:flex;flex-direction:column;position:relative;flex-shrink:0}
30         .sidebar::before{content:'';position:absolute;top:0;left:0;right:0;height:1px;background:linear-
gradient(90deg,transparent,var(--accent),transparent)}
31         .logo{padding:24px 20px;border-bottom:1px solid var(--border)}
32         .logo-mark{display:flex;align-items:center;gap:10px;margin-bottom:6px}
33         .logo-icon{width:32px;height:32px;background:linear-gradient(135deg,var(--accent),#0ea5e9);border-
radius:8px;display:flex;align-items:center;justify-content:center;font-weight:700;font-size...
34         .logo h2{font-size:16px;font-weight:600;color:var(--text);letter-spacing:0.5px}
35         .logo p{font-size:11px;color:var(--text3);font-family:var(--mono);margin-top:2px}
36         .role-badge{display:inline-flex;align-items:center;gap:5px;padding:4px 10px;background:var(--accent-dim);border:1px solid rgba(56,189,248,0.25);border-radius:20px;font-size:10px;font-weight...
37         .role-badge::before{content:'';width:6px;height:6px;background:var(--accent);border-
radius:50%;animation:pulse 2s infinite}
38         @keyframes pulse{0%,100%{opacity:1}50%{opacity:0.4}}
39         .nav{flex:1;padding:12px 0;overflow-y:auto}
40         .nav-section{padding:4px 20px 8px;font-size:10px;font-weight:600;color:var(--text3);letter-
spacing:1.5px;text-transform:uppercase;margin-top:8px}
41         .nav-item{display:flex;align-items:center;gap:10px;padding:10px 20px;margin:2px 8px;border-
radius:8px;cursor:pointer;transition:all 0.2s;color:var(--text2);font-size:13px;font-weight:500}
42         .nav-item:hover{background:var(--accent-dim);color:var(--accent)}
43         .nav-item.active{background:var(--accent-dim);color:var(--accent);border:1px solid
rgba(56,189,248,0.2)}
44         .nav-item .icon{width:20px;text-align:center;font-size:14px;flex-shrink:0}
45         .sidebar-footer{padding:16px 20px;border-top:1px solid var(--border)}
46         .user-card{display:flex;align-items:center;gap:10px}
47         .avatar{width:32px;height:32px;background:linear-gradient(135deg,#0ea5e9,var(--accent));border-
radius:50%;display:flex;align-items:center;justify-content:center;font-size:13px;font-weight:6...
48         .user-info-side .name{font-size:12px;font-weight:600;color:var(--text)}
49         .user-info-side .email{font-size:10px;color:var(--text3);font-family:var(--mono);white-
space:nowrap;overflow:hidden;text-overflow:ellipsis;max-width:140px}
50
51         /* MAIN */
52         .main{flex:1;display:flex;flex-direction:column;overflow:hidden}
53         .topbar{background:var(--bg2);padding:14px 24px;display:flex;justify-content:space-between;align-
items:center;border-bottom:1px solid var(--border);flex-shrink:0}
54         .page-title{font-size:16px;font-weight:600;color:var(--text)}
55         .topbar-right{display:flex;align-items:center;gap:12px}
56         .logout-btn{padding:7px 16px;background:transparent;border:1px solid var(--border2);border-
radius:6px;color:var(--text2);font-family:var(--font);font-size:12px;cursor:pointer;transition:all...
57         .logout-btn:hover{background:var(--red-dim);border-color:var(--red);color:var(--red)}

```

```

58     .notif-wrapper { position: relative; cursor: pointer; }
59
60     .notif-badge {
61         position: absolute; top: -5px; right: -6px;
62         background: #e53e3e; color: white;
63         font-size: 10px; font-weight: 600;
64         min-width: 16px; height: 16px;
65         border-radius: 50%; display: flex;
66         align-items: center; justify-content: center;
67         padding: 0 3px;
68     }
69
70     .notif-panel {
71         position: absolute; top: 48px; right: 16px;
72         width: 320px; max-height: 400px; overflow-y: auto;
73         background: white; border: 1px solid #e2e8f0;
74         border-radius: 10px; box-shadow: 0 4px 16px rgba(0,0,0,0.1);
75         z-index: 999;
76     }
77
78     .notif-header {
79         padding: 12px 16px; font-weight: 600;
80         border-bottom: 1px solid #e2e8f0;
81     }
82
83     .notif-item {
84         padding: 12px 16px; font-size: 14px;
85         border-bottom: 1px solid #f1f5f9;
86     }
87
88     .notif-item.unread { background: #eff6ff; }
89     .notif-time { font-size: 11px; color: #94a3b8; margin-top: 4px; }
90
91     /* CONTENT */
92     .content{flex:1;overflow-y:auto;padding:24px}
93     .view{display:none}
94     .view.active{display:block;animation:fadeIn 0.3s ease}
95     @keyframes fadeIn{from{opacity:0;transform:translateY(8px)}to{opacity:1;transform:translateY(0)}}
96
97     /* CARDS */
98     .stats-grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(160px,1fr));gap:16px;margin-
bottom:24px}
99     .stat-card{background:var(--panel);border:1px solid var(--border);border-
radius:12px;padding:20px;position:relative;overflow:hidden;transition:border-color 0.2s}
100     .stat-card:hover{border-color:var(--border2)}
101     .stat-card::after{content:'';position:absolute;top:0;left:0;right:0;height:2px}
102     .stat-card.blue::after{background:var(--accent)}
103     .stat-card.green::after{background:var(--green)}
104     .stat-card.amber::after{background:var(--amber)}
105     .stat-card.red::after{background:var(--red)}
106     .stat-label{font-size:11px;color:var(--text3);text-transform:uppercase;letter-spacing:1px;margin-
bottom:10px;font-weight:500}
107     .stat-value{font-size:28px;font-weight:700;color:var(--text);font-family:var(--mono)}
108     .stat-sub{font-size:11px;color:var(--text3);margin-top:4px}
109
110     /* SECTION */
111     .section{background:var(--panel);border:1px solid var(--border);border-radius:12px;margin-
bottom:20px;overflow:hidden}
112     .section-head{padding:16px 20px;border-bottom:1px solid var(--border);display:flex;justify-
content:space-between;align-items:center}
113     .section-title{font-size:14px;font-weight:600;color:var(--text)}
114     .section-sub{font-size:12px;color:var(--text3)}
115
116     /* TABLE */
117     .tablewrap{overflow-x:auto}
118     table{width:100%;border-collapse:collapse}
119     th{padding:10px 16px;text-align:left;font-size:11px;color:var(--text3);font-weight:600;letter-
spacing:0.8px;text-transform:uppercase;background:var(--bg3);border-bottom:1px solid var(--bord...
120     td{padding:12px 16px;border-bottom:1px solid var(--border);font-size:13px;color:var(--text2)}
121     tr:last-child td{border-bottom:none}
122     tr:hover td{background:rgba(56,189,248,0.03);color:var(--text)}
123
124     /* CHIPS */
125     .chip{display:inline-flex;align-items:center;padding:3px 10px;border-radius:20px;font-
size:11px;font-weight:600;letter-spacing:0.5px}
126     .chip.pending{background:var(--amber-dim);color:var(--amber);border:1px solid rgba(251,191,36,0.2)}

```

```

127     .chip.open{background:var(--accent-dim);color:var(--accent)}
128     .chip.approved{background:var(--green-dim);color:var(--green);border:1px solid
rgba(52,211,153,0.2)}
129     .chip.rejected{background:var(--red-dim);color:var(--red);border:1px solid rgba(248,113,113,0.2)}
130     .chip.active{background:var(--green-dim);color:var(--green);border:1px solid rgba(52,211,153,0.2)}
131     .chip.sick{background:var(--red-dim);color:var(--red)}
132     .chip.casual{background:var(--amber-dim);color:var(--amber)}
133     .chip.maternity{background:var(--accent-dim);color:var(--accent)}
134     .chip.under_repair { background: var(--amber-dim); color: var(--amber); }
135     .chip.unpaid { background: rgba(139,92,246,0.15); color: #a78bfa; border: 1px solid
rgba(139,92,246,0.2); }
136     .chip.assigned{background:var(--green-dim);color:var(--green)}
137     .chip.condemned{background:var(--red-dim);color:var(--red)}
138
139     /* FORM */
140     .form-card{padding:24px}
141     .form-grid{display:grid;grid-template-columns:1fr 1fr;gap:16px}
142     .form-group{display:flex;flex-direction:column;gap:6px}
143     .form-group.full{grid-column:1/-1}
144     label{font-size:11px;font-weight:600;color:var(--text3);text-transform:uppercase;letter-
spacing:0.8px}
145     input,select,textarea{background:var(--bg3);border:1px solid var(--border);border-
radius:8px;padding:10px 14px;color:var(--text);font-family:var(--font);font-size:13px;transition:border-col...
146     input:focus,select:focus,textarea:focus{outline:none;border-color:var(--accent);background:var(--
bg2)}
147     input::placeholder,textarea::placeholder{color:var(--text3)}
148     select option{background:var(--bg2)}
149
150     /* BUTTONS */
151     .btn{padding:9px 18px;border-radius:8px;border:none;font-family:var(--font);font-size:13px;font-
weight:600;cursor:pointer;transition:all 0.2s}
152     .btn-primary{background:var(--accent);color:#fff}
153     .btn-primary:hover{background:#0ea5e9;transform:translateY(-1px);box-shadow:0 4px 12px
rgba(56,189,248,0.3)}
154     .btn-secondary{background:var(--panel2);color:var(--text2);border:1px solid var(--border)}
155     .btn-secondary:hover{border-color:var(--border2);color:var(--text)}
156     .btn-sm{padding:6px 12px;font-size:12px}
157     .btn-danger{background:var(--red-dim);color:var(--red);border:1px solid rgba(248,113,113,0.2)}
158     .btn-danger:hover{background:var(--red);color:#fff}
159     .form-actions{display:flex;gap:10px;margin-top:20px}
160
161     /* ALERT */
162     .alert{position:fixed;top:20px;right:20px;padding:12px 20px;border-radius:10px;z-
index:1100;animation:slideIn 0.3s ease;font-size:13px;font-weight:500}
163     .alert-success{background:var(--green-dim);color:var(--green);border:1px solid
rgba(52,211,153,0.3)}
164     .alert-error{background:var(--red-dim);color:var(--red);border:1px solid rgba(248,113,113,0.3)}
165     @keyframes slideIn{from{transform:translateX(120%);opacity:0}to{transform:translateX(0);opacity:1}}
166
167     .empty-state{text-align:center;color:var(--text3);font-style:italic;padding:30px}
168     .profile-grid{display:grid;grid-template-columns:1fr 1fr;gap:16px;padding:24px}
169     .profile-field{display:flex;flex-direction:column;gap:4px}
170     .profile-field .field-label{font-size:10px;color:var(--text3);text-transform:uppercase;letter-
spacing:0.8px;font-weight:600}
171     .profile-field .field-value{font-size:14px;color:var(--text);padding:10px 14px;background:var(--
bg3);border-radius:8px;border:1px solid var(--border);font-family:var(--mono)}
172
173     .subscription-card{display:grid;grid-template-columns:repeat(auto-
fill,minmax(200px,1fr));gap:12px;padding:20px}
174     .sub-item{background:var(--bg3);border:1px solid var(--border);border-
radius:10px;padding:16px;transition:border-color 0.2s}
175     .sub-item:hover{border-color:var(--border2)}
176     .sub-item .meal-name{font-weight:600;color:var(--text);margin-bottom:6px}
177     .sub-item .meal-meta{font-size:11px;color:var(--text3)}
178
179     .chip.paused{background:var(--amber-dim);color:var(--amber);border:1px solid rgba(251,191,36,0.2)}
180     .chip.expired{background:var(--red-dim);color:var(--red);border:1px solid rgba(248,113,113,0.2)}
181
182     /* ===== */
183     /* CALENDAR VIEW STYLES */
184     /* ===== */
185
186     /* Leave balance summary cards above the calendar */
187     .leave-balance-grid {
188         display: grid;
189         grid-template-columns: repeat(auto-fit, minmax(160px, 1fr));

```

```

190         gap: 14px;
191         padding: 20px;
192         border-bottom: 1px solid var(--border);
193     }
194     .leave-bal-card {
195         background: var(--bg3);
196         border: 1px solid var(--border);
197         border-radius: 10px;
198         padding: 16px 18px;
199         display: flex;
200         flex-direction: column;
201         gap: 6px;
202         position: relative;
203         overflow: hidden;
204         transition: border-color 0.2s;
205     }
206     .leave-bal-card:hover { border-color: var(--border2); }
207     .leave-bal-card::before {
208         content: '';
209         position: absolute;
210         left: 0; top: 0; bottom: 0;
211         width: 3px;
212         border-radius: 3px 0 0 3px;
213     }
214     .leave-bal-card.sick::before { background: var(--red); }
215     .leave-bal-card.casual::before { background: var(--amber); }
216     .leave-bal-card.mat::before { background: var(--accent); }
217     .leave-bal-card.total::before { background: var(--green); }
218     .lbc-icon { font-size: 18px; }
219     .lbc-label { font-size: 10px; color: var(--text3); text-transform: uppercase; letter-spacing: 1px;
font-weight: 600; }
220     .lbc-count { font-size: 24px; font-weight: 700; color: var(--text); font-family: var(--mono); line-
height: 1; }
221     .lbc-sub { font-size: 11px; color: var(--text3); }
222
223     /* Legend row */
224     .cal-legend {
225         display: flex;
226         align-items: center;
227         gap: 20px;
228         padding: 12px 20px;
229         border-bottom: 1px solid var(--border);
230         flex-wrap: wrap;
231     }
232     .legend-label { font-size: 11px; color: var(--text3); font-weight: 600; text-transform: uppercase;
letter-spacing: 0.8px; margin-right: 4px; }
233     .legend-item { display: flex; align-items: center; gap: 6px; font-size: 12px; color: var(--text2);
}
234     .legend-dot { width: 10px; height: 10px; border-radius: 50%; flex-shrink: 0; }
235
236     /* FullCalendar dark-theme overrides */
237     #employee-calendar {
238         padding: 16px 20px 20px;
239     }
240     /* Toolbar */
241     #employee-calendar .fc .fc-toolbar-title {
242         font-family: var(--font);
243         font-size: 15px;
244         font-weight: 600;
245         color: var(--text);
246     }
247     #employee-calendar .fc .fc-button {
248         background: var(--panel2);
249         border: 1px solid var(--border2);
250         color: var(--text2);
251         font-family: var(--font);
252         font-size: 12px;
253         font-weight: 600;
254         border-radius: 6px;
255         padding: 5px 12px;
256         transition: all 0.2s;
257         box-shadow: none;
258         text-transform: capitalize;
259     }
260     #employee-calendar .fc .fc-button:hover,
261     #employee-calendar .fc .fc-button-active,

```

```

262     #employee-calendar .fc .fc-button:not(:disabled):active {
263         background: var(--accent-dim);
264         border-color: var(--accent);
265         color: var(--accent);
266         box-shadow: none;
267     }
268     #employee-calendar .fc .fc-button:focus { box-shadow: none; outline: none; }
269
270     /* Grid */
271     #employee-calendar .fc-theme-standard td,
272     #employee-calendar .fc-theme-standard th,
273     #employee-calendar .fc-theme-standard .fc-scrollgrid {
274         border-color: var(--border);
275     }
276     #employee-calendar .fc .fc-col-header-cell-cushion {
277         font-family: var(--font);
278         font-size: 11px;
279         font-weight: 600;
280         color: var(--text3);
281         text-transform: uppercase;
282         letter-spacing: 0.8px;
283         padding: 8px 4px;
284     }
285     #employee-calendar .fc .fc-daygrid-day-number {
286         font-family: var(--mono);
287         font-size: 12px;
288         color: var(--text3);
289         padding: 6px 8px;
290     }
291     #employee-calendar .fc .fc-daygrid-day.fc-day-today {
292         background: var(--accent-dim) !important;
293     }
294     #employee-calendar .fc .fc-daygrid-day.fc-day-today .fc-daygrid-day-number {
295         color: var(--accent);
296         font-weight: 700;
297     }
298     #employee-calendar .fc .fc-daygrid-day:hover {
299         background: rgba(56,189,248,0.03);
300     }
301     #employee-calendar .fc-daygrid-day-frame {
302         background: var(--bg3);
303     }
304     #employee-calendar .fc .fc-day-other .fc-daygrid-day-frame {
305         background: var(--bg2);
306     }
307
308     /* Events */
309     #employee-calendar .fc-event {
310         border: none !important;
311         border-radius: 4px;
312         font-family: var(--font);
313         font-size: 11px;
314         font-weight: 600;
315         padding: 2px 6px;
316         cursor: pointer;
317     }
318     #employee-calendar .fc-event-title { font-weight: 600; }
319
320     /* Event tooltip */
321     .cal-tooltip {
322         position: fixed;
323         background: var(--panel2);
324         border: 1px solid var(--border2);
325         border-radius: 8px;
326         padding: 10px 14px;
327         font-size: 12px;
328         color: var(--text);
329         z-index: 9999;
330         pointer-events: none;
331         max-width: 200px;
332         box-shadow: 0 8px 24px rgba(0,0,0,0.4);
333     }
334     .cal-tooltip .tt-title { font-weight: 600; margin-bottom: 4px; font-size: 13px; }
335     .cal-tooltip .tt-row { color: var(--text2); margin-top: 2px; }
336
337     /* balance loading state */

```

```
338         .lbc-count.loading { color: var(--text3); font-size: 16px; }
339     </style>
340 </head>
341 <body>
342     <div class="app">
343         <!-- SIDEBAR -->
344         <div class="sidebar">
345             <div class="logo">
346                 <div class="logo-mark">
347                     <div class="logo-icon">E</div>
348                     <h2>EMS Portal</h2>
349                 </div>
350                 <p>Employee Management</p>
351                 <div class="role-badge">Employee</div>
352             </div>
353             <nav class="nav">
354                 <div class="nav-section">Overview</div>
355                 <div class="nav-item active" onclick="showView('dashboard',this)"><span class="icon"></span>
Dashboard</div>
356                 <div class="nav-section">Leave</div>
357                 <div class="nav-item" onclick="showView('apply-leave',this)"><span class="icon"></span> Apply
Leave</div>
358                 <div class="nav-item" onclick="showView('my-leaves',this)"><span class="icon"></span> My Leave
History</div>
359                 <div class="nav-item" onclick="showView('my-calendar',this)"><span class="icon"></span> My
Calendar</div>
360                 <div class="nav-section">Meals</div>
361                 <div class="nav-item" onclick="showView('restaurants',this)"><span class="icon"></span>
Restaurants</div>
362                 <div class="nav-item" onclick="showView('subscriptions',this)"><span class="icon"></span> My
Subscriptions</div>
363                 <div class="nav-item" onclick="showView('add-subscription',this)"><span class="icon"></span>
Add Subscription</div>
364                 <div class="nav-section">Account</div>
365                 <div class="nav-item" onclick="showView('profile',this)"><span class="icon"></span> My
Profile</div>
366                 <div class="nav-item" onclick="showView('my-devices',this)"><span class="icon"></span> My
Devices</div>
367                 <div class="nav-item" onclick="showView('service-requests',this)"><span class="icon"></span>
Service Requests</div>
368             </nav>
369             <div class="sidebar-footer">
370                 <div class="user-card">
371                     <div class="avatar" id="avatar-initials">?</div>
372                     <div class="user-info-side">
373                         <div class="name" id="sidebar-name">Loading...</div>
374                         <div class="email" id="sidebar-email"></div>
375                     </div>
376                 </div>
377             </div>
378         </div>
379
380         <!-- MAIN -->
381         <div class="main">
382             <div class="topbar">
383                 <div class="page-title" id="pageTitle">Dashboard</div>
384                 <!-- Bell icon with badge -->
385                 <div class="notif-wrapper" onclick="toggleNotifPanel()">
386                     <svg width="22" height="22" viewBox="0 0 24 24" fill="none"
387                         stroke="currentColor" stroke-width="2" stroke-linecap="round">
388                         <path d="M18 8A6 6 0 0 0 6 8c0 7-3 9-3 9h18s-3-2-3-9"/>
389                         <path d="M13.73 21a2 2 0 0 1-3.46 0"/>
390                     </svg>
391                     <span id="notif-badge" class="notif-badge" style="display:none"></span>
392                 </div>
393
394                 <!-- Dropdown panel -->
395                 <div id="notif-panel" class="notif-panel" style="display:none">
396                     <div class="notif-header">Notifications</div>
397                     <div id="notif-list"></div>
398                 </div>
399                 <div class="topbar-right">
400                     <span style="font-size:12px;color:var(--text3);font-family:var(--mono)" id="topbar-
email"></span>
401                     <button class="logout-btn" onclick="logout()">Sign Out</button>
402                 </div>

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403     </div>
404     <div class="content">
405
406         <!-- DASHBOARD -->
407         <div id="dashboard-view" class="view active">
408             <div class="stats-grid">
409                 <div class="stat-card blue">
410                     <div class="stat-label">Leave Balance</div>
411                     <div class="stat-value" id="stat-balance">--</div>
412                     <div class="stat-sub">days remaining</div>
413                 </div>
414                 <div class="stat-card amber">
415                     <div class="stat-label">Pending Leaves</div>
416                     <div class="stat-value" id="stat-pending">--</div>
417                     <div class="stat-sub">awaiting approval</div>
418                 </div>
419                 <div class="stat-card green">
420                     <div class="stat-label">Approved Leaves</div>
421                     <div class="stat-value" id="stat-approved">--</div>
422                     <div class="stat-sub">this year</div>
423                 </div>
424                 <div class="stat-card red">
425                     <div class="stat-label">Rejected Leaves</div>
426                     <div class="stat-value" id="stat-rejected">--</div>
427                     <div class="stat-sub">this year</div>
428                 </div>
429             </div>
430             <div class="section">
431                 <div class="section-head">
432                     <div>
433                         <div class="section-title">Recent Leave Requests</div>
434                         <div class="section-sub">Your last 5 requests</div>
435                     </div>
436                     <button class="btn btn-secondary btn-sm" onclick="loadDashboard()">
Refresh</button>
437                 </div>
438                 <div class="tablewrap">
439                     <table>
440
441                         <thead><tr><th>Type</th><th>From</th><th>To</th><th>Reason</th><th>Status</th></tr></thead>
442                         <tbody id="recent-leaves"><tr><td colspan="5" class="empty-
state">Loading...</td></tr></tbody>
443                     </table>
444                 </div>
445                 <div class="section">
446                     <div class="section-head">
447                         <div class="section-title">Active Subscriptions</div>
448                     </div>
449                     <div id="dashboard-subs" class="subscription-card"></div>
450                 </div>
451             </div>
452
453             <!-- APPLY LEAVE -->
454             <div id="apply-leave-view" class="view">
455                 <div class="section">
456                     <div class="section-head"><div class="section-title">Apply for Leave</div></div>
457                     <div class="form-card">
458                         <div class="form-grid">
459                             <div class="form-group">
460                                 <label>Leave Type</label>
461                                 <select id="leave-type">
462                                     <option value="SICK">Sick Leave</option>
463                                     <option value="CASUAL">Casual Leave</option>
464                                     <option value="UNPAID">Unpaid Leave</option>
465                                 </select>
466                                 <div id="sick-leave-hint" style="display:none;margin-top:6px;font-
size:11px;color:var(--amber);background:var(--amber-dim);border:1px solid rgba(251,191,36,0.2);bord...
467                                     Sick leave is limited to <strong>1 day per month</strong>. If already
used this month, please select Unpaid Leave.
468                                 </div>
469                                 <div id="casual-leave-hint" style="display:none; margin-top:6px; font-
size:11px; color:var(--amber); background:var(--amber-dim); border:1px solid rgba(251,191,36,0...
470                                     Casual leave is limited to <strong>1 day per month</strong>. If already
used this month, it will be automatically recorded as <strong>Unpaid Leave</strong>.
471                                 </div>

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472                 </div>
473                 <div class="form-group">
474                     <label>Start Date</label>
475                     <input type="date" id="leave-start">
476                 </div>
477                 <div class="form-group">
478                     <label>End Date</label>
479                     <input type="date" id="leave-end">
480                 </div>
481                 <div class="form-group full">
482                     <label>Reason</label>
483                     <textarea id="leave-reason" rows="3" placeholder="Please describe the
reason for your leave..."></textarea>
484                 </div>
485             </div>
486             <div class="form-actions">
487                 <button class="btn btn-primary" onclick="applyLeave()">Submit Request</button>
488                 <button class="btn btn-secondary" onclick="clearLeaveForm()">Clear</button>
489             </div>
490         </div>
491     </div>
492 </div>
493
494     <!-- MY LEAVES -->
495     <div id="my-leaves-view" class="view">
496         <div class="section">
497             <div class="section-head">
498                 <div><div class="section-title">My Leave History</div><div class="section-sub">All
your leave requests</div></div>
499                 <button class="btn btn-secondary btn-sm" onclick="loadMyLeaves()"> Refresh</button>
500             </div>
501             <div class="tablewrap">
502                 <table>
503
504                     <thead><tr><th>Type</th><th>From</th><th>To</th><th>Days</th><th>Reason</th><th>Status</th><th>Action</th></tr><
/thead>
505                     <tbody id="my-leaves-list"><tr><td colspan="7" class="empty-
state">Loading...</td></tr></tbody>
506                 </table>
507             </div>
508         </div>
509
510     <!-- ===== -->
511     <!-- MY CALENDAR VIEW (NEW) -->
512     <!-- ===== -->
513     <div id="my-calendar-view" class="view">
514         <div class="section">
515             <div class="section-head">
516                 <div>
517                     <div class="section-title">My Leave Calendar</div>
518                     <div class="section-sub">All approved leaves and public holidays</div>
519                 </div>
520                 <button class="btn btn-secondary btn-sm" onclick="refreshCalendar()">
Refresh</button>
521             </div>
522
523     <!-- Leave Balance Summary Cards -->
524     <div class="leave-balance-grid">
525         <!-- Sick Leave Card -->
526         <div class="leave-bal-card sick">
527             <span class="lbc-icon"></span>
528             <span class="lbc-label">Sick Leave</span>
529             <span class="lbc-count" id="cal-sick-remaining">--</span>
530             <span class="lbc-sub" id="cal-sick-sub">loading...</span>
531         </div>
532         <!-- Casual Leave Card -->
533         <div class="leave-bal-card casual">
534             <span class="lbc-icon"></span>
535             <span class="lbc-label">Casual Leave</span>
536             <span class="lbc-count" id="cal-casual-remaining">--</span>
537             <span class="lbc-sub" id="cal-casual-sub">loading...</span>
538         </div>
539         <!-- Maternity Leave Card -->
540         <div class="leave-bal-card mat">
541             <span class="lbc-icon"></span>
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542         <span class="lbc-label">Maternity Leave</span>
543         <span class="lbc-count" id="cal-mat-remaining">--</span>
544         <span class="lbc-sub" id="cal-mat-sub">loading...</span>
545     </div>
546     <!-- Total Used Card -->
547     <div class="leave-bal-card total">
548         <span class="lbc-icon"></span>
549         <span class="lbc-label">Total Used</span>
550         <span class="lbc-count" id="cal-total-used">--</span>
551         <span class="lbc-sub" id="cal-total-sub">loading...</span>
552     </div>
553 </div>
554
555     <!-- Legend -->
556     <div class="cal-legend">
557         <span class="legend-label">Legend:</span>
558         <div class="legend-item">
559             <div class="legend-dot" style="background:#34d399"></div>
560             Approved Leave
561         </div>
562         <div class="legend-item">
563             <div class="legend-dot" style="background:#fbbf24"></div>
564             Pending Leave
565         </div>
566         <div class="legend-item">
567             <div class="legend-dot" style="background:#f87171"></div>
568             Rejected Leave
569         </div>
570         <div class="legend-item">
571             <div class="legend-dot" style="background:#38bdf8"></div>
572             Public Holiday
573         </div>
574     </div>
575
576     <!-- FullCalendar Mount Point -->
577     <div id="employee-calendar"></div>
578 </div>
579 </div>
580 <!-- END CALENDAR VIEW -->
581
582 <!-- RESTAURANTS -->
583 <div id="restaurants-view" class="view">
584     <div class="section">
585         <div class="section-head">
586             <div class="section-title">Available Restaurants</div>
587             <button class="btn btn-secondary btn-sm" onclick="loadRestaurants()">
Refresh</button>
588         </div>
589         <div class="tablewrap">
590             <table>
591                 <thead><tr><th>Name</th><th>Meal
Slot</th><th>Status</th><th>Action</th></tr></thead>
592                 <tbody id="restaurants-list"><tr><td colspan="4" class="empty-
state">Loading...</td></tr></tbody>
593             </table>
594         </div>
595     </div>
596 </div>
597
598 <!-- SUBSCRIPTIONS -->
599 <div id="subscriptions-view" class="view">
600     <div class="section">
601         <div class="section-head">
602             <div class="section-title">My Meal Subscriptions</div>
603             <button class="btn btn-secondary btn-sm" onclick="loadSubscriptions()">
Refresh</button>
604         </div>
605         <div class="tablewrap">
606             <table>
607                 <thead>
608                     <tr>
609                         <th>Restaurant</th><th>Meal Slot</th><th>Schedule</th>
610                         <th>Next Delivery</th><th>Status</th><th>Action</th>
611                     </tr>
612                 </thead>
613                 <tbody id="subscriptions-list"><tr><td colspan="6" class="empty-

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state">Loading...</td></tr></tbody>
614         </table>
615     </div>
616 </div>
617 </div>
618
619 <!-- ADD SUBSCRIPTION -->
620 <div id="add-subscription-view" class="view">
621     <div class="section">
622         <div class="section-head"><div class="section-title">Subscribe to Meal Plan</div></div>
623         <div class="form-card">
624             <div class="form-grid">
625                 <div class="form-group">
626                     <label>Meal Slot</label>
627                     <select id="sub-meal-slot">
628                         <option value="BREAKFAST">Breakfast</option>
629                         <option value="LUNCH">Lunch</option>
630                         <option value="DINNER">Dinner</option>
631                     </select>
632                 </div>
633                 <div class="form-group">
634                     <label>Restaurant</label>
635                     <select id="sub-restaurant-id">
636                         <option value="">Select a restaurant</option>
637                     </select>
638                 </div>
639                 <div class="form-group">
640                     <label>Schedule Type</label>
641                     <select id="sub-schedule-type">
642                         <option value="DAILY">Daily</option>
643                         <option value="WEEKLY">Weekly</option>
644                     </select>
645                 </div>
646                 <div class="form-group" id="day-of-week-group" style="display:none">
647                     <label>Day of Week</label>
648                     <select id="sub-day-of-week">
649                         <option value="MONDAY">Monday</option>
650                         <option value="TUESDAY">Tuesday</option>
651                         <option value="WEDNESDAY">Wednesday</option>
652                         <option value="THURSDAY">Thursday</option>
653                         <option value="FRIDAY">Friday</option>
654                         <option value="SATURDAY">Saturday</option>
655                         <option value="SUNDAY">Sunday</option>
656                     </select>
657                 </div>
658             </div>
659             <div class="form-actions">
660                 <button class="btn btn-primary" onclick="addSubscription()">Subscribe</button>
661             </div>
662         </div>
663     </div>
664 </div>
665
666 <!-- MY DEVICES -->
667 <div id="my-devices-view" class="view">
668     <div class="section">
669         <div class="section-head">
670             <div>
671                 <div class="section-title">My Assigned Devices</div>
672                 <div class="section-sub">Devices currently assigned to you</div>
673             </div>
674             <button class="btn btn-secondary btn-sm" onclick="loadMyAssignedDevices()">
Refresh</button>
675         </div>
676         <div class="tablewrap">
677             <table>
678                 <thead>
679                     <tr>
680                         <th>ID</th><th>Device Name</th><th>Brand</th>
681                         <th>Type</th><th>Warranty</th><th>Assigned On</th><th>Status</th>
682                     </tr>
683                 </thead>
684                 <tbody id="my-devices-list"></tbody>
685             </table>
686         </div>
687     </div>
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688         </div>
689
690         <!-- SERVICE REQUESTS -->
691         <div id="service-requests-view" class="view">
692             <div class="section" style="margin-bottom:20px">
693                 <div class="section-head">
694                     <div>
695                         <div class="section-title">My Service Requests</div>
696                         <div class="section-sub">All requests you have raised</div>
697                     </div>
698                     <button class="btn btn-secondary btn-sm" onclick="loadServiceRequestsList()">
Refresh</button>
699                 </div>
700                 <div class="tablewrap">
701                     <table>
702                         <thead>
703                             <tr>
704                                 <th>ID</th><th>Device</th><th>Type</th>
705                                 <th>Description</th><th>Urgent</th><th>Status</th><th>Raised At</th>
706                             </tr>
707                         </thead>
708                         <tbody id="service-requests-list">
709                             <tr><td colspan="7" class="empty-state">Loading...</td></tr>
710                         </tbody>
711                     </table>
712                 </div>
713             </div>
714             <div class="section">
715                 <div class="section-head"><div class="section-title">Raise a New Request</div></div>
716                 <div class="form-card">
717                     <div class="form-grid">
718                         <div class="form-group full">
719                             <label>Device</label>
720                             <select id="sr-device-select">
721                                 <option value="">-- Select your device --</option>
722                             </select>
723                         </div>
724                         <div class="form-group">
725                             <label>Request Type</label>
726                             <select id="sr-type">
727                                 <option value="REPAIR">Repair</option>
728                                 <option value="REPLACEMENT">Replacement</option>
729                                 <option value="UPGRADE">Upgrade</option>
730                                 <option value="MAINTENANCE">Maintenance</option>
731                                 <option value="RETURN">Return</option>
732                             </select>
733                         </div>
734                         <div class="form-group" style="display:flex;align-
items:center;gap:10px;padding-top:22px">
735                             <input type="checkbox" id="sr-urgent"
style="width:16px;height:16px;padding:0;flex-shrink:0">
736                             <label for="sr-urgent" style="text-transform:none;font-size:13px;font-
weight:500;color:var(--text);cursor:pointer;letter-spacing:0">Mark as Urgent</label>
737                         </div>
738                         <div class="form-group full">
739                             <label>Issue Description</label>
740                             <textarea id="sr-description" rows="3" placeholder="Describe the issue in
detail..."></textarea>
741                         </div>
742                     </div>
743                     <div class="form-actions">
744                         <button class="btn btn-primary" onclick="submitServiceRequest()">Submit
Request</button>
745                         <button class="btn btn-secondary"
onclick="clearServiceRequestForm()">Clear</button>
746                     </div>
747                 </div>
748             </div>
749         </div>
750
751         <!-- PROFILE -->
752         <div id="profile-view" class="view">
753             <div class="section">
754                 <div class="section-head"><div class="section-title">My Profile</div></div>
755                 <div class="profile-grid" id="profile-data">
756                     <div class="profile-field"><div class="field-label">Name</div><div class="field-

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value" id="p-name"></div></div>
757         <div class="profile-field"><div class="field-label">Email</div><div class="field-
value" id="p-email"></div></div>
758         <div class="profile-field"><div class="field-label">Department</div><div
class="field-value" id="p-dept"></div></div>
759         <div class="profile-field"><div class="field-label">Role</div><div class="field-
value" id="p-role"></div></div>
760         <div class="profile-field"><div class="field-label">Status</div><div class="field-
value" id="p-status"></div></div>
761         <div class="profile-field"><div class="field-label">Timezone</div><div
class="field-value" id="p-tz"></div></div>
762         </div>
763     </div>
764
765     <div class="section" id="maternity-apply-btn" style="display:none">
766         <div class="section-head">
767             <div>
768                 <div class="section-title"> Maternity Leave</div>
769                 <div class="section-sub">Apply for 180 days maternity leave</div>
770             </div>
771         </div>
772         <div class="form-card">
773             <div class="form-grid">
774                 <div class="form-group">
775                     <label>Start Date</label>
776                     <input type="date" id="mat-start-date">
777                 </div>
778                 <div class="form-group">
779                     <label>End Date (Auto)</label>
780                     <input type="text" id="mat-end-date"
781                         placeholder="Auto: start + 180 days"
782                         disabled
783                         style="color:var(--text3);cursor:not-allowed">
784                 </div>
785                 <div class="form-group full">
786                     <label>Reason / Notes</label>
787                     <textarea id="mat-reason" rows="3"
788                         placeholder="Optional notes..."></textarea>
789                 </div>
790                 <!-- Info box -->
791                 <div class="form-group full">
792                     <div style="background:var(--accent-dim);border:1px solid
rgba(56,189,248,0.2);
793                         border-radius:8px;padding:12px 16px;font-size:12px;color:var(--text2)">
794                         Maternity leave is fixed at <strong style="color:var(--accent)">180
days</strong>.
795                         The end date will be automatically calculated.
796                         You can apply <strong style="color:var(--accent)">once per
year</strong>.
797                     </div>
798                 </div>
799             </div>
800             <div class="form-actions">
801                 <button class="btn btn-primary" onclick="applyMaternityLeave()">
802                     Submit Maternity Leave
803                 </button>
804             </div>
805         </div>
806     </div>
807
808 </div>
809
810 </div><!-- /content -->
811 </div><!-- /main -->
812 </div><!-- /app -->
813
814 <!-- Tooltip for calendar events -->
815 <div class="cal-tooltip" id="cal-tooltip" style="display:none"></div>
816
817 <script>
818     const API = 'http://localhost:8080';
819     let availableRestaurants = [];
820
821     /* =====
822     CALENDAR GLOBALS
823     ===== */

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824     let employeeCalendar = null;    // FullCalendar instance
825     let calendarInitialized = false;
826     let currentEmployeeGender = null;
827     let currentEmployeeId = null;
828
829     // Event color map
830     const STATUS_COLORS = {
831         APPROVED : { bg:'rgba(52,211,153,0.25)', border:'#34d399', text:'#34d399' },
832         PENDING  : { bg:'rgba(251,191,36,0.25)',  border:'#fbbf24', text:'#fbbf24' },
833         REJECTED : { bg:'rgba(248,113,113,0.25)', border:'#f87171', text:'#f87171' },
834         HOLIDAY  : { bg:'rgba(56,189,248,0.25)', border:'#38bdf8', text:'#38bdf8' },
835     };
836
837     const TYPE_LABEL = { SICK:' Sick', CASUAL:' Casual', MATERNITY:' Maternity', UNPAID:' Unpaid'
838     };
839
840     /* =====
841     AUTH / INIT
842     ===== */
843     function extractRoleFromToken(token) {
844         try {
845             const payload = JSON.parse(atob(token.split('.')[1]));
846             return payload.role || payload.roles || payload.authorities || '';
847         } catch(e) { return ''; }
848     }
849
850     (function(){
851         const token = localStorage.getItem('jwt_token');
852         const authType = localStorage.getItem('auth_type');
853         if(!token && authType !== 'session') { window.location.href = '/google.html'; return; }
854         if(token) {
855             try {
856                 const roles = String(extractRoleFromToken(token));
857                 if(roles.includes('ROLE_MANAGER')) { window.location.href='/manager-dashboard.html';
858 return; }
859                 if(roles.includes('ROLE_VENDOR')) { window.location.href='/vendor-dashboard.html';
860 return; }
861                 if(roles.includes('ROLE_ADMIN') || roles.includes('ADMIN')) {
862 window.location.href='/dashboard.html'; return; }
863             } catch(e) { console.warn('Could not decode JWT for role check'); }
864         }
865         initUser();
866     })();
867
868     function getAuthHeader(){
869         const token = localStorage.getItem('jwt_token');
870         return token ? {'Authorization':`Bearer ${token}`} : {};
871     }
872
873     async function apiFetch(path, opts={}){
874         const token = localStorage.getItem('jwt_token');
875         const authType = localStorage.getItem('auth_type');
876         const res = await fetch(`${API}${path}`, {
877             ...opts,
878             headers: {'Content-Type': 'application/json', ...getAuthHeader(), ...(opts.headers||{})},
879             credentials: (!token && authType==='session') ? 'include' : 'omit'
880         });
881         if(res.status===401){
882             localStorage.removeItem('jwt_token'); localStorage.removeItem('auth_type');
883             showAlert('Session expired. Please login again.','error');
884             setTimeout(()=>window.location.href='/google.html',1500);
885             throw new Error('Unauthorized');
886         }
887         if(res.status===403) throw new Error('ACCESS_DENIED');
888         if(!res.ok){
889             const errText = await res.text();
890             throw new Error(errText || 'Request failed');
891         }
892         const text = await res.text();
893         try { return text ? JSON.parse(text) : null; }
894         catch { return text; }
895     }
896
897     function asArray(value){ return Array.isArray(value) ? value : []; }
898
899     function daysBetween(start,end){

```

```

897     if(!start||!end) return '-';
898     const d=(new Date(end)-new Date(start))/86400000+1;
899     return d>0?d:'-';
900 }
901
902 async function initUser(){
903     try {
904         let email='', name='';
905         const token = localStorage.getItem('jwt_token');
906         if(token){
907             try{ const p=JSON.parse(atob(token.split('.')[1])); email=p.sub||p.email||''; }catch(e){}
908         }
909         try{
910             const me = await apiFetch('/employee/me');
911             if(me){
912                 name=me.name||email; email=me.email||email;
913                 currentEmployeeGender = me.gender; // store gender
914                 currentEmployeeId = me.employeeId;
915                 document.getElementById('p-name').textContent=me.name||'-';
916                 document.getElementById('p-email').textContent=me.email||'-';
917                 document.getElementById('p-dept').textContent=me.dept||'-';
918                 document.getElementById('p-role').textContent=me.role||'EMPLOYEE';
919                 document.getElementById('p-status').textContent=me.status||'-';
920                 document.getElementById('p-tz').textContent=me.timezone||'-';
921
922                 const maternityBtn = document.getElementById('maternity-apply-btn');
923                 if(maternityBtn) {
924                     maternityBtn.style.display = me.gender === 'F' ? 'block' : 'none';
925                 }
926             }
927             }catch(e){ console.warn('Could not load profile'); }
928             const initials=(name||email||'?').charAt(0).toUpperCase();
929             document.getElementById('avatar-initials').textContent=initials;
930             document.getElementById('sidebar-name').textContent=name||email||'Employee';
931             document.getElementById('sidebar-email').textContent=email;
932             document.getElementById('topbar-email').textContent=email;
933         }catch(e){ console.error(e); }
934         loadDashboard();
935     }
936
937     /* =====
938     VIEW SWITCHING
939     ===== */
940     function showView(name, el){
941         document.querySelectorAll('.view').forEach(v=>v.classList.remove('active'));
942         document.getElementById(`${name}-view`).classList.add('active');
943         document.querySelectorAll('.nav-item').forEach(n=>n.classList.remove('active'));
944         if(el) el.classList.add('active');
945         document.getElementById('pageTitle').textContent=
946             name.split('-').map(w=>w.charAt(0).toUpperCase()+w.slice(1)).join(' ');
947
948         const loaders={
949             dashboard: loadDashboard,
950             'my-leaves': loadMyLeaves,
951             'apply-leave': ()=>{},
952             'my-calendar': loadMyCalendar,
953             restaurants: loadRestaurants,
954             subscriptions: loadSubscriptions,
955             'add-subscription': ()=>{},
956             profile: ()=>{},
957             'my-devices': loadMyAssignedDevices,
958             'service-requests': loadServiceRequests
959         };
960         if(loaders[name]) loaders[name]();
961         if(name==='add-subscription') loadRestaurantChoices();
962     }
963
964     /* =====
965     DASHBOARD
966     ===== */
967     async function loadDashboard(){
968         const [leavesResult, dashboardResult] = await Promise.allSettled([
969             apiFetch('/leave_requests/employee'),
970             apiFetch('/api/employees/me/leave-dashboard')
971         ]);
972         const leavesPayload = leavesResult.status==='fulfilled' ? leavesResult.value : null;

```



```

973     const dashboard      = dashboardResult.status==='fulfilled' ? dashboardResult.value : null;
974     const arr = asArray(leavesPayload?.requests || leavesPayload);
975     const pending  = arr.filter(l=>l.status==='PENDING').length;
976     const approved = arr.filter(l=>l.status==='APPROVED').length;
977     const rejected = arr.filter(l=>l.status==='REJECTED').length;
978     document.getElementById('stat-pending').textContent=pending;
979     document.getElementById('stat-approved').textContent=approved;
980     document.getElementById('stat-rejected').textContent=rejected;
981     const balances  = dashboard?.leaveBalances ? Object.values(dashboard.leaveBalances) : [];
982     const available = balances.reduce((t,b)=>t+Number(b.available||0),0);
983     document.getElementById('stat-balance').textContent=available;
984     const recent=arr.slice(-5).reverse();
985     document.getElementById('recent-leaves').innerHTML=recent.length
986       ? recent.map(l=>`<tr>
987         <td><span class="chip ${l.leaveType||''}.toLowerCase()>${l.leaveType||'N/A'}</span></td>
988         <td>${l.startDate||'-'}</td><td>${l.endDate||'-'}</td>
989         <td>${(l.reason||'-').substring(0,30)}</td>
990         <td><span class="chip ${l.status||''}.toLowerCase()>${l.status||'-'}</span></td>
991       </tr>`).join('')
992       : leavesResult.status==='rejected'
993       ? '<tr><td colspan="5" class="empty-state">Could not load leave data</td></tr>'
994       : '<tr><td colspan="5" class="empty-state">No leave requests yet</td></tr>';
995     try{
996       const subs = await apiFetch('/subscription/user');
997       const arr2 = subs || [];
998       document.getElementById('dashboard-sub').innerHTML = arr2.length
999       ? arr2.filter(s=>s.status==='ACTIVE').slice(0,4).map(s=>`
1000         <div class="sub-item">
1001           <div class="meal-name">${s.restaurantName||'Restaurant'}</div>
1002           <div class="meal-meta">${s.mealSlot||''} · ${s.scheduleType||''}${s.dayOfWeek?' ·
1003             +s.dayOfWeek:''}</div>
1004           <div class="meal-meta" style="margin-top:4px;color:var(--accent)">Next:
1005             ${s.nextDeliveryTime?new Date(s.nextDeliveryTime).toLocaleString():'N/A'}</div>
1006         </div>`).join('')
1007       : '<div style="padding:16px;color:var(--text3);font-style:italic">No active
subscriptions</div>';
1008     }catch(e){
1009       document.getElementById('dashboard-sub').innerHTML='<div style="padding:16px;color:var(--
text3)">Could not load subscriptions</div>';
1010     }
1011   }
1012   /* =====
1013   MY LEAVES
1014   ===== */
1015   async function loadMyLeaves(){
1016     try{
1017       const data=await apiFetch('/leave_requests/employee');
1018       const leaves=data?.requests||data||[];
1019       document.getElementById('my-leaves-list').innerHTML=leaves.length
1020       ? leaves.map(l=>`<tr>
1021         <td><span class="chip
1022           ${l.leaveType||''}.toLowerCase()>${l.leaveType||'N/A'}</span></td>
1023         <td>${l.startDate||'-'}</td><td>${l.endDate||'-'}</td>
1024         <td>${l.numberOfDays||daysBetween(l.startDate,l.endDate)}</td>
1025         <td>${(l.reason||'-').substring(0,30)}</td>
1026         <td><span class="chip ${l.status||''}.toLowerCase()>${l.status}</span></td>
1027         <td>${l.status==='PENDING'?`<button class="btn btn-danger btn-sm"
onclick="cancelLeave(${l.leaveRequestId})">Cancel</button>`:'-'}</td>
1028       </tr>`).join('')
1029       : '<tr><td colspan="7" class="empty-state">No leave requests found</td></tr>';
1030     }catch(e){
1031       document.getElementById('my-leaves-list').innerHTML='<tr><td colspan="7" class="empty-
state">Could not load leave requests</td></tr>';
1032     }
1033   }
1034   async function applyLeave(){
1035     const leaveType=document.getElementById('leave-type').value;
1036     const startDate=document.getElementById('leave-start').value;
1037     const endDate  =document.getElementById('leave-end').value;
1038     const reason =document.getElementById('leave-reason').value.trim();
1039     if(!startDate||!endDate){showAlert('Please fill all required fields','error');return;}
1040     try{
1041       const data = await apiFetch('/leave_requests', {
1042         method: 'POST',

```

```

1042         body: JSON.stringify({ leaveType, startDate, endDate, reason })
1043     });
1044     if (data && data.warning) {
1045         showAlert(data.warning, 'warning'); // amber toast
1046     } else {
1047         showAlert('Leave request submitted successfully!', 'success');
1048     }
1049     clearLeaveForm();
1050     loadDashboard();
1051     loadMyLeaves();
1052 }catch(e){showAlert('Failed to submit: '+e.message,'error');}
1053 }
1054 document.getElementById('leave-type').addEventListener('change', function() {
1055     document.getElementById('sick-leave-hint').style.display =
1056     this.value === 'SICK' ? 'block' : 'none';
1057     document.getElementById('casual-leave-hint').style.display =
1058     this.value === 'CASUAL' ? 'block' : 'none';
1059 });
1060
1061 async function cancelLeave(id){
1062     if(!confirm('Cancel this leave request?')) return;
1063     try{
1064         await apiFetch(`/leave_requests/cancel/${id}`, {method: 'PUT'});
1065         showAlert('Request cancelled successfully', 'success');
1066         loadMyLeaves();
1067     }catch(e){
1068         if(e.message==='ACCESS_DENIED') showAlert('You do not have permission to cancel this
leave','error');
1069         else showAlert('Failed to cancel: '+e.message,'error');
1070     }
1071 }
1072
1073 function clearLeaveForm(){
1074     ['leave-start', 'leave-end', 'leave-reason'].forEach(id=>document.getElementById(id).value='');
1075 }
1076
1077 /* =====
1078 MY CALENDAR (NEW)
1079 ===== */
1080 async function loadMyCalendar(){
1081     // 1. Load balance cards
1082     await loadCalendarBalanceCards();
1083     // 2. Build or refresh FullCalendar
1084     if(!calendarInitialized){
1085         initFullCalendar();
1086     } else {
1087         await refreshCalendarEvents();
1088     }
1089 }
1090
1091 async function loadCalendarBalanceCards(){
1092     // Default state
1093     ['cal-sick-remaining', 'cal-casual-remaining', 'cal-mat-remaining', 'cal-total-used']
1094     .forEach(id=>{ document.getElementById(id).textContent='--'; });
1095
1096     try{
1097         const dashboard = await apiFetch('/api/employees/me/leave-dashboard');
1098         const balances = dashboard?.leaveBalances || {};
1099
1100         let totalUsed=0, totalAllowed=0;
1101
1102         // SICK
1103         const sick = balances['SICK'] || balances['sick'] || {};
1104         const sickRem = Number(sick.available || 0);
1105         const sickTotal = Number(sick.total || 0);
1106         const sickUsed = sickTotal - sickRem;
1107         document.getElementById('cal-sick-remaining').textContent = sickRem;
1108         document.getElementById('cal-sick-sub').textContent = `${sickUsed} used of ${sickTotal}
days`;
1109
1110         // CASUAL
1111         const cas = balances['CASUAL'] || balances['casual'] || {};
1112         const casRem = Number(cas.available || 0);
1113         const casTotal = Number(cas.total || 0);
1114         const casUsed = casTotal - casRem;
1115         document.getElementById('cal-casual-remaining').textContent = casRem;

```

```

1116         document.getElementById('cal-casual-sub').textContent      = `${casUsed} used of ${casTotal}
days`;
1117
1118         // MATERNITY
1119         const mat = balances['MATERNITY'] || balances['maternity'] || {};
1120         const matRem = Number(mat.available || 0);
1121         const matTotal = Number(mat.total || 0);
1122         const matUsed = matTotal - matRem;
1123         document.getElementById('cal-mat-remaining').textContent = matRem;
1124         document.getElementById('cal-mat-sub').textContent      = `${matUsed} used of ${matTotal}
days`;
1125
1126         // TOTAL
1127         Object.values(balances).forEach(b=>{
1128             totalUsed += (Number(b.total||0) - Number(b.available||0));
1129             totalAllowed += Number(b.total||0);
1130         });
1131         document.getElementById('cal-total-used').textContent = totalUsed;
1132         document.getElementById('cal-total-sub').textContent  = `out of ${totalAllowed} days total`;
1133
1134     }catch(e){
1135         console.error('Could not load leave balance for calendar:', e);
1136         ['cal-sick-sub', 'cal-casual-sub', 'cal-mat-sub', 'cal-total-sub']
1137             .forEach(id=>{ document.getElementById(id).textContent='unavailable'; });
1138     }
1139 }
1140
1141 async function buildCalendarEvents(){
1142     const events = [];
1143     try{
1144         // - Employee leaves -
1145         const data = await apiFetch('/leave_requests/employee');
1146         const leaves = data?.requests || data || [];
1147         leaves.forEach(l=>{
1148             const s = (l.status||'PENDING').toUpperCase();
1149             const c = STATUS_COLORS[s] || STATUS_COLORS.PENDING;
1150             const typeLabel = TYPE_LABEL[l.leaveType] || l.leaveType || 'Leave';
1151
1152             // FullCalendar end date is exclusive, so add 1 day for multi-day events
1153             let endDate = l.endDate;
1154             if(endDate){
1155                 const d = new Date(endDate);
1156                 d.setDate(d.getDate()+1);
1157                 endDate = d.toISOString().split('T')[0];
1158             }
1159
1160             events.push({
1161                 id      : `leave-${l.leaveRequestId}`,
1162                 title   : `${typeLabel} (${s.charAt(0)+s.slice(1).toLowerCase()})`,
1163                 start   : l.startDate,
1164                 end     : endDate,
1165                 allDay   : true,
1166                 backgroundColor : c.bg,
1167                 borderColor : c.border,
1168                 textColor  : c.text,
1169                 extendedProps : {
1170                     type    : 'leave',
1171                     status   : s,
1172                     reason  : l.reason || '-',
1173                     leaveType: l.leaveType
1174                 }
1175             });
1176         });
1177     }catch(e){ console.warn('Could not load leaves for calendar:', e); }
1178
1179     try{
1180         // - Public holidays (admin-added) -
1181         const holidays = await apiFetch('/holidays');
1182         (holidays||[]).forEach(h=>{
1183             const c = STATUS_COLORS.HOLIDAY;
1184             events.push({
1185                 id      : `holiday-${h.id||h.holidayId||Math.random()}`,
1186                 title   : `${h.name||h.holidayName||'Holiday'}`,
1187                 start   : h.date||h.holidayDate,
1188                 allDay   : true,
1189                 backgroundColor : c.bg,

```

```

1190         borderColor      : c.border,
1191         textColor         : c.text,
1192         extendedProps     : { type:'holiday', description: h.description||'' }
1193     });
1194 });
1195 }catch(e){ console.warn('Could not load holidays for calendar:', e); }
1196
1197     return events;
1198 }
1199
1200 function initFullCalendar(){
1201     const calendarEl = document.getElementById('employee-calendar');
1202
1203     employeeCalendar = new FullCalendar.Calendar(calendarEl, {
1204         initialView: 'dayGridMonth',
1205         height      : 'auto',
1206         headerToolbar:{
1207             left : 'prev,next today',
1208             center: 'title',
1209             right : 'dayGridMonth,dayGridWeek,listMonth'
1210         },
1211         buttonText:{
1212             today      : 'Today',
1213             month       : 'Month',
1214             week        : 'Week',
1215             listMonth   : 'List'
1216         },
1217         dayMaxEvents: 3,      // "+N more" link if overflowing
1218         navLinks      : true, // click day number day view
1219         events         : [],   // start empty, populate below
1220
1221         // Event hover tooltip
1222         eventMouseEnter: function(info){
1223             const ep = info.event.extendedProps;
1224             const tooltip = document.getElementById('cal-tooltip');
1225             let html = `<div class="tt-title">${info.event.title}</div>`;
1226             if(ep.type==='leave'){
1227                 html += `<div class="tt-row"> ${info.event.startStr}
1228 ${info.event.endStr||info.event.startStr}</div>`;
1229                 if(ep.reason && ep.reason!=='-'){
1230                     html += `<div class="tt-row"> ${ep.reason.substring(0,60)}</div>`;
1231                     html += `<div class="tt-row">Status: <strong
1232 style="color:${STATUS_COLORS[ep.status]?.text||'#fff'}">${ep.status}</strong></div>`;
1233                 } else {
1234                     if(ep.description) html += `<div class="tt-row">${ep.description}</div>`;
1235                 }
1236                 tooltip.innerHTML = html;
1237                 tooltip.style.display = 'block';
1238             },
1239             eventMouseLeave: function(){
1240                 document.getElementById('cal-tooltip').style.display='none';
1241             },
1242             eventClick: function(info){
1243                 // Click navigates to leave history
1244                 const ep = info.event.extendedProps;
1245                 if(ep.type==='leave'){
1246                     showView('my-leaves', document.querySelector('.nav-item:nth-child(4)'));
1247                 }
1248             },
1249         });
1250
1251     employeeCalendar.render();
1252     calendarInitialized = true;
1253
1254     // Load events asynchronously after render
1255     refreshCalendarEvents();
1256 }
1257
1258 async function refreshCalendarEvents(){
1259     if(!employeeCalendar) return;
1260     const events = await buildCalendarEvents();
1261     // Remove all existing events and re-add
1262     employeeCalendar.removeAllEvents();
1263     events.forEach(e=>employeeCalendar.addEvent(e));
1264 }

```

```

1264     async function refreshCalendar(){
1265         await loadCalendarBalanceCards();
1266         await refreshCalendarEvents();
1267         showAlert('Calendar refreshed!', 'success');
1268     }
1269
1270     // Move tooltip with mouse
1271     document.addEventListener('mousemove', function(e){
1272         const tt = document.getElementById('cal-tooltip');
1273         if(tt.style.display==='block'){
1274             tt.style.left = (e.clientX+14)+'px';
1275             tt.style.top = (e.clientY+14)+'px';
1276             // Keep tooltip on screen
1277             const rect = tt.getBoundingClientRect();
1278             if(rect.right > window.innerWidth)
1279                 tt.style.left = (e.clientX - rect.width - 10)+'px';
1280             if(rect.bottom > window.innerHeight)
1281                 tt.style.top = (e.clientY - rect.height - 10)+'px';
1282         }
1283     });
1284
1285     /* =====
1286     RESTAURANTS
1287     ===== */
1288     async function loadRestaurants(){
1289         try{
1290             const data=await apiFetch('/restaurants');
1291             availableRestaurants=data||[];
1292             document.getElementById('restaurants-list').innerHTML=data?.length
1293                 ? data.map(r=>`<tr>
1294                     <td>${r.name}||'-'}</td>
1295                     <td>${r.supportedMealSlots?.join(', ')}||'-'}</td>
1296                     <td><span class="chip
1297 $${r.active?'active':'rejected'}>${r.active?'Active':'Inactive'}</span></td>
1298                     <td><button class="btn btn-secondary btn-sm"
1299 onclick='openSubscriptionForRestaurant(${r.restaurantId}||r.id},${JSON.stringify(r.supportedMealSlots)}||[]))'>
1300             Subscribe</button></td>
1301             </tr>`).join('')
1302             : `<tr><td colspan="4" class="empty-state">No restaurants available</td></tr>`;
1303         }catch(e){
1304             document.getElementById('restaurants-list').innerHTML=`<tr><td colspan="4" class="empty-
1305 state">Could not load restaurants</td></tr>`;
1306         }
1307     }
1308
1309     async function loadRestaurantChoices(){
1310         const select=document.getElementById('sub-restaurant-id');
1311         try{
1312             const data=await apiFetch('/restaurants');
1313             availableRestaurants=data||[];
1314             select.innerHTML=availableRestaurants.length
1315                 ? `<option value="">Select a restaurant</option>+
1316             availableRestaurants.map(r=>`<option
1317 value="${r.restaurantId}||r.id">${r.name}</option>`).join('')
1318             : `<option value="">No restaurants available</option>`;
1319         }catch(e){ select.innerHTML=`<option value="">Could not load restaurants</option>`; }
1320     }
1321
1322     function openSubscriptionForRestaurant(restaurantId, supportedMealSlots){
1323         showView('add-subscription',null);
1324         if(supportedMealSlots?.length) document.getElementById('sub-meal-
1325 slot').value=supportedMealSlots[0];
1326         loadRestaurantChoices().then(()=>{ document.getElementById('sub-restaurant-
1327 id').value=String(restaurantId); });
1328     }
1329
1330     /* =====
1331     SUBSCRIPTIONS
1332     ===== */
1333     async function loadSubscriptions(){
1334         try{
1335             const data=await apiFetch('/subscription/user');
1336             const arr=data||[];
1337             document.getElementById('subscriptions-list').innerHTML=arr.length
1338                 ? arr.map(s=>`<tr>

```

```

1334         <td>${s.restaurantName||s.restaurantId||'-'}</td>
1335         <td>${s.mealSlot||'-'}</td>
1336         <td>${s.scheduleType||'-'}${s.dayOfWeek?' ' +s.dayOfWeek:''}</td>
1337         <td>${s.nextDeliveryTime?new Date(s.nextDeliveryTime).toLocaleString():'-'}</td>
1338         <td><span class="chip
${(s.status||'').toLowerCase()}>${s.status||'ACTIVE'}</span></td>
1339         <td><button class="btn btn-danger btn-sm"
onclick="cancelSub(${s.subscriptionId})">Cancel</button></td>
1340     </tr>` ).join('')
1341     : `<tr><td colspan="6" class="empty-state">No subscriptions found</td></tr>`;
1342     }catch(e){
1343         document.getElementById('subscriptions-list').innerHTML=`<tr><td colspan="6" class="empty-
state">Could not load subscriptions</td></tr>`;
1344     }
1345 }
1346
1347 document.getElementById('sub-schedule-type').addEventListener('change',function(){
1348     document.getElementById('day-of-week-group').style.display=this.value==='WEEKLY'?'block':'none';
1349 });
1350 document.getElementById('sub-meal-slot').addEventListener('change', loadRestaurantChoices);
1351
1352 async function addSubscription(){
1353     const restaurantId=parseInt(document.getElementById('sub-restaurant-id').value);
1354     const mealSlot =document.getElementById('sub-meal-slot').value;
1355     const scheduleType=document.getElementById('sub-schedule-type').value;
1356     const dayOfWeek =document.getElementById('sub-day-of-week')?.value;
1357     if(!restaurantId||!mealSlot){showAlert('Please fill all fields','error');return;}
1358     const body={scheduleType,mealSlots:[mealSlot],slotOrders:[{mealSlot,restaurantId}]};
1359     if(scheduleType==='WEEKLY'&&dayOfWeek) body.dayOfWeek=dayOfWeek;
1360     try{
1361         await apiFetch('/subscription',{method:'POST',body:JSON.stringify(body)});
1362         showAlert('Subscribed successfully!','success');
1363         loadSubscriptions(); loadDashboard();
1364     }catch(e){ showAlert('Subscription failed: '+e.message,'error'); }
1365 }
1366
1367 async function cancelSub(id){
1368     if(!confirm('Cancel this subscription?')) return;
1369     try{
1370         await apiFetch(`/subscription/${id}/expire`,{method:'PUT'});
1371         showAlert('Subscription cancelled','success');
1372         loadSubscriptions();
1373     }catch(e){ showAlert('Failed to cancel: '+e.message,'error'); }
1374 }
1375
1376 /* =====
1377 MY DEVICES
1378 ===== */
1379 async function loadMyAssignedDevices(){
1380     try{
1381         const devices=await apiFetch('/devices/my');
1382         const tbody=document.getElementById('my-devices-list');
1383         if(!devices||!devices.length){
1384             tbody.innerHTML=`<tr><td colspan="7" class="empty-state">No devices assigned to
you</td></tr>`;
1385             return;
1386         }
1387         const today=new Date();
1388         tbody.innerHTML=devices.map(d=>{
1389             const we=d.warrantyExpiryDate?new Date(d.warrantyExpiryDate):null;
1390             const wd=we?(we>today
1391                 ?`<span style="color:var(--green);font-size:11px"> ${d.warrantyExpiryDate}</span>`
1392                 :`<span style="color:var(--red);font-size:11px"> Expired
${d.warrantyExpiryDate}</span>`)
1393             :`<span style="color:var(--text3);font-size:11px">—</span>`;
1394             return `<tr>
1395                 <td>#${d.id}</td>
1396                 <td><strong>${d.deviceName||'-'}</strong></td>
1397                 <td>${d.brand||'-'}</td>
1398                 <td><span class="chip">${(d.deviceType||'').toLowerCase()}</span></td>
1399                 <td>${wd}</td>
1400                 <td style="font-size:12px;color:var(--text2)">${d.assignedDate||'-'}</td>
1401                 <td><span class="chip
${(d.deviceStatus||'').toLowerCase()}>${d.deviceStatus||'-'}</span></td>
1402             </tr>`;
1403         }).join('');

```

```

1404         }catch(e){ console.error(e); }
1405     }
1406
1407     /* =====
1408     SERVICE REQUESTS
1409     ===== */
1410     async function loadServiceRequests(){
1411         await Promise.all([loadServiceRequestsList(), loadMyDevicesForSR()]);
1412     }
1413
1414     async function loadServiceRequestsList(){
1415         try{
1416             const requests=await apiFetch('/service-requests/my');
1417             const tbody=document.getElementById('service-requests-list');
1418             if(!requests||!requests.length){
1419                 tbody.innerHTML=<tr><td colspan="7" class="empty-state">No service requests
yet</td></tr>';
1420                 return;
1421             }
1422             tbody.innerHTML=requests.map(r=>`<tr>
1423                 <td style="color:var(--text3);font-family:var(--mono)">#${r.id}</td>
1424                 <td>${r.deviceName||'-'}</td>
1425                 <td><span class="chip pending">${r.requestType}</span></td>
1426                 <td style="max-width:200px;overflow:hidden;text-overflow:ellipsis;white-
space:nowrap">${r.issueDescription||'-'}</td>
1427                 <td>${r.urgent?'<span class="chip rejected">YES</span>':'<span style="color:var(--
text3)">No</span>'}</td>
1428                 <td><span class="chip ${r.status||''}.toLowerCase())">${r.status||'-'}</span></td>
1429                 <td style="font-size:12px;color:var(--text3)">${r.raisedAt?new
Date(r.raisedAt).toLocaleDateString():'-'}</td>
1430                 </tr>`).join('');
1431         }catch(e){ console.error(e); }
1432     }
1433
1434     async function loadMyDevicesForSR(){
1435         try{
1436             const devices=await apiFetch('/devices/my');
1437             const select=document.getElementById('sr-device-select');
1438             select.innerHTML=<option value="">-- Select your device --</option>;
1439             (devices||[]).forEach(d=>{
1440                 const opt=document.createElement('option');
1441                 opt.value=d.id;
1442                 opt.textContent=`${d.deviceName} (${d.brand||d.deviceType||'Device'})`;
1443                 select.appendChild(opt);
1444             });
1445         }catch(e){ console.error('Failed to load devices for SR',e); }
1446     }
1447
1448     async function submitServiceRequest(){
1449         const deviceId      =parseInt(document.getElementById('sr-device-select').value);
1450         const requestType    =document.getElementById('sr-type').value;
1451         const issueDescription=document.getElementById('sr-description').value.trim();
1452         const urgent         =document.getElementById('sr-urgent').checked;
1453         if(!deviceId){showAlert('Please select a device','error');return;}
1454         if(!issueDescription){showAlert('Please describe the issue','error');return;}
1455         try{
1456             await apiFetch('/service-
requests',{method:'POST',body:JSON.stringify({deviceId,requestType,issueDescription,urgent})});
1457             showAlert('Service request submitted!','success');
1458             clearServiceRequestForm();
1459             loadServiceRequestsList();
1460         }catch(e){ showAlert('Failed: '+e.message,'error'); }
1461     }
1462
1463     function clearServiceRequestForm(){
1464         document.getElementById('sr-device-select').value='';
1465         document.getElementById('sr-type').value='REPAIR';
1466         document.getElementById('sr-description').value='';
1467         document.getElementById('sr-urgent').checked=false;
1468     }
1469
1470     /* =====
1471     LOGOUT + ALERT
1472     ===== */
1473     function logout(){
1474         localStorage.removeItem('jwt_token');

```

```

1475     localStorage.removeItem('refresh_token');
1476     localStorage.removeItem('auth_type');
1477     localStorage.removeItem('last_role');
1478     fetch(`${API}/session/logout`, {method: 'POST', credentials: 'include'}).finally(()=>{
1479         document.cookie='JSESSIONID=; expires=Thu, 01 Jan 1970 00:00:00 UTC; path=/;';
1480         window.location.href='/google.html';
1481     });
1482 }
1483 async function loadNotifCount() {
1484     try {
1485         const data = await apiFetch('/notifications/unread-count');
1486         const badge = document.getElementById('notif-badge');
1487         if (data && data.count > 0) {
1488             badge.textContent = data.count;
1489             badge.style.display = 'flex';
1490         } else {
1491             badge.style.display = 'none';
1492         }
1493     } catch(e) {
1494         console.warn('Could not load notification count:', e);
1495     }
1496 }
1497
1498 async function toggleNotifPanel() {
1499     const panel = document.getElementById('notif-panel');
1500     if (panel.style.display === 'none') {
1501         panel.style.display = 'block';
1502         try {
1503             await loadNotifications();
1504             await apiFetch('/notifications/mark-read', { method: 'POST' });
1505             document.getElementById('notif-badge').style.display = 'none';
1506         } catch(e) {
1507             console.warn('Notification panel error:', e);
1508         }
1509     } else {
1510         panel.style.display = 'none';
1511     }
1512 }
1513
1514 async function loadNotifications() {
1515     try {
1516         const data = await apiFetch('/notifications');
1517         const list = document.getElementById('notif-list');
1518         if (!data || data.length === 0) {
1519             list.innerHTML = '<div class="notif-item">No notifications yet.</div>';
1520             return;
1521         }
1522         list.innerHTML = data.map(n => `
1523         <div class="notif-item ${n.read ? '' : 'unread'}">
1524             <div>${n.message}</div>
1525             <div class="notif-time">${new Date(n.createdAt).toLocaleString()}</div>
1526         </div>
1527     `).join('');
1528     } catch(e) {
1529         document.getElementById('notif-list').innerHTML =
1530             '<div class="notif-item">Could not load notifications.</div>';
1531         console.warn('loadNotifications failed:', e);
1532     }
1533 }
1534
1535 // Load badge count on page load
1536 loadNotifCount();
1537 // Refresh count every 30 seconds
1538 setInterval(loadNotifCount, 30000);
1539
1540 document.getElementById('mat-start-date').addEventListener('change', function(){
1541     if(this.value) {
1542         const start = new Date(this.value);
1543         const end = new Date(start);
1544         end.setDate(end.getDate() + 179); // 180 days inclusive
1545         document.getElementById('mat-end-date').value =
1546             end.toISOString().split('T')[0];
1547     } else {
1548         document.getElementById('mat-end-date').value = '';
1549     }
1550 });

```



```

1551
1552     async function applyMaternityLeave() {
1553         const startDate = document.getElementById('mat-start-date').value;
1554         const reason     = document.getElementById('mat-reason').value.trim();
1555
1556         if(!startDate) {
1557             showAlert('Please select a start date', 'error');
1558             return;
1559         }
1560
1561         try {
1562             await apiFetch('/leave_requests/maternity/apply', {
1563                 method: 'POST',
1564                 body: JSON.stringify({ startDate, reason })
1565             });
1566             showAlert('Maternity leave request submitted successfully!', 'success');
1567             // Clear form
1568             document.getElementById('mat-start-date').value = '';
1569             document.getElementById('mat-end-date').value = '';
1570             document.getElementById('mat-reason').value = '';
1571             // Refresh dashboard
1572             loadDashboard();
1573         } catch(e) {
1574             showAlert('Failed to submit: ' + e.message, 'error');
1575         }
1576     }
1577
1578     // Close panel when clicking outside
1579     document.addEventListener('click', e => {
1580         const panel = document.getElementById('notif-panel');
1581         const wrapper = document.querySelector('.notif-wrapper');
1582         if (!panel.contains(e.target) && !wrapper.contains(e.target)) {
1583             panel.style.display = 'none';
1584         }
1585     });
1586
1587     function showAlert(msg,type){
1588         const el=document.createElement('div');
1589         el.className=`alert alert-${type}`;
1590         el.textContent=msg;
1591         document.body.appendChild(el);
1592         setTimeout(()=>el.remove(),3000);
1593     }
1594 </script>
1595 </body>
1596 </html>

```

```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>EMS – Sign In</title>
7      <link
href="https://fonts.googleapis.com/css2?family=Outfit:wght@300;400;500;600;700&family=JetBrains+Mono:wght@400;50
0&display=swap" rel="stylesheet">
8      <style>
9          *{margin:0;padding:0;box-sizing:border-box}
10         :root{
11             --bg:#070910;
12             --panel:#0e1118;
13             --panel2:#131821;
14             --border:rgba(255,255,255,0.07);
15             --border-lit:rgba(255,255,255,0.14);
16             --text:#dde4f0;--text2:#6b7591;--text3:#3d4560;
17             --blue:#4f8cff;--blue-dim:rgba(79,140,255,0.12);
18             --violet:#a78bfa;--violet-dim:rgba(167,139,250,0.12);
19             --red:#f87171;--red-dim:rgba(248,113,113,0.12);
20             --orange:#f59e0b;--orange-dim:rgba(245,158,11,0.12);
21             --green:#34d399;--green-dim:rgba(52,211,153,0.12);
22             --font:'Outfit', sans-serif;--mono:'JetBrains Mono', monospace;
23         }
24
25         body{
26             font-family:var(--font);
27             background:var(--bg);
28             color:var(--text);
29             min-height:100vh;
30             display:flex;
31             overflow-y:auto;
32         }
33
34         /* ---- BACKGROUND ---- */
35         .bg-layer{
36             position:fixed;inset:0;z-index:0;
37             background:
38                 radial-gradient(ellipse 60% 50% at 20% 50%, rgba(79,140,255,0.06) 0%, transparent 70%),
39                 radial-gradient(ellipse 40% 60% at 80% 20%, rgba(167,139,250,0.05) 0%, transparent 60%),
40                 radial-gradient(ellipse 50% 40% at 70% 80%, rgba(52,211,153,0.04) 0%, transparent 60%);
41         }
42         .grid-overlay{
43             position:fixed;inset:0;z-index:0;
44             background-image:
45                 linear-gradient(rgba(255,255,255,0.025) 1px, transparent 1px),
46                 linear-gradient(90deg, rgba(255,255,255,0.025) 1px, transparent 1px);
47             background-size:40px 40px;
48             mask-image:radial-gradient(ellipse 80% 80% at center, black 20%, transparent 80%);
49         }
50
51         #oauth-loading {
52             display: none;
53             position: fixed; inset: 0;
54             background: linear-gradient(135deg, #0d0f14 0%, #131720 100%);
55             z-index: 9999;
56             justify-content: center;
57             align-items: center;
58             flex-direction: column;
59             gap: 20px;
60         }
61         #oauth-loading.visible { display: flex; }
62         .spinner {
63             width: 48px; height: 48px;
64             border: 3px solid #2a2f3f;
65             border-top-color: #4f8cff;
66             border-radius: 50%;
67             animation: spin 0.9s linear infinite;
68         }
69         @keyframes spin { to { transform: rotate(360deg); } }
70

```

```

71      /* ---- SPLIT LAYOUT ---- */
72      .left-panel{
73          flex:1;display:flex;flex-direction:column;justify-content:center;
74          padding:60px;position:relative;z-index:1;
75          animation:fadeLeft 0.8s ease both;
76      }
77      @keyframes fadeLeft{from{opacity:0;transform:translateX(24px)}to{opacity:1;transform:translateX(0)}}
78      .brand{margin-bottom:48px}
79      .brand-logo{display:flex;align-items:center;gap:12px;margin-bottom:20px}
80      .brand-icon{
81          width:44px;height:44px;background:linear-gradient(135deg,#4f8cff,#6366f1);
82          border-radius:12px;display:flex;align-items:center;justify-content:center;
83          font-weight:700;font-size:18px;color:#fff;
84          box-shadow:0 0 30px rgba(79,140,255,0.35);
85          flex-shrink:0;
86      }
87      .brand-name{font-size:22px;font-weight:700;color:var(--text);letter-spacing:-0.3px}
88      .brand-tagline{font-size:13px;color:var(--text2)}
89      .hero-text h1{
90          font-size:clamp(36px,4vw,52px);font-weight:700;
91          line-height:1.1;letter-spacing:-1px;color:var(--text);margin-bottom:16px;
92      }
93      .hero-text h1 .hl-blue{color:var(--blue)}
94      .hero-text h1 .hl-violet{color:var(--violet)}
95      .hero-text h1 .hl-orange{color:var(--orange)}
96      .hero-text p{font-size:15px;color:var(--text2);line-height:1.6;max-width:380px}
97      .feature-pills{display:flex;flex-wrap:wrap;gap:8px;margin-top:32px}
98      .pill{
99          display:inline-flex;align-items:center;gap:6px;
100         padding:6px 14px;border-radius:20px;font-size:12px;font-weight:500;
101         border:1px solid var(--border);background:var(--panel);color:var(--text2);
102     }
103     .pill .dot{width:6px;height:6px;border-radius:50%;flex-shrink:0}
104     .pill.blue .dot{background:var(--blue)} .pill.blue{border-
color:rgba(79,140,255,0.2);color:rgba(79,140,255,0.8)}
105     .pill.violet .dot{background:var(--violet)} .pill.violet{border-
color:rgba(167,139,250,0.2);color:rgba(167,139,250,0.8)}
106     .pill.orange .dot{background:var(--orange)} .pill.orange{border-
color:rgba(245,158,11,0.2);color:rgba(245,158,11,0.8)}
107
108     /* ---- RIGHT PANEL (FORM) ---- */
109     .right-panel{
110         width:460px;flex-shrink:0;
111         display:flex;align-items:flex-start;justify-content:center;
112         overflow-y:auto;
113         padding:32px 40px;position:relative;z-index:1;
114         border-left:1px solid var(--border);
115         background:rgba(14,17,24,0.8);
116         backdrop-filter:blur(20px);
117         animation:fadeRight 0.8s ease both;
118     }
119     @keyframes
fadeRight{from{opacity:0;transform:translateX(24px)}to{opacity:1;transform:translateX(0)}}
120     .form-box{width:100%}
121
122     /* ---- ROLE SELECTOR — 3 roles only: Employee | Manager | Admin ---- */
123     .role-label{font-size:10px;font-weight:600;color:var(--text3);letter-spacing:1.5px;text-
transform:uppercase;font-family:var(--mono);margin-bottom:10px}
124     .role-tabs{display:grid;grid-template-columns:repeat(3,1fr);gap:6px;margin-bottom:28px}
125     .role-tab{
126         padding:10px 8px;border-radius:10px;border:1.5px solid var(--border);
127         background:var(--bg);color:var(--text2);font-family:var(--font);
128         font-size:12px;font-weight:600;cursor:pointer;transition:all 0.2s;
129         display:flex;flex-direction:column;align-items:center;gap:4px;
130     }
131     .role-tab .role-icon{font-size:16px}
132     .role-tab:hover{border-color:var(--border-lit);color:var(--text)}
133     .role-tab.active-emp{border-color:var(--blue);background:var(--blue-dim);color:var(--blue)}
134     .role-tab.active-mgr{border-color:var(--violet);background:var(--violet-dim);color:var(--violet)}
135     .role-tab.active-adm{border-color:var(--orange);background:var(--orange-dim);color:var(--orange)}
136
137     /* ---- AUTH TABS ---- */
138     .auth-tabs{display:flex;background:var(--bg);border-radius:10px;padding:4px;gap:4px;margin-
bottom:24px;border:1px solid var(--border)}
139     .auth-tab{
140         flex:1;padding:9px;border:none;background:transparent;

```

```

141         color:var(--text2);font-family:var(--font);font-weight:600;
142         font-size:13px;cursor:pointer;border-radius:7px;transition:all 0.2s;
143     }
144     .auth-tab.active{background:var(--panel2);color:var(--text);box-shadow:0 1px 4px rgba(0,0,0,0.3)}
145
146     /* ---- FORM ELEMENTS ---- */
147     .form-group{margin-bottom:16px}
148     .form-label{display:block;font-size:10px;font-weight:600;color:var(--text2);text-
transform:uppercase;letter-spacing:0.8px;font-family:var(--mono);margin-bottom:7px}
149     .form-input{
150         width:100%;padding:11px 14px;
151         background:var(--bg);border:1px solid var(--border);
152         border-radius:9px;color:var(--text);
153         font-family:var(--font);font-size:14px;
154         transition:all 0.2s;
155     }
156     .form-input:focus{outline:none;border-color:var(--blue);box-shadow:0 0 3px rgba(79,140,255,0.08)}
157     .form-input::placeholder{color:var(--text3)}
158     .form-input.focus-violet:focus{border-color:var(--violet);box-shadow:0 0 3px
159     rgba(167,139,250,0.08)}
160     .form-input.focus-orange:focus{border-color:var(--orange);box-shadow:0 0 3px
161     rgba(245,158,11,0.08)}
162
163     /* ---- BUTTONS ---- */
164     .btn-login{
165         width:100%;padding:12px;border:none;border-radius:9px;
166         font-family:var(--font);font-size:14px;font-weight:600;
167         cursor:pointer;transition:all 0.2s;margin-top:4px;
168         position:relative;overflow:hidden;
169     }
170     .btn-login.blue{background:var(--blue);color:#fff}
171     .btn-login.blue:hover{background:#6ba0ff;box-shadow:0 6px 20px
172     rgba(79,140,255,0.35);transform:translateY(-1px)}
173     .btn-login.violet{background:var(--violet);color:#fff}
174     .btn-login.violet:hover{background:#b8a0fc;box-shadow:0 6px 20px
175     rgba(167,139,250,0.35);transform:translateY(-1px)}
176     .btn-login.orange{background:var(--orange);color:#fff}
177     .btn-login.orange:hover{background:#f7b23b;box-shadow:0 6px 20px
178     rgba(245,158,11,0.35);transform:translateY(-1px)}
179     .btn-login.active{transform:translateY(0)}
180     .btn-login.loading{opacity:0.7;pointer-events:none}
181
182     .divider{display:flex;align-items:center;gap:12px;margin:20px 0;color:var(--text3);font-
183     size:11px;font-family:var(--mono)}
184     .divider::before,.divider::after{content:'';flex:1;height:1px;background:var(--border)}
185
186     .btn-google{
187         width:100%;padding:11px;background:transparent;
188         border:1px solid var(--border-lit);border-radius:9px;
189         color:var(--text2);font-family:var(--font);font-size:13px;font-weight:500;
190         cursor:pointer;transition:all 0.2s;
191         display:flex;align-items:center;justify-content:center;gap:10px;
192     }
193     .btn-google:hover{background:rgba(255,255,255,0.04);border-color:rgba(255,255,255,0.2);color:var(--
194     text)}
195
196     /* ---- GOOGLE ROLE SELECT ---- */
197     .google-role-row{display:flex;gap:8px;margin-bottom:10px;align-items:center}
198     .google-role-label{font-size:11px;color:var(--text3);font-family:var(--mono);white-space:nowrap}
199     .google-role-select{
200         flex:1;padding:8px 10px;background:var(--bg);
201         border:1px solid var(--border);border-radius:8px;
202         color:var(--text);font-family:var(--font);font-size:13px;cursor:pointer;
203     }
204     .google-role-select:focus{outline:none;border-color:var(--border-lit)}
205     .google-role-select option{background:var(--panel)}
206
207     /* ---- VENDOR LINK ---- */
208     .vendor-link-row{
209         margin-top:18px;padding-top:16px;
210         border-top:1px solid var(--border);
211         display:flex;align-items:center;justify-content:space-between;
212     }
213     .vendor-link-row p{font-size:12px;color:var(--text3);}
214     .btn-vendor-link{
215         padding:7px 16px;border-radius:8px;

```

```

209         border:1px solid rgba(248,113,113,0.3);
210         background:var(--red-dim);color:var(--red);
211         font-family:var(--font);font-size:12px;font-weight:600;
212         cursor:pointer;text-decoration:none;transition:all 0.2s;
213         white-space:nowrap;
214     }
215     .btn-vendor-link:hover{background:rgba(248,113,113,0.2);border-color:var(--red);}
216
217     /* ---- ERROR / STATUS ---- */
218     .status-box{
219         border-radius:9px;padding:10px 14px;margin-top:14px;
220         font-size:13px;font-weight:500;display:none;
221         animation:fadeIn 0.3s ease;
222     }
223     @keyframes fadeIn{from{opacity:0;transform:translateY(-4px)}to{opacity:1;transform:translateY(0)}}
224     .status-box.error{background:var(--red-dim);color:var(--red);border:1px solid
225     rgba(248,113,113,0.2)}
226     .status-box.success{background:var(--green-dim);color:var(--green);border:1px solid
227     rgba(52,211,153,0.2)}
228     .status-box.loading{background:var(--blue-dim);color:var(--blue);border:1px solid
229     rgba(79,140,255,0.2)}
230
231     /* ---- FOOTER ---- */
232     .form-footer{margin-top:20px;padding-top:18px;border-top:1px solid var(--border)}
233     .tz-row{display:flex;align-items:center;gap:8px;margin-bottom:8px}
234     .tz-label{font-size:10px;color:var(--text3);font-family:var(--mono);flex-shrink:0}
235     .tz-select{
236         flex:1;padding:6px 8px;background:var(--bg);border:1px solid var(--border);
237         border-radius:7px;color:var(--text2);font-family:var(--mono);font-size:11px;cursor:pointer;
238     }
239     .tz-select:focus{outline:none;border-color:var(--border-lit)}
240     .tz-select option{background:var(--panel)}
241     .tz-time{font-size:10px;color:var(--blue);font-family:var(--mono)}
242
243     /* ---- RESPONSIVE ---- */
244     @media(max-width:900px){
245         .left-panel{display:none}
246         .right-panel{width:100%;border-left:none;background:var(--bg)}
247     }
248 </style>
249 </head>
250 <body>
251 <div id="oauth-loading">
252     <div class="spinner"></div>
253     <div style="text-align: center;">
254         <p style="color: #e0e0e0; font-size: 16px; font-weight: 600; margin: 0 0 6px; font-family:
255         'Outfit', sans-serif;">Signing you in...</p>
256         <p style="color: #8b92a8; font-size: 13px; margin: 0; font-family: 'Outfit', sans-serif;">Verifying
257         your Google account</p>
258     </div>
259 </div>
260 <div class="bg-layer"></div>
261 <div class="grid-overlay"></div>
262
263 <!-- LEFT HERO -->
264 <div class="left-panel">
265     <div class="brand">
266         <div class="brand-logo">
267             <div class="brand-icon">E</div>
268             <div>
269                 <div class="brand-name">EMS Portal</div>
270                 <div class="brand-tagline">Employee Management System</div>
271             </div>
272         </div>
273     </div>
274     <div class="hero-text">
275         <h1>Your workspace,<br><span class="hl-blue">one login</span><br>away.</h1>
276         <p>Employees, Managers, and Admins each get their own tailored experience. Sign in to access your
277         personalized dashboard.</p>
278     </div>
279     <div class="feature-pills">
280         <div class="pill blue"><span class="dot"></span>Employee Portal</div>
281         <div class="pill violet"><span class="dot"></span>Manager Console</div>
282         <div class="pill orange"><span class="dot"></span>Admin Control Center</div>
283         <div class="pill blue"><span class="dot"></span>Leave Management</div>
284         <div class="pill violet"><span class="dot"></span>Meal Subscriptions</div>

```

```

279     <div class="pill orange"><span class="dot"></span>Delivery Tracking</div>
280   </div>
281 </div>
282
283 <!-- RIGHT FORM -->
284 <div class="right-panel">
285   <div class="form-box">
286
287     <!-- ROLE SELECTOR: 3 roles only – Vendor removed -->
288     <div class="role-label">Sign in as</div>
289     <div class="role-tabs">
290       <button class="role-tab active-emp" id="role-emp" onclick="selectRole('EMPLOYEE')">
291         <span class="role-icon"></span>Employee
292       </button>
293       <button class="role-tab" id="role-mgr" onclick="selectRole('MANAGER')">
294         <span class="role-icon"></span>Manager
295       </button>
296       <button class="role-tab" id="role-adm" onclick="selectRole('ADMIN')">
297         <span class="role-icon"></span>Admin
298       </button>
299     </div>
300
301     <!-- AUTH METHOD TABS -->
302     <div class="auth-tabs">
303       <button class="auth-tab active" id="tab-jwt" onclick="switchTab('jwt')">JWT Login</button>
304       <button class="auth-tab" id="tab-session" onclick="switchTab('session')">Session Login</button>
305     </div>
306
307     <!-- JWT FORM -->
308     <div id="jwt-form">
309       <div class="form-group">
310         <label class="form-label">Enter your Email</label>
311         <input class="form-input" id="jwt-username" type="text"
placeholder="your.email@company.com" autocomplete="off">
312       </div>
313       <div class="form-group">
314         <label class="form-label">Password</label>
315         <input class="form-input" id="jwt-password" type="password" placeholder="....."
autocomplete="current-password">
316       </div>
317       <button class="btn-login blue" id="jwt-btn" onclick="jwtLogin()">Sign In</button>
318     </div>
319
320     <!-- SESSION FORM -->
321     <div id="session-form" style="display:none">
322       <div class="form-group">
323         <label class="form-label">Enter your Email</label>
324         <input class="form-input" id="session-username" type="text"
placeholder="your.email@company.com">
325       </div>
326       <div class="form-group">
327         <label class="form-label">Password</label>
328         <input class="form-input" id="session-password" type="password" placeholder=".....">
329       </div>
330       <button class="btn-login blue" id="session-btn" onclick="sessionLogin()">Sign In with
Session</button>
331     </div>
332
333     <!-- OAUTH – Vendor removed from Google role select -->
334     <div class="divider">or continue with</div>
335     <div class="google-role-row">
336       <span class="google-role-label">Sign in as</span>
337       <select class="google-role-select" id="oauth-role">
338         <option value="EMPLOYEE">Employee</option>
339         <option value="MANAGER">Manager</option>
340         <option value="ADMIN">Admin</option>
341       </select>
342     </div>
343     <button class="btn-google" onclick="googleLogin()">
344       <svg width="18" height="18" viewBox="0 0 24 24">
345         <path fill="#4285F4" d="M22.56 12.25c0-.78-.07-1.53-.2-2.25H12v4.26h5.92c-.26 1.37-1.04
2.53-2.21 3.31v2.77h3.57c2.08-1.92 3.28-4.74 3.28-8.09z"/>
346         <path fill="#34A853" d="M12 23c2.97 0 5.46-.98 7.28-2.66l-3.57-2.77c-.98.66-2.23 1.06-3.71
1.06-2.86 0-5.29-1.93-6.16-4.53H2.18v2.84C3.99 20.53 7.7 23 12 23z"/>
347         <path fill="#FBBC05" d="M5.84 14.09c-.22-.66-.35-1.36-.35-2.09s.13-1.43.35-
2.09V7.07H2.18c1.43 8.55 1 10.22 1 12s.43 3.45 1.18 4.93l2.85-2.22.81-.62z"/>

```

```

348         <path fill="#EA4335" d="M12 5.38c1.62 0 3.06.56 4.21 1.64l3.15-3.15C17.45 2.09 14.97 1 12 1
7.7 1 3.99 3.47 2.18 7.07l3.66 2.84c.87-2.6 3.3-4.53 6.16-4.53z"/>
349     </svg>
350     Continue with Google
351 </button>
352
353 <div class="status-box" id="status-box"></div>
354
355 <!-- Vendor Login Link -->
356 <div class="vendor-link-row">
357     <p>Are you a vendor?</p>
358     <a class="btn-vendor-link" href="/vendor-login.html">Vendor Login </a>
359 </div>
360
361 <!-- FOOTER -->
362 <div class="form-footer">
363     <div class="tz-row">
364         <span class="tz-label">TZ</span>
365         <select class="tz-select" id="tz-select" onchange="updateTz()">
366             <option value="Asia/Kolkata">India – IST UTC+5:30</option>
367             <option value="UTC">UTC</option>
368             <option value="America/New_York">New York – EST UTC-5</option>
369             <option value="America/Los_Angeles">Los Angeles – PST UTC-8</option>
370             <option value="Europe/London">London – GMT UTC+0</option>
371             <option value="Europe/Paris">Paris – CET UTC+1</option>
372             <option value="Asia/Tokyo">Tokyo – JST UTC+9</option>
373             <option value="Asia/Dubai">Dubai – GST UTC+4</option>
374             <option value="Australia/Sydney">Sydney – AEDT UTC+11</option>
375             <option value="Asia/Singapore">Singapore – SGT UTC+8</option>
376         </select>
377         <span class="tz-time" id="tz-time">--:--:--</span>
378     </div>
379 </div>
380
381 </div>
382 <!-- TOTP MODAL OVERLAY -->
383 <div id="totp-overlay" style="display:none;position:fixed;inset:0;z-
index:999;background:rgba(7,9,16,0.85);backdrop-filter:blur(8px);display:none;align-items:center;justify-
content:center;">
384     <div style="background:var(--panel);border:1px solid var(--border-lit);border-
radius:16px;padding:36px;width:340px;text-align:center;animation:fadeIn 0.25s ease;">
385         <div style="font-size:28px;margin-bottom:8px;"></div>
386         <h3 style="font-size:17px;font-weight:700;color:var(--text);margin-bottom:6px;">Two-Factor
Verification</h3>
387         <p style="font-size:13px;color:var(--text2);margin-bottom:24px;">Enter the 6-digit code from
your authenticator app.</p>
388         <input id="totp-code" class="form-input" type="text" inputmode="numeric" maxlength="6"
placeholder="000000"
389             style="text-align:center;font-size:22px;font-family:var(--mono);letter-
spacing:8px;padding:14px;"
390             oninput="this.value=this.value.replace(/\D/g,'')">
391         <div id="totp-status" style="display:none;margin-top:10px;font-size:13px;color:var(--
red);"></div>
392         <button id="totp-verify-btn" class="btn-login blue" style="margin-top:16px;"
onclick="submitTotp()">Verify</button>
393         <button onclick="cancelTotp()" style="margin-
top:10px;width:100%;padding:10px;border:none;background:transparent;color:var(--text2);font-family:var(--
font);font-size:13px;cursor:pointer...
394     </div>
395 </div>
396
397 </div>
398
399 <script>
400     const API = 'http://localhost:8080';
401     const GOOGLE_CLIENT_ID = '525324511556-s9eugd1o761n6cjje2nr60cv8ho6joq2.apps.googleusercontent.com';
402     const REDIRECT_URI = 'http://localhost:8080/google.html';
403
404     // Role dashboard mapping – VENDOR removed from this page
405     const DASHBOARDS = {
406         EMPLOYEE: '/employee-dashboard.html',
407         MANAGER: '/manager-dashboard.html',
408         ADMIN: '/dashboard.html',
409         USER: '/pending.html'
410     };
411
412     let currentRole = 'EMPLOYEE';

```

```

413
414     function buildAuthOptions(token, authType) {
415         const headers = {};
416         if (token) headers.Authorization = `Bearer ${token}`;
417         return {
418             headers,
419             credentials: authType === 'session' ? 'include' : 'same-origin'
420         };
421     }
422
423     async function resolveRoleFromServer(token, authType) {
424         const authOptions = buildAuthOptions(token, authType);
425
426         // Check employee/manager/admin endpoint – vendors are not expected here
427         const employeeResponse = await fetch(`${API}/employee/me`, authOptions);
428         if (employeeResponse.ok) {
429             const employee = await employeeResponse.json();
430             return employee.role || 'EMPLOYEE';
431         }
432
433         throw new Error('Unable to determine authenticated role');
434     }
435
436     async function redirectByRole(token, authType = localStorage.getItem('auth_type')) {
437         let role;
438
439         if (token) {
440             role = getRoleFromJWT(token);
441         } else {
442             try {
443                 const res = await fetch(`${API}/employee/me`, { credentials: 'include' });
444                 if (res.ok) {
445                     const emp = await res.json();
446                     role = emp.role || 'USER';
447                 } else {
448                     role = 'USER';
449                 }
450             } catch(e) {
451                 role = localStorage.getItem('last_role') || 'USER';
452             }
453         }
454
455         console.log('Final role for redirect:', role);
456         localStorage.setItem('last_role', role);
457
458         // Block ALL vendor types from using the main login page
459         if (role === 'FOOD_VENDOR' || role === 'TECH_VENDOR') {
460             localStorage.removeItem('jwt_token');
461             localStorage.removeItem('auth_type');
462             localStorage.removeItem('last_role');
463             // Reset button state
464             const jwtBtn = document.getElementById('jwt-btn');
465             const sesBtn = document.getElementById('session-btn');
466             if (jwtBtn) { jwtBtn.classList.remove('loading'); jwtBtn.textContent = 'Sign In'; }
467             if (sesBtn) { sesBtn.classList.remove('loading'); sesBtn.textContent = 'Sign In with Session'; }
468             showStatus('error', 'Vendors must use the Vendor Login page. ');
469             return;
470         }
471
472         const destination = DASHBOARDS[role];
473         if (!destination) {
474             // Unknown role – don't redirect anywhere, just show error
475             localStorage.removeItem('jwt_token');
476             localStorage.removeItem('auth_type');
477             showStatus('error', 'Unknown role: ' + role + '. Please contact your administrator. ');
478             const jwtBtn = document.getElementById('jwt-btn');
479             if (jwtBtn) { jwtBtn.classList.remove('loading'); jwtBtn.textContent = 'Sign In'; }
480             return;
481         }
482
483         console.log('Redirecting to:', destination);
484         window.location.href = destination;
485     }
486
487     // ---- Role selector (3 tabs: Employee | Manager | Admin) ----
488     function selectRole(role) {

```



```

489     currentRole = role;
490     document.getElementById('role-emp').className = 'role-tab';
491     document.getElementById('role-mgr').className = 'role-tab';
492     document.getElementById('role-adm').className = 'role-tab';
493
494     const classMap = { EMPLOYEE: 'active-emp', MANAGER: 'active-mgr', ADMIN: 'active-adm' };
495     const idMap = { EMPLOYEE: 'role-emp', MANAGER: 'role-mgr', ADMIN: 'role-adm' };
496     document.getElementById(idMap[role]).classList.add(classMap[role]);
497
498     document.getElementById('oauth-role').value = role;
499
500     const colorMap = { EMPLOYEE: 'blue', MANAGER: 'violet', ADMIN: 'orange' };
501     const col = colorMap[role];
502     document.getElementById('jwt-btn').className = `btn-login ${col}`;
503     document.getElementById('session-btn').className = `btn-login ${col}`;
504
505     document.querySelectorAll('.form-input').forEach(inp => {
506         inp.className = (`form-input focus-${col === 'blue' ? '' : col}`).trim();
507     });
508 }
509
510 // ---- Auth tab switching ----
511 function switchTab(type) {
512     document.getElementById('jwt-form').style.display = type === 'jwt' ? 'block' : 'none';
513     document.getElementById('session-form').style.display = type === 'session' ? 'block' : 'none';
514     document.getElementById('tab-jwt').classList.toggle('active', type === 'jwt');
515     document.getElementById('tab-session').classList.toggle('active', type === 'session');
516     hideStatus();
517 }
518
519 // ---- Status display ----
520 function showStatus(type, msg) {
521     const box = document.getElementById('status-box');
522     box.className = `status-box ${type}`;
523     box.textContent = msg;
524     box.style.display = 'block';
525 }
526 function hideStatus() {
527     document.getElementById('status-box').style.display = 'none';
528 }
529
530 // ---- JWT Login ----
531 let _preAuthToken = null; // holds token between step 1 and step 2
532
533 async function jwtLogin() {
534     const username = document.getElementById('jwt-username').value.trim();
535     const password = document.getElementById('jwt-password').value;
536     if (!username || !password) {
537         showStatus('error', 'Please enter your username and password');
538         return;
539     }
540
541     const btn = document.getElementById('jwt-btn');
542     btn.classList.add('loading'); btn.textContent = 'Signing in...';
543     showStatus('loading', 'Authenticating...');
544
545     try {
546         const res = await fetch(`${API}/Authenticate`, {
547             method: 'POST',
548             headers: { 'Content-Type': 'application/json' },
549             body: JSON.stringify({ username, password })
550         });
551
552         if (res.ok) {
553             const data = await res.json();
554
555             // --- TOTP required: show step-2 modal ---
556             if (data.requiresTotp) {
557                 _preAuthToken = data.preAuthToken;
558                 btn.classList.remove('loading'); btn.textContent = 'Sign In';
559                 hideStatus();
560                 openTotpModal();
561                 return;
562             }
563
564             // --- No TOTP: issue JWT directly ---

```

```

565         localStorage.setItem('jwt_token', data.accessToken);
566         localStorage.setItem('refresh_token', data.refreshToken || '');
567         localStorage.setItem('auth_type', 'jwt');
568         localStorage.setItem('last_role', currentRole);
569         localStorage.setItem('user_timezone', document.getElementById('tz-select').value);
570         showStatus('success', 'Login successful! Redirecting...');
571         setTimeout(() => redirectByRole(data.accessToken, 'jwt'), 600);
572
573     } else {
574         const err = await res.text();
575         showStatus('error', 'Login failed: ' + (err || 'Invalid credentials'));
576         btn.classList.remove('loading'); btn.textContent = 'Sign In';
577     }
578 } catch (e) {
579     showStatus('error', 'Connection error: ' + e.message);
580     btn.classList.remove('loading'); btn.textContent = 'Sign In';
581 }
582 }
583
584 function openTotpModal() {
585     const overlay = document.getElementById('totp-overlay');
586     overlay.style.display = 'flex';
587     document.getElementById('totp-code').value = '';
588     document.getElementById('totp-status').style.display = 'none';
589     setTimeout(() => document.getElementById('totp-code').focus(), 100);
590 }
591
592 function cancelTotp() {
593     _preAuthToken = null;
594     document.getElementById('totp-overlay').style.display = 'none';
595 }
596
597 async function submitTotp() {
598     const code = document.getElementById('totp-code').value.trim();
599     if (code.length !== 6) {
600         document.getElementById('totp-status').textContent = 'Please enter a 6-digit code.';
601         document.getElementById('totp-status').style.display = 'block';
602         return;
603     }
604
605     const verifyBtn = document.getElementById('totp-verify-btn');
606     verifyBtn.classList.add('loading'); verifyBtn.textContent = 'Verifying...';
607     document.getElementById('totp-status').style.display = 'none';
608
609     try {
610         const res = await fetch(`${API}/Authenticate/totp`, {
611             method: 'POST',
612             headers: { 'Content-Type': 'application/json' },
613             body: JSON.stringify({ preAuthToken: _preAuthToken, code })
614         });
615
616         if (res.ok) {
617             const data = await res.json();
618             _preAuthToken = null;
619             document.getElementById('totp-overlay').style.display = 'none';
620             localStorage.setItem('jwt_token', data.accessToken);
621             localStorage.setItem('refresh_token', data.refreshToken || '');
622             localStorage.setItem('auth_type', 'jwt');
623             localStorage.setItem('last_role', currentRole);
624             localStorage.setItem('user_timezone', document.getElementById('tz-select').value);
625             showStatus('success', 'Verified! Redirecting...');
626             setTimeout(() => redirectByRole(data.accessToken, 'jwt'), 600);
627         } else {
628             const err = await res.json().catch(() => ({}));
629             document.getElementById('totp-status').textContent = err.error || 'Invalid code. Try
again.';
630             document.getElementById('totp-status').style.display = 'block';
631             document.getElementById('totp-code').value = '';
632             document.getElementById('totp-code').focus();
633         }
634     } catch (e) {
635         document.getElementById('totp-status').textContent = 'Connection error: ' + e.message;
636         document.getElementById('totp-status').style.display = 'block';
637     } finally {
638         verifyBtn.classList.remove('loading'); verifyBtn.textContent = 'Verify';
639     }

```

```

640     }
641
642     // ---- Session Login ----
643     async function sessionLogin() {
644         const username = document.getElementById('session-username').value.trim();
645         const password = document.getElementById('session-password').value;
646         if (!username || !password) { showStatus('error', 'Please enter your username and password');
return; }
647
648         const btn = document.getElementById('session-btn');
649         btn.classList.add('loading'); btn.textContent = 'Signing in...';
650         showStatus('loading', 'Authenticating...');
651
652         try {
653             const res = await fetch(`${API}/session/login`, {
654                 method: 'POST',
655                 headers: { 'Content-Type': 'application/json' },
656                 body: JSON.stringify({ username, password }),
657                 credentials: 'include'
658             });
659             if (res.ok) {
660                 localStorage.setItem('auth_type', 'session');
661                 localStorage.setItem('last_role', currentRole);
662                 localStorage.setItem('user_timezone', document.getElementById('tz-select').value);
663                 showStatus('success', 'Login successful! Redirecting...');
664                 setTimeout(() => redirectByRole(null, 'session'), 600);
665             } else {
666                 const err = await res.json().catch(() => ({}));
667                 showStatus('error', 'Login failed: ' + (err.error || 'Invalid credentials'));
668                 btn.classList.remove('loading'); btn.textContent = 'Sign In with Session';
669             }
670         } catch(e) {
671             showStatus('error', 'Connection error: ' + e.message);
672             btn.classList.remove('loading'); btn.textContent = 'Sign In with Session';
673         }
674     }
675
676     // ---- Google OAuth – goes through backend /auth/google/init ----
677     function googleLogin() {
678         const role = document.getElementById('oauth-role').value;
679         const timezone = document.getElementById('tz-select').value;
680         window.location.href =
681             `/auth/google/init` +
682             `?role=${encodeURIComponent(role)}` +
683             `&timezone=${encodeURIComponent(timezone)}`;
684     }
685
686     function getRoleFromJWT(token) {
687         try {
688             const payload = JSON.parse(atob(token.split('.')[1]));
689             console.log('JWT payload:', payload);
690             const role = payload.role || 'USER';
691             // Strip ROLE_ prefix if accidentally present
692             return typeof role === 'string' ? role.replace(/^ROLE_/, '') : 'USER';
693         } catch (e) {
694             console.error('Failed to parse JWT:', e);
695             return 'USER';
696         }
697     }
698
699     // ---- Spring redirects back here with tokens as query params ----
700     async function handleOAuthCallback() {
701         const params = new URLSearchParams(window.location.search);
702         const accessToken = params.get('accessToken');
703         const refreshToken = params.get('refreshToken');
704
705         if (!accessToken) return;
706
707         window.history.replaceState({}, document.title, window.location.pathname);
708
709         localStorage.setItem('jwt_token', accessToken);
710         localStorage.setItem('refresh_token', refreshToken || '');
711         localStorage.setItem('auth_type', 'jwt');
712
713         const role = getRoleFromJWT(accessToken);
714         console.log('Authenticated role:', role);

```

```

715     localStorage.setItem('last_role', role);
716
717     showStatus('success', 'Signed in! Redirecting...');
718     setTimeout(() => {
719         window.location.href = DASHBOARDS[role] || '/google.html';
720     }, 800);
721 }
722
723 // ---- Timezone ----
724 function updateTz() {
725     localStorage.setItem('user_timezone', document.getElementById('tz-select').value);
726     refreshTzClock();
727 }
728 function refreshTzClock() {
729     const tz = document.getElementById('tz-select').value;
730     document.getElementById('tz-time').textContent =
731         new Date().toLocaleTimeString('en-GB', { timeZone: tz, hour12: false });
732 }
733
734 // ---- Enter key support ----
735 document.addEventListener('keydown', e => {
736     if (e.key !== 'Enter') return;
737     // If TOTP modal is open, submit TOTP
738     if (document.getElementById('totp-overlay').style.display === 'flex') {
739         submitTotp();
740         return;
741     }
742     if (document.getElementById('jwt-form').style.display !== 'none') jwtLogin();
743     else sessionLogin();
744 });
745 // ---- Init ----
746 window.onload = () => {
747     const savedTz = localStorage.getItem('user_timezone') || 'Asia/Kolkata';
748     document.getElementById('tz-select').value = savedTz;
749     refreshTzClock();
750     setInterval(refreshTzClock, 1000);
751
752     const params = new URLSearchParams(window.location.search);
753     const accessToken = params.get('accessToken');
754     if (accessToken) {
755         document.getElementById('oauth-loading').classList.add('visible');
756         document.querySelector('.right-panel').style.display = 'none';
757         document.querySelector('.left-panel').style.display = 'none';
758         handleOAuthCallback();
759     }
760 };
761 </script>
762 </body>
763 </html>
764

```

```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>EMS - Leave Management</title>
7      <link
href="https://fonts.googleapis.com/css2?family=DM+Sans:wght@400;500;600;700&family=DM+Mono:wght@400;500&display=
swap" rel="stylesheet">
8      <style>
9          * { margin: 0; padding: 0; box-sizing: border-box; }
10         :root {
11             --bg: #0d0f14; --bg2: #131720; --bg3: #1a2030;
12             --panel: #1e2436; --panel2: #252c40;
13             --border: rgba(255,255,255,0.07); --border2: rgba(255,255,255,0.12);
14             --text: #e0e0e0; --text2: #8b92a8; --text3: #5557fa;
15             --accent: #4f8cff; --accent-dim: rgba(79,140,255,0.12);
16             --green: #36d399; --green-dim: rgba(54,211,153,0.12);
17             --amber: #f9c74f; --amber-dim: rgba(249,199,79,0.12);
18             --red: #f87171; --red-dim: rgba(248,113,113,0.12);
19             --violet: #a78bfa; --violet-dim: rgba(167,139,250,0.12);
20             --font: 'DM Sans', sans-serif; --mono: 'DM Mono', monospace;
21         }
22         body { font-family: var(--font); background: var(--bg); color: var(--text); font-size: 14px; line-
height: 1.5; overflow: hidden; }
23         .app { display: flex; height: 100vh; }
24
25         /* Sidebar */
26         .sidebar { width: 260px; background: var(--bg2); border-right: 1px solid var(--border); display:
flex; flex-direction: column; overflow-y: auto; }
27         .logo { padding: 20px; border-bottom: 1px solid var(--border); }
28         .logo h2 { font-size: 20px; color: var(--text); }
29         .logo span { color: var(--accent); }
30         .logo p { font-size: 11px; color: var(--text3); margin-top: 4px; }
31         .nav-section { padding: 12px 20px 4px; font-size: 10px; font-weight: 600; color: var(--text2); text-
transform: uppercase; letter-spacing: 0.08em; }
32         .nav-item { padding: 10px 20px; margin: 2px 12px; border-radius: 8px; cursor: pointer; transition:
all 0.2s; color: var(--text2); display: flex; align-items: center; gap: 10px; font-size: 1.1em; }
33         .nav-item svg { opacity: 0.6; flex-shrink: 0; }
34         .nav-item:hover, .nav-item.active { background: var(--accent-dim); color: var(--accent); }
35         .nav-item:hover svg, .nav-item.active svg { opacity: 1; }
36         .nav-item .badge-count { margin-left: auto; background: var(--amber-dim); color: var(--amber); font-
size: 10px; padding: 2px 6px; border-radius: 10px; font-weight: 600; }
37
38         /* Main */
39         .main-content { flex: 1; display: flex; flex-direction: column; overflow: hidden; }
40         .topbar { background: var(--bg2); padding: 16px 24px; display: flex; justify-content: space-between;
align-items: center; border-bottom: 1px solid var(--border); }
41         .page-title { font-size: 20px; font-weight: 600; }
42         .topbar-right { display: flex; align-items: center; gap: 16px; }
43         .user-pill { display: flex; align-items: center; gap: 8px; background: var(--bg3); border: 1px solid
var(--border2); border-radius: 20px; padding: 6px 12px; }
44         .user-avatar { width: 24px; height: 24px; border-radius: 50%; background: var(--accent-dim);
display: flex; align-items: center; justify-content: center; font-size: 10px; font-weight: 600; ... }
45         .user-name { font-size: 12px; color: var(--text2); }
46         .logout-btn { padding: 8px 16px; background: var(--red-dim); border: 1px solid
rgba(248,113,113,0.3); border-radius: 6px; color: var(--red); cursor: pointer; font-family: var(--font); font-
... }
47         .logout-btn:hover { background: rgba(248,113,113,0.2); }
48         .content { flex: 1; overflow-y: auto; padding: 24px; }
49
50         /* Views */
51         .view { display: none; }
52         .view.active { display: block; }
53
54         /* Stats */
55         .stats-grid { display: grid; grid-template-columns: repeat(4, 1fr); gap: 16px; margin-bottom: 24px;
}
56         .stat-card { background: var(--panel); padding: 20px; border-radius: 12px; border: 1px solid var(--
border); position: relative; overflow: hidden; }
57         .stat-card::before { content: ''; position: absolute; top: 0; left: 0; right: 0; height: 2px; }
58         .stat-card.blue::before { background: var(--accent); }

```

```

59     .stat-card.green::before { background: var(--green); }
60     .stat-card.amber::before { background: var(--amber); }
61     .stat-card.red::before { background: var(--red); }
62     .stat-label { font-size: 11px; color: var(--text2); text-transform: uppercase; letter-spacing:
0.05em; margin-bottom: 8px; }
63     .stat-value { font-size: 28px; font-weight: 600; }
64     .stat-value.blue { color: var(--accent); }
65     .stat-value.green { color: var(--green); }
66     .stat-value.amber { color: var(--amber); }
67     .stat-value.red { color: var(--red); }
68
69     /* Section header */
70     .section-head { display: flex; justify-content: space-between; align-items: center; margin-bottom:
16px; flex-wrap: wrap; gap: 10px; }
71     .section-title { font-size: 16px; font-weight: 600; }
72     .section-actions { display: flex; gap: 8px; align-items: center; }
73
74     /* Table */
75     .table-wrap { background: var(--panel); border-radius: 12px; border: 1px solid var(--border);
overflow: hidden; }
76     table { width: 100%; border-collapse: collapse; }
77     th { text-align: left; padding: 12px 16px; background: var(--bg3); color: var(--text2); font-size:
11px; font-weight: 600; text-transform: uppercase; letter-spacing: 0.04em; border-bottom: ...
78     td { padding: 12px 16px; border-top: 1px solid var(--border); font-size: 13px; }
79     tr:hover td { background: rgba(255,255,255,0.02); }
80     .empty-state { text-align: center; padding: 48px; color: var(--text2); font-size: 13px; }
81
82     /* Chips */
83     .chip { padding: 4px 10px; border-radius: 20px; font-size: 11px; font-weight: 600; display: inline-
block; }
84     .chip.pending, .chip.PENDING { background: var(--amber-dim); color: var(--amber); }
85     .chip.approved, .chip.APPROVED { background: var(--green-dim); color: var(--green); }
86     .chip.rejected, .chip.REJECTED { background: var(--red-dim); color: var(--red); }
87     .chip.cancelled, .chip.CANCELLED { background: rgba(139,146,168,0.12); color: var(--text2); }
88     .chip.sick, .chip.SICK { background: var(--red-dim); color: var(--red); }
89     .chip.casual, .chip.CASUAL { background: var(--accent-dim); color: var(--accent); }
90     .chip.maternity, .chip.MATERNITY { background: var(--violet-dim); color: var(--violet); }
91
92     /* Buttons */
93     .btn { padding: 8px 16px; border-radius: 6px; border: none; font-family: var(--font); font-size:
13px; font-weight: 500; cursor: pointer; transition: all 0.2s; }
94     .btn-primary { background: var(--accent); color: white; }
95     .btn-primary:hover { background: #6ba0ff; }
96     .btn-secondary { background: var(--bg3); color: var(--text2); border: 1px solid var(--border2); }
97     .btn-secondary:hover { color: var(--text); }
98     .btn-danger { background: var(--red-dim); color: var(--red); border: 1px solid
rgba(248,113,113,0.3); }
99     .btn-danger:hover { background: rgba(248,113,113,0.2); }
100    .btn-success { background: var(--green-dim); color: var(--green); border: 1px solid
rgba(54,211,153,0.3); }
101    .btn-success:hover { background: rgba(54,211,153,0.2); }
102    .btn-sm { padding: 4px 10px; font-size: 11px; }
103    .action-row { display: flex; gap: 6px; flex-wrap: wrap; }
104
105    /* Form */
106    .form-card { background: var(--panel); border-radius: 12px; border: 1px solid var(--border);
padding: 24px; max-width: 640px; }
107    .form-grid { display: grid; grid-template-columns: 1fr 1fr; gap: 16px; }
108    .form-group { display: flex; flex-direction: column; gap: 6px; }
109    .form-group.full { grid-column: 1 / -1; }
110    label { font-size: 11px; font-weight: 600; color: var(--text2); text-transform: uppercase; letter-
spacing: 0.04em; }
111    input, select, textarea { padding: 10px 12px; background: var(--bg3); border: 1px solid var(--
border2); border-radius: 8px; color: var(--text); font-family: var(--font); font-size: 13px; tr...
112    input:focus, select:focus, textarea:focus { outline: none; border-color: var(--accent); }
113    input[readonly] { opacity: 0.5; cursor: not-allowed; }
114    select option { background: var(--bg3); }
115
116    /* Balance cards */
117    .balance-grid { display: grid; grid-template-columns: repeat(auto-fill, minmax(200px, 1fr)); gap:
16px; margin-bottom: 24px; }
118    .balance-card { background: var(--panel); border-radius: 12px; border: 1px solid var(--border);
padding: 20px; }
119    .balance-name { font-size: 12px; color: var(--text2); text-transform: uppercase; letter-spacing:
0.05em; margin-bottom: 4px; }
120    .balance-row { display: flex; align-items: baseline; gap: 6px; margin-bottom: 10px; }

```

```

121     .balance-big { font-size: 28px; font-weight: 600; }
122     .balance-of { font-size: 13px; color: var(--text2); }
123     .balance-bar { height: 4px; background: var(--border2); border-radius: 2px; overflow: hidden; }
124     .balance-fill { height: 100%; border-radius: 2px; transition: width 0.4s ease; }
125
126     /* Tabs */
127     .tab-bar { display: flex; gap: 2px; background: var(--bg3); border-radius: 8px; padding: 4px;
margin-bottom: 20px; width: fit-content; }
128     .tab-btn { padding: 7px 16px; border-radius: 6px; font-size: 12px; font-weight: 500; cursor:
pointer; color: var(--text2); border: none; background: transparent; font-family: var(--font); t...
129     .tab-btn.active { background: var(--panel); color: var(--text); }
130     .tab-btn:hover:not(.active) { color: var(--text); }
131
132     /* Modal */
133     .modal-overlay { display: none; position: fixed; top: 0; left: 0; right: 0; bottom: 0; background:
rgba(0,0,0,0.7); z-index: 1000; align-items: center; justify-content: center; }
134     .modal-overlay.open { display: flex; }
135     .modal { background: var(--panel); border-radius: 12px; border: 1px solid var(--border2); padding:
24px; width: 460px; max-width: 90%; }
136     .modal-title { font-size: 18px; font-weight: 600; margin-bottom: 16px; }
137     .modal-actions { display: flex; gap: 12px; margin-top: 20px; justify-content: flex-end; }
138
139     /* Alert */
140     .alert { position: fixed; top: 20px; right: 20px; padding: 12px 20px; border-radius: 8px; z-index:
1100; animation: slideIn 0.3s ease; font-size: 13px; font-weight: 500; }
141     .alert-success { background: var(--green-dim); color: var(--green); border: 1px solid
rgba(54,211,153,0.3); }
142     .alert-error { background: var(--red-dim); color: var(--red); border: 1px solid
rgba(248,113,113,0.3); }
143     @keyframes slideIn { from { transform: translateX(100%); opacity: 0; } to { transform:
translateX(0); opacity: 1; } }
144
145     /* Pending count indicator */
146     .pending-dot { width: 7px; height: 7px; border-radius: 50%; background: var(--amber); display:
inline-block; margin-left: 4px; animation: pulse 2s infinite; }
147     @keyframes pulse { 0%,100% { opacity: 1; } 50% { opacity: 0.4; } }
148 </style>
149 </head>
150 <body>
151 <div class="app">
152
153     <!-- Sidebar -->
154     <div class="sidebar">
155         <div class="logo">
156             <h2>E<span>M</span>S</h2>
157             <p>Leave Management</p>
158         </div>
159         <div style="flex:1; padding: 8px 0;">
160             <div class="nav-section">Overview</div>
161             <div class="nav-item active" onclick="showView('dashboard',this)">
162                 <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-
width="2"><rect x="3" y="3" width="7" height="7"/><rect x="14" y="3" width="7" height="7"/><...
163                 Dashboard
164             </div>
165             <div class="nav-section">My Leave</div>
166             <div class="nav-item" onclick="showView('apply',this)">
167                 <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-
width="2"><circle cx="12" cy="12" r="10"/><line x1="12" y1="8" x2="12" y2="16"/><line x1="8"...
168                 Apply for Leave
169             </div>
170             <div class="nav-item" onclick="showView('my-leaves',this)">
171                 <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-
width="2"><path d="M14 2H6a2 2 0 0 0 2 2V16a2 2 0 0 0 2 2h12a2 2 0 0 0 2 2V8z"/><polyline po...
172                 My Requests
173             </div>
174             <div class="nav-item" onclick="showView('balance',this)">
175                 <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-
width="2"><line x1="18" y1="20" x2="18" y2="10"/><line x1="12" y1="20" x2="12" y2="4"/><line...
176                 Leave Balance
177             </div>
178             <div class="nav-section">Admin</div>
179             <div class="nav-item" id="nav-pending" onclick="showView('pending',this)">
180                 <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-
width="2"><polyline points="9 11 12 14 22 4"/><path d="M21 12v7a2 2 0 0 1 2 2H5a2 2 0 0 1 2-...
181                 Pending Approvals
182                 <span class="badge-count" id="pending-count" style="display:none">0</span>

```

[illegible]


```

250         <select id="leave-priority" style="color:var(--text2);">
251             <option value=""> Optional </option>
252             <option value="NORMAL">Normal</option>
253             <option value="URGENT">Urgent</option>
254         </select>
255     </div>
256     <div class="form-group">
257         <label>Start Date</label>
258         <input type="date" id="leave-start">
259     </div>
260     <div class="form-group">
261         <label>End Date</label>
262         <input type="date" id="leave-end">
263     </div>
264     <div class="form-group full">
265         <label>Reason</label>
266         <textarea id="leave-reason" rows="3" placeholder="Please provide a brief reason
for your leave request..."></textarea>
267     </div>
268 </div>
269     <div style="display:flex;gap:12px;margin-top:20px;">
270         <button class="btn btn-primary" onclick="applyLeave()">Submit Leave
Request</button>
271         <button class="btn btn-secondary" onclick="clearLeaveForm()">Clear</button>
272     </div>
273 </div>
274 </div>
275
276 <!-- MY LEAVES VIEW -->
277 <div id="my-leaves-view" class="view">
278     <div class="section-head">
279         <div class="section-title">My Leave Requests</div>
280         <button class="btn btn-secondary btn-sm" onclick="loadMyLeaves()"> Refresh</button>
281     </div>
282     <div class="table-wrap">
283         <table>
284             <thead><tr><th>ID</th><th>Type</th><th>Start Date</th><th>End
Date</th><th>Days</th><th>Status</th><th>Reason</th><th>Action</th></tr></thead>
285             <tbody id="my-leaves-list"></tbody>
286         </table>
287     </div>
288 </div>
289
290 <!-- LEAVE BALANCE VIEW -->
291 <div id="balance-view" class="view">
292     <div class="section-head">
293         <div class="section-title">Leave Balance</div>
294         <button class="btn btn-secondary btn-sm" onclick="loadBalance()"> Refresh</button>
295     </div>
296     <div class="balance-grid" id="balance-grid">
297         <div class="balance-card" style="color:var(--text2);text-
align:center;padding:40px;">Loading balance...</div>
298     </div>
299     <div class="section-head" style="margin-top:8px;">
300         <div class="section-title">Leave History</div>
301     </div>
302     <div class="table-wrap">
303         <table>
304             <thead><tr><th>ID</th><th>Type</th><th>Start Date</th><th>End
Date</th><th>Days</th><th>Status</th><th>Remarks</th></tr></thead>
305             <tbody id="balance-history-list"></tbody>
306         </table>
307     </div>
308 </div>
309
310 <!-- PENDING APPROVALS VIEW -->
311 <div id="pending-view" class="view">
312     <div class="section-head">
313         <div class="section-title">Pending Approvals</div>
314         <button class="btn btn-secondary btn-sm" onclick="loadPendingLeaves()">
Refresh</button>
315     </div>
316     <div class="table-wrap">
317         <table>
318             <thead><tr><th>ID</th><th>Employee
ID</th><th>Type</th><th>Start</th><th>End</th><th>Days</th><th>Reason</th><th>Actions</th></tr></thead>

```

```

319         <tbody id="pending-leaves-list"></tbody>
320     </table>
321 </div>
322 </div>
323
324 <!-- ALL LEAVES VIEW -->
325 <div id="all-leaves-view" class="view">
326     <div class="section-head">
327         <div class="section-title">All Leave Requests</div>
328         <div class="section-actions">
329             <select id="status-filter" onchange="filterAllLeaves()" style="padding:6px
10px;font-size:12px;border-radius:6px;">
330                 <option value="">All Statuses</option>
331                 <option value="PENDING">Pending</option>
332                 <option value="APPROVED">Approved</option>
333                 <option value="REJECTED">Rejected</option>
334                 <option value="CANCELLED">Cancelled</option>
335             </select>
336             <button class="btn btn-secondary btn-sm" onclick="loadAllLeaves()">
Refresh</button>
337         </div>
338     </div>
339     <div class="table-wrap">
340         <table>
341             <thead><tr><th>ID</th><th>Employee
ID</th><th>Type</th><th>Start</th><th>End</th><th>Days</th><th>Status</th><th>Actions</th></tr></thead>
342             <tbody id="all-leaves-list"></tbody>
343         </table>
344     </div>
345 </div>
346
347 </div><!-- /content -->
348 </div><!-- /main-content -->
349 </div><!-- /app -->
350
351 <!-- Approval Modal -->
352 <div class="modal-overlay" id="action-modal">
353     <div class="modal">
354         <div class="modal-title" id="action-modal-title">Review Leave Request</div>
355         <input type="hidden" id="action-leave-id">
356         <input type="hidden" id="action-type">
357         <div class="form-group" style="margin-bottom:12px;">
358             <label>Remarks (optional)</label>
359             <textarea id="action-remarks" rows="3" placeholder="Add remarks for the
employee..."></textarea>
360         </div>
361         <div class="modal-actions">
362             <button class="btn btn-secondary" onclick="closeModal('action-modal')">Cancel</button>
363             <button class="btn btn-primary" id="action-submit-btn"
onclick="submitApproval()">Approve</button>
364         </div>
365     </div>
366 </div>
367
368 <!-- Cancel Modal -->
369 <div class="modal-overlay" id="cancel-modal">
370     <div class="modal">
371         <div class="modal-title">Cancel Leave Request</div>
372         <p style="color:var(--text2);font-size:13px;margin-bottom:16px;">Are you sure you want to cancel
this leave request? This action cannot be undone.</p>
373         <input type="hidden" id="cancel-leave-id">
374         <div class="form-group" style="margin-bottom:12px;">
375             <label>Reason for cancellation (optional)</label>
376             <textarea id="cancel-remarks" rows="2" placeholder="Reason..."></textarea>
377         </div>
378         <div class="modal-actions">
379             <button class="btn btn-secondary" onclick="closeModal('cancel-modal')">Keep Request</button>
380             <button class="btn btn-danger" onclick="confirmCancel()">Yes, Cancel</button>
381         </div>
382     </div>
383 </div>
384
385 <script>
386     const API = '';
387     let currentUser = null;
388     let allLeavesCache = [];

```

```

389
390 // Auth
391 function getAuthHeader() {
392     const token = localStorage.getItem('jwt_token');
393     return token ? { 'Authorization': `Bearer ${token}` } : {};
394 }
395
396 async function apiFetch(path, opts = {}) {
397     const response = await fetch(`${API}${path}`, {
398         ...opts,
399         headers: { 'Content-Type': 'application/json', ...getAuthHeader(), ...opts.headers },
400         credentials: 'include'
401     });
402     if (response.status === 401) {
403         localStorage.removeItem('jwt_token');
404         localStorage.removeItem('auth_type');
405         showAlert('Session expired. Please login again.', 'error');
406         setTimeout(() => window.location.href = '/google.html', 1500);
407         throw new Error('Unauthorized');
408     }
409     if (!response.ok) {
410         const err = await response.json().catch(() => ({ message: response.statusText }));
411         throw new Error(err.message || `Error ${response.status}`);
412     }
413     const text = await response.text();
414     return text ? JSON.parse(text) : {};
415 }
416
417 function logout() {
418     localStorage.removeItem('jwt_token');
419     localStorage.removeItem('auth_type');
420     window.location.href = '/google.html';
421 }
422
423 (async function initAuth() {
424     const token = localStorage.getItem('jwt_token');
425     const authType = localStorage.getItem('auth_type');
426     if (!token && authType !== 'session') { window.location.href = '/google.html'; return; }
427     try {
428         const employee = await apiFetch('/employee/me');
429         currentUser = employee;
430         const initials = (employee.name || employee.email || 'U').split(' ').map(w =>
w[0]).join('').substring(0,2).toUpperCase();
431         document.getElementById('user-initials').textContent = initials;
432         document.getElementById('username-display').textContent = employee.name || employee.email ||
'User';
433         document.getElementById('leave-email').value = employee.email || '';
434         loadDashboard();
435         loadPendingCount();
436     } catch(e) {
437         if (e.message !== 'Unauthorized') window.location.href = '/google.html';
438     }
439 })();
440
441 // Navigation
442 const viewTitles = {
443     dashboard: 'Leave Dashboard',
444     apply: 'Apply for Leave',
445     'my-leaves': 'My Leave Requests',
446     balance: 'Leave Balance',
447     pending: 'Pending Approvals',
448     'all-leaves': 'All Leave Requests'
449 };
450 const viewLoaders = {
451     dashboard: loadDashboard,
452     apply: () => {},
453     'my-leaves': loadMyLeaves,
454     balance: loadBalance,
455     pending: loadPendingLeaves,
456     'all-leaves': loadAllLeaves
457 };
458
459 function showView(name, el) {
460     document.querySelectorAll('.view').forEach(v => v.classList.remove('active'));
461     document.querySelectorAll('.nav-item').forEach(n => n.classList.remove('active'));
462     document.getElementById(name + '-view').classList.add('active');

```

```

463     if (el) el.classList.add('active');
464     document.getElementById('pageTitle').textContent = viewTitles[name] || name;
465     if (viewLoaders[name]) viewLoaders[name]();
466 }
467
468 // Helpers
469 function daysBetween(start, end) {
470     if (!start || !end) return '-';
471     const d = (new Date(end) - new Date(start)) / 86400000 + 1;
472     return d > 0 ? d : '-';
473 }
474
475 function chip(val) {
476     if (!val) return '-';
477     const cls = val.toLowerCase();
478     return `${val}</span>`;
479 }
480
481 function showAlert(msg, type = 'success') {
482     const el = document.createElement('div');
483     el.className = `alert alert-${type}`;
484     el.textContent = msg;
485     document.body.appendChild(el);
486     setTimeout(() => el.remove(), 3000);
487 }
488
489 function closeModal(id) {
490     document.getElementById(id).classList.remove('open');
491 }
492
493 document.querySelectorAll('.modal-overlay').forEach(m => {
494     m.addEventListener('click', e => { if (e.target === m) m.classList.remove('open'); });
495 });
496
497 // Dashboard
498 async function loadDashboard() {
499     try {
500         const [all, mine] = await Promise.all([
501             apiFetch('/leave_requests').catch(() => []),
502             apiFetch('/leave_requests/employee').catch(() => ({ requests: [] }))
503         ]);
504         const leaves = Array.isArray(all) ? all : [];
505         document.getElementById('s-total').textContent = leaves.length;
506         document.getElementById('s-pending').textContent = leaves.filter(l => l.status ===
507 'PENDING').length;
507         document.getElementById('s-approved').textContent = leaves.filter(l => l.status ===
508 'APPROVED').length;
508         document.getElementById('s-rejected').textContent = leaves.filter(l => l.status ===
509 'REJECTED').length;
509         const recent = leaves.slice(-8).reverse();
510         const tbody = document.getElementById('recent-leaves-body');
511         tbody.innerHTML = recent.length
512             ? recent.map(l => `|
513 <td>${l.employeeId || '-'}</td>
514 <td>${chip(l.leaveType)}</td>
515 <td>${l.startDate || '-'}</td>
516 <td>${l.endDate || '-'}</td>
517 <td>${daysBetween(l.startDate, l.endDate)}</td>
518 <td>${chip(l.status)}</td>
519 </tr>`).join('')
520             : `|<td colspan="6" class="empty-state">No leave requests yet</td></tr>`;
521     } catch(e) { showAlert('Failed to load dashboard: ' + e.message, 'error'); }
522 }
523
524 async function loadPendingCount() {
525     try {
526         const leaves = await apiFetch('/leave_requests/pending').catch(() => []);
527         const count = Array.isArray(leaves) ? leaves.filter(l => l.status === 'PENDING').length : 0;
528         const el = document.getElementById('pending-count');
529         if (count > 0) { el.textContent = count; el.style.display = 'inline'; }
530         else el.style.display = 'none';
531     } catch(e) {}
532 }
533
534 // Apply Leave
535 async function applyLeave() {

|  |

|  |

```

```

536     const leaveType = document.getElementById('leave-type').value;
537     const startDate = document.getElementById('leave-start').value;
538     const endDate = document.getElementById('leave-end').value;
539     const reason = document.getElementById('leave-reason').value.trim();
540     if (!startDate || !endDate) { showAlert('Please select both start and end dates.', 'error');
return; }
541     if (new Date(endDate) < new Date(startDate)) { showAlert('End date cannot be before start date.',
'error'); return; }
542     try {
543         await apiFetch('/leave_requests', {
544             method: 'POST',
545             body: JSON.stringify({ leaveType, startDate, endDate, reason })
546         });
547         showAlert('Leave request submitted successfully!', 'success');
548         clearLeaveForm();
549         loadPendingCount();
550     } catch(e) { showAlert('Failed to submit: ' + e.message, 'error'); }
551 }
552
553 function clearLeaveForm() {
554     document.getElementById('leave-start').value = '';
555     document.getElementById('leave-end').value = '';
556     document.getElementById('leave-reason').value = '';
557 }
558
559 // My Leaves
560 async function loadMyLeaves() {
561     try {
562         const data = await apiFetch('/leave_requests/employee');
563         const leaves = data.requests || (Array.isArray(data) ? data : []);
564         const tbody = document.getElementById('my-leaves-list');
565         tbody.innerHTML = leaves.length
566             ? leaves.map(l => `<tr>
567 <td>${l.leaveRequestId}</td>
568 <td>${chip(l.leaveType)}</td>
569 <td>${l.startDate || '-'}</td>
570 <td>${l.endDate || '-'}</td>
571 <td>${daysBetween(l.startDate, l.endDate)}</td>
572 <td>${chip(l.status)}</td>
573 <td style="max-width:180px;overflow:hidden;text-overflow:ellipsis;white-space:nowrap;"
title="${l.reason || ''}">${l.reason || '-'}</td>
574 <td>
575     ${l.status === 'PENDING'
576         ? `<button class="btn btn-danger btn-sm"
onclick="openCancelModal('${l.leaveRequestId}')>Cancel</button>`
577         : '-'}
578     </td>
579 </tr>`).join('')
580         : `<tr><td colspan="8" class="empty-state">No leave requests found</td></tr>`;
581     } catch(e) { showAlert('Failed to load your leaves: ' + e.message, 'error'); }
582 }
583
584 function openCancelModal(id) {
585     document.getElementById('cancel-leave-id').value = id;
586     document.getElementById('cancel-remarks').value = '';
587     document.getElementById('cancel-modal').classList.add('open');
588 }
589
590 async function confirmCancel() {
591     const id = document.getElementById('cancel-leave-id').value;
592     try {
593         await apiFetch(`/leave_requests/cancel/${id}`, { method: 'PUT' });
594         showAlert('Leave request cancelled.', 'success');
595         closeModal('cancel-modal');
596         loadMyLeaves();
597         loadPendingCount();
598     } catch(e) { showAlert('Failed to cancel: ' + e.message, 'error'); }
599 }
600
601 // Leave Balance
602 const balanceColors = {
603     SICK: '#f87171', CASUAL: '#4f8cff', MATERNITY: '#a78bfa', UNPAID: '#8b92a8'
604 };
605
606 function normalizeDashboardBalances(data) {
607     if (!data) return [];

```

```

608     if (data.leaveBalances) {
609         return Object.values(data.leaveBalances).map(balance => ({
610             type: balance.leaveType,
611             opening: Number(balance.opening ?? 0),
612             accrued: Number(balance.accrued ?? 0),
613             used: Number(balance.used ?? 0),
614             available: Number(balance.available ?? 0)
615         }));
616     }
617     return ['sick', 'casual', 'maternity']
618         .map(key => data[key])
619         .filter(Boolean)
620         .map(balance => ({
621             type: balance.leaveType,
622             opening: Number(balance.opening ?? 0),
623             accrued: Number(balance.accrued ?? 0),
624             used: Number(balance.used ?? 0),
625             available: Number(balance.available ?? 0)
626         }));
627 }
628
629 async function loadBalance() {
630     try {
631         const data = await apiFetch('/api/employees/me/leave-dashboard');
632         const grid = document.getElementById('balance-grid');
633         const balances = normalizeDashboardBalances(data);
634
635         if (balances.length) {
636             grid.innerHTML = balances.map(info => {
637                 const type = info.type;
638                 const used = info.used;
639                 const total = info.opening + info.accrued;
640                 const left = info.available;
641                 const pct = total ? Math.min(100, Math.round((used / total) * 100)) : 0;
642                 const color = balanceColors[type] || '#4f8cff';
643                 return `<div class="balance-card">
644                     <div class="balance-name">${type.replace(/_/g, ' ')}</div>
645                     <div class="balance-row">
646                         <span class="balance-big" style="color:${color}">${left}</span>
647                         <span class="balance-of"> / ${total} days left</span>
648                     </div>
649                     <div class="balance-bar"><div class="balance-fill"
650 style="width:${pct}%;background:${color}"></div></div>
651                 </div>`;
652             }).join('');
653         } else {
654             grid.innerHTML = `<div class="balance-card" style="color:var(--text2);grid-column:1/-
655 1;text-align:center;padding:32px;">
656                 Leave balance data not available from the API. Check <code>api/employees/me/leave-dashboard</code>
657                 endpoint.
658             </div>`;
659         }
660     } catch (e) {
661         showAlert('Failed to load balance: ' + e.message, 'error');
662     }
663 }
664
665 // Pending Approvals
666 async function loadPendingLeaves() {
667     try {
668         const leaves = await apiFetch('/leave_requests/pending');
669         const arr = Array.isArray(leaves) ? leaves : [];
670         const count = arr.filter(l => l.status === 'PENDING').length;

```

```

681         const el = document.getElementById('pending-count');
682         if (count > 0) { el.textContent = count; el.style.display = 'inline'; }
683         else el.style.display = 'none';
684         const tbody = document.getElementById('pending-leaves-list');
685         tbody.innerHTML = arr.length
686             ? arr.map(l => `<tr>
687                 <td>${l.leaveRequestId}</td>
688                 <td>${l.employeeName || '-'}</td>
689                 <td>${chip(l.leaveType)}</td>
690                 <td>${l.startDate || '-'}</td>
691                 <td>${l.endDate || '-'}</td>
692                 <td>${daysBetween(l.startDate, l.endDate)}</td>
693                 <td style="max-width:140px;overflow:hidden;text-overflow:ellipsis;white-space:nowrap;"
title="${l.reason||''}">${l.reason || '-'}</td>
694                 <td class="action-row">
695                     <button class="btn btn-success btn-sm"
onclick="openActionModal(${l.leaveRequestId}, 'APPROVED')">Approve</button>
696                     <button class="btn btn-danger btn-sm"
onclick="openActionModal(${l.leaveRequestId}, 'REJECTED')">Reject</button>
697                 </td>
698             </tr>`).join('')
699         : `<tr><td colspan="8" class="empty-state">No pending leave requests</td></tr>`;
700     } catch(e) { showAlert('Failed to load pending: ' + e.message, 'error'); }
701 }
702
703 function openActionModal(id, action) {
704     document.getElementById('action-leave-id').value = id;
705     document.getElementById('action-type').value = action;
706     document.getElementById('action-remarks').value = '';
707     document.getElementById('action-modal-title').textContent = `${action === 'APPROVED' ? 'Approve' :
'Reject'} Leave Request #${id}`;
708     const btn = document.getElementById('action-submit-btn');
709     btn.textContent = action === 'APPROVED' ?
710
711     'Approve' : 'Reject';
712     btn.className = action === 'APPROVED' ? 'btn btn-success' : 'btn btn-danger';
713     document.getElementById('action-modal').classList.add('open');
714 }
715
716 async function submitApproval() {
717     const leaveRequestId = parseInt(document.getElementById('action-leave-id').value);
718     const action = document.getElementById('action-type').value;
719     const remarks = document.getElementById('action-remarks').value;
720     try {
721         await apiFetch('/leave_requests/approval', {
722             method: 'PUT',
723             body: JSON.stringify({ leaveRequestId, action, remarks })
724         });
725         showAlert(`Leave request ${action.toLowerCase()} successfully!`, 'success');
726         closeModal('action-modal');
727         loadPendingLeaves();
728         loadDashboard();
729     } catch(e) { showAlert('Failed: ' + e.message, 'error'); }
730 }
731
732 // All Leaves
733 async function loadAllLeaves() {
734     try {
735         const leaves = await apiFetch('/leave_requests');
736         allLeavesCache = Array.isArray(leaves) ? leaves : [];
737         renderAllLeaves(allLeavesCache);
738     } catch(e) { showAlert('Failed to load all leaves: ' + e.message, 'error'); }
739 }
740
741 function filterAllLeaves() {
742     const status = document.getElementById('status-filter').value;
743     const filtered = status ? allLeavesCache.filter(l => l.status === status) : allLeavesCache;
744     renderAllLeaves(filtered);
745 }
746
747 function renderAllLeaves(leaves) {
748     const tbody = document.getElementById('all-leaves-list');
749     tbody.innerHTML = leaves.length
750         ? leaves.map(l => `<tr>
751             <td>${l.leaveRequestId}</td>
752             <td>${l.employeeName || '-'}</td>

```

```

753         <td>${chip(l.leaveType)}</td>
754         <td>${l.startDate || '-'}</td>
755         <td>${l.endDate || '-'}</td>
756         <td>${daysBetween(l.startDate, l.endDate)}</td>
757         <td>${chip(l.status)}</td>
758         <td class="action-row">
759             ${l.status === 'PENDING'
760                 ? `<button class="btn btn-success btn-sm"
onclick="openActionModal(${l.leaveRequestId}, 'APPROVED')">Approve</button>
761                     <button class="btn btn-danger btn-sm"
onclick="openActionModal(${l.leaveRequestId}, 'REJECTED')">Reject</button>`
762                 : '-'}
763         </td>
764     </tr>`.join('')
765     : `<tr><td colspan="8" class="empty-state">No leave requests found</td></tr>`;
766 }
767 </script>
768 </body>
769 </html>
770

```



```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4     <meta charset="UTF-8">
5     <title>Title</title>
6 </head>
7 <body>
8 <h1>hi</h1>
9 </body>
10 </html>
```

```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>EMS – Manager Console</title>
7      <link
href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@300;400;500;600;700&family=IBM+Plex+Mono:w
ght@400;500&display=swap" rel="stylesheet">
8      <link href="https://cdn.jsdelivr.net/npm/fullcalendar@6.1.8/index.global.min.css" rel="stylesheet">
9      <script src="https://cdn.jsdelivr.net/npm/fullcalendar@6.1.8/index.global.min.js"></script>
10     <style>
11         *{margin:0;padding:0;box-sizing:border-box}
12         :root{
13             --bg:#0a0b0f;--bg2:#0f1117;--bg3:#14171f;
14             --panel:#171c28;--panel2:#1d2336;
15             --border:rgba(167,139,250,0.08);--border2:rgba(167,139,250,0.18);
16             --text:#e8e0f5;--text2:#8b82a8;--text3:#504d68;
17             --accent:#a78bfa;--accent-dim:rgba(167,139,250,0.1);
18             --green:#34d399;--green-dim:rgba(52,211,153,0.1);
19             --amber:#fbbf24;--amber-dim:rgba(251,191,36,0.1);
20             --red:#f87171;--red-dim:rgba(248,113,113,0.1);
21             --blue:#60a5fa;--blue-dim:rgba(96,165,250,0.1);
22             --font:'Plus Jakarta Sans',sans-serif;--mono:'IBM Plex Mono',monospace;
23             --sidebar:250px;
24         }
25         body{font-family:var(--font);background:var(--bg);color:var(--text);font-
size:14px;overflow:hidden;height:100vh}
26         .app{display:flex;height:100vh}
27
28         /* SIDEBAR */
29         .sidebar{width:var(--sidebar);background:var(--bg2);border-right:1px solid var(--
border);display:flex;flex-direction:column;flex-shrink:0;position:relative;overflow:hidden}
30         .sidebar::after{content:'';position:absolute;bottom:0;left:0;right:0;height:200px;background:linear-
gradient(to top,rgba(167,139,250,0.04),transparent);pointer-events:none}
31         .logo{padding:22px 20px;border-bottom:1px solid var(--border);position:relative}
32         .logo-row{display:flex;align-items:center;gap:10px;margin-bottom:4px}
33         .logo-icon{width:34px;height:34px;background:linear-gradient(135deg,var(--accent),#7c3aed);border-
radius:10px;display:flex;align-items:center;justify-content:center;font-weight:700;font-siz...
34         .logo h2{font-size:16px;font-weight:700;color:var(--text)}
35         .logo p{font-size:11px;color:var(--text3);font-family:var(--mono)}
36         .role-badge{display:inline-flex;align-items:center;gap:6px;padding:4px 12px;background:linear-
gradient(90deg,rgba(167,139,250,0.15),rgba(124,58,237,0.08));border:1px solid rgba(167,139,250,...
37         .role-dot{width:6px;height:6px;background:var(--accent);border-radius:50%;animation:glow 2s
infinite}
38         @keyframes glow{0%,100%{box-shadow:0 0 4px var(--accent)}50%{box-shadow:0 0 10px var(--accent),0 0
20px rgba(167,139,250,0.4)}}
39         .nav{flex:1;padding:10px 0;overflow-y:auto}
40         .nav-section{padding:12px 20px 6px;font-size:9px;font-weight:700;color:var(--text3);letter-
spacing:2px;text-transform:uppercase}
41         .nav-item{display:flex;align-items:center;gap:10px;padding:10px 20px;margin:1px 8px;border-
radius:8px;cursor:pointer;transition:all 0.2s;color:var(--text2);font-size:13px;font-weight:500}
42         .nav-item:hover{background:var(--accent-dim);color:var(--accent)}
43         .nav-item.active{background:linear-gradient(90deg,var(--accent-
dim),rgba(124,58,237,0.06));color:var(--accent);border:1px solid rgba(167,139,250,0.2)}
44         .nav-badge{margin-left:auto;background:var(--red);color:#fff;font-size:10px;font-
weight:700;padding:2px 7px;border-radius:10px;min-width:18px;text-align:center}
45         .nav-badge.hidden{display:none}
46         .sidebar-footer{padding:16px 20px;border-top:1px solid var(--border)}
47         .user-card{display:flex;align-items:center;gap:10px}
48         .avatar{width:34px;height:34px;background:linear-gradient(135deg,#7c3aed,var(--accent));border-
radius:50%;display:flex;align-items:center;justify-content:center;font-size:14px;font-weight:7...
49         .user-info-side .name{font-size:12px;font-weight:600;color:var(--text)}
50         .user-info-side .role-text{font-size:10px;color:var(--accent);font-family:var(--mono);text-
transform:uppercase;letter-spacing:0.8px}
51
52         /* MAIN */
53         .main{flex:1;display:flex;flex-direction:column;overflow:hidden}
54         .topbar{background:var(--bg2);padding:14px 24px;display:flex;justify-content:space-between;align-
items:center;border-bottom:1px solid var(--border);flex-shrink:0}
55         .page-title{font-size:16px;font-weight:700;color:var(--text)}
56         .topbar-right{display:flex;align-items:center;gap:12px}

```

```

57     .logout-btn{padding:7px 16px;background:transparent;border:1px solid var(--border2);border-
radius:6px;color:var(--text2);font-family:var(--font);font-size:12px;cursor:pointer;transition:all...
58     .logout-btn:hover{background:var(--red-dim);border-color:var(--red);color:var(--red)}
59     .content{flex:1;overflow-y:auto;padding:24px}
60     .view{display:none}
61     .view.active{display:block;animation:fadeIn 0.3s ease}
62     @keyframes fadeIn{from{opacity:0;transform:translateY(8px)}to{opacity:1;transform:translateY(0)}}
63     .notif-wrapper { position: relative; cursor: pointer; }
64
65     .notif-badge {
66         position: absolute; top: -5px; right: -6px;
67         background: #e53e3e; color: white;
68         font-size: 10px; font-weight: 600;
69         min-width: 16px; height: 16px;
70         border-radius: 50%; display: flex;
71         align-items: center; justify-content: center;
72         padding: 0 3px;
73     }
74
75     .notif-panel {
76         position: absolute; top: 48px; right: 16px;
77         width: 320px; max-height: 400px; overflow-y: auto;
78         background: white; border: 1px solid #e2e8f0;
79         border-radius: 10px; box-shadow: 0 4px 16px rgba(0,0,0,0.1);
80         z-index: 999;
81     }
82
83     .notif-header {
84         padding: 12px 16px; font-weight: 600;
85         border-bottom: 1px solid #e2e8f0;
86     }
87
88     .notif-item {
89         padding: 12px 16px; font-size: 14px;
90         border-bottom: 1px solid #f1f5f9;
91     }
92
93     .notif-item.unread { background: #eff6ff; }
94     .notif-time { font-size: 11px; color: #94a3b8; margin-top: 4px; }
95
96     /* STATS */
97     .stats-grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(160px,1fr));gap:16px;margin-
bottom:24px}
98     .stat-card{background:var(--panel);border:1px solid var(--border);border-
radius:14px;padding:20px;position:relative;overflow:hidden;transition:all 0.2s;cursor:default}
99     .stat-card:hover{border-color:var(--border2);transform:translateY(-1px)}
100     .stat-card::before{content:'';position:absolute;top:0;left:0;right:0;height:3px}
101     .stat-card.purple::before{background:linear-gradient(90deg,#7c3aed,var(--accent))}
102     .stat-card.green::before{background:var(--green)}
103     .stat-card.amber::before{background:var(--amber)}
104     .stat-card.blue::before{background:var(--blue)}
105     .stat-label{font-size:11px;color:var(--text3);text-transform:uppercase;letter-spacing:0.8px;margin-
bottom:10px;font-weight:600}
106     .stat-value{font-size:30px;font-weight:700;color:var(--text);font-family:var(--mono);line-height:1}
107     .stat-sub{font-size:11px;color:var(--text3);margin-top:6px}
108
109     /* SECTION */
110     .section{background:var(--panel);border:1px solid var(--border);border-radius:14px;margin-
bottom:20px;overflow:hidden}
111     .section-head{padding:16px 20px;border-bottom:1px solid var(--border);display:flex;justify-
content:space-between;align-items:center}
112     .section-title{font-size:14px;font-weight:700;color:var(--text)}
113     .section-sub{font-size:12px;color:var(--text3);margin-top:2px}
114
115     /* TABLE */
116     .tablewrap{overflow-x:auto}
117     table{width:100%;border-collapse:collapse}
118     th{padding:10px 16px;text-align:left;font-size:10px;color:var(--text3);font-weight:700;letter-
spacing:1px;text-transform:uppercase;background:var(--bg3);border-bottom:1px solid var(--border)}
119     td{padding:12px 16px;border-bottom:1px solid var(--border);font-size:13px;color:var(--text2)}
120     tr:last-child td{border-bottom:none}
121     tr:hover td{background:rgba(167,139,250,0.03);color:var(--text)}
122
123     /* CHIPS */
124     .chip{display:inline-flex;align-items:center;padding:3px 10px;border-radius:20px;font-
size:11px;font-weight:600}

```

```

125     .chip.pending{background:var(--amber-dim);color:var(--amber);border:1px solid rgba(251,191,36,0.2)}
126     .chip.approved{background:var(--green-dim);color:var(--green);border:1px solid
rgba(52,211,153,0.2)}
127     .chip.rejected{background:var(--red-dim);color:var(--red);border:1px solid rgba(248,113,113,0.2)}
128     .chip.active{background:var(--green-dim);color:var(--green);border:1px solid rgba(52,211,153,0.2)}
129     .chip.open { background: var(--accent-dim); color: var(--accent); }
130     .chip.sick{background:var(--red-dim);color:var(--red)}
131     .chip.casual{background:var(--amber-dim);color:var(--amber)}
132     .chip.maternity{background:var(--accent-dim);color:var(--accent)}
133     .chip.employee{background:var(--blue-dim);color:var(--blue)}
134     .chip.manager{background:var(--accent-dim);color:var(--accent)}
135     .chip.under_repair { background: var(--amber-dim); color: var(--amber); }
136     .chip.unpaid { background: rgba(139,92,246,0.15); color: #a78bfa; border: 1px solid
rgba(139,92,246,0.2); }
137     .chip.assigned { background: var(--green-dim); color: var(--green);}
138     .chip.condemned { background: var(--red-dim); color: var(--red);}
139
140     /* BUTTONS */
141     .btn{padding:8px 16px;border-radius:8px;border:none;font-family:var(--font);font-size:12px;font-
weight:600;cursor:pointer;transition:all 0.2s}
142     .btn-primary{background:var(--accent);color:#fff}
143     .btn-primary:hover{background:#7c3aed;box-shadow:0 4px 12px rgba(167,139,250,0.3)}
144     .btn-approve{background:var(--green-dim);color:var(--green);border:1px solid rgba(52,211,153,0.2)}
145     .btn-approve:hover{background:var(--green);color:#fff}
146     .btn-reject{background:var(--red-dim);color:var(--red);border:1px solid rgba(248,113,113,0.2)}
147     .btn-reject:hover{background:var(--red);color:#fff}
148     .btn-secondary{background:var(--panel2);color:var(--text2);border:1px solid var(--border)}
149     .btn-secondary:hover{border-color:var(--border2);color:var(--text)}
150     .btn-sm{padding:5px 12px;font-size:11px}
151     .action-row{display:flex;gap:6px}
152
153     /* FORM */
154     .form-card{padding:24px}
155     .form-grid{display:grid;grid-template-columns:1fr 1fr;gap:16px}
156     .form-group{display:flex;flex-direction:column;gap:6px}
157     .form-group.full{grid-column:1/-1}
158     label{font-size:10px;font-weight:700;color:var(--text3);text-transform:uppercase;letter-
spacing:0.8px}
159     input,select,textarea{background:var(--bg3);border:1px solid var(--border);border-
radius:8px;padding:10px 14px;color:var(--text);font-family:var(--font);font-size:13px;transition:border-col...
160     input:focus,select:focus,textarea:focus{outline:none;border-color:var(--accent)}
161     input::placeholder{color:var(--text3)}
162     select option{background:var(--bg2)}
163     .form-actions{display:flex;gap:10px;margin-top:20px}
164
165     /* ALERT */
166     .alert{position:fixed;top:20px;right:20px;padding:12px 20px;border-radius:10px;z-
index:1100;animation:slideIn 0.3s ease;font-size:13px;font-weight:500}
167     .alert-success{background:var(--green-dim);color:var(--green);border:1px solid
rgba(52,211,153,0.3)}
168     .alert-error{background:var(--red-dim);color:var(--red);border:1px solid rgba(248,113,113,0.3)}
169     .alert-warning{background:var(--amber-dim);color:var(--amber);border:1px solid
rgba(251,191,36,0.3)}
170     @keyframes slideIn{from{transform:translateX(120%);opacity:0}to{transform:translateX(0);opacity:1}}
171     .empty-state{text-align:center;color:var(--text3);font-style:italic;padding:30px}
172
173     /* GRID DASHBOARD */
174     .dash-grid{display:grid;grid-template-columns:1fr 1fr;gap:20px;margin-bottom:0}
175
176     /* SEARCH */
177     .search-bar{display:flex;align-items:center;gap:10px;margin-bottom:16px}
178     .search-input{flex:1;background:var(--bg3);border:1px solid var(--border);border-
radius:8px;padding:9px 14px;color:var(--text);font-family:var(--font);font-size:13px}
179     .search-input:focus{outline:none;border-color:var(--accent)}
180     .search-input::placeholder{color:var(--text3)}
181
182     /* ===== */
183     /* CALENDAR VIEW STYLES */
184     /* ===== */
185
186     /* Leave balance summary cards above the calendar */
187     .leave-balance-grid {
188         display: grid;
189         grid-template-columns: repeat(auto-fit, minmax(160px, 1fr));
190         gap: 14px;
191         padding: 20px;

```

```

192         border-bottom: 1px solid var(--border);
193     }
194     .leave-bal-card {
195         background: var(--bg3);
196         border: 1px solid var(--border);
197         border-radius: 10px;
198         padding: 16px 18px;
199         display: flex;
200         flex-direction: column;
201         gap: 6px;
202         position: relative;
203         overflow: hidden;
204         transition: border-color 0.2s;
205     }
206     .leave-bal-card:hover { border-color: var(--border2); }
207     .leave-bal-card::before {
208         content: '';
209         position: absolute;
210         left: 0; top: 0; bottom: 0;
211         width: 3px;
212         border-radius: 3px 0 0 3px;
213     }
214     .leave-bal-card.sick::before { background: var(--red); }
215     .leave-bal-card.casual::before { background: var(--amber); }
216     .leave-bal-card.mat::before { background: var(--accent); }
217     .leave-bal-card.total::before { background: var(--green); }
218     .lbc-icon { font-size: 18px; }
219     .lbc-label { font-size: 10px; color: var(--text3); text-transform: uppercase; letter-spacing: 1px;
font-weight: 600; }
220     .lbc-count { font-size: 24px; font-weight: 700; color: var(--text); font-family: var(--mono); line-
height: 1; }
221     .lbc-sub { font-size: 11px; color: var(--text3); }
222
223     /* Legend row */
224     .cal-legend {
225         display: flex;
226         align-items: center;
227         gap: 20px;
228         padding: 12px 20px;
229         border-bottom: 1px solid var(--border);
230         flex-wrap: wrap;
231     }
232     .legend-label { font-size: 11px; color: var(--text3); font-weight: 600; text-transform: uppercase;
letter-spacing: 0.8px; margin-right: 4px; }
233     .legend-item { display: flex; align-items: center; gap: 6px; font-size: 12px; color: var(--text2);
}
234     .legend-dot { width: 10px; height: 10px; border-radius: 50%; flex-shrink: 0; }
235
236     /* FullCalendar dark-theme overrides */
237     #employee-calendar {
238         padding: 16px 20px 20px;
239     }
240     /* Toolbar */
241     #employee-calendar .fc .fc-toolbar-title {
242         font-family: var(--font);
243         font-size: 15px;
244         font-weight: 600;
245         color: var(--text);
246     }
247     #employee-calendar .fc .fc-button {
248         background: var(--panel2);
249         border: 1px solid var(--border2);
250         color: var(--text2);
251         font-family: var(--font);
252         font-size: 12px;
253         font-weight: 600;
254         border-radius: 6px;
255         padding: 5px 12px;
256         transition: all 0.2s;
257         box-shadow: none;
258         text-transform: capitalize;
259     }
260     #employee-calendar .fc .fc-button:hover,
261     #employee-calendar .fc .fc-button-active,
262     #employee-calendar .fc .fc-button:not(:disabled):active {
263         background: var(--accent-dim);

```

```

264         border-color: var(--accent);
265         color: var(--accent);
266         box-shadow: none;
267     }
268     #employee-calendar .fc .fc-button:focus { box-shadow: none; outline: none; }
269
270     /* Grid */
271     #employee-calendar .fc-theme-standard td,
272     #employee-calendar .fc-theme-standard th,
273     #employee-calendar .fc-theme-standard .fc-scrollgrid {
274         border-color: var(--border);
275     }
276     #employee-calendar .fc .fc-col-header-cell-cushion {
277         font-family: var(--font);
278         font-size: 11px;
279         font-weight: 600;
280         color: var(--text3);
281         text-transform: uppercase;
282         letter-spacing: 0.8px;
283         padding: 8px 4px;
284     }
285     #employee-calendar .fc .fc-daygrid-day-number {
286         font-family: var(--mono);
287         font-size: 12px;
288         color: var(--text3);
289         padding: 6px 8px;
290     }
291     #employee-calendar .fc .fc-daygrid-day.fc-day-today {
292         background: var(--accent-dim) !important;
293     }
294     #employee-calendar .fc .fc-daygrid-day.fc-day-today .fc-daygrid-day-number {
295         color: var(--accent);
296         font-weight: 700;
297     }
298     #employee-calendar .fc .fc-daygrid-day:hover {
299         background: rgba(56,189,248,0.03);
300     }
301     #employee-calendar .fc-daygrid-day-frame {
302         background: var(--bg3);
303     }
304     #employee-calendar .fc .fc-day-other .fc-daygrid-day-frame {
305         background: var(--bg2);
306     }
307
308     /* Events */
309     #employee-calendar .fc-event {
310         border: none !important;
311         border-radius: 4px;
312         font-family: var(--font);
313         font-size: 11px;
314         font-weight: 600;
315         padding: 2px 6px;
316         cursor: pointer;
317     }
318     #employee-calendar .fc-event-title { font-weight: 600; }
319
320     /* Event tooltip */
321     .cal-tooltip {
322         position: fixed;
323         background: var(--panel2);
324         border: 1px solid var(--border2);
325         border-radius: 8px;
326         padding: 10px 14px;
327         font-size: 12px;
328         color: var(--text);
329         z-index: 9999;
330         pointer-events: none;
331         max-width: 200px;
332         box-shadow: 0 8px 24px rgba(0,0,0,0.4);
333     }
334     .cal-tooltip .tt-title { font-weight: 600; margin-bottom: 4px; font-size: 13px; }
335     .cal-tooltip .tt-row { color: var(--text2); margin-top: 2px; }
336
337     /* balance loading state */
338     .lbc-count.loading { color: var(--text3); font-size: 16px; }
339 </style>

```

```
340 </head>
341 <body>
342 <div class="app">
343   <!-- SIDEBAR -->
344   <div class="sidebar">
345     <div class="logo">
346       <div class="logo-row">
347         <div class="logo-icon">M</div>
348         <h2>EMS Console</h2>
349       </div>
350       <p>Management Portal</p>
351       <div class="role-badge"><span class="role-dot"></span>Manager</div>
352     </div>
353     <nav class="nav">
354       <div class="nav-section">Overview</div>
355       <div class="nav-item active" onclick="showView('dashboard',this)">
356         <span></span> Dashboard
357       </div>
358       <div class="nav-section">Team</div>
359       <div class="nav-item" onclick="showView('employees',this)"><span></span> All Employees</div>
360       <div class="nav-item" onclick="showView('add-employee',this)"><span></span> Add Employee</div>
361       <div class="nav-item" onclick="showView('promote-users',this)"><span></span> Promote Users
362       <span class="nav-badge" id="users-badge">0</span></div>
363       <div class="nav-section">Leave Management</div>
364       <div class="nav-item" onclick="showView('pending-leaves',this)">
365         <span></span> Pending Approvals
366         <span class="nav-badge" id="pending-badge">0</span>
367       </div>
368       <div class="nav-item" onclick="showView('all-leaves',this)"><span></span> All Leave
369       Requests</div>
370       <div class="nav-section">Personal</div>
371       <div class="nav-item" onclick="showView('apply-leave',this)"><span></span> Apply Leave</div>
372       <div class="nav-item" onclick="showView('my-leaves',this)"><span></span> My Leaves</div>
373       <div class="nav-item" onclick="showView('my-calendar',this)"><span class="icon"></span> My
374       Calendar</div>
375       <div class="nav-item" onclick="showView('profile',this)"><span></span> My Profile</div>
376       <div class="nav-item" onclick="showView('my-subscriptions',this)"><span></span> My
377       Subscriptions</div>
378       <div class="nav-item" onclick="showView('add-subscription',this)"><span></span> Subscribe to
379       Meal</div>
380       <div class="nav-item" onclick="showView('restaurants',this)"><span></span> Restaurants</div>
381       <div class="nav-item" onclick="showView('my-devices',this)"><span></span> My Devices</div>
382       <div class="nav-item" onclick="showView('service-requests', this)"><span class="icon"></span>
383       Service Requests</div>
384       <div class="nav-item" onclick="showView('search-employee',this)"><span></span> Search
385       Employee</div>
386     </nav>
387     <div class="sidebar-footer">
388       <div class="user-card">
389         <div class="avatar" id="avatar-initials">M</div>
390         <div class="user-info-side">
391           <div class="name" id="sidebar-name">Manager</div>
392           <div class="role-text">Manager</div>
393         </div>
394       </div>
395     </div>
396   </div>
397   <!-- MAIN -->
398   <div class="main">
399     <div class="topbar">
400       <div class="page-title" id="pageTitle">Dashboard</div>
401       <!-- Bell icon with badge -->
402       <div class="notif-wrapper" onclick="toggleNotifPanel()">
403         <svg width="22" height="22" viewBox="0 0 24 24" fill="none"
404           stroke="currentColor" stroke-width="2" stroke-linecap="round">
405           <path d="M18 8A6 6 0 0 0 6 8c0 7-3 9-3 9h18s-3-2-3-9"/>
406           <path d="M13.73 21a2 2 0 0 1-3.46 0"/>
407         </svg>
408         <span id="notif-badge" class="notif-badge" style="display:none"></span>
409       </div>
410       <!-- Dropdown panel -->
411       <div id="notif-panel" class="notif-panel" style="display:none">
412         <div class="notif-header">Notifications</div>
413         <div id="notif-list"></div>
414       </div>
415     </div>
416   </div>
417 </body>
418 </html>
```

```

409         </div>
410         <div class="topbar-right">
411             <span style="font-size:12px;color:var(--text3);font-family:var(--mono)" id="topbar-
email"></span>
412             <button class="logout-btn" onclick="logout()">Sign Out</button>
413         </div>
414     </div>
415     <div class="content">
416
417         <!-- DASHBOARD -->
418         <div id="dashboard-view" class="view active">
419             <div class="stats-grid">
420                 <div class="stat-card purple">
421                     <div class="stat-label">Total Employees</div>
422                     <div class="stat-value" id="stat-total">--</div>
423                     <div class="stat-sub">in your organization</div>
424                 </div>
425                 <div class="stat-card green">
426                     <div class="stat-label">Active Now</div>
427                     <div class="stat-value" id="stat-active">--</div>
428                     <div class="stat-sub">currently working</div>
429                 </div>
430                 <div class="stat-card amber">
431                     <div class="stat-label">Pending Approvals</div>
432                     <div class="stat-value" id="stat-pending">--</div>
433                     <div class="stat-sub">need your action</div>
434                 </div>
435                 <div class="stat-card blue">
436                     <div class="stat-label">On Leave Today</div>
437                     <div class="stat-value" id="stat-on-leave">--</div>
438                     <div class="stat-sub">employees absent</div>
439                 </div>
440             </div>
441             <div class="dash-grid">
442                 <div class="section">
443                     <div class="section-head">
444                         <div><div class="section-title"> Needs Attention</div><div class="section-
sub">Pending leave requests</div></div>
445                         <button class="btn btn-secondary btn-sm" onclick="showView('pending-
leaves',null)">View All </button>
446                     </div>
447                     <div class="tablewrap">
448                         <table>
449 <thead><tr><th>Employee</th><th>Type</th><th>Dates</th><th>Action</th></tr></thead>
450                             <tbody id="dash-pending"><tr><td colspan="4" class="empty-
state">Loading...</td></tr></tbody>
451                         </table>
452                     </div>
453                 </div>
454                 <div class="section">
455                     <div class="section-head">
456                         <div><div class="section-title"> Team Overview</div><div class="section-
sub">Recent employees</div></div>
457                         <button class="btn btn-secondary btn-sm"
onclick="showView('employees',null)">View All </button>
458                     </div>
459                     <div class="tablewrap">
460                         <table>
461                             <thead><tr><th>Name</th><th>Dept</th><th>Status</th></tr></thead>
462                             <tbody id="dash-employees"><tr><td colspan="3" class="empty-
state">Loading...</td></tr></tbody>
463                         </table>
464                     </div>
465                 </div>
466             </div>
467         </div>
468
469         <!-- ALL EMPLOYEES -->
470         <div id="employees-view" class="view">
471             <div class="section">
472                 <div class="section-head">
473                     <div><div class="section-title">All Employees</div><div class="section-sub">Manage
your team</div></div>
474                     <button class="btn btn-secondary btn-sm" onclick="loadEmployees()">
Refresh</button>

```



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475         </div>
476         <div style="padding:16px 20px;border-bottom:1px solid var(--border)">
477             <div class="search-bar">
478                 <input class="search-input" id="emp-search" placeholder="Search by name, email
or department..." oninput="filterEmployees()">
479             </div>
480         </div>
481         <div class="tablewrap">
482             <table>
483 <thead><tr><th>ID</th><th>Name</th><th>Email</th><th>Department</th><th>Role</th><th>Status</th><th>Actions</th>
</tr></thead>
484             <tbody id="employees-list"><tr><td colspan="7" class="empty-
state">Loading...</td></tr></tbody>
485             </table>
486         </div>
487     </div>
488 </div>
489
490     <!-- ADD EMPLOYEE -->
491     <div id="add-employee-view" class="view">
492         <div class="section">
493             <div class="section-head"><div class="section-title">Add New Employee</div></div>
494             <div class="form-card">
495                 <div class="form-grid">
496                     <div class="form-group"><label>Full Name</label><input id="emp-name"
placeholder="John Doe"></div>
497                     <div class="form-group"><label>Email</label><input type="email" id="emp-email"
placeholder="john@company.com"></div>
498                     <div class="form-group"><label>Password</label><input type="password" id="emp-
pass" placeholder="Temporary password"></div>
499                     <div class="form-group"><label>Gender</label>
500                     <select id="emp-gender">
501                         <option value="M">Male</option>
502                         <option value="F">Female</option>
503                         <option value="O">Other</option>
504                     </select>
505                     </div>
506                     <div class="form-group"><label>Timezone</label>
507                     <select id="emp-timezone">
508                         <option value="UTC">UTC</option>
509                         <option value="Asia/Kolkata">Asia/Kolkata (IST)</option>
510                         <option value="America/New_York">America/New_York (EST)</option>
511                         <option value="America/Los_Angeles">America/Los_Angeles (PST)</option>
512                         <option value="Europe/London">Europe/London (GMT)</option>
513                         <option value="Asia/Singapore">Asia/Singapore (SGT)</option>
514                     </select>
515                     </div>
516                 </div>
517                 <div class="form-actions">
518                     <button class="btn btn-primary" onclick="addEmployee()">Add Employee</button>
519                     <button class="btn btn-secondary" onclick="clearEmpForm()">Clear</button>
520                 </div>
521             </div>
522         </div>
523     </div>
524
525     <!-- SEARCH EMPLOYEE -->
526     <div id="search-employee-view" class="view">
527         <div class="section">
528             <div class="section-head">
529                 <div><div class="section-title"> Search Employee</div><div class="section-sub">Look
up any employee across the organisation</div></div>
530             </div>
531             <div style="padding:20px;border-bottom:1px solid var(--border)">
532                 <div style="display:flex;gap:10px">
533                     <input class="search-input" id="org-search-input" placeholder="Enter exact
employee name..."
534                         style="flex:1" onkeydown="if(event.key==='Enter')searchEmployee()">
535                     <button class="btn btn-primary" onclick="searchEmployee()">Search</button>
536                 </div>
537                 <div id="org-search-error" style="font-size:12px;color:var(--red);margin-
top:8px;display:none"></div>
538             </div>
539             <div id="org-search-results" style="padding:20px;display:none">
540

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541         <!-- Employee Info Card -->
542         <div style="display:grid;grid-template-columns:1fr 1fr;gap:16px;margin-
bottom:20px">
543             <div style="background:var(--bg3);border:1px solid var(--border);border-
radius:10px;padding:16px">
544                 <div style="font-size:10px;font-weight:700;color:var(--text3);text-
transform:uppercase;letter-spacing:0.8px;margin-bottom:12px">Employee Info</div>
545                 <div style="display:flex;flex-direction:column;gap:8px">
546                     <div style="display:flex;justify-content:space-between">
547                         <span style="font-size:12px;color:var(--text3)">Name</span>
548                         <span style="font-size:13px;font-weight:600;color:var(--text)"
id="sr-name"></span>
549                     </div>
550                     <div style="display:flex;justify-content:space-between">
551                         <span style="font-size:12px;color:var(--text3)">ID</span>
552                         <span style="font-size:12px;font-family:var(--mono);color:var(--
accent)" id="sr-id"></span>
553                     </div>
554                     <div style="display:flex;justify-content:space-between">
555                         <span style="font-size:12px;color:var(--text3)">Email</span>
556                         <span style="font-size:12px;font-family:var(--mono);color:var(--
text2)" id="sr-email"></span>
557                     </div>
558                     <div style="display:flex;justify-content:space-between">
559                         <span style="font-size:12px;color:var(--text3)">Department</span>
560                         <span style="font-size:13px;color:var(--text)" id="sr-
dept"></span>
561                     </div>
562                 </div>
563             </div>
564             <div style="background:var(--bg3);border:1px solid var(--border);border-
radius:10px;padding:16px">
565                 <div style="font-size:10px;font-weight:700;color:var(--text3);text-
transform:uppercase;letter-spacing:0.8px;margin-bottom:12px">Status</div>
566                 <div style="display:flex;flex-direction:column;gap:8px">
567                     <div style="display:flex;justify-content:space-between;align-
items:center">
568                         <span style="font-size:12px;color:var(--text3)">Work Status</span>
569                         <span id="sr-status"></span>
570                     </div>
571                     <div style="display:flex;justify-content:space-between;align-
items:center">
572                         <span style="font-size:12px;color:var(--text3)">Attendance</span>
573                         <span style="font-size:13px;font-weight:600;color:var(--text)"
id="sr-attendance"></span>
574                     </div>
575                 </div>
576             </div>
577         </div>
578
579         <!-- Devices -->
580         <div style="background:var(--bg3);border:1px solid var(--border);border-
radius:10px;overflow:hidden">
581             <div style="padding:12px 16px;border-bottom:1px solid var(--border)">
582                 <span style="font-size:12px;font-weight:700;color:var(--text3);text-
transform:uppercase;letter-spacing:0.8px">Assigned Devices</span>
583             </div>
584             <table style="width:100%;border-collapse:collapse">
585                 <thead>
586                     <tr>
587                         <th style="padding:10px 16px;text-align:left;font-size:10px;color:var(-
-text3);font-weight:700;letter-spacing:1px;text-transform:uppercase;background:var(--bg3)"...
588                         <th style="padding:10px 16px;text-align:left;font-size:10px;color:var(-
-text3);font-weight:700;letter-spacing:1px;text-transform:uppercase;background:var(--bg3)"...
589                         <th style="padding:10px 16px;text-align:left;font-size:10px;color:var(-
-text3);font-weight:700;letter-spacing:1px;text-transform:uppercase;background:var(--bg3)"...
590                         <th style="padding:10px 16px;text-align:left;font-size:10px;color:var(-
-text3);font-weight:700;letter-spacing:1px;text-transform:uppercase;background:var(--bg3)"...
591                         <th style="padding:10px 16px;text-align:left;font-size:10px;color:var(-
-text3);font-weight:700;letter-spacing:1px;text-transform:uppercase;background:var(--bg3)"...
592                         <th style="padding:10px 16px;text-align:left;font-size:10px;color:var(-
-text3);font-weight:700;letter-spacing:1px;text-transform:uppercase;background:var(--bg3)"...
593                     </tr>
594                 </thead>
595                 <tbody id="sr-devices"></tbody>
596             </table>

```

```

597             </div>
598         </div>
599
600         <!-- Empty / initial state -->
601         <div id="org-search-empty" style="padding:40px;text-align:center;color:var(--
text3);font-style:italic">
602             Enter a name above to look up an employee
603         </div>
604     </div>
605 </div>
606
607 <!-- PROMOTE USERS -->
608 <div id="promote-users-view" class="view">
609     <div class="section">
610         <div class="section-head">
611             <div><div class="section-title"> Promote Users</div><div class="section-
sub">Onboard pending user registrations to your department</div></div>
612             <button class="btn btn-secondary btn-sm" onclick="loadUsers()"> Refresh</button>
613         </div>
614         <div class="tablewrap">
615             <table>
616                 <thead><tr><th>Name</th><th>Email</th><th>Timezone</th><th>Temp
Password</th><th>Action</th></tr></thead>
617                 <tbody id="users-list"><tr><td colspan="5" class="empty-
state">Loading...</td></tr></tbody>
618             </table>
619         </div>
620     </div>
621 </div>
622
623 <!-- PENDING LEAVE APPROVALS -->
624 <div id="pending-leaves-view" class="view">
625     <div class="section">
626         <div class="section-head">
627             <div><div class="section-title"> Pending Leave Approvals</div><div class="section-
sub">Review and take action</div></div>
628             <button class="btn btn-secondary btn-sm" onclick="loadPendingLeaves()">
Refresh</button>
629         </div>
630         <div class="tablewrap">
631             <table>
632
633                 <thead><tr><th>Employee</th><th>Type</th><th>From</th><th>To</th><th>Days</th><th>Reason</th><th>Actions</th></tr></thead>
634                 <tbody id="pending-leaves-list"><tr><td colspan="7" class="empty-
state">Loading...</td></tr></tbody>
635             </table>
636         </div>
637     </div>
638
639
640
641 <!-- ALL LEAVES -->
642 <div id="all-leaves-view" class="view">
643     <div class="section">
644         <div class="section-head">
645             <div><div class="section-title">All Leave Requests</div></div>
646             <button class="btn btn-secondary btn-sm" onclick="loadAllLeaves()">
Refresh</button>
647         </div>
648         <div class="tablewrap">
649             <table>
650
651                 <thead><tr><th>Employee</th><th>Type</th><th>From</th><th>To</th><th>Reason</th><th>Status</th></tr></thead>
652                 <tbody id="all-leaves-list"><tr><td colspan="7" class="empty-
state">Loading...</td></tr></tbody>
653             </table>
654         </div>
655     </div>
656
657 <!-- APPLY LEAVE -->
658 <div id="apply-leave-view" class="view">
659     <div class="section">
660         <div class="section-head">

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661         <div>
662             <div class="section-title"> Apply for Leave</div>
663             <div class="section-sub">Submit a new leave request</div>
664         </div>
665     </div>
666     <div class="form-card">
667         <div class="form-grid">
668             <div class="form-group">
669                 <label>Leave Type</label>
670                 <select id="leave-type">
671                     <option value="SICK">Sick Leave</option>
672                     <option value="CASUAL">Casual Leave</option>
673                     <option value="UNPAID">Unpaid Leave</option>
674                 </select>
675                 <div id="sick-leave-hint" style="display:none;margin-top:6px;font-
size:11px;color:var(--amber);background:var(--amber-dim);border:1px solid rgba(251,191,36,0.2);bord...
676                     Sick leave is limited to <strong>1 day per month</strong>. If already
used this month, please select Unpaid Leave.
677                 </div>
678                 <div id="casual-leave-hint" style="display:none; margin-top:6px; font-
size:11px; color:var(--amber); background:var(--amber-dim); border:1px solid rgba(251,191,36,0...
679                     Casual leave is limited to <strong>1 day per month</strong>. If already
used this month, it will be automatically recorded as <strong>Unpaid Leave</strong>.
680                 </div>
681             </div>
682             <div class="form-group">
683                 <label>Reason</label>
684                 <input id="leave-reason" placeholder="Brief reason for leave">
685             </div>
686             <div class="form-group">
687                 <label>Start Date</label>
688                 <input type="date" id="leave-start">
689             </div>
690             <div class="form-group">
691                 <label>End Date</label>
692                 <input type="date" id="leave-end">
693             </div>
694         </div>
695         <div class="form-actions">
696             <button class="btn btn-primary" onclick="applyLeave()">Submit Request</button>
697             <button class="btn btn-secondary" onclick="clearLeaveForm()">Clear</button>
698         </div>
699     </div>
700 </div>
701 </div>
702
703 <!-- MY LEAVES -->
704 <div id="my-leaves-view" class="view">
705     <div class="section">
706         <div class="section-head">
707             <div class="section-title">My Leave Requests</div>
708             <button class="btn btn-secondary btn-sm" onclick="loadMyLeaves()"> Refresh</button>
709         </div>
710         <div class="tablewrap">
711             <table>
712
713 <thead><tr><th>Type</th><th>From</th><th>To</th><th>Days</th><th>Status</th></tr></thead>
714 <tbody id="my-leaves-list"><tr><td colspan="5" class="empty-
state">Loading...</td></tr></tbody>
715         </table>
716     </div>
717 </div>
718
719 <!-- ===== -->
720 <!-- MY CALENDAR VIEW (NEW) -->
721 <!-- ===== -->
722 <div id="my-calendar-view" class="view">
723     <div class="section">
724         <div class="section-head">
725             <div>
726                 <div class="section-title">My Leave Calendar</div>
727                 <div class="section-sub">All approved leaves and public holidays</div>
728             </div>
729             <button class="btn btn-secondary btn-sm" onclick="refreshCalendar()">
Refresh</button>

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730         </div>
731
732         <!-- Leave Balance Summary Cards -->
733         <div class="leave-balance-grid">
734             <!-- Sick Leave Card -->
735             <div class="leave-bal-card sick">
736                 <span class="lbc-icon"></span>
737                 <span class="lbc-label">Sick Leave</span>
738                 <span class="lbc-count" id="cal-sick-remaining">--</span>
739                 <span class="lbc-sub" id="cal-sick-sub">loading...</span>
740             </div>
741             <!-- Casual Leave Card -->
742             <div class="leave-bal-card casual">
743                 <span class="lbc-icon"></span>
744                 <span class="lbc-label">Casual Leave</span>
745                 <span class="lbc-count" id="cal-casual-remaining">--</span>
746                 <span class="lbc-sub" id="cal-casual-sub">loading...</span>
747             </div>
748             <!-- Maternity Leave Card -->
749             <div class="leave-bal-card mat">
750                 <span class="lbc-icon"></span>
751                 <span class="lbc-label">Maternity Leave</span>
752                 <span class="lbc-count" id="cal-mat-remaining">--</span>
753                 <span class="lbc-sub" id="cal-mat-sub">loading...</span>
754             </div>
755             <!-- Total Used Card -->
756             <div class="leave-bal-card total">
757                 <span class="lbc-icon"></span>
758                 <span class="lbc-label">Total Used</span>
759                 <span class="lbc-count" id="cal-total-used">--</span>
760                 <span class="lbc-sub" id="cal-total-sub">loading...</span>
761             </div>
762         </div>
763
764         <!-- Legend -->
765         <div class="cal-legend">
766             <span class="legend-label">Legend:</span>
767             <div class="legend-item">
768                 <div class="legend-dot" style="background:#34d399"></div>
769                 Approved Leave
770             </div>
771             <div class="legend-item">
772                 <div class="legend-dot" style="background:#fbbf24"></div>
773                 Pending Leave
774             </div>
775             <div class="legend-item">
776                 <div class="legend-dot" style="background:#f87171"></div>
777                 Rejected Leave
778             </div>
779             <div class="legend-item">
780                 <div class="legend-dot" style="background:#38bdf8"></div>
781                 Public Holiday
782             </div>
783         </div>
784
785         <!-- FullCalendar Mount Point -->
786         <div id="employee-calendar"></div>
787     </div>
788 </div>
789 <!-- END CALENDAR VIEW -->
790
791 <!-- MY SUBSCRIPTIONS -->
792 <div id="my-subscriptions-view" class="view">
793     <div class="section">
794         <div class="section-head">
795             <div><div class="section-title"> My Subscriptions</div><div class="section-
796 sub">Your active meal subscriptions</div></div>
797             <button class="btn btn-secondary btn-sm" onclick="loadMySubscriptions()">
Refresh</button>
798         </div>
799         <div class="tablewrap">
800             <table>
801                 <thead><tr><th>Slot</th><th>Restaurant</th><th>Address</th><th>Schedule</th><th>Day</th><th>Status</th><th>Next
Delivery</th></tr></thead>
802                 <tbody id="my-subscriptions-list"><tr><td colspan="7" class="empty-
```

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state">Loading...</td></tr></tbody>
802         </table>
803     </div>
804 </div>
805 </div>
806
807 <!-- ADD SUBSCRIPTION -->
808 <div id="add-subscription-view" class="view">
809     <div class="section">
810         <div class="section-head">
811             <div><div class="section-title"> Subscribe to Meal</div><div class="section-
812 sub">Add a new meal slot subscription</div></div>
813         </div>
814         <div class="form-card">
815             <div class="form-grid">
816                 <div class="form-group">
817                     <label>Schedule Type</label>
818                     <select id="sub-schedule" onchange="toggleDayOfWeek()">
819                         <option value="DAILY">Daily</option>
820                         <option value="WEEKLY">Weekly</option>
821                     </select>
822                 </div>
823                 <div class="form-group" id="day-of-week-group" style="display:none">
824                     <label>Day of Week</label>
825                     <select id="sub-day">
826                         <option value="MONDAY">Monday</option>
827                         <option value="TUESDAY">Tuesday</option>
828                         <option value="WEDNESDAY">Wednesday</option>
829                         <option value="THURSDAY">Thursday</option>
830                         <option value="FRIDAY">Friday</option>
831                         <option value="SATURDAY">Saturday</option>
832                         <option value="SUNDAY">Sunday</option>
833                     </select>
834                 </div>
835             </div>
836
837 <!-- Slot Orders -->
838 <div style="margin-top:20px">
839     <div style="display:flex;justify-content:space-between;align-
840 items:center;margin-bottom:12px">
841         <label style="font-size:11px;font-weight:700;color:var(--text3);letter-
842 spacing:0.8px;text-transform:uppercase">Meal Slots & Restaurants</label>
843         <button class="btn btn-secondary btn-sm" onclick="addSlotRow()"> Add
844 Slot</button>
845     </div>
846     <div id="slot-rows"></div>
847 </div>
848
849 <div class="form-actions">
850     <button class="btn btn-primary"
851 onclick="submitSubscription()">Subscribe</button>
852     <button class="btn btn-secondary"
853 onclick="clearSubscriptionForm()">Clear</button>
854 </div>
855 </div>
856 </div>
857 <!-- RESTAURANTS -->
858 <div id="restaurants-view" class="view">
859     <div class="section">
860         <div class="section-head">
861             <div><div class="section-title"> Restaurants</div><div class="section-sub">All
862 available restaurants</div></div>
863             <button class="btn btn-secondary btn-sm" onclick="loadRestaurants()">
864 Refresh</button>
865         </div>
866         <div class="tablewrap">
867             <table>
868                 <thead><tr><th>ID</th><th>Name</th><th>Address</th><th>Status</th></tr></thead>
869                 <tbody id="restaurants-list"><tr><td colspan="4" class="empty-
870 state">Loading...</td></tr></tbody>
871             </table>
872         </div>
873     </div>
874 </div>
```

```

868
869     <!-- MY DEVICES -->
870     <div id="my-devices-view" class="view">
871         <div class="section">
872             <div class="section-head">
873                 <div>
874                     <div class="section-title"> My Assigned Devices</div>
875                     <div class="section-sub">Devices currently assigned to you</div>
876                 </div>
877                 <button class="btn btn-secondary btn-sm" onclick="loadMyAssignedDevices()">
Refresh</button>
878             </div>
879             <div class="tablewrap">
880                 <table>
881                     <thead>
882                         <tr>
883                             <th>ID</th>
884                             <th>Device Name</th>
885                             <th>Brand</th>
886                             <th>Type</th>
887                             <th>Warranty</th>
888                             <th>Assigned On</th>
889                             <th>Status</th>
890                         </tr>
891                     </thead>
892                     <tbody id="my-devices-list">
893                         <tr><td colspan="7" class="empty-state">Loading...</td></tr>
894                     </tbody>
895                 </table>
896             </div>
897         </div>
898     </div>
899
900     <!-- SERVICE REQUESTS -->
901     <div id="service-requests-view" class="view">
902
903         <!-- List -->
904         <div class="section" style="margin-bottom:20px">
905             <div class="section-head">
906                 <div>
907                     <div class="section-title">My Service Requests</div>
908                     <div class="section-sub">All requests you have raised</div>
909                 </div>
910                 <button class="btn btn-secondary btn-sm" onclick="loadServiceRequestsList()">
Refresh</button>
911             </div>
912             <div class="tablewrap">
913                 <table>
914                     <thead>
915                         <tr>
916                             <th>ID</th>
917                             <th>Device</th>
918                             <th>Type</th>
919                             <th>Description</th>
920                             <th>Urgent</th>
921                             <th>Status</th>
922                             <th>Raised At</th>
923                         </tr>
924                     </thead>
925                     <tbody id="service-requests-list">
926                         <tr><td colspan="7" class="empty-state">Loading...</td></tr>
927                     </tbody>
928                 </table>
929             </div>
930         </div>
931
932         <!-- Raise form -->
933         <div class="section">
934             <div class="section-head"><div class="section-title">Raise a New Request</div></div>
935             <div class="form-card">
936                 <div class="form-grid">
937                     <div class="form-group full">
938                         <label>Device</label>
939                         <select id="sr-device-select">
940                             <option value="">-- Select your device --</option>
941                         </select>

```

```

942         </div>
943         <div class="form-group">
944             <label>Request Type</label>
945             <select id="sr-type">
946                 <option value="REPAIR">Repair</option>
947                 <option value="REPLACEMENT">Replacement</option>
948                 <option value="UPGRADE">Upgrade</option>
949                 <option value="MAINTENANCE">Maintenance</option>
950                 <option value="RETURN">Return</option>
951             </select>
952         </div>
953         <div class="form-group" style="display:flex;align-
items:center;gap:10px;padding-top:22px">
954             <input type="checkbox" id="sr-urgent"
style="width:16px;height:16px;padding:0;flex-shrink:0">
955             <label for="sr-urgent" style="text-transform:none;font-size:13px;font-
weight:500;color:var(--text);cursor:pointer;letter-spacing:0">
956                 Mark as Urgent
957             </label>
958         </div>
959         <div class="form-group full">
960             <label>Issue Description</label>
961             <textarea id="sr-description" rows="3" placeholder="Describe the issue in
detail..."></textarea>
962         </div>
963     </div>
964     <div class="form-actions">
965         <button class="btn btn-primary" onclick="submitServiceRequest()">Submit
Request</button>
966         <button class="btn btn-secondary"
onclick="clearServiceRequestForm()">Clear</button>
967     </div>
968 </div>
969 </div>
970 </div>
971 <!-- PROFILE -->
972 <div id="profile-view" class="view">
973     <div class="section">
974         <div class="section-head">
975             <div class="section-title">My Profile</div>
976         </div>
977         <div style="display:grid;grid-template-columns:1fr 1fr;gap:16px;padding:24px">
978             <div style="display:flex;flex-direction:column;gap:4px">
979                 <div style="font-size:10px;color:var(--text3);text-transform:uppercase;
letter-spacing:0.8px;font-weight:700">Name</div>
980                 <div style="font-size:14px;color:var(--text);padding:10px 14px;
background:var(--bg3);border-radius:8px;border:1px solid var(--border);
font-family:var(--mono)" id="p-name"></div>
981             </div>
982             <div style="display:flex;flex-direction:column;gap:4px">
983                 <div style="font-size:10px;color:var(--text3);text-transform:uppercase;
letter-spacing:0.8px;font-weight:700">Email</div>
984                 <div style="font-size:14px;color:var(--text);padding:10px 14px;
background:var(--bg3);border-radius:8px;border:1px solid var(--border);
font-family:var(--mono)" id="p-email"></div>
985             </div>
986             <div style="display:flex;flex-direction:column;gap:4px">
987                 <div style="font-size:10px;color:var(--text3);text-transform:uppercase;
letter-spacing:0.8px;font-weight:700">Department</div>
988                 <div style="font-size:14px;color:var(--text);padding:10px 14px;
background:var(--bg3);border-radius:8px;border:1px solid var(--border);
font-family:var(--mono)" id="p-dept"></div>
989             </div>
990             <div style="display:flex;flex-direction:column;gap:4px">
991                 <div style="font-size:10px;color:var(--text3);text-transform:uppercase;
letter-spacing:0.8px;font-weight:700">Role</div>
992                 <div style="font-size:14px;color:var(--text);padding:10px 14px;
background:var(--bg3);border-radius:8px;border:1px solid var(--border);
font-family:var(--mono)" id="p-role"></div>
993             </div>
994             <div style="display:flex;flex-direction:column;gap:4px">
995                 <div style="font-size:10px;color:var(--text3);text-transform:uppercase;
letter-spacing:0.8px;font-weight:700">Status</div>
996                 <div style="font-size:14px;color:var(--text);padding:10px 14px;
background:var(--bg3);border-radius:8px;border:1px solid var(--border);
font-family:var(--mono)" id="p-status"></div>
997             </div>
998         </div>
999     </div>
1000 </div>

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1012         font-family:var(--mono)" id="p-status"></div>
1013     </div>
1014     <div style="display:flex;flex-direction:column;gap:4px">
1015         <div style="font-size:10px;color:var(--text3);text-transform:uppercase;
1016             letter-spacing:0.8px;font-weight:700">Timezone</div>
1017         <div style="font-size:14px;color:var(--text);padding:10px 14px;
1018             background:var(--bg3);border-radius:8px;border:1px solid var(--border);
1019             font-family:var(--mono)" id="p-tz"></div>
1020     </div>
1021 </div>
1022 </div>
1023
1024 <!-- MATERNITY SECTION – only visible for female managers -->
1025 <div class="section" id="maternity-apply-btn" style="display:none;margin-top:0">
1026     <div class="section-head">
1027         <div>
1028             <div class="section-title"> Maternity Leave</div>
1029             <div class="section-sub">Apply for 180 days maternity leave</div>
1030         </div>
1031     </div>
1032     <div class="form-card">
1033         <div class="form-grid">
1034             <div class="form-group">
1035                 <label>Start Date</label>
1036                 <input type="date" id="mat-start-date">
1037             </div>
1038             <div class="form-group">
1039                 <label>End Date (Auto)</label>
1040                 <input type="text" id="mat-end-date"
1041                     placeholder="Auto: start + 180 days"
1042                     disabled
1043                     style="color:var(--text3);cursor:not-allowed">
1044             </div>
1045             <div class="form-group full">
1046                 <label>Reason / Notes</label>
1047                 <textarea id="mat-reason" rows="3"
1048                     placeholder="Optional notes..."></textarea>
1049             </div>
1050             <div class="form-group full">
1051                 <div style="background:var(--accent-dim);
1052                     border:1px solid rgba(167,139,250,0.2);
1053                     border-radius:8px;padding:12px 16px;
1054                     font-size:12px;color:var(--text2)">
1055                     Maternity leave is fixed at
1056                     <strong style="color:var(--accent)">180 days</strong>.
1057                     End date is calculated automatically.
1058                     You can apply <strong style="color:var(--accent)">once per
1059 year</strong>.
1060                 </div>
1061             </div>
1062         </div>
1063         <div class="form-actions">
1064             <button class="btn btn-primary" onclick="applyMaternityLeave()">
1065                 Submit Maternity Leave
1066             </button>
1067         </div>
1068     </div>
1069 </div>
1070
1071 </div>
1072 </div>
1073 </div>
1074 <!-- Tooltip for calendar events -->
1075 <div class="cal-tooltip" id="cal-tooltip" style="display:none"></div>
1076 <script>
1077     const API='http://localhost:8080';
1078     let allEmployeesCache=[];
1079
1080     // Add these near the top of your <script>, right after: let allEmployeesCache=[];
1081
1082     let employeeCalendar = null;
1083     let calendarInitialized = false;
1084
1085     const STATUS_COLORS = {
1086         APPROVED : { bg:'rgba(52,211,153,0.25)', border:'#34d399', text:'#34d399' },

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1087     PENDING : { bg:'rgba(251,191,36,0.25)', border:'#fbbf24', text:'#fbbf24' },
1088     REJECTED : { bg:'rgba(248,113,113,0.25)', border:'#f87171', text:'#f87171' },
1089     HOLIDAY : { bg:'rgba(167,139,250,0.25)', border:'#a78bfa', text:'#a78bfa' },
1090 };
1091
1092 const TYPE_LABEL = { SICK:' Sick', CASUAL:' Casual', MATERNITY:' Maternity', UNPAID:' Unpaid'
1093 };
1094
1095 (function(){
1096     const token=localStorage.getItem('jwt_token');
1097     const authType=localStorage.getItem('auth_type');
1098     if(!token&&authType!=='session'){window.location.href='/login.html';return;}
1099     if(token){
1100         try{
1101             const p=JSON.parse(atob(token.split('.')[1]));
1102             const rawRole = p.role || p.roles || p.authorities || '';
1103             const normalizedRoles = Array.isArray(rawRole)
1104                 ? rawRole.map(v => String(v).replace(/^ROLE_/, ''))
1105                 : [String(rawRole).replace(/^ROLE_/, '')];
1106             if(normalizedRoles.includes('MANAGER')) { initUser(); return; }
1107             if(normalizedRoles.includes('ADMIN')){window.location.href='/dashboard.html';return;}
1108             if(normalizedRoles.includes('EMPLOYEE')){window.location.href='/employee-
dashboard.html';return;}
1109             if(normalizedRoles.includes('VENDOR')){window.location.href='/vendor-
dashboard.html';return;}
1110         }catch(e){}
1111     }
1112     initUser();
1113 })();
1114
1115 function getAuthHeader(){
1116     const t=localStorage.getItem('jwt_token');
1117     return t?{'Authorization':`Bearer ${t}`}:{};
1118 }
1119 async function apiFetch(path, opts={}){
1120     const res = await fetch(`${API}${path}`, {
1121         ...opts,
1122         headers: {'Content-Type':'application/json',...getAuthHeader(),...(opts.headers||{})},
1123         credentials: 'include'
1124     });
1125     if(res.status===401){
1126         localStorage.removeItem('jwt_token');
1127         localStorage.removeItem('auth_type');
1128         showAlert('Session expired','error');
1129         setTimeout(()=>window.location.href='/google.html',1500);
1130         throw new Error('Unauthorized');
1131     }
1132     const text = await res.text();
1133     if(!res.ok){
1134         let message = res.statusText || `Error ${res.status}`;
1135         try {
1136             const body = text ? JSON.parse(text) : null;
1137             message = body?.message || body?.error || message;
1138         } catch {
1139             message = text || message;
1140         }
1141         throw new Error(message);
1142     }
1143     if(!text) return null;
1144     try { return JSON.parse(text); } // try JSON first
1145     catch { return text; } // fall back to plain string
1146 }
1147
1148 function asArray(value){
1149     return Array.isArray(value) ? value : [];
1150 }
1151
1152 async function initUser(){
1153     const token=localStorage.getItem('jwt_token');
1154     let email='';
1155     if(token){try{const p=JSON.parse(atob(token.split('.')[1]));email=p.sub||p.email||'';}catch(e){}}
1156     document.getElementById('avatar-initials').textContent=(email||'M').charAt(0).toUpperCase();
1157     document.getElementById('sidebar-name').textContent=email||'Manager';
1158     document.getElementById('topbar-email').textContent=email;
1159     loadDashboard();
1160     loadUsers();

```

```

1161     }
1162
1163     function showView(name,el){
1164         document.querySelectorAll('.view').forEach(v=>v.classList.remove('active'));
1165         document.getElementById(`${name}-view`).classList.add('active');
1166         document.querySelectorAll('.nav-item').forEach(n=>n.classList.remove('active'));
1167         if(el)el.classList.add('active');
1168         document.getElementById('pageTitle').textContent=name.split('-
1169         ').map(w=>w.charAt(0).toUpperCase()+w.slice(1)).join(' ');
1170         const loaders = {
1171             dashboard: loadDashboard,
1172             employees: loadEmployees,
1173             'pending-leaves': loadPendingLeaves,
1174             'all-leaves': loadAllLeaves,
1175             'my-leaves': loadMyLeaves,
1176             'my-calendar': loadMyCalendar,
1177             'add-employee': () => {},
1178             'promote-users': loadUsers,
1179             'my-subscriptions': loadMySubscriptions,
1180             'add-subscription': () => {},
1181             restaurants: loadRestaurants,
1182             'my-devices': loadMyAssignedDevices,
1183             'service-requests': loadServiceRequests,
1184             'search-employee': () => {},
1185             'profile': loadProfile, // ADD THIS
1186         };
1187         if(loaders[name])loaders[name]();
1188     }
1189
1190     async function loadDashboard(){
1191         const [empsResult, leavesResult] = await Promise.allSettled([
1192             apiFetch('/manager/employees'),
1193             apiFetch('/manager/pending')
1194         ]);
1195
1196         const emps = asArray(empsResult.status === 'fulfilled' ? empsResult.value : []);
1197         const leaves = asArray(leavesResult.status === 'fulfilled' ? leavesResult.value : []);
1198
1199         if (empsResult.status === 'rejected') {
1200             console.error('Failed to load manager employees:', empsResult.reason);
1201             document.getElementById('dash-employees').innerHTML =
1202                 '<tr><td colspan="3" class="empty-state">Could not load employees</td></tr>';
1203         }
1204         if (leavesResult.status === 'rejected') {
1205             console.error('Failed to load pending leaves:', leavesResult.reason);
1206             document.getElementById('dash-pending').innerHTML =
1207                 '<tr><td colspan="4" class="empty-state">Could not load pending requests</td></tr>';
1208         }
1209
1210         // Stats
1211         document.getElementById('stat-total').textContent = emps.length;
1212         document.getElementById('stat-active').textContent = emps.filter(e => e.status ===
1213         'ACTIVE').length;
1214         document.getElementById('stat-on-leave').textContent = emps.filter(e => e.status ===
1215         'ON_LEAVE').length;
1216         const pending = leaves.filter(l => l.status === 'PENDING');
1217         document.getElementById('stat-pending').textContent = pending.length;
1218         document.getElementById('pending-badge').textContent = pending.length;
1219         document.getElementById('pending-badge').classList.toggle('hidden', pending.length === 0);
1220
1221         // Build employeeId name lookup from employees list
1222         const empMap = {};
1223         emps.forEach(e => { empMap[e.employeeId] = e.name || '-'; });
1224
1225         // Dash pending
1226         if (leavesResult.status === 'fulfilled') {
1227             document.getElementById('dash-pending').innerHTML = pending.length
1228             ? pending.slice(0, 5).map(l => {
1229                 const leaveId = l.leaveRequestId || 0;
1230                 const empName = empMap[l.employeeId] || `#${l.employeeId}`;
1231                 return `<tr>
1232                     <td>${empName}</td>
1233                     <td><span class="chip ${l.leaveType||''}.toLowerCase()">${l.leaveType}</span></td>
1234                     <td>${l.startDate} ${l.endDate}</td>
1235                     <td><div class="action-row">
1236                         <button class="btn btn-approve btn-sm"

```

```

onclick="approveLeave(${leaveId})"></button>
1234         <button class="btn btn-reject btn-sm" onclick="rejectLeave(${leaveId})"></button>
1235     </div></td>
1236 </tr>`;
1237 }).join('')
1238 : '<tr><td colspan="4" class="empty-state">No pending requests </td></tr>';
1239 }
1240
1241 // Dash employees
1242 if (empsResult.status === 'fulfilled') {
1243     document.getElementById('dash-employees').innerHTML = emps.length
1244     ? emps.slice(0, 6).map(e => `<tr>
1245         <td>${e.name || '-'}</td>
1246         <td>${e.dept || '-'}</td>
1247         <td><span class="chip ${e.status || ''}.toLowerCase()>${e.status || '-'}</span></td>
1248     </tr>`).join('')
1249     : '<tr><td colspan="3" class="empty-state">No employees</td></tr>';
1250 }
1251 }
1252
1253 async function loadEmployees(){
1254     try{
1255         const emps=await apiFetch('/manager/employees');
1256         allEmployeesCache=emps||[];
1257         renderEmployees(allEmployeesCache);
1258     }catch(e){document.getElementById('employees-list').innerHTML='<tr><td colspan="7" class="empty-
state">Could not load employees</td></tr>';}
1259 }
1260
1261 function renderEmployees(emps){
1262     document.getElementById('employees-list').innerHTML=emps.length
1263     ? emps.map(e=>`<tr><td style="font-family:var(--mono);font-
size:12px">#${e.employeeId}</td><td>${e.name||'-'}</td><td style="font-family:var(--mono);font-
size:12px">${e.email||'-'}</td>...
1264     : '<tr><td colspan="7" class="empty-state">No employees found</td></tr>';
1265 }
1266
1267 function filterEmployees(){
1268     const q=document.getElementById('emp-search').value.toLowerCase();
1269     renderEmployees(allEmployeesCache.filter(e=>(e.name||'').toLowerCase().includes(q)||e.email||'').toLowerCase().
includes(q)||e.dept||'').toLowerCase().includes(q)));
1270 }
1271
1272 async function addEmployee(){
1273     const name = document.getElementById('emp-name').value.trim();
1274     const email = document.getElementById('emp-email').value.trim();
1275     const password = document.getElementById('emp-pass').value;
1276     const gender = document.getElementById('emp-gender').value;
1277     const timezone = document.getElementById('emp-timezone').value;
1278     if(!name || !email || !password){
1279         showAlert('Please fill all required fields','error');
1280         return;
1281     }
1282     try{
1283         await apiFetch('/manager/employees', {
1284             method: 'POST',
1285             body: JSON.stringify({ name, email, password, timezone, gender })
1286         });
1287         showAlert('Employee added successfully!','success');
1288         clearEmpForm();
1289         loadEmployees();
1290     }catch(e){ showAlert('Failed: '+e.message,'error'); }
1291 }
1292
1293 function clearEmpForm(){
1294     ['emp-name', 'emp-email', 'emp-pass'].forEach(id=>document.getElementById(id).value='');
1295     document.getElementById('emp-timezone').value = 'UTC';
1296     document.getElementById('emp-gender').selectedIndex = 0;
1297 }
1298
1299 async function deleteEmployee(id){
1300     if(!confirm('Remove this employee?'))return;
1301     try{await apiFetch(`/employee/${id}`, {method:'DELETE'});showAlert('Employee
removed', 'success');loadEmployees();}
1302     catch(e){
1303         console.log(e);

```

```

1303     showAlert('Failed to remove employee','error');
1304     }
1305 }
1306
1307 async function loadPendingLeaves(){
1308     try{
1309         const leaves = await apiFetch('/manager/pending');
1310         console.log('Leave sample:', leaves?.[0]); // remove after confirming
1311         const pending=(leaves||[]).filter(l=>l.status==='PENDING');
1312         document.getElementById('pending-leaves-list').innerHTML=pending.length
1313             ? pending.map(l=>{
1314                 // Try all possible ID field names your DTO might use
1315                 const leaveId = l.leaveRequestId || l.leaveId || l.id || 0;
1316                 return `<tr>
1317                     <td>${l.employeeName||l.employeeId||'-'}</td>
1318                     <td><span class="chip ${l.leaveType||''}.toLowerCase()">${l.leaveType}</span></td>
1319                     <td>${l.startDate}</td>
1320                     <td>${l.endDate}</td>
1321                     <td>${l.numberOfDays||'-'}</td>
1322                     <td>${(l.reason||'-').substring(0,40)}</td>
1323                     <td><div class="action-row">
1324                         <button class="btn btn-approve btn-sm" onclick="approveLeave(${leaveId})">
Approve</button>
1325                         <button class="btn btn-reject btn-sm" onclick="rejectLeave(${leaveId})">
Reject</button>
1326                     </div></td>
1327                 </tr>`;
1328             }).join('')
1329             : '<tr><td colspan="7" class="empty-state">No pending approvals </td></tr>';
1330     }catch(e){
1331         document.getElementById('pending-leaves-list').innerHTML='<tr><td colspan="7" class="empty-
state">Could not load</td></tr>';
1332     }
1333 }
1334
1335 async function approveLeave(id) {
1336     console.log('Approving leaveRequestId:', id); // verify this is not 0
1337     try {
1338         const res = await fetch(`${API}/leave_requests/approval`, {
1339             method: 'PUT',
1340             headers: { 'Content-Type': 'application/json', ...getAuthHeader() },
1341             credentials: 'include',
1342             body: JSON.stringify({
1343                 leaveRequestId: id, // matches ActionDTO exactly
1344                 action: 'APPROVED', // check if your service expects uppercase or lowercase
1345                 remarks: ''
1346             })
1347         });
1348         const text = await res.text();
1349         if (res.ok) {
1350             showAlert('Leave approved ', 'success');
1351             loadPendingLeaves(); loadDashboard(); loadAllLeaves();
1352         } else {
1353             showAlert('Error: ' + text, 'error');
1354         }
1355     } catch(e) { showAlert(e.message, 'error'); }
1356 }
1357
1358 async function rejectLeave(id) {
1359     console.log('Rejecting leaveRequestId:', id); // verify this is not 0
1360     try {
1361         const res = await fetch(`${API}/leave_requests/approval`, {
1362             method: 'PUT',
1363             headers: { 'Content-Type': 'application/json', ...getAuthHeader() },
1364             credentials: 'include',
1365             body: JSON.stringify({
1366                 leaveRequestId: id, // matches ActionDTO exactly
1367                 action: 'REJECTED', // check if your service expects uppercase or lowercase
1368                 remarks: ''
1369             })
1370         });
1371         const text = await res.text();
1372         if (res.ok) {
1373             showAlert('Leave rejected', 'success');
1374             loadPendingLeaves(); loadDashboard(); loadAllLeaves();
1375         } else {
1376             showAlert('Error: ' + text, 'error');

```

```

1376     }
1377   } catch(e) { showAlert('Failed: ' + e.message, 'error'); }
1378 }
1379
1380 async function loadAllLeaves(){
1381   try{
1382     const leaves=await apiFetch('/manager/requests');
1383     document.getElementById('all-leaves-list').innerHTML=(leaves||[]).length
1384       ? (leaves).map(l=><tr>
1385         <td>${l.employeeName||'-'}</td>
1386         <td><span class="chip ${l.leaveType||''}.toLowerCase()>${l.leaveType||'-'}</span></td>
1387         <td>${l.startDate||'-'}</td>
1388         <td>${l.endDate||'-'}</td>
1389         <td>${l.reason||'-'}.substring(0,50}</td>
1390         <td><span class="chip ${l.status||''}.toLowerCase()>${l.status||'-'}</span></td>
1391       </tr>`)
1392     : <tr><td colspan="7" class="empty-state">No leave requests</td></tr>;
1393   }catch(e){document.getElementById('all-leaves-list').innerHTML=<tr><td colspan="7" class="empty-
state">Could not load</td></tr>;}
1394 }
1395
1396 async function applyLeave() {
1397   const leaveType = document.getElementById('leave-type').value;
1398   const reason = document.getElementById('leave-reason').value.trim();
1399   const startDate = document.getElementById('leave-start').value;
1400   const endDate = document.getElementById('leave-end').value;
1401
1402   if (!reason || !startDate || !endDate) {
1403     showAlert('Please fill all fields', 'error');
1404     return;
1405   }
1406   if (endDate < startDate) {
1407     showAlert('End date cannot be before start date', 'error');
1408     return;
1409   }
1410
1411   try {
1412     const res = await fetch(`${API}/leave_requests`, {
1413       method: 'POST',
1414       headers: { 'Content-Type': 'application/json', ...getAuthHeader() },
1415       credentials: 'include',
1416       body: JSON.stringify({ leaveType, reason, startDate, endDate })
1417     });
1418
1419     const text = await res.text();
1420
1421     if (res.ok) {
1422       let data = null;
1423       try { data = JSON.parse(text); } catch {}
1424
1425       if (data && data.warning) {
1426         showAlert(data.warning, 'warning');
1427       } else {
1428         showAlert('Leave request submitted successfully!', 'success');
1429       }
1430       clearLeaveForm();
1431       loadMyLeaves();
1432     } else {
1433       showAlert('Failed: ' + text, 'error');
1434     }
1435   } catch(e) {
1436     showAlert('Error: ' + e.message, 'error');
1437   }
1438 }
1439
1440 document.getElementById('leave-type').addEventListener('change', function() {
1441   document.getElementById('sick-leave-hint').style.display =
1442     this.value === 'SICK' ? 'block' : 'none';
1443   document.getElementById('casual-leave-hint').style.display =
1444     this.value === 'CASUAL' ? 'block' : 'none';
1445 });
1446
1447 function clearLeaveForm() {
1448   document.getElementById('leave-type').value = 'SICK';
1449   document.getElementById('leave-reason').value = '';
1450   document.getElementById('leave-start').value = '';
1451   document.getElementById('leave-end').value = '';

```

```

1451 }
1452
1453 async function loadMyLeaves(){
1454     try{
1455         const data=await apiFetch('/leave_requests/employee');
1456         const leaves=data?.requests||data||[];
1457         document.getElementById('my-leaves-list').innerHTML=leaves.length
1458             ? leaves.map(l=><tr><td><span class="chip
1459 ${l.leaveType||''}.toLowerCase()">${l.leaveType}</span></td><td>${l.startDate}</td><td>${l.endDate}</td><td>${
1460 l.numberOfDays||'-'}</td><td><...
1461 : <tr><td colspan="5" class="empty-state">No leave requests</td></tr>';
1462     }catch(e){document.getElementById('my-leaves-list').innerHTML='<tr><td colspan="5" class="empty-
1463 state">Could not load</td></tr>';}
1464 }
1465
1466 async function loadUsers(){
1467     try{
1468         const users = await apiFetch('/manager/Users');
1469         const badge = document.getElementById('users-badge');
1470         if(badge){ badge.textContent = (users||[]).length;
1471 badge.classList.toggle('hidden',(users||[]).length===0); }
1472         document.getElementById('users-list').innerHTML = (users||[]).length
1473             ? users.map(u=><tr>
1474                 <td>${u.name||'-'}</td>
1475                 <td style="font-family:var(--mono);font-size:12px">${u.email||'-'}</td>
1476                 <td style="font-family:var(--mono);font-size:12px">${u.timezone||'-'}</td>
1477                 <td><input type="password" placeholder="Set password" id="pwd-${CSS.escape(u.email)}"
1478                     style="padding:6px 10px;font-size:12px;width:160px"></td>
1479                 <td><button class="btn btn-approve btn-sm" onclick="promoteUser('${u.email}')">
1480 Promote</button></td>
1481                 </tr>`).join('')
1482             : <tr><td colspan="5" class="empty-state">No pending user registrations</td></tr>';
1483     }catch(e){
1484         document.getElementById('users-list').innerHTML='<tr><td colspan="5" class="empty-state">Could not
1485 load</td></tr>';
1486     }
1487 }
1488
1489 async function promoteUser(email){
1490     const pwdInput = document.getElementById('pwd-'+CSS.escape(email));
1491     const password = pwdInput?.value?.trim();
1492     if(!password){ showAlert('Please set a password before promoting','error'); return; }
1493     try{
1494         const res = await fetch(`${API}/manager/promote`,{
1495             method:'PUT',
1496             headers:{'Content-Type':'application/json',...getAuthHeader()},
1497             credentials:'include',
1498             body: JSON.stringify({ email, password })
1499         });
1500         const text = await res.text();
1501         if(res.ok){
1502             showAlert(`${email} promoted to Employee `, 'success');
1503             loadUsers();
1504         } else {
1505             showAlert('Error: '+text, 'error');
1506         }
1507     }catch(e){ showAlert('Failed: '+e.message, 'error'); }
1508 }
1509
1510 async function loadRestaurants(){
1511     try{
1512         const data = await apiFetch('/restaurants');
1513         document.getElementById('restaurants-list').innerHTML = (data||[]).length
1514             ? data.map(r=><tr>
1515                 <td style="font-family:var(--mono);font-size:12px">#${r.restaurantId||r.id}</td>
1516                 <td>${r.name||'-'}</td>
1517                 <td style="font-size:12px;color:var(--text3)">${r.address||'-'}</td>
1518                 <td><span class="chip active">${r.status||'ACTIVE'}</span></td>
1519                 </tr>`).join('')
1520             : <tr><td colspan="4" class="empty-state">No restaurants found</td></tr>';
1521     }catch(e){
1522         document.getElementById('restaurants-list').innerHTML='<tr><td colspan="4" class="empty-
1523 state">Could not load</td></tr>';
1524     }
1525 }
1526
1527 // SUBSCRIPTIONS

```

```

1520
1521 async function loadMySubscriptions(){
1522     try{
1523         const subs = await apiFetch('/subscription/user');
1524         document.getElementById('my-subscriptions-list').innerHTML = (subs||[]).length
1525             ? subs.map(s=><tr>
1526                 <td><span class="chip pending">${s.mealSlot||'-'}</span></td>
1527                 <td>${s.restaurantName||'-'}</td>
1528                 <td style="font-size:12px;color:var(--text3)">${s.restaurantAddress||'-'}</td>
1529                 <td>${s.scheduleType||'-'}</td>
1530                 <td>${s.dayOfWeek||'-'}</td>
1531                 <td><span class="chip ${s.status||''}.toLowerCase()">${s.status||'-'}</span></td>
1532                 <td style="font-family:var(--mono);font-size:11px">${s.nextDeliveryTime ? new
Date(s.nextDeliveryTime).toLocaleString() : '-'}</td>
1533             </tr>`).join('')
1534             : '<tr><td colspan="7" class="empty-state">No subscriptions yet</td></tr>';
1535     }catch(e){
1536         document.getElementById('my-subscriptions-list').innerHTML='<tr><td colspan="7" class="empty-
state">Could not load</td></tr>';
1537     }
1538 }
1539
1540 let slotRowCount = 0;
1541 async function addSlotRow(){
1542     slotRowCount++;
1543     const id = slotRowCount;
1544     const div = document.createElement('div');
1545     div.id = `slot-row-${id}`;
1546     div.style.cssText = 'display:grid;grid-template-columns:1fr 1fr auto;gap:12px;margin-
bottom:10px;align-items:end';
1547
1548     // Fetch restaurants for the dropdown
1549     let options = '<option value="">Loading...</option>';
1550     try{
1551         const data = await apiFetch('/restaurants');
1552         options = data.map(r => `<option
value="${r.restaurantId||r.id}">${r.name}</option>`).join('');
1553     }catch(e){ options = '<option value="">Could not load</option>'; }
1554
1555     div.innerHTML = `
1556     <div class="form-group" style="margin:0">
1557         <label>Meal Slot</label>
1558         <select id="slot-meal-${id}">
1559             <option value="BREAKFAST">Breakfast</option>
1560             <option value="LUNCH">Lunch</option>
1561             <option value="DINNER">Dinner</option>
1562         </select>
1563     </div>
1564     <div class="form-group" style="margin:0">
1565         <label>Restaurant</label>
1566         <select id="slot-restaurant-${id}">${options}</select>
1567     </div>
1568     <button class="btn btn-reject btn-sm" onclick="removeSlotRow(${id})" style="margin-
bottom:1px"></button>
1569 `;
1570     document.getElementById('slot-rows').appendChild(div);
1571 }
1572
1573 function removeSlotRow(id){
1574     document.getElementById(`slot-row-${id}`)?.remove();
1575 }
1576
1577 function toggleDayOfWeek(){
1578     const isWeekly = document.getElementById('sub-schedule').value === 'WEEKLY';
1579     document.getElementById('day-of-week-group').style.display = isWeekly ? 'flex' : 'none';
1580 }
1581
1582 async function submitSubscription(){
1583     const scheduleType = document.getElementById('sub-schedule').value;
1584     const dayOfWeek = scheduleType === 'WEEKLY' ? document.getElementById('sub-day').value : null;
1585
1586     const slotOrders = [];
1587     let valid = true;
1588
1589     document.querySelectorAll('[id^="slot-row-"]').forEach(row => {
1590         const idNum = row.id.split('-')[2];

```



```

1591         const mealSlot      = document.getElementById(`slot-meal-${idNum}`)?.value;
1592         const restaurantVal = document.getElementById(`slot-restaurant-${idNum}`)?.value;
1593
1594         if(!mealSlot || !restaurantVal){ valid = false; return; }
1595
1596         const restaurantId = parseInt(restaurantVal);
1597         if(isNaN(restaurantId)){ valid = false; return; }
1598
1599         slotOrders.push({ mealSlot, restaurantId });
1600     });
1601
1602     if(!valid || slotOrders.length === 0){
1603         showAlert('Please add at least one valid slot','error');
1604         return;
1605     }
1606
1607     // Derive mealSlots from slotOrders to satisfy @NotEmpty
1608     const mealSlots = slotOrders.map(s => s.mealSlot);
1609
1610     try{
1611         const res = await fetch(`${API}/subscription`, {
1612             method: 'POST',
1613             headers: {'Content-Type':'application/json', ...getAuthHeader()},
1614             credentials: 'include',
1615             body: JSON.stringify({ scheduleType, dayOfWeek, mealSlots, slotOrders })
1616         });
1617         const text = await res.text();
1618         if(res.ok){
1619             showAlert('Subscription added successfully! ','success');
1620             clearSubscriptionForm();
1621             loadMySubscriptions();
1622         } else {
1623             showAlert('Error: '+text,'error');
1624         }
1625     }catch(e){ showAlert('Failed: '+e.message,'error'); }
1626 }
1627
1628 function clearSubscriptionForm(){
1629     document.getElementById('sub-schedule').value = 'DAILY';
1630     document.getElementById('day-of-week-group').style.display = 'none';
1631     document.getElementById('slot-rows').innerHTML = '';
1632     slotRowCount = 0;
1633 }
1634
1635 /* =====
1636 MY CALENDAR (NEW)
1637 ===== */
1638 async function loadMyCalendar(){
1639     // 1. Load balance cards
1640     await loadCalendarBalanceCards();
1641     // 2. Build or refresh FullCalendar
1642     if(!calendarInitialized){
1643         initFullCalendar();
1644     } else {
1645         await refreshCalendarEvents();
1646     }
1647 }
1648
1649 async function loadCalendarBalanceCards(){
1650     // Default state
1651     ['cal-sick-remaining','cal-casual-remaining','cal-mat-remaining','cal-total-used']
1652     .forEach(id=>{ document.getElementById(id).textContent='--'; });
1653
1654     try{
1655         const dashboard = await apiFetch('/api/employees/me/leave-dashboard');
1656         const balances  = dashboard?.leaveBalances || {};
1657
1658         let totalUsed=0, totalAllowed=0;
1659
1660         // SICK
1661         const sick = balances['SICK'] || balances['sick'] || {};
1662         const sickRem  = Number(sick.available || 0);
1663         const sickTotal = Number(sick.total    || 0);
1664         const sickUsed  = sickTotal - sickRem;
1665         document.getElementById('cal-sick-remaining').textContent = sickRem;
1666         document.getElementById('cal-sick-sub').textContent      = `${sickUsed} used of ${sickTotal}

```

```

days`;
1667
1668         // CASUAL
1669         const cas = balances['CASUAL'] || balances['casual'] || {};
1670         const casRem = Number(cas.available || 0);
1671         const casTotal = Number(cas.total || 0);
1672         const casUsed = casTotal - casRem;
1673         document.getElementById('cal-casual-remaining').textContent = casRem;
1674         document.getElementById('cal-casual-sub').textContent = `${casUsed} used of ${casTotal}
days`;
1675
1676         // MATERNITY
1677         const mat = balances['MATERNITY'] || balances['maternity'] || {};
1678         const matRem = Number(mat.available || 0);
1679         const matTotal = Number(mat.total || 0);
1680         const matUsed = matTotal - matRem;
1681         document.getElementById('cal-mat-remaining').textContent = matRem;
1682         document.getElementById('cal-mat-sub').textContent = `${matUsed} used of ${matTotal}
days`;
1683
1684         // TOTAL
1685         Object.values(balances).forEach(b=>{
1686             totalUsed += (Number(b.total||0) - Number(b.available||0));
1687             totalAllowed += Number(b.total||0);
1688         });
1689         document.getElementById('cal-total-used').textContent = totalUsed;
1690         document.getElementById('cal-total-sub').textContent = `out of ${totalAllowed} days total`;
1691
1692     }catch(e){
1693         console.error('Could not load leave balance for calendar:', e);
1694         ['cal-sick-sub', 'cal-casual-sub', 'cal-mat-sub', 'cal-total-sub']
1695             .forEach(id=>{ document.getElementById(id).textContent='unavailable'; });
1696     }
1697 }
1698
1699 async function buildCalendarEvents(){
1700     const events = [];
1701     try{
1702         // - Employee leaves -
1703         const data = await apiFetch('/leave_requests/employee');
1704         const leaves = data?.requests || data || [];
1705         leaves.forEach(l=>{
1706             const s = (l.status||'PENDING').toUpperCase();
1707             const c = STATUS_COLORS[s] || STATUS_COLORS.PENDING;
1708             const typeLabel = TYPE_LABEL[l.leaveType] || l.leaveType || 'Leave';
1709
1710             // FullCalendar end date is exclusive, so add 1 day for multi-day events
1711             let endDate = l.endDate;
1712             if(endDate){
1713                 const d = new Date(endDate);
1714                 d.setDate(d.getDate()+1);
1715                 endDate = d.toISOString().split('T')[0];
1716             }
1717
1718             events.push({
1719                 id : `leave-${l.leaveRequestId}`,
1720                 title : `${typeLabel} (${s.charAt(0)+s.slice(1).toLowerCase()})`,
1721                 start : l.startDate,
1722                 end : endDate,
1723                 allDay : true,
1724                 backgroundColor : c.bg,
1725                 borderColor : c.border,
1726                 textColor : c.text,
1727                 extendedProps : {
1728                     type : 'leave',
1729                     status : s,
1730                     reason : l.reason || '-',
1731                     leaveType: l.leaveType
1732                 }
1733             });
1734         });
1735     }catch(e){ console.warn('Could not load leaves for calendar:', e); }
1736
1737     try{
1738         // - Public holidays (admin-added) -
1739         const holidays = await apiFetch('/holidays');

```

```

1740         (holidays||[]).forEach(h=>{
1741             const c = STATUS_COLORS.HOLIDAY;
1742             events.push({
1743                 id      : `holiday-${h.id}|h.holidayId|Math.random()`,
1744                 title   : `${h.name}|h.holidayName|'Holiday'}`,
1745                 start   : h.date|h.holidayDate,
1746                 allDay   : true,
1747                 backgroundColor : c.bg,
1748                 borderColor : c.border,
1749                 textColor  : c.text,
1750                 extendedProps : { type:'holiday', description: h.description||'' }
1751             });
1752         });
1753     }catch(e){ console.warn('Could not load holidays for calendar:', e); }
1754
1755     return events;
1756 }
1757
1758 function initFullCalendar(){
1759     const calendarEl = document.getElementById('employee-calendar');
1760
1761     employeeCalendar = new FullCalendar.Calendar(calendarEl, {
1762         initialView: 'dayGridMonth',
1763         height     : 'auto',
1764         headerToolbar:{
1765             left : 'prev,next today',
1766             center: 'title',
1767             right : 'dayGridMonth,dayGridWeek,listMonth'
1768         },
1769         buttonText:{
1770             today : 'Today',
1771             month : 'Month',
1772             week  : 'Week',
1773             listMonth: 'List'
1774         },
1775         dayMaxEvents: 3,      // "+N more" link if overflowing
1776         navLinks     : true,  // click day number day view
1777         events       : [],    // start empty, populate below
1778
1779         // Event hover tooltip
1780         eventMouseEnter: function(info){
1781             const ep = info.event.extendedProps;
1782             const tooltip = document.getElementById('cal-tooltip');
1783             let html = `
```

```

1814     async function refreshCalendarEvents(){
1815         if(!employeeCalendar) return;
1816         const events = await buildCalendarEvents();
1817         // Remove all existing events and re-add
1818         employeeCalendar.removeAllEvents();
1819         events.forEach(e=>employeeCalendar.addEvent(e));
1820     }
1821
1822     async function refreshCalendar(){
1823         await loadCalendarBalanceCards();
1824         await refreshCalendarEvents();
1825         showAlert('Calendar refreshed!', 'success');
1826     }
1827
1828     // Move tooltip with mouse
1829     document.addEventListener('mousemove', function(e){
1830         const tt = document.getElementById('cal-tooltip');
1831         if(tt.style.display==='block'){
1832             tt.style.left = (e.clientX+14)+'px';
1833             tt.style.top = (e.clientY+14)+'px';
1834             // Keep tooltip on screen
1835             const rect = tt.getBoundingClientRect();
1836             if(rect.right > window.innerWidth)
1837                 tt.style.left = (e.clientX - rect.width - 10)+'px';
1838             if(rect.bottom > window.innerHeight)
1839                 tt.style.top = (e.clientY - rect.height - 10)+'px';
1840         }
1841     });
1842
1843     // ===== MY DEVICES =====
1844     async function loadMyAssignedDevices() {
1845         try {
1846             const devices = await apiFetch('/devices/my');
1847             const tbody = document.getElementById('my-devices-list');
1848             if (!devices || !devices.length) {
1849                 tbody.innerHTML = '<tr><td colspan="7" class="empty-state">No devices assigned to
you</td></tr>';
1850                 return;
1851             }
1852             const today = new Date();
1853             tbody.innerHTML = devices.map(d => {
1854                 const warrantyExpiry = d.warrantyExpiryDate ? new Date(d.warrantyExpiryDate) : null;
1855                 const warrantyDisplay = warrantyExpiry
1856                     ? (warrantyExpiry > today
1857                       ? '<span style="color:var(--green);font-size:11px"> ${d.warrantyExpiryDate}</span>'
1858                       : '<span style="color:var(--red);font-size:11px"> Expired
${d.warrantyExpiryDate}</span>')
1859                     : '<span style="color:var(--text3);font-size:11px"></span>';
1860                 return `<tr>
1861 <td>#${d.id}</td>
1862 <td><strong>${d.deviceName || '-'}</strong></td>
1863 <td>${d.brand || '-'}</td>
1864 <td><span class="chip">${(d.deviceType || '').toLowerCase()}</span></td>
1865 <td>${warrantyDisplay}</td>
1866 <td style="font-size:12px;color:var(--text2)">${d.assignedDate || '-'}</td>
1867 <td><span class="chip">${(d.deviceStatus || '').toLowerCase()}>${d.deviceStatus || '-'}</span></td>
1868 </tr>`;
1869             }).join('');
1870         } catch (e) {
1871             document.getElementById('my-devices-list').innerHTML =
1872                 '<tr><td colspan="7" class="empty-state">Could not load devices</td></tr>';
1873         }
1874     }
1875
1876     // ===== SERVICE REQUESTS =====
1877     async function loadServiceRequests() {
1878         await Promise.all([loadServiceRequestsList(), loadMyDevicesForSR()]);
1879     }
1880
1881     async function loadServiceRequestsList() {
1882         try {
1883             const requests = await apiFetch('/service-requests/my');
1884             const tbody = document.getElementById('service-requests-list');
1885             if (!requests || !requests.length) {
1886                 tbody.innerHTML = '<tr><td colspan="7" class="empty-state">No service requests yet</td></tr>';
1887                 return;

```

```

1888     }
1889     tbody.innerHTML = requests.map(r => `<tr>
1890         <td style="color:var(--text3);font-family:var(--mono)">#${r.id}</td>
1891         <td>${r.deviceName || '-'}</td>
1892         <td><span class="chip pending">${r.requestType}</span></td>
1893         <td style="max-width:200px;overflow:hidden;text-overflow:ellipsis;white-space:nowrap">
1894             ${r.issueDescription || '-'}
1895         </td>
1896         <td>${r.urgent
1897             ? '<span class="chip rejected">YES</span>'
1898             : '<span style="color:var(--text3)">No</span>'}
1899         </td>
1900         <td><span class="chip ${r.status || ''}.toLowerCase()">${r.status || '-'}</span></td>
1901         <td style="font-size:12px;color:var(--text3)">
1902             ${r.raisedAt ? new Date(r.raisedAt).toLocaleDateString() : '-'}
1903         </td>
1904     </tr>`).join('');
1905 } catch (e) { console.error(e); }
1906 }
1907
1908 async function loadMyDevicesForSR() {
1909     try {
1910         const devices = await apiFetch('/devices/my');
1911         const select = document.getElementById('sr-device-select');
1912         select.innerHTML = '<option value="">-- Select your device --</option>';
1913         (devices || []).forEach(d => {
1914             const opt = document.createElement('option');
1915             opt.value = d.id;
1916             opt.textContent = `${d.deviceName} (${d.brand || d.deviceType || 'Device'})`;
1917             select.appendChild(opt);
1918         });
1919     } catch (e) { console.error('Failed to load devices for SR', e); }
1920 }
1921
1922 async function submitServiceRequest() {
1923     const deviceId = parseInt(document.getElementById('sr-device-select').value);
1924     const requestType = document.getElementById('sr-type').value;
1925     const issueDescription = document.getElementById('sr-description').value.trim();
1926     const urgent = document.getElementById('sr-urgent').checked;
1927
1928     if (!deviceId) { showAlert('Please select a device', 'error'); return; }
1929     if (!issueDescription) { showAlert('Please describe the issue', 'error'); return; }
1930
1931     try {
1932         await apiFetch('/service-requests', {
1933             method: 'POST',
1934             body: JSON.stringify({ deviceId, requestType, issueDescription, urgent })
1935         });
1936         showAlert('Service request submitted!', 'success');
1937         clearServiceRequestForm();
1938         loadServiceRequestsList();
1939     } catch (e) { showAlert('Failed: ' + e.message, 'error'); }
1940 }
1941
1942 function clearServiceRequestForm() {
1943     document.getElementById('sr-device-select').value = '';
1944     document.getElementById('sr-type').value = 'REPAIR';
1945     document.getElementById('sr-description').value = '';
1946     document.getElementById('sr-urgent').checked = false;
1947 }
1948
1949 async function searchEmployee() {
1950     const name = document.getElementById('org-search-input').value.trim();
1951     const errorEl = document.getElementById('org-search-error');
1952     const resultsEl = document.getElementById('org-search-results');
1953     const emptyEl = document.getElementById('org-search-empty');
1954
1955     errorEl.style.display = 'none';
1956     resultsEl.style.display = 'none';
1957     emptyEl.style.display = 'block';
1958
1959     if (!name) {
1960         errorEl.textContent = 'Please enter a name to search.';
1961         errorEl.style.display = 'block';
1962         return;
1963     }

```

```

1964
1965     try {
1966         const emp = await apiFetch(`/manager/employees/search?name=${encodeURIComponent(name)}`);
1967
1968         // Populate info card
1969         document.getElementById('sr-name').textContent = emp.name || '-';
1970         document.getElementById('sr-id').textContent = `#${emp.employeeId}`;
1971         document.getElementById('sr-email').textContent = emp.email || '-';
1972         document.getElementById('sr-dept').textContent = emp.dept || '-';
1973         document.getElementById('sr-attendance').textContent = emp.attendance || '-';
1974
1975         // Status chip
1976         const statusMap = { ACTIVE: 'approved', ON_LEAVE: 'pending', INACTIVE: 'rejected' };
1977         const statusKey = (emp.status || '').toString();
1978         document.getElementById('sr-status').innerHTML =
1979             `
1997                     <td style="padding:10px 16px;border-top:1px solid var(--border);font-
1998 size:13px;color:var(--text)"><strong>${d.deviceName || '-'}</strong></td>
1999                     <td style="padding:10px 16px;border-top:1px solid var(--border);font-
2000 size:13px;color:var(--text2)">${d.brand || '-'}</td>
2001                     <td style="padding:10px 16px;border-top:1px solid var(--border)"><span class="chip
2002 pending">${(d.deviceType || '').toLowerCase()}</span></td>
2003                     <td style="padding:10px 16px;border-top:1px solid var(--border)"><span class="chip
2004 ${statusCss[ds] || ''}">${ds || '-'}</span></td>
2005                     <td style="padding:10px 16px;border-top:1px solid var(--border)">${warranty}</td>
2006                     <td style="padding:10px 16px;border-top:1px solid var(--border);font-
2007 size:12px;color:var(--text3)">${d.assignedDate || '-'}</td>
2008                 </tr>;
2009             }).join('')
2010             : `

```

```

2032     });
2033 }
2034 async function loadNotifCount() {
2035     try {
2036         const data = await apiFetch('/notifications/unread-count');
2037         const badge = document.getElementById('notif-badge');
2038         if (data.count > 0) {
2039             badge.textContent = data.count;
2040             badge.style.display = 'flex';
2041         } else {
2042             badge.style.display = 'none';
2043         }
2044     } catch(e) { console.warn('Could not load notif count', e); }
2045 }
2046
2047 async function toggleNotifPanel() {
2048     const panel = document.getElementById('notif-panel');
2049     if (panel.style.display === 'none') {
2050         panel.style.display = 'block';
2051         await loadNotifications();
2052         try { await apiFetch('/notifications/mark-read', { method: 'POST' }); } catch(e) {}
2053         document.getElementById('notif-badge').style.display = 'none';
2054     } else {
2055         panel.style.display = 'none';
2056     }
2057 }
2058
2059 async function loadNotifications() {
2060     try {
2061         const data = await apiFetch('/notifications');
2062         const list = document.getElementById('notif-list');
2063         list.innerHTML = data.length
2064             ? data.map(n => `
2065                 <div class="notif-item ${n.read ? '' : 'unread'}">
2066                     <div>${n.message}</div>
2067                     <div class="notif-time">${new Date(n.createdAt).toLocaleString()}</div>
2068                 </div>`).join('')
2069             : '<div class="notif-item">No notifications yet.</div>';
2070     } catch(e) { console.warn('Could not load notifications', e); }
2071 }
2072
2073 loadNotifCount();
2074 setInterval(loadNotifCount, 30000);
2075
2076 document.addEventListener('click', e => {
2077     const panel = document.getElementById('notif-panel');
2078     const wrapper = document.querySelector('.notif-wrapper');
2079     if (panel && wrapper && !panel.contains(e.target) && !wrapper.contains(e.target)) {
2080         panel.style.display = 'none';
2081     }
2082 });
2083 function showAlert(msg,type){
2084     const el=document.createElement('div');
2085     el.className=`alert alert-${type}`;el.textContent=msg;
2086     document.body.appendChild(el);setTimeout(()=>el.remove(),3000);
2087 }
2088
2089 // Gender store
2090 let currentManagerGender = null;
2091
2092 // Load profile data
2093 async function loadProfile() {
2094     try {
2095         const me = await apiFetch('/employee/me');
2096         if(me) {
2097             currentManagerGender = me.gender;
2098
2099             document.getElementById('p-name').textContent = me.name || '-';
2100             document.getElementById('p-email').textContent = me.email || '-';
2101             document.getElementById('p-dept').textContent = me.dept || '-';
2102             document.getElementById('p-role').textContent = me.role || 'MANAGER';
2103             document.getElementById('p-status').textContent = me.status || '-';
2104             document.getElementById('p-tz').textContent = me.timezone || '-';
2105
2106             // Show maternity section only for female managers
2107             const maternityBtn = document.getElementById('maternity-apply-btn');

```

```

2108         if(maternityBtn) {
2109             maternityBtn.style.display = me.gender === 'F' ? 'block' : 'none';
2110         }
2111
2112         // Also update sidebar name with real name
2113         document.getElementById('sidebar-name').textContent = me.name || '-';
2114         document.getElementById('avatar-initials').textContent =
2115             (me.name || me.email || 'M').charAt(0).toUpperCase();
2116     }
2117 } catch(e) {
2118     console.error('Could not load profile:', e);
2119 }
2120 }
2121
2122 // Auto calculate maternity end date
2123 document.getElementById('mat-start-date').addEventListener('change', function(){
2124     if(this.value) {
2125         const start = new Date(this.value);
2126         const end = new Date(start);
2127         end.setDate(end.getDate() + 179);
2128         document.getElementById('mat-end-date').value =
2129             end.toISOString().split('T')[0];
2130     } else {
2131         document.getElementById('mat-end-date').value = '';
2132     }
2133 });
2134
2135 // Submit maternity leave
2136 async function applyMaternityLeave() {
2137     const startDate = document.getElementById('mat-start-date').value;
2138     const reason = document.getElementById('mat-reason').value.trim();
2139
2140     if(!startDate) {
2141         showAlert('Please select a start date', 'error');
2142         return;
2143     }
2144
2145     try {
2146         await apiFetch('/leave_requests/maternity/apply', {
2147             method: 'POST',
2148             body: JSON.stringify({ startDate, reason })
2149         });
2150         showAlert('Maternity leave submitted successfully!', 'success');
2151         document.getElementById('mat-start-date').value = '';
2152         document.getElementById('mat-end-date').value = '';
2153         document.getElementById('mat-reason').value = '';
2154         loadMyLeaves();
2155     } catch(e) {
2156         showAlert('Failed: ' + e.message, 'error');
2157     }
2158 }
2159 </script>
2160 </body>
2161 </html>
2162

```



```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <title>Account Pending</title>
6      <style>
7          * { margin: 0; padding: 0; box-sizing: border-box; }
8          body {
9              font-family: 'Outfit', sans-serif;
10             background: #070910;
11             color: #dde4f0;
12             min-height: 100vh;
13             display: flex;
14             align-items: center;
15             justify-content: center;
16         }
17         .card {
18             background: #0e1118;
19             border: 1px solid rgba(255,255,255,0.07);
20             border-radius: 16px;
21             padding: 48px;
22             max-width: 420px;
23             text-align: center;
24         }
25         .icon { font-size: 48px; margin-bottom: 20px; }
26         h1 { font-size: 22px; font-weight: 700; margin-bottom: 12px; }
27         p { font-size: 14px; color: #6b7591; line-height: 1.6; margin-bottom: 24px; }
28         .email {
29             background: #131821;
30             border: 1px solid rgba(79,140,255,0.2);
31             border-radius: 8px;
32             padding: 10px 16px;
33             font-size: 13px;
34             color: #4f8cff;
35             margin-bottom: 28px;
36             font-family: monospace;
37         }
38         .btn {
39             padding: 10px 24px;
40             background: transparent;
41             border: 1px solid rgba(255,255,255,0.1);
42             border-radius: 8px;
43             color: #6b7591;
44             font-size: 13px;
45             cursor: pointer;
46         }
47         .btn:hover { border-color: rgba(255,255,255,0.2); color: #dde4f0; }
48     </style>
49 </head>
50 <body>
51 <div class="card">
52     <div class="icon"></div>
53     <h1>Account Pending Approval</h1>
54     <p>You've successfully signed in with Google. Your account is currently awaiting approval from an admin
or manager.</p>
55     <div class="email" id="user-email">Loading...</div>
56     <p>Once approved, you'll have full access to the portal. Please contact your manager or admin.</p>
57     <button class="btn" onclick="logout()">Sign out</button>
58 </div>
59
60 <script>
61     // Show the user their email from JWT
62     const token = localStorage.getItem('jwt_token');
63     if (token) {
64         try {
65             const payload = JSON.parse(atob(token.split('.')[1]));
66             document.getElementById('user-email').textContent = payload.sub || 'Unknown';
67         } catch(e) {}
68     }
69
70     function logout() {
71         localStorage.clear();

```

```
72         window.location.href = '/google.html';
73     }
74 </script>
75 </body>
76 </html>
```

```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>TechVendor Hub</title>
7      <link
href="https://fonts.googleapis.com/css2?family=Syne:wght@400;500;600;700;800&family=DM+Mono:wght@400;500&display
=swap" rel="stylesheet">
8      <style>
9          *{margin:0;padding:0;box-sizing:border-box}
10         :root{
11             --bg:#080b10;--bg2:#0c1018;--bg3:#111620;--bg4:#161d2a;
12             --panel:#131a26;--panel2:#1a2235;
13             --border:rgba(0,210,255,0.08);--border2:rgba(0,210,255,0.18);--border3:rgba(0,210,255,0.35);
14             --text:#d4e8f5;--text2:#6b8aaa;--text3:#3a5270;
15             --cyan:#00d2ff;--cyan-dim:rgba(0,210,255,0.08);--cyan-glow:rgba(0,210,255,0.15);
16             --amber:#f59e0b;--amber-dim:rgba(245,158,11,0.1);
17             --green:#10d48e;--green-dim:rgba(16,212,142,0.1);
18             --red:#f05454;--red-dim:rgba(240,84,84,0.1);
19             --violet:#a78bfa;--violet-dim:rgba(167,139,250,0.1);
20             --orange:#fb923c;--orange-dim:rgba(251,146,60,0.1);
21             --font:'Syne',sans-serif;--mono:'DM Mono',monospace;
22             --sidebar:260px;
23         }
24         body{font-family:var(--font);background:var(--bg);color:var(--text);font-
size:14px;overflow:hidden;height:100vh}
25         .app{display:flex;height:100vh}
26
27         body::before{content:'';position:fixed;inset:0;background:repeating-linear-
gradient(0deg,transparent,transparent 2px,rgba(0,0,0,0.03) 2px,rgba(0,0,0,0.03) 4px);pointer-events:none;z-
index:9...
28
29         .sidebar{width:var(--sidebar);background:var(--bg2);border-right:1px solid var(--
border);display:flex;flex-direction:column;flex-shrink:0;overflow:hidden;position:relative}
30         .sidebar::after{content:'';position:absolute;bottom:0;left:0;right:0;height:300px;background:linear-
gradient(to top,rgba(0,210,255,0.03),transparent);pointer-events:none}
31
32         .logo{padding:24px 20px 20px;border-bottom:1px solid var(--border)}
33         .logo-mark{display:flex;align-items:center;gap:10px;margin-bottom:6px}
34         .logo-hex{width:36px;height:36px;background:linear-gradient(135deg,#00d2ff,#0066ff);clip-
path:polygon(50% 0%,100% 25%,100% 75%,50% 100%,0% 75%,0% 25%);display:flex;align-items:center;justifyf...
35         .logo-name{font-size:17px;font-weight:800;color:var(--text);letter-spacing:-0.5px}
36         .logo-sub{font-size:10px;color:var(--text3);font-family:var(--mono);letter-spacing:1px;text-
transform:uppercase}
37         .vendor-badge{display:inline-flex;align-items:center;gap:6px;margin-top:10px;padding:5px
12px;background:var(--cyan-dim);border:1px solid var(--border2);border-radius:4px;font-size:10px;font...
38         .badge-dot{width:5px;height:5px;background:var(--cyan);border-radius:50%;animation:pulse 2s
infinite}
39         @keyframes pulse{0%,100%{opacity:1;box-shadow:0 0 0 0 var(--cyan-glow)}50%{opacity:0.7;box-shadow:0
0 0 4px transparent}}
40
41         .nav{flex:1;padding:12px 0;overflow-y:auto}
42         .nav::-webkit-scrollbar{width:3px}
43         .nav::-webkit-scrollbar-thumb{background:var(--border2);border-radius:2px}
44         .nav-section{padding:14px 20px 5px;font-size:9px;font-weight:700;color:var(--text3);letter-
spacing:2.5px;text-transform:uppercase;font-family:var(--mono)}
45         .nav-item{display:flex;align-items:center;gap:10px;padding:10px 20px;margin:1px 8px;border-
radius:6px;cursor:pointer;transition:all 0.15s;color:var(--text2);font-size:13px;font-weight:500;p...
46         .nav-item:hover{background:var(--cyan-dim);color:var(--cyan)}
47         .nav-item.active{background:var(--cyan-dim);color:var(--cyan);border:1px solid var(--border2)}
48         .nav-item.active::before{content:'';position:absolute;left:-8px;top:50%;transform:translateY(-
50%);width:3px;height:60%;background:var(--cyan);border-radius:0 2px 2px 0}
49         .nav-icon{width:16px;height:16px;opacity:0.7;flex-shrink:0}
50         .nav-item.active .nav-icon,.nav-item:hover .nav-icon{opacity:1}
51         .nav-badge{margin-left:auto;background:var(--red);color:#fff;font-size:9px;font-
weight:700;padding:2px 6px;border-radius:3px;font-family:var(--mono)}
52         .nav-badge.amber{background:var(--amber)}
53
54         .sidebar-footer{padding:16px 20px;border-top:1px solid var(--border)}
55         .user-row{display:flex;align-items:center;gap:10px}
56         .avatar{width:34px;height:34px;background:linear-gradient(135deg,#00d2ff22,#0066ff22);border:1px

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solid var(--border2);border-radius:6px;display:flex;align-items:center;justify-content:cente...
57 .user-name{font-size:12px;font-weight:600;color:var(--text)}
58 .user-role{font-size:10px;color:var(--cyan);font-family:var(--mono);text-transform:uppercase;letter-
spacing:0.8px}
59
60 .main{flex:1;display:flex;flex-direction:column;overflow:hidden}
61 .topbar{background:var(--bg2);padding:0 24px;height:56px;display:flex;justify-content:space-
between;align-items:center;border-bottom:1px solid var(--border);flex-shrink:0}
62 .page-title{font-size:15px;font-weight:700;color:var(--text);display:flex;align-
items:center;gap:8px}
63 .page-title-prefix{color:var(--cyan);font-family:var(--mono);font-size:12px;font-weight:400}
64 .topbar-right{display:flex;align-items:center;gap:10px}
65 .topbar-email{font-size:11px;color:var(--text3);font-family:var(--mono)}
66 .signout-btn{padding:6px 14px;background:transparent;border:1px solid var(--border2);border-
radius:5px;color:var(--text2);font-family:var(--font);font-size:11px;font-weight:600;cursor:point...
67 .signout-btn:hover{background:var(--red-dim);border-color:var(--red);color:var(--red)}
68
69 .content{flex:1;overflow-y:auto;padding:24px}
70 .content::-webkit-scrollbar{width:4px}
71 .content::-webkit-scrollbar-thumb{background:var(--border2)}
72 .view{display:none}
73 .view.active{display:block;animation:viewIn 0.25s ease}
74 @keyframes viewIn{from{opacity:0;transform:translateY(6px)}to{opacity:1;transform:translateY(0)}}
75
76 .stats-row{display:grid;grid-template-columns:repeat(4,1fr);gap:14px;margin-bottom:22px}
77 .stat{background:var(--panel);border:1px solid var(--border);border-radius:8px;padding:18px
20px;position:relative;overflow:hidden;cursor:default;transition:border-color 0.2s}
78 .stat:hover{border-color:var(--border2)}
79 .stat::after{content:'';position:absolute;top:0;left:0;right:0;height:2px}
80 .stat.c{--c:var(--cyan)} .stat.g{--c:var(--green)} .stat.a{--c:var(--amber)} .stat.r{--c:var(--red)}
.stat.v{--c:var(--violet)} .stat.o{--c:var(--orange)}
81 .stat::after{background:var(--c)}
82 .stat-label{font-size:10px;color:var(--text3);text-transform:uppercase;letter-spacing:1px;font-
weight:600;margin-bottom:8px;font-family:var(--mono)}
83 .stat-val{font-size:28px;font-weight:800;color:var(--text);line-height:1;font-family:var(--mono)}
84 .stat-sub{font-size:11px;color:var(--text3);margin-top:5px}
85 .stat-icon{position:absolute;right:16px;top:50%;transform:translateY(-50%);font-
size:24px;opacity:0.12}
86
87 .card{background:var(--panel);border:1px solid var(--border);border-radius:10px;margin-
bottom:18px;overflow:hidden}
88 .card-head{padding:16px 20px;border-bottom:1px solid var(--border);display:flex;justify-
content:space-between;align-items:center}
89 .card-title{font-size:13px;font-weight:700;color:var(--text);letter-spacing:0.2px}
90 .card-sub{font-size:11px;color:var(--text3);margin-top:2px;font-family:var(--mono)}
91 .card-body{padding:20px}
92 .dash-grid{display:grid;grid-template-columns:1fr 1fr;gap:18px}
93
94 .tablewrap{overflow-x:auto}
95 table{width:100%;border-collapse:collapse}
96 th{padding:9px 16px;text-align:left;font-size:9px;color:var(--text3);font-weight:700;letter-
spacing:1.5px;text-transform:uppercase;background:var(--bg3);border-bottom:1px solid var(--border...
97 td{padding:12px 16px;border-bottom:1px solid var(--border);font-size:12px;color:var(--text2);font-
family:var(--mono)}
98 tr:last-child td{border-bottom:none}
99 tr:hover td{background:rgba(0,210,255,0.02);color:var(--text)}
100 td.main-cell{font-family:var(--font);font-size:13px;color:var(--text);font-weight:500}
101
102 .chip{display:inline-flex;align-items:center;gap:5px;padding:3px 9px;border-radius:3px;font-
size:10px;font-weight:700;letter-spacing:0.8px;text-transform:uppercase;font-family:var(--mono)}
103 .chip::before{content:'';width:4px;height:4px;border-radius:50%;flex-shrink:0}
104 .chip.available{background:var(--green-dim);color:var(--green);border:1px solid
rgba(16,212,142,0.2)} .chip.available::before{background:var(--green)}
105 .chip.assigned{background:var(--cyan-dim);color:var(--cyan);border:1px solid var(--border2)}
.chip.assigned::before{background:var(--cyan)}
106 .chip.under_repair{background:var(--amber-dim);color:var(--amber);border:1px solid
rgba(245,158,11,0.2)} .chip.under_repair::before{background:var(--amber)}
107 .chip.condemned{background:var(--red-dim);color:var(--red);border:1px solid rgba(240,84,84,0.2)}
.chip.condemned::before{background:var(--red)}
108 .chip.returned_to_vendor{background:var(--violet-dim);color:var(--violet);border:1px solid
rgba(167,139,250,0.2)} .chip.returned_to_vendor::before{background:var(--violet)}
109 .chip.open{background:var(--amber-dim);color:var(--amber);border:1px solid rgba(245,158,11,0.2)}
.chip.open::before{background:var(--amber)}
110 .chip.acknowledged{background:var(--cyan-dim);color:var(--cyan);border:1px solid var(--border2)}
.chip.acknowledged::before{background:var(--cyan)}
111 .chip.sent_for_repair{background:var(--orange-dim);color:var(--orange);border:1px solid

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111 rgba(251,146,60,0.2)} .chip.sent_for_repair::before{background:var(--orange)}
112 .chip.repair_done{background:var(--green-dim);color:var(--green);border:1px solid
113 rgba(16,212,142,0.2)} .chip.repair_done::before{background:var(--green)}
114 .chip.resolved{background:var(--green-dim);color:var(--green);border:1px solid
115 rgba(16,212,142,0.2)} .chip.resolved::before{background:var(--green)}
116 .chip.rejected{background:var(--red-dim);color:var(--red);border:1px solid rgba(240,84,84,0.2)}
117 .chip.rejected::before{background:var(--red)}
118 .chip.sent{background:var(--amber-dim);color:var(--amber);border:1px solid rgba(245,158,11,0.2)}
119 .chip.sent::before{background:var(--amber)}
120 .chip.in_progress{background:var(--orange-dim);color:var(--orange);border:1px solid
121 rgba(251,146,60,0.2)} .chip.in_progress::before{background:var(--orange)}
122 .chip.fixed{background:var(--green-dim);color:var(--green);border:1px solid rgba(16,212,142,0.2)}
123 .chip.fixed::before{background:var(--green)}
124 .chip.irreparable{background:var(--red-dim);color:var(--red);border:1px solid rgba(240,84,84,0.2)}
125 .chip.irreparable::before{background:var(--red)}
126 .chip.laptop{background:var(--cyan-dim);color:var(--cyan);border:1px solid var(--border2)}
127 .chip.laptop::before{background:var(--cyan)}
128 .chip.phone{background:var(--violet-dim);color:var(--violet);border:1px solid
129 rgba(167,139,250,0.2)} .chip.phone::before{background:var(--violet)}
130 .chip.tablet{background:var(--green-dim);color:var(--green);border:1px solid rgba(16,212,142,0.2)}
131 .chip.tablet::before{background:var(--green)}
132 .chip.monitor,.chip.other,.chip.desktop,.chip.keyboard,.chip.mouse,.chip.headset{background:var(--
133 amber-dim);color:var(--amber);border:1px solid rgba(245,158,11,0.2)} .chip.monitor::before,...
134 .chip.active_assign{background:var(--green-dim);color:var(--green);border:1px solid
135 rgba(16,212,142,0.2)} .chip.active_assign::before{background:var(--green)}
136 .chip.returned{background:var(--text3);color:var(--text2);border:1px solid var(--border)}
137 .chip.returned::before{background:var(--text2)}
138 .chip.not_fixable{background:var(--red-dim);color:var(--red);border:1px solid rgba(240,84,84,0.2)}
139 .chip.not_fixable::before{background:var(--red)}
140
141
142 .btn{padding:8px 16px;border-radius:6px;border:none;font-family:var(--font);font-size:12px;font-
143 weight:700;cursor:pointer;transition:all 0.2s;letter-spacing:0.3px}
144 .btn-primary{background:var(--cyan);color:#080b10}
145 .btn-primary:hover{background:#33dbff;box-shadow:0 0 20px var(--cyan-glow)}
146 .btn-ghost{background:transparent;color:var(--text2);border:1px solid var(--border2)}
147 .btn-ghost:hover{border-color:var(--border3);color:var(--text)}
148 .btn-danger{background:var(--red-dim);color:var(--red);border:1px solid rgba(240,84,84,0.2)}
149 .btn-danger:hover{background:var(--red);color:#fff}
150 .btn-warn{background:var(--amber-dim);color:var(--amber);border:1px solid rgba(245,158,11,0.2)}
151 .btn-warn:hover{background:var(--amber);color:#080b10}
152 .btn-success{background:var(--green-dim);color:var(--green);border:1px solid rgba(16,212,142,0.2)}
153 .btn-success:hover{background:var(--green);color:#080b10}
154 .btn-sm{padding:5px 10px;font-size:11px}
155 .btn-xs{padding:3px 8px;font-size:10px}
156 .action-row{display:flex;gap:6px;flex-wrap:wrap}
157
158
159 .form-grid{display:grid;grid-template-columns:1fr 1fr;gap:14px}
160 .form-group{display:flex;flex-direction:column;gap:6px}
161 .form-group.full{grid-column:1/-1}
162 .form-label{font-size:9px;font-weight:700;color:var(--text3);text-transform:uppercase;letter-
163 spacing:1.5px;font-family:var(--mono)}
164 .form-input{background:var(--bg3);border:1px solid var(--border);border-radius:6px;padding:10px
165 13px;color:var(--text);font-family:var(--mono);font-size:12px;transition:border-color 0.2s;wi...
166 .form-input:focus{outline:none;border-color:var(--cyan);box-shadow:0 0 3px var(--cyan-dim)}
167 .form-input:placeholder{color:var(--text3)}
168 select.form-input option{background:var(--bg2)}
169 .form-actions{display:flex;gap:10px;margin-top:18px}
170
171
172 .empty{text-align:center;padding:40px 20px;color:var(--text3);font-family:var(--mono);font-
173 size:12px}
174
175 .empty-icon{font-size:32px;margin-bottom:10px;opacity:0.4}
176
177 .alert{position:fixed;top:20px;right:20px;padding:12px 20px;border-radius:6px;z-
178 index:2000;animation:alertIn 0.25s ease;font-size:12px;font-weight:600;font-family:var(--mono);display:flex;a...
179 .alert-success{background:var(--green-dim);color:var(--green);border:1px solid
180 rgba(16,212,142,0.3)}
181 .alert-error{background:var(--red-dim);color:var(--red);border:1px solid rgba(240,84,84,0.3)}
182 @keyframes alertIn{from{transform:translateX(120%);opacity:0}to{transform:translateX(0);opacity:1}}
183
184
185 .search-row{display:flex;gap:10px;padding:14px 16px;border-bottom:1px solid var(--border)}
186 .search-input{flex:1;background:var(--bg3);border:1px solid var(--border);border-
187 radius:6px;padding:8px 13px;color:var(--text);font-family:var(--mono);font-size:12px}
188 .search-input:focus{outline:none;border-color:var(--cyan)}
189 .search-input:placeholder{color:var(--text3)}
190
191
192 .timeline{padding:20px;display:flex;flex-direction:column;gap:0}

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166     .tl-item{display:flex;gap:14px;padding-bottom:20px;position:relative}
167     .tl-item:last-child{padding-bottom:0}
168     .tl-item:not(:last-
child)::after{content:'';position:absolute;left:14px;top:28px;bottom:0;width:1px;background:var(--border)}
169     .tl-dot{width:28px;height:28px;border-radius:50%;border:2px solid var(--border2);background:var(--
bg3);display:flex;align-items:center;justify-content:center;flex-shrink:0;font-size:11px;z-...
170     .tl-dot.cyan{border-color:var(--cyan);background:var(--cyan-dim)}
171     .tl-dot.amber{border-color:var(--amber);background:var(--amber-dim)}
172     .tl-dot.green{border-color:var(--green);background:var(--green-dim)}
173     .tl-dot.red{border-color:var(--red);background:var(--red-dim)}
174     .tl-content{flex:1;padding-top:4px}
175     .tl-title{font-size:12px;font-weight:600;color:var(--text);margin-bottom:2px}
176     .tl-meta{font-size:10px;color:var(--text3);font-family:var(--mono)}
177
178     /* Outcome hint box */
179     .outcome-hint{margin:12px 0 0;padding:10px 14px;border-radius:6px;font-size:11px;font-family:var(--
mono);transition:all 0.2s}
180     .outcome-hint.success{background:var(--green-dim);color:var(--green);border:1px solid
rgba(16,212,142,0.2)}
181     .outcome-hint.danger{background:var(--red-dim);color:var(--red);border:1px solid
rgba(240,84,84,0.2)}
182
183     /* Inline repair panel */
184     .repair-inline{background:var(--bg3);border:1px solid var(--border2);border-radius:8px;padding:18px
20px;margin:0 16px 16px;animation:viewIn 0.2s ease}
185
186     .modal-overlay{display:none;position:fixed;inset:0;background:rgba(8,11,16,0.85);z-
index:1500;align-items:center;justify-content:center;backdrop-filter:blur(4px)}
187     .modal-overlay.open{display:flex}
188     .modal{background:var(--panel2);border:1px solid var(--border2);border-radius:12px;width:560px;max-
width:95%;max-height:85vh;overflow-y:auto;animation:modalIn 0.2s ease}
189     @keyframes modalIn{from{opacity:0;transform:scale(0.96)}to{opacity:1;transform:scale(1)}}
190     .modal-head{padding:20px 24px;border-bottom:1px solid var(--border);display:flex;justify-
content:space-between;align-items:center}
191     .modal-title{font-size:16px;font-weight:700;color:var(--text)}
192     .modal-close{background:none;border:none;color:var(--text3);font-size:20px;cursor:pointer;padding:0
4px;transition:color 0.2s}
193     .modal-close:hover{color:var(--text)}
194     .modal-body{padding:24px}
195     .detail-grid{display:grid;grid-template-columns:1fr 1fr;gap:16px;margin-bottom:20px}
196     .detail-item{display:flex;flex-direction:column;gap:4px}
197     .detail-label{font-size:9px;font-weight:700;color:var(--text3);text-transform:uppercase;letter-
spacing:1.5px;font-family:var(--mono)}
198     .detail-value{font-size:13px;color:var(--text);font-family:var(--mono)}
199 </style>
200 </head>
201 <body>
202 <div class="app">
203
204     <!-- SIDEBAR -->
205     <div class="sidebar">
206         <div class="logo">
207             <div class="logo-mark">
208                 <div class="logo-hex">TV</div>
209                 <div>
210                     <div class="logo-name">TechVendor</div>
211                     <div class="logo-sub">Asset Hub</div>
212                 </div>
213             </div>
214             <div class="vendor-badge"><span class="badge-dot"></span>Tech Vendor</div>
215         </div>
216
217         <nav class="nav">
218             <div class="nav-section">Overview</div>
219             <div class="nav-item active" onclick="showView('dashboard',this)">
220                 <svg class="nav-icon" fill="none" viewBox="0 0 16 16"><rect x="1" y="1" width="6"
height="6" rx="1" stroke="currentColor" stroke-width="1.5"/><rect x="9" y="1" width="6" height="6" ...
221                 Dashboard
222             </div>
223
224             <div class="nav-section">Devices</div>
225             <div class="nav-item" onclick="showView('my-devices',this)">
226                 <svg class="nav-icon" fill="none" viewBox="0 0 16 16"><rect x="1" y="3" width="14"
height="10" rx="2" stroke="currentColor" stroke-width="1.5"/><path d="M5 13v2M11 13v2M3 15h10" str...
227                 My Devices
228             </div>

```

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229         <div class="nav-item" onclick="showView('add-device',this)">
230             <svg class="nav-icon" fill="none" viewBox="0 0 16 16"><circle cx="8" cy="8" r="7"
stroke="currentColor" stroke-width="1.5"/><path d="M8 5v6M5 8h6" stroke="currentColor" stroke-width...
231             Add Device
232         </div>
233
234         <div class="nav-section">Service</div>
235         <div class="nav-item" onclick="showView('service-requests',this)">
236             <svg class="nav-icon" fill="none" viewBox="0 0 16 16"><path d="M8 11l1.5 4.5H14l-3.5 2.5 1.5 1.5
4.5L8 10l-4 2.5 1.5-4.5L2 5.5h4.5L8 1z" stroke="currentColor" stroke-width="1.5" stroke-l...
237             Service Requests
238             <span class="nav-badge amber" id="sr-badge" style="display:none">0</span>
239         </div>
240
241         <div class="nav-section">Assignments</div>
242         <div class="nav-item" onclick="showView('assignments',this)">
243             <svg class="nav-icon" fill="none" viewBox="0 0 16 16"><circle cx="6" cy="5" r="3"
stroke="currentColor" stroke-width="1.5"/><path d="M1 14c0-3 2-5 5-5" stroke="currentColor" stroke-...
244             Assignments
245         </div>
246
247         <div class="nav-section">Account</div>
248         <div class="nav-item" onclick="showView('profile',this)">
249             <svg class="nav-icon" fill="none" viewBox="0 0 16 16"><circle cx="8" cy="5" r="3"
stroke="currentColor" stroke-width="1.5"/><path d="M2 14c0-3.3 2.7-6 6-6s6 2.7 6 6" stroke="current...
250             My Profile
251         </div>
252     </nav>
253
254     <div class="sidebar-footer">
255         <div class="user-row">
256             <div class="avatar" id="sidebar-avatar">TV</div>
257             <div>
258                 <div class="user-name" id="sidebar-name">TechVendor</div>
259                 <div class="user-role">Tech Vendor</div>
260             </div>
261         </div>
262     </div>
263 </div>
264
265 <!-- MAIN -->
266 <div class="main">
267     <div class="topbar">
268         <div class="page-title">
269             <span class="page-title-prefix">// </span>
270             <span id="pageTitle">dashboard</span>
271         </div>
272         <div class="topbar-right">
273             <span class="topbar-email" id="topbar-email"></span>
274             <button class="signout-btn" onclick="logout()">Sign Out</button>
275         </div>
276     </div>
277
278     <div class="content">
279
280         <!-- DASHBOARD -->
281         <div id="dashboard-view" class="view active">
282             <div class="stats-row">
283                 <div class="stat c">
284                     <div class="stat-label">Total Devices</div>
285                     <div class="stat-val" id="stat-total">--</div>
286                     <div class="stat-sub">in catalog</div>
287                     <div class="stat-icon"></div>
288                 </div>
289                 <div class="stat g">
290                     <div class="stat-label">Available</div>
291                     <div class="stat-val" id="stat-avail">--</div>
292                     <div class="stat-sub">ready to assign</div>
293                     <div class="stat-icon"></div>
294                 </div>
295                 <div class="stat a">
296                     <div class="stat-label">Pending Repairs</div>
297                     <div class="stat-val" id="stat-repair">--</div>
298                     <div class="stat-sub">awaiting your update</div>
299                     <div class="stat-icon"></div>
300             </div>

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301         <div class="stat r">
302             <div class="stat-label">Open Requests</div>
303             <div class="stat-val" id="stat-open">--</div>
304             <div class="stat-sub">sent for repair</div>
305             <div class="stat-icon">!</div>
306         </div>
307     </div>
308
309     <div class="dash-grid">
310         <div class="card">
311             <div class="card-head">
312                 <div>
313                     <div class="card-title">Repair Queue</div>
314                     <div class="card-sub">devices sent to you for repair</div>
315                 </div>
316                 <button class="btn btn-ghost btn-sm" onclick="showView('service-
requests',null)">View all </button>
317             </div>
318             <div class="tablewrap">
319                 <table>
320                     <thead><tr><th>Device</th><th>Issue</th><th>Status</th></tr></thead>
321                     <tbody id="dash-sr"></tbody>
322                 </table>
323             </div>
324         </div>
325
326         <div class="card">
327             <div class="card-head">
328                 <div>
329                     <div class="card-title">Active Repairs</div>
330                     <div class="card-sub">currently being serviced</div>
331                 </div>
332                 <button class="btn btn-ghost btn-sm" onclick="showView('repair-
logs',null)">View all </button>
333             </div>
334             <div class="tablewrap">
335                 <table>
336                     <thead><tr><th>Device</th><th>Sent</th><th>Status</th></tr></thead>
337                     <tbody id="dash-repairs"></tbody>
338                 </table>
339             </div>
340         </div>
341     </div>
342
343     <div class="card">
344         <div class="card-head">
345             <div class="card-title">Device Status Overview</div>
346         </div>
347         <div class="tablewrap">
348             <table>
349                 <thead><tr><th>Device</th><th>Type</th><th>Status</th><th>Assigned
To</th></tr></thead>
350                 <tbody id="dash-devices"></tbody>
351             </table>
352         </div>
353     </div>
354 </div>
355
356 <!-- MY DEVICES -->
357 <div id="my-devices-view" class="view">
358     <div class="card">
359         <div class="card-head">
360             <div>
361                 <div class="card-title">My Device Catalog</div>
362                 <div class="card-sub">all devices supplied by you</div>
363             </div>
364             <div class="action-row">
365                 <select class="form-input btn-sm" id="device-filter" onchange="filterDevices()"
style="width:140px;padding:5px 10px;font-size:11px">
366                     <option value="">All Status</option>
367                     <option value="AVAILABLE">Available</option>
368                     <option value="ASSIGNED">Assigned</option>
369                     <option value="UNDER_REPAIR">Under Repair</option>
370                     <option value="CONDEMNED">Condemned</option>
371                     <option value="RETURNED_TO_VENDOR">Returned</option>
372                 </select>

```



```

373         <button class="btn btn-ghost btn-sm" onclick="loadMyDevices()">
Refresh</button>
374     </div>
375 </div>
376 <div class="search-row">
377     <input class="search-input" id="device-search" placeholder="Search by name, serial,
brand..." oninput="filterDevices()">
378 </div>
379 <div class="tablewrap">
380     <table>
381         <thead><tr><th>ID</th><th>Device
Name</th><th>Brand</th><th>Type</th><th>Warranty</th><th>Status</th><th>Actions</th></tr></thead>
382         <tbody id="devices-list"></tbody>
383     </table>
384 </div>
385 </div>
386 </div>
387
388 <!-- ADD DEVICE -->
389 <div id="add-device-view" class="view">
390     <div class="card" style="max-width:700px">
391         <div class="card-head">
392             <div>
393                 <div class="card-title">Add New Device</div>
394                 <div class="card-sub">register a device to your catalog</div>
395             </div>
396         </div>
397         <div class="card-body">
398             <div class="form-grid">
399                 <div class="form-group">
400                     <label class="form-label">Device Name</label>
401                     <input class="form-input" id="d-name" placeholder="Dell XPS 15 9520">
402                 </div>
403                 <div class="form-group">
404                     <label class="form-label">Brand</label>
405                     <input class="form-input" id="d-brand" placeholder="Dell">
406                 </div>
407                 <div class="form-group">
408                     <label class="form-label">Device Type</label>
409                     <select class="form-input" id="d-type">
410                         <option value="LAPTOP">Laptop</option>
411                         <option value="DESKTOP">Desktop</option>
412                         <option value="PHONE">Phone</option>
413                         <option value="TABLET">Tablet</option>
414                         <option value="MONITOR">Monitor</option>
415                         <option value="KEYBOARD">Keyboard</option>
416                         <option value="MOUSE">Mouse</option>
417                         <option value="HEADSET">Headset</option>
418                         <option value="OTHER">Other</option>
419                     </select>
420                 </div>
421                 <div class="form-group">
422                     <label class="form-label">Warranty Expiry</label>
423                     <input class="form-input" type="date" id="d-warranty">
424                 </div>
425             </div>
426             <div class="form-actions">
427                 <button class="btn btn-primary" onclick="addDevice()">Register Device</button>
428                 <button class="btn btn-ghost" onclick="clearDeviceForm()">Clear</button>
429             </div>
430         </div>
431     </div>
432 </div>
433
434 <!--
435     SERVICE REQUESTS - vendor repair queue
436     GET /service-requests/vendor/me     SENT_FOR_REPAIR list
437     PUT /service-requests/vendor       RepairDTO { requestId, resolution, status }
438     status=REPAIR_DONE   device becomes ASSIGNED
439     status=anything else device becomes CONDEMNED
440 -->
441 <div id="service-requests-view" class="view">
442     <div class="card">
443         <div class="card-head">
444             <div>
445                 <div class="card-title">Repair Queue</div>

```

```

446         <div class="card-sub">devices sent to you – mark as repaired or condemned</div>
447     </div>
448     <button class="btn btn-ghost btn-sm" onclick="loadServiceRequests()">
Refresh</button>
449 </div>
450 <div class="tablewrap">
451     <table>
452         <thead>
453             <tr>
454                 <th>ID</th>
455                 <th>Device</th>
456                 <th>Issue Type</th>
457                 <th>Description</th>
458                 <th>Urgent</th>
459                 <th>Raised</th>
460                 <th>Actions</th>
461             </tr>
462         </thead>
463         <tbody id="sr-list"></tbody>
464     </table>
465 </div>
466 </div>
467
468 <!-- INLINE REPAIR UPDATE PANEL -->
469 <div class="card" id="vendor-repair-panel" style="max-width:640px;display:none">
470     <div class="card-head">
471         <div>
472             <div class="card-title">Update Repair Result</div>
473             <div class="card-sub" id="vr-panel-subtitle">Request #</div>
474         </div>
475         <button class="btn btn-ghost btn-sm" onclick="closeRepairPanel()"> Cancel</button>
476     </div>
477     <div class="card-body">
478         <input type="hidden" id="vr-request-id">
479
480         <div class="form-grid">
481             <!-- Outcome selector -->
482             <div class="form-group full">
483                 <label class="form-label">Repair Outcome</label>
484                 <select class="form-input" id="vr-status" onchange="onOutcomeChange()">
485                     <option value="REPAIR_DONE"> Repair Done – device returns to
employee</option>
486                     <option value="NOT_FIXABLE"> Could Not Repair – device will be
condemned</option>
487                 </select>
488             </div>
489
490             <!-- Resolution notes -->
491             <div class="form-group full">
492                 <label class="form-label">Resolution Notes</label>
493                 <input class="form-input" id="vr-resolution"
494                     placeholder="Describe what was repaired, parts replaced, or why it
could not be fixed...">
495             </div>
496         </div>
497
498         <!-- Dynamic outcome hint -->
499         <div id="vr-outcome-hint" class="outcome-hint success">
500             Device status <strong>ASSIGNED</strong> – returned to the employee
501         </div>
502
503         <div class="form-actions">
504             <button class="btn btn-primary" onclick="submitRepairUpdate()">Submit
Update</button>
505             <button class="btn btn-ghost" onclick="closeRepairPanel()">Cancel</button>
506         </div>
507     </div>
508 </div>
509 </div>
510
511 <!-- ASSIGNMENTS -->
512 <div id="assignments-view" class="view">
513     <div class="card">
514         <div class="card-head">
515             <div>
516                 <div class="card-title">Device Assignments</div>

```

```

517         <div class="card-sub">who has your devices</div>
518     </div>
519     <button class="btn btn-ghost btn-sm" onclick="loadAssignments()"> Refresh</button>
520 </div>
521 <div class="tablewrap">
522     <table>
523         <thead>
524             <tr>
525                 <th>Device</th>
526                 <th>Type</th>
527                 <th>Employee</th>
528                 <th>Assigned Date</th>
529                 <th>Status</th>
530             </tr>
531         </thead>
532         <tbody id="assignments-list"></tbody>
533     </table>
534 </div>
535 </div>
536 </div>
537
538 <!-- PROFILE -->
539 <div id="profile-view" class="view">
540     <div class="card" style="max-width:600px">
541         <div class="card-head">
542             <div class="card-title">My Profile</div>
543         </div>
544         <div class="card-body">
545             <div style="display:flex;align-items:center;gap:16px;margin-bottom:24px;padding-
bottom:20px;border-bottom:1px solid var(--border)">
546                 <div style="width:56px;height:56px;background:var(--cyan-dim);border:2px solid
var(--border2);border-radius:10px;display:flex;align-items:center;justify-content:center;f...
547             </div>
548                 <div style="font-size:18px;font-weight:700;color:var(--text)" id="profile-
name">--</div>
549                 <div style="font-size:11px;color:var(--cyan);font-family:var(--
mono);margin-top:3px">TECH_VENDOR</div>
550             </div>
551         </div>
552         <div class="detail-grid">
553             <div class="detail-item"><div class="detail-label">Email</div><div
class="detail-value" id="profile-email">--</div></div>
554             <div class="detail-item"><div class="detail-label">Phone</div><div
class="detail-value" id="profile-phone">--</div></div>
555             <div class="detail-item"><div class="detail-label">Registered</div><div
class="detail-value" id="profile-reg">--</div></div>
556             <div class="detail-item"><div class="detail-label">Status</div><div
class="detail-value"><span class="chip active_assign">ACTIVE</span></div></div>
557         </div>
558     </div>
559 </div>
560 </div>
561
562 </div>
563 </div>
564 </div>
565
566 <!-- DEVICE DETAIL MODAL -->
567 <div class="modal-overlay" id="device-modal">
568     <div class="modal">
569         <div class="modal-head">
570             <div class="modal-title" id="modal-device-name">Device Details</div>
571             <button class="modal-close" onclick="closeModal()"></button>
572         </div>
573         <div class="modal-body">
574             <div class="detail-grid" id="modal-detail-grid"></div>
575             <div style="margin-top:16px">
576                 <div class="card-title" style="margin-bottom:12px;font-size:12px;color:var(--text3);font-
family:var(--mono);text-transform:uppercase;letter-spacing:1px">Repair History</div>
577                 <div id="modal-repair-history"></div>
578             </div>
579         </div>
580     </div>
581 </div>
582
583 <script>

```

```

584     const API = 'http://localhost:8080';
585     let devicesCache = [], srCache = [], repairsCache = [], assignmentsCache = [];
586
587     // AUTH
588     (function(){
589         const token = localStorage.getItem('jwt_token');
590         if(!token){ window.location.href='/login.html'; return; }
591         try{
592             const p = JSON.parse(atob(token.split('.')[1]));
593             const role = (p.role||'').replace(/^ROLE_/, '');
594             if(role !== 'TECH_VENDOR'){ window.location.href='/google.html'; return; }
595             const email = p.sub||p.email||'TechVendor';
596             document.getElementById('topbar-email').textContent = email;
597             document.getElementById('sidebar-name').textContent = email.split('@')[0];
598             document.getElementById('sidebar-avatar').textContent = email.charAt(0).toUpperCase();
599             document.getElementById('profile-avatar').textContent = email.charAt(0).toUpperCase();
600             document.getElementById('profile-email').textContent = email;
601         }catch(e){ window.location.href='/login.html'; }
602         initDashboard();
603     })();
604
605     function getAuthHeader(){
606         const t = localStorage.getItem('jwt_token');
607         return t ? {'Authorization': `Bearer ${t}`} : {};
608     }
609
610     async function apiFetch(path, opts={}){
611         const res = await fetch(`${API}${path}`, {
612             ...opts,
613             headers: {'Content-Type': 'application/json', ...getAuthHeader(), ...(opts.headers||{})},
614             credentials: 'include'
615         });
616         if(res.status===401){
617             localStorage.removeItem('jwt_token');
618             window.location.href='/login.html';
619             throw new Error('Unauthorized');
620         }
621         const text = await res.text();
622         if(!res.ok) throw new Error(text || 'Request failed');
623         return text ? JSON.parse(text) : null;
624     }
625
626     // NAV
627     function showView(name, el){
628         document.querySelectorAll('.view').forEach(v=>v.classList.remove('active'));
629         document.getElementById(`${name}-view`).classList.add('active');
630         document.querySelectorAll('.nav-item').forEach(n=>n.classList.remove('active'));
631         if(el) el.classList.add('active');
632         document.getElementById('pageTitle').textContent = name.replace(/-/g, ' ');
633         const loaders = {
634             'dashboard': loadDashboard,
635             'my-devices': loadMyDevices,
636             'service-requests': loadServiceRequests,
637             'repair-logs': loadRepairLogs,
638             'assignments': loadAssignments,
639             'profile': loadProfile,
640             'add-device': ()=>{}
641         };
642         if(loaders[name]) loaders[name]();
643     }
644
645     // INIT
646     async function initDashboard(){
647         await Promise.all([loadMyDevices(), loadServiceRequests(), loadRepairLogs(), loadAssignments()]);
648         loadDashboard();
649     }
650
651     // DASHBOARD
652     function loadDashboard(){
653         document.getElementById('stat-total').textContent = devicesCache.length || 0;
654         document.getElementById('stat-avail').textContent =
655         devicesCache.filter(d=>d.deviceStatus==='AVAILABLE').length || 0;
656         document.getElementById('stat-repair').textContent = srCache.length || 0;
657         document.getElementById('stat-open').textContent = srCache.length || 0;
658
659         const badge = document.getElementById('sr-badge');

```

```

659     if(srCache.length){ badge.textContent=srCache.length; badge.style.display=''; }
660     else badge.style.display='none';
661
662     document.getElementById('dash-sr').innerHTML = srCache.slice(0,5).map(s=>`
663         <tr>
664             <td class="main-cell">${s.deviceName||'Device #' +s.deviceId}</td>
665             <td>${(s.requestType||'').toLowerCase().replace('_', ' ')}</td>
666             <td>${chip('SENT_FOR_REPAIR')}</td>
667         </tr>`).join('') ||
668         `<tr><td colspan="3" class="empty">No pending repairs</td></tr>`;
669
670     document.getElementById('dash-repairs').innerHTML =
repairsCache.filter(r=>r.repairStatus!=='FIXED'&&r.repairStatus!=='IRREPARABLE').slice(0,5).map(r=>`
671         <tr>
672             <td class="main-cell">${r.deviceName||'Device #' +r.deviceId}</td>
673             <td>${r.sentDate||'-'}</td>
674             <td>${chip(r.repairStatus)}</td>
675         </tr>`).join('') ||
676         `<tr><td colspan="3" class="empty">No active repairs</td></tr>`;
677
678     document.getElementById('dash-devices').innerHTML = devicesCache.slice(0,8).map(d=>`
679         <tr>
680             <td class="main-cell">${d.deviceName||'-'}</td>
681             <td>${chip(d.deviceType)}</td>
682             <td>${chip(d.deviceStatus)}</td>
683             <td>${d.assignedEmployeeName||'-'}</td>
684         </tr>`).join('') ||
685         `<tr><td colspan="4" class="empty">No devices</td></tr>`;
686     }
687
688     // DEVICES
689     async function loadMyDevices(){
690         try{
691             const data = await apiFetch('/devices/vendor/me');
692             devicesCache = data || [];
693             renderDevices(devicesCache);
694         }catch(e){ renderDevices([]); }
695     }
696
697     function renderDevices(list){
698         document.getElementById('devices-list').innerHTML = list.length
699         ? list.map(d=>`<tr>
700             <td style="font-family:var(--mono);font-size:11px;color:var(--text3)">#${d.id}</td>
701             <td class="main-cell">${d.deviceName||'-'}</td>
702             <td>${d.brand||'-'}</td>
703             <td>${chip(d.deviceType)}</td>
704             <td style="font-size:11px;font-family:var(--mono)">
705                 ${d.warrantyExpiryDate
706                 ? (new Date(d.warrantyExpiryDate) > new Date()
707                 ? `<span style="color:var(--green)"> ${d.warrantyExpiryDate}</span>`
708                 : `<span style="color:var(--red)"> ${d.warrantyExpiryDate}</span>`)
709                 : `<span style="color:var(--text3)">—</span>`}
710             </td>
711             <td>${chip(d.deviceStatus)}</td>
712             <td><div class="action-row">
713                 <button class="btn btn-ghost btn-xs" onclick="viewDevice(${d.id})">Details</button>
714             </div></td>
715         </tr>`).join('')
716         : `<tr><td colspan="7"><div class="empty"><div class="empty-icon"></div>No devices registered
yet</div></tr>`;
717     }
718
719     function filterDevices(){
720         const q = document.getElementById('device-search').value.toLowerCase();
721         const f = document.getElementById('device-filter').value;
722         renderDevices(devicesCache.filter(d=>{
723             const matchQ = !q ||
(d.deviceName||'').toLowerCase().includes(q)||d.serialNumber||'').toLowerCase().includes(q)||d.brand||'').toLo
werCase().includes(q);
724             const matchF = !f || d.deviceStatus === f;
725             return matchQ && matchF;
726         }));
727     }
728
729     async function addDevice(){
730         const body = {

```

```

731     deviceName: document.getElementById('d-name').value.trim(),
732     brand: document.getElementById('d-brand').value.trim(),
733     deviceType: document.getElementById('d-type').value,
734     warrantyExpiryDate: document.getElementById('d-warranty').value || null,
735 };
736 if(!body.deviceName || !body.brand || !body.deviceType){
737     showAlert('Device name, brand and type are required','error'); return;
738 }
739 try{
740     await apiFetch('/devices', {method:'POST', body:JSON.stringify(body)});
741     showAlert('Device registered successfully!','success');
742     clearDeviceForm();
743     loadMyDevices();
744 }catch(e){ showAlert('Failed: '+e.message,'error'); }
745 }
746
747 function clearDeviceForm(){
748     ['d-name','d-brand','d-warranty'].forEach(id=>document.getElementById(id).value='');
749     document.getElementById('d-type').value='LAPTOP';
750 }
751
752 // SERVICE REQUESTS (vendor repair queue)
753 // GET /service-requests/vendor/me list of SENT_FOR_REPAIR requests
754 // PUT /service-requests/vendor { requestId, resolution, status }
755 // REPAIR_DONE device becomes ASSIGNED (back to employee)
756 // anything else device becomes CONDEMNED
757 //
758 async function loadServiceRequests(){
759     try{
760         const data = await apiFetch('/service-requests/vendor/me');
761         srCache = data || [];
762         renderServiceRequests(srCache);
763         // refresh badge
764         const badge = document.getElementById('sr-badge');
765         if(srCache.length){ badge.textContent=srCache.length; badge.style.display=''; }
766         else badge.style.display='none';
767     }catch(e){ renderServiceRequests([]); }
768 }
769
770 function renderServiceRequests(list){
771     document.getElementById('sr-list').innerHTML = list.length
772     ? list.map(s=><tr>
773         <td style="font-family:var(--mono);font-size:11px;color:var(--text3)">#${s.id}</td>
774         <td class="main-cell">${s.deviceName}||'Device #'+s.deviceId</td>
775         <td>${(s.requestType||'').toLowerCase().replace(/_/g,' ')}</td>
776         <td style="max-width:200px;overflow:hidden;text-overflow:ellipsis;white-
space:nowrap;color:var(--text2)">${s.issueDescription}||'-'</td>
777         <td>${s.urgent
778             ? '<span style="color:var(--red);font-family:var(--mono);font-size:10px;font-weight:700">
URGENT</span>'
779             : '<span style="color:var(--text3);font-family:var(--mono);font-size:10px">--</span>'}</td>
780         <td style="font-size:10px;font-family:var(--mono);color:var(--text3)">${s.raisedAt} ?
s.raisedAt.split('T')[0] : '-'</td>
781         <td>
782             <button class="btn btn-success btn-xs" onclick="openRepairPanel(${s.id},
'${(s.deviceName}||'Device #'+s.deviceId).replace(/'/g,'\\'')}'">
783                 Update Repair
784             </button>
785         </td>
786     </tr>` ).join('')
787     : `<tr><td colspan="7"><div class="empty"><div class="empty-icon"></div>No devices in repair
queue</div></td></tr>`;
788 }
789
790 // REPAIR PANEL
791 function openRepairPanel(requestId, deviceName){
792     document.getElementById('vr-request-id').value = requestId;
793     document.getElementById('vr-panel-subtitle').textContent = `Request #${requestId} - ${deviceName}`;
794     document.getElementById('vr-status').value = 'REPAIR_DONE';
795     document.getElementById('vr-resolution').value = '';
796     onOutcomeChange(); // set correct hint
797     const panel = document.getElementById('vendor-repair-panel');
798     panel.style.display = 'block';
799     panel.scrollIntoView({behavior:'smooth', block:'start'});
800 }
801

```

```

802     function closeRepairPanel(){
803         document.getElementById('vendor-repair-panel').style.display = 'none';
804     }
805
806     function onOutcomeChange(){
807         const status = document.getElementById('vr-status').value;
808         const hint = document.getElementById('vr-outcome-hint');
809         if(status === 'REPAIR_DONE'){
810             hint.className = 'outcome-hint success';
811             hint.innerHTML = ' Device status <strong>ASSIGNED</strong> – returned to the employee';
812         } else {
813             hint.className = 'outcome-hint danger';
814             hint.innerHTML = ' Device status <strong>CONDEMNED</strong> – device will be marked unusable';
815         }
816     }
817
818     async function submitRepairUpdate(){
819         const requestId = document.getElementById('vr-request-id').value;
820         const status = document.getElementById('vr-status').value;
821         const resolution = document.getElementById('vr-resolution').value.trim();
822
823         if(!resolution){
824             showAlert('Please enter resolution notes','error'); return;
825         }
826
827         // RepairDTO: { requestId, resolution, status }
828         const body = {
829             requestId: parseInt(requestId),
830             resolution,
831             status // "REPAIR_DONE" ASSIGNED | anything else CONDEMNED
832         };
833
834         try{
835             await apiFetch('/service-requests/vendor', {method:'PUT', body:JSON.stringify(body)});
836             const label = status === 'REPAIR_DONE' ? 'Repair marked done – device returned to employee' :
837             'Device condemned';
838             showAlert(label, 'success');
839             closeRepairPanel();
840             // Refresh everything that could have changed
841             await Promise.all([loadServiceRequests(), loadMyDevices(), loadRepairLogs()]);
842             loadDashboard();
843         }catch(e){ showAlert('Failed: '+e.message,'error'); }
844
845         // REPAIR LOGS
846         async function loadRepairLogs(){
847             try{
848                 const data = await apiFetch('/tech-vendor/repair-logs');
849                 repairsCache = data || [];
850                 renderRepairLogs(repairsCache);
851             }catch(e){ renderRepairLogs([]); }
852         }
853
854         function renderRepairLogs(list){
855             document.getElementById('repair-list').innerHTML = list.length
856             ? list.map(r=><tr>
857                 <td style="font-family:var(--mono);font-size:11px;color:var(--text3)">#${r.id}</td>
858                 <td class="main-cell">${r.deviceName}||Device #'+r.deviceId}</td>
859                 <td style="max-width:140px;overflow:hidden;text-overflow:ellipsis;white-
860 space:nowrap">${r.issueReported}||'-'}</td>
861                 <td>${r.technicianName}||'-'}</td>
862                 <td style="font-size:11px;font-family:var(--mono)">${r.sentDate}||'-'}</td>
863                 <td style="font-size:11px;font-family:var(--mono)">${r.expectedReturnDate}||'-'}</td>
864                 <td style="font-family:var(--mono)">${r.repairCost ? ''+r.repairCost : '-'}</td>
865                 <td><chip(r.repairStatus)</td>
866                 <td><button class="btn btn-ghost btn-xs"
867 onclick="openRepairUpdate(${r.id})">Update</button></td>
868             </tr>` ).join('')
869             : `<tr><td colspan="9"><div class="empty"><div class="empty-icon"></div>No repair
870 logs</div></td></tr>`;
871         }
872
873         function openRepairUpdate(id){
874             document.getElementById('repair-id').value = id;
875             const form = document.getElementById('repair-update-form');
876             form.style.display = 'block';

```

```

874     form.scrollIntoView({behavior: 'smooth'});
875 }
876
877 async function updateRepair(){
878     const id = document.getElementById('repair-id').value;
879     const body = {
880         repairStatus: document.getElementById('repair-status-input').value,
881         technicianName: document.getElementById('repair-tech').value.trim(),
882         repairCost: parseFloat(document.getElementById('repair-cost').value)||null,
883         invoiceNumber: document.getElementById('repair-invoice').value.trim(),
884         actualReturnDate: document.getElementById('repair-return-date').value||null,
885         fixedSuccessfully: document.getElementById('repair-fixed').value==='true',
886         repairNotes: document.getElementById('repair-notes').value.trim(),
887         postRepairAction: document.getElementById('repair-action').value
888     };
889     try{
890         await apiFetch(`/tech-vendor/repair-logs/${id}`, {method: 'PUT', body: JSON.stringify(body)});
891         showAlert('Repair log updated!', 'success');
892         document.getElementById('repair-update-form').style.display='none';
893         loadRepairLogs(); loadMyDevices(); loadDashboard();
894     }catch(e){ showAlert('Failed: '+e.message, 'error'); }
895 }
896
897 // ASSIGNMENTS
898 async function loadAssignments(){
899     try{
900         const data = await apiFetch('/devices/assignments');
901         assignmentsCache = data || [];
902         document.getElementById('assignments-list').innerHTML = assignmentsCache.length
903             ? assignmentsCache.map(a=><tr>
904                 <td class="main-cell">${a.deviceName||'-'}</td>
905                 <td>${chip(a.deviceType)}</td>
906                 <td>${a.employeeName||'-'}</td>
907                 <td style="font-size:11px;font-family:var(--mono)">${a.assignedDate||'-'}</td>
908                 <td>${chip(a.status==='ACTIVE'? 'active_assign':a.status?.toLowerCase())}</td>
909             </tr>`).join('')
910             : `<tr><td colspan="5"><div class="empty"><div class="empty-icon"></div>No assignments
found</div></td></tr>`;
911     }catch(e){}
912 }
913
914 // PROFILE
915 async function loadProfile(){
916     try{
917         const data = await apiFetch('/vendors/me');
918         document.getElementById('profile-name').textContent = data.name || '--';
919         document.getElementById('profile-email').textContent = data.email || '--';
920         document.getElementById('profile-phone').textContent = data.phone || '--';
921         document.getElementById('profile-reg').textContent = data.registeredAt || '--';
922         document.getElementById('profile-avatar').textContent = (data.name||'T').charAt(0).toUpperCase();
923         document.getElementById('sidebar-name').textContent = data.name || 'TechVendor';
924     }catch(e){}
925 }
926
927 // DEVICE DETAIL MODAL
928 async function viewDevice(id){
929     const d = devicesCache.find(x=>x.id===id);
930     if(!d) return;
931     document.getElementById('modal-device-name').textContent = d.deviceName;
932
933     const today = new Date();
934     const warrantyExpiry = d.warrantyExpiryDate ? new Date(d.warrantyExpiryDate) : null;
935     const warrantyDisplay = warrantyExpiry
936         ? (warrantyExpiry > today
937             ? `<span style="color:var(--green);font-family:var(--mono)"> Valid until
938 ${d.warrantyExpiryDate}</span>`
939             : `<span style="color:var(--red);font-family:var(--mono)"> Expired
940 ${d.warrantyExpiryDate}</span>`)
941         : `<span style="color:var(--text3);font-family:var(--mono)">—</span>`;
942
943     document.getElementById('modal-detail-grid').innerHTML = `
944         <div class="detail-item"><div class="detail-label">Brand</div><div class="detail-
value">${d.brand||'-'}</div></div>
945         <div class="detail-item"><div class="detail-label">Type</div><div class="detail-
value">${chip(d.deviceType)}</div></div>
946         <div class="detail-item"><div class="detail-label">Status</div><div class="detail-

```



```

value">${chip(d.deviceStatus)}</div></div>
945     <div class="detail-item"><div class="detail-label">Warranty</div><div class="detail-
value">${warrantyDisplay}</div></div>
946     <div class="detail-item"><div class="detail-label">Assigned To</div><div class="detail-
value">${d.assignedEmployeeName||'<span style="color:var(--text3)">Unassigned</span>'}</div></div>
947     <div class="detail-item"><div class="detail-label">Added On</div><div class="detail-
value">${d.createdAt ? d.createdAt.split('T')[0] : '-'}</div></div>
948 `;
949
950     const repairs = repairsCache.filter(r=>r.deviceId===id);
951     document.getElementById('modal-repair-history').innerHTML = repairs.length
952     ? `<div class="timeline">${repairs.map(r=>`
953         <div class="tl-item">
954             <div class="tl-dot
955 ${r.repairStatus=== 'FIXED'? 'green': r.repairStatus=== 'IRREPARABLE'? 'red': r.repairStatus=== 'IN_PROGRESS'? 'amber': '
cyan'}"></div>
956                 <div class="tl-content">
957                     <div class="tl-title">${r.issueReported||'Repair'}</div>
958                     <div class="tl-meta">${chip(r.repairStatus)} · ${r.sentDate||'-'}
959 ${r.actualReturnDate||'pending'} · ${r.repairCost ? ''+r.repairCost : 'cost pending'}</div>
960                     </div>
961                 </div>`}).join('')}</div>`
962     : `<div class="empty" style="padding:16px">No repair history</div>`;
963
964     document.getElementById('device-modal').classList.add('open');
965 }
966
967 function closeModal(){
968     document.getElementById('device-modal').classList.remove('open');
969 }
970
971 // HELPERS
972 function chip(val){
973     if(!val) return '-';
974     const cls = val.toLowerCase().replace(/ /g, '_');
975     return `<span class="chip ${cls}">${val.replace(/_/g, ' ')}</span>`;
976 }
977
978 function showAlert(msg, type){
979     const el = document.createElement('div');
980     el.className = `alert alert-${type}`;
981     el.innerHTML = `<span>${type==='success'? '' : ''}</span>${msg}`;
982     document.body.appendChild(el);
983     setTimeout(()=>el.remove(), 3500);
984 }
985
986 function logout(){
987     localStorage.removeItem('jwt_token');
988     localStorage.removeItem('auth_type');
989     fetch(`${API}/session/logout`, {method: 'POST'}).finally(()=>window.location.href='/vendor-
login.html');
990 }
991 </script>
992 </body>
993 </html>

```

```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>EMS – Two-Factor Authentication Setup</title>
7      <link
href="https://fonts.googleapis.com/css2?family=Outfit:wght@300;400;500;600;700&family=JetBrains+Mono:wght@400;50
0&display=swap" rel="stylesheet">
8      <!-- qrcode.js for QR rendering -->
9      <script src="https://cdnjs.cloudflare.com/ajax/libs/qrcodejs/1.0.0/qrcode.min.js"></script>
10     <style>
11         *{margin:0;padding:0;box-sizing:border-box}
12         :root{
13             --bg:#070910;--panel:#0e1118;--panel2:#131821;
14             --border:rgba(255,255,255,0.07);--border-lit:rgba(255,255,255,0.16);
15             --text:#dde4f0;--text2:#6b7591;--text3:#3d4560;
16             --blue:#4f8cff;--blue-dim:rgba(79,140,255,0.13);
17             --green:#34d399;--green-dim:rgba(52,211,153,0.12);
18             --red:#f87171;--red-dim:rgba(248,113,113,0.12);
19             --amber:#f9c74f;
20             --font:'Outfit',sans-serif;--mono:'JetBrains Mono',monospace;
21         }
22         body{font-family:var(--font);background:var(--bg);color:var(--text);min-height:100vh;}
23
24         .bg-layer{position:fixed;inset:0;z-index:0;
25             background:
26                 radial-gradient(ellipse 50% 50% at 20% 30%, rgba(79,140,255,0.06) 0%,transparent 70%),
27                 radial-gradient(ellipse 40% 40% at 80% 70%, rgba(52,211,153,0.04) 0%,transparent 60%);
28         }
29         .grid-overlay{position:fixed;inset:0;z-index:0;
30             background-image:linear-gradient(rgba(255,255,255,0.015) 1px,transparent 1px),linear-
gradient(90deg,rgba(255,255,255,0.015) 1px,transparent 1px);
31             background-size:40px 40px;
32         }
33
34         /* Nav */
35         .nav{position:relative;z-index:10;display:flex;align-items:center;justify-content:space-
between;padding:18px 40px;border-bottom:1px solid var(--border);}
36         .nav-brand{display:flex;align-items:center;gap:10px;}
37         .nav-icon{width:36px;height:36px;background:linear-gradient(135deg,#4f8cff,#818cf8);border-
radius:9px;display:flex;align-items:center;justify-content:center;font-weight:700;font-size:14px;c...
38         .nav-name{font-size:16px;font-weight:700;}
39         .nav-back{font-size:13px;color:var(--text2);text-decoration:none;display:flex;align-
items:center;gap:6px;transition:color .2s;}
40         .nav-back:hover{color:var(--blue);}
41
42         /* Page */
43         .page{position:relative;z-index:1;max-width:680px;margin:0 auto;padding:48px 24px 80px;}
44
45         .page-header{margin-bottom:36px;}
46         .page-header h1{font-size:28px;font-weight:700;letter-spacing:-.4px;margin-bottom:6px;}
47         .page-header p{font-size:14px;color:var(--text2);line-height:1.6;}
48
49         /* Status badge */
50         .status-bar{display:flex;align-items:center;justify-content:space-between;background:var(--
panel);border:1px solid var(--border);border-radius:14px;padding:18px 22px;margin-bottom:28px;}
51         .status-left{display:flex;align-items:center;gap:14px;}
52         .status-icon{width:44px;height:44px;border-radius:12px;display:flex;align-items:center;justify-
content:center;font-size:22px;flex-shrink:0;}
53         .status-icon.on{background:var(--green-dim);}
54         .status-icon.off{background:var(--border);}
55         .status-title{font-size:15px;font-weight:600;margin-bottom:2px;}
56         .status-desc{font-size:12px;color:var(--text2);}
57         .badge{font-size:11px;font-weight:700;padding:4px 10px;border-radius:20px;letter-spacing:.04em;text-
transform:uppercase;}
58         .badge-on{background:var(--green-dim);color:var(--green);border:1px solid rgba(52,211,153,0.25);}
59         .badge-off{background:var(--border);color:var(--text2);border:1px solid var(--border-lit);}
60
61         /* Cards */
62         .card{background:var(--panel);border:1px solid var(--border);border-radius:14px;padding:24px;margin-
bottom:20px;}

```

```

63     .card-title{font-size:15px;font-weight:600;margin-bottom:4px;display:flex;align-
items:center;gap:8px;}
64     .card-subtitle{font-size:13px;color:var(--text2);margin-bottom:20px;}
65
66     /* Steps list */
67     .steps-list{display:flex;flex-direction:column;gap:16px;margin-bottom:24px;}
68     .step-row{display:flex;gap:14px;align-items:flex-start;}
69     .step-num{width:28px;height:28px;border-radius:50%;background:var(--blue-dim);border:1px solid
rgba(79,140,255,0.3);color:var(--blue);font-size:13px;font-weight:700;display:flex;align-items...
70     .step-text strong{font-size:13px;color:var(--text);display:block;margin-bottom:2px;}
71     .step-text span{font-size:12px;color:var(--text2);line-height:1.5;}
72
73     /* QR area */
74     .qr-area{display:none;flex-direction:column;align-
items:center;gap:20px;padding:24px;background:var(--panel2);border:1px solid var(--border);border-
radius:12px;margin-bottom:20px;}
75     .qr-area.show{display:flex;}
76     #qr-container{background:#fff;padding:14px;border-radius:10px;display:inline-block;}
77     .qr-container canvas, .qr-container img{display:block;}
78     .secret-box{width:100%;background:var(--bg);border:1px solid var(--border);border-
radius:9px;padding:12px 16px;text-align:center;}
79     .secret-label{font-size:11px;color:var(--text3);text-transform:uppercase;letter-
spacing:.06em;margin-bottom:6px;}
80     .secret-val{font-family:var(--mono);font-size:14px;color:var(--amber);letter-spacing:.12em;word-
break:break-all;}
81     .copy-btn{font-size:11px;color:var(--blue);background:none;border:none;cursor:pointer;font-
family:var(--font);padding:0;margin-top:4px;}
82     .copy-btn:hover{text-decoration:underline;}
83
84     /* OTP row */
85     .otp-row{display:flex;gap:10px;justify-content:center;margin:0 0 8px;}
86     .otp-digit{width:48px;height:58px;text-align:center;font-size:24px;font-family:var(--mono);font-
weight:500;background:var(--panel2);border:1px solid var(--border);border-radius:10px;color:v...
87     .otp-digit:focus{border-color:var(--blue);box-shadow:0 0 3px var(--blue-dim);}
88     .otp-digit.filled{border-color:rgba(79,140,255,0.4);}
89
90     /* Alerts */
91     .alert{border-radius:9px;padding:11px 14px;font-size:13px;margin-bottom:14px;display:none;align-
items:flex-start;gap:9px;}
92     .alert.show{display:flex;}
93     .alert-err{background:var(--red-dim);border:1px solid rgba(248,113,113,0.25);color:var(--red);}
94     .alert-ok{background:var(--green-dim);border:1px solid rgba(52,211,153,0.25);color:var(--green);}
95     .alert-icon{font-size:15px;flex-shrink:0;}
96
97     /* Buttons */
98     .btn{padding:12px 22px;border-radius:10px;border:none;font-family:var(--font);font-size:14px;font-
weight:600;cursor:pointer;transition:all .2s;}
99     .btn-primary{background:var(--blue);color:#fff;}
100    .btn-primary:hover{background:#6ba1ff;box-shadow:0 4px 16px rgba(79,140,255,0.3);}
101    .btn-primary.disabled{opacity:.5;cursor:not-allowed;box-shadow:none;}
102    .btn-danger{background:var(--red-dim);color:var(--red);border:1px solid rgba(248,113,113,0.25);}
103    .btn-danger:hover{background:rgba(248,113,113,0.2);}
104    .btn-ghost{background:transparent;color:var(--text2);border:1px solid var(--border);}
105    .btn-ghost:hover{border-color:var(--border-lit);color:var(--text);}
106
107    .btn-row{display:flex;gap:10px;flex-wrap:wrap;}
108
109    .spinner{display:inline-block;width:14px;height:14px;border:2px solid rgba(255,255,255,0.2);border-
top-color:#fff;border-radius:50%;animation:spin .6s linear infinite;vertical-align:middle;...
110    @keyframes spin{to{transform:rotate(360deg)}}
111
112    /* Danger zone */
113    .danger-card{background:var(--red-dim);border:1px solid rgba(248,113,113,0.2);border-
radius:14px;padding:20px 24px;display:flex;align-items:center;justify-content:space-between;gap:16px;}
114    .danger-text strong{font-size:14px;color:var(--red);display:block;margin-bottom:3px;}
115    .danger-text span{font-size:12px;color:rgba(248,113,113,0.7);}
116
117    .loading{text-align:center;padding:60px 0;color:var(--text2);}
118    .loading .spinner{border-top-color:var(--blue);border-color:var(--border-
lit);width:24px;height:24px;}
119    </style>
120 </head>
121 <body>
122 <div class="bg-layer"></div>
123 <div class="grid-overlay"></div>
124

```

```

125 <nav class="nav">
126     <div class="nav-brand">
127         <div class="nav-icon">EMS</div>
128         <div class="nav-name">EmployeeMS</div>
129     </div>
130     <a class="nav-back" href="javascript:history.back()"> Back to dashboard</a>
131 </nav>
132
133 <div class="page">
134     <div class="page-header">
135         <h1>Two-Factor Authentication</h1>
136         <p>Add an extra layer of security to your account. Once enabled, you'll need to provide a 6-digit
code from your authenticator app every time you sign in.</p>
137     </div>
138
139     <!-- Loading state -->
140     <div class="loading" id="loading">
141         <span class="spinner"></span>
142     </div>
143
144     <!-- Main content (shown after status loads) -->
145     <div id="main-content" style="display:none;">
146
147         <!-- Status bar -->
148         <div class="status-bar">
149             <div class="status-left">
150                 <div class="status-icon" id="status-icon"></div>
151                 <div>
152                     <div class="status-title">Two-Factor Authentication</div>
153                     <div class="status-desc" id="status-desc">Loading...</div>
154                 </div>
155             </div>
156             <span class="badge" id="status-badge"></span>
157         </div>
158
159         <!-- ===== SETUP SECTION (shown when disabled) ===== -->
160         <div id="setup-section">
161             <div class="card">
162                 <div class="card-title"> Enable 2FA</div>
163                 <div class="card-subtitle">Follow these steps to add two-factor authentication to your
account.</div>
164
165                 <div class="steps-list">
166                     <div class="step-row">
167                         <div class="step-num">1</div>
168                         <div class="step-text">
169                             <strong>Install an authenticator app</strong>
170                             <span>Download <a href="https://support.google.com/accounts/answer/1066447"
target="_blank" style="color:var(--blue)">Google Authenticator</a> or <a href="https://authy...
171                         </div>
172                     </div>
173                     <div class="step-row">
174                         <div class="step-num">2</div>
175                         <div class="step-text">
176                             <strong>Scan the QR code</strong>
177                             <span>Click "Generate QR Code" below, then scan it with your authenticator app.
You can also enter the secret key manually.</span>
178                         </div>
179                     </div>
180                     <div class="step-row">
181                         <div class="step-num">3</div>
182                         <div class="step-text">
183                             <strong>Verify the setup</strong>
184                             <span>Enter the 6-digit code shown in your app to confirm everything is working
correctly.</span>
185                         </div>
186                     </div>
187                 </div>
188
189                 <!-- QR area -->
190                 <div class="qr-area" id="qr-area">
191                     <div id="qr-container"></div>
192                     <div class="secret-box">
193                         <div class="secret-label">Manual entry key (backup)</div>
194                         <div class="secret-val" id="secret-display"></div>
195                         <button class="copy-btn" onclick="copySecret()">Copy key</button>

```

```

196         </div>
197     </div>
198
199     <!-- Alerts -->
200     <div class="alert alert-err" id="setup-err"><span class="alert-icon"></span><span
id="setup-err-msg"></span></div>
201     <div class="alert alert-ok" id="setup-ok"><span class="alert-icon"></span><span id="setup-
ok-msg"></span></div>
202
203     <!-- Verify code input (shown after QR generated) -->
204     <div id="verify-area" style="display:none;">
205         <p style="font-size:13px;color:var(--text2);margin-bottom:16px;">Enter the code shown
in your authenticator app to activate 2FA:</p>
206         <div class="otp-row" id="setup-otp-row">
207             <input class="otp-digit" type="text" inputmode="numeric" maxlength="1" id="s0">
208             <input class="otp-digit" type="text" inputmode="numeric" maxlength="1" id="s1">
209             <input class="otp-digit" type="text" inputmode="numeric" maxlength="1" id="s2">
210             <input class="otp-digit" type="text" inputmode="numeric" maxlength="1" id="s3">
211             <input class="otp-digit" type="text" inputmode="numeric" maxlength="1" id="s4">
212             <input class="otp-digit" type="text" inputmode="numeric" maxlength="1" id="s5">
213         </div>
214     </div>
215
216     <div class="btn-row" style="margin-top:8px;">
217         <button class="btn btn-ghost" id="btn-generate" onclick="generateQr()"> Generate QR
Code</button>
218         <button class="btn btn-primary" id="btn-verify" style="display:none;" disabled
onclick="verifySetup()">Activate 2FA</button>
219     </div>
220 </div>
221 </div>
222
223     <!-- ===== ACTIVE SECTION (shown when enabled) ===== -->
224     <div id="active-section" style="display:none;">
225         <div class="card">
226             <div class="card-title"> 2FA is active</div>
227             <div class="card-subtitle">Your account is protected. Every sign-in requires your
authenticator app code.</div>
228             <p style="font-size:13px;color:var(--text2);line-height:1.6;">
229                 If you lose access to your authenticator app, contact your admin to reset 2FA. We
recommend also saving backup codes.
230             </p>
231         </div>
232
233         <div class="danger-card">
234             <div class="danger-text">
235                 <strong>Disable Two-Factor Authentication</strong>
236                 <span>This will remove the extra protection from your account.</span>
237             </div>
238             <button class="btn btn-danger" onclick="disableTotp()">Disable 2FA</button>
239         </div>
240     </div>
241
242 </div>
243 </div>
244
245 <script>
246     const token = () => localStorage.getItem('jwt_token');
247     const authHeader = () => ({ 'Authorization': 'Bearer ' + token(), 'Content-Type': 'application/json'
});
248
249     // ---- Load status on page open ----
250     async function loadStatus() {
251         try {
252             const res = await fetch('/totp/status', { headers: authHeader() });
253             if (res.status === 401) { window.location.href = '/google.html'; return; }
254             const data = await res.json();
255             renderStatus(data.totpEnabled);
256         } catch (e) {
257             console.error(e);
258         } finally {
259             document.getElementById('loading').style.display = 'none';
260             document.getElementById('main-content').style.display = 'block';
261         }
262     }
263

```

```

264     function renderStatus(enabled) {
265         const icon = document.getElementById('status-icon');
266         const desc = document.getElementById('status-desc');
267         const badge = document.getElementById('status-badge');
268         const setup = document.getElementById('setup-section');
269         const active = document.getElementById('active-section');
270
271         if (enabled) {
272             icon.textContent = '';
273             icon.className = 'status-icon on';
274             desc.textContent = 'Your account requires a 6-digit code on every sign-in.';
275             badge.textContent = 'Enabled';
276             badge.className = 'badge badge-on';
277             setup.style.display = 'none';
278             active.style.display = 'block';
279         } else {
280             icon.textContent = '';
281             icon.className = 'status-icon off';
282             desc.textContent = 'Not enabled – only your password protects your account.';
283             badge.textContent = 'Disabled';
284             badge.className = 'badge badge-off';
285             setup.style.display = 'block';
286             active.style.display = 'none';
287         }
288     }
289
290     // ---- Generate QR ----
291     let currentSecret = null;
292
293     async function generateQr() {
294         const btn = document.getElementById('btn-generate');
295         btn.disabled = true;
296         btn.innerHTML = '<span class="spinner"></span>Generating...';
297         clearAlerts();
298
299         try {
300             const res = await fetch('/totp/setup', { method: 'POST', headers: authHeader() });
301             const data = await res.json();
302             if (!res.ok) { showAlert('setup-err', data.error || 'Failed to generate QR code'); return; }
303
304             currentSecret = data.secret;
305
306             // Render QR code
307             const qrContainer = document.getElementById('qr-container');
308             qrContainer.innerHTML = '';
309             new QRCode(qrContainer, {
310                 text: data.qrCodeUri,
311                 width: 180,
312                 height: 180,
313                 colorDark: '#000000',
314                 colorLight: '#ffffff',
315                 correctLevel: QRCode.CorrectLevel.M
316             });
317
318             document.getElementById('secret-display').textContent = formatSecret(data.secret);
319             document.getElementById('qr-area').classList.add('show');
320             document.getElementById('verify-area').style.display = 'block';
321             document.getElementById('btn-verify').style.display = 'inline-block';
322             btn.textContent = 'Regenerate';
323
324             // Focus first digit
325             setTimeout(() => document.getElementById('s0').focus(), 200);
326         } catch (e) {
327             showAlert('setup-err', 'Network error. Please try again.');
```

```

340     navigator.clipboard.writeText(currentSecret || '');
341     const btn = document.querySelector('.copy-btn');
342     btn.textContent = 'Copied!';
343     setTimeout(() => btn.textContent = 'Copy key', 1500);
344 }
345
346 // ---- OTP inputs for setup ----
347 const setupDigits = Array.from({ length: 6 }, (_, i) => document.getElementById('s' + i));
348 const btnVerify = document.getElementById('btn-verify');
349
350 setupDigits.forEach((d, i) => {
351     d.addEventListener('input', () => {
352         const val = d.value.replace(/\D/g, '');
353         d.value = val;
354         d.classList.toggle('filled', val.length > 0);
355         if (val && i < 5) setupDigits[i + 1].focus();
356         btnVerify.disabled = setupDigits.map(x => x.value).join('').length < 6;
357     });
358     d.addEventListener('keydown', e => {
359         if (e.key === 'Backspace' && !d.value && i > 0) {
360             setupDigits[i - 1].focus();
361             setupDigits[i - 1].value = '';
362             setupDigits[i - 1].classList.remove('filled');
363             btnVerify.disabled = true;
364         }
365         if (e.key === 'Enter' && !btnVerify.disabled) verifySetup();
366     });
367     d.addEventListener('paste', e => {
368         const pasted = (e.clipboardData || window.clipboardData).getData('text').replace(/\D/g,
369             '').slice(0, 6);
370         if (pasted.length === 6) {
371             pasted.split('').forEach((ch, idx) => { setupDigits[idx].value = ch;
372             setupDigits[idx].classList.add('filled'); });
373             setupDigits[5].focus();
374             btnVerify.disabled = false;
375             e.preventDefault();
376         }
377     });
378 });
379
380 // ---- Verify setup ----
381 async function verifySetup() {
382     clearAlerts();
383     const code = setupDigits.map(d => d.value).join('');
384     if (code.length < 6) return;
385
386     btnVerify.disabled = true;
387     btnVerify.innerHTML = '<span class="spinner"></span>Activating...';
388
389     try {
390         const res = await fetch('/totp/verify-setup', {
391             method: 'POST',
392             headers: authHeader(),
393             body: JSON.stringify({ code })
394         });
395         const data = await res.json();
396         if (!res.ok) {
397             showAlert('setup-err', data.error || 'Invalid code. Try again.');
```

398 setupDigits.forEach(d => { d.value = ''; d.classList.remove('filled'); });
399 setupDigits[0].focus();
400 btnVerify.disabled = true;
401 return;
402 }
403 showAlert('setup-ok', '2FA activated! Your account is now protected.');

404 setTimeout(() => renderStatus(true), 1200);
405 } catch (e) {
406 showAlert('setup-err', 'Network error. Please try again.');

407 } finally {
408 btnVerify.disabled = false;
409 btnVerify.textContent = 'Activate 2FA';
410 }
411 }
412
413 // ---- Disable TOTP ----
414 async function disableTotp() {
415 if (!confirm('Are you sure you want to disable two-factor authentication? This will make your

```
account less secure.')) return;
414
415     try {
416         const res = await fetch('/totp/disable', { method: 'DELETE', headers: authHeader() });
417         if (!res.ok) { alert('Failed to disable 2FA.');
```

return; }

```
418         renderStatus(false);
419     } catch (e) {
420         alert('Network error.');
```

}

```
421     }
422 }
423
424 // ---- Alerts ----
425 function showAlert(id, msg) {
426     const el = document.getElementById(id);
427     el.classList.add('show');
```

document.getElementById(id + '-msg').textContent = msg;

```
429 }
430 function clearAlerts() {
431     ['setup-err', 'setup-ok'].forEach(id => document.getElementById(id).classList.remove('show'));
```

}

```
433
434 // ---- Boot ----
435 if (!localStorage.getItem('jwt_token')) {
436     window.location.href = '/google.html';
437 } else {
438     loadStatus();
439 }
440 </script>
441 </body>
442 </html>
```



```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>EMS – Vendor Hub</title>
7      <link
href="https://fonts.googleapis.com/css2?family=DM+Sans:wght@300;400;500;600;700&family=Space+Mono:wght@400;700&
isplay=swap" rel="stylesheet">
8      <style>
9          *{margin:0;padding:0;box-sizing:border-box}
10         :root{
11             --bg:#08100a;--bg2:#0c160e;--bg3:#101c12;
12             --panel:#131f15;--panel2:#182219;
13             --border:rgba(52,211,153,0.08);--border2:rgba(52,211,153,0.18);
14             --text:#d4edd9;--text2:#6b9474;--text3:#3d5e44;
15             --accent:#34d399;--accent-dim:rgba(52,211,153,0.1);
16             --amber:#fbbf24;--amber-dim:rgba(251,191,36,0.1);
17             --red:#f87171;--red-dim:rgba(248,113,113,0.1);
18             --blue:#60a5fa;--blue-dim:rgba(96,165,250,0.1);
19             --font:'DM Sans',sans-serif;--mono:'Space Mono',monospace;
20             --sidebar:245px;
21         }
22         body{font-family:var(--font);background:var(--bg);color:var(--text);font-
size:14px;overflow:hidden;height:100vh}
23         .app{display:flex;height:100vh}
24
25         /* SIDEBAR */
26         .sidebar{width:var(--sidebar);background:var(--bg2);border-right:1px solid var(--
border);display:flex;flex-direction:column;flex-shrink:0;position:relative}
27         .sidebar::before{content:'';position:absolute;left:0;top:0;bottom:0;width:2px;background:linear-
gradient(to bottom,transparent,var(--accent),transparent)}
28         .logo{padding:22px 20px;border-bottom:1px solid var(--border)}
29         .logo-row{display:flex;align-items:center;gap:10px;margin-bottom:4px}
30         .logo-icon{width:34px;height:34px;background:linear-gradient(135deg,#16a34a,var(--accent));border-
radius:10px;display:flex;align-items:center;justify-content:center;font-weight:700;font-siz...
31         .logo h2{font-size:16px;font-weight:700;color:var(--text)}
32         .logo p{font-size:10px;color:var(--text3);font-family:var(--mono)}
33         .role-badge{display:inline-flex;align-items:center;gap:6px;padding:4px 12px;background:var(--accent-
dim);border:1px solid rgba(52,211,153,0.2);border-radius:4px;font-size:10px;font-weight:7...
34         .nav{flex:1;padding:10px 0;overflow-y:auto}
35         .nav-section{padding:12px 20px 6px;font-size:9px;font-weight:700;color:var(--text3);letter-
spacing:2px;text-transform:uppercase;font-family:var(--mono)}
36         .nav-item{display:flex;align-items:center;gap:10px;padding:10px 20px;margin:2px 8px;border-
radius:6px;cursor:pointer;transition:all 0.2s;color:var(--text2);font-size:13px;font-weight:500}
37         .nav-item:hover{background:var(--accent-dim);color:var(--accent)}
38         .nav-item.active{background:var(--accent-dim);color:var(--accent);border-left:2px solid var(--
accent);margin-left:6px}
39         .sidebar-footer{padding:16px 20px;border-top:1px solid var(--border)}
40         .user-card{display:flex;align-items:center;gap:10px}
41         .avatar{width:34px;height:34px;background:linear-gradient(135deg,#16a34a,var(--accent));border-
radius:6px;display:flex;align-items:center;justify-content:center;font-size:14px;font-weight:7...
42         .user-info-side .name{font-size:12px;font-weight:600;color:var(--text)}
43         .user-info-side .role-text{font-size:9px;color:var(--accent);font-family:var(--mono);text-
transform:uppercase;letter-spacing:1px}
44
45         /* MAIN */
46         .main{flex:1;display:flex;flex-direction:column;overflow:hidden}
47         .topbar{background:var(--bg2);padding:14px 24px;display:flex;justify-content:space-between;align-
items:center;border-bottom:1px solid var(--border);flex-shrink:0}
48         .page-title{font-size:16px;font-weight:700;color:var(--text);font-family:var(--mono)}
49         .topbar-right{display:flex;align-items:center;gap:12px}
50         .logout-btn{padding:7px 16px;background:transparent;border:1px solid var(--border2);border-
radius:4px;color:var(--text2);font-family:var(--mono);font-size:11px;cursor:pointer;transition:all...
51         .logout-btn:hover{background:var(--red-dim);border-color:var(--red);color:var(--red)}
52         .content{flex:1;overflow-y:auto;padding:24px}
53         .view{display:none}
54         .view.active{display:block;animation:fadeIn 0.25s ease}
55         @keyframes fadeIn{from{opacity:0;transform:translateY(6px)}to{opacity:1;transform:translateY(0)}}
56
57         /* STATS */
58         .stats-grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(150px,1fr));gap:14px;margin-

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bottom:24px}
59     .stat-card{background:var(--panel);border:1px solid var(--border);border-
radius:10px;padding:18px;position:relative;transition:border-color 0.2s}
60     .stat-card:hover{border-color:var(--border2)}
61     .stat-card::before{content:'';position:absolute;top:0;left:0;right:0;height:2px;border-radius:10px
10px 0 0}
62     .stat-card.green::before{background:var(--accent)}
63     .stat-card.amber::before{background:var(--amber)}
64     .stat-card.blue::before{background:var(--blue)}
65     .stat-card.red::before{background:var(--red)}
66     .stat-label{font-size:10px;color:var(--text3);text-transform:uppercase;letter-spacing:1.2px;margin-
bottom:10px;font-weight:600;font-family:var(--mono)}
67     .stat-value{font-size:28px;font-weight:700;color:var(--text);font-family:var(--mono)}
68     .stat-sub{font-size:11px;color:var(--text3);margin-top:4px}
69
70     /* SECTION */
71     .section{background:var(--panel);border:1px solid var(--border);border-radius:10px;margin-
bottom:18px;overflow:hidden}
72     .section-head{padding:14px 20px;border-bottom:1px solid var(--border);display:flex;justify-
content:space-between;align-items:center}
73     .section-title{font-size:13px;font-weight:700;color:var(--text);font-family:var(--mono)}
74     .section-sub{font-size:11px;color:var(--text3);margin-top:2px}
75
76     /* TABLE */
77     .tablewrap{overflow-x:auto}
78     table{width:100%;border-collapse:collapse}
79     th{padding:10px 16px;text-align:left;font-size:10px;color:var(--text3);font-weight:700;letter-
spacing:1px;text-transform:uppercase;background:var(--bg3);border-bottom:1px solid var(--border...
80     td{padding:11px 16px;border-bottom:1px solid var(--border);font-size:13px;color:var(--text2)}
81     tr:last-child td{border-bottom:none}
82     tr:hover td{background:rgba(52,211,153,0.03);color:var(--text)}
83
84     /* CHIPS */
85     .chip{display:inline-flex;align-items:center;padding:3px 8px;border-radius:4px;font-size:10px;font-
weight:700;font-family:var(--mono);letter-spacing:0.5px}
86     .chip.active{background:var(--accent-dim);color:var(--accent);border:1px solid rgba(52,211,153,0.2)}
87     .chip.inactive{background:var(--red-dim);color:var(--red);border:1px solid rgba(248,113,113,0.2)}
88     .chip.pending{background:var(--amber-dim);color:var(--amber)}
89     .chip.delivered{background:var(--accent-dim);color:var(--accent)}
90     .chip.processing{background:var(--blue-dim);color:var(--blue)}
91     .chip.breakfast{background:var(--amber-dim);color:var(--amber)}
92     .chip.lunch{background:var(--accent-dim);color:var(--accent)}
93     .chip.dinner{background:var(--blue-dim);color:var(--blue)}
94
95     /* BUTTONS */
96     .btn{padding:8px 16px;border-radius:6px;border:none;font-family:var(--mono);font-size:11px;font-
weight:700;cursor:pointer;transition:all 0.2s;letter-spacing:0.5px;text-transform:uppercase}
97     .btn-primary{background:var(--accent);color:#08100a}
98     .btn-primary:hover{background:#22c77a;box-shadow:0 4px 12px rgba(52,211,153,0.3)}
99     .btn-secondary{background:var(--panel2);color:var(--text2);border:1px solid var(--border)}
100    .btn-secondary:hover{border-color:var(--border2);color:var(--text)}
101    .btn-danger{background:var(--red-dim);color:var(--red);border:1px solid rgba(248,113,113,0.2)}
102    .btn-danger:hover{background:var(--red);color:#fff}
103    .btn-warning{background:var(--amber-dim);color:var(--amber);border:1px solid rgba(251,191,36,0.2)}
104    .btn-warning:hover{background:var(--amber);color:#000}
105    .btn-sm{padding:5px 10px;font-size:10px}
106    .action-row{display:flex;gap:6px}
107
108    /* FORM */
109    .form-card{padding:22px}
110    .form-grid{display:grid;grid-template-columns:1fr 1fr;gap:14px}
111    .form-group{display:flex;flex-direction:column;gap:5px}
112    .form-group.full{grid-column:1/-1}
113    label{font-size:10px;font-weight:700;color:var(--text3);text-transform:uppercase;letter-
spacing:0.8px;font-family:var(--mono)}
114    input,select,textarea{background:var(--bg3);border:1px solid var(--border);border-
radius:6px;padding:9px 12px;color:var(--text);font-family:var(--font);font-size:13px;transition:border-colo...
115    input:focus,select:focus,textarea:focus{outline:none;border-color:var(--accent)}
116    input::placeholder,textarea::placeholder{color:var(--text3)}
117    select option{background:var(--bg2)}
118    .form-actions{display:flex;gap:10px;margin-top:18px}
119
120    /* ALERT */
121    .alert{position:fixed;top:20px;right:20px;padding:12px 20px;border-radius:8px;z-
index:1100;animation:slideIn 0.3s ease;font-size:12px;font-weight:700;font-family:var(--mono)}
122    .alert-success{background:var(--accent-dim);color:var(--accent);border:1px solid

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```

rgba(52,211,153,0.3)}
123     .alert-error{background:var(--red-dim);color:var(--red);border:1px solid rgba(248,113,113,0.3)}
124     @keyframes slideIn{from{transform:translateX(120%);opacity:0}to{transform:translateX(0);opacity:1}}
125     .empty-state{text-align:center;color:var(--text3);font-style:italic;padding:28px;font-family:var(--
mono);font-size:12px}
126
127     /* RESTAURANT CARDS */
128     .restaurant-grid{display:grid;grid-template-columns:repeat(auto-
fill,minmax(240px,1fr));gap:14px;padding:20px}
129     .restaurant-card{background:var(--bg3);border:1px solid var(--border);border-
radius:10px;padding:18px;transition:all 0.2s;position:relative;overflow:hidden}
130     .restaurant-
card::before{content:'';position:absolute;top:0;left:0;bottom:0;width:3px;background:var(--accent)}
131     .restaurant-card:hover{border-color:var(--border2);transform:translateX(2px)}
132     .restaurant-card.inactive::before{background:var(--red)}
133     .rest-name{font-weight:700;color:var(--text);margin-bottom:6px;font-size:14px}
134     .rest-meta{font-size:11px;color:var(--text3);font-family:var(--mono);margin-bottom:12px}
135     .rest-actions{display:flex;gap:8px}
136   </style>
137 </head>
138 <body>
139 <div class="app">
140   <!-- SIDEBAR -->
141   <div class="sidebar">
142     <div class="logo">
143       <div class="logo-row">
144         <div class="logo-icon">V</div>
145         <h2>Vendor Hub</h2>
146       </div>
147       <p>EMS Vendor Portal</p>
148       <div class="role-badge">// FOOD VENDOR</div>
149     </div>
150     <nav class="nav">
151       <div class="nav-section">// overview</div>
152       <div class="nav-item active" onclick="showView('dashboard',this)"><span></span> Dashboard</div>
153       <div class="nav-section">// restaurants</div>
154       <div class="nav-item" onclick="showView('my-restaurants',this)"><span></span> My
Restaurants</div>
155       <div class="nav-item" onclick="showView('add-restaurant',this)"><span></span> Add
Restaurant</div>
156       <div class="nav-section">// operations</div>
157       <div class="nav-item" onclick="showView('subscriptions',this)"><span></span>
Subscriptions</div>
158       <div class="nav-item" onclick="showView('deliveries',this)"><span></span> Deliveries</div>
159       <div class="nav-section">// account</div>
160       <div class="nav-item" onclick="showView('profile',this)"><span></span> My Profile</div>
161     </nav>
162     <div class="sidebar-footer">
163       <div class="user-card">
164         <div class="avatar" id="avatar-initials">V</div>
165         <div class="user-info-side">
166           <div class="name" id="sidebar-name">Vendor</div>
167           <div class="role-text">// vendor</div>
168         </div>
169       </div>
170     </div>
171   </div>
172
173   <!-- MAIN -->
174   <div class="main">
175     <div class="topbar">
176       <div class="page-title" id="pageTitle">// dashboard</div>
177       <div class="topbar-right">
178         <span style="font-size:11px;color:var(--text3);font-family:var(--mono)" id="topbar-
email"></span>
179         <button class="logout-btn" onclick="logout()">SIGN OUT</button>
180       </div>
181     </div>
182     <div class="content">
183
184     <!-- DASHBOARD -->
185     <div id="dashboard-view" class="view active">
186       <div class="stats-grid">
187         <div class="stat-card green">
188           <div class="stat-label">Restaurants</div>
189           <div class="stat-value" id="stat-restaurants">--</div>

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190         <div class="stat-sub">active listings</div>
191     </div>
192     <div class="stat-card amber">
193         <div class="stat-label">Subscriptions</div>
194         <div class="stat-value" id="stat-subs">--</div>
195         <div class="stat-sub">active customers</div>
196     </div>
197     <div class="stat-card blue">
198         <div class="stat-label">Deliveries</div>
199         <div class="stat-value" id="stat-deliveries">--</div>
200         <div class="stat-sub">total today</div>
201     </div>
202     <div class="stat-card red">
203         <div class="stat-label">Pending</div>
204         <div class="stat-value" id="stat-pending">--</div>
205         <div class="stat-sub">deliveries pending</div>
206     </div>
207 </div>
208 <div style="display:grid;grid-template-columns:1fr 1fr;gap:18px">
209     <div class="section">
210         <div class="section-head">
211             <div><div class="section-title">// my_restaurants</div></div>
212             <button class="btn btn-secondary btn-sm" onclick="showView('my-
restaurants',null)">view </button>
213         </div>
214         <div id="dash-restaurants" style="padding:4px 0"></div>
215     </div>
216     <div class="section">
217         <div class="section-head">
218             <div><div class="section-title">// recent_deliveries</div></div>
219             <button class="btn btn-secondary btn-sm"
onclick="showView('deliveries',null)">view </button>
220         </div>
221         <div class="tablewrap">
222             <table>
223                 <thead><tr><th>Meal</th><th>Slot</th><th>Status</th></tr></thead>
224                 <tbody id="dash-deliveries"><tr><td colspan="3" class="empty-
state">loading...</td></tr></tbody>
225             </table>
226         </div>
227     </div>
228 </div>
229 </div>
230
231 <!-- MY RESTAURANTS -->
232 <div id="my-restaurants-view" class="view">
233     <div class="section">
234         <div class="section-head">
235             <div><div class="section-title">// my_restaurants</div><div class="section-
sub">Manage your listings</div></div>
236             <button class="btn btn-secondary btn-sm" onclick="loadMyRestaurants()">
refresh</button>
237         </div>
238         <div class="restaurant-grid" id="restaurants-grid">
239             <div class="empty-state">loading...</div>
240         </div>
241     </div>
242 </div>
243
244 <!-- ADD RESTAURANT -->
245 <div id="add-restaurant-view" class="view">
246     <div class="section">
247         <div class="section-head"><div class="section-title">// add_restaurant</div></div>
248         <div class="form-card">
249             <div class="form-grid">
250                 <div class="form-group full">
251                     <label>Restaurant Name</label>
252                     <input id="rest-name" placeholder="My Restaurant Name">
253                 </div>
254
255                 <div class="form-group full">
256                     <label>Address</label>
257                     <input id="rest-address" placeholder="Street, area, city">
258                 </div>
259
260                 <div class="form-group full">

```

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261             <label>Meal Slots</label>
262             <div style="display:flex;gap:14px;flex-wrap:wrap;padding:10px
12px;background:var(--bg3);border:1px solid var(--border);border-radius:6px">
263                 <label style="display:flex;align-items:center;gap:8px;color:var(--
text);font-family:var(--font);text-transform:none;letter-spacing:0;font-size:13px">
264                     <input type="checkbox" class="rest-slot-check" value="BREAKFAST"
style="width:auto"> Breakfast
265                 </label>
266                 <label style="display:flex;align-items:center;gap:8px;color:var(--
text);font-family:var(--font);text-transform:none;letter-spacing:0;font-size:13px">
267                     <input type="checkbox" class="rest-slot-check" value="LUNCH"
style="width:auto"> Lunch
268                 </label>
269                 <label style="display:flex;align-items:center;gap:8px;color:var(--
text);font-family:var(--font);text-transform:none;letter-spacing:0;font-size:13px">
270                     <input type="checkbox" class="rest-slot-check" value="DINNER"
style="width:auto"> Dinner
271                 </label>
272             </div>
273         </div>
274
275         <div class="form-group full">
276             <label>Description (optional)</label>
277             <textarea id="rest-desc" rows="2" placeholder="Brief description of your
restaurant..."></textarea>
278         </div>
279     </div>
280     <div class="form-actions">
281         <button class="btn btn-primary" onclick="addRestaurant()">CREATE
RESTAURANT</button>
282         <button class="btn btn-secondary" onclick="clearRestForm()">CLEAR</button>
283     </div>
284 </div>
285 </div>
286 </div>
287
288 <!-- SUBSCRIPTIONS -->
289 <div id="subscriptions-view" class="view">
290     <div class="section">
291         <div class="section-head">
292             <div><div class="section-title">// subscriptions</div><div class="section-
sub">Customers subscribed to your restaurants</div></div>
293             <button class="btn btn-secondary btn-sm" onclick="loadSubscriptions()">
refresh</button>
294         </div>
295         <div class="tablewrap">
296             <table>
297                 <thead><tr><th>Employee</th><th>Restaurant</th><th>Meal
Slot</th><th>From</th><th>To</th><th>Status</th></tr></thead>
298                 <tbody id="subscriptions-list"><tr><td colspan="6" class="empty-
state">loading...</td></tr></tbody>
299             </table>
300         </div>
301     </div>
302 </div>
303
304 <!-- DELIVERIES -->
305 <div id="deliveries-view" class="view">
306     <div class="section">
307         <div class="section-head">
308             <div><div class="section-title">// deliveries</div></div>
309             <button class="btn btn-secondary btn-sm" onclick="loadDeliveries()">
refresh</button>
310         </div>
311         <div class="tablewrap">
312             <table>
313                 <thead><tr><th>ID</th><th>Restaurant</th><th>Meal
Slot</th><th>Date</th><th>Status</th><th>Action</th></tr></thead>
314                 <tbody id="deliveries-list"><tr><td colspan="6" class="empty-
state">loading...</td></tr></tbody>
315             </table>
316         </div>
317     </div>
318 </div>
319
320 <!-- PROFILE -->

```

```

321         <div id="profile-view" class="view">
322             <div class="section">
323                 <div class="section-head"><div class="section-title">// my_profile</div></div>
324                 <div style="padding:24px;display:grid;grid-template-columns:1fr 1fr;gap:16px"
id="vendor-profile">
325                     <div><div style="font-size:10px;color:var(--text3);font-family:var(--mono);text-
transform:uppercase;letter-spacing:0.8px;margin-bottom:6px">NAME</div><div style="font-family...
326                     <div><div style="font-size:10px;color:var(--text3);font-family:var(--mono);text-
transform:uppercase;letter-spacing:0.8px;margin-bottom:6px">EMAIL</div><div style="font-famil...
327                     <div><div style="font-size:10px;color:var(--text3);font-family:var(--mono);text-
transform:uppercase;letter-spacing:0.8px;margin-bottom:6px">VENDOR ID</div><div style="font-f...
328                     <div><div style="font-size:10px;color:var(--text3);font-family:var(--mono);text-
transform:uppercase;letter-spacing:0.8px;margin-bottom:6px">ROLE</div><div style="font-family...
329                     </div>
330                 </div>
331             </div>
332
333         </div>
334     </div>
335 </div>
336
337 <script>
338     const API='http://localhost:8080';
339     let vendorId=null;
340
341     (function(){
342         const token=localStorage.getItem('jwt_token');
343         const authType=localStorage.getItem('auth_type');
344         if(!token&&authType!=='session'){window.location.href='/login.html';return;}
345         if(token){
346             try{
347                 const p=JSON.parse(atob(token.split('.')[1]));
348                 const roles=p.roles||p.authorities||'';
349                 if(roles.includes('ROLE_VENDOR')) { initUser(); return; }
350                 if(roles.includes('ROLE_EMPLOYEE')){window.location.href='/employee-
dashboard.html';return;}
351                 if(roles.includes('ROLE_MANAGER')){window.location.href='/manager-dashboard.html';return;}
352                 if(roles.includes('ROLE_ADMIN')){window.location.href='/dashboard.html';return;}
353             }catch(e){}
354         }
355         initUser();
356     })();
357
358     function getAuthHeader(){const t=localStorage.getItem('jwt_token');return t?{'Authorization':`Bearer
${t}`}:{};};
359     async function apiFetch(path,opts={}){
360         const res=await fetch(`${API}${path}`,{...opts,headers:{'Content-
Type':'application/json',...getAuthHeader(),...(opts.headers||{})},credentials:'include'});
361         if(res.status===401){localStorage.removeItem('jwt_token');localStorage.removeItem('auth_type');showAlert('SESSIO
N EXPIRED','error');setTimeout(()=>window.location.href='/login.html',1500);t...
362         const text=await res.text();return text?JSON.parse(text):null;
363     }
364
365     async function initUser(){
366         const token=localStorage.getItem('jwt_token');
367         let email='';
368         if(token){try{const p=JSON.parse(atob(token.split('.')[1]));email=p.sub||p.email||'';}catch(e){}}
369         document.getElementById('avatar-initials').textContent=(email||'V').charAt(0).toUpperCase();
370         document.getElementById('sidebar-name').textContent=(email.split('@')[0]||'Vendor');
371         document.getElementById('topbar-email').textContent=email;
372         document.getElementById('p-email').textContent=email;
373
374         try{
375             const me = await apiFetch('/vendors/me');
376             if(me){
377                 vendorId = me.id || me.vendorId || null;
378                 document.getElementById('p-name').textContent = me.name || '-';
379                 document.getElementById('p-id').textContent = vendorId || '-';
380             }
381         }catch(e){
382             console.error('Could not load vendor profile', e);
383         }
384
385         loadDashboard();
386     }

```

```

387
388     function showView(name,el){
389         document.querySelectorAll('.view').forEach(v=>v.classList.remove('active'));
390         document.getElementById(`${name}-view`).classList.add('active');
391         document.querySelectorAll('.nav-item').forEach(n=>n.classList.remove('active'));
392         if(el)el.classList.add('active');
393         document.getElementById('pageTitle').textContent='// '+name.replace(/-/g, '_');
394         const loaders={dashboard:loadDashboard,'my-restaurants':loadMyRestaurants,'add-
restaurant':()=>{},'subscriptions':loadSubscriptions,'deliveries':loadDeliveries,'profile':()=>{}};
395         if(loaders[name])loaders[name]();
396     }
397
398     async function loadDashboard(){
399         try{
400             const [rests,subs,deliveries]=await Promise.all([
401                 apiFetch('/restaurants/vendor').catch(()=>[]),
402                 apiFetch('/subscriptions').catch(()=>[]),
403                 apiFetch('/deliveries').catch(()=>[])
404             ]);
405
406             const restArr=rests||[];
407             const subArr=subs||[];
408             const delArr=deliveries||[];
409
410             document.getElementById('stat-
restaurants').textContent=restArr.filter(r=>r.active!==false).length;
411             document.getElementById('stat-subs').textContent=subArr.length;
412             document.getElementById('stat-deliveries').textContent=delArr.length;
413             document.getElementById('stat-
pending').textContent=delArr.filter(d=>d.status==='PENDING').length;
414
415             document.getElementById('dash-restaurants').innerHTML=restArr.length
416                 ? restArr.slice(0,4).map(r=><div style="padding:12px 20px;border-bottom:1px solid var(--
border);display:flex;justify-content:space-between;align-items:center"><div><div style="font...
417                 : '<div class="empty-state">no restaurants yet</div>';
418
419             document.getElementById('dash-deliveries').innerHTML=delArr.length
420                 ? delArr.slice(0,5).map(d=><tr><td>${d.restaurantName||d.restaurantId||'-'}</td><td><span
class="chip ${((d.mealSlot||'')).toLowerCase()}">${d.mealSlot||'-'}</span></td><td><span cla...
421                 : '<tr><td colspan="3" class="empty-state">no deliveries</td></tr>';
422             }catch(e){console.error(e);}
423         }
424
425
426     async function loadMyRestaurants(){
427         try{
428             const rests=await apiFetch('/restaurants/vendor');
429             const arr=rests||[];
430             document.getElementById('restaurants-grid').innerHTML=arr.length
431                 ? arr.map(r=><div class="restaurant-card ${r.active===false?'inactive':''}>
432                 <div class="rest-name">${r.name||'-'}</div>
433                 <div class="rest-meta">${r.supportedMealSlots?.join(', ')}||'-'} · ID:
${r.id||r.restaurantId||'-'}</div>
434                 <div style="font-size:12px;color:var(--text2);margin-bottom:8px">${r.address||'-'}</div>
435                 <div style="font-size:11px;color:var(--text3);margin-bottom:12px">${r.description||''}</div>
436                 <div class="rest-actions">
437                     ${r.active!==false
438                     ? <button class="btn btn-warning btn-sm"
onclick="deactivateRestaurant(${r.id}|r.restaurantId)">DEACTIVATE</button>`
439                     : `<span style="font-size:10px;color:var(--red);font-family:var(--
mono)">INACTIVE</span>`}
440                 </div>
441             </div>`).join('')
442                 : '<div class="empty-state">no restaurants found. create one!</div>';
443             }catch(e){
444                 document.getElementById('restaurants-grid').innerHTML='<div class="empty-state">could not load
restaurants</div>';
445             }
446         }
447
448
449     async function addRestaurant(){
450         const name=document.getElementById('rest-name').value.trim();
451         const address=document.getElementById('rest-address').value.trim();
452         const description=document.getElementById('rest-desc').value.trim();
453         const supportedMealSlots=Array.from(document.querySelectorAll('.rest-slot-

```

```

check:checked')).map(el=>el.value);
454
455     if(!name || !address || supportedMealSlots.length===0){
456         showAlert('NAME, ADDRESS AND AT LEAST ONE MEAL SLOT REQUIRED','error');
457         return;
458     }
459
460     try{
461         await apiFetch('/restaurants',{
462             method:'POST',
463             body:JSON.stringify({name,address,description,supportedMealSlots})
464         });
465         showAlert('RESTAURANT CREATED!','success');
466         clearRestForm();
467         loadMyRestaurants();
468         loadDashboard();
469     }catch(e){
470         showAlert('FAILED: '+e.message,'error');
471     }
472 }
473
474
475 function clearRestForm(){
476     ['rest-name','rest-address','rest-desc'].forEach(id=>document.getElementById(id).value='');
477     document.querySelectorAll('.rest-slot-check').forEach(el=>el.checked=false);
478 }
479
480 async function loadSubscriptions(){
481     try{
482         const data=await apiFetch('/subscriptions');
483         document.getElementById('subscriptions-list').innerHTML=(data||[]).length
484             ? data.map(s=><tr><td style="font-family:var(--mono);font-size:12px">${s.employeeEmail}|${s.employeeId}|-</td><td>${s.restaurantName}|${s.restaurantId}|-</td><td><span class="chi...
485             : '<tr><td colspan="6" class="empty-state">no subscriptions found</td></tr>';
486     }catch(e){document.getElementById('subscriptions-list').innerHTML='<tr><td colspan="6" class="empty-state">could not load subscriptions</td></tr>';}
487 }
488
489 async function loadDeliveries(){
490     try{
491         const data=await apiFetch('/deliveries');
492         document.getElementById('deliveries-list').innerHTML=(data||[]).length
493             ? data.map(d=><tr><td style="font-family:var(--mono);font-size:11px">#${d.id}|-</td><td>${d.restaurantName}|${d.restaurantId}|-</td><td><span class="chip
494             : '<tr><td colspan="6" class="empty-state">no deliveries found</td></tr>';
495     }catch(e){document.getElementById('deliveries-list').innerHTML='<tr><td colspan="6" class="empty-state">could not load deliveries</td></tr>';}
496 }
497
498 async function markDelivered(id){
499     try{await apiFetch(`/deliveries/${id}/delivered`,{method:'PUT'});showAlert('MARKED AS
DELIVERED','success');loadDeliveries();}
500     catch(e){showAlert('FAILED','error');}
501 }
502
503 function logout(){
504     // Clear all localStorage
505     localStorage.removeItem('jwt_token');
506     localStorage.removeItem('refresh_token');
507     localStorage.removeItem('auth_type');
508     localStorage.removeItem('last_role');
509
510     // Call server logout to invalidate session
511     fetch(`${API}/session/logout`, {
512         method: 'POST',
513         credentials: 'include'
514     }).finally(() => {
515         // Force expire the session cookie on client side
516         document.cookie = 'JSESSIONID=; expires=Thu, 01 Jan 1970 00:00:00 UTC; path=/;';
517         window.location.href = '/google.html';
518     });
519 }
520
521 function showAlert(msg,type){

```



```
522     const el=document.createElement('div');
523     el.className=`alert alert-${type}`;el.textContent=msg;
524     document.body.appendChild(el);setTimeout(()=>el.remove(),3000);
525 }
526 </script>
527 </body>
528 </html>
529
```

```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>EMS – Vendor Login</title>
7      <link
href="https://fonts.googleapis.com/css2?family=Outfit:wght@300;400;500;600;700&family=JetBrains+Mono:wght@400;50
0&display=swap" rel="stylesheet">
8      <style>
9          *{margin:0;padding:0;box-sizing:border-box}
10         :root{
11             --bg:#070910;
12             --panel:#0e1118;
13             --panel2:#131821;
14             --border:rgba(255,255,255,0.07);
15             --border-lit:rgba(255,255,255,0.14);
16             --text:#dde4f0;--text2:#6b7591;--text3:#3d4560;
17             --red:#f87171;--red-dim:rgba(248,113,113,0.12);
18             --blue:#4f8cff;--blue-dim:rgba(79,140,255,0.12);
19             --green:#34d399;--green-dim:rgba(52,211,153,0.12);
20             --amber:#f9c74f;--amber-dim:rgba(249,199,79,0.12);
21             --font:'Outfit',sans-serif;--mono:'JetBrains Mono',monospace;
22         }
23         body{font-family:var(--font);background:var(--bg);color:var(--text);min-
height:100vh;display:flex;overflow:auto;}
24
25         .bg-layer{position:fixed;inset:0;z-index:0;
26             background:
27                 radial-gradient(ellipse 60% 50% at 20% 50%, rgba(248,113,113,0.06) 0%,transparent 70%),
28                 radial-gradient(ellipse 40% 60% at 80% 20%, rgba(249,199,79,0.05) 0%,transparent 60%),
29                 radial-gradient(ellipse 50% 40% at 70% 80%, rgba(248,113,113,0.04) 0%,transparent 60%);
30         }
31         .grid-overlay{position:fixed;inset:0;z-index:0;background-image:linear-
gradient(rgba(255,255,255,0.02) 1px,transparent 1px),linear-gradient(90deg,rgba(255,255,255,0.02)
1px,transparent 1px)...
32
33         /* LEFT */
34         .left-panel{flex:1;display:flex;flex-direction:column;justify-
content:center;padding:60px;position:relative;z-index:1;animation:fadeLeft 0.8s ease both;}
35         @keyframes fadeLeft{from{opacity:0;transform:translateX(-
24px)}to{opacity:1;transform:translateX(0)}}
36         .brand-logo{display:flex;align-items:center;gap:12px;margin-bottom:32px}
37         .brand-icon{width:44px;height:44px;background:linear-gradient(135deg,#f87171,#fb923c);border-
radius:12px;display:flex;align-items:center;justify-content:center;font-weight:700;font-size:18p...
38         .brand-name{font-size:22px;font-weight:700;color:var(--text);letter-spacing:-0.3px}
39         .brand-tagline{font-size:13px;color:var(--text2)}
40         .hero-text h1{font-size:clamp(36px,4vw,50px);font-weight:700;line-height:1.1;letter-spacing:-
1px;color:var(--text);margin-bottom:16px;}
41         .hl-red{color:var(--red)}
42         .hero-text p{font-size:15px;color:var(--text2);line-height:1.6;max-width:360px}
43
44         .info-cards{display:flex;flex-direction:column;gap:12px;margin-top:40px;max-width:360px;}
45         .info-card{background:var(--panel);border:1px solid var(--border);border-radius:10px;padding:14px
18px;display:flex;align-items:flex-start;gap:12px;}
46         .info-card-icon{font-size:18px;flex-shrink:0;margin-top:1px}
47         .info-card-text strong{font-size:13px;color:var(--text);display:block;margin-bottom:2px}
48         .info-card-text span{font-size:12px;color:var(--text2)}
49
50         .back-link{display:inline-flex;align-items:center;gap:6px;font-size:12px;color:var(--text3);text-
decoration:none;margin-top:36px;transition:color 0.2s;}
51         .back-link:hover{color:var(--text2)}
52
53         /* RIGHT */
54         .right-panel{width:460px;flex-shrink:0;display:flex;align-items:flex-start;overflow-y:auto;justify-
content:center;padding:32px 40px;position:relative;z-index:1;border-left:1px solid var(--b...
55         @keyframes
fadeRight{from{opacity:0;transform:translateX(24px)}to{opacity:1;transform:translateX(0)}}
56         .form-box{width:100%}
57
58         .vendor-badge{display:inline-flex;align-items:center;gap:8px;background:var(--red-dim);border:1px
solid rgba(248,113,113,0.25);border-radius:20px;padding:6px 14px;font-size:12px;font-weight...

```

```

59     .vendor-badge .dot{width:7px;height:7px;border-radius:50%;background:var(--red);animation:pulse 1.4s
infinite;}}
60     @keyframes pulse{0%,100%{opacity:1}50%{opacity:0.4}}
61
62     .form-title{font-size:22px;font-weight:700;color:var(--text);margin-bottom:6px;}
63     .form-subtitle{font-size:13px;color:var(--text2);margin-bottom:28px;}
64
65     /* AUTH TABS */
66     .auth-tabs{display:flex;background:var(--bg);border-radius:10px;padding:4px;gap:4px;margin-
bottom:24px;border:1px solid var(--border)}
67     .auth-tab{flex:1;padding:9px;border:none;background:transparent;color:var(--text2);font-family:var(-
font);font-weight:600;font-size:13px;cursor:pointer;border-radius:7px;transition:all 0.2s;}
68     .auth-tab.active{background:var(--panel2);color:var(--text);box-shadow:0 1px 4px rgba(0,0,0,0.3)}
69
70     /* FORM ELEMENTS */
71     .form-group{margin-bottom:16px}
72     .form-label{display:block;font-size:10px;font-weight:600;color:var(--text2);text-
transform:uppercase;letter-spacing:0.8px;font-family:var(--mono);margin-bottom:7px}
73     .form-input{width:100%;padding:11px 14px;background:var(--bg);border:1px solid var(--border);border-
radius:9px;color:var(--text);font-family:var(--font);font-size:14px;transition:all 0.2s;}
74     .form-input:focus{outline:none;border-color:var(--red);box-shadow:0 0 3px rgba(248,113,113,0.08)}
75     .form-input::placeholder{color:var(--text3)}
76
77     /* BUTTONS */
78     .btn-login{width:100%;padding:12px;border:none;border-radius:9px;font-family:var(--font);font-
size:14px;font-weight:600;cursor:pointer;transition:all 0.2s;margin-top:4px;background:var(--re...
79     .btn-login:hover{background:#f98a8a;box-shadow:0 6px 20px
rgba(248,113,113,0.35);transform:translateY(-1px)}
80     .btn-login.active{transform:translateY(0)}
81     .btn-login.loading{opacity:0.7;pointer-events:none}
82
83     /* STATUS */
84     .status-box{border-radius:9px;padding:10px 14px;margin-top:14px;font-size:13px;font-
weight:500;display:none;animation:fadeIn 0.3s ease;}
85     @keyframes fadeIn{from{opacity:0;transform:translateY(4px)}to{opacity:1;transform:translateY(0)}}
86     .status-box.error{background:var(--red-dim);color:var(--red);border:1px solid rgba(248,113,113,0.2)}
87     .status-box.success{background:var(--green-dim);color:var(--green);border:1px solid
rgba(52,211,153,0.2)}
88     .status-box.loading{background:var(--blue-dim);color:var(--blue);border:1px solid
rgba(79,140,255,0.2)}
89
90     .notice{background:rgba(249,199,79,0.06);border:1px solid rgba(249,199,79,0.15);border-
radius:9px;padding:12px 14px;margin-top:20px;font-size:12px;color:var(--text2);line-height:1.5;}
91     .notice strong{color:var(--amber)}
92
93     /* FOOTER */
94     .form-footer{margin-top:20px;padding-top:18px;border-top:1px solid var(--border)}
95     .tz-row{display:flex;align-items:center;gap:8px;margin-bottom:8px}
96     .tz-label{font-size:10px;color:var(--text3);font-family:var(--mono);flex-shrink:0}
97     .tz-select{flex:1;padding:6px 8px;background:var(--bg);border:1px solid var(--border);border-
radius:7px;color:var(--text2);font-family:var(--mono);font-size:11px;cursor:pointer;}
98     .tz-select:focus{outline:none;border-color:var(--border-lit)}
99     .tz-time{font-size:10px;color:var(--red);font-family:var(--mono)}
100
101     @media(max-width:900px){
102         .left-panel{display:none}
103         .right-panel{width:100%;border-left:none;background:var(--bg)}
104     }
105 </style>
106 </head>
107 <body>
108 <div class="bg-layer"></div>
109 <div class="grid-overlay"></div>
110
111 <!-- LEFT -->
112 <div class="left-panel">
113     <div class="brand-logo">
114         <div class="brand-icon">V</div>
115         <div>
116             <div class="brand-name">Vendor Hub</div>
117             <div class="brand-tagline">Employee Management System</div>
118         </div>
119     </div>
120     <div class="hero-text">
121         <h1>Manage your<br><span class="hl-red">vendor account</span><br>securely.</h1>
122         <p>Access your deliveries, restaurants, and subscriptions. Your account is created and managed by

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the system admin.</p>
123     </div>
124     <div class="info-cards">
125         <div class="info-card">
126             <span class="info-card-icon"></span>
127             <div class="info-card-text">
128                 <strong>Delivery Tracking</strong>
129                 <span>Monitor your active and scheduled meal deliveries in real-time.</span>
130             </div>
131         </div>
132         <div class="info-card">
133             <span class="info-card-icon"></span>
134             <div class="info-card-text">
135                 <strong>Restaurant Management</strong>
136                 <span>View and update restaurant details and meal offerings.</span>
137             </div>
138         </div>
139         <div class="info-card">
140             <span class="info-card-icon"></span>
141             <div class="info-card-text">
142                 <strong>Account by Admin Only</strong>
143                 <span>Vendor accounts are created by the system admin. Contact your admin if you don't have
credentials.</span>
144             </div>
145         </div>
146     </div>
147     <a class="back-link" href="/google.html"> Back to main login</a>
148 </div>
149
150 <!-- RIGHT FORM -->
151 <div class="right-panel">
152     <div class="form-box">
153
154         <div class="vendor-badge"><span class="dot"></span>Vendor Portal</div>
155         <div class="form-title">Vendor Sign In</div>
156         <div class="form-subtitle">Enter the credentials provided by your system admin.</div>
157
158         <!-- AUTH TABS -->
159         <div class="auth-tabs">
160             <button class="auth-tab active" id="tab-jwt" onclick="switchTab('jwt')">JWT Login</button>
161             <button class="auth-tab" id="tab-session" onclick="switchTab('session')">Session Login</button>
162         </div>
163
164         <!-- JWT FORM -->
165         <div id="jwt-form">
166             <div class="form-group">
167                 <label class="form-label">Email / Username</label>
168                 <input class="form-input" id="jwt-username" type="text" placeholder="vendor@company.com"
autocomplete="off">
169             </div>
170             <div class="form-group">
171                 <label class="form-label">Password</label>
172                 <input class="form-input" id="jwt-password" type="password" placeholder="....."
autocomplete="current-password">
173             </div>
174             <button class="btn-login" id="jwt-btn" onclick="jwtLogin()">Sign In as Vendor</button>
175         </div>
176
177         <!-- SESSION FORM -->
178         <div id="session-form" style="display:none">
179             <div class="form-group">
180                 <label class="form-label">Email / Username</label>
181                 <input class="form-input" id="session-username" type="text"
placeholder="vendor@company.com">
182             </div>
183             <div class="form-group">
184                 <label class="form-label">Password</label>
185                 <input class="form-input" id="session-password" type="password" placeholder=".....">
186             </div>
187             <button class="btn-login" id="session-btn" onclick="sessionLogin()">Sign In with
Session</button>
188         </div>
189
190     <div class="status-box" id="status-box"></div>
191
192     <div class="notice">
```

```

193         <strong>Don't have an account?</strong> Vendor accounts are created exclusively by the system
admin. Please contact your administrator to set up access.
194     </div>
195
196     <!-- FOOTER -->
197     <div class="form-footer">
198         <div class="tz-row">
199             <span class="tz-label">TZ</span>
200             <select class="tz-select" id="tz-select" onchange="updateTz()">
201                 <option value="Asia/Kolkata">India – IST UTC+5:30</option>
202                 <option value="UTC">UTC</option>
203                 <option value="America/New_York">New York – EST UTC-5</option>
204                 <option value="America/Los_Angeles">Los Angeles – PST UTC-8</option>
205                 <option value="Europe/London">London – GMT UTC+0</option>
206                 <option value="Europe/Paris">Paris – CET UTC+1</option>
207                 <option value="Asia/Tokyo">Tokyo – JST UTC+9</option>
208                 <option value="Asia/Dubai">Dubai – GST UTC+4</option>
209                 <option value="Australia/Sydney">Sydney – AEDT UTC+11</option>
210                 <option value="Asia/Singapore">Singapore – SGT UTC+8</option>
211             </select>
212             <span class="tz-time" id="tz-time">--:--:--</span>
213         </div>
214     </div>
215
216 </div>
217 </div>
218
219 <script>
220     const API = 'http://localhost:8080';
221
222     const DASHBOARDS = {
223         FOOD_VENDOR: '/vendor-dashboard.html',
224         TECH_VENDOR: '/tech-vendor-dashboard.html'
225     };
226
227     // ALREADY LOGGED IN?
228     (function(){
229         const token = localStorage.getItem('jwt_token');
230         if(!token) return;
231         try{
232             const role = getRoleFromJWT(token);
233             if(DASHBOARDS[role]) window.location.href = DASHBOARDS[role];
234         }catch(e){}
235     })();
236
237     // HELPERS
238     function getRoleFromJWT(token){
239         const payload = JSON.parse(atob(token.split('.')[1]));
240         const raw = payload.role || payload.roles || payload.authorities || payload.scope || '';
241         const role = Array.isArray(raw) ? raw[0] : raw;
242         return String(role).replace(/^ROLE_/, '').toUpperCase().trim();
243     }
244
245     function showStatus(type, msg){
246         const box = document.getElementById('status-box');
247         box.className = `status-box ${type}`;
248         box.textContent = msg;
249         box.style.display = 'block';
250     }
251     function hideStatus(){
252         document.getElementById('status-box').style.display = 'none';
253     }
254
255     function switchTab(type){
256         document.getElementById('jwt-form').style.display = type === 'jwt' ? 'block' : 'none';
257         document.getElementById('session-form').style.display = type === 'session' ? 'block' : 'none';
258         document.getElementById('tab-jwt').classList.toggle('active', type === 'jwt');
259         document.getElementById('tab-session').classList.toggle('active', type === 'session');
260         hideStatus();
261     }
262
263     function redirectByRole(role){
264         const dest = DASHBOARDS[role];
265         if(!dest){
266             showStatus('error', 'This portal is for vendors only. Please use the main login page.');
```

```

268     }
269     showStatus('success', `Login successful! Redirecting to ${role === 'TECH_VENDOR' ? 'Tech Vendor' :
'Food Vendor'} dashboard...`);
270     setTimeout(() => window.location.href = dest, 700);
271     return true;
272 }
273
274 // JWT LOGIN
275 async function jwtLogin(){
276     const username = document.getElementById('jwt-username').value.trim();
277     const password = document.getElementById('jwt-password').value;
278     if(!username || !password){ showStatus('error', 'Please enter your email and password'); return; }
279
280     const btn = document.getElementById('jwt-btn');
281     btn.classList.add('loading'); btn.textContent = 'Signing in...';
282     showStatus('loading', 'Authenticating...');
283
284     try{
285         const res = await fetch(`${API}/Authenticate`, {
286             method: 'POST',
287             headers: {'Content-Type': 'application/json'},
288             body: JSON.stringify({username, password})
289         });
290
291         if(res.ok){
292             const data = await res.json();
293             const token = data.accessToken || data.token;
294             const role = getRoleFromJWT(token);
295
296             if(!DASHBOARDS[role]){
297                 showStatus('error', 'This portal is for vendors only. Please use the main login page.');
```

btn.classList.remove('loading'); btn.textContent = 'Sign In as Vendor';

```

299                 return;
300             }
301
302             localStorage.setItem('jwt_token', token);
303             localStorage.setItem('refresh_token', data.refreshToken || '');
304             localStorage.setItem('auth_type', 'jwt');
305             localStorage.setItem('last_role', role);
306             localStorage.setItem('user_timezone', document.getElementById('tz-select').value);
307
308             btn.textContent = 'Signed In ';
309             redirectByRole(role);
310
311         }else{
312             const err = await res.text();
313             showStatus('error', 'Login failed: ' + (err || 'Invalid credentials'));
314             btn.classList.remove('loading'); btn.textContent = 'Sign In as Vendor';
315         }
316     }catch(e){
317         showStatus('error', 'Connection error: ' + e.message);
318         btn.classList.remove('loading'); btn.textContent = 'Sign In as Vendor';
319     }
320 }
321
322 // SESSION LOGIN
323 async function sessionLogin(){
324     const username = document.getElementById('session-username').value.trim();
325     const password = document.getElementById('session-password').value;
326     if(!username || !password){ showStatus('error', 'Please enter your email and password'); return; }
327
328     const btn = document.getElementById('session-btn');
329     btn.classList.add('loading'); btn.textContent = 'Signing in...';
330     showStatus('loading', 'Authenticating...');
331
332     try{
333         const res = await fetch(`${API}/session/login`, {
334             method: 'POST',
335             headers: {'Content-Type': 'application/json'},
336             body: JSON.stringify({username, password}),
337             credentials: 'include'
338         });
339
340         if(res.ok){
341             // Session login – fetch /employee/me or /vendor/me to get role
342             try{

```

```

343         const meRes = await fetch(`${API}/vendor/me`, {credentials: 'include'});
344         if(meRes.ok){
345             const me = await meRes.json();
346             const role = String(me.role || '').replace(/^ROLE_/, '').toUpperCase();
347             localStorage.setItem('auth_type', 'session');
348             localStorage.setItem('last_role', role);
349             localStorage.setItem('user_timezone', document.getElementById('tz-select').value);
350             btn.textContent = 'Signed In ';
351             redirectByRole(role);
352             return;
353         }
354     }catch(e){}
355
356     // Fallback if /vendor/me not available
357     showStatus('error', 'Could not determine vendor type. Please use JWT login.');
```

358 btn.classList.remove('loading'); btn.textContent = 'Sign In with Session';

359

```

360     }else{
361         const err = await res.json().catch(()=>({}));
362         showStatus('error', 'Login failed: ' + (err.error || 'Invalid credentials'));
363         btn.classList.remove('loading'); btn.textContent = 'Sign In with Session';
364     }
365     }catch(e){
366         showStatus('error', 'Connection error: ' + e.message);
367         btn.classList.remove('loading'); btn.textContent = 'Sign In with Session';
368     }
369 }
370
371 // TIMEZONE
372 function updateTz(){
373     localStorage.setItem('user_timezone', document.getElementById('tz-select').value);
374     refreshTzClock();
375 }
376 function refreshTzClock(){
377     const tz = document.getElementById('tz-select').value;
378     document.getElementById('tz-time').textContent =
379         new Date().toLocaleTimeString('en-GB', {timeZone: tz, hour12: false});
380 }
381
382 document.addEventListener('keydown', e => {
383     if(e.key !== 'Enter') return;
384     if(document.getElementById('jwt-form').style.display !== 'none') jwtLogin();
385     else sessionLogin();
386 });
387
388 window.onload = () => {
389     const savedTz = localStorage.getItem('user_timezone') || 'Asia/Kolkata';
390     document.getElementById('tz-select').value = savedTz;
391     refreshTzClock();
392     setInterval(refreshTzClock, 1000);
393 };
394 </script>
395 </body>
396 </html>

```

169 Current:

src\test\java\com\example\EmployeeManagementSystem\EmployeeManagementSystemApplication

```
1 package com.example.EmployeeManagementSystem;
2
3 import org.junit.jupiter.api.Test;
4 import org.springframework.boot.test.context.SpringBootTest;
5
6 @SpringBootTest
7 class EmployeeManagementSystemApplicationTests {
8
9     @Test
10     void contextLoads() {
11     }
12
13 }
14
```



```

1  package com.example.EmployeeManagementSystem.service;
2
3
4  import com.example.EmployeeManagementSystem.Entity.Employee;
5  import com.example.EmployeeManagementSystem.Entity.LeaveEntitlement;
6  import com.example.EmployeeManagementSystem.Entity.LeaveType;
7  import com.example.EmployeeManagementSystem.Enum.Gender;
8  import com.example.EmployeeManagementSystem.Enum.Status;
9  import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
10 import com.example.EmployeeManagementSystem.Repository.LeaveEntitlementRepository;
11 import com.example.EmployeeManagementSystem.Repository.LeaveTypeRepository;
12 import com.example.EmployeeManagementSystem.Service.LeaveAccrualService;
13 import org.junit.jupiter.api.BeforeEach;
14 import org.junit.jupiter.api.Test;
15 import org.mockito.*;
16
17 import java.math.BigDecimal;
18 import java.time.LocalDate;
19 import java.util.List;
20 import java.util.Optional;
21
22 import static org.assertj.core.api.Assertions.assertThat;
23 import static org.mockito.ArgumentMatchers.any;
24 import static org.mockito.Mockito.*;
25
26 class LeaveAccrualServiceTest {
27
28     @Mock
29     EmployeeRepo employeeRepo;
30     @Mock
31     LeaveTypeRepository leaveTypeRepo;
32     @Mock
33     LeaveEntitlementRepository entitlementRepo;
34
35     @InjectMocks
36     LeaveAccrualService service;
37
38     @BeforeEach void setup() { MockitoAnnotations.openMocks(this); }
39
40     @Test
41     void maleEmployee_doesNotAccrueMaternity() {
42         Employee emp = employee(Gender.M);
43         LeaveType maternity = leaveType("MATERNITY", true, Gender.F);
44
45         when(employeeRepo.findByStatus(Status.ACTIVE)).thenReturn(List.of(emp));
46         when(leaveTypeRepo.findAll()).thenReturn(List.of(maternity));
47
48         service.runMonthlyAccrual(LocalDate.of(2025, 6, 1));
49
50         verify(entitlementRepo, never()).save(any());
51     }
52
53     @Test
54     void femaleEmployee_accruesMaternity() {
55         Employee emp = employee(Gender.F);
56         LeaveType maternity = leaveType("MATERNITY", true, Gender.F);
57
58         when(employeeRepo.findByStatus(Status.ACTIVE)).thenReturn(List.of(emp));
59         when(leaveTypeRepo.findAll()).thenReturn(List.of(maternity));
60         when(entitlementRepo.findByEmployeeIdAndLeaveTypeIdAndYear(any(), any(), any()))
61             .thenReturn(Optional.empty());
62
63         // Simulate new entitlement creation
64         when(entitlementRepo.save(any())).thenAnswer(inv -> inv.getArgument(0));
65
66         service.runMonthlyAccrual(LocalDate.of(2025, 6, 1));
67
68         ArgumentCaptor<LeaveEntitlement> captor = ArgumentCaptor.forClass(LeaveEntitlement.class);
69         verify(entitlementRepo, atLeastOnce()).save(captor.capture());
70
71         LeaveEntitlement saved = captor.getAllValues().stream()

```

```

72         .filter(e -> e.getAccruedThisYear().compareTo(BigDecimal.ONE) == 0)
73         .findFirst().orElseThrow();
74
75         assertThat(saved.getAccruedThisYear()).isEqualToComparingTo("1");
76     }
77
78     @Test
79     void employeeJoiningAfterAccrualDate_isSkipped() {
80         Employee emp = employee(Gender.M);
81         emp.setJoined_at(LocalDate.of(2025, 6, 15)); // joins mid-month
82         LeaveType casual = leaveType("CASUAL", false, null);
83
84         when(employeeRepo.findByStatus(Status.ACTIVE)).thenReturn(List.of(emp));
85         when(leaveTypeRepo.findAll()).thenReturn(List.of(casual));
86
87         service.runMonthlyAccrual(LocalDate.of(2025, 6, 1));
88
89         verify(entitlementRepo, never()).save(any());
90     }
91
92     // --- helpers ---
93     private Employee employee(Gender gender) {
94         Employee e = new Employee();
95         e.setEmployeeId(1L);
96         e.setGender(gender);
97         e.setStatus(Status.ACTIVE);
98         e.setJoined_at(LocalDate.of(2023, 1, 1));
99         return e;
100     }
101
102     private LeaveType leaveType(String name, boolean genderRestricted, Gender restriction) {
103         LeaveType lt = new LeaveType();
104         lt.setId(1);
105         lt.setName(name);
106         lt.setGenderRestricted(genderRestricted);
107         lt.setGenderRestriction(restriction);
108         lt.setCarriesForward(!name.equals("SICK"));
109         lt.setMonthlyReset(name.equals("SICK"));
110         return lt;
111     }
112 }

```

```

1  package com.example.EmployeeManagementSystem.service;
2
3  import com.example.EmployeeManagementSystem.Entity.Employee;
4  import com.example.EmployeeManagementSystem.Entity.LeaveEntitlement;
5  import com.example.EmployeeManagementSystem.Entity.LeaveRequest;
6  import com.example.EmployeeManagementSystem.Enum.LeaveStatus;
7  import com.example.EmployeeManagementSystem.Enum.LeaveType;
8  import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
9  import com.example.EmployeeManagementSystem.Repository.LeaveEntitlementRepository;
10 import com.example.EmployeeManagementSystem.Repository.LeaveRequestRepo;
11 import com.example.EmployeeManagementSystem.Repository.LeaveTypeRepository;
12 import com.example.EmployeeManagementSystem.Service.LeaveRequestService;
13 import org.junit.jupiter.api.BeforeEach;
14 import org.junit.jupiter.api.Test;
15 import org.mockito.InjectMocks;
16 import org.mockito.Mock;
17 import org.mockito.MockitoAnnotations;
18
19 import java.math.BigDecimal;
20 import java.time.LocalDate;
21 import java.util.Optional;
22
23 import static org.assertj.core.api.Assertions.assertThat;
24 import static org.mockito.ArgumentMatchers.any;
25 import static org.mockito.Mockito.when;
26
27 class LeaveRequestServiceTest {
28
29     @Mock
30     LeaveRequestRepo leaveRequestRepo;
31     @Mock
32     EmployeeRepo employeeRepo;
33     @Mock
34     LeaveTypeRepository leaveTypeRepository;
35     @Mock
36     LeaveEntitlementRepository leaveEntitlementRepository;
37
38     @InjectMocks
39     LeaveRequestService service;
40
41     @BeforeEach
42     void setup() {
43         MockitoAnnotations.openMocks(this);
44     }
45
46     @Test
47     void approveLeave_updatesUsedEntitlementDays() {
48         Employee employee = new Employee();
49         employee.setEmployeeId(1L);
50
51         LeaveRequest request = new LeaveRequest();
52         request.setId(10L);
53         request.setEmployee(employee);
54         request.setLeaveType(LeaveType.CASUAL);
55         request.setStartDate(LocalDate.of(2026, 5, 4));
56         request.setEndDate(LocalDate.of(2026, 5, 6));
57         request.setStatus(LeaveStatus.PENDING);
58
59         com.example.EmployeeManagementSystem.Entity.LeaveType casual =
60             new com.example.EmployeeManagementSystem.Entity.LeaveType();
61         casual.setId(2);
62         casual.setName("CASUAL");
63         casual.setCarriesForward(true);
64
65         LeaveEntitlement entitlement = new LeaveEntitlement();
66         entitlement.setEmployee(employee);
67         entitlement.setLeaveType(casual);
68         entitlement.setYear(2026);
69         entitlement.setOpeningBalance(BigDecimal.ZERO);
70         entitlement.setAccruedThisYear(BigDecimal.TEN);
71         entitlement.setUsedThisYear(BigDecimal.ONE);

```

```
72     entitlement.setClosingBalance(BigDecimal.ZERO);
73
74     when(leaveRequestRepo.findById(10L)).thenReturn(Optional.of(request));
75     when(leaveRequestRepo.save(any())).thenReturn(inv -> inv.getArgument(0));
76     when(leaveTypeRepository.findByName("CASUAL")).thenReturn(Optional.of(casual));
77     when(leaveEntitlementRepository.findByIdAndLeaveTypeIdAndYear(1L, 2, 2026))
78         .thenReturn(Optional.of(entitlement));
79     when(leaveEntitlementRepository.save(any())).thenReturn(inv -> inv.getArgument(0));
80
81     service.approveLeave(10L);
82
83     assertThat(entitlement.getUsedThisYear()).isEqualByComparingTo("4");
84 }
85 }
86
```

```

1  package com.example.EmployeeManagementSystem.service;
2
3  import com.example.EmployeeManagementSystem.Repository.LeaveEntitlementRepository;
4  import com.example.EmployeeManagementSystem.Repository.LeaveTypeRepository;
5  import org.junit.jupiter.api.BeforeEach;
6  import org.junit.jupiter.api.Test;
7  import org.mockito.*;
8  import com.example.EmployeeManagementSystem.Service.SickLeaveResetService;
9  import com.example.EmployeeManagementSystem.Entity.LeaveType;
10 import com.example.EmployeeManagementSystem.Entity.LeaveEntitlement;
11 import com.example.EmployeeManagementSystem.Entity.Employee;
12
13 import java.math.BigDecimal;
14 import java.time.LocalDate;
15 import java.time.YearMonth;
16 import java.util.List;
17
18 import static org.assertj.core.api.Assertions.assertThat;
19 import static org.mockito.Mockito.*;
20
21 class SickLeaveResetServiceTest {
22
23     @Mock
24     LeaveTypeRepository leaveTypeRepo;
25     @Mock
26     LeaveEntitlementRepository entitlementRepo;
27     @InjectMocks
28     SickLeaveResetService service;
29
30     @BeforeEach
31     void setup() {
32         MockitoAnnotations.openMocks(this);
33     }
34
35     @Test
36     void unusedSickDays_areZeroedAtMonthEnd() {
37         LeaveType sick = sickLeaveType();
38         LeaveEntitlement entitlement = entitlement(sick, BigDecimal.valueOf(3), BigDecimal.ONE);
39         // accrued=3, used=1, available=2 should be wiped
40
41         when(leaveTypeRepo.findByMonthlyResetTrue()).thenReturn(List.of(sick));
42         when(entitlementRepo.findByLeaveTypeIdAndYear(sick.getId(), 2025))
43             .thenReturn(List.of(entitlement));
44         when(entitlementRepo.save(any())).thenAnswer(inv -> inv.getArgument(0));
45
46         service.runMonthlyReset(LocalDate.of(2025, 6, 30));
47
48         // After reset: accrued == used, available == 0
49         assertThat(entitlement.getAccruedThisYear())
50             .isEqualToComparingTo(entitlement.getUsedThisYear());
51         assertThat(entitlement.getAvailableBalance())
52             .isEqualToComparingTo(BigDecimal.ZERO);
53     }
54
55     @Test
56     void fullyUsedSickLeave_noChangeNeeded() {
57         LeaveType sick = sickLeaveType();
58         LeaveEntitlement entitlement = entitlement(sick, BigDecimal.ONE, BigDecimal.ONE);
59         // accrued=1, used=1, available=0 nothing to wipe
60
61         when(leaveTypeRepo.findByMonthlyResetTrue()).thenReturn(List.of(sick));
62         when(entitlementRepo.findByLeaveTypeIdAndYear(sick.getId(), 2025))
63             .thenReturn(List.of(entitlement));
64         when(entitlementRepo.save(any())).thenAnswer(inv -> inv.getArgument(0));
65
66         service.runMonthlyReset(LocalDate.of(2025, 6, 30));
67
68         // accrued unchanged – still equals used
69         assertThat(entitlement.getAccruedThisYear()).isEqualToComparingTo("1");
70     }
71 }

```

```

72 // FIX: New test – ensures reset does NOT alter a negative balance (over-used edge case)
73 @Test
74 void negativeBalance_isNotTouched() {
75     LeaveType sick = sickLeaveType();
76     // accrued=0, used=1, available=-1 (negative – already over-consumed)
77     LeaveEntitlement entitlement = entitlement(sick, BigDecimal.ZERO, BigDecimal.ONE);
78
79     when(leaveTypeRepo.findByMonthlyResetTrue()).thenReturn(List.of(sick));
80     when(entitlementRepo.findByLeaveTypeIdAndYear(sick.getId(), 2025))
81         .thenReturn(List.of(entitlement));
82     when(entitlementRepo.save(any())).thenAnswer(inv -> inv.getArgument(0));
83
84     service.runMonthlyReset(LocalDate.of(2025, 6, 30));
85
86     // available is <= 0 so no reset should happen; accrued stays 0
87     assertThat(entitlement.getAccruedThisYear()).isEqualByComparingTo(BigDecimal.ZERO);
88 }
89
90 // FIX: New test – verifies correct year/month is used when resetDate is end-of-month
91 // (guards against the midnight-fire fragility in the scheduler)
92 @Test
93 void resetDate_usesCorrectYearFromPassedDate_notFromSystemClock() {
94     LeaveType sick = sickLeaveType();
95     LeaveEntitlement entitlement = entitlement(sick, BigDecimal.valueOf(2), BigDecimal.ZERO);
96
97     LocalDate endOfJune2025 = YearMonth.of(2025, 6).atEndOfMonth(); // 2025-06-30
98
99     when(leaveTypeRepo.findByMonthlyResetTrue()).thenReturn(List.of(sick));
100    when(entitlementRepo.findByLeaveTypeIdAndYear(sick.getId(), 2025))
101        .thenReturn(List.of(entitlement));
102    when(entitlementRepo.save(any())).thenAnswer(inv -> inv.getArgument(0));
103
104    service.runMonthlyReset(endOfJune2025);
105
106    // Should query year=2025, not system year – verified by mock setup using 2025 only
107    verify(entitlementRepo).findByLeaveTypeIdAndYear(sick.getId(), 2025);
108    assertThat(entitlement.getAvailableBalance()).isEqualByComparingTo(BigDecimal.ZERO);
109 }
110
111 // --- helpers ---
112
113 private LeaveType sickLeaveType() {
114     LeaveType lt = new LeaveType();
115     lt.setId(1);
116     lt.setName("SICK");
117     lt.setMonthlyReset(true);
118     lt.setCarriesForward(false);
119     return lt;
120 }
121
122 private LeaveEntitlement entitlement(LeaveType type, BigDecimal accrued, BigDecimal used) {
123     Employee emp = new Employee();
124     emp.setEmployeeId(1L);
125     LeaveEntitlement e = new LeaveEntitlement();
126     e.setEmployee(emp);
127     e.setLeaveType(type);
128     e.setYear(2025);
129     e.setOpeningBalance(BigDecimal.ZERO);
130     e.setAccruedThisYear(accrued);
131     e.setUsedThisYear(used);
132     e.setClosingBalance(BigDecimal.ZERO);
133     return e;
134 }
135 }

```

```

1  package com.example.EmployeeManagementSystem.service;
2
3  import com.example.EmployeeManagementSystem.Entity.Employee;
4  import com.example.EmployeeManagementSystem.Entity.LeaveEntitlement;
5  import com.example.EmployeeManagementSystem.Entity.LeaveType;
6  import com.example.EmployeeManagementSystem.Enum.Status;
7  import com.example.EmployeeManagementSystem.Repository.EmployeeRepo;
8  import com.example.EmployeeManagementSystem.Repository.LeaveEntitlementRepository;
9  import com.example.EmployeeManagementSystem.Repository.LeaveWarningRepository;
10 import com.example.EmployeeManagementSystem.Repository.YearEndCarryForwardLogRepository;
11 import com.example.EmployeeManagementSystem.Service.YearEndCarryForwardService;
12
13 import org.junit.jupiter.api.BeforeEach;
14 import org.junit.jupiter.api.Test;
15
16 import org.mockito.InjectMocks;
17 import org.mockito.Mock;
18 import org.mockito.MockitoAnnotations;
19
20 import java.math.BigDecimal;
21 import java.util.List;
22 import java.util.Optional;
23
24 import static org.mockito.ArgumentMatchers.any;
25 import static org.mockito.ArgumentMatchers.eq;
26
27 import static org.mockito.Mockito.when;
28 import static org.mockito.Mockito.verify;
29 import static org.mockito.Mockito.never;
30 import static org.mockito.Mockito.atLeastOnce;
31
32 import static org.assertj.core.api.Assertions.assertThat;
33 class YearEndCarryForwardServiceTest {
34
35     @Mock
36     EmployeeRepo employeeRepo;
37     @Mock
38     LeaveEntitlementRepository entitlementRepo;
39     @Mock
40     YearEndCarryForwardLogRepository logRepo;
41     @Mock
42     LeaveWarningRepository warningRepo;
43     @InjectMocks
44     YearEndCarryForwardService service;
45
46     @BeforeEach
47     void setup() { MockitoAnnotations.openMocks(this); }
48
49     @Test
50     void whenTotalUnderCap_noCapApplied() {
51         Employee emp = employee();
52         LeaveType casual = carryForwardType("CASUAL");
53
54         // casual: opening=0, accrued=8, used=2 closing=6
55         LeaveEntitlement ent = entitlement(emp, casual, 0, 8, 2);
56
57         when(logRepo.existsByProcessedYear(2025)).thenReturn(false);
58         when(employeeRepo.findById(Status.ACTIVE)).thenReturn(List.of(emp));
59         when(entitlementRepo.findByIdEmployeeEmployeeIdAndYear(emp.getEmployeeId(), 2025))
60             .thenReturn(List.of(ent));
61         when(entitlementRepo.findByIdEmployeeEmployeeIdAndLeaveTypeIdAndYear(any(), any(), eq(2026)))
62             .thenReturn(Optional.empty());
63         when(entitlementRepo.save(any())).thenAnswer(inv -> inv.getArgument(0));
64         when(logRepo.save(any())).thenAnswer(inv -> inv.getArgument(0));
65
66         service.runYearEnd(2025);
67
68         // closing balance = 6, no cap needed, nothing wasted
69         assertThat(ent.getClosingBalance()).isEqualToComparingTo("6");
70         verify(warningRepo, never()).save(any());
71     }

```

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```
@Test
void whenTotalExceedsCap_capAppliedProportionally() {
    Employee emp = employee();
    LeaveType casual = carryForwardType("CASUAL");
    LeaveType maternity = carryForwardType("MATERNITY");

    // casual: closing=20, maternity: closing=16 total=36, over cap by 6
    LeaveEntitlement casualEnt = entitlement(emp, casual, 0, 20, 0);
    LeaveEntitlement maternityEnt = entitlement(emp, maternity, 0, 16, 0);

    when(logRepo.existsByProcessedYear(2025)).thenReturn(false);
    when(employeeRepo.findByStatus(Status.ACTIVE)).thenReturn(List.of(emp));
    when(entitlementRepo.findByEmployeeEmployeeIdAndYear(emp.getEmployeeId(), 2025))
        .thenReturn(List.of(casualEnt, maternityEnt));
    when(entitlementRepo.findByEmployeeEmployeeIdAndLeaveTypeIdAndYear(any(), any(), eq(2026)))
        .thenReturn(Optional.empty());
    when(entitlementRepo.save(any())).thenAnswer(inv -> inv.getArgument(0));
    when(logRepo.save(any())).thenAnswer(inv -> inv.getArgument(0));

    service.runYearEnd(2025);

    // casual: 20/36 * 30 = 16.67, maternity: 16/36 * 30 = 13.33, total = 30
    BigDecimal casualCapped = casualEnt.getClosingBalance();
    BigDecimal maternityCapped = maternityEnt.getClosingBalance();
    BigDecimal total = casualCapped.add(maternityCapped);

    assertThat(total).isEqualByComparingTo("30.00");
    verify(warningRepo, atLeastOnce()).save(any()); // wasted warning issued
}

@Test
void idempotencyGuard_preventsDoubleRun() {
    when(logRepo.existsByProcessedYear(2025)).thenReturn(true);

    service.runYearEnd(2025);

    verify(employeeRepo, never()).findByStatus(any());
}

// --- helpers ---
private Employee employee() {
    Employee e = new Employee(); e.setEmployeeId(1L); return e;
}

private LeaveType carryForwardType(String name) {
    LeaveType lt = new LeaveType();
    lt.setId(name.equals("CASUAL") ? 2 : 3);
    lt.setName(name);
    lt.setCarriesForward(true);
    lt.setMonthlyReset(false);
    return lt;
}

private LeaveEntitlement entitlement(Employee emp, LeaveType type,
    int opening, int accrued, int used) {
    LeaveEntitlement e = new LeaveEntitlement();
    e.setEmployee(emp); e.setLeaveType(type); e.setYear(2025);
    e.setOpeningBalance(BigDecimal.valueOf(opening));
    e.setAccruedThisYear(BigDecimal.valueOf(accrued));
    e.setUsedThisYear(BigDecimal.valueOf(used));
    e.setClosingBalance(BigDecimal.ZERO);
    return e;
}
}
```